Analysis II

Several Variables — Exercise Class Notes & Transcript

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PART I

Foundations

Analysis II begins where **Analysis I** stopped, and the first thing to settle is how much of it carries over. The definitions of convergence, of continuity, and of a set being open survive the move to several variables essentially verbatim. What does not survive is every statement that quietly used the ordering of $\mathbb{R}$, and each of those has to be rebuilt on a new hypothesis. This part draws that line, and then rehearses the one skill every later chapter assumes without ever teaching it: turning a set-builder description into a picture, and a picture back into equations. The three families of level sets used for that turn out to be the unit balls of the three standard metrics, so the drawing done here is the metric-space theory of the next part in disguise. Young's inequality is proved along the way, together with the Hölder inequality that follows from it in three lines, because both are wanted as soon as p-norms appear.

Prerequisites and a First Look at $\mathbb{R}^n$ (Chapter 1)

This chapter collects the **Analysis I** material the course takes for granted, together with the $\mathbb{R}^n$ warm-up that precedes any of its theory. Two things are worth settling before the theory starts. The first is which definitions survive the passage to several variables unchanged and which statements do not, since the two look alike and behave very differently. The second is fluency at turning a set-builder description into a picture and back again, a skill every later chapter assumes without ever teaching. Young's inequality (Proposition 1.b.3) is proved here as well, slightly out of sequence: it belongs to no chapter in particular, and it is wanted as soon as p-norms appear.

From Analysis I to several variables (Section 1.a)

Several of the definitions you will meet in the coming chapters are exactly the definitions you already know from **Analysis I**, only stripped of their reliance on the real line. A "*sequence*" (German: „*Folge*“) $(x_k)_{k\in\mathbb{N}}$ in a set X converging to $x \in X$, a function being continuous at a point, a set being open or closed: all of these notions carry over to $\mathbb{R}^n$ and, as Section 2.a will show, to spaces far more general than $\mathbb{R}^n$.

What does **not** carry over automatically is anything that relied on the **order** of $\mathbb{R}$: there is no "$\leq$" on $\mathbb{R}^n$ for $n \geq 2$, so statements like the intermediate value theorem or *"every bounded sequence has a convergent subsequence"* need new proofs, or new hypotheses, once we leave $\mathbb{R}$.

Remark 1.a.1: The second of those statements is worth reading carefully, because one word carries it. Bounded does not imply convergent (Exercise 1.a.2**(a)** below has a counterexample). What is true is that a bounded sequence has a convergent **subsequence**, which is the Bolzano–Weierstrass theorem. In $\mathbb{R}^n$ it survives, by extracting a convergent subsequence one coordinate at a time. In a general metric space it fails, and the property that replaces it is compactness, introduced in Section 7.a.

Exercise 1.a.2 (*Recap Questions*):

(a) Let $\{x_k\}_{k\in\mathbb{N}} \subset [1,2] \subset \mathbb{R}$ be a sequence. Is it necessarily convergent?

(b) Describe all differentiable functions $f : \mathbb{R}\setminus\{0\} \to \mathbb{R}$ such that $f'(x) = 1/x$ for all $x \neq 0$.

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and bounded. Is it true that

$$\int_0^{\pi/2} f(\cos(x))\,dx = \int_0^1 \frac{f(t)}{\sqrt{1-t^2}}\,dt\,?$$

(d) Let $f : [0,\infty) \to \mathbb{R}$ have $f''(x) > 0$ for all $x \geq 0$. Can f have a global maximum point?

(e) Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map and let $Q := [0,1] \times [0,1]$ be the unit square. How does $\phi(Q)$ look like? What are the possibilities?

(f) Consider the vector space V of polynomials of degree less than 4, and the function $D : V \to V$ which associates to a polynomial p its derivative p'. Is D a linear map? If so, what is the dimension of its null space?

Info: *This exercise is Problem 1.1 of Problem Sheet 1, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 13*

Young's inequality (Section 1.b)

The following inequality is not part of the lecture's main line of development, but it is the cleanest route to comparing different norms on $\mathbb{R}^n$ (see Section 2.c) and to the Hölder-type estimates that

appear whenever two L^p-type quantities are played off against each other. It is worth proving once and citing from then on.

Proposition 1.b.3 (*Young's inequality*): Let $a, b > 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Remark 1.b.4: The case $p = q = 2$ recovers the familiar inequality $ab \le \frac{a^2}{2} + \frac{b^2}{2}$, which is just a rewriting of $(a-b)^2 \ge 0$.

Proof:

Fix conjugate exponents $p, q > 1$, and define $g : (0, \infty) \to \mathbb{R}$ by $g(t) := \frac{t}{p} + \frac{1}{q} - t^{1/p}$. We first show that $g \ge 0$, with equality only at $t = 1$. Differentiating,

$$g'(t) = \frac{1}{p}\left(1 - t^{-(p-1)/p}\right).$$

Since $p > 1$, the exponent $-(p-1)/p$ is negative, so $t^{-(p-1)/p} > 1$ for $t \in (0, 1)$ and $t^{-(p-1)/p} < 1$ for $t > 1$. Hence $g' < 0$ on $(0, 1)$ and $g' > 0$ on $(1, \infty)$, so g attains a global minimum at $t = 1$. As $g(1) = \frac{1}{p} + \frac{1}{q} - 1 = 0$, we obtain

$$t^{1/p} \le \frac{t}{p} + \frac{1}{q} \qquad \text{for all } t > 0. \tag{1.1}$$

Now fix $a, b > 0$ and apply (1.1) with $t := a^p / b^q$. Multiplying both sides by $b^q > 0$ gives

$$\left(\frac{a^p}{b^q}\right)^{1/p} b^q \;\le\; \frac{a^p}{p} + \frac{b^q}{q}.$$

It remains to identify the left-hand side, and this is where the conjugacy of p and q enters. Using $1 - \frac{1}{p} = \frac{1}{q}$,

$$\left(\frac{a^p}{b^q}\right)^{1/p} b^q \;=\; a\, b^{-q/p}\, b^q \;=\; a\, b^{q\left(1-\frac{1}{p}\right)} \;=\; a\, b^{q \cdot \frac{1}{q}} \;=\; ab,$$

which is exactly the asserted inequality. Q.E.D.

Remark 1.b.5 (*From Young to Hölder*): The exercise sheet also asks for the vector version

$$\langle a, b \rangle \le \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q}, \qquad a, b \in \mathbb{R}^n, \tag{1.2}$$

where $\|x\|_p := \left(\sum_i |x_i|^p\right)^{1/p}$. This is **not** yet Hölder's inequality: it is the scalar case applied coordinatewise and summed (see the solution of Exercise 1.b.6 below), and the right-hand side is a sum of p-th and q-th powers rather than the product $\|a\|_p \|b\|_q$. The step from one to the other is a normalisation, and it is worth seeing once, because the same trick recurs whenever an inequality is homogeneous.

Assume $a, b \neq 0$ and apply (1.2) to the rescaled vectors $a/\|a\|_p$ and $b/\|b\|_q$, each of which has p- resp. q-norm equal to 1:

$$\frac{\langle a, b \rangle}{\|a\|_p \, \|b\|_q} \;\le\; \frac{1}{p} + \frac{1}{q} \;=\; 1,$$

which is exactly "*Hölder's inequality*" (German: „*Höldersche Ungleichung*") $\langle a, b \rangle \le \|a\|_p \|b\|_q$. The conjugacy condition $\frac{1}{p} + \frac{1}{q} = 1$, which so far looked like an arbitrary bookkeeping device, is what makes the two constants add up to exactly 1 here.

This is also a first hint of why p-norms other than $p = 2$ are worth naming: Section 2.c only treats $p = q = 2$, where Hölder's inequality is Cauchy–Schwarz, but the same machinery bounds $\langle a, b \rangle$ for every conjugate pair (p, q).

Exercise 1.b.6 (*Proving Young's inequality, and its vector form*)**:** Let $a, b > 0$, and let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Prove that $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$.

(a) Prove that for all $t > 0$, $t^{1/p} \leq \frac{t}{p} + \frac{1}{q}$.

(b) Multiply the inequality by b^q and choose an appropriate value of t to conclude.

A bonus: show that $\langle a, b\rangle \leq \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q}$ for all $a, b \in \mathbb{R}^n$.

Info: *This exercise is Problem 1.4 of Problem Sheet 1, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 16*

Visualizing sets in $\mathbb{R}^2$ and $\mathbb{R}^3$ (Section 1.c)

Much of the geometric intuition for this course (open sets, level sets, submanifolds) rests on a skill that has nothing to do with any theorem: reading a set-builder description and producing an accurate picture, or going the other way. Two techniques are worth naming explicitly, both used repeatedly in the solutions below.

- **Slicing.** To sketch a set $S \subseteq \mathbb{R}^3$ defined by an equation involving z, fix $z = t$ and sketch the resulting planar slice $S \cap \{z = t\}$ as t varies. Stacking the slices gives the full picture. Two questions settle most cases: for which t is the slice non-empty, and how does its size depend on t?
- **Level sets.** If a set is defined by $f(x, y) = t$ for a fixed t, first work out the level sets $\{(x, y) : f(x, y) = c\}$ of f for varying c. Many natural functions have level sets one can simply name: those of $f(x, y) = \sqrt{x^2 + y^2}$ are circles, those of $f(x, y) = \max\{|x|, |y|\}$ are squares, and those of $f(x, y) = |x| + |y|$ are diamonds. Naming them turns a three-dimensional sketch into a stack of familiar plane figures.

Those last three functions are not a random selection. They are the Euclidean, supremum and Manhattan norms of (x, y), and their level sets at height 1 are the boundaries of the three unit balls drawn in Section 3.c. It is worth carrying the picture forward: a great deal of what looks like metric-space abstraction in the next two chapters is this one diagram in disguise.

Both techniques are applied to Exercises 1.c.7 and 1.c.8 below, and worked out in Section 1.d.

Exercise 1.c.7 (*From equations to drawings*)**:** Sketch the following subsets of $\mathbb{R}^2$, $\mathbb{R}^3$.

(a) $A := \{(x, y) : x^2 + y^2 < 1,\ y \geq x^2\}$

(b) $B := \{(x, y) : y^2 = x^3 - x\}$

(c) $C := \{(x, y) : \frac{\pi}{6} < \arctan(y/x) < \frac{\pi}{3},\ 0 < x < y\}$

(d) $D := \{(x, y, z) : \sqrt{x^2 + y^2} = 1 - z,\ 0 < z < 1\}$

(e) $E := \{(x, y, z) : z = \sqrt{1 + x^2 + y^2}\}$

(f) $F := \{(x, y, z) : z = \max\{|x|, |y|\}\}$

Info: *This exercise is Problem 1.2 of Problem Sheet 1, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 14*

Exercise 1.c.8 (*From drawings to equations*)**:** Express in terms of functions and equations the following sets.

(a) The coordinates of a point in the plane that goes endlessly back and forth from the origin to the point $(-1, 1)$.

(b) A planar domain with two holes in it.

(c) A bi-dimensional disk sitting inside the plane $\{x + y + z = 0\}$ in $\mathbb{R}^3$.

(d) A Pac-Man shaped domain in the plane.

(e) A pyramid, whose base is a square, sitting in $\mathbb{R}^3$.

Info: *This exercise is Problem 1.3 of Problem Sheet 1, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 15*

Solutions (Section 1.d)

Solution of Exercise 1.a.2:

(a) No. The sequence $x_k := \frac{3}{2} + \frac{1}{2}(-1)^k$ lies in $[1,2]$ but oscillates between 1 and 2 without settling, so it does not converge (it has two distinct accumulation points, 1 and 2). What **is** true, by the Bolzano–Weierstrass theorem, is that every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

(b) On each of the intervals $(-\infty, 0)$ and $(0, \infty)$ separately, $f'(x) = 1/x$ forces $f(x) = \log|x| + c$ for some constant c, since two antiderivatives on a common interval differ by a constant. The subtlety is that $\mathbb{R} \setminus \{0\}$ is **disconnected** into these two intervals, so the two constants need not agree:

$$f(x) = \begin{cases} \alpha + \log(x) & x > 0, \\ \beta + \log(-x) & x < 0, \end{cases} \qquad \alpha, \beta \in \mathbb{R} \text{ arbitrary.}$$

The answer $f(x) = \log|x| + c$ for a **single** constant c is wrong; connectedness of the domain is exactly what would be needed to force $\alpha = \beta$, and it is precisely what Section 8.a formalizes.

(c) Yes. Substituting $t = \cos(x)$, $dt = -\sin(x)\,dx$, and $\sin(x) = \sqrt{1-t^2}$ for $x \in [0, \pi/2]$ turns the left integral into the right one; this is the one-variable change of variables formula, which the change-of-variables theorem (Section 20.a) generalizes to $\mathbb{R}^n$.

(d) Yes: take $f(x) := e^{-x}$. Then $f''(x) = e^{-x} > 0$ for all x, yet f attains its global maximum on $[0, \infty)$ at the boundary point $x = 0$. Everywhere-positive second derivative (strict convexity) rules out an **interior** local maximum, but says nothing whatever about the boundary of the domain.

(e) $\phi(Q)$ is the parallelogram spanned by $\phi(1,0)$ and $\phi(0,1)$, with vertices at 0, $\phi(1,0)$, $\phi(0,1)$, and $\phi(1,0)+\phi(0,1)$. Two degenerate cases occur: if $\phi(1,0)$ and $\phi(0,1)$ are linearly dependent (parallel or one is zero) but not both zero, the parallelogram collapses to a line segment; if $\phi \equiv 0$, it collapses to the single point $\{0\}$.

(f) Yes: differentiation is linear on functions in general, hence in particular on the subspace of polynomials of degree < 4. With respect to the basis $\mathcal{B} = \{1, X, X^2, X^3\}$, the map D sends $1 \mapsto 0$, $X \mapsto 1$, $X^2 \mapsto 2X$, $X^3 \mapsto 3X^2$, giving matrix

$$[D]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This matrix has rank 3 (its three nonzero rows are independent), so by the rank–nullity theorem the null space has dimension $4-3 = 1$, and it is spanned by the constant polynomials.

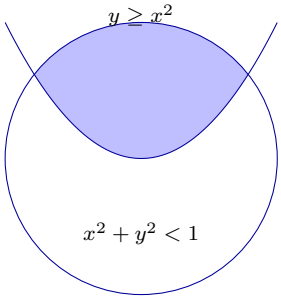
Q.E.D.

Remark 1.d.9 (*The two parts most often answered wrongly*): Two of the parts above account for most of the wrong answers to this sheet. Part **(b)** is one: the single-constant answer $f(x) = \log|x| + c$ is the usual first guess, and it fails for the reason just given, namely that the domain falls apart into two intervals. Part **(d)** is the other: a function with everywhere positive second derivative is easily assumed to admit no global maximum at all, when in fact it may attain one at a boundary

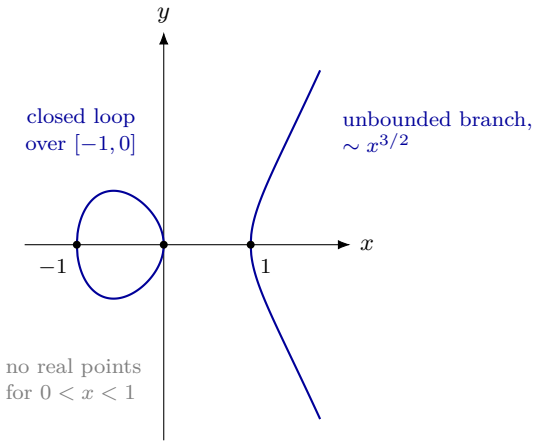
point. Both mistakes have the same shape. A hypothesis that controls the **interior** of a domain was read as though it controlled the domain.

Solution of Exercise 1.c.7:

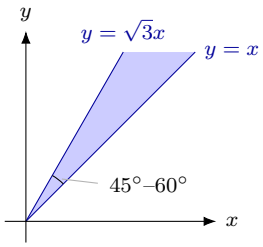
(a) A is the intersection of the open unit disk with the region on or above the parabola $y = x^2$, a lens-shaped region:



(b) Since $x^3 - x = x(x+1)(x-1) \geq 0$ exactly for $x \in [-1, 0] \cup [1, \infty)$, the curve $y = \pm\sqrt{x^3 - x}$ is only real on these two intervals; B is the union of the graphs of $f(x) := \sqrt{x^3 - x}$ and $-f(x)$, giving a closed loop over $[-1, 0]$ and an unbounded branch over $[1, \infty)$ growing like $x^{3/2}$. Note that the two branches meet exactly where $f = 0$, i.e. at the three roots $x = -1, 0, 1$, which is why the loop closes up and the right branch starts with a vertical tangent.



(c) Since $0 < x < y$, points of C lie in the first quadrant with $y > x$; the condition $\pi/6 < \arctan(y/x) < \pi/3$ says the angle to the positive x-axis is between $30°$ and $60°$. So C is the open angular sector between the lines $y = x$ and $y = \sqrt{3}\,x$:

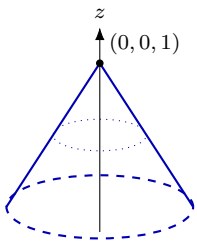


(d) Slicing at height $z = t \in (0, 1)$: the equation $\sqrt{x^2 + y^2} = 1 - t$ describes a circle of radius $1 - t$ centered on the z-axis. As t ranges over $(0, 1)$, the radius shrinks linearly from 1 to 0, so D is the lateral surface of a right circular cone of height 1 over the unit circle, apex at $(0, 0, 1)$, base circle excluded (the inequalities are strict).

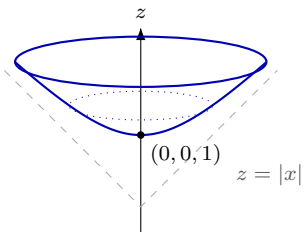
(e) Slicing at height $z = t$: the equation $\sqrt{1+x^2+y^2} = t$ has solutions only for $t \geq 1$, and the slice is then a circle of radius $\sqrt{t^2-1}$. Note that the radius is $\sqrt{t^2-1}$ and not t; this is the single most common slip on this part, and it turns the answer into a cone. E is instead a bowl-shaped surface of revolution with vertex at $(0,0,1)$, obtained by rotating the curve $z = \sqrt{1+x^2}$ of the xz-plane about the z-axis. It approaches slope 1 for large x, y without ever attaining it, which is exactly how it differs from the cone $z = \sqrt{x^2+y^2}$ that it is so often mistaken for.

(f) The level sets $\{\max\{|x|,|y|\} = t\}$ are the boundaries of the squares $[-t,t]^2$. Stacking these square outlines at height $z = t$ as t grows from 0 upward sweeps out four planar faces, unbounded wedges lying in the planes $z = \pm x$ and $z = \pm y$, meeting at the apex $(0,0,0)$. So F is an upward-opening **square cone**: the exact analogue of the circular cone of part **(d)**, with the round level circles of $\sqrt{x^2+y^2}$ replaced by the square level curves of $\max\{|x|,|y|\}$.

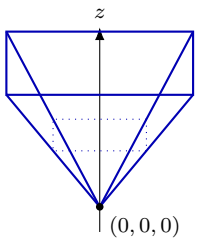
Parts **(d)**, **(e)** and **(f)** are the same construction carried out three times, stacking level sets, and are best compared side by side. In each panel the dotted curve is one intermediate level set:



(d) $\sqrt{x^2+y^2} = 1-z$
radius shrinks $1 \to 0$



(e) $z = \sqrt{1+x^2+y^2}$
radius $\sqrt{z^2-1}$, **not** z



(f) $z = \max\{|x|,|y|\}$
squares instead of circles

Q.E.D.

Solution of Exercise 1.c.8:

(a) A point moving back and forth along the segment from $(0,0)$ to $(-1,1)$ can be parametrized as $t \mapsto (-f(t), f(t))$ for $t \geq 0$, where $f : [0,\infty) \to [0,1]$ oscillates endlessly; $f(t) := \frac{1}{2}(1+\sin t)$ is one valid choice.

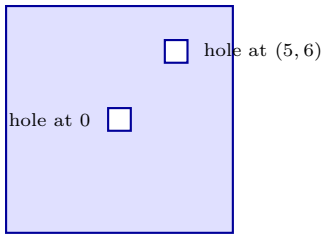
(b) Remove two small disjoint squares from a large square, e.g. $\{10 > \max\{|x|,|y|\} > 1\} \setminus \{\max\{|x-5|, |y-6|\} \leq 1\}$.

(c) Intersect the closed unit ball with the plane: $\{x^2+y^2+z^2 \leq 1\} \cap \{x+y+z = 0\}$.

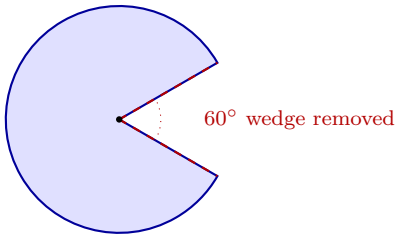
(d) Remove a $60°$ wedge from a disk: $\{x^2+y^2 \leq 1\} \setminus \{|\arctan(y/x)| \leq \pi/6,\ x > 0\}$.

(e) Combining the square level sets from Exercise 1.c.7 part **(f)** with the conical slicing from part **(d)**: $\{1 - z \geq \max\{|x|,|y|\},\ 0 \leq z \leq 1\}$ is the solid pyramid of height 1 over the square $[-1,1]^2$, apex at $(0,0,1)$. Writing an equality instead, $\{1 - z = \max\{|x|,|y|\},\ 0 \leq z \leq 1\}$, keeps only the lateral surface: the four triangular faces, with the base square excluded.

The two planar answers are worth drawing, since in both cases the description *"remove a piece"* translates directly into a set difference:



2. a domain with two holes: $\{10 > \max\{|x|, |y|\} > 1\}$ minus a unit square at $(5, 6)$



4. Pac-Man: the closed unit disk minus the wedge $\{|\arctan(y/x)| \le \pi/6,\ x > 0\}$

Q.E.D.

Solution of Exercise 1.b.6:

Parts **(a)** and **(b)** are exactly the proof of Proposition 1.b.3 above. For the bonus vector version, apply the scalar inequality coordinatewise, with $|a_i|$ in place of a and $|b_i|$ in place of b, and sum over i:

$$\begin{aligned}\langle a, b\rangle \ &\le\ \sum_{i=1}^{n} |a_i|\,|b_i| \ \le\ \sum_{i=1}^{n} \left(\frac{|a_i|^p}{p} + \frac{|b_i|^q}{q}\right) \\ &= \ \frac{1}{p}\sum_{i=1}^{n} |a_i|^p + \frac{1}{q}\sum_{i=1}^{n} |b_i|^q \ = \ \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q}.\end{aligned}$$

Note that the two sums separate only because the right-hand side of Young's inequality is a **sum** of powers rather than a product; this is precisely the feature that Remark 1.b.5 then removes by rescaling.

Q.E.D.

PART II

Metric Spaces and Topology

Very little of one-variable analysis actually used the arithmetic of $\mathbb{R}$. Convergence, continuity and compactness need only a notion of distance, and this part takes that observation seriously by developing all of them for an arbitrary metric space. The payoff is not generality for its own sake. Function spaces such as $C^0([0,1],\mathbb{R})$ are metric spaces too, so the same theorems apply to a sequence of functions and to a sequence of points alike, and the last compactness criterion proved here is one for a family of functions rather than for a set of points. Three properties then carry the remaining five parts. Completeness manufactures limits nobody can write down, and through the Banach fixed point theorem it becomes the engine of both the inverse function theorem and the existence theory for differential equations. Compactness replaces an infinite problem by a finite one, which is what guarantees that a continuous function on a closed bounded set attains its maximum at all. Connectedness forbids a space from falling into pieces, and its refinement to simple connectedness decides, five parts later, whether a curl-free field has a potential.

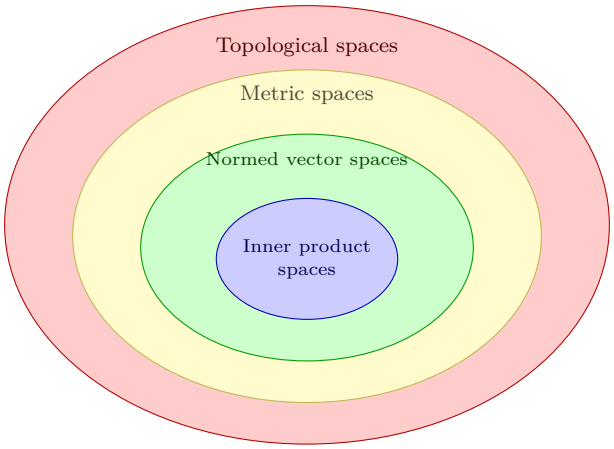
Metric, Normed and Inner-Product Spaces (Chapter 2)

How much structure does a set need before one can do analysis on it? This chapter answers that by taking $\mathbb{R}^n$ apart into layers. Distance comes first, and distance alone already carries convergence and continuity. Adding a linear structure gives a norm, which measures the length of one vector rather than the gap between two. Adding an inner product on top of that gives angles, and with them orthogonal projection. The layers nest, each one strictly inside the last, and the inequality that makes the innermost layer work at all is Cauchy–Schwarz. A last section settles the question all of this raises: on $\mathbb{R}^n$ the choice of norm turns out not to matter for anything topological, and it matters a great deal that this is a theorem rather than an assumption.

Structured spaces (Section 2.a)

This semester, you will be introduced to certain structures that can be given to a set, which in this context is also called a "*space*" (German: „*Raum*“) and which we will denote by X:

- **0. Topological spaces:** fourth semester.
- **1. Metric spaces** $(X,\ d : X \times X \to \mathbb{R}_{\geq 0})$
  - Defines a **distance** between points.
  - X is **not** necessarily a vector space.
  - Covered in *Analysis*.
- **2. Normed metric spaces** $(X,\ \|\cdot\| : X \to \mathbb{R}_{\geq 0})$
  - Defines the **length** of a vector.
  - Presupposes a linear structure: X must be a **vector space**.
  - Covered in *Linalg / Analysis*.
- **3. Inner product spaces** $(X,\ \langle\cdot,\cdot\rangle : X \times X \to \mathbb{R})$
  - Defines **both** lengths **and** angles!
  - Super important!
  - Covered in *Linalg* (and sometimes used in *Analysis*). X is a **vector space**.



You will see that an **inner product** $\langle\cdot,\cdot\rangle$ induces a **norm** by the formula

$$\|x\| := \sqrt{\langle x, x\rangle}, \qquad x \in X.$$

Furthermore, a **norm** induces a **metric** by

$$d(x, y) := \|x - y\|, \qquad x, y \in X.$$

Definition 2.a.1 (*Inner product / scalar product*): Let V be a **vector space** over $\mathbb{R}$. An "*inner product*" (German: „*Skalarprodukt*“) on V is a map

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$$

satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

- **(a) Bilinearity:** $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$, and equivalently in the second variable.
- **(b) Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$.
- **(c) Definiteness:** $\langle u, u \rangle \geq 0$, and $\langle u, u \rangle = 0 \iff u = 0$.

Definition 2.a.2 (*Norm*): Let V be a **real** vector space. A "*norm*" (German: „*Norm*“) on V is a map $\| \cdot \| : V \to [0, \infty)$ satisfying, for all $u, v \in V$ and $\alpha \in \mathbb{R}$:

- **(a) Definiteness:** $\|v\| \geq 0$, and $\|v\| = 0 \iff v = 0$.
- **(b) Homogeneity:** $\|\alpha v\| = |\alpha| \|v\|$.
- **(c) Δ-inequality:** $\|u + v\| \leq \|u\| + \|v\|$.

AI-Note 2.1 (*Which bars mean which norm*): From p. 26 onward the lecture notes fix a notational convention that is worth stating once, because it is used everywhere afterwards and never repeated: single bars $|x|$ denote the **Euclidean** norm of $x \in \mathbb{R}^n$, and the double bars $\| \cdot \|$ are reserved for every **other** norm. This document follows that convention. Concretely:

- $|x|$ is the Euclidean norm on $\mathbb{R}^n$. Nothing is lost by overloading the absolute-value bars, since for $n = 1$ the Euclidean norm **is** the absolute value, and for $n = 2$ read through $\mathbb{R}^2 \cong \mathbb{C}$ it is the complex modulus. All three agree wherever two of them are defined.
- $\| \cdot \|$ is a norm that is not the Euclidean one, or a norm on a vector space that carries no Euclidean structure to begin with: the p-norms $\| \cdot \|_p$, the supremum norm $\| \cdot \|_\infty$, the Hilbert–Schmidt norm of Example 2.a.3, the operator norm $\| \cdot \|_{\mathrm{op}}$, and the abstract $\| \cdot \|$ of Definition 2.a.2.

The present chapter is the one place where double bars also cover the Euclidean case, and for the same reason: everything here is proved for an arbitrary normed space, so writing $|x|$ would assert a particular norm that is not meant.

Example 2.a.3 (*Hilbert–Schmidt norm on matrix spaces*): A very important example of a norm on the space of real $n \times m$ matrices $\mathbb{R}^{n \times m}$ is the "*Hilbert–Schmidt norm*" (also known as the Frobenius norm). It is defined by

$$\|M\|_2 := \sqrt{\mathrm{Tr}(M^{\mathsf{T}} M)},$$

where Tr denotes the trace of a matrix. Only the diagonal of $M^{\mathsf{T}} M$ is needed, so the off-diagonal entries are left unevaluated below. For a 2×3 matrix $M = \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right)$, we have

$$M^{\mathsf{T}} M = \begin{pmatrix} a & d \\ b & e \\ c & f \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} = \begin{pmatrix} a^2 + d^2 & * & * \\ * & b^2 + e^2 & * \\ * & * & c^2 + f^2 \end{pmatrix}.$$

The trace is the sum of the diagonal entries, so $\|M\|_2 = \sqrt{a^2 + b^2 + c^2 + d^2 + e^2 + f^2}$. This norm is therefore nothing but the standard Euclidean norm of the matrix entries read as a single vector in $\mathbb{R}^6$, or in general in $\mathbb{R}^{nm}$. The inner product inducing it is $\langle A, B \rangle := \mathrm{Tr}(A^{\mathsf{T}} B)$.

Lemma 2.a.4 (*The Hilbert–Schmidt norm bounds lengths*): For every linear map $M : \mathbb{R}^n \to \mathbb{R}^m$,

$$|My| \leq \|M\|_2 \, |y| \qquad \text{for all } y \in \mathbb{R}^n.$$

Proof:

Write $y = \sum_{i=1}^{n} y_i e_i$ in the standard basis. By linearity and the triangle inequality,

$$|My| = \Big| \sum_{i=1}^{n} y_i \, Me_i \Big| \le \sum_{i=1}^{n} |y_i| \, |Me_i|,$$

and Cauchy–Schwarz applied to the two vectors $(|y_i|)_i$ and $(|Me_i|)_i$ of $\mathbb{R}^n$ bounds the right-hand side by

$$\Big(\sum_{i=1}^{n} |y_i|^2 \Big)^{1/2} \Big(\sum_{i=1}^{n} |Me_i|^2 \Big)^{1/2} = |y| \, \|M\|_2,$$

the last equality being the definition of the Hilbert–Schmidt norm read column by column. Q.E.D.

Remark 2.a.5 (*What this is for*): This one inequality is the reason the Hilbert–Schmidt norm is worth introducing this early. It is what turns a statement about a **matrix** into a statement about **lengths**, and it is the step behind every later estimate of the form *"the image of a short vector is short"*: the proof that a C^1 function with bounded derivative is Lipschitz (Remark 11.a.4), the Lipschitz perturbation lemma driving the inverse function theorem (Lemma 15.a.3), and the bound that makes a C^1 map send null sets to null sets in Chapter 19. Note that $\|M\|_2$ is not the smallest constant that works, since the operator norm is, but it is the one that can be read straight off the entries, which is usually what one wants.

The two lists of axioms just given already line up with the metric axioms of the next section (Definition 2.b.10). A norm's definiteness and triangle inequality are the metric's, restated for a single vector instead of a pair of points. The metric's symmetry is the special case $\alpha = -1$ of homogeneity, and homogeneity for other α has no metric analogue at all, since stating it requires scalar multiplication. An inner product goes further and replaces the triangle inequality by bilinearity, which is far stronger, and then recovers the triangle inequality as a **theorem** (Proposition 2.a.8) rather than assuming it. That is the precise sense in which each layer carries more structure than the one outside it.

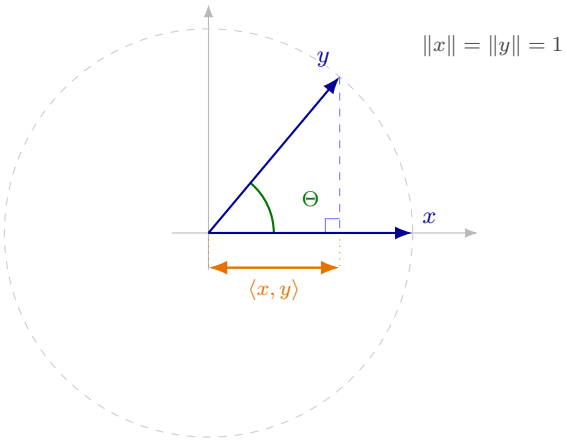
Remark 2.a.6 (*Geometric angle and projection in inner product spaces*): In an inner product space $(X, \langle \cdot, \cdot \rangle)$ over $\mathbb{R}$, the Cauchy–Schwarz inequality $|\langle x, y \rangle| \le \|x\| \, \|y\|$ guarantees that for any non-zero vectors $x, y \in X$,

$$-1 \le \frac{\langle x, y \rangle}{\|x\| \, \|y\|} \le 1.$$

This allows us to formally define the "*angle*" $\Theta(x, y) \in [0, \pi]$ between x and y via

$$\Theta(x, y) := \arccos \left(\frac{\langle x, y \rangle}{\|x\| \, \|y\|} \right).$$

Geometrically, when $\|x\| = \|y\| = 1$, the inner product $\langle x, y \rangle$ is the signed length of the orthogonal projection of y onto the direction of x. In that reading Cauchy–Schwarz becomes visually obvious: a projection of a unit vector cannot be longer than the unit vector itself, so $\langle x, y \rangle \le 1$, with equality exactly when y already points along x.



Proposition 2.a.7 (*Geometric derivation of the Cauchy–Schwarz inequality*): Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space. Then for all $v, w \in V$,

$$|\langle v, w\rangle| \le \|v\|\,\|w\|.$$

Proof:

If $w = 0$, both sides vanish and the inequality holds trivially. For $w \neq 0$, consider the squared distance from v to the points of the line $L := \operatorname{span}_{\mathbb{R}}\{w\}$, parametrized by $\lambda \mapsto \lambda w$:

$$f(\lambda) := \|v - \lambda w\|^2 = \langle v - \lambda w, v - \lambda w\rangle = \|v\|^2 - 2\lambda\langle v, w\rangle + \lambda^2\|w\|^2.$$

By definiteness $f(\lambda) \ge 0$ for all $\lambda \in \mathbb{R}$, and f is a quadratic polynomial in λ with positive leading coefficient $\|w\|^2 > 0$. Differentiating or completing the square shows that f attains its global minimum at $\lambda_0 := \frac{\langle v,w\rangle}{\|w\|^2}$. Evaluating f at λ_0 yields

$$0 \le f(\lambda_0) = \|v\|^2 - 2\frac{\langle v, w\rangle^2}{\|w\|^2} + \frac{\langle v, w\rangle^2}{\|w\|^2} = \|v\|^2 - \frac{\langle v, w\rangle^2}{\|w\|^2}.$$

Multiplying by $\|w\|^2$ and taking square roots gives $|\langle v, w\rangle| \le \|v\|\,\|w\|$. Geometrically, $\pi_L(v) := \lambda_0 w = \frac{\langle v,w\rangle}{\|w\|^2} w$ is the unique orthogonal projection of v onto the line L, achieving the minimal distance $d(v, L) = \sqrt{f(\lambda_0)}$. Q.E.D.

Proposition 2.a.8 (*Inner product ⇒ norm ⇒ metric*): Let V be a real vector space.

(a) If $\langle\cdot,\cdot\rangle$ is an inner product on V (Definition 2.a.1), then $\|v\| := \sqrt{\langle v, v\rangle}$ is a norm on V (Definition 2.a.2).

(b) If $\|\cdot\|$ is a norm on V, then $d(x, y) := \|x - y\|$ is a metric on V (Definition 2.b.10).

Proof:

(a) Definiteness of the norm is definiteness of the inner product, and homogeneity follows from bilinearity: $\|\alpha v\|^2 = \langle \alpha v, \alpha v\rangle = \alpha^2\|v\|^2$. For the triangle inequality, expand and apply Proposition 2.a.7:

$$\begin{aligned}\|u + v\|^2 &= \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \\ &\le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \\ &\le \|u\|^2 + 2\|u\|\,\|v\| + \|v\|^2 \;=\; \big(\|u\| + \|v\|\big)^2,\end{aligned}$$

and taking square roots gives $\|u + v\| \le \|u\| + \|v\|$.

(b) Definiteness: $d(x, y) = 0$ means $\|x - y\| = 0$, hence $x = y$. Symmetry: homogeneity with $\alpha = -1$ gives $\|x - y\| = \|-(y - x)\| = \|y - x\|$. Triangle inequality: writing $x - z = (x - y) + (y - z)$,

$$d(x, z) = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\| = d(x, y) + d(y, z).$$ Q.E.D.

Remark 2.a.9 (*Neither arrow reverses*): The nesting drawn above is strict at both steps.

- **Not every metric comes from a norm.** The discrete metric (Example 2.b.13**(a)**) on a vector space cannot: a norm-induced metric satisfies $d(\alpha x, 0) = |\alpha|\, d(x, 0)$, which fails as soon as $\alpha \neq 0, \pm 1$.
- **Not every norm comes from an inner product.** A norm induced by an inner product obeys the "*parallelogram law*" $\|u+v\|^2+\|u-v\|^2 = 2\|u\|^2+2\|v\|^2$ (expand both sides by bilinearity). The supremum norm on $\mathbb{R}^2$ violates it: for $u := (1, 0)$ and $v := (0, 1)$ the left side is $1+1 = 2$ and the right side is $2 + 2 = 4$.

Consequently, each layer of the hierarchy really does carry strictly more structure than the one outside it. The picture at the start of this section is a genuine nesting of four distinct classes, not a chain of equivalent descriptions of the same thing.

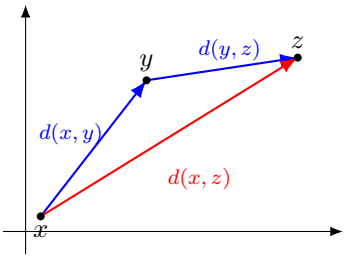
Metric spaces (Section 2.b)

Definition 2.b.10 (*Metric space*): A "*metric space*" (German: „*metrischer Raum*“) (X, d) is any set X with a "*metric*" (German: „*Metrik*“) $d : X \times X \to \mathbb{R}_{\geq 0}$ such that for all $x, y, z \in X$:

(a) Definiteness: $d(x, y) \geq 0$, with equality if and only if $x = y$. In words: the distance is non-negative, and the only point at distance zero from x is x itself.

(b) Symmetry: $d(x, y) = d(y, x)$, that is, x has the same distance to y as y has to x.

(c) Triangle inequality (Δ-inequality): $d(x, z) \leq d(x, y) + d(y, z)$. In words, *"detours make the way longer"*.



$d(x, z) \leq d(x, y) + d(y, z)$

Exercise 2.b.11 (*Polygon inequality*): Let (X, d) be a metric space and $N \geq 3$. Prove, by induction on N, that

$$d(x_1, x_N) \leq \sum_{i=1}^{N-1} d(x_i, x_{i+1})$$

for all $x_1, \ldots, x_N \in X$.

Solution: *See solution on Page 36*

AI-Example 2.b.12 (*Each axiom rules out something, and none is redundant*): Three functions on $\mathbb{R}$, each failing exactly one axiom of Definition 2.b.10. It is worth checking these once: a definition only becomes memorable once you know what it is keeping out.

(a) Definiteness fails: $d(x, y) := |x^2 - y^2|$. This is symmetric and satisfies the triangle inequality (it is $|\cdot|$ composed with $x \mapsto x^2$), but $d(-1, 1) = 0$ while $-1 \neq 1$. Distinct points at distance zero: the space cannot tell -1 from 1, and no sequence could ever have a unique limit.

(b) Symmetry fails: $d(x, y) := \max\{\, y - x,\ 2(x - y)\,\}$, i.e. moving to the right costs 1 per unit and moving to the left costs 2. Here $d(0, 1) = 1$ but $d(1, 0) = 2$, so the return trip is more expensive than the outward one. Definiteness still holds, since $d(x, y) = 0$ forces both $y - x \leq 0$

and $x - y \leq 0$, hence $x = y$. So does the triangle inequality: writing $d(x,y) = \varphi(y-x)$ for $\varphi(t) := \max\{t, -2t\}$, the function φ is a maximum of two linear functions vanishing at 0, hence convex and positively homogeneous, and therefore subadditive. Such a d is called a "*quasi-metric*" and is genuinely useful elsewhere, though note that $B_r(x)$ would then depend on which argument the centre sits in.

(c) Triangle inequality fails: $d(x,y) := (x-y)^2$. Definiteness and symmetry are clear, but

$$d(0,2) = 4 \quad > \quad d(0,1) + d(1,2) = 1 + 1 = 2,$$

so the detour through 1 is **shorter** than going direct. This is exactly the *"detours make the way longer"* slogan running backwards.

Examples (Subsection 2.b.1)

Three axioms are few enough that a great many things qualify as metrics, and the point of the list below is exactly that breadth. Two of the examples are not what one would draw on a blackboard as *"distance"*: the discrete metric measures nothing but *"same or different"*, and the supremum metric measures the distance between two **functions**. Both are legitimate, and it is worth letting them stretch the intuition now. The second in particular is the setting in which uniform convergence turns out to be nothing more than an ordinary instance of Definition 4.a.1; see Exercise 4.a.14.

Example 2.b.13 (*Standard metrics on $\mathbb{R}^n$*):

(a) Let X be a set. The "*discrete metric*" is given by

$$d(x,y) := \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

Important for counterexamples!

(b) Let $X := \mathbb{R}^n$. Then we have many choices for a metric; some common ones are:

(i) Manhattan metric:

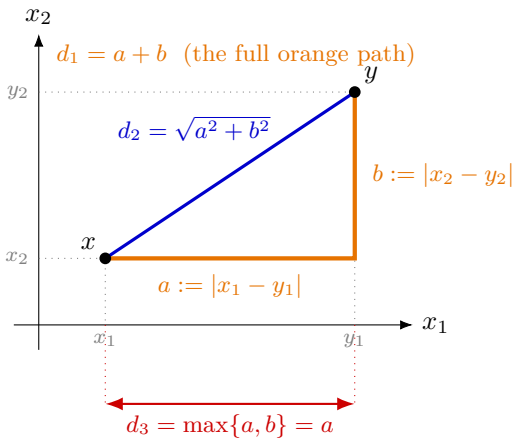
$$d_1(x,y) := |x_1 - y_1| + \cdots + |x_n - y_n| = \sum_{k=1}^{n} |x_k - y_k|$$

(ii) Standard metric:

$$d_2(x,y) := \sqrt{(x_1-y_1)^2 + \cdots + (x_n-y_n)^2} = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

(iii) Supremum metric:

$$d_3(x,y) := \sup_{k=1,\ldots,n} |x_k - y_k|$$



Note that d_1 is the **sum** of the two legs and d_3 the **larger** of them. Neither is a single leg on its own. Only d_2 is the straight-line distance one would draw by reflex, and it is worth remembering that the other two are just as legitimate.

Remark 2.b.14 (*Why d_1 is the "Manhattan" metric*): The official script motivates the name explicitly. In a grid of streets running only north–south and east–west, as in much of Manhattan, a taxi cannot cut across a block diagonally. To travel from $x = (x_1, x_2)$ to $y = (y_1, y_2)$ it moves horizontally from x to (y_1, x_2) and then vertically to y (or the two legs in the other order), covering exactly $|x_1 - y_1| + |x_2 - y_2| = d_1(x, y)$ either way. This is precisely the orange path in the figure above.

In general,

$$d_m(x, y) := \sqrt[m]{\sum_{k=1}^{n} |x_k - y_k|^m}$$

is a metric on $\mathbb{R}^n$ for every $m \geq 1$, with d_1, d_2 above as the cases $m = 1, 2$. Definiteness and symmetry are clear; the triangle inequality is the "*Minkowski inequality*", proved by a convexity argument in Exercise 2.c.42**(e)**.

Remark 2.b.15 (*Why the absolute values, and why $m \geq 1$*): The powers must be taken of **absolute values**, since for odd m the radicand could otherwise be negative. And the result is a metric only for $m \geq 1$: for $m \in (0, 1)$ the triangle inequality fails, as part **(f)** of Exercise 2.c.42 shows with $x = e_1$ and $y = e_2$, where $|x + y|_m = 2^{1/m} > 2 = |x|_m + |y|_m$.

Proposition 2.b.16 (*Concave reshaping preserves the triangle inequality*): Let (X, d) be a metric space and let $h : [0, \infty) \to [0, \infty)$ satisfy

- **(a)** $h(0) = 0$ and $h(t) > 0$ for $t > 0$;
- **(b)** h is non-decreasing;
- **(c)** h is concave.

Then $h \circ d$ is again a metric on X.

Proof:

Definiteness is **(a)** together with definiteness of d, and symmetry is inherited from d directly. Only the triangle inequality needs an argument, and the key step is that **(a)** and **(c)** together force h to be "*subadditive*": $h(a + b) \leq h(a) + h(b)$ for all $a, b \geq 0$. Indeed, for $a, b > 0$ concavity applied to the convex combination writing a as a blend of $a + b$ and 0 gives

$$h(a) = h\Big(\tfrac{a}{a+b}(a+b) + \tfrac{b}{a+b} \cdot 0\Big) \geq \tfrac{a}{a+b} h(a+b) + \tfrac{b}{a+b} \underbrace{h(0)}_{=0} = \tfrac{a}{a+b} h(a+b),$$

and symmetrically $h(b) \geq \frac{b}{a+b} h(a+b)$. Adding the two gives $h(a) + h(b) \geq h(a+b)$. Now, for $x, y, z \in X$, monotonicity **(b)** applied to the triangle inequality for d and then subadditivity give

$$h(d(x, z)) \leq h\big(d(x, y) + d(y, z)\big) \leq h(d(x, y)) + h(d(y, z)).$$ Q.E.D.

Remark 2.b.17 (*Concave is safe, convex is not*): This gives a quick reflex for the standard exercise *"is this a metric?"* when the candidate has the form $h(|x - y|)$ on $\mathbb{R}$: **sketch h and look at its curvature.** A concave h bends **towards** the axis, so distances grow ever more slowly and long jumps become relatively cheap, which is exactly the direction the triangle inequality wants. A convex h does the opposite, making one long jump more expensive than two short ones, and that is precisely what the triangle inequality forbids.

The two textbook cases sit on either side.

- $h(t) := \log(1+t)$ is increasing and concave with $h(0) = 0$, so $d(x,y) := \log(1+|x-y|)$ **is** a metric on $\mathbb{R}$ by Proposition 2.b.16.
- $h(t) := e^t - 1$ is increasing with $h(0) = 0$ but **convex**, and $d(x,y) := e^{|x-y|} - 1$ is **not** a metric: taking $x = 0$, $y = 1$, $z = 2$,

$$d(0,2) = e^2 - 1 \approx 6.389 \quad > \quad 2(e-1) \approx 3.437 = d(0,1) + d(1,2).$$

The heuristic runs in one direction only, and is worth stating carefully. Concavity **guarantees** a metric, whereas convexity merely **suggests** failure. Convexity is not a proof of anything, and a convex h must still be refuted by an explicit triple, as above.

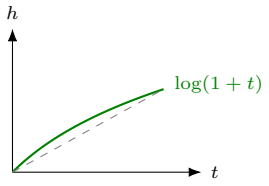
The m-metrics of the preceding Example 2.b.13 fail for $m \in (0,1)$, and the curvature picture locates the failure, though not along the coordinate axes: two points on one axis have $d_m(x,y) = |x_k - y_k|$ for every m, so nothing is reshaped there. The reshaping happens in two stages.

- Each coordinate distance first passes through $t \mapsto t^m$. For $m \in (0,1)$ this profile is concave, so $(a,b) \mapsto |a-b|^m$ is a metric on $\mathbb{R}$ by Proposition 2.b.16, and summing over the coordinates gives a metric on $\mathbb{R}^n$,

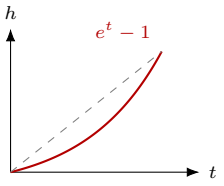
$$\rho_m(x,y) := \sum_{k=1}^{n} |x_k - y_k|^m.$$

- Then $d_m = \rho_m^{1/m}$ passes ρ_m through $s \mapsto s^{1/m}$, which is convex for $m < 1$, and this is the stage that breaks the triangle inequality: $d_m(e_1,e_2) = 2^{1/m} > 2 = d_m(e_1,0) + d_m(0,e_2)$.

So for $m \in (0,1)$ it is the root that has to go, and ρ_m is the metric to use. For $m \geq 1$ the two curvatures swap, t^m convex and $s^{1/m}$ concave, and the proposition decides nothing. The triangle inequality for d_m is then Minkowski's inequality, which needs a proof of its own; Exercise 2.c.42**(e)** gives one.



concave: curve **above** chord
⇒ subadditive ⇒ metric



convex: curve **below** chord
⇒ triangle inequality fails

Exercise 2.b.18 (*Reshaping a metric does not change its convergent sequences*): In the setting of Proposition 2.b.16, assume in addition that h is continuous at 0. Show that a sequence $(x_n)_n$ in X converges to x with respect to d if and only if it converges to x with respect to $h \circ d$.

Solution: *See solution on Page 36*

Example 2.b.19 (*continued*):

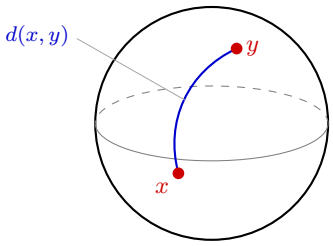
(c) Let $X := C^0([0,1],\mathbb{R})$ be the set of continuous functions $[0,1] \to \mathbb{R}$. Then we can define the "*supremum metric*" for $f, g \in X$:

$$d(f,g) := \sup_{x \in [0,1]} |f(x) - g(x)|$$

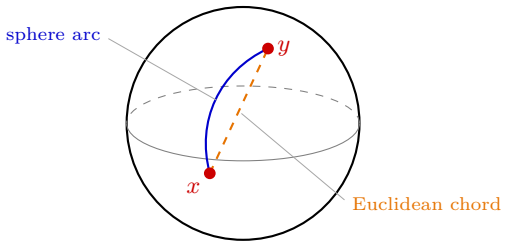
This is well defined, meaning that the supremum is a real number rather than $+\infty$, because $|f-g|$ is continuous on the compact interval $[0,1]$ and therefore bounded, by the extreme value theorem from **Analysis I** (revisited in this generality as Lemma 7.a.7 below). Compactness of the domain is doing real work here: on $C^0(\mathbb{R},\mathbb{R})$ the same formula would produce ∞ for $f(x) := x$, $g := 0$.

(d) Let $X := S^2 := \{(x_1,x_2,x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$. S^n is called the "*n-dimensional sphere*", or "*n-sphere*". The shortest path between two points on the sphere is an **arc of a**

great circle (a *geodesic*). This defines a metric, similar to how straight lines define a metric on $\mathbb{R}^n$.



Distance on sphere



Sphere arc vs. Euclidean chord

Remark 2.b.20 (*Two metrics on the sphere, made explicit*): The two pictures above compare two different metrics that S^2 inherits or acquires, and it is worth writing both down. Restricting the ambient Euclidean metric of $\mathbb{R}^3$ gives the "*chord metric*"

$$d_0(x,y) := |x-y|, \qquad x,y \in S^2,$$

the orange dashed segment in the right-hand panel; since $|x| = |y| = 1$, expanding $|x-y|^2 = \langle x-y, x-y\rangle$ gives the closed form $d_0(x,y) = \sqrt{2-2\langle x,y\rangle}$. The geodesic metric described above measures arc length along a great circle instead, and is the solid blue curve in both panels. It is given by

$$d_1(x,y) := \arccos\big(\langle x,y\rangle\big) \in [0,\pi].$$

Both are genuine metrics on S^2 (Example 2.b.13**(d)** already noted d_1, which measures the distance a great-circle flight would fly; d_0 appears here for the first time). They disagree as soon as $x \neq y$, since the chord always cuts inside the sphere. Therefore, $d_0(x,y) \leq d_1(x,y)$, with equality only when $x = y$, although the ratio d_0/d_1 tends to 1 as $y \to x$. This is the sharpest instance in the chapter of a general warning. A subset inherits a metric from its ambient space, but that inherited metric need not be the one a geometer studying the subset **intrinsically** would choose.

Example 2.b.21 (*The British Rail metric*): Let $(X, \|\cdot\|)$ be a normed space and define

$$d(x,y) := \begin{cases} 0, & x = y, \\ \|x\| + \|y\|, & x \neq y. \end{cases}$$

The name records the joke that to travel between any two towns you must change trains in London: every journey is routed through the origin, and its cost is the distance in plus the distance out.

This is a metric. Definiteness holds by construction and symmetry is evident. For the triangle inequality, take $x \neq z$, so $d(x,z) = \|x\| + \|z\|$. If y equals x or z then one of the two terms on the right vanishes and the other **is** $d(x,z)$, so the inequality holds with equality; otherwise

$$d(x,y) + d(y,z) = \|x\| + 2\|y\| + \|z\| \geq \|x\| + \|z\| = d(x,z).$$

Remark 2.b.22 (*The topology it generates*): This metric is not induced by any norm, and the quickest way to see that something strange is happening is to look at small balls. For $x \neq 0$ and any radius $r \leq \|x\|$, a point $y \neq x$ has $d(y,x) = \|x\| + \|y\| \geq \|x\| \geq r$, so

$$B_r(x) = \{x\}.$$

Every non-zero point is therefore **isolated**, and $\{x\}$ is open. The origin is the sole exception: $d(y,0) = \|y\|$ for $y \neq 0$, so $B_r(0)$ is the ordinary ball of radius r. The space is thus discrete everywhere except at one point, where it looks entirely normal. That is a useful shape to keep in mind whenever a proof seems to be using more about a metric space than the three axioms actually give.

Remark 2.b.23 (*Naming points and their components*): A practical point, and a common source of avoidable errors in written solutions. Once a proof involves several points, each with several

components, two independent naming decisions have to be made, and mixing the two conventions is what causes the damage:

- **Subscripts for components**, different letters or primes for points: $x = (x_1, x_2)$ and $y = (y_1, y_2)$, or $x = (x_1, x_2)$ and $x' = (x'_1, x'_2)$.
- **Subscripts for points**, letters for components: (x_1, y_1) and (x_2, y_2).

Either is internally consistent, and the recommendation is the first: reserve subscripts for components, and distinguish points by x, y, z or by x, x', x''. When many points are needed at once, index them as x_k and write $x_{k,i}$ for the i-th component of the k-th point. For a single point of $\mathbb{R}^2$ or $\mathbb{R}^3$, plain (x, y) or (x, y, z) is of course fine.

The reason this matters more here than in **Analysis I** is that the product metrics of Exercise 2.b.30 and the continuity proofs that follow genuinely need both indices at once. An expression such as $d(x_1, y_1) + d(x_2, y_2)$ is therefore ambiguous until the convention is fixed, and means quite different things under the two readings.

Exercise 2.b.24 (*Supremum metric*): Show that the supremum metric from Example 2.b.13 **(c)** is a metric.

Proof of Exercise 2.b.24:

Clearly, symmetry and definiteness hold. For the triangle inequality, let $f, g, h \in C^0([0,1])$ and set $p := f - h$ and $q := h - g$. Then for all $x \in [0,1]$, we have:

$$|p(x)| \leq \sup |p| \qquad \text{and} \qquad |q(x)| \leq \sup |q|.$$

Adding the inequalities, we obtain for all $x \in [0,1]$:

$$\begin{aligned} |f(x) - g(x)| &= |p(x) + q(x)| \\ &\leq |p(x)| + |q(x)| \\ &\leq \sup |p| + \sup |q| \\ &= \sup |f - h| + \sup |h - g|. \end{aligned}$$

The right-hand side does not depend on x, so it is an upper bound for all the values $|f(x) - g(x)|$, and therefore for their supremum: $d(f, g) \leq d(f, h) + d(h, g)$. Q.E.D.

Exercise 2.b.25 (L^1-*metric*): Let $X := C^0([0,1], \mathbb{R})$ be the set of continuous functions $[0,1] \to \mathbb{R}$. For $f, g \in X$, define

$$d(f, g) := \int_0^1 |f(t) - g(t)| \, dt.$$

Is (X, d) a metric space? Provide a proof.

Proof of Exercise 2.b.25:

Yes, it is a metric space. We verify the three axioms of a metric:

(a) Definiteness: Clearly $d(f, g) \geq 0$ since the integrand $|f(t) - g(t)|$ is non-negative. If $f = g$, then $d(f, f) = \int_0^1 0 \, dt = 0$. Conversely, suppose $d(f, g) = 0$. Since $|f - g|$ is a continuous, non-negative function, its integral over $[0,1]$ can only be zero if the integrand is identically zero. Thus $f(t) = g(t)$ for all $t \in [0,1]$, which means $f = g$.

(b) Symmetry: $d(f, g) = \int_0^1 |f(t) - g(t)| \, dt = \int_0^1 |g(t) - f(t)| \, dt = d(g, f)$.

(c) Triangle inequality: Let $f, g, h \in X$. The triangle inequality for real numbers yields $|f(t) - h(t)| \leq |f(t) - g(t)| + |g(t) - h(t)|$ for all $t \in [0,1]$. Integrating both sides, we obtain:

$$\begin{aligned} d(f,h) &= \int_0^1 |f(t) - h(t)|\, dt \\ &\leq \int_0^1 \left(|f(t) - g(t)| + |g(t) - h(t)|\right) dt \\ &= \int_0^1 |f(t) - g(t)|\, dt + \int_0^1 |g(t) - h(t)|\, dt \\ &= d(f,g) + d(g,h). \end{aligned}$$

Q.E.D.

AI-Note 2.2 (*Where the exercise-sheet problems live*): The official problem sheets are not reproduced as blocks anywhere in this document. Each problem is quoted verbatim (from `exercises/ExN_Analysis2_eng.pdf`; attribution: Prof. Joaquim Serra, D-MATH, ETH Zürich) and placed with the material it tests, so a problem from sheet 3 may well appear several chapters away from a problem from sheet 2. Its solution is then in the **Solutions** section of that same chapter. The priority tags (**important**, **semi-important**, **optional**; see Notation D.1) and the official **(*)** difficulty marker are carried along in each problem's title.

Notation 2.b.26: Some problems in this document have a closed-answer format, similar to what you might find on the final exam. *"Multiple Choice"* means that zero, one or more answers can be correct. Questions marked with (*) are a bit more complex, you might want to skip them at the first read.

AI-Exercise 2.b.27 (*Equivalence of Manhattan and Supremum Metrics*): Let $x := (x_1, x_2) \in \mathbb{R}^2$. Show that the Manhattan metric $d_1(x,0) := |x_1| + |x_2|$ and the supremum metric $d_3(x,0) := \max\{|x_1|, |x_2|\}$ satisfy the sharp inequalities:

$$d_3(x,0) \leq d_1(x,0) \leq 2\, d_3(x,0).$$

Solution: *See solution on Page 36*

Exercise 2.b.28 (*Examples and Non-examples of Metric spaces*): Which of the following pairs are metric spaces? Prove it or provide a counterexample.

(a) $(B(X), d)$, where $B(X)$ denotes the set of all bounded functions from a non-empty set X to $\mathbb{R}$ and

$$d(f,g) := \sup_{x \in X} |f(x) - g(x)|.$$

(b) $(\mathbb{Q}_+, d)$, where $\mathbb{Q}_+$ are the positive rational numbers and $d(x,y) := \left|\frac{1}{x} - \frac{1}{y}\right|$.

(c) $(\mathbb{R}^2, d)$, where $d(x,y) := (x_1 - y_1)^2 + |x_2 - y_2|$.

(d) $(\mathbb{R}^2, d)$, where $d(x,y) := |x_1 - y_1|^{1/2} + |x_2 - y_2|$.

(e) $(\mathbb{R}^{n\times n}, d)$ with $d(X,Y) := \left(\mathrm{Tr}\{(X-Y)^{\mathsf{T}}(X-Y)\}\right)^{1/2}$.

(f) (*) $(\mathbb{R}^{n\times n}, d)$ with

$$d(X,Y) := \sup\{|v^{\mathsf{T}}(X-Y)v| : v \in \mathbb{R}^n, |v| = 1\},$$

and $\mathbb{R}^{n\times n}$ denoting the set of square matrices.

(g) (*) $(\mathbb{R}^2/\mathbb{Z}^2, d)$ where the flat 2-dimensional torus $\mathbb{R}^2/\mathbb{Z}^2$ is the set of equivalence classes of pairs of real numbers under the equivalence relation

$$x, y \in \mathbb{R}^2, \quad x \sim y \iff x_1 - y_1 \in \mathbb{Z},\ x_2 - y_2 \in \mathbb{Z},$$

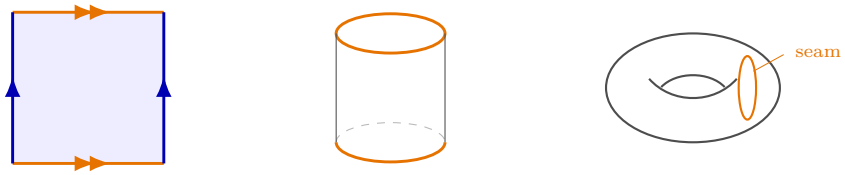
and $d([x],[y]) := \inf_{k,h\in\mathbb{Z}} |x - y + (k,h)|$, with $|\cdot|$ denoting the Euclidean distance and $[x] \in \mathbb{R}^2/\mathbb{Z}^2$ denoting the equivalence class of $x \in \mathbb{R}^2$.

Official hints: ***2.1.3*** *and* ***2.1.4:*** *ignore what happens in the second variable, it is there only to distract you.* ***2.1.5:*** *rewrite, for a general matrix* $A := X - Y$, *what* $\mathrm{Tr}\{A^\mathsf{T}A\}^{1/2}$ *actually means; it should look familiar.* ***2.1.6:*** *see what happens if* $(X - Y)$ *is anti-symmetric.* ***2.1.7:*** *convince yourself first that the "inf" is in fact a "min".*

Info: *This exercise is Problem 2.1 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 37*

Remark 2.b.29 (*The flat torus* $\mathbb{R}^2/\mathbb{Z}^2$*, folded up*): Part **(g)** is much easier to think about with a picture. Every equivalence class in $\mathbb{R}^2/\mathbb{Z}^2$ has a representative in the unit square $[0,1]\times[0,1]$, so the torus is that square with its opposite edges identified: $(0,t)\sim(1,t)$ glues left to right, and $(t,0)\sim(t,1)$ glues bottom to top. Performing the two gluings in turn folds the square first into a cylinder and then into a torus $\mathbb{T}^2 := \mathbb{R}^2/\mathbb{Z}^2$.



(a) the square $[0,1]^2$ opposite edges identified

(b) glue left to right: a cylinder, two orange rims

(c) glue the two rims: the torus $\mathbb{T}^2$

With the picture in hand the formula reads off directly. The distance $d([x],[y])$ is the shortest Euclidean distance between representatives, which is to say the length of the shortest path on the folded torus. The infimum over $k,h\in\mathbb{Z}$ in the problem statement is just the search over which copy of y in the tiling of the plane by unit squares lies nearest to x, and since only finitely many copies can be candidates, that infimum is attained. This is what the official hint means when it says to convince yourself the infimum is really a minimum.

Exercise 2.b.30 (*Product of metric spaces*): Let (X,d_X) and (Y,d_Y) be a pair of metric spaces. Recall that the set of ordered pairs (x,y) with $x\in X$ and $y\in Y$ is denoted by $X\times Y$. Consider the following functions $X\times Y\to[0,\infty)$:

$$d_1((x,y),(x',y')) := \max\{d_X(x,x'),d_Y(y,y')\}$$
$$d_2((x,y),(x',y')) := d_X(x,x')+d_Y(y,y')$$
$$d_3((x,y),(x',y')) := \sqrt{d_X(x,x')^2+d_Y(y,y')^2}.$$

(a) Show that they are all valid distance functions on $X\times Y$.

(b) Show that they are all equivalent, i.e. there is a number $C>0$ such that

$$d_1((x,y),(x',y')) \le C\,d_2((x,y),(x',y')) \le C^2 d_3((x,y),(x',y')) \le C^3 d_1((x,y),(x',y'))$$

for all $x,x'\in X$, $y,y'\in Y$.

(c) Show that a sequence $(x_n,y_n)\to(x,y)$ with respect to $(X\times Y,d_3)$ if and only if $x_n\to x$ with respect to d_X and $y_n\to y$ with respect to d_Y.

Official hint: *Don't get distracted by the abstract set-up, since you already know all these things for* $\mathbb{R}\times\mathbb{R}$*; start from there, then rewrite those arguments in this general framework.*

Info: *This exercise is Problem 2.5 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 38*

Cauchy–Schwarz (Section 2.c)

Recap: inner products and norms (Subsection 2.c.1)

Both definitions were stated in Section 2.a, where the hierarchy of structured spaces was drawn. Rather than repeat them, here is what to have in mind:

- An **inner product** (Definition 2.a.1) is **bilinear**, **symmetric** and **definite**. It measures length **and** angle.
- A **norm** (Definition 2.a.2) is **definite**, **homogeneous** and satisfies the **triangle inequality**. It measures length only.
- Each induces the next, via $\|v\| := \sqrt{\langle v, v\rangle}$ and $d(x, y) := \|x - y\|$, and neither implication reverses (Proposition 2.a.8, Remark 2.a.9).

What this section adds is the **geometry**: why $\langle x, y\rangle$ deserves to be read as an angle at all, and the resulting second proof of Cauchy–Schwarz.

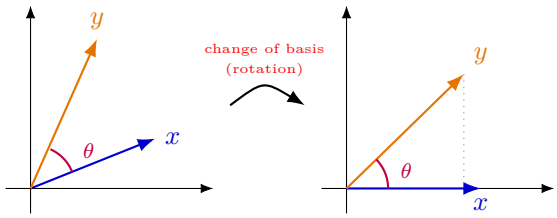
Geometric meaning of the inner product (Subsection 2.c.2)

In $\mathbb{R}^2$, the following identity holds:

$$\langle x, y\rangle := x_1y_1 + x_2y_2 = \|x\|\|y\| \cos\theta$$

where θ is the angle between $x, y \in \mathbb{R}^2$.

To see this, conveniently choose our basis vectors such that $x = (x_1, 0)$, $y = (y_1, y_2)$.



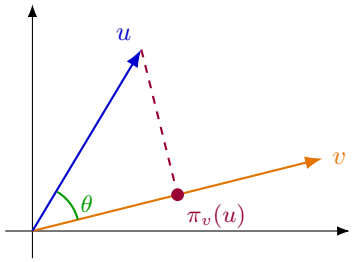
Then

$$\langle x, y\rangle = x_1y_1 = x_1\|y\| \cos\theta = \|x\|\|y\| \cos\theta.$$

For more rigour, you will show in Linalg that the rotations in $\mathbb{R}^2$ are given by matrices R satisfying $\langle Rx, Ry\rangle = \langle x, y\rangle$.

In a general inner product space V, this becomes the **definition** of the angle θ. It then makes sense to define the "*projection onto* v", π_v:

$$\pi_v(u) := \frac{\langle u, v\rangle}{\|v\|^2} v = \|u\| \cos\theta \frac{v}{\|v\|}.$$



In words, $\pi_v(u)$ is the part of u which points along v. In particular,

$$\|\pi_v(u)\| = \|u\|\,|\cos\theta| = \frac{|\langle u, v\rangle|}{\|v\|}.$$

In this sense, Cauchy–Schwarz says something visually obvious. Reading the picture above, all three of the following are the same statement:

$$\frac{|\langle u, v\rangle|}{\|v\|} \le \|u\| \iff \|\pi_v(u)\| \le \|u\| \iff \|u\|\,|\cos\theta| \le \|u\|.$$

Remark 2.c.31 (*Hint for the proof of Cauchy–Schwarz*): Use definiteness and expand the term

$$\langle u - \pi_v(u),\ u - \pi_v(u)\rangle = \dots$$

where $u - \pi_v(u)$ is the **perpendicular part** of u to v.

The Cauchy–Schwarz inequality (Subsection 2.c.3)

Theorem 2.c.32 (*Cauchy–Schwarz inequality*): Let V be an inner product space (Definition 2.a.1) with induced norm $\|\cdot\| := \sqrt{\langle\cdot,\cdot\rangle}$ (Definition 2.a.2). Then for all $u, v \in V$,

$$|\langle u, v\rangle| \le \|u\|\,\|v\|,$$

with equality if and only if u, v are linearly dependent.

Proof:

If $v = 0$, both sides vanish and the claim holds trivially, so assume $v \neq 0$. Following Remark 2.c.31, definiteness (Definition 2.a.1) gives

$$0 \ \le\ \langle u - \pi_v(u),\ u - \pi_v(u)\rangle \ =\ \langle u, u\rangle - 2\langle u, \pi_v(u)\rangle + \langle \pi_v(u), \pi_v(u)\rangle.$$

Using $\pi_v(u) = \frac{\langle u,v\rangle}{\|v\|^2} v$ and bilinearity, the two right-hand terms are

$$\langle u, \pi_v(u)\rangle = \frac{\langle u, v\rangle^2}{\|v\|^2},$$

$$\langle \pi_v(u), \pi_v(u)\rangle = \frac{\langle u, v\rangle^2}{\|v\|^4}\langle v, v\rangle = \frac{\langle u, v\rangle^2}{\|v\|^2}.$$

Note that the two are equal, so substituting them leaves a single copy behind:

$$0 \ \le\ \|u\|^2 - \frac{2\langle u, v\rangle^2}{\|v\|^2} + \frac{\langle u, v\rangle^2}{\|v\|^2} \ =\ \|u\|^2 - \frac{\langle u, v\rangle^2}{\|v\|^2}.$$

This rearranges to $\langle u, v\rangle^2 \le \|u\|^2\|v\|^2$, that is, $|\langle u, v\rangle| \le \|u\|\,\|v\|$. By definiteness, equality holds if and only if $u - \pi_v(u) = 0$, which says precisely that u is a scalar multiple of v. **Q.E.D.**

Remark 2.c.33 (*The same proof twice*): This is the second proof of Cauchy–Schwarz in these notes; Proposition 2.a.7 gave another. The two look different at first. One minimises the quadratic $f(\lambda) := \|v - \lambda w\|^2$ over λ, while the other expands $\|u - \pi_v(u)\|^2 \ge 0$. But the minimiser found in the first is $\lambda_0 = \frac{\langle v,w\rangle}{\|w\|^2}$, which is exactly the coefficient defining $\pi_w(v)$. Consequently, the first proof **derives** the projection by optimisation, and the second **postulates** it and checks that it works. Keeping both is worthwhile precisely because the pair shows where the projection comes from. It is not a lucky guess, but the closest point of the line to v.

Exercise 2.c.34 (*An integral inequality from Cauchy–Schwarz*): Let $a < b$ and let $f : [a, b] \to \mathbb{R}_{>0}$ be continuous and strictly positive. Show that

$$\int_a^b f(x)\,dx \int_a^b \frac{1}{f(x)}\,dx \ \ge\ (b - a)^2,$$

and determine for which f equality holds.

Hint: *The space $C([a,b],\mathbb{R})$ carries the inner product $\langle g,h\rangle := \int_a^b g(x)h(x)\,dx$. Only the choice of the two vectors is left to find.*

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *(undated) in* `old_exams/fs2023/`*, Problem 11. That paper names no lecturer, and asks only for the inequality; the equality case is added here.*

Solution: *See solution on Page 39*

Exercise 2.c.35 (*Cauchy–Schwarz*): Let V be a vector space over $\mathbb{R}$, let $\langle\cdot,\cdot\rangle$ be an inner product on V, and let $\|\cdot\| : V \to \mathbb{R}$ be given by $\|v\| = \sqrt{\langle v,v\rangle}$. Then the inequality

$$|\langle v,w\rangle| \le \|v\|\|w\| \tag{1}$$

holds for all $v,w \in V$. Furthermore, equality in (1) holds if and only if v and w are linearly dependent.

Info: *This exercise is Problem 3.7 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 39*

Exercise 2.c.36 (*Inner product and norm*): Let V be a vector space over $\mathbb{R}$, let $\langle -,-\rangle$ be an inner product on V. The map defined by

$$\|\cdot\| : V \to \mathbb{R}, \qquad \|v\| = \sqrt{\langle v,v\rangle}$$

satisfies the triangular inequality and is a norm.

Info: *This exercise is Problem 3.8 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 40*

Exercise 2.c.37 (*All norms are equivalent in $\mathbb{R}^n$*): Let $|\cdot|$ denote the standard Euclidean norm in $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to [0,\infty)$ be another norm (that is, a function satisfying the properties of *Definition 9.89*).

- **(a)** Expressing x in a basis and using the "abstract" properties that f must have, show that there is a constant $C_1 > 0$ such that $f(x) \le C_1|x|$ for all $x \in \mathbb{R}^n$.
- **(b)** Show that f is continuous in $\mathbb{R}^n$ (with respect to the standard distance of $\mathbb{R}^n$!).
- **(c)** Show that there is a number $c_2 > 0$ such that $f(x) \ge c_2$ for all $|x| = 1$.
- **(d)** Conclude that $f(x) \ge c_2|x|$ for all $x \in \mathbb{R}^n$.
- **(e)** Show that if $\tilde f$ is yet another norm, then there is $C > 0$ such that $C^{-1}f(x) \le \tilde f(x) \le Cf(x)$ for all $x \in \mathbb{R}^n$.

Official hints: ***3.9.2:*** *recall that from the definition it follows that $|f(x)-f(y)| \le f(x-y)$, and that Lipschitz functions are always continuous.* ***3.9.3:*** *apply the Weierstrass theorem to f on $S := \{x \in \mathbb{R}^n : |x| = 1\}$ and use that f, being a norm, is nondegenerate.* ***3.9.5:*** *if f is equivalent to $|\cdot|$ and $\tilde f$ is equivalent to $|\cdot|$, it follows by transitivity that f is equivalent to $\tilde f$.*

Info: *This exercise is Problem 3.9 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 40*

Remark 2.c.38: Hint 3.9.3 is the Weierstrass extreme value theorem (Item **(b)**) applied on the compact unit sphere, which is one of the main reasons to care about compactness at all.

Exercise 2.c.39 (*The polynomial ring $\mathbb{R}[x]$ is infinite-dimensional*): Show that the vector space $\mathbb{R}[x]$ of polynomials with real coefficients is infinite dimensional, and exhibit a basis.

Solution: *See solution on Page 41*

Remark 2.c.40: This is exactly the hypothesis Exercise 2.c.37 needs and cannot do without. Remark 2.e.54, with the solution of that exercise, gives $C^0([0,1],\mathbb{R})$ as an infinite-dimensional space where norm equivalence fails. Note that $\mathbb{R}[x]$ is the algebraically simplest such example, with an explicit basis and no analysis required at all.

Exercise 2.c.41 (*Hilbert–Schmidt norm of the composition*): Take two linear functions $\varphi : \mathbb{R}^d \to \mathbb{R}^n$ and $\psi : \mathbb{R}^n \to \mathbb{R}^m$, and denote with Φ, Ψ the matrices that represent them in the canonical bases. Recall that the linear map $\psi \circ \varphi : \mathbb{R}^d \to \mathbb{R}^m$ is represented in these bases by the matrix $\Psi \cdot \Phi$. Show that

$$\|\Psi \cdot \Phi\| \leq \|\Psi\|\|\Phi\|,$$

where $\|\cdot\|$ is the Hilbert–Schmidt norm of a matrix (see *Definition 9.96* in the notes).

Official hint: *Recall that $\|M\|^2$ is the sum of the squares of the entries of M. Notice that $(\Psi \cdot \Phi)^i_j$ (i-th row, j-th column) is the scalar product of Ψ^i and Φ_j, which are vectors of $\mathbb{R}^n$. Apply Cauchy–Schwarz to each of them.*

Info: *This exercise is Problem 3.10 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 41*

Exercise 2.c.42 (*p-norms* (semi-important; parts (d) and (f) important)): For $p \geq 1$ and $x \in \mathbb{R}^n$ define the *p-norm* of x as

$$|x|_p := \Big(\sum_{i=1}^n |x_i|^p\Big)^{1/p}.$$

(a) For $n = 2$ and $p = 1, 2, 10$ sketch the sets $\{x \in \mathbb{R}^2 : |x|_p \leq 1\}$.

(b) For a given $x \in \mathbb{R}^n$, compute the limit $|x|_\infty := \lim_{p\to\infty} |x|_p$.

(c) Using an appropriate inequality that you have seen in class, prove that

$$\Big(\sum_{i=1}^n a_i^{p-1} b_i\Big)^2 \leq \Big(\sum_{i=1}^n a_i^p\Big)\Big(\sum_{i=1}^n a_i^{p-2} b_i^2\Big),$$

whenever a_i, b_i are n-tuples of positive numbers.

(d) Fix $x, y \in \mathbb{R}^n$ and consider the function $f : [0,1] \to \mathbb{R}$ defined as

$$f(t) := |tx + (1-t)y|_p = \Big(\sum_{i=1}^n |tx_i + (1-t)y_i|^p\Big)^{1/p}. \tag{1}$$

Show that f is convex. You may assume that the coordinates of x and y are all strictly positive and use the inequality of the previous point.

(e) Deduce from the previous point that the triangular inequality holds, i.e. $|x+y|_p \leq |x|_p + |y|_p$ for all $x, y \in \mathbb{R}^n$.

(f) What happens for $p \in (0,1)$?

Official hints: ***4.3.3:*** *use Cauchy–Schwarz and the fact that $a_i^{p-1} b_i = a_i^{p/2} \cdot a_i^{(p-2)/2} b_i$.* ***4.3.4:*** *show that $f''(t) \geq 0$; it is not the simplest derivative, but if you get it right the inequality in 4.3.3 will be just what you need.* ***4.3.5:*** *write the convexity inequality between x, y and the middle point $(x+y)/2$. To get the general case from the one with positive coordinates, observe that for $x \in \mathbb{R}^n$, denoting $\hat{x} := (|x_1|, \dots, |x_n|)$, we have $|x+y|_p \leq |\hat{x}+\hat{y}|_p$ and $|\hat{x}|_p = |x|_p$ for all $x, y \in \mathbb{R}^n$.*

Info: *This exercise is Problem 4.3 of Problem Sheet 4, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 41*

Exercise 2.c.43 (*p-means* (optional)): For $x \in \mathbb{R}^n$ with positive coordinates and $p \neq 0$ define the *p*-**mean** as

$$\mu_p(x) := \left(\frac{x_1^p + \cdots + x_n^p}{n} \right)^{1/p}.$$

(a) Compute the limits $p \to \pm\infty$, $p \to 0$ and define accordingly $\mu_{-\infty}(x)$, $\mu_0(x)$, $\mu_{+\infty}(x)$.

(b) For any n-tuple of numbers $a_i > 0$ show that

$$\sum_{i=1}^{n} \frac{a_i \log(a_i)}{a_1 + \cdots + a_n} \geq \log\left(\frac{a_1 + \cdots + a_n}{n} \right).$$

(c) For a fixed x, show that the function $f : \mathbb{R} \to \mathbb{R}$, given by $f(t) := \mu_t(x)$, is continuous and increasing.

(d) Prove the Arithmetic–Geometric inequality and Arithmetic–Quadratic inequality:

$$n(x_1 x_2 \cdots x_n)^{1/n} \leq x_1 + \cdots + x_n, \qquad (x_1 + \cdots + x_n)^2 \leq n(x_1^2 + \cdots + x_n^2).$$

(e) (*) Is f continuously differentiable in the whole $\mathbb{R}$?

Official hints: ***4.4.2:*** *combine the concavity of* $\log(\cdot)$ *and the Cauchy–Schwarz inequality; use the generalisation of the concavity inequality: for any concave* $f : \mathbb{R} \to \mathbb{R}$, $f(\lambda_1 x_1 + \cdots + \lambda_n x_n) \geq \lambda_1 f(x_1) + \cdots + \lambda_n f(x_n)$ *for all* $x_i \in \mathbb{R}$, $0 \leq \lambda_i \leq 1$ *with* $\lambda_1 + \cdots + \lambda_n = 1$. ***4.4.3:*** *it might be more convenient to work with* $\log f(t)$.

Info: *This exercise is Problem 4.4 of Problem Sheet 4, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 43*

Equivalence of norms (Section 2.d)

The previous sections put several norms on the same vector space: the Euclidean norm, the p-norms, the supremum norm. A natural worry follows. Convergence, openness, continuity and compactness were all defined from a metric, and on a normed space that metric came from the norm. So does changing the norm change which sequences converge and which sets are open?

On $\mathbb{R}^n$ the answer is no, and the reason is worth isolating, because it is one of the few places in this course where finite-dimensionality is doing essential work.

Definition 2.d.44 (*Equivalent norms*): Let V be a vector space over $\mathbb{R}$ and let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V. They are called "*equivalent*" (German: „*äquivalent*“), or **comparable**, if there are constants $A, B > 0$ with

$$\|v\|_1 \leq A\|v\|_2 \qquad \text{and} \qquad \|v\|_2 \leq B\|v\|_1 \qquad \text{for all } v \in V.$$

Remark 2.d.45 (*Why two inequalities and not one*): A single inequality $\|v\|_1 \leq A\|v\|_2$ says only that $\|\cdot\|_2$-convergence implies $\|\cdot\|_1$-convergence: it makes the first norm the coarser of the two. Equivalence is the two-sided statement, and it is exactly what is needed for the two norms to have the same null sequences, and hence the same convergent sequences, open sets and compact sets. That equivalence of norms really is an equivalence relation is part **(e)** of Exercise 2.c.37.

Example 2.d.46 (*The 1-norm against the maximum norm*): On $\mathbb{R}^n$,

$$\|v\|_\infty \leq \|v\|_1 \qquad \text{and} \qquad \|v\|_1 \leq n\|v\|_\infty,$$

so the two are equivalent with $A = 1$ and $B = n$. The left inequality holds because the largest $|v_i|$ is one of the terms of the sum $\sum_i |v_i|$; the right because each of the n terms of that sum is at most $\max_i |v_i|$.

The constant n on the right is sharp, as $v = (1, 1, \ldots, 1)$ shows, and it is the first sign of what goes wrong later: the constants in Theorem 2.d.47 are allowed to depend on the dimension, and there is no dimension to depend on in an infinite-dimensional space.

Theorem 2.d.47 (*All norms on $\mathbb{R}^n$ are equivalent*): Any two norms on $\mathbb{R}^n$ are equivalent in the sense of Definition 2.d.44.

Proof:

This is Exercise 2.c.37, whose five parts are exactly the five steps of the argument, and it is proved in full in the solutions to that exercise. In outline, with $|\cdot|$ the Euclidean norm and f an arbitrary norm: expanding x in the standard basis and applying the triangle inequality, homogeneity and Cauchy–Schwarz gives $f(x) \le C_1|x|$; that bound makes f Lipschitz, hence continuous, for the Euclidean metric; Weierstrass then gives f a positive minimum c_2 on the Euclidean unit sphere, which is compact by Heine–Borel (Theorem 7.b.28); and homogeneity spreads that bound from the sphere to all of $\mathbb{R}^n$, giving $f(x) \ge c_2|x|$. Q.E.D.

Remark 2.d.48 (*Where each hypothesis is used*): Both halves of the proof use finite-dimensionality, and they use it differently. The first inequality expands x in a **finite** basis, so the sum of n terms it produces is a sum and not a series. The second applies Weierstrass on the unit sphere, which needs that sphere to be **compact**, and by Heine–Borel it is, in $\mathbb{R}^n$ and nowhere else. Remark 2.e.54 exhibits the failure in $C^0([0,1],\mathbb{R})$, and Exercise 2.c.39 gives the algebraically simplest infinite-dimensional space to have in mind.

Corollary 2.d.49 (*Any finite-dimensional real vector space*): Let V be a real vector space of finite dimension n. Then any two norms on V are equivalent.

Proof:

Fix a basis $\mathcal{B} = \{e_1, \dots, e_n\}$ of V. The coordinate map $\iota_{\mathcal{B}} : \mathbb{R}^n \to V$, $(x_1, \dots, x_n) \mapsto x_1e_1 + \cdots + x_ne_n$, is a linear isomorphism, so a norm $\|\cdot\|$ on V pulls back to the norm $x \mapsto \|\iota_{\mathcal{B}}(x)\|$ on $\mathbb{R}^n$, and the two-sided inequalities of Definition 2.d.44 are preserved in both directions under this identification. Applying Theorem 2.d.47 on $\mathbb{R}^n$ and transporting back gives the claim. Q.E.D.

Corollary 2.d.50 (*All norms give the same topology*): Topological properties of a subset $E \subseteq \mathbb{R}^n$ (openness, closedness, convergence of sequences, compactness, connectedness) do not depend on which norm was used to define the metric.

Proof:

Let d_1, d_2 be the metrics induced by two norms. By Theorem 2.d.47 there is a constant $C > 0$ with $C^{-1}d_1(x,y) \le d_2(x,y) \le Cd_1(x,y)$ for all x, y, since $d_i(x,y) = \|x - y\|_i$ and the norms are equivalent. Hence every d_1-ball contains a d_2-ball about the same centre and conversely, which by Definition 3.a.6 makes a set open for d_1 exactly when it is open for d_2. The two metrics therefore generate the same topology, and every property defined purely in terms of the open sets agrees. Q.E.D.

AI-Note 2.3 (*What this licenses, and what it does not*): Corollary 2.d.50 is the reason this document is free to say *"open"*, *"compact"* or *"convergent"* in $\mathbb{R}^n$ without ever naming a norm, and it is worth knowing that the licence has been earned rather than assumed.

Three things it does **not** give you.

- It is about **topological** properties only. Metric quantities are not preserved: the unit ball genuinely changes shape with the norm (see the three drawn in Section 3.c), distances change, and so do Lipschitz constants. A 1-Lipschitz map for one norm need not be 1-Lipschitz for another, only Lipschitz with some other constant.
- It says nothing in infinite dimensions, where the conclusion is false (Remark 2.d.48).

- It does not make the choice of norm irrelevant in practice. Estimates are still written in whichever norm makes the constants come out cleanly, which is why the supremum norm is the one used on $C(K,\mathbb{R}^m)$ in Section 7.d even though that space is infinite-dimensional and the theorem above does not reach it.

Solutions (Section 2.e)

Not every solution in this chapter is collected here. Exercise 2.b.24 (Page 27) and Exercise 2.b.25 (Page 27) are solved where they appear.

Proof of Exercise 2.b.11:

For $N=3$ this is the triangle inequality itself. For the inductive step, assume the claim holds for some $N\geq 3$ and let $x_1,\dots,x_{N+1}\in X$. Applying the triangle inequality to the triple (x_1,x_N,x_{N+1}) and then the inductive hypothesis to $x_1,\dots,x_N$,

$$d(x_1,x_{N+1}) \;\leq\; d(x_1,x_N)+d(x_N,x_{N+1}) \;\leq\; \sum_{i=1}^{N-1} d(x_i,x_{i+1})+d(x_N,x_{N+1}) \;=\; \sum_{i=1}^{N} d(x_i,x_{i+1}),$$

which is the claim for $N+1$. Q.E.D.

Proof of Exercise 2.b.18:

Suppose $x_n\to x$ with respect to d, so $d(x_n,x)\to 0$. By hypothesis h is continuous at 0, and $h(0)=0$, so $h(d(x_n,x))\to 0$, i.e. $x_n\to x$ with respect to $h\circ d$.

Conversely, suppose $x_n\to x$ with respect to $h\circ d$. Fix $\varepsilon>0$; by hypothesis **(a)** of Proposition 2.b.16, $\delta:=h(\varepsilon)>0$. Choose N such that $h(d(x_n,x))<\delta$ for all $n\geq N$. If $d(x_n,x)\geq\varepsilon$ held for some such n, monotonicity **(b)** would give $h(d(x_n,x))\geq h(\varepsilon)=\delta$, a contradiction. So $d(x_n,x)<\varepsilon$ for all $n\geq N$, i.e. $x_n\to x$ with respect to d.

Remark 2.e.51 (*Why continuity at 0 has to be assumed separately*): Concavity does not supply that hypothesis, and it is worth seeing why not. A concave function is automatically continuous on the **interior** of its domain, but $[0,\infty)$ has an endpoint, and at an endpoint a concave function may jump. The value at the endpoint can only lie **below** the limit from inside, since concavity bounds the function below by its chords. Take

$$h(0):=0, \qquad h(t):=1+t \quad \text{for } t>0.$$

All three hypotheses of Proposition 2.b.16 hold: h vanishes exactly at 0, it is non-decreasing, and it is concave, the only case needing a look being a blend of 0 with some $t>0$, where the value $1+(1-\lambda)t$ does exceed the chord's $(1-\lambda)(1+t)$. So $h\circ d$ is a perfectly good metric. It is just a discrete one: $h(d(x,y))>1$ whenever $x\neq y$, so the sequences converging in $h\circ d$ are exactly the eventually constant ones, and on $\mathbb{R}$ with the standard metric $\frac{1}{n}\to 0$ is lost.

Note which half of the proof breaks. The converse direction uses only **(a)** and **(b)**, never continuity, and remains true for this h, because an eventually constant sequence converges in every metric. It is the forward direction that needs the hypothesis.

Q.E.D.

Proof of AI-Exercise 2.b.27:

For any $x=(x_1,x_2)\in\mathbb{R}^2$, we have $\max\{|x_1|,|x_2|\}\leq|x_1|+|x_2|$, giving $d_3(x,0)\leq d_1(x,0)$. Conversely, since $|x_1|\leq\max\{|x_1|,|x_2|\}$ and $|x_2|\leq\max\{|x_1|,|x_2|\}$, summing these yields $|x_1|+$

$|x_2| \le 2\max\{|x_1|,|x_2|\}$, i.e., $d_1(x,0) \le 2\,d_3(x,0)$. Both bounds are sharp: equality on the left holds for $x=(1,0)$, and equality on the right holds for $x=(1,1)$. Q.E.D.

Solution to Exercise 2.b.28:

Throughout, non-negativity and symmetry are immediate in every part and are not belaboured; the interest is in **definiteness** and the **triangle inequality**.

(a) **Yes.** Boundedness of f and g makes the supremum finite, so d really maps into $\mathbb{R}$. Definiteness: $d(f,g)=0$ forces $|f(x)-g(x)|=0$ for every x, i.e. $f=g$. For the triangle inequality, estimate pointwise and only then take the supremum:

$$|f(x)-h(x)| \le |f(x)-g(x)| + |g(x)-h(x)| \le d(f,g)+d(g,h) \quad \text{for all } x,$$

so the right-hand side is an upper bound for $|f-h|$ and hence dominates $d(f,h)$. This is Exercise 2.b.24 verbatim, with $[0,1]$ replaced by an arbitrary set X and continuity replaced by boundedness. Note that it was boundedness, and not continuity, that the argument there actually used.

(b) **Yes.** Let $\varphi(x) := 1/x$, an injective map $\mathbb{Q}_+ \to \mathbb{Q}_+$, and observe that $d(x,y) = |\varphi(x)-\varphi(y)|$. Symmetry and the triangle inequality are inherited from $|\cdot|$ without any work, and definiteness is exactly injectivity of φ: from $d(x,y)=0$ we get $1/x = 1/y$, hence $x=y$. This is the same pullback construction as Exercise 6.a.10, and the same warning applies. Pulling a metric back along an injection **always** produces a metric, which is why the interesting question about such a d is never whether it is one, but whether it is complete.

(c) **No:** the triangle inequality fails. Ignore the second coordinate, as the official hint suggests, and take three collinear points $x := (0,0)$, $y := (1,0)$ and $z := (2,0)$:

$$d(x,z) = 2^2 = 4 \quad > \quad d(x,y)+d(y,z) = 1+1 = 2.$$

Squaring produces the same failure as in AI-Example 2.b.12**(c)**: a function growing faster than linearly makes one long step more expensive than two short ones.

(d) **Yes.** A sum of two metrics acting on separate coordinates is a metric, so everything reduces to the first coordinate, where the claim is that $(s,t) \mapsto |s-t|^{1/2}$ satisfies the triangle inequality on $\mathbb{R}$. That follows from **subadditivity of the square root**, $\sqrt{a+b} \le \sqrt{a}+\sqrt{b}$ for $a,b \ge 0$, which holds because squaring the right-hand side gives $a+b+2\sqrt{ab} \ge a+b$. Hence

$$|x_1-z_1|^{1/2} \le \big(|x_1-y_1|+|y_1-z_1|\big)^{1/2} \le |x_1-y_1|^{1/2} + |y_1-z_1|^{1/2}.$$

Contrast this with part **(c)**: $\sqrt{\cdot}$ is concave and vanishes at 0, hence subadditive, while $(\cdot)^2$ is convex, hence super-additive. That single distinction settles both parts.

(e) **Yes.** Write $A := X-Y$. Then

$$\operatorname{Tr}(A^{\mathsf{T}}A) = \sum_{i=1}^{n} (A^{\mathsf{T}}A)_{ii} = \sum_{i,j=1}^{n} A_{ji}^2,$$

so $d(X,Y)$ is nothing but the Euclidean distance between X and Y regarded as vectors of $\mathbb{R}^{n^2}$, and it is a metric by Proposition 2.a.8**(b)**. The associated norm is the "*Frobenius norm*" (German: „*Frobeniusnorm*"), also called the Hilbert–Schmidt norm.

(f) **No**, and here it is **definiteness** that fails, not the triangle inequality. Following the hint, suppose $A := X-Y$ is **anti-symmetric**, $A^{\mathsf{T}} = -A$. For any v the number $v^{\mathsf{T}}Av$ is a 1×1 matrix, hence equal to its own transpose:

$$v^{\mathsf{T}}Av = (v^{\mathsf{T}}Av)^{\mathsf{T}} = v^{\mathsf{T}}A^{\mathsf{T}}v = -\,v^{\mathsf{T}}Av,$$

so $v^{\mathsf{T}}Av = 0$ for **every** v. Taking $n=2$ with

$$X := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Y := \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

gives $X \neq Y$ but $d(X,Y) = 0$. Every other axiom does hold, so d is a "*pseudo-metric*" (German: „*Pseudometrik*"): it cannot separate two matrices differing by an anti-symmetric one. Equivalently, it restricts to a genuine metric on the symmetric matrices.

(g) Yes. Three things need checking, and the first is the one that is easy to skip.

Well defined. Replacing x by $x+(m,n)$ and y by $y+(m',n')$ shifts $x-y$ by the integer vector $(m-m',\, n-n')$, which merely re-indexes the infimum. So $d([x],[y])$ does not depend on the representatives chosen.

The infimum is a minimum. Only finitely many integer vectors (k,h) satisfy $|x-y+(k,h)| \le |x-y|$, because $\mathbb{Z}^2$ is discrete; the infimum is therefore taken over a finite non-empty set of candidates and is attained. This is what the official hint asks you to settle first, and it is what makes the next step legitimate.

Definiteness. If $d([x],[y]) = 0$ then, the minimum being attained, $x-y+(k,h) = 0$ for some integers k,h, that is $x \sim y$ and $[x] = [y]$. Note exactly where *"the infimum is a minimum"* was needed: an infimum equal to 0 but not attained would say nothing at all.

Triangle inequality. Choose minimisers (k_1,h_1) for the pair (x,y) and (k_2,h_2) for (y,z). Their sum is a legitimate candidate for the pair (x,z), so

$$\begin{aligned} d([x],[z]) &\le \big|x - z + (k_1{+}k_2,\ h_1{+}h_2)\big| \\ &= \big|\big(x-y+(k_1,h_1)\big) + \big(y-z+(k_2,h_2)\big)\big| \\ &\le d([x],[y]) + d([y],[z]). \end{aligned}$$

Q.E.D.

AI-Note 2.4: Parts **(c)**, **(d)** and **(f)** are the ones worth revisiting, because each breaks (or rescues) a different axiom and together they cover every way the definition can fail. Part **(c)** kills the triangle inequality by growing too fast; part **(d)** rescues it by growing slowly enough; part **(f)** satisfies the triangle inequality perfectly and fails on definiteness instead. Read alongside AI-Example 2.b.12, they make the case that the three axioms are genuinely independent.

Solution to Exercise 2.b.30:

Write $s := d_X(x,x')$ and $t := d_Y(y,y')$ throughout, so that the three candidates are $d_1 = \max\{s,t\}$, $d_2 = s+t$ and $d_3 = \sqrt{s^2+t^2}$. These are precisely the supremum, Manhattan and standard metrics of Example 2.b.13, applied to the pair (s,t). Almost everything below is that one observation.

(a) All three are metrics. Each vanishes exactly when $s = t = 0$, i.e. when $x = x'$ and $y = y'$, which is definiteness; symmetry is inherited from d_X and d_Y. For the triangle inequality, set

$$a := \big(d_X(x,x'),\, d_Y(y,y')\big), \quad b := \big(d_X(x',x''),\, d_Y(y',y'')\big), \quad c := \big(d_X(x,x''),\, d_Y(y,y'')\big),$$

all lying in $[0,\infty)^2$. The triangle inequalities in X and in Y say precisely that $c \le a+b$ **componentwise**. Each of $\max$, the sum and $\|\cdot\|_2$ is monotone on $[0,\infty)^2$ and subadditive, so

$$N(c) \;\le\; N(a+b) \;\le\; N(a) + N(b)$$

for $N \in \{\max,\ \textstyle\sum,\ \|\cdot\|_2\}$, which is the required inequality in all three cases at once.

(b) Equivalence. For all $s,t \ge 0$,

$$\max\{s,t\} \;\le\; \sqrt{s^2+t^2} \;\le\; s+t \;\le\; 2\max\{s,t\},$$

i.e. $d_1 \le d_3 \le d_2 \le 2d_1$; the middle inequality holds because $(s+t)^2 = s^2+t^2+2st \ge s^2+t^2$. Hence $C := 2$ satisfies the chain in the statement: $d_1 \le 2d_2$, then $2d_2 \le 4d_3$ (since $d_2 \le \sqrt{2}\,d_3$ by Cauchy–Schwarz, and $\sqrt{2} \le 2$), and finally $4d_3 \le 8d_1$ (since $d_3 \le \sqrt{2}\,d_1$).

(c) Convergence is coordinatewise. From part **(b)**,

$$\max\big\{d_X(x_n,x),\ d_Y(y_n,y)\big\} \;\le\; d_3\big((x_n,y_n),(x,y)\big) \;\le\; d_X(x_n,x) + d_Y(y_n,y).$$

If the middle term tends to 0, the left inequality forces both coordinates to converge; if both coordinates converge, the right inequality forces the middle term to 0. This is Proposition 6.a.5**(a)** with $n = 2$ and the two factors allowed to be arbitrary metric spaces. By part **(b)** the same conclusion holds for d_1 and d_2, since equivalent metrics have the same convergent sequences.

Q.E.D.

Proof of Exercise 2.c.34:

Work in $V := C([a,b],\mathbb{R})$ with $\langle g,h\rangle := \int_a^b gh$, and take

$$g := \sqrt{f}, \qquad h := \frac{1}{\sqrt{f}}.$$

Both are continuous, because f is continuous and strictly positive, so the square root and the reciprocal are applied only where they are smooth. The three quantities Cauchy–Schwarz relates are then exactly the three appearing in the problem:

$$\langle g,h\rangle = \int_a^b \sqrt{f}\cdot\frac{1}{\sqrt{f}} = \int_a^b 1 = b-a, \qquad \|g\|^2 = \int_a^b f, \qquad \|h\|^2 = \int_a^b \frac{1}{f}.$$

Theorem 2.c.32 gives $\langle g,h\rangle^2 \le \|g\|^2\|h\|^2$, which is

$$(b-a)^2 \;\le\; \int_a^b f(x)\,dx \int_a^b \frac{1}{f(x)}\,dx,$$

as claimed.

Equality. Theorem 2.c.32 attains equality exactly when g and h are linearly dependent, that is when $\sqrt{f} = \lambda/\sqrt{f}$ for some $\lambda \in \mathbb{R}$, which says $f \equiv \lambda$. Since $f > 0$ the constant is positive, and conversely every positive constant gives equality, both sides being $(b-a)^2$. So equality holds precisely for the constant functions.

Remark 2.e.52 (*What the inner product bought*): Nothing in the statement mentions a vector space, and the inequality is about two ordinary one-variable integrals. What makes it a two-line proof rather than a calculation is choosing the right two **vectors** in a space of functions, and the choice is forced once one notices that the product of the two integrands must be 1, so that the pairing collapses to the length of the interval.

This is the first place in these notes where the abstract inner product does something the Euclidean one could not, and it is worth registering why: Theorem 2.c.32 was proved for **any** inner product space, so it applies verbatim to $C([a,b],\mathbb{R})$, where the vectors are functions and $\|g\|^2$ is an integral. The generality was not decoration. The discrete analogue, obtained the same way from the standard inner product on $\mathbb{R}^n$, is $\big(\sum_i x_i\big)\big(\sum_i 1/x_i\big) \ge n^2$ for positive x_i.

Q.E.D.

Proof of Exercise 2.c.35:

This is Theorem 2.c.32, proved in Section 2.c above by expanding $\|u - \pi_v(u)\|^2 \ge 0$, including the equality case. A second, independent proof, by minimising $\lambda \mapsto \|v - \lambda w\|^2$, is Proposition 2.a.7. Remark 2.c.33 explains why the two are the same computation approached from opposite ends.

Remark 2.e.53 (*A third route: the discriminant*): The official solution gives a third proof, worth recording because it is the one that generalises most mechanically. By definiteness, the quadratic polynomial

$$\lambda \;\longmapsto\; \langle \lambda v + w,\ \lambda v + w\rangle = \lambda^2\langle v,v\rangle + 2\lambda\langle v,w\rangle + \langle w,w\rangle$$

is non-negative for every real λ. A real quadratic with positive leading coefficient is non-negative exactly when its discriminant is ≤ 0, i.e.

$$\big(2\langle v, w\rangle\big)^2 - 4\langle v, v\rangle\langle w, w\rangle \leq 0,$$

which rearranges to the inequality. Equality forces the discriminant to vanish, so the polynomial has a real root λ_0, and definiteness turns $\langle \lambda_0 v + w, \lambda_0 v + w\rangle = 0$ into $\lambda_0 v + w = 0$, which is the asserted linear dependence. Note that this is the previous proof with the geometry stripped out. Here λ_0 is again the projection coefficient, now arriving as a root rather than as a minimiser.

Q.E.D.

Proof of Exercise 2.c.36:

This is Proposition 2.a.8**(a)**, proved immediately after Proposition 2.a.7, which is exactly the input the triangle inequality needs and the reason the two problems appear adjacently on this sheet. In one line: expanding $\|u + v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$ and bounding the middle term by $2\|u\|\,\|v\|$ turns the right-hand side into $(\|u\| + \|v\|)^2$. **Q.E.D.**

Solution to Exercise 2.c.37:

Write $e_1, \dots, e_n$ for the standard basis and $|\cdot|$ for the Euclidean norm.

(a) Expand $x = \sum_{i=1}^n x_i e_i$ and use the triangle inequality and homogeneity of f (Definition 2.a.2):

$$f(x) \;\leq\; \sum_{i=1}^n f(x_i e_i) \;=\; \sum_{i=1}^n |x_i|\, f(e_i) \;\leq\; \Big(\max_i f(e_i)\Big) \sum_{i=1}^n |x_i| \;\leq\; \Big(\max_i f(e_i)\Big)\sqrt{n}\,|x|,$$

the last step by Cauchy–Schwarz applied to $(|x_1|, \dots, |x_n|)$ and $(1, \dots, 1)$. So $C_1 := \sqrt{n}\max_i f(e_i)$ works, and $C_1 > 0$ because $e_i \neq 0$ forces $f(e_i) > 0$ by definiteness.

(b) The reverse triangle inequality for f, namely $f(x) \leq f(y) + f(x - y)$ together with its mirror image, gives $|f(x) - f(y)| \leq f(x - y)$. Combined with part **(a)** this yields

$$|f(x) - f(y)| \;\leq\; f(x - y) \;\leq\; C_1 |x - y|.$$

So f is C_1-Lipschitz with respect to the **Euclidean** distance (Definition 5.b.20), hence continuous (Remark 5.b.21). Note which distance is which here: the claim is continuity of f as a function on the metric space $(\mathbb{R}^n, |\cdot|)$, and part **(a)** is what makes that meaningful.

(c) The unit sphere $S := \{x : |x| = 1\}$ is closed (preimage of $\{1\}$ under the continuous $|\cdot|$) and bounded, hence **compact** by Theorem 7.b.28. By part **(b)** the function f is continuous on S, so it attains a minimum there (Item **(b)**), say $c_2 := f(x_0)$ with $|x_0| = 1$. Since $x_0 \neq 0$, definiteness gives $c_2 = f(x_0) > 0$.

(d) For $x \neq 0$ the vector $x/|x|$ lies in S, so $f(x/|x|) \geq c_2$, and homogeneity gives

$$f(x) = |x|\, f\!\left(\frac{x}{|x|}\right) \;\geq\; c_2\,|x|.$$

For $x = 0$ both sides vanish. Together with part **(a)**, $c_2|x| \leq f(x) \leq C_1|x|$.

(e) Apply parts **(a)** and **(d)** to both norms: $c_2|x| \leq f(x) \leq C_1|x|$ and $\tilde{c}_2|x| \leq \tilde{f}(x) \leq \tilde{C}_1|x|$, all four constants positive. Then

$$\tilde{f}(x) \;\leq\; \tilde{C}_1|x| \;\leq\; \frac{\tilde{C}_1}{c_2}\, f(x), \qquad\qquad \tilde{f}(x) \;\geq\; \tilde{c}_2|x| \;\geq\; \frac{\tilde{c}_2}{C_1}\, f(x),$$

so $C := \max\big\{\tilde{C}_1/c_2,\; C_1/\tilde{c}_2\big\}$ satisfies $C^{-1} f \leq \tilde{f} \leq C f$. Equivalence of norms is thus an equivalence relation, and the Euclidean norm serves as the common reference point.

Remark 2.e.54 (*Where finite dimension is indispensable*): The result is false in infinite dimensions, and the proof shows exactly which step breaks: part **(a)** expands x in a **finite** basis and part **(c)** uses compactness of the unit sphere. In an infinite-dimensional normed space the closed unit ball is never compact, and one can write down inequivalent norms. Take for instance $\|\cdot\|_\infty$ and $\int_0^1 |\cdot|$ on $C^0([0,1])$, where a tall thin spike has supremum norm 1 and integral norm arbitrarily small. Every appearance of *"all norms on $\mathbb{R}^n$ are equivalent"* later in the course silently uses $n < \infty$.

Q.E.D.

Proof of Exercise 2.c.39:

The monomials $\mathcal{B} := \{1, x, x^2, x^3, \dots\}$ form a basis: every polynomial is by definition a **finite** linear combination $\sum_{k=0}^n a_k x^k$ of them, so $\mathcal{B}$ spans $\mathbb{R}[x]$; and $\sum_{k=0}^n a_k x^k = 0$ as a polynomial forces every coefficient $a_k = 0$ (a nonzero polynomial of degree n has at most n roots, so it cannot vanish identically on all of $\mathbb{R}$), so $\mathcal{B}$ is linearly independent.

To see $\mathbb{R}[x]$ is infinite-dimensional, suppose for contradiction it had finite dimension d. Then $1, x, \dots, x^d$, a set of $d+1$ vectors of $\mathbb{R}[x]$, would have to be linearly dependent, since any $d+1$ vectors in a d-dimensional space are. But they are linearly independent by the argument above (restricted to $n \le d$): a contradiction. So no finite basis exists, and the (necessarily infinite) $\mathcal{B}$ above is one. Q.E.D.

Proof of Exercise 2.c.41:

Write $\Psi^i \in \mathbb{R}^n$ for the i-th **row** of Ψ and $\Phi_j \in \mathbb{R}^n$ for the j-th **column** of Φ, so that by the definition of matrix multiplication

$$(\Psi\Phi)_{ij} = \langle \Psi^i,\ \Phi_j \rangle.$$

Cauchy–Schwarz (Theorem 2.c.32) applied to this pair of vectors of $\mathbb{R}^n$ gives $\left|(\Psi\Phi)_{ij}\right|^2 \le \|\Psi^i\|^2 \, \|\Phi_j\|^2$. Summing over all entries and factorising the resulting double sum,

$$\|\Psi\Phi\|^2 = \sum_{i=1}^m \sum_{j=1}^d \left|(\Psi\Phi)_{ij}\right|^2 \le \sum_{i=1}^m \sum_{j=1}^d \|\Psi^i\|^2 \|\Phi_j\|^2 = \left(\sum_{i=1}^m \|\Psi^i\|^2\right)\left(\sum_{j=1}^d \|\Phi_j\|^2\right) = \|\Psi\|^2 \, \|\Phi\|^2,$$

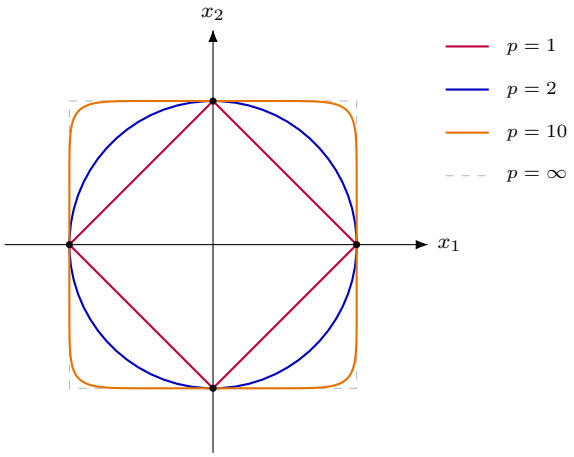
using that the Hilbert–Schmidt norm squared is the sum of the squares of all entries, read row by row for Ψ and column by column for Φ. Taking square roots gives $\|\Psi\Phi\| \le \|\Psi\| \, \|\Phi\|$.

Remark 2.e.55 (*Submultiplicativity, and what it later buys*): The Hilbert–Schmidt norm is the same object as the Frobenius norm met in Exercise 2.b.28 part **(e)** on the previous sheet, where it was shown to be the Euclidean norm of the matrix read as a vector of $\mathbb{R}^{n^2}$. The inequality just proved says that this norm is "*submultiplicative*" (German: „*submultiplikativ*"), which is exactly the property that makes it useful for estimating compositions of linear maps. It is also what will later let $\|Df\|$ be handled like an absolute value in the mean value inequality (Important remark 11.a.2).

Q.E.D.

Solution to Exercise 2.c.42:

(a) For $n = 2$, $\{x : |x|_1 \le 1\}$ is the diamond with vertices $(\pm 1, 0)$, $(0, \pm 1)$; $\{x : |x|_2 \le 1\}$ is the Euclidean unit disc; $\{x : |x|_{10} \le 1\}$ is a region close to the square $[-1,1]^2$, bounded by a curve whose rounded corners sharpen towards the square's corners as p grows. All three contain the four points $(\pm 1, 0)$, $(0, \pm 1)$, since $|e_i|_p = 1$ for every p, and they increase with p, which is the geometric content of part **(b)**.



(b) Let $M := \max_i |x_i|$. Then

$$M \le |x|_p = M\Big(\sum_{i=1}^{n}\Big(\frac{|x_i|}{M}\Big)^p\Big)^{1/p} \le M\, n^{1/p}.$$

Since $n^{1/p} \to 1$ as $p \to \infty$, the squeeze theorem gives $|x|_\infty = \lim_{p\to\infty} |x|_p = M = \max_i |x_i|$.

(c) Set $u_i := a_i^{p/2}$ and $v_i := a_i^{(p-2)/2} b_i$, so that $u_i v_i = a_i^{p-1} b_i$, $u_i^2 = a_i^p$, and $v_i^2 = a_i^{p-2} b_i^2$. Cauchy–Schwarz in $\mathbb{R}^n$ gives

$$\Big(\sum_{i=1}^{n} a_i^{p-1} b_i\Big)^2 = \Big(\sum_{i=1}^{n} u_i v_i\Big)^2 \le \Big(\sum_{i=1}^{n} u_i^2\Big)\Big(\sum_{i=1}^{n} v_i^2\Big) = \Big(\sum_{i=1}^{n} a_i^p\Big)\Big(\sum_{i=1}^{n} a_i^{p-2} b_i^2\Big).$$

(d) Fix x, y with all coordinates strictly positive, let $d := x - y$, and write $z(t) := tx + (1-t)y = y + td$, so $z_i(t) > 0$ for all $t \in [0,1]$. Let $g(t) := f(t)^p = \sum_i z_i(t)^p$, so $f = g^{1/p}$. Since $z_i'(t) = d_i$,

$$g'(t) = p \sum_i z_i(t)^{p-1} d_i.$$

Differentiating $f = g^{1/p}$ twice and simplifying,

$$f''(t) = (p-1)\, g(t)^{1/p-2} \left[g(t) \sum_i z_i(t)^{p-2} d_i^2 - \Big(\sum_i z_i(t)^{p-1} d_i\Big)^2 \right].$$

Applying the inequality of the previous point with $a_i = z_i(t) > 0$ and $b_i = d_i$ (any sign is allowed, since the inequality only uses b_i^2 on the right and squares the left) shows the bracket is ≥ 0. Since $g(t) > 0$ and $p \ge 1$, we get $f''(t) \ge 0$ for $p > 1$ (for $p = 1$, f is affine in t and hence convex). Thus f is convex on $[0,1]$.

(e) Convexity of f at $t = \frac{1}{2}$ gives $f(\frac{1}{2}) \le \frac{1}{2}(f(0) + f(1))$, i.e.

$$\Big|\frac{x+y}{2}\Big|_p \le \frac{|x|_p + |y|_p}{2} \quad\Longrightarrow\quad |x+y|_p \le |x|_p + |y|_p$$

for x, y with strictly positive coordinates. For general $x, y \in \mathbb{R}^n$, write $\hat{x} := (|x_1|, \dots, |x_n|)$, $\hat{y} := (|y_1|, \dots, |y_n|)$; these have non-negative coordinates (the strictly positive case above extends to this case by continuity), and $|x+y|_p \le |\hat{x} + \hat{y}|_p \le |\hat{x}|_p + |\hat{y}|_p = |x|_p + |y|_p$.

(f) For $p \in (0,1)$, the triangle inequality fails, so $|\cdot|_p$ is not a norm: taking $x = e_1$, $y = e_2$ gives $|x|_p = |y|_p = 1$ but $|x+y|_p = 2^{1/p} > 2 = |x|_p + |y|_p$.

Q.E.D.

Solution to Exercise 2.c.43:

Throughout write $S(t) := \sum_{i=1}^{n} x_i^t$, so that $\mu_t(x) = \big(S(t)/n\big)^{1/t}$ for $t \neq 0$, and set

$$g(t) := \log \mu_t(x) = \frac{h(t)}{t}, \qquad h(t) := \log \frac{S(t)}{n}.$$

Note $h(0) = \log(n/n) = 0$; this single fact does most of the work below.

(a) Let $M := \max_i x_i$ and $m := \min_i x_i$, and let k be the number of indices attaining M. Factoring out the largest term,

$$\mu_t(x) = M \left(\frac{1}{n} \sum_{i=1}^{n} \left(\frac{x_i}{M} \right)^t \right)^{1/t},$$

and as $t \to +\infty$ the bracket tends to k/n, whose t-th root tends to 1. Hence $\mu_{+\infty}(x) = M = \max_i x_i$. Replacing t by $-t$ turns the maximum into the minimum, so $\mu_{-\infty}(x) = m = \min_i x_i$.

For $t \to 0$ the exponent and the logarithm both degenerate, which is exactly the $\frac{0}{0}$ that $g = h/t$ resolves: h is differentiable with $h(0) = 0$, so

$$\lim_{t \to 0} g(t) = h'(0) = \frac{S'(0)}{S(0)} = \frac{1}{n} \sum_{i=1}^{n} \log x_i,$$

using $S'(t) = \sum_i x_i^t \log x_i$ and $S(0) = n$. Exponentiating,

$$\mu_0(x) = \big(x_1 x_2 \cdots x_n\big)^{1/n},$$

the "*geometric mean*" (German: „*geometrisches Mittel*").

(b) The inequality is Jensen applied to $\varphi(s) := s \log s$, which is convex on $(0, \infty)$ because $\varphi''(s) = 1/s > 0$. With the uniform weights $\frac{1}{n}$,

$$\frac{1}{n} \sum_{i=1}^{n} a_i \log a_i \;\geq\; \varphi\left(\frac{1}{n} \sum_{i=1}^{n} a_i \right) = \frac{A}{n} \log \frac{A}{n}, \qquad A := a_1 + \cdots + a_n.$$

Multiplying by $n/A > 0$ gives exactly

$$\sum_{i=1}^{n} \frac{a_i \log a_i}{a_1 + \cdots + a_n} \;\geq\; \log\left(\frac{a_1 + \cdots + a_n}{n} \right).$$

(c) *Continuity.* On $\mathbb{R} \setminus \{0\}$, $g = h/t$ is a quotient of smooth functions with non-vanishing denominator. At $t = 0$ the value $g(0) := h'(0)$ computed in part **(a)** is precisely the limit, so g, and hence also $f = e^g$, is continuous everywhere.

Monotonicity. Differentiating $g = h/t$ for $t \neq 0$,

$$t^2 g'(t) = t\, h'(t) - h(t) = \frac{t\, S'(t)}{S(t)} - \log \frac{S(t)}{n}.$$

Now substitute $a_i := x_i^t$. Then $\log a_i = t \log x_i$, so $t\, S'(t) = \sum_i a_i \log a_i$ and $S(t) = \sum_i a_i$, and the displayed quantity becomes

$$\sum_{i=1}^{n} \frac{a_i \log a_i}{a_1 + \cdots + a_n} \;-\; \log\left(\frac{a_1 + \cdots + a_n}{n} \right) \;\geq\; 0$$

by part **(b)**. Hence $g' \geq 0$ on $t \neq 0$, and by continuity g is non-decreasing on all of $\mathbb{R}$; so is $f = e^g$.

This is why part **(b)** was placed where it is: it is not a separate exercise but the derivative computation of part **(c)** in disguise.

(d) Both inequalities are instances of $\mu_s(x) \leq \mu_t(x)$ for $s \leq t$, i.e. of part **(c)**.

- **Arithmetic–geometric:** $\mu_0(x) \leq \mu_1(x)$ reads $(x_1 \cdots x_n)^{1/n} \leq \frac{1}{n}(x_1 + \cdots + x_n)$; multiplying by n gives $n(x_1 x_2 \cdots x_n)^{1/n} \leq x_1 + \cdots + x_n$.

- **Arithmetic–quadratic:** $\mu_1(x) \le \mu_2(x)$ reads $\frac{1}{n}\sum_i x_i \le \left(\frac{1}{n}\sum_i x_i^2\right)^{1/2}$. Both sides are positive, so squaring preserves the inequality: $\frac{1}{n^2}\left(\sum_i x_i\right)^2 \le \frac{1}{n}\sum_i x_i^2$, and multiplying by n^2 gives $(x_1+\cdots+x_n)^2 \le n(x_1^2+\cdots+x_n^2)$.

(e) **(*) Yes**, and considerably more than C^1. Each $x_i^t = e^{t\log x_i}$ is real-analytic in t, hence so is S, and $S>0$ everywhere, so $h=\log(S/n)$ is real-analytic too. Because $h(0)=0$, the quotient $h(t)/t$ has a **removable** singularity at the origin: writing $h(t)=\sum_{k\ge1}c_kt^k$ gives $g(t)=\sum_{k\ge1}c_kt^{k-1}$, a convergent power series near 0. So g, and therefore $f=e^g$, is real-analytic on all of $\mathbb{R}$, in particular C^∞, and so certainly C^1.

The value at the origin is worth recording. From $g(t)=h'(0)+\frac12h''(0)t+O(t^2)$,

$$f'(0)=f(0)\,g'(0)=\mu_0(x)\cdot\tfrac12h''(0)=\mu_0(x)\cdot\frac12\left(\frac1n\sum_{i=1}^n(\log x_i)^2-\left(\frac1n\sum_{i=1}^n\log x_i\right)^2\right),$$

using $h''=(S''S-(S')^2)/S^2$ at $t=0$ with $S(0)=n$, $S'(0)=\sum_i\log x_i$ and $S''(0)=\sum_i(\log x_i)^2$. The bracket is the variance of the numbers $\log x_i$, so $f'(0)\ge0$, with equality exactly when all x_i agree. In that case $\mu_t(x)$ is that common value for every t, and f is constant.

Q.E.D.

Remark 2.e.56 (*Why parts **(b)** and **(c)** are one statement*)**:** Jensen's inequality, used above without being stated, is available as Proposition 14.a.8 (Chapter 14). The official hint for 4.4.2 routes through the concavity of log together with Cauchy–Schwarz. The solution above instead applies Jensen once, to the convex function $\varphi(s)=s\log s$. Both arrive at the same inequality, but the second makes visible why the two are really a single statement: substituting $a_i:=x_i^t$ turns the inequality of part **(b)** into the sign of $g'(t)$ in part **(c)**. That is worth seeing, because monotonicity of the power means is the whole point of the problem, and the classical AM–GM inequality falls out of it as the single comparison $\mu_0\le\mu_1$.

Open and Closed Sets (Chapter 3)

With a distance in hand, the next question is which subsets a metric singles out. The answer is built up from a single object, the open ball, and everything in this chapter is defined in terms of it: open and closed sets first, then the interior, closure and boundary of an arbitrary set.

The chapter ends by drawing the unit ball of each of the three standard metrics on $\mathbb{R}^2$, and the result is worth anticipating. The three balls are a square, a diamond and a disk, so they are as visibly different as three convex sets can be. Yet they induce exactly the same open sets. That gap between what a metric looks like and what it decides is the first genuinely topological phenomenon in the course.

Open balls, open and closed sets (Section 3.a)

Definition 3.a.1 (*Open ball*): Let (X, d) be a metric space, let $x \in X$, and let $r \in \mathbb{R}$. Then we define the "*open ball*" of radius r as

$$B_r(x) := \{\, \boxed{y \in X} : d(y, x) < r \,\}$$

Important remark 3.a.2: The boxed part is an important subtlety: the ambient set X in $y \in X$ is what makes openness relative to the space, and it is exactly what the common misconception in Important remark 4.a.13 turns on.

AI-Note 3.1 (*Two things to know about this definition*): First, Definition 3.a.1 is stated for $r \in \mathbb{R}$ rather than $r > 0$. Harmless here, but loose; the two degenerate cases are spelled out in the remark below.

Second, and more likely to trip you up when reading alongside the lecture: **the lecture notes put the arguments the other way round**, writing $B(x, r)$ where this document writes $B_r(x)$ (Definition 9.38, p. 11). The two mean exactly the same set, centre x and radius r; only the typography differs. This document keeps $B_r(x)$, which is what every later chapter here is written in. The one thing to watch is that in the lecture's ordering the centre comes **first**, so $B(0, 1)$ and $B_1(0)$ are the same ball while $B(1, 0)$ and $B_1(0)$ are not.

AI-Example 3.a.3 (*Open Balls in the Discrete Metric*): Let (X, d_{disc}) be a discrete metric space. The open ball $B_r(x)$ centered at $x \in X$ with radius $r > 0$ is

$$B_r(x) = \begin{cases} \{x\} & \text{if } 0 < r \leq 1, \\ X & \text{if } r > 1. \end{cases}$$

Hence every singleton $\{x\}$ is open in the discrete metric, and consequently every subset of X is both open and closed ("*clopen*"). This is exactly what Exercise 3.b.16 below asks you to verify for X and $\emptyset$ in a general metric space.

Remark 3.a.4 (*Two degenerate balls worth knowing*): Taking Definition 3.a.1 literally at the edges of its range is a cheap way to test whether you have read it correctly.

- If $r \leq 0$, then $B_r(x) = \emptyset$, and **not** $\{x\}$. The condition is $d(y, x) < r$, strictly, and $d(x, x) = 0 \not< 0$. So even the centre is missing from $B_0(x)$.
- In the discrete metric $B_1(x) = \{x\}$, while $B_{1.01}(x) = X$: a ball can jump from a single point to the entire space without passing through anything in between. Radii do not have to behave like radii.

These are the two cases that make *"shrink the ball a little"* arguments go wrong if applied without thinking.

AI-Exercise 3.a.5 (*Open balls are open*): The name promises it, but Definition 3.a.1 does not say it. Show that in any metric space (X, d) the ball $B_r(x)$ is an open set in the sense of Definition 3.a.6.

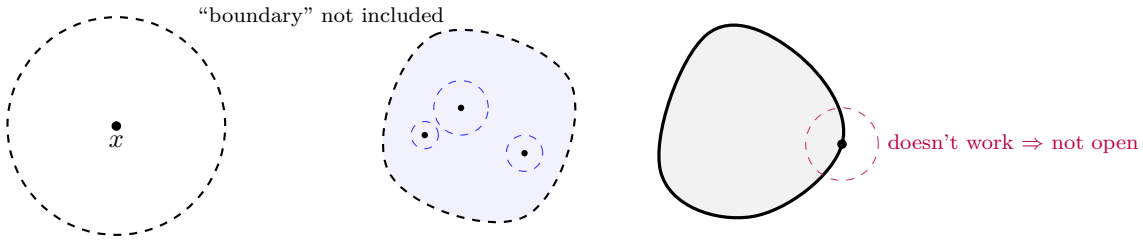
Hint: *Given $y \in B_r(x)$, the number $\varepsilon := r - d(y, x)$ is strictly positive. Draw the picture, then let the triangle inequality do the work.*

Solution: *See solution on Page 51*

Definition 3.a.6 (*Open and closed sets*): Then, we call a subset $U \subseteq X$ "*open*" (German: „*offen*“) if and only if

$$\forall y \in U \ \exists \varepsilon > 0 \text{ such that } B_\varepsilon(y) \subseteq U.$$

And we call $A \subseteq X$ "*closed*" (German: „*abgeschlossen*“) if and only if $X \setminus A$ is open.



Example 3.a.7 (*Clopen sets other than X and $\emptyset$*): Exercise 3.b.16 shows X and $\emptyset$ are always clopen, but nothing in Definition 3.a.6 forces these to be the **only** clopen subsets of X. Take $X := (0,1) \cup (2,3) \subset \mathbb{R}$ with the standard metric: both $(0,1)$ and $(2,3)$ are clopen **in** X. Each is open in $\mathbb{R}$, hence open in X; and each is the complement in X of the other, so each is also closed in X. Whether a space has clopen subsets besides X and $\emptyset$ turns out to be exactly the question Chapter 8 is organised around. A space with none is called "*connected*".

Proposition 3.a.8 (*How open and closed sets combine*): Let (X, d) be a metric space.

(a) **Arbitrary** unions of open sets are open: if $\{U_i\}_{i \in I}$ are open, so is $\bigcup_{i \in I} U_i$.

(b) **Finite** intersections of open sets are open: if $U_1, \dots, U_k$ are open, so is $\bigcap_{i=1}^{k} U_i$.

(c) **Arbitrary** intersections of closed sets are closed, and **finite** unions of closed sets are closed.

Proof:

(a) Let $x \in \bigcup_i U_i$. Then $x \in U_{i_0}$ for some i_0, and since U_{i_0} is open there is $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U_{i_0} \subseteq \bigcup_i U_i$.

(b) Let $x \in \bigcap_{i=1}^{k} U_i$. For each i pick $\varepsilon_i > 0$ with $B_{\varepsilon_i}(x) \subseteq U_i$, and set $\varepsilon := \min\{\varepsilon_1, \dots, \varepsilon_k\}$. Then $\varepsilon > 0$, and **this is the only place finiteness is used**: a minimum of finitely many positive numbers is positive, whereas an infimum of infinitely many can be 0. Since $B_\varepsilon(x) \subseteq U_i$ for every i, we conclude that $B_\varepsilon(x) \subseteq \bigcap_i U_i$.

(c) Take complements and apply **(a)**, **(b)** via De Morgan: $X \setminus \bigcap_i A_i = \bigcup_i (X \setminus A_i)$ and $X \setminus \bigcup_{i=1}^{k} A_i = \bigcap_{i=1}^{k} (X \setminus A_i)$.

Q.E.D.

Important remark 3.a.9 (*The asymmetry is real, not an artefact of the proof*): It is tempting to read *"finite"* in **(b)** as laziness. It is not: an infinite intersection of open sets genuinely need not be open. In $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty} \left(-\tfrac{1}{n}, \tfrac{1}{n}\right) = \{0\},$$

an intersection of open intervals which is not open. The proof shows exactly where it breaks: $\varepsilon := \min_i \varepsilon_i$ is the whole argument, and an infimum of infinitely many positive numbers can be 0. Dually, $\bigcup_{n\geq 1} \left[\frac{1}{n}, 1\right] = (0,1]$ is an infinite union of closed sets that is not closed.

AI-Note 3.2: Parts **(a)** and **(b)**, together with the fact that X and $\emptyset$ are open (Exercise 3.b.16), are exactly the axioms of a "*topology*", which is how the subject is set up when one drops the metric altogether (Section 2.a, *"fourth semester"*); part **(c)** is their translation to closed sets. That is why they are worth isolating: they are the only facts about open sets that survive into the general theory.

Example 3.a.10 (*Open and closed sets in* $\mathbb{R}$): Consider the following subsets of $\mathbb{R}$ with the standard metric:

- **(a)** **Is $\mathbb{N} = \{0, 1, 2, \dots\}$ open, closed, or neither?** It is **closed**, since its complement $\mathbb{R} \setminus \mathbb{N} = (-\infty, 0) \cup \bigcup_{n=0}^{\infty} (n, n+1)$ is a union of open intervals, which is open by Proposition 3.a.8**(a)**. It is not open, because no open ball around any $n \in \mathbb{N}$ is contained in $\mathbb{N}$.
- **(b)** **Let $A := \{\frac{1}{n} \mid n = 1, 2, \dots\}$.** The set A is **neither** open nor closed. It is not open for the same reason as $\mathbb{N}$. It is not closed because 0 is an accumulation point of A, but $0 \notin A$, meaning its complement $\mathbb{R} \setminus A$ is not open (any open ball around 0 intersects A).
- **(c)** **Let $F := \{\frac{1}{n} \mid n = 1, 2, \dots\} \cup \{0\}$.** The set F is **closed**. By adding the limit point 0, the complement $\mathbb{R} \setminus F$ becomes a union of open intervals, which is open.

Exercise 3.a.11 (*A longer self-test*): The following list makes a good self-test. Each subset of $\mathbb{R}$ carries the standard metric; decide for each whether it is **open**, **closed**, **both**, or **neither**.

- **(a)** $[0, \infty)$
- **(b)** $\mathbb{Q}$
- **(c)** $\{x - \lfloor x \rfloor : x \in \mathbb{R}\}$
- **(d)** $S := \{x \in \mathbb{R} : x \cdot 2^p \in [1, 2] \text{ for some integer } p \geq 0\}$
- **(e)** $T := \{x \in \mathbb{R} : x \cdot 2^p \in (1, 2) \text{ for some integer } p \geq 0\}$
- **(f)** $\{x_n : n \in \mathbb{N}\}$, where $x_0 := 1$ and $x_n := \frac{1}{2}(x_{n-1} + x_{n-1}^2)$
- **(g)** $\{\mathrm{Re}(c) : c \in \mathbb{C},\ \zeta(c) = 0\}$, where ζ is the Riemann zeta function

Compare **(d)** with **(e)** before reading the solution: the two differ only in one bracket.

Info: *This exercise is taken from Linus Lüchinger's slides (Week 2, slide 20).*

Solution: *See solution on Page 54*

Exercise 3.a.12 (*Relative openness*): Let $X := (\{0\} \times \mathbb{R}) \cup (0, 1)^2$ be equipped with the subspace metric from $\mathbb{R}^2$. Consider the following subsets of X:

$$A := \{(x, y) \in \mathbb{R}^2 \mid x = 0, y \in (2, 3)\}$$
$$B := \left\{(x, y) \in \mathbb{R}^2 \;\middle|\; \left(x - \tfrac{1}{2}\right)^2 + \left(y - \tfrac{1}{2}\right)^2 < \tfrac{1}{16}\right\}$$

Are A and B open with respect to X? Are they open with respect to $\mathbb{R}^2$?

Info: *This exercise is taken from Jérôme Paschoud's class notes (Week 2.1, p. 5).*

Proof of Exercise 3.a.12:

- A is **open with respect to X**, but **not open with respect to $\mathbb{R}^2$**. It is open in X because $A = X \cap U$ where $U = (-\varepsilon, \varepsilon) \times (2, 3)$ is an open set in $\mathbb{R}^2$. It is not open in $\mathbb{R}^2$ because any open ball around a point in A will contain points with $x \neq 0$, which are not in A.
- B is **open with respect to both X and $\mathbb{R}^2$**. B is exactly the open ball $B_{1/4}((\frac{1}{2}, \frac{1}{2}))$ in $\mathbb{R}^2$, which is open in $\mathbb{R}^2$. Since $B \subseteq (0, 1)^2 \subset X$, we have $B = X \cap B$, so B is also open relative to X.

Q.E.D.

Interior, closure and boundary (Section 3.b)

Definition 3.b.13 (*Interior, closure, and boundary*): Let (X, d) be a metric space and $A \subseteq X$. The "*interior*" (German: „*Inneres*“) of A is

$$\operatorname{int}(A) := \{x \in A : \exists\, r > 0 \text{ such that } B_r(x) \subseteq A\},$$

the largest open subset of X contained in A. The "*closure*" (German: „*Abschluss*“) of A is

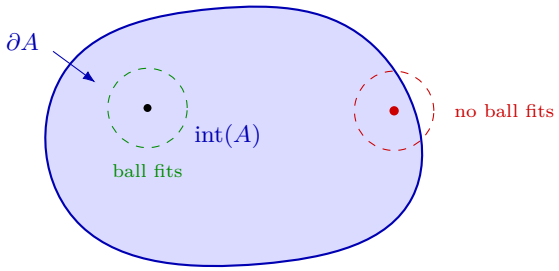
$$\overline{A} := \{x \in X : B_r(x) \cap A \neq \emptyset \text{ for all } r > 0\},$$

the smallest closed subset of X containing A. The "*boundary*" (German: „*Rand*“) of A is $\partial A := \overline{A} \setminus \operatorname{int}(A)$.

AI-Example 3.b.14 (*Interior, Closure, and Boundary of an Interval*): Applying Definition 3.b.13 to $A := (0, 1] \subset \mathbb{R}$ under the standard Euclidean metric:

- The **interior** is $\operatorname{int}(A) = (0, 1)$, the largest open set contained in A.
- The **closure** is $\overline{A} = [0, 1]$, the smallest closed set containing A.
- The **boundary** is $\partial A = \overline{A} \setminus \operatorname{int}(A) = \{0, 1\}$.

Note that $1 \in A$ while $0 \notin A$: the boundary is not required to lie inside the set, nor outside it. That is the point of defining it as a difference of two other sets rather than as *"the edge of A"*.



$\operatorname{int}(A)$: points with a ball inside A · ∂A: points every ball of which meets both A and $X \setminus A$ · $\overline{A} = \operatorname{int}(A) \cup \partial A$

Remark 3.b.15 (*Interior and closure are dual*): The two constructions are not independent: taking complements swaps them,

$$X \setminus \operatorname{int}(A) = \overline{X \setminus A}, \qquad X \setminus \overline{A} = \operatorname{int}(X \setminus A).$$

Both read off Definition 3.b.13 directly, since x fails to have a ball inside A exactly when every ball around x meets $X \setminus A$. In particular $\partial A = \partial(X \setminus A)$: a set and its complement share their boundary, which is why the boundary of the open disk and of its closed complement are the same circle.

Proof of the two extremal properties in Definition 3.b.13:

$\operatorname{int}(A)$ is the largest open subset of A. Let $x \in \operatorname{int}(A)$, with $B_r(x) \subseteq A$. Every $y \in B_r(x)$ has a ball $B_\varepsilon(y) \subseteq B_r(x) \subseteq A$, because open balls are open (AI-Exercise 3.a.5); so $B_r(x) \subseteq \operatorname{int}(A)$, and $\operatorname{int}(A)$ is open. If $U \subseteq A$ is open, each $x \in U$ has a ball inside $U \subseteq A$, so $x \in \operatorname{int}(A)$. Hence every open subset of A lies in $\operatorname{int}(A)$.

$\overline{A}$ is the smallest closed superset of A. By Remark 3.b.15, $X \setminus \overline{A} = \operatorname{int}(X \setminus A)$, which is open by the first part, so $\overline{A}$ is closed, and it contains A since every ball around a point of A meets A.

If $F \supseteq A$ is closed, then $X \setminus F$ is an open subset of $X \setminus A$, so $X \setminus F \subseteq \operatorname{int}(X \setminus A) = X \setminus \overline{A}$ by the first part, that is, $\overline{A} \subseteq F$. Q.E.D.

Exercise 3.b.16 (*X and $\emptyset$ are clopen*): Let (X, d) be a metric space. Show that X and $\emptyset$ are both open and closed.

Info: *This exercise is taken from Corsin Nick's class notes (Week 2, p. 7).*

Proof of Exercise 3.b.16:

X is open, since $B_r(x) \subseteq X$ for all $r > 0$ and $x \in X$ (Definition 3.a.1). The empty set contains no points and is therefore vacuously open. Since X and $\emptyset$ are each other's complement in X, they are also closed. Q.E.D.

Example 3.b.17 (*Interior, closure, and boundary in $\mathbb{R}$ and $\mathbb{R}^2$*): Applying Definition 3.b.13 to various sets:

(a) Let $\Omega := \mathbb{Z} \subset \mathbb{R}$. Then $\operatorname{int}(\Omega) = \emptyset$ (no interval fits inside $\mathbb{Z}$), $\overline{\Omega} = \mathbb{Z}$ (it is already closed), and $\partial\Omega = \mathbb{Z} \setminus \emptyset = \mathbb{Z}$.

(b) Let $\Omega := B_1(0) \subset \mathbb{R}^2$. Then $\operatorname{int}(\Omega) = B_1(0)$ (it is already open), the closure is the closed disk $\overline{\Omega} = \{x \in \mathbb{R}^2 \mid d(x, 0) \leq 1\}$, and the boundary is the unit circle $\partial\Omega = \{x \in \mathbb{R}^2 \mid d(x, 0) = 1\}$.

(c) Let $\Omega := \mathbb{Q} \subset \mathbb{R}$. Then $\operatorname{int}(\Omega) = \emptyset$ (every open interval contains irrationals), $\overline{\Omega} = \mathbb{R}$ (every open interval contains rationals, so every real number is a closure point), and $\partial\Omega = \mathbb{R} \setminus \emptyset = \mathbb{R}$.

Info: *This example is taken from Sascha Brack's class notes (Week 2, p. 2).*

Exercise 3.b.18 (*Closure and boundary of a ball in $\mathbb{R}^n$*): For balls $B_r(x)$ in $\mathbb{R}^n$ with the Euclidean metric, prove that

$$\overline{B_r(x)} = \{y \in \mathbb{R}^n : d(x, y) \leq r\}$$

and deduce $\partial B_r(x) = \{y \in \mathbb{R}^n : d(x, y) = r\}$.

Info: *This exercise is taken from the Analysis II lecture script, p. 15.*

Solution: *See solution on Page 51*

Remark 3.b.19: Item **(b)** of Example 3.b.17 above is exactly the case $r = 1$, $x = 0$, $n = 2$ of this exercise, worked here for general x, r and n.

Exercise 3.b.20 (*Interior, closure, and boundary of a mixed set*): Let $E := (\{0\} \times \mathbb{R}) \cup (0, 1)^2 \subset \mathbb{R}^2$. Find the interior $\operatorname{int}(E)$, the closure $\overline{E}$, and the boundary ∂E with respect to the standard metric on $\mathbb{R}^2$.

Info: *This exercise is taken from Jérôme Paschoud's class notes (Week 2.1, p. 6).*

Proof of Exercise 3.b.20:

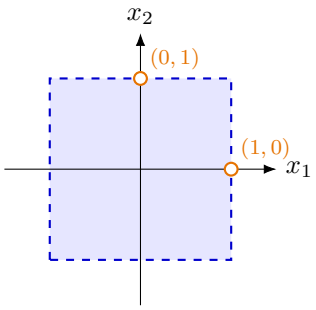
- **Interior:** $\operatorname{int}(E) = (0, 1)^2$. The line $\{0\} \times \mathbb{R}$ contains no 2D open balls, so it contributes nothing to the interior. The open square $(0, 1)^2$ is already open in $\mathbb{R}^2$.
- **Closure:** $\overline{E} = (\{0\} \times \mathbb{R}) \cup [0, 1]^2$. The line $\{0\} \times \mathbb{R}$ is closed. The closure of the open square is the closed square $[0, 1]^2$. The closure of a finite union is the union of the closures.
- **Boundary:** $\partial E = \overline{E} \setminus \operatorname{int}(E) = (\{0\} \times \mathbb{R}) \cup \left([0, 1]^2 \setminus (0, 1)^2\right)$. The second term is the perimeter of the square.

Q.E.D.

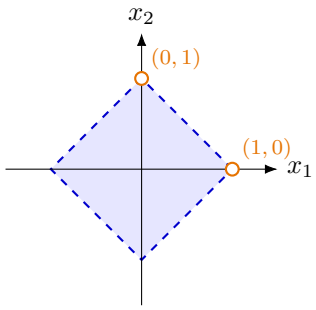
Unit balls of the three standard metrics (Section 3.c)

Warm up 3.c.21: What is the shape of $B_1(0) \subseteq \mathbb{R}^2$ with the supremum metric? With the Manhattan metric?

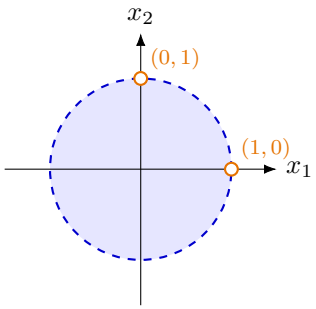
Answer 3.c.22: Supremum: the axis-aligned **open square** $(-1,1)^2$. Manhattan: the **open diamond** $\{|x_1| + |x_2| < 1\}$. In both cases the strict inequality in Definition 3.a.1 matters: the points $(\pm 1, 0)$ and $(0, \pm 1)$ sit on the boundary and are **not** in the ball. They are the corners of the diamond and the edge midpoints of the square, and are drawn hollow below.



Supremum metric d_3: open square $(-1,1)^2$



Manhattan metric d_1: open diamond



Standard metric d_2: open disk

Remark 3.c.23 (*The unit ball remembers which metric it came from*): Three metrics, three unmistakably different unit balls: the square for d_3, the disk for d_2, the diamond for d_1. This is the quickest way to tell the three apart, and it also makes AI-Exercise 2.b.27 visible, since the inclusions

$$B_1^{d_1}(0) \subseteq B_1^{d_2}(0) \subseteq B_1^{d_3}(0)$$

are precisely the inequalities $d_3 \leq d_2 \leq d_1$ read backwards. The larger the metric, the harder it is to be within distance 1, and hence the smaller the ball. The same family, continued to all $p \geq 1$, is sketched in part **(a)** of Exercise 2.c.42, where the square reappears as the limiting case $p \to \infty$.

None of this changes which sets are open. All three metrics have the same open sets: each ball of one metric contains a ball of the same centre for each of the others, as the inclusions above already show for radius 1, and Corollary 2.d.50 turns this into a theorem about any two norms on $\mathbb{R}^n$. The balls look different, and the topology does not notice.

AI-Exercise 3.c.24 (*Boundary of the Open Unit Disk*): Let $D := \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$ be the open unit disk in $\mathbb{R}^2$. Show that the boundary of D is the unit circle $\partial D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$.

Solution: *See solution on Page 52*

Exercise 3.c.25 (*Multiple choice (closure)*): Let (X,d) be a metric space, and $Y_1, Y_2 \subset X$ subsets. Select all the statements below that are necessarily true.

- **(a)** $\overline{Y_1 \cup Y_2} = \overline{Y_1} \cup \overline{Y_2}$
- **(b)** $\overline{Y_1} \cap \overline{Y_2} \subset \overline{Y_1 \cap Y_2}$
- **(c)** $\overline{Y_1 \cap Y_2} \subset \overline{Y_1} \cap \overline{Y_2}$
- **(d)** $\overline{Y_1 \cap Y_2} = \overline{Y_1} \cap \overline{Y_2}$

Official hints: ***(b):*** *compare with* ***(c)****.* ***(d):*** *play with a set of three points (a triangle).*

Info: *This exercise is Problem 2.2 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 52*

Exercise 3.c.26 (*Multiple choice (distance from a set)*): Let (X, d) be a metric space, and $A \subset X$ a non-empty subset. We define the function "distance from A" as

$$d(\cdot, A) : X \to [0, \infty), \qquad d(x, A) := \inf_{a \in A} d(x, a).$$

Select all the statements below that are necessarily true.

(a) If A is closed and $x \in A^c$, then $d(x, A) > 0$.
(b) The set $M := \{x \in X : d(x, A) \geq 1\}$ is closed in X.
(c) For $x, y \in X$, $d(x, A) \leq d(x, y) + d(y, A)$ holds.
(d) If A° is non-empty and $x \in X$, then $d(x, A) = d(x, A^\circ)$.

Official hint: *First convince yourself with a drawing that the name of this function is appropriate; then use the characterization of open/closed sets with sequences.*

Info: *This exercise is Problem 2.3 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 52*

Exercise 3.c.27 (*Boundary, Interior etc.*): Determine the interior, closure, and boundary of the following subsets Y of $\mathbb{R}$, for the standard topology on $\mathbb{R}$. No need to justify the answer.

(a)	$Y = [0, 1]$	**(b)**	$Y = \mathbb{Q}$
(c)	$Y = \emptyset$	**(d)**	$Y = (0, 1)$
(e)	$Y = [-1, 1) \setminus \{0\}$	**(f)**	$Y = [0, \infty)$
(g)	$Y = \{0\}$	**(h)**	$Y = \left\{\frac{1}{n} \mid n \in \mathbb{N} \setminus \{0\}\right\}$

Info: *This exercise is Problem 2.4 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 53*

Solutions (Section 3.d)

Not every solution in this chapter is collected here. Exercise 3.a.12 (Page 47), Exercise 3.b.16 (Page 49) and Exercise 3.b.20 (Page 49) are solved where they appear.

Proof of AI-Exercise 3.a.5:

Let $y \in B_r(x)$, so $d(y, x) < r$, and set $\varepsilon := r - d(y, x) > 0$. We claim that $B_\varepsilon(y) \subseteq B_r(x)$, which is exactly what Definition 3.a.6 requires.

Let $z \in B_\varepsilon(y)$, so $d(z, y) < \varepsilon$. By the triangle inequality (Definition 2.b.10),

$$d(z, x) \leq d(z, y) + d(y, x) < \varepsilon + d(y, x) = \big(r - d(y, x)\big) + d(y, x) = r,$$

so $z \in B_r(x)$. Since $y \in B_r(x)$ was arbitrary, $B_r(x)$ is open.

Remark 3.d.28: The entire content is the choice $\varepsilon := r - d(y, x)$: the radius left over after travelling from the centre out to y. Note that it shrinks to 0 as y approaches the edge of the ball, so no **single** ε works for every y at once. That is the same *"the constant depends on the point"* phenomenon which separates continuity from uniform continuity in Important remark 5.a.5, met here for the first time.

Q.E.D.

Proof of Exercise 3.b.18:

Write $C := \{y \in \mathbb{R}^n : d(x, y) \leq r\}$ for the target set.

$\overline{B_r(x)} \subseteq C$. C is closed (it is $d(x,\cdot)^{-1}([0,r])$, the preimage of a closed set under the continuous map $d(x,\cdot)$) and contains $B_r(x)$, and the closure is the **smallest** closed set containing $B_r(x)$ (Definition 3.b.13).

$C \subseteq \overline{B_r(x)}$. Let $y \in C$, so $d(x,y) \le r$; we must show every ball $B_\varepsilon(y)$ meets $B_r(x)$. If $d(x,y) < r$ then $y \in B_r(x)$ itself, so this is immediate. If $d(x,y) = r$, consider the points $z_t := x + t(y-x)$ for $t \in (0,1)$, which lie on the segment from x to y: since $\mathbb{R}^n$ has a vector space structure (unlike a general metric space, where this argument would not make sense), $d(x,z_t) = t\,d(x,y) = tr < r$, so $z_t \in B_r(x)$, and $d(z_t,y) = (1-t)d(x,y) = (1-t)r \to 0$ as $t \to 1$. So every ball around y eventually contains some z_t, hence meets $B_r(x)$, and $y \in \overline{B_r(x)}$.

For the boundary, $\operatorname{int}(B_r(x)) = B_r(x)$ since $B_r(x)$ is itself open (AI-Exercise 3.a.5), so

$$\partial B_r(x) = \overline{B_r(x)} \setminus \operatorname{int}(B_r(x)) = C \setminus B_r(x) = \{y \in \mathbb{R}^n : d(x,y) = r\}.$$ Q.E.D.

Proof of AI-Exercise 3.c.24:

Since D is open, its interior is $\operatorname{int}(D) = D$. The closure of D is $\overline{D} = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1\}$. By Definition 3.b.13, the boundary is:

$$\begin{aligned}\partial D &= \overline{D} \setminus \operatorname{int}(D) \\ &= \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1\} \setminus \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\} \\ &= \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.\end{aligned}$$

Q.E.D.

Solution to Exercise 3.c.25:

(a) and (c) are necessarily true; (b) and (d) are not.

(a) True. Both inclusions come from general properties of the closure. Since $Y_i \subseteq Y_1 \cup Y_2$ and the closure is monotone, $\overline{Y_i} \subseteq \overline{Y_1 \cup Y_2}$ for $i = 1, 2$, which gives "$\supseteq$". Conversely $\overline{Y_1} \cup \overline{Y_2}$ is a **finite** union of closed sets, hence closed (Proposition 3.a.8**(c)**), and it contains $Y_1 \cup Y_2$; since $\overline{Y_1 \cup Y_2}$ is the smallest closed set containing $Y_1 \cup Y_2$ (Definition 3.b.13), "$\subseteq$" follows.

(c) True, and it is the easy half: $Y_1 \cap Y_2 \subseteq Y_i$ gives $\overline{Y_1 \cap Y_2} \subseteq \overline{Y_i}$ for each i by monotonicity, hence $\overline{Y_1 \cap Y_2} \subseteq \overline{Y_1} \cap \overline{Y_2}$.

(b) False, and therefore **(d) False** as well, (d) being (b) and (c) combined. In $\mathbb{R}$ take

$$Y_1 := (0,1), \qquad Y_2 := (1,2).$$

Then $\overline{Y_1} \cap \overline{Y_2} = [0,1] \cap [1,2] = \{1\}$, whereas $Y_1 \cap Y_2 = \emptyset$ and so $\overline{Y_1 \cap Y_2} = \emptyset$. The two sets meet only in the limit, and the closure of the intersection cannot see a point belonging to neither set.

Remark 3.d.29 (*Where finiteness enters, again*): Part **(a)** is false for infinite unions, for exactly the reason given in Important remark 3.a.9. Taking $Y_q := \{q\}$ for each $q \in \mathbb{Q}$,

$$\bigcup_{q \in \mathbb{Q}} \overline{Y_q} = \mathbb{Q} \qquad \text{but} \qquad \overline{\bigcup_{q \in \mathbb{Q}} Y_q} = \overline{\mathbb{Q}} = \mathbb{R}.$$

So the honest statement is: the closure commutes with **finite** unions, and only one-directionally with intersections. That is the same asymmetry that runs through the whole of Proposition 3.a.8.

Q.E.D.

Solution to Exercise 3.c.26:

(a), (b) and (c) are necessarily true; (d) is not. It is efficient to prove **(c)** first, since **(b)** then follows from it.

(c) True. Fix $x, y \in X$. For every $a \in A$ the triangle inequality gives $d(x,a) \le d(x,y) + d(y,a)$, and the left-hand side is bounded below by $d(x,A)$, so

$$d(x,A) \le d(x,y) + d(y,a) \qquad \text{for every } a \in A.$$

Taking the infimum over a on the right yields $d(x,A) \le d(x,y) + d(y,A)$. Exchanging x and y and combining,

$$\big|d(x,A) - d(y,A)\big| \le d(x,y),$$

so $d(\cdot, A)$ is 1-**Lipschitz** (Definition 5.b.20), and in particular continuous. That is what the remaining parts use.

(a) True. If A is closed then A^c is open, so for $x \in A^c$ there is $r > 0$ with $B_r(x) \subseteq A^c$. No point of A lies in that ball, i.e. $d(x,a) \ge r$ for every $a \in A$, and taking the infimum gives $d(x,A) \ge r > 0$.

(b) True. By **(c)** the function $d(\cdot, A)$ is continuous, and $M = d(\cdot,A)^{-1}\big([1,\infty)\big)$ is the preimage of a closed set under a continuous map, hence closed (Definition 5.a.1**(c)**, applied to complements).

(d) False. In $\mathbb{R}$ take $A := [0,1] \cup \{2\}$, whose interior $A^\circ = (0,1)$ is non-empty as required. At the point $x := 2$,

$$d(x,A) = 0 \quad (\text{because } 2 \in A), \qquad\qquad d(x, A^\circ) = \inf_{a \in (0,1)} |2 - a| = 1.$$

The isolated point 2 of A contributes to $d(\cdot, A)$ but is invisible to A°. The hypothesis *"A° is non-empty"* is far too weak, because it constrains A nowhere near x; the statement would become true under a hypothesis such as $A = \overline{A^\circ}$.

Q.E.D.

Solution to Exercise 3.c.27:

All with respect to the standard metric on $\mathbb{R}$, using Definition 3.b.13.

	Y	$\operatorname{int}(Y)$	$\overline{Y}$	∂Y
(a)	$[0,1]$	$(0,1)$	$[0,1]$	$\{0,1\}$
(b)	$\mathbb{Q}$	$\emptyset$	$\mathbb{R}$	$\mathbb{R}$
(c)	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
(d)	$(0,1)$	$(0,1)$	$[0,1]$	$\{0,1\}$
(e)	$[-1,1) \setminus \{0\}$	$(-1,0) \cup (0,1)$	$[-1,1]$	$\{-1,0,1\}$
(f)	$[0,\infty)$	$(0,\infty)$	$[0,\infty)$	$\{0\}$
(g)	$\{0\}$	$\emptyset$	$\{0\}$	$\{0\}$
(h)	$\{\frac{1}{n} : n \ge 1\}$	$\emptyset$	$\{\frac{1}{n} : n \ge 1\} \cup \{0\}$	$\{\frac{1}{n} : n \ge 1\} \cup \{0\}$

Three rows repay a second look.

- **(a)** versus **(d)**. The closed interval and the open interval have the same interior, the same closure and the same boundary. These three operations simply cannot distinguish a set from any other set squeezed between its interior and its closure.
- **(e)**. Removing the single point 0 from $[-1,1)$ adds 0 to the boundary without changing the closure: a removed interior point becomes a boundary point. Note also that $1 \in \partial Y$ although $1 \notin Y$, while $-1 \in \partial Y$ and $-1 \in Y$. The boundary is indifferent to membership.
- **(b)** and **(h)**. Both have empty interior, so $\partial Y = \overline{Y}$: once a set has no interior, the boundary swallows the entire closure. That is the general reason for $\partial\mathbb{Q} = \mathbb{R}$ in Example 3.b.17**(c)**, and not an accident of $\mathbb{Q}$.

Q.E.D.

Solution to Exercise 3.a.11:

(a) Closed. The complement is $(-\infty, 0)$, which is open, so $[0, \infty)$ is closed by Definition 3.a.6. It is not open: every ball $B_\varepsilon(0)$ contains negative numbers.

(b) Neither. Not open, since every ball around a rational contains irrationals. Not closed, since $\sqrt{2}$ is a limit of rationals without lying in $\mathbb{Q}$; indeed $\overline{\mathbb{Q}} = \mathbb{R}$.

(c) Neither. The map $x \mapsto x - \lfloor x \rfloor$ is the fractional part, whose range is exactly $[0, 1)$. This is not open (0 has no ball inside it) and not closed (1 is a limit point that is missing). The two defects sit at the two ends.

(d) Neither. Multiplying by 2^{-p} turns the condition into a union of closed blocks,

$$S = \bigcup_{p \geq 0} \left[2^{-p},\, 2^{-p+1}\right] = [1, 2] \cup \left[\tfrac{1}{2}, 1\right] \cup \left[\tfrac{1}{4}, \tfrac{1}{2}\right] \cup \cdots = (0, 2],$$

since consecutive blocks share an endpoint and their left ends decrease to 0. Now $(0, 2]$ is not open (no ball around 2 fits) and not closed (0 is a missing limit point).

(e) Open, and this is the point of putting **(d)** next to it. The same computation with open blocks gives

$$T = \bigcup_{p \geq 0} \left(2^{-p},\, 2^{-p+1}\right) = (0, 2) \setminus \{1, \tfrac{1}{2}, \tfrac{1}{4}, \dots\},$$

because now the shared endpoints 2^{-k} are excluded by **both** neighbouring blocks and so belong to neither. An arbitrary union of open sets is open (Proposition 3.a.8), so T is open. It is not closed: 0 and each removed point 2^{-k} are limit points of T outside T.

The lesson is that the union in **(d)** was closed-looking but glued into a half-open interval, whereas the union in **(e)** punctures itself at every join. Being a union of closed sets buys nothing; being a union of open sets buys everything.

(f) Closed. Compute one step before doing anything else: $x_1 = \frac{1}{2}(1 + 1^2) = 1 = x_0$, so the recursion is stationary and $x_n = 1$ for every n. The set is therefore $\{1\}$, a single point, which is closed (finite sets are closed) and not open.

(g) Not known. The trivial zeros of ζ are $-2, -4, -6, \dots$, contributing the real parts $\{-2, -4, -6, \dots\}$; the non-trivial zeros all lie in the critical strip $0 < \mathrm{Re}(c) < 1$. If the Riemann hypothesis holds, every non-trivial zero has $\mathrm{Re}(c) = \frac{1}{2}$, so the set is

$$\{\tfrac{1}{2}\} \cup \{-2n : n \geq 1\},$$

which is closed: its points are isolated and the only way to escape is towards $-\infty$, which is not a real accumulation point. If the Riemann hypothesis fails, the real parts of the off-line zeros are not understood well enough to say. So this entry is a joke with a sharp edge. It is a perfectly well-posed question about a perfectly explicit set, and nobody can currently answer it.

Q.E.D.

Sequences and Convergence (Chapter 4)

Open and closed sets describe a space statically. They say which points sit comfortably inside a set and which sit on its edge, but they say it all at once, with no notion of approach. Sequences make the same information dynamic, and in a metric space the two descriptions turn out to be equivalent: a set is closed exactly when it catches the limits of its own convergent sequences.

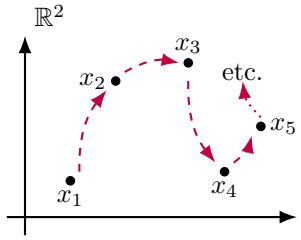
That equivalence is what makes this short chapter worth its own place. Almost every later argument that needs to produce a point produces it as a limit, and the sequential criterion is what converts that limit back into a statement about open or closed sets. The chapter also isolates a fact easy to pass over, namely that convergence depends only on the topology and not on the metric generating it, so two visibly different metrics can have exactly the same convergent sequences.

Sequences and convergence (Section 4.a)

Having both descriptions, the static one by open and closed sets and the dynamic one by sequences, is what makes metric spaces pleasant to work in: one is usually easier to verify, the other easier to apply, and Proposition 4.a.7 says they agree. Note that the definition below is word for word the one from **Analysis I**, with $|x_n - x|$ replaced by $d(x_n, x)$. That substitution is the pattern for the whole chapter.

Definition 4.a.1 (*Convergent sequence*): A "*sequence*" $(x_n)_{n=0}^{\infty}$ in a metric space (X, d) is a function $x : \mathbb{N} \to X$. We say that $(x_n)_{n=0}^{\infty}$ "*converges*" to $x \in X$ if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ such that } d(x_n, x) < \varepsilon \quad \forall n \geq N.$$



Notation 4.a.2 ("*Eventually*"): Let $(x_n)_{n \geq 0}$ be a sequence in a set X and let $\mathcal{P}$ be a property that a point of X may or may not have. One says x_n satisfies $\mathcal{P}$ "*eventually*" (German: „*schliesslich*“) if there exists $N \in \mathbb{N}$ such that $\mathcal{P}(x_n)$ holds for all $n \geq N$, or equivalently if $\mathcal{P}(x_n)$ fails for at most finitely many n. In this vocabulary, Definition 4.a.1 reads: $(x_n)_n$ converges to x if, for every $\varepsilon > 0$, x_n eventually lies in $B_\varepsilon(x)$. The same compression applies throughout the rest of this chapter and the next three.

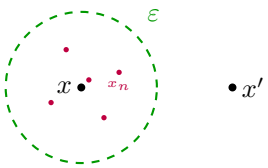
Example 4.a.3 (*A convergent sequence in* $\mathbb{R}^3$): Define a sequence $(x_n)_{n=1}^{\infty}$ in $\mathbb{R}^3$ by $x_n := \left(\frac{1}{n}, \frac{2}{n}, 1\right)$ for all $n \geq 1$. Does a limit exist? If yes, what is it? Yes, the sequence converges. Its limit can be computed coordinate-wise:

$$\lim_{n\to\infty} x_n = \left(\lim_{n\to\infty} \frac{1}{n}, \lim_{n\to\infty} \frac{2}{n}, \lim_{n\to\infty} 1 \right) = (0, 0, 1).$$

Proposition 4.a.4 (*Uniqueness of limits*): Let (X, d) be a metric space. If a sequence $(x_n)_{n=0}^{\infty}$ in X converges to both $x \in X$ and $x' \in X$, then $x = x'$.

Proof:

The idea is that two different limits could be told apart by a small enough ε. Past some index every x_n lies inside the ball of radius ε about x; if x' is outside that ball, the tail of the sequence never comes close to x' and cannot converge to it as well.



Fix $\varepsilon > 0$. By Definition 4.a.1, there exist $N, N' \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon/2$ for all $n \geq N$ and $d(x_n, x') < \varepsilon/2$ for all $n \geq N'$. For any $n \geq \max\{N, N'\}$, the triangle inequality (Definition 2.b.10) gives

$$d(x, x') \leq d(x, x_n) + d(x_n, x') < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $d(x, x') = 0$, so $x = x'$ by definiteness (Definition 2.b.10). **Q.E.D.**

This justifies speaking of *"the"* limit of a convergent sequence, as we do implicitly throughout.

Definition 4.a.5 (*Accumulation point*): Let $(x_n)_{n=0}^{\infty}$ be a sequence in a metric space (X, d). A point $x \in X$ is an "*accumulation point*" (German: „*Häufungspunkt*“) of $(x_n)_{n=0}^{\infty}$ if some subsequence $(x_{n_k})_{k=0}^{\infty}$ converges to x.

Remark 4.a.6 (*A claim in the lecture script that is not fully correct*): The official lecture script states that a sequence converges to x if and only if x is its **only** accumulation point (Definition 4.a.5). This is false in general, and two counterexamples are worth recording. First, a sequence can have a unique accumulation point without converging: on $[0, 1)$, let $x_k := \frac{1}{k}$ for odd $k \geq 3$ and $x_k := 1 - \frac{1}{k}$ for even $k \geq 2$, both of which lie in $[0, 1)$; the only accumulation point is 0, yet (x_k) does not converge, since the even terms tend to $1 \notin [0, 1)$ and so leave every neighbourhood of 0 infinitely often. Second, even in $\mathbb{R}$ itself, the sequence with $x_n := \frac{1}{n}$ for odd n and $x_n := n$ for even n has 0 as its only accumulation point but is unbounded, hence divergent. What **is** true is the easier direction: if (x_n) converges to x, then x is its unique accumulation point (this follows directly from Proposition 4.a.4 together with the fact that every subsequence of a convergent sequence converges to the same limit). The converse needs one extra hypothesis and no more: (sequential) compactness of the ambient space (Section 7.a). If (x_k) has x as its only accumulation point but does not converge to it, some $\varepsilon > 0$ leaves a subsequence outside $B_\varepsilon(x)$; compactness gives that subsequence a convergent sub-subsequence, whose limit y satisfies $d(y, x) \geq \varepsilon$ and is a second accumulation point, a contradiction. Both counterexamples above are simply non-compact: $[0, 1)$ in the first, $\mathbb{R}$ in the second.

Proposition 4.a.7 (*Open/closed sets via sequences*):

(a) $U \subseteq X$ is **open** if and only if, for every convergent sequence (Definition 4.a.1) in X with limit in U, the sequence eventually lies in U.

(b) $A \subseteq X$ is **closed** if and only if every convergent sequence in A has its limit in A.

Proof:

(a) **(⇒)** Let U be open and let $x_n \to x \in U$. Choose $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U$ (Definition 3.a.6). By Definition 4.a.1 there is N with $d(x_n, x) < \varepsilon$ for all $n \geq N$, so $x_n \in B_\varepsilon(x) \subseteq U$ from N onwards.

(⇐) By contraposition. If U is not open, some $x \in U$ has $B_{1/n}(x) \not\subseteq U$ for every $n \geq 1$, so we may pick $x_n \in B_{1/n}(x) \setminus U$. Then $d(x_n, x) < \frac{1}{n} \to 0$, so $x_n \to x \in U$, and yet **no** term of the sequence lies in U.

(b) **(⇒)** Let A be closed, (x_n) a sequence in A with $x_n \to x$. If $x \notin A$, then x lies in the open set $X \setminus A$, so by **(a)** the sequence eventually lies in $X \setminus A$, contradicting $x_n \in A$ for all n. Hence $x \in A$.

(⇐) We show $X \setminus A$ is open. If it were not, the same construction as in **(a)** yields $x \in X \setminus A$ and points $x_n \in B_{1/n}(x) \cap A$. Then (x_n) is a sequence in A converging to x, so the hypothesis forces $x \in A$, which is the required contradiction.

Q.E.D.

Remark 4.a.8 (*The shrinking-ball construction*): Both *"⇐"* directions used the same device: negating *"some ball fits inside U"* gives a bad point in $B_{1/n}(x)$ for **every** n, and collecting those points produces a sequence converging to x. It is worth recognising, because it is the standard bridge from a statement about balls to a statement about sequences, and it is exactly what makes Definition 3.a.6 and Proposition 4.a.7 interchangeable in practice. In the usual division of labour one proves closedness by the sequence criterion and openness by the ball criterion. Note that it needs the radii $\frac{1}{n}$ to shrink to 0, which is where the metric enters; in a general topological space the equivalence can fail.

Lemma 4.a.9 (*Convergence is a topological notion*): Let (X, d) be a metric space. A sequence $(x_n)_n$ in X converges to x if and only if, for every open set $U \ni x$, $(x_n)_n$ eventually (Notation 4.a.2) lies in U.

Proof:

If $x_n \to x$ and $U \ni x$ is open, some $\varepsilon > 0$ has $B_\varepsilon(x) \subseteq U$ (Definition 3.a.6), and x_n eventually lies in $B_\varepsilon(x)$ (Definition 4.a.1), hence in U. Conversely, taking $U := B_\varepsilon(x)$ itself for each $\varepsilon > 0$ recovers exactly Definition 4.a.1. **Q.E.D.**

Corollary 4.a.10 (*Two metrics with the same topology have the same convergent sequences*): Let X be a set endowed with two metrics d_1 and d_2. Then (X, d_1) and (X, d_2) have the same convergent sequences if and only if the topologies they generate coincide.

Proof:

By Lemma 4.a.9, whether $x_n \to x$ depends only on the collection of open sets, that is, only on the topology d_1 or d_2 generates. Consequently, equal topologies give equal convergent sequences.

For the converse, suppose $U \subseteq X$ is open with respect to d_1 but not with respect to d_2; we exhibit a sequence distinguishing the two. Since U is not d_2-open, some $x \in U$ has, for every $k \geq 0$, a point $x_k \in B^{d_2}_{2^{-k}}(x) \setminus U$. Then $x_k \to x$ with respect to d_2 (Definition 4.a.1, since $2^{-k} \to 0$), so by hypothesis also $x_k \to x$ with respect to d_1. But U is d_1-open and contains x, so Lemma 4.a.9 forces $(x_k)_k$ to eventually lie in U, contradicting $x_k \notin U$ for every k. So every d_1-open set is d_2-open, and by symmetry the two topologies coincide. **Q.E.D.**

Remark 4.a.11: This is the precise sense in which *"only the topology matters, not the metric itself"* for convergence. Two visibly different metrics, even one bounded and one unbounded, generate the same convergent sequences exactly when they generate the same open sets, and one never has to compare the metrics directly. The three product metrics of Exercise 2.b.30, shown equivalent in its part **(b)**, are one instance of this; so is Proposition 2.b.16 together with Exercise 2.b.18.

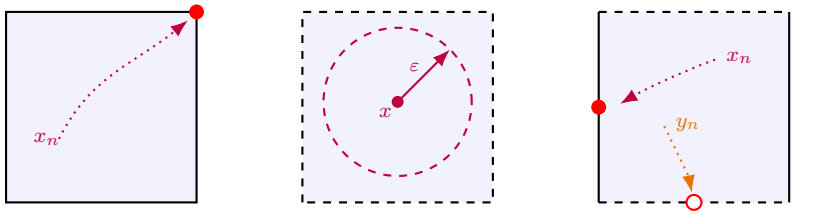
Example 4.a.12 (*Open, closed, and neither in $\mathbb{R}^2$*): In the euclidean metric space $\mathbb{R}^2$, and with Proposition 4.a.7 in hand, each answer is a one-line sequence argument:

(a) $[0,1]^2$ is **closed**: a sequence in $[0,1]^2$ converges coordinatewise (Proposition 6.a.5), and each coordinate limit stays in $[0,1]$ because weak inequalities survive limits.

(b) $(0,1)^2$ is **open**: given $x \in (0,1)^2$, the distance from x to the nearest edge is some $\varepsilon > 0$, and $B_\varepsilon(x) \subseteq (0,1)^2$.

(c) $[0,1] \times (0,1)$ is **neither open nor closed**: it is not open because no ball around $(0, \frac{1}{2})$ fits inside it, and it is not closed because $(\frac{1}{2}, \frac{1}{n}) \to (\frac{1}{2}, 0)$ is a convergent sequence in the set whose limit is not.

The picture below shows exactly these three witnesses. In **(a)** the escaping sequence stays trapped, whereas in **(c)** one sequence escapes through the open edge while another does not.



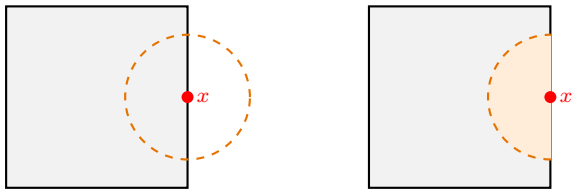
(a) $[0,1]^2$ (**closed**) (b) $(0,1)^2$ (**open**) (c) $[0,1] \times (0,1)$ (**neither**)

● limit lies in the set ○ limit lies outside it

Important remark 4.a.13 (*Common misconception!*)**:** In the euclidean metric space $X = [0,1]^2$, the set $[0,1]^2$ **is open**!

$[0,1]^2 \subseteq \mathbb{R}^2$
Not open, since $B_\varepsilon(x)$ is not contained in $[0,1]^2$ if x is on the boundary.

$[0,1]^2 \subseteq [0,1]^2$
Open, since the full space is always open, as shown before.



$[0,1]^2 \subseteq \mathbb{R}^2$ (**not open**) $[0,1]^2 \subseteq [0,1]^2$ (**open**)

Exercise 4.a.14 (*Uniform convergence*)**:**

(a) Write down the definition of convergence of a sequence $(f_n)_{n=0}^\infty$ to f in the metric space X of bounded functions $[0,1] \to \mathbb{R}$ with the supremum metric.

(b) You know the resulting expression from Analysis 1. What is it called?

(c) *(Optional)* Use the fact that $C^0([0,1]) \subseteq X$ is closed and Proposition 4.a.7 on *"sequentially closed"* to conclude a big result from Analysis 1.

Solution to Exercise 4.a.14:

(a) $\forall \varepsilon > 0 \; \exists N \in \mathbb{N} : \sup_{x \in [0,1]} |f_n(x) - f(x)| < \varepsilon \quad \forall n \geq N$.

(b) This is "*uniform convergence*" (German: „*gleichmässige Konvergenz*“)!

(c) $C^0([0,1])$ is closed in the space of bounded functions with the supremum metric.

Proof. Let $f \in X \setminus C^0([0,1])$, i.e., $f : [0,1] \to \mathbb{R}$ is bounded and has at least one discontinuity at a point $x_0 \in [0,1]$. That is, we find $\varepsilon > 0$ such that for all $\delta > 0$ there exists $y \in [0,1] \cap (x_0 - \delta, x_0 + \delta)$ with $|f(x_0) - f(y)| \geq \varepsilon$. Then the $\frac{\varepsilon}{3}$-ball in X,

$$B_{\varepsilon/3}(f) := \Big\{ g \in X : \sup_{x \in [0,1]} |f(x) - g(x)| < \tfrac{\varepsilon}{3} \Big\},$$

is contained in $X \setminus C^0([0,1])$.

Why: let $g \in B_{\varepsilon/3}(f)$ and let $\delta > 0$ be arbitrary. Choose $y \in (x_0 - \delta, x_0 + \delta)$ with $|f(x_0) - f(y)| \geq \varepsilon$. Then

$$\varepsilon \leq |f(x_0) - f(y)| \leq \underbrace{|f(x_0) - g(x_0)|}_{< \varepsilon/3} + |g(x_0) - g(y)| + \underbrace{|g(y) - f(y)|}_{< \varepsilon/3},$$

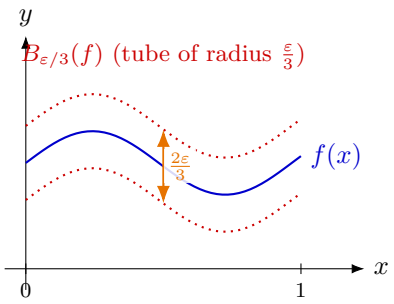
so $|g(x_0) - g(y)| > \varepsilon/3$. As δ was arbitrary, g is discontinuous at x_0 too, i.e. $g \notin C^0([0,1])$. The whole point of the $\frac{\varepsilon}{3}$ is that two of the three terms are spent on moving from f to g and back, leaving a third of the jump behind.

Given this fact, Proposition 4.a.7 ensures that the uniform limit f of $(f_n)_{n=0}^{\infty}$ in $C^0([0,1])$ is also in $C^0([0,1])$, i.e., continuous, given that $f : [0,1] \to \mathbb{R}$ is bounded. This is the case, since f_n is bounded for all $n \in \mathbb{N}$, and with $\varepsilon = 1$ we find $N \in \mathbb{N}$ such that

$$\sup_{x \in [0,1]} |f_N(x) - f(x)| < 1,$$

which implies that f is bounded.

Q.E.D.



Continuity (Chapter 5)

With distance, shape and convergence in place, we can finally ask the question the whole apparatus was built for. What does it mean for a map **between** two metric spaces to respect that structure? Three answers are given, and they are equivalent: the ε–δ condition inherited from **Analysis I**, a sequential condition, and a topological one asking that preimages of open sets be open. Having all three is not redundancy. The first is what one verifies for a concrete function, the second is what combines with the sequential arguments of the previous chapter, and the third is the only one that survives when the metric is taken away. The chapter then adds the two strengthenings needed later, uniform continuity and Lipschitz continuity, and locates them in a hierarchy of increasingly demanding conditions.

Continuity (Section 5.a)

Definition 5.a.1 (*Continuous map*): For (X, d_X) and (Y, d_Y) metric spaces, a function $f : X \to Y$ is called "*continuous*" (German: „*stetig*") if one of the following equivalent conditions holds:

(a) **(ε–δ):** $\forall x_0 \in X, \varepsilon > 0\ \exists \delta > 0$ such that for all $x \in X$:

$$d(x_0, x) < \delta \implies d(f(x_0), f(x)) < \varepsilon$$

(b) **(Sequentially continuous):** for any sequence $(x_n)_{n=0}^{\infty}$ in X with limit $x \in X$, $\lim_{n\to\infty} f(x_n) = f(x)$ (and the limit exists).

(c) **(Topological):** for all $U \subseteq Y$ open, $f^{-1}(U) \subseteq X$ is open (Definition 3.a.6).

Proposition 5.a.2 (*The three definitions of continuity agree*): The conditions **(a)**, **(b)** and **(c)** of Definition 5.a.1 are equivalent, so Definition 5.a.1 is well posed.

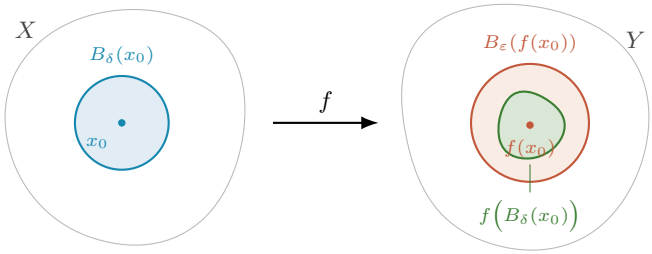
Proof:

We prove the cycle **(a)** ⇒ **(b)** ⇒ **(c)** ⇒ **(a)**.

(a) ⇒ (b). Let $x_n \to x$ and let $\varepsilon > 0$. Pick $\delta > 0$ as in **(a)** for the point x, then N with $d_X(x_n, x) < \delta$ for $n \geq N$. For those n, $d_Y(f(x_n), f(x)) < \varepsilon$, so $f(x_n) \to f(x)$.

(b) ⇒ (c). Let $U \subseteq Y$ be open and suppose $f^{-1}(U)$ were not. By the shrinking-ball construction (Remark 4.a.8) there are $x \in f^{-1}(U)$ and points $x_n \in B_{1/n}(x)$ with $x_n \notin f^{-1}(U)$. Then $x_n \to x$, so $f(x_n) \to f(x) \in U$ by **(b)**, and since U is open, Proposition 4.a.7**(a)** forces $f(x_n) \in U$ for large n. That is, $x_n \in f^{-1}(U)$, which is a contradiction.

(c) ⇒ (a). Fix $x_0 \in X$ and $\varepsilon > 0$. The ball $B_\varepsilon(f(x_0)) \subseteq Y$ is open (AI-Exercise 3.a.5), so $f^{-1}\big(B_\varepsilon(f(x_0))\big)$ is open by **(c)** and contains x_0. Hence there is $\delta > 0$ with $B_\delta(x_0) \subseteq f^{-1}\big(B_\varepsilon(f(x_0))\big)$, which says precisely that $d_X(x_0, x) < \delta$ implies $d_Y(f(x_0), f(x)) < \varepsilon$. **Q.E.D.**



Condition **(a)** is the statement that the green region fits inside the red ball, and that a δ making it fit can be found however small the red ball is made. Condition **(c)** is the same picture read from right to left: $B_\delta(x_0)$ lies inside $f^{-1}\big(B_\varepsilon(f(x_0))\big)$, which is exactly the containment produced in the **(c)** $\Rightarrow$ **(a)** step above. Note what the drawing does not claim: the image need not be round, need not be centred at $f(x_0)$, and need not fill the ball.

Remark 5.a.3 (*Which of the three to reach for*): Condition **(c)** is the only one that never mentions the metric. That is why it, and not the ε–δ formulation, is what survives into general topology (Section 2.a). Condition **(b)** is usually the fastest way to prove a function is **dis**continuous, since a single badly chosen sequence suffices; this is exactly how AI-Example 9.b.8 works. Condition **(a)** is the one to use when a quantitative modulus is wanted, as in Definition 5.a.4 immediately below.

Definition 5.a.4 (*Uniform continuity*): For (X, d_X) and (Y, d_Y) metric spaces, a function $f : X \to Y$ is "*uniformly continuous*" (German: „*gleichmässig stetig*“) if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, x' \in X : \quad d_X(x, x') < \delta \ \text{ implies } \ d_Y(f(x), f(x')) < \varepsilon.$$

Important remark 5.a.5 (*Where the quantifiers sit*): Compare Definition 5.a.1 **(a)** with Definition 5.a.4: the **formulas** differ only in the order of two quantifiers, and that order is the whole content.

- **Continuity:** $\forall x_0\ \forall \varepsilon\ \exists \delta$. The point x_0 is chosen **before** δ, so δ may depend on it, and a different δ is allowed at every point.
- **Uniform continuity:** $\forall \varepsilon\ \exists \delta\ \forall x$. Now δ is chosen **before** seeing the point, so one and the same δ must work everywhere at once.

Uniform continuity is therefore strictly stronger. The standard witness is $f(x) := 1/x$ on $(0, 1)$: it is continuous, but near 0 the graph steepens without bound, so the δ that works at $x_0 = \frac{1}{2}$ is hopeless at $x_0 = \frac{1}{1000}$, and no single δ survives. Proposition 7.a.17 below says that on a **compact** space this can never happen. The gap between the two notions closes exactly when there is no room left to run off towards a bad point.

Example 5.a.6 ($f(x) = x^2$): Let $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$. Let $a < b \in \mathbb{R}$. Then

- for $a < 0 < b$: $\quad f^{-1}((a, b)) = (-\sqrt{b}, \sqrt{b})$, which is open.
- for $a < b < 0$: $\quad f^{-1}((a, b)) = \emptyset$.
- for $0 < a < b$: $\quad f^{-1}((a, b)) = (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b})$.

Why is it enough to check only these cases? Hint: $f^{-1}(U_1 \cup U_2) = f^{-1}(U_1) \cup f^{-1}(U_2)$.

Example 5.a.7 (*Limits in $\mathbb{R}^2$ depend on the path*): When computing limits in $\mathbb{R}^n$, one must ensure that the limit exists and is identical along **all** possible paths approaching the point. For example, consider the function $f : \mathbb{R}^2 \setminus \{(0, 0)\} \to \mathbb{R}$ given by $f(x, y) = \frac{x^2}{x^2+y^2}$. If we approach the origin along the x-axis (setting $y = 0$ first), we get

$$\lim_{x \to 0} f(x, 0) = \lim_{x \to 0} \frac{x^2}{x^2} = 1.$$

However, if we approach along the y-axis (setting $x = 0$ first), we get

$$\lim_{y \to 0} f(0, y) = \lim_{y \to 0} \frac{0}{y^2} = 0.$$

Since these limits disagree, $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist. A useful method for checking limits at the origin is to switch to "*polar coordinates*" ($x = r \cos \varphi, y = r \sin \varphi$). What must be produced is a bound

$$\big|f(r \cos \varphi,\, r \sin \varphi) - L\big| \ \le \ g(r) \qquad \text{with } g(r) \to 0 \text{ as } r \downarrow 0,$$

where g does **not** depend on φ; then $\lim_{(x,y)\to(0,0)} f(x, y) = L$. (This is explored further in Exercise 5.a.15).

AI-Note 5.1: The uniformity is the whole content of the method, and dropping it is the standard trap. It is **not** enough that $\lim_{r\to 0} f(r\cos\varphi, r\sin\varphi)$ exist for each fixed φ and take the same value throughout: that only says the limit is L along every straight ray, and a function can approach the origin along a curve without ever following a ray. For

$$f(x,y) := \frac{x^2 y}{x^4+y^2}, \qquad f(r\cos\varphi, r\sin\varphi) = \frac{r\cos^2\varphi\sin\varphi}{r^2\cos^4\varphi+\sin^2\varphi},$$

the limit as $r \downarrow 0$ is 0 for every fixed φ, including $\sin\varphi = 0$, where f vanishes identically. Yet along the parabola $y = x^2$ one gets $f = x^4/(2x^4) = \frac{1}{2}$, so the limit does not exist. The denominator is what defeats the ray test: it is small precisely where $\sin\varphi$ is comparable to r, which no fixed φ ever sees.

Question 5.a.8: Let (X, d_X) and (Y, d_Y) be metric spaces, where d_Y is the discrete metric. Is **every** function $f : (X, d_X) \to (Y, d_Y)$ continuous?

Answer 5.a.9: No. For example, take the spaces $X = Y = \mathbb{R}$, d_X the standard metric, and $f = \mathrm{id} : X \to Y, z \mapsto z$. Then $\{0\} \subseteq Y$ is open, but $\{0\} = \mathrm{id}^{-1}(\{0\})$ is not open in (X, d_X).

Question 5.a.10: And what if the roles are reversed, with d_X the discrete metric and d_Y arbitrary?

Answer 5.a.11: Any subset of X is open; therefore, $f^{-1}(U)$ is open for any open $U \subseteq Y$. So every such f is continuous.

Exercise 5.a.12 (*Continuity of the distance*): Let (X, d) be a metric space, and endow $X \times X$ with the product distance $d_2(x, y) := d(x_1, y_1) + d(x_2, y_2)$. Show that the distance function $d : X \times X \to \mathbb{R}$ is continuous with respect to d_2 and, more precisely, that it is 1-Lipschitz. Is d also continuous with respect to the distance $d_1(x, y) := \max\{d(x_1, y_1), d(x_2, y_2)\}$? Is d also Lipschitz continuous with respect to the distance $d_3(x, y) := \sqrt{d(x_1, y_1)^2 + d(x_2, y_2)^2}$? (Notation consistent with Exercise 2.b.30.)

Info: *This exercise is Problem 2.6 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 65*

Exercise 5.a.13 (*Continuity of the composition*): Let X, Y, Z be metric spaces and let $f : X \to Y$, $g : Y \to Z$ be continuous functions. Show that $g \circ f : X \to Z$ is continuous using at least two of the three equivalent definitions of continuity seen in class.

Info: *This exercise is Problem 2.7 of Problem Sheet 2, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 66*

Remark 5.a.14: The three equivalent definitions of continuity referenced in Exercise 5.a.13 are those of Definition 5.a.1 (ε–δ, sequential, topological), shown to agree in Proposition 5.a.2.

Exercise 5.a.15 (*Problems at the origin*): Show that there is no continuous function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $f(x, y) = xy/(x^2 + y^2)$ for all $(x, y) \neq (0, 0)$. On the other hand, show that there is exactly one continuous function $g : \mathbb{R}^2 \to \mathbb{R}$ such that $g(x, y) = xy/\sqrt{x^2 + y^2}$ for all $(x, y) \neq (0, 0)$.

Official hint: *If a function is continuous at 0 and $x_k \to 0$ and $y_k \to 0$, then $f(x_k)$ and $f(y_k)$ must converge to the same value.*

Info: *This exercise is Problem 3.2 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 67*

Exercise 5.a.16 (*Distance to a point is continuous*): Let (X, d) be a metric space and fix $a \in X$. Show that the function $f : X \to [0, \infty)$ defined by $f(x) := d(x, a)$ is continuous.

Proof of Exercise 5.a.16:

We use ε–δ continuity. Let $x \in X$ and $\varepsilon > 0$. We need to find $\delta > 0$ such that for all $x' \in X$ with $d(x, x') < \delta$, we have $|f(x) - f(x')| < \varepsilon$. Using the reverse triangle inequality, we have

$$|f(x) - f(x')| = |d(x, a) - d(x', a)| \leq d(x, x').$$

Therefore, we can simply choose $\delta := \varepsilon$. Then $d(x, x') < \delta$ directly implies $|f(x) - f(x')| < \varepsilon$, meaning f is continuous at x. Since x was arbitrary, f is continuous everywhere. Q.E.D.

Corollary 5.a.17 (*Closed sets are exactly the zero sets of a distance function*): Let (X, d) be a metric space and $\emptyset \neq E \subseteq X$. The function $f_E(x) := d(x, E) = \inf_{z \in E} d(x, z)$ is 1-Lipschitz, which is Exercise 3.c.26**(c)**, proved there by the same reverse-triangle-inequality argument as Exercise 5.a.16 above. Moreover,

$$E \text{ is closed} \quad \Longleftrightarrow \quad E = \{x \in X : f_E(x) = 0\}.$$

Proof:

The inclusion $E \subseteq \{f_E = 0\}$ always holds, since $z \in E$ gives $f_E(z) \leq d(z, z) = 0$. So only the reverse inclusion carries content, and it is exactly where closedness is needed.

(⇒) Let E be closed and $f_E(x) = 0$. By definition of the infimum, choose $z_n \in E$ with $d(x, z_n) \to 0$; then $z_n \to x$, and closedness of E gives $x \in E$ (Proposition 4.a.7**(b)**).

(⇐) Suppose $E = \{f_E = 0\}$ and let (z_n) be a sequence in E with $z_n \to x$. Then $f_E(x) \leq d(x, z_n) \to 0$, so $f_E(x) = 0$, i.e. $x \in E$ by hypothesis. Proposition 4.a.7**(b)** then gives that E is closed. Q.E.D.

Remark 5.a.18: This sharpens Exercise 3.c.26**(a)**, which showed only one direction (E closed $\Rightarrow$ $f_E > 0$ off E) for a general subset. It is exactly closedness, and nothing weaker, that lets f_E detect membership in E by vanishing.

Exercise 5.a.19 (*Evaluation map is continuous*): Let $X := C^0([0, 1])$ equipped with the supremum metric $d(f, g) := \sup_{t \in [0,1]} |f(t) - g(t)|$. Define the function $F : X \to \mathbb{R}$ by $F(f) := f(0)$. Show that F is continuous.

Proof of Exercise 5.a.19:

Let $f \in X$ and $\varepsilon > 0$. Choose $\delta := \varepsilon$. If $g \in X$ satisfies $d(f, g) < \delta$, then by the definition of the supremum metric, we have $|f(t) - g(t)| < \delta$ for all $t \in [0, 1]$. In particular, for $t = 0$, we get

$$|F(f) - F(g)| = |f(0) - g(0)| \leq \sup_{t \in [0,1]} |f(t) - g(t)| = d(f, g) < \delta = \varepsilon.$$

Thus F is continuous at f. Since f was arbitrary, F is continuous on all of X. Q.E.D.

Lipschitz continuity (Section 5.b)

Definition 5.b.20 (*Lipschitz continuity*): Let (X, d_X), (Y, d_Y) be metric spaces. A function $f : X \to Y$ is "*Lipschitz continuous*" (German: „*Lipschitz-stetig*“) if there exists $L \geq 0$, a "*Lipschitz constant*" for f, such that

$$d_Y(f(x), f(x')) \leq L \cdot d_X(x, x') \quad \text{for all } x, x' \in X.$$

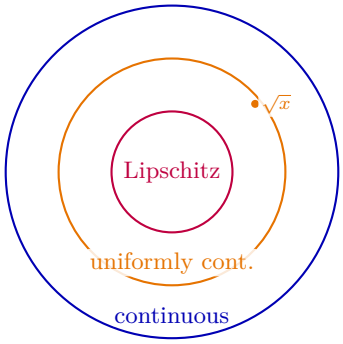
Remark 5.b.21 (*Hierarchy of continuity notions*): Lipschitz continuity is stronger than uniform continuity (Definition 5.a.4), which in turn is stronger than continuity (Definition 5.a.1). For the

first implication, assume $L > 0$ (for $L = 0$ the function is constant and there is nothing to prove); then $\delta := \varepsilon/L$ satisfies the definition of uniform continuity, since

$$d_X(x, x') < \frac{\varepsilon}{L} \quad \text{gives} \quad d_Y(f(x), f(x')) \leq L\, d_X(x, x') < \varepsilon$$

for **every** pair x, x'. This is one and the same δ at every point, which is exactly what uniform continuity asks for. The second implication is immediate: a δ that works at all points in particular works at each one.

Both converses fail. Every continuously differentiable function on a compact **interval** is Lipschitz continuous, with $L := \sup |f'|$ by the mean value theorem. The word **interval** matters here, since the mean value theorem needs the segment between the two points to lie in the domain. But $f(x) := \sqrt{x}$ on $[0, \infty)$ is uniformly continuous and **not** Lipschitz. A Lipschitz constant would force δ to shrink only linearly in ε, whereas here $|\sqrt{x} - \sqrt{x'}| < \varepsilon$ near 0 already requires δ of order ε^2. The reason is visible in the graph, since the tangent at 0 is vertical and no finite slope L can dominate it.



Exercise 5.b.22 (*Classifying continuity*)**:** For each of the following functions, determine whether it is continuous, uniformly continuous, and/or Lipschitz continuous on the given domain:

Function	Domain
$\sqrt{x}$	$[0, \infty)$
$1/x$	$(0, 1)$
$\operatorname{sgn}(x)$	$(-1, 1)$
$\lvert x\rvert^{3/2}$	$(-1, 1)$

Solution: *See solution on Page 66*

Exercise 5.b.23 (*Lipschitz or not*)**:** Give an example of:

(a) A $\frac{1}{2}$-Lipschitz function $f : [0, 1) \to [0, 1)$ that has no fixed point.

(b) A map $f : \mathbb{R}^2 \to \mathbb{R}^2$ that has no fixed point, but is an isometry, i.e. $|f(x) - f(x')| = |x - x'|$ for all $x, x' \in \mathbb{R}^2$.

Info: *This exercise is Problem 3.1 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 67*

Exercise 5.b.24 (*Multiple choice (uniform continuity)*)**:** Select all the statements below that are true.

(a) The function $(x, y) \mapsto x + y$ is uniformly continuous in $\mathbb{R}^2$.

(b) The function $(x, y) \mapsto xy$ is uniformly continuous in $\mathbb{R}^2$.

(c) The function $(x, y) \mapsto x + y$ is uniformly continuous in $[0, 1]^2$.

(d) The function $(x, y) \mapsto xy$ is uniformly continuous in $[0, 1]^2$.

Info: *This exercise is Problem 3.6 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 67*

Remark 5.b.25: A continuous function on a compact space is uniformly continuous (Item **(c)**), and that settles Exercise 5.b.24**(c)** and **(d)** at once.

Exercise 5.b.26 (*(*) Uniform continuity and moduli of continuity*): Let (X, d_X) and (Y, d_Y) be metric spaces and let $f : X \to Y$.

A **modulus of continuity** is a function $\omega : [0, \infty) \to [0, \infty]$ such that

$$\omega(0) = 0, \qquad \omega \text{ is nondecreasing}, \qquad \lim_{t \downarrow 0} \omega(t) = 0.$$

(Here $[0, \infty]$ means we also allow the value $+\infty$.) We say that f **has modulus of continuity** ω if

$$d_Y\big(f(x), f(y)\big) \le \omega\big(d_X(x, y)\big) \qquad \forall x, y \in X.$$

(a) Prove that f is uniformly continuous on X if and only if f has a modulus of continuity.

(b) *(Bonus 1)* Assume $X \subset \mathbb{R}$ is an interval with the usual distance $d_X(x, y) = |x - y|$. Show that the modulus constructed in (1) can be chosen **subadditive**, i.e. $\omega(t_1 + t_2) \le \omega(t_1) + \omega(t_2)$ for all $t_1, t_2 \ge 0$.

(c) *(Bonus 2)* Assume $X \subset \mathbb{R}^n$ is **convex**, meaning: for all $x_1, x_2 \in X$ and all $\lambda \in [0, 1]$, $(1 - \lambda)x_1 + \lambda x_2 \in X$. With $d_X(x, y) = |x - y|$ (Euclidean distance), show again that the modulus from (1) can be chosen subadditive.

Remark 5.b.27 (*official*): In general metric spaces a modulus of continuity may have infinite value for finite $t > 0$. Example: $X = \mathbb{Z}$ with the usual distance, $Y = \mathbb{R}$, $f(n) = n^2$. This f is uniformly continuous on $\mathbb{Z}$ (because points closer than 1 must be equal), but no finite $\omega(1)$ can bound $|f(n+1) - f(n)| = 2n + 1$ for all n. Allowing the value $+\infty$ avoids this issue, and in many familiar settings (e.g. intervals/convex sets in $\mathbb{R}^n$) the resulting $\omega_f(t)$ is finite for all t.

Official hints: ***3.11.1:*** *define the "worst oscillation at scale t"* $\omega_f(t) := \sup\{d_Y(f(x), f(y)) : d_X(x, y) \le t\}$. ***3.11.2 / 3.11.3:*** *if $d_X(x, y) \le t_1 + t_2$, choose an intermediate point z on the segment from x to y so that $d_X(x, z) \le t_1$ and $d_X(z, y) \le t_2$, then use the triangle inequality in Y.*

Info: *This exercise is Problem 3.11 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 68*

Solutions (Section 5.c)

Not every solution in this chapter is collected here. Exercise 5.a.16 (Page 62) and Exercise 5.a.19 (Page 63) are solved where they appear.

Solution to Exercise 5.a.12:

d is 1-Lipschitz with respect to d_2. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be points of $X \times X$. Applying the triangle inequality in X twice,

$$d(x_1, x_2) \;\le\; d(x_1, y_1) + d(y_1, y_2) + d(y_2, x_2),$$

so $d(x_1, x_2) - d(y_1, y_2) \le d(x_1, y_1) + d(x_2, y_2) = d_2(x, y)$. Exchanging the roles of x and y bounds the opposite difference in the same way, hence

$$\big|d(x_1, x_2) - d(y_1, y_2)\big| \;\le\; d_2(x, y),$$

which is 1-Lipschitz continuity (Definition 5.b.20) and therefore continuity (Remark 5.b.21).

For d_1 and d_3: yes, still Lipschitz, but the constant changes. By Exercise 2.b.30 part **(b)** we have $d_2 \le 2d_1$ and $d_2 \le \sqrt{2}\, d_3$, so

$$\big|d(x_1,x_2) - d(y_1,y_2)\big| \le 2\, d_1(x,y) \qquad \text{and} \qquad \big|d(x_1,x_2) - d(y_1,y_2)\big| \le \sqrt{2}\, d_3(x,y).$$

This is a general principle rather than a computation: **Lipschitz continuity survives passage to an equivalent metric**, with the constant multiplied by the comparison factor. Continuity survives too, being topological, and by Exercise 2.b.30 the three product metrics induce the same topology.

Remark 5.c.28 (*The constant 2 cannot be improved*): The constant 2 for d_1 is sharp, so d is genuinely **not** 1-Lipschitz with respect to d_1. Take $X = \mathbb{R}$ with $x := (0,0)$ and $y := (-1,1)$. Then $d(x_1,x_2) = 0$ and $d(y_1,y_2) = 2$, while $d_1(x,y) = \max\{1,1\} = 1$, so the difference equals exactly $2 = 2\, d_1(x,y)$. This is worth checking, because the problem asks only **whether** d is Lipschitz for d_1 and d_3, and it is easy to answer *"yes, with the same constant"* without noticing that the constant has in fact changed.

Q.E.D.

Solution to Exercise 5.a.13:

All three arguments are short; the point of giving more than one is to see how differently the same fact reads through the three definitions of Definition 5.a.1.

Topological (Definition 5.a.1(c)). This is the shortest, because preimages compose. Let $W \subseteq Z$ be open. Then

$$(g \circ f)^{-1}(W) = f^{-1}\big(g^{-1}(W)\big),$$

and $g^{-1}(W)$ is open in Y by continuity of g, so its preimage under f is open in X by continuity of f. Hence $g \circ f$ is continuous.

Sequential (Definition 5.a.1(b)). Let $x_n \to x$ in X. Continuity of f gives $f(x_n) \to f(x)$ in Y, and continuity of g applied to **that** sequence gives $g(f(x_n)) \to g(f(x))$ in Z.

ε–δ (Definition 5.a.1(a)). Fix $x_0 \in X$ and $\varepsilon > 0$. Continuity of g at $f(x_0)$ supplies $\eta > 0$ such that $d_Y(f(x_0), y) < \eta$ implies $d_Z\big(g(f(x_0)), g(y)\big) < \varepsilon$. Continuity of f at x_0, applied to **that** η, supplies $\delta > 0$ such that $d_X(x_0, x) < \delta$ implies $d_Y(f(x_0), f(x)) < \eta$. Chaining the two, $d_X(x_0,x) < \delta$ implies $d_Z\big((g \circ f)(x_0), (g \circ f)(x)\big) < \varepsilon$.

Remark 5.c.29 (*Why the topological proof is the short one*): All three arguments have the same shape, applying the hypothesis for g first and then feeding the result to f. Only the topological version needs no quantifiers at all, because the set identity $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ does the bookkeeping the other two proofs carry out by hand. This is a fair advertisement for Definition 5.a.1**(c)**, which can look like the least intuitive of the three on first meeting: it is the formulation in which composition is trivial, and that is exactly why general topology takes it as **the** definition.

Q.E.D.

Solution to Exercise 5.b.22:

- $\sqrt{x}$ on $[0,\infty)$: **uniformly continuous, not Lipschitz.** Uniform continuity follows from continuity on $[0,1]$ (compact, Item **(c)**) together with $\frac{1}{2}$-Lipschitz behaviour on $[1,\infty)$ (since $|\sqrt{x} - \sqrt{y}| \le \frac{1}{2}|x - y|$ there by the mean value theorem, as $|(\sqrt{\cdot})'| = \frac{1}{2\sqrt{x}} \le \frac{1}{2}$ for $x \ge 1$). Not Lipschitz near 0: for $x_n := 1/n^2 \to 0$, $\dfrac{|\sqrt{x_n} - \sqrt{0}|}{|x_n - 0|} = n \to \infty$, so no global L works.
- $1/x$ on $(0,1)$: **continuous, neither uniformly nor Lipschitz continuous.** For $x_n := 1/n$, $y_n := 1/(2n)$, $|x_n - y_n| \to 0$ but $|1/x_n - 1/y_n| = n \to \infty$, so no uniform $\delta(\varepsilon)$ (and hence no Lipschitz constant) can work near 0.

- $\operatorname{sgn}(x)$ on $(-1,1)$: **none of the three**, being discontinuous at 0.
- $|x|^{3/2}$ on $(-1,1)$: **Lipschitz, hence also uniformly continuous and continuous.** $|x|^{3/2}$ is C^1 on $(-1,1)\setminus\{0\}$ with $\left|\frac{d}{dx}|x|^{3/2}\right| = \frac{3}{2}|x|^{1/2} \le \frac{3}{2}$, and the bound extends across 0 by direct estimation, so $L := \frac{3}{2}$ works globally.

Q.E.D.

Solution to Exercise 5.b.23:

Both parts ask for a map with no fixed point, and each defeats one hypothesis of Theorem 6.c.19 while keeping the other, which is exactly what Remark 6.c.20 predicts must be possible.

(a) Take $f(x) := \frac{x+1}{2}$. It maps $[0,1)$ into $[\frac{1}{2},1) \subseteq [0,1)$, and $|f(x)-f(y)| = \frac{1}{2}|x-y|$, so it is $\frac{1}{2}$-Lipschitz. A fixed point would satisfy $x = \frac{x+1}{2}$, i.e. $x = 1$, which is not in $[0,1)$. **What fails is completeness:** $[0,1)$ is not a complete metric space, the iterates $x_{n+1} = f(x_n)$ march towards the missing endpoint 1, and Theorem 6.c.19 does not apply.

(b) Take the translation $f(x) := x + (1,0)$. It is an isometry, since $|f(x)-f(x')| = |x-x'|$ exactly, and $f(x) = x$ would force $(1,0) = 0$. **What fails is the strict contraction:** an isometry has Lipschitz constant $L = 1$, not $L < 1$, and $\mathbb{R}^2$ is perfectly complete. Together with part **(a)** this shows neither hypothesis of the fixed point theorem can be dropped.

Q.E.D.

Solution to Exercise 5.a.15:

No continuous extension exists for $xy/(x^2+y^2)$. Suppose f were continuous on $\mathbb{R}^2$ and agreed with $xy/(x^2+y^2)$ off the origin. Approach the origin along two different lines:

$$f\left(\tfrac{1}{k},\tfrac{1}{k}\right) = \frac{1/k^2}{2/k^2} = \frac{1}{2}, \qquad f\left(\tfrac{1}{k},0\right) = 0 \quad \text{for every } k.$$

Both sequences tend to $(0,0)$, so continuity would force $f(0,0) = \frac{1}{2}$ and $f(0,0) = 0$ simultaneously. Hence no such f exists. (This is the same function, and the same argument, as AI-Example 9.b.8, where it reappears to show that the existence of both partial derivatives implies nothing.)

Exactly one continuous extension exists for $xy/\sqrt{x^2+y^2}$. Write $(x,y) = (r\cos\theta, r\sin\theta)$ with $r > 0$. Then

$$g(x,y) = \frac{r^2\cos\theta\sin\theta}{r} = r\cos\theta\sin\theta, \qquad \text{so} \qquad |g(x,y)| \le \frac{r}{2} = \frac{|(x,y)|}{2}.$$

The bound is uniform in θ, so $g(x,y) \to 0$ as $(x,y) \to (0,0)$, and setting $g(0,0) := 0$ produces a continuous function on all of $\mathbb{R}^2$.

Uniqueness is the part worth spelling out, and it needs no computation at all: the value at the origin is **forced**. Since $(0,0)$ lies in the closure of $\mathbb{R}^2\setminus\{(0,0)\}$, any two continuous extensions agree on a dense subset of $\mathbb{R}^2$, and two continuous functions agreeing on a dense set agree everywhere. To see the latter, take a sequence from the dense set converging to the point in question and apply Definition 5.a.1**(b)** to both. So the extension is unique whenever it exists, which is also why the first half of the problem had only to exhibit **two** conflicting limits. Q.E.D.

Solution to Exercise 5.b.24:

(a), (c) and (d) are true; only (b) is false.

(a) True. Addition is Lipschitz, hence uniformly continuous (Remark 5.b.21):

$$\big|(x+y)-(x'+y')\big| \le |x-x'| + |y-y'| \le \sqrt{2}\,\big|(x,y)-(x',y')\big|,$$

the last step by Cauchy–Schwarz. So $\delta := \varepsilon/\sqrt{2}$ works everywhere at once.

(b) False. Multiplication is continuous but not uniformly so on the unbounded domain $\mathbb{R}^2$. Take

$$p_k := \big(k,\ \tfrac{1}{k}\big), \qquad q_k := (k,\ 0).$$

Then $|p_k - q_k| = \frac{1}{k} \to 0$, while the values are $k \cdot \frac{1}{k} = 1$ and 0, a constant gap of 1. So no δ can work for $\varepsilon < 1$. The failure is exactly the one described in Important remark 5.a.5: the function steepens without bound as one runs off to infinity, so the local δ shrinks without a positive floor.

(c) and (d) True, and both for free: $[0,1]^2$ is closed and bounded, hence **compact** by Theorem 7.b.28, and a continuous function on a compact metric space is automatically uniformly continuous by Proposition 7.a.17 (see also Item **(c)**). No estimate is needed for either.

Remark 5.c.30: The pair **(b)** versus **(d)** is the whole lesson: the **same** function xy is uniformly continuous on $[0,1]^2$ and not on $\mathbb{R}^2$. Uniform continuity is therefore not a property of a formula but of a formula **together with its domain**, and compactness of the domain is what removes the room to run off to a bad point.

Q.E.D.

Solution to Exercise 5.b.26:

(a) ($\Leftarrow$) A modulus of continuity forces uniform continuity. Suppose $d_Y(f(x), f(y)) \le \omega(d_X(x,y))$ for all x, y, with ω as in the statement. Let $\varepsilon > 0$. Since $\omega(t) \to 0$ as $t \downarrow 0$, there is $\delta > 0$ with $\omega(\delta) < \varepsilon$. If $d_X(x,y) < \delta$ then, ω being nondecreasing,

$$d_Y\big(f(x), f(y)\big) \le \omega\big(d_X(x,y)\big) \le \omega(\delta) < \varepsilon.$$

The δ did not depend on x or y, which is precisely Definition 5.a.4.

($\Rightarrow$) Uniform continuity produces one. Following the official hint, define the "*modulus of continuity of f*" as its worst oscillation at scale t:

$$\omega_f(t) := \sup\big\{\, d_Y(f(x), f(y)) \ :\ x, y \in X,\ d_X(x,y) \le t \,\big\} \ \in\ [0, \infty].$$

It has all three required properties. $\omega_f(0) = 0$, because $d_X(x,y) \le 0$ forces $x = y$ by definiteness and then $d_Y(f(x), f(y)) = 0$. It is nondecreasing, because enlarging t enlarges the set over which the supremum is taken. And $\omega_f(t) \to 0$ as $t \downarrow 0$: given $\varepsilon > 0$, uniform continuity supplies $\delta > 0$ with $d_X(x,y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon$, so every $t < \delta$ has $\omega_f(t) \le \varepsilon$. Finally f has modulus ω_f by construction, since $d_Y(f(x), f(y))$ is one of the numbers the supremum defining $\omega_f(d_X(x,y))$ ranges over.

(b) (Bonus 1) On an interval, ω_f is already subadditive. Let $t_1, t_2 \ge 0$ and take any $x, y \in X$ with $|x - y| \le t_1 + t_2$. Choose an intermediate point z on the segment from x to y as follows: if $|x - y| \le t_1$ put $z := y$; otherwise let z be the point of that segment at distance exactly t_1 from x, so that $|z - y| = |x - y| - t_1 \le t_2$. Either way $z \in X$, because X is an interval and z lies between x and y, and $|x - z| \le t_1$, $|z - y| \le t_2$. The triangle inequality in Y now gives

$$d_Y\big(f(x), f(y)\big) \ \le\ d_Y\big(f(x), f(z)\big) + d_Y\big(f(z), f(y)\big) \ \le\ \omega_f(t_1) + \omega_f(t_2).$$

Taking the supremum over all admissible x, y yields $\omega_f(t_1 + t_2) \le \omega_f(t_1) + \omega_f(t_2)$.

(c) (Bonus 2) Convexity is exactly what the argument needed. Repeat part **(b)** verbatim, now reading $|x - y|$ as the Euclidean norm on $\mathbb{R}^n$ rather than the absolute value on $\mathbb{R}$: the only property of an interval that was used is that the segment from x to y lies inside X, and that is the definition of convexity. The intermediate point is $z := x + \min\{t_1, |x - y|\}\, \frac{y - x}{|y - x|}$ for $x \ne y$, which lies on the segment and satisfies the same two estimates.

Remark 5.c.31 (*Why the value $+\infty$ has to be allowed*): The official remark accompanying this problem gives the reason, and it is worth restating: on $X = \mathbb{Z} \subseteq \mathbb{R}$ the map $f(n) := n^2$ is uniformly continuous for the trivial reason that any two distinct points are at distance at least 1. Take $\delta := \frac{1}{2}$, and the implication holds vacuously. Yet $|f(n+1) - f(n)| = 2n + 1$ is unbounded, so

$\omega_f(1) = +\infty$. Parts **(b)** and **(c)** rule this out by hypothesis rather than by luck: on a convex set the intermediate-point construction lets a large displacement be broken into many small ones, so $\omega_f(t) \le \lceil t/s \rceil \, \omega_f(s)$ is finite as soon as some $\omega_f(s)$ is. A space like $\mathbb{Z}$ has no intermediate points to subdivide with.

Q.E.D.

Completeness and the Banach Fixed Point Theorem (Chapter 6)

Having examined how maps behave between metric spaces, we return to a property of a single space. Does every sequence that *"looks like"* it ought to converge, because its terms eventually crowd arbitrarily close together, actually converge in the sense of Definition 4.a.1? Such sequences are called Cauchy, and a space in which every one of them converges is called complete.

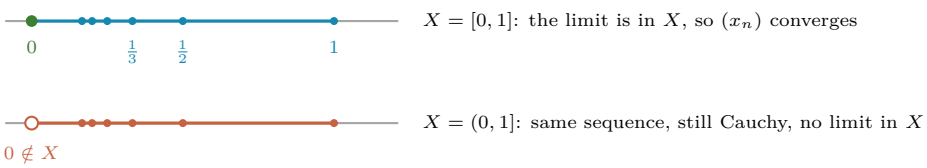
Completeness is the first genuinely **metric** rather than topological property met in this part, and the distinction is sharp: two metrics can generate exactly the same open sets while only one of them is complete. What completeness is usually good for is producing objects one cannot write down. The Banach fixed point theorem at the end of the chapter is the standard instrument for that, and it reappears as the engine of both the inverse function theorem and the Picard–Lindelöf existence theorem later on.

Completeness (Section 6.a)

Definition 6.a.1 (*Cauchy sequence and complete metric space*): Let (X, d) be a metric space and $(x_n)_{n=0}^{\infty}$ a sequence in X. We say that $(x_n)_{n=0}^{\infty}$ is "*Cauchy*" if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ such that } d(x_n, x_m) < \varepsilon \quad \forall n, m \geq N.$$

(X, d) is a "*complete metric space*" (German: „*vollständiger metrischer Raum*“) if every Cauchy sequence in X converges with respect to d.



The two rows carry the same points, at the same spacing, and (x_n) is Cauchy in both: whether the terms bunch together is decided by the terms alone. Whether they bunch together **around something** is not. Completeness is therefore a property of the space, and a sequence cannot be inspected on its own to determine it.

Exercise 6.a.2 (*Elementary facts about Cauchy sequences*): Let (X, d) be a metric space and $(x_n)_{n=0}^{\infty}$ a sequence in X. Prove:

(a) Every Cauchy sequence is bounded (meaning $\{d(x_n, x_0) : n \geq 0\} \subset \mathbb{R}$ is a bounded set).

(b) Every convergent sequence is Cauchy.

(c) A Cauchy sequence converges if and only if it has a convergent subsequence.

Solution: *See solution on Page 78*

Remark 6.a.3: Part **(c)** is the standard route to proving completeness of a space **without** guessing the limit in advance: instead of constructing a candidate limit directly, one only has to extract **some** convergent subsequence, and **(c)** upgrades that to convergence of the whole sequence. This is exactly the shape of the argument used throughout Chapter 7.

Remark 6.a.4 (*The converse construction*): Incompleteness is a defect that can always be repaired. Given **any** metric space (X, d), complete or not, one can build a complete space $(\overline{X}, \overline{d})$ containing an isometric copy of X as a dense subset. This is the "*completion*" of (X, d), and $\mathbb{R}$ as the completion of $\mathbb{Q}$ is the motivating case. The construction is carried out in Section 6.b below.

That specific route to $\mathbb{R}$ itself is not repeated here, the course script marking it as extra material belonging to **Grundstrukturen**. What is worth taking from the general construction is that no metric space is irreparably incomplete, so the interesting question is never whether a completion exists, but whether the one you already have is complete.

Proposition 6.a.5 (*Convergence in $\mathbb{R}^n$ is coordinatewise; $\mathbb{R}^n$ is complete*): Equip $\mathbb{R}^n$ with the standard metric d_2, and write $x_m = (x_{m,1}, \dots, x_{m,n})$.

(a) $x_m \to x$ in $\mathbb{R}^n$ if and only if $x_{m,i} \to x_i$ in $\mathbb{R}$ for every $i = 1, \dots, n$.

(b) $(\mathbb{R}^n, d_2)$ is complete.

Proof:

Everything follows from the two-sided estimate, valid for every j:

$$|x_{m,j} - x_j| \;\le\; |x_m - x| \;=\; \sqrt{\sum_{i=1}^{n} |x_{m,i} - x_i|^2} \;\le\; \sqrt{n}\, \max_i |x_{m,i} - x_i|.$$

(a) If $|x_m - x| \to 0$, the left inequality forces each coordinate to converge. Conversely, if every coordinate converges then $\max_i |x_{m,i} - x_i| \to 0$ (a maximum of **finitely** many null sequences), so the right inequality forces $|x_m - x| \to 0$.

(b) Let (x_m) be Cauchy in $\mathbb{R}^n$. The same left inequality gives $|x_{m,j} - x_{k,j}| \le |x_m - x_k|$, so each coordinate sequence $(x_{m,j})_m$ is Cauchy in $\mathbb{R}$. Now $\mathbb{R}$ is complete, this being the completeness axiom from **Analysis I** and the only input the proof has, so each coordinate sequence converges, say $x_{m,j} \to x_j$. Setting $x := (x_1, \dots, x_n)$, part **(a)** gives $x_m \to x$.

Q.E.D.

Remark 6.a.6 (*How little was actually used*): Completeness of $\mathbb{R}^n$ is completeness of $\mathbb{R}$, applied n times. Nothing else enters, which is worth noticing because it shows exactly where the argument would break in infinite dimensions. There $\max_i$ runs over infinitely many coordinates and need not tend to 0 even when every individual coordinate does. That failure is the same phenomenon behind the ε-net argument in Lemma 7.a.12.

Example 6.a.7 (*Completeness examples*):

- The Euclidean spaces $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_{\text{eucl}})$ are **complete**.
- Any **non-closed** subset $U \subseteq \mathbb{R}^n$ (standard metric) and the rationals $\mathbb{Q}$ are **not complete**.
- $\big(C^0([0,1]), \sup_{x\in[0,1]} |\cdot|\big)$ is **complete** (Zorich, p. 22).
- Every set X equipped with the **discrete metric** is **complete** (e.g., the finite grid $X := \{(x,y) \in \mathbb{Z}^2 \mid |x|, |y| \le 3\}$). In such a space, every Cauchy sequence is eventually constant, hence convergent.

Remark 6.a.8 (*Completeness is not the same as closedness*): The second bullet is easy to over-read as *"complete means closed"*. It does not, and the difference is worth pinning down now (because it recurs).

Closedness is relative, completeness is intrinsic. *"Closed"* only means anything once you say **closed in what**, since it is a statement about how a set sits inside an ambient space. Completeness asks nothing about an ambient space, but only whether the Cauchy sequences of the space converge **within it**. So:

- $\mathbb{Q}$ is **closed in** $\mathbb{Q}$, every space being closed in itself (Exercise 3.b.16), and yet **not complete**: the decimal approximations 1, 1.4, 1.41, 1.414, … form a Cauchy sequence in $\mathbb{Q}$ that converges to nothing there. (Alternatively, the sequence defined recursively by $x_0 := 1$ and $x_{n+1} := \frac{x_n}{2} + \frac{1}{x_n}$ is Cauchy in $\mathbb{Q}$, but its limit $\sqrt{2}$ is not in $\mathbb{Q}$.)

- $(0,1)$ is **not closed in** $\mathbb{R}$ and also not complete, which is the second bullet's case. The two failures nevertheless have different causes, and the first does not imply the second in general.

What **is** true comes in two halves. In **any** metric space X, a complete subset $A \subseteq X$ is closed in X, since a sequence in A converging in X is Cauchy, so it converges in A, and limits are unique. The converse needs the ambient space to be complete: if X is complete and $A \subseteq X$, then

$$A \text{ closed in } X \iff A \text{ complete.}$$

This is why the second bullet is correct **as stated for** $\mathbb{R}^n$. It silently uses completeness of the ambient $\mathbb{R}^n$. Drop that hypothesis and the equivalence dies, as $\mathbb{Q}$ inside $\mathbb{Q}$ shows.

The same relative-versus-intrinsic distinction lies behind Important remark 4.a.13 ($[0,1]^2$ is open in itself, not in $\mathbb{R}^2$), and it returns as Remark 7.b.35. Compactness, by contrast, is intrinsic in the way completeness is: a subset K is compact exactly when the metric space $(K, d|_{K\times K})$ is, with no reference to any ambient space. That is precisely the reduction used in Definition 7.a.6.

Remark 6.a.9: The second bullet of Remark 6.a.8's example, in which $(0,1) \subset \mathbb{R}$ is open and not complete, is not a quirk of that particular interval: script Exercise 9.76 asks for exactly this in general, and the fact is already in this document as Exercise 7.c.42**(b)** below. **Every** proper, non-empty open subset of $\mathbb{R}^n$ fails to be complete, and for the same underlying reason. That solution gives the topological proof via connectedness, and a remark alongside it records the script's more elementary, connectedness-free alternative.

Exercise 6.a.10 (*Completeness is not a topological property*): Does completeness survive a change of metric that leaves the open sets alone? This exercise settles it. Let $X := (0,1]$, let $d_{\text{eucl}}(x,y) := |x-y|$ be the standard metric on X, and define

$$d(x,y) := \left|\frac{1}{x} - \frac{1}{y}\right|, \qquad x, y \in X.$$

(a) Show that d is a metric on X.
(b) Show that (X, d_{eucl}) is **not** complete.
(c) Show that (X, d) **is** complete.
(d) Show that d_{eucl} and d induce the **same topology** on X.

Diego Torres Tejeda's hint for (c): *Let (x_n) be d-Cauchy. Then $(1/x_n)$ is Cauchy for the standard metric, so it has a limit $\ell \in \mathbb{R}$; show $\ell \geq 1$ and that $x_n \to 1/\ell$.*

Info: *This exercise is adapted from Diego Torres Tejeda's class notes.*

Solution: *See solution on Page 78*

Compactness will turn out to be a strictly stronger finiteness-type property than completeness: every compact metric space is complete (Exercise 7.a.24), but the converse fails. Note that $\mathbb{R}^n$ itself is complete and yet, being unbounded, not compact, as we will see next (Theorem 7.a.3, Section 7.b).

Exercise 6.a.11 (*A metric space that is not complete*): Give an explicit example of a metric space (X, d) that is **not** complete, specifying both the set X and the distance function d.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), Box Problem 1, where only the final answer was graded.*

Solution: *See solution on Page 82*

Exercise 6.a.12 (*Single choice: when does a Cauchy sequence converge?*): Let $(a_i)_{i\in\mathbb{N}}$ be a Cauchy sequence in a metric space X. Exactly one of the following holds. Which?

(a) This sequence is necessarily convergent in X.

(b) This sequence has a limit in X if X is closed.
(c) This sequence converges in X if $X \subseteq \mathbb{R}$ is an interval.
(d) This sequence converges in X if X is topologically compact.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 1, the opening single-choice question of that paper.*

Solution: *See solution on Page 83*

The completion of a metric space (Section 6.b)

Remark 6.a.8 left one question open. We know $\mathbb{Q}$ is not complete and $\mathbb{R}$ is, and that $\mathbb{R}$ is built to repair exactly that defect. Is the repair a one-off trick special to the reals, or can any incomplete space be fixed the same way?

It can, and the construction below is the general answer. Every metric space embeds isometrically, and densely, into a complete one, obtained by taking the space of its own Cauchy sequences and declaring two of them equal when they ought to have the same limit. The reals from the rationals is then just the first instance.

Definition 6.b.13 (*Completion of a metric space*): Let (X, d) be a metric space. Write $\mathcal{C}_X$ for the set of all Cauchy sequences in X, and define an equivalence relation on $\mathcal{C}_X$ by

$$(x_n)_{n=0}^{\infty} \sim (y_n)_{n=0}^{\infty} \quad :\Longleftrightarrow \quad \lim_{n\to\infty} d(x_n, y_n) = 0.$$

The quotient set $\overline{X} := \mathcal{C}_X/\!\sim$, equipped with the map

$$\overline{d}\big([(x_n)_{n=0}^{\infty}],\, [(y_n)_{n=0}^{\infty}]\big) := \lim_{n\to\infty} d(x_n, y_n),$$

is called the "*completion*" (German: „*Vervollständigung*“) of (X, d). The injection $\iota : X \to \overline{X}$ sending x to the class of the constant sequence $(x, x, x, \dots)$ is called the "*canonical embedding*". For all $x, y \in X$,

$$d(x, y) = \overline{d}\big(\iota(x), \iota(y)\big),$$

which in particular shows ι is injective.

Exercise 6.b.14 (*Well-definedness of the completion*): Show that the objects introduced in Definition 6.b.13 are well defined. In particular, verify that $\overline{d}$ is indeed a metric on $\overline{X}$.

Solution: *See solution on Page 79*

Exercise 6.b.15 (*The completion is complete*): As the name suggests, the completion $(\overline{X}, \overline{d})$ of a metric space is complete, meaning every Cauchy sequence in $\overline{X}$ converges. A sequence in $\overline{X}$ is essentially a sequence of sequences, i.e. $\big[(x_{m,n})_{n=0}^{\infty}\big]_{m=0}^{\infty}$. Show that the representatives may be chosen so that $\big[(x_{n,n})_{n=0}^{\infty}\big]$ is a limit of this sequence. (Some such choice is needed: for arbitrary representatives the diagonal need not be Cauchy at all.)

Solution: *See solution on Page 79*

Exercise 6.b.16 (*Universal property of the completion*): Let (X, d) be a metric space with completion $(\overline{X}, \overline{d})$. Let (Y, d_Y) be a complete metric space, and let $f : X \to Y$ be a function such that

$$d(x, y) = d_Y\big(f(x), f(y)\big) \qquad \text{for all } x, y \in X.$$

Show that there exists a unique function $\overline{f} : \overline{X} \to Y$ such that $f = \overline{f} \circ \iota_X$ and

$$\overline{d}(\overline{x}, \overline{y}) = d_Y\big(\overline{f}(\overline{x}), \overline{f}(\overline{y})\big) \qquad \text{for all } \overline{x}, \overline{y} \in \overline{X}.$$

This can be interpreted as: "$\overline{X}$ *is the* ***smallest*** *complete metric space containing* X*.*"

Solution: *See solution on Page 80*

Remark 6.b.17: Exercise 6.b.16 is what makes *"the"* completion well-posed rather than just *"a"* completion: any other complete space that X embeds into isometrically receives a unique isometric copy of $\overline{X}$ as well, via $\overline{f}$. This is the same uniqueness-up-to-isomorphism pattern behind *"the"* real numbers as *"the"* complete ordered field, and it is exactly how the script uses this construction in §9.1.4, taking $X = \mathbb{Q}$.

Contraction mappings and the Banach fixed point theorem (Section 6.c)

Completeness guarantees that certain limits exist, but on its own it does not say what they are. The following theorem is the standard way of turning that guarantee into a construction: it identifies a single point, proves it exists, proves it is unique, and supplies an algorithm that converges to it. Almost every existence proof in the second half of this course goes through it.

Definition 6.c.18 (*Contraction mapping*): Let (X, d) be a metric space. A map $f : X \to X$ is a "*contraction*" (German: „*Kontraktion*“) if it is Lipschitz continuous (Definition 5.b.20) with some Lipschitz constant $L \in [0, 1)$, i.e.,

$$d(f(x), f(y)) \leq L \cdot d(x, y) \quad \text{for all } x, y \in X.$$

Theorem 6.c.19 (*Banach fixed point theorem*): Let (X, d) be a nonempty complete metric space (Definition 6.a.1) and let $f : X \to X$ be a contraction (Definition 6.c.18). Then f has a unique "*fixed point*" (German: „*Fixpunkt*“) $x^* \in X$, i.e., $f(x^*) = x^*$; moreover, for any $x_0 \in X$, the iterates $x_{n+1} := f(x_n)$ converge to x^*.

Proof idea:

The core idea is to create a Cauchy sequence by repeatedly applying the contraction mapping f. Fix an arbitrary starting point $x_0 \in X$, and define the sequence by $x_{n+1} := f(x_n)$ for all $n \geq 0$. Comparing consecutive terms gives:

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq L \cdot d(x_n, x_{n-1}).$$

Iterating this inequality yields $d(x_{n+1}, x_n) \leq L^n d(x_1, x_0)$. Since $L < 1$, one can use the triangle inequality and the geometric series to show that the distance between any two terms x_m and x_n (for $m > n$) becomes arbitrarily small as $n \to \infty$. Thus, the sequence is Cauchy. Because X is a complete metric space, this sequence converges to some limit $x^* \in X$. Since contraction mappings are continuous, one can pass the limit inside the function to conclude $f(x^*) = x^*$. Q.E.D.

Proof of existence and convergence:

Fix $x_0 \in X$ and let $x_{n+1} := f(x_n)$. As in the sketch, $d(x_{k+1}, x_k) \leq L^k\, d(x_1, x_0)$ for every k. For $m > n$ the triangle inequality and the geometric series give

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq d(x_1, x_0) \sum_{k=n}^{\infty} L^k = \frac{L^n}{1 - L}\, d(x_1, x_0),$$

and the right-hand side tends to 0 as $n \to \infty$ because $0 \leq L < 1$. So (x_n) is Cauchy, and by completeness it converges to some $x^* \in X$. A contraction is Lipschitz, hence continuous (Remark 5.b.21), so

$$f(x^*) = f\Big(\lim_{n\to\infty} x_n\Big) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = x^*.$$

Letting $m \to \infty$ in the displayed estimate, with n fixed, also gives the "*a-priori error bound*"

$$d(x_n, x^*) \leq \frac{L^n}{1 - L}\, d(x_1, x_0),$$

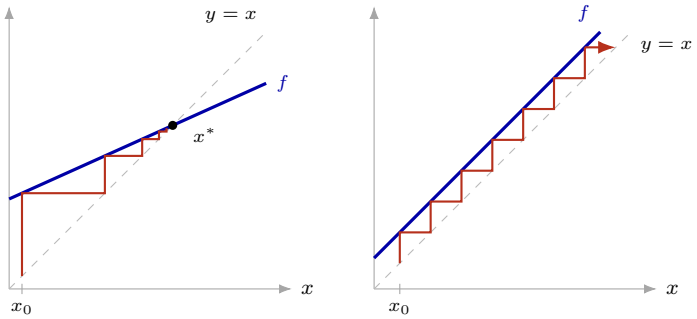
which says, before any iteration is carried out, how many steps guarantee a prescribed accuracy. Q.E.D.

Uniqueness of the fixed point:

Suppose $f(x) = x$ and $f(y) = y$. Then

$$d(x, y) = d\big(f(x), f(y)\big) \leq L\, d(x, y),$$

so $(1 - L)\, d(x, y) \leq 0$. Since $L < 1$ we have $1 - L > 0$, forcing $d(x, y) \leq 0$ and hence $d(x, y) = 0$, i.e. $x = y$. Q.E.D.



contraction, $L = 0.45$ iterates climb to the unique x^*

$L = 1$: graph parallel to $y = x$ no fixed point, iterates escape

The left panel is the proof in miniature. Each step moves vertically to the graph of f and then horizontally back to the diagonal, and because the graph is shallower than the diagonal, the steps shrink by a factor of at least L every time. That is precisely the geometric-series estimate the proof runs on. The right panel shows what a single missing hypothesis costs: with slope exactly 1 the graph never meets the diagonal, so no fixed point exists and the same iteration marches off instead of converging.

Remark 6.c.20 (*Where each hypothesis is used*): Both hypotheses of Theorem 6.c.19 are indispensable, and each fails in a different way.

- **Completeness** is what supplies the limit. On the incomplete space $X := (0, 1]$, the map $f(x) := x/2$ is a contraction with $L = \frac{1}{2}$ and has no fixed point in X, since the iterates march towards 0, which is exactly the point missing.
- $L < 1$ **strictly** is what forces the iterates together. The map $f(x) := x + \frac{1}{1+e^x}$ on the complete space $\mathbb{R}$ satisfies the strict inequality $|f(x) - f(y)| < |x - y|$ for $x \neq y$, yet has no fixed point, since $f(x) > x$ everywhere. A supremum of 1 that is never attained is still not good enough.

The uniqueness proof above shows the second point cleanly: the whole argument is the division by $1 - L$, which is exactly what $L = 1$ forbids.

Exercise 6.c.21 (*Both hypotheses of Banach's theorem are sharp*): Find examples for:

(a) A Lipschitz contraction $T : X \to X$ on a non-complete metric space X that has no fixed point.

(b) A complete metric space (X, d) and an "*isometry*" (a mapping $T : X \to X$ with $d(T(x_1), T(x_2)) = d(x_1, x_2)$ for all x_1, x_2) that has no fixed point.

Solution: *See solution on Page 80*

Remark 6.c.22: Part **(b)** sharpens Remark 6.c.20's second bullet: that example already showed a Lipschitz constant of exactly 1 can defeat the theorem, but its map T was not an isometry

($|T(x) - T(y)| < |x - y|$ strictly, just with no uniform gap). An isometry is the most rigid failure imaginable, not shrinking distances even a little, and it **still** need not have a fixed point, even on a complete space. Note that unboundedness is doing real work in the counterexample: on a **compact** space an isometry into itself is automatically surjective, and even that is not enough to force a fixed point. An irrational rotation of the circle S^1, which is compact and an isometry, has none either. Fixed points of isometries are a genuinely separate question from Theorem 6.c.19, not one it settles as a special case.

Exercise 6.c.23 (*A fixed point without a complete domain*): Let $E \subseteq X$ be a non-empty subset of a **complete** metric space X, and let $f : E \to X$ be a contraction, so that

$$d\big(f(x), f(y)\big) \leq L\, d(x, y) \qquad \text{for all } x, y \in E, \text{ with } L < 1.$$

Suppose further that there are $x_0 \in E$ and a fixed $r > 0$ such that the **closed** ball $\overline{B}_r(x_0)$ is contained in E, and that

$$d\big(f(x_0), x_0\big) \leq (1 - L)\, r.$$

Show that f has a fixed point in E: there exists $\bar{x} \in E$ with $f(\bar{x}) = \bar{x}$.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Problem 15 a).*

Solution: *See solution on Page 80*

Remark 6.c.24 (*What this buys over Theorem 6.c.19*): Read the hypotheses against the theorem's and the exercise stops looking like a technical variant. Three things are given up. The domain E is **not** assumed complete, nor even closed. The map does **not** send E into E; its codomain is all of X. And E may be far too large for the iteration to stay inside it.

What replaces them is one inequality. The condition $d(f(x_0), x_0) \leq (1 - L)r$ says the very first step is short relative to the room available, and the geometric series then guarantees that the **whole** orbit never leaves $\overline{B}_r(x_0)$: the total distance travelled is at most $\sum_{n \geq 0} L^n\, d(f(x_0), x_0) \leq r$. So the theorem is applied not to E but to $\overline{B}_r(x_0)$, which is closed in the complete space X and therefore complete, and which f does map into itself. This is how the fixed-point theorem is used in practice. In the proof of Theorem 15.a.1, for instance, one never has a globally invariant complete set to hand, only a small ball on which the first step is controlled.

AI-Exercise 6.c.25 (*Contraction Mapping on $[0,1]$*): Let $f\colon [0,1] \to [0,1]$ be defined by $f(x) := \frac{1}{2}\cos(x)$.

(a) Show that f is a contraction mapping (Definition 6.c.18) with Lipschitz constant $L = \frac{1}{2}\sin(1) < 1$.

(b) Deduce, using Theorem 6.c.19, that f has a unique fixed point $x^* \in [0,1]$.

Solution: *See solution on Page 81*

Exercise 6.c.26 (*Does the fixed-point theorem apply?*): Remark 6.c.20 shows that each hypothesis is needed by exhibiting one failure apiece. The checklist below drills the same point the other way round: for each row, decide whether Theorem 6.c.19 **applies** to $T : X \to X$, and separately determine the set of fixed points of T in X. The two questions have different answers more often than one expects.

$T(x)$	X	**Applies? / fixed points**
$\sin(x)$	$\mathbb{R}$	
$\frac{9}{10}\cos(x)$	$\mathbb{R}$	
x^2	$\mathbb{R}$	
$\frac{1}{2}x^2$	$(-1, 1)$	
$\frac{1}{2}\big(x + \frac{2}{x}\big)$	$\mathbb{Q} \cap (1, 2)$	

Info: *This exercise is taken from Linus Lüchinger's slides (2 March, slide 16, and 5 March, slide 3).*

Solution: *See solution on Page 82*

Exercise 6.c.27 (*Multiple choice (fixed points)*): Select all the statements below that are true.

(a) If you spread a map of Zurich on your desk, then one point of the desk will coincide with its representation on the map (in an ideal world).

(b) If $f : [0,1] \to [0,2]$ is $\frac{1}{2}$-Lipschitz, then f has a fixed point.

(c) If $f : [0,2] \to [0,1]$ is $\frac{1}{2}$-Lipschitz, then f has a fixed point.

(d) If $f : [0,1] \cup [2,3] \to [0,1] \cup [2,3]$ is continuously differentiable with $|f'(x)| < 1$ for all $x \in [0,1] \cup [2,3]$, then it has a fixed point.

(e) **(*)** If $f : [0,1] \to [0,1]$ is differentiable with $|f'(x)| < 1$ for all $x \in (0,1)$, then it has a fixed point.

Info: *This exercise is Problem 3.5 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 81*

Exercise 6.c.28 (*Contraction mappings and the equation $f(x) = \lambda x$*): Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be L-Lipschitz with respect to the Euclidean metric, for some $L > 0$, and let $\lambda > L$.

(a) Show that $g : \mathbb{R}^n \to \mathbb{R}^n$, $g(x) := \lambda^{-1} f(x)$, is a contraction.

(b) Conclude that the equation $f(x) = \lambda x$ has a unique solution $x \in \mathbb{R}^n$.

Info: *This exercise is taken from the Analysis I & II exam of 24 January 2022 (Prof. Giovanni Felder), Part B, Problem 2**(d)**. The original asks only for the conclusion, and only for all sufficiently large λ; the split into two parts, and the explicit threshold $\lambda > L$, are added here.*

Solution: *See solution on Page 84*

Exercise 6.c.29 (*True or False* (the hypotheses of Banach's theorem)): Let $X := [0,1]$ carry the Euclidean metric, and let $f : X \to X$ be $f(x) := x^2$. Decide whether each of the following is true or false, with justification.

(a) f has Lipschitz constant 2.

(b) X and f satisfy the hypotheses of the Banach fixed point theorem.

(c) f has fixed points.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the second block of true/false questions.*

Solution: *See solution on Page 83*

Exercise 6.c.30 (*True or False* (the hypotheses of Banach's theorem, part II)): Let $X := [-\pi, \pi]$ carry the Euclidean metric, and let $f : X \to X$ be $f(x) := \cos(x)$. Decide whether each of the following is true or false.

(a) f has Lipschitz constant 1.

(b) X and f satisfy the hypotheses of the Banach fixed point theorem.

(c) f has a fixed point.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the second block of true/false questions.*

Solution: *See solution on Page 84*

Solutions (Section 6.d)

Proof of Exercise 6.a.2:

(a) Bounded. Take $\varepsilon = 1$: there is N with $d(x_n, x_m) < 1$ for all $n, m \geq N$; in particular $d(x_n, x_N) < 1$ for $n \geq N$. Setting $M := \max\{d(x_0, x_N), \dots, d(x_{N-1}, x_N), 1\}$, we get $d(x_n, x_N) \leq M$ for **every** n: by the choice of M for the finitely many $n < N$, and because $d(x_n, x_N) < 1 \leq M$ for $n \geq N$. Since $d(x_N, x_0) \leq M$ as well, the triangle inequality gives $d(x_n, x_0) \leq d(x_n, x_N) + d(x_N, x_0) \leq 2M$ for every n, so $(x_n)_n \subseteq B_{2M+1}(x_0)$.

(b) Convergent ⇒ Cauchy. If $x_n \to x$, fix $\varepsilon > 0$ and choose N with $d(x_n, x) < \varepsilon/2$ for $n \geq N$. Then for $n, m \geq N$, $d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon$.

(c) Cauchy + convergent subsequence ⇒ convergent. Let (x_n) be Cauchy and $x_{n_k} \to x$. The *"only if"* direction is part **(b)** applied to the whole sequence. For *"if"*: fix $\varepsilon > 0$. Cauchyness gives N with $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$; convergence of the subsequence gives K with $d(x_{n_k}, x) < \varepsilon/2$ for $k \geq K$. Since $n_k \to \infty$, pick $k \geq K$ with $n_k \geq N$. Then for every $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon,$$

so $x_n \to x$. **Q.E.D.**

Solution to Exercise 6.a.10:

Write $\varphi : (0, 1] \to [1, \infty)$, $\varphi(x) := 1/x$. It is a bijection, and by construction $d(x, y) = |\varphi(x) - \varphi(y)|$. Consequently, d is nothing but the Euclidean metric of $[1, \infty)$ transported back to $(0, 1]$ along φ. Every part follows from that.

(a) Symmetry and the triangle inequality are inherited from $|\cdot|$ directly. For definiteness, $d(x, y) = 0$ forces $\varphi(x) = \varphi(y)$, and φ is injective, so $x = y$.

(b) The sequence $x_n := \frac{1}{n}$ lies in $(0, 1]$ and is Cauchy for d_{eucl}, being Cauchy in $\mathbb{R}$. Its limit in $\mathbb{R}$ is $0 \notin X$, and limits in $\mathbb{R}$ are unique (Proposition 4.a.4), so it has no limit in X.

(c) Let (x_n) be d-Cauchy. Then $\big(\varphi(x_n)\big)$ is Cauchy in $\mathbb{R}$, hence converges to some $\ell \in \mathbb{R}$. Since $\varphi(x_n) \geq 1$ for every n, also $\ell \geq 1$; in particular $\ell \neq 0$, so $x := 1/\ell$ lies in $(0, 1]$. Then

$$d(x_n, x) = \big|\varphi(x_n) - \varphi(x)\big| = \big|\varphi(x_n) - \ell\big| \longrightarrow 0,$$

so $x_n \to x$ in (X, d). Note where the boundedness of X entered: it is what forces $\ell \geq 1$ rather than merely $\ell \geq 0$.

(d) φ and $\varphi^{-1}(t) = 1/t$ are continuous, so φ is a homeomorphism. A set is d-open exactly when its φ-image is open in $[1, \infty)$, exactly when it is d_{eucl}-open. Concretely, the same two-sided-ball argument as in Exercise 7.b.34**(b)** works: on $(0, 1]$ one has $|x - y| \leq d(x, y) \leq \frac{1}{xy}\,|x - y|$ with $\frac{1}{xy}$ locally bounded.

Remark 6.d.31: So d_{eucl} and d have the **same open sets**, the same convergent sequences, the same continuous functions and the same compact sets, and yet one is complete and the other is not. **Completeness is a property of the metric, not of the topology it generates.**

This completes a trio worth holding together, all three proved by the same device of comparing two metrics that induce one topology:

- **Compactness is topological.** It is defined by open covers alone (Definition 7.a.1), so it cannot distinguish two metrics with the same open sets.
- **Boundedness is not:** see Remark 7.b.35.
- **Completeness is not:** this exercise.

That is exactly why *"compact if and only if closed and bounded"* needs the standard metric, and why the correct general replacement is *"complete and totally bounded"* (Remark 7.a.14). Both ingredients of the replacement are non-topological, and only their combination is.

Q.E.D.

Proof of Exercise 6.b.14:

$\sim$ is an equivalence relation on $\mathcal{C}_X$. Reflexivity is $d(x_n, x_n) = 0 \to 0$. Symmetry is symmetry of d. For transitivity, if $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$, the triangle inequality gives $0 \le d(x_n, z_n) \le d(x_n, y_n) + d(y_n, z_n) \to 0 + 0 = 0$, so $(x_n) \sim (z_n)$.

$\overline{d}$ is well-defined. First, the limit defining $\overline{d}$ exists: for $(x_n), (y_n) \in \mathcal{C}_X$, the reverse triangle inequality gives

$$\big|d(x_n, y_n) - d(x_m, y_m)\big| \le d(x_n, x_m) + d(y_n, y_m),$$

and the right-hand side $\to 0$ as $n, m \to \infty$ since (x_n) and (y_n) are each Cauchy, so $\big(d(x_n, y_n)\big)_n$ is a Cauchy sequence of **real numbers**, hence convergent. Note that this is the completeness axiom for $\mathbb{R}$ itself from **Analysis I**, and not an instance of Definition 6.a.1 being proved circularly. Second, the limit does not depend on the choice of representative: if $(x_n) \sim (x'_n)$ and $(y_n) \sim (y'_n)$, the same reverse triangle inequality gives $\big|d(x_n, y_n) - d(x'_n, y'_n)\big| \le d(x_n, x'_n) + d(y_n, y'_n) \to 0$, so the two limits agree.

$\overline{d}$ is a metric. Fix representatives $(x_n) \in \overline{x}$, $(y_n) \in \overline{y}$, $(z_n) \in \overline{z}$.

- **Definiteness.** $\overline{d}(\overline{x}, \overline{y}) = 0$ means $\lim_n d(x_n, y_n) = 0$, i.e. exactly $(x_n) \sim (y_n)$, i.e. $\overline{x} = \overline{y}$.
- **Symmetry.** Immediate from $d(x_n, y_n) = d(y_n, x_n)$ for every n.
- **Triangle inequality.** Take $n \to \infty$ in $d(x_n, z_n) \le d(x_n, y_n) + d(y_n, z_n)$, using that limits preserve weak inequalities.

Finally $d(x, y) = \overline{d}(\iota(x), \iota(y))$ holds because the constant sequences $(x, x, \dots)$ and $(y, y, \dots)$ give $d(x_n, y_n) = d(x, y)$ for every n, so the limit is $d(x, y)$ itself; and this identity makes ι injective, since $d(x, y) = 0 \iff x = y$. Q.E.D.

Proof of Exercise 6.b.15:

Let $(\xi_m)_{m=0}^{\infty}$ be Cauchy in $\overline{X}$, with $\xi_m = [(x_{m,n})_{n=0}^{\infty}]$ for some Cauchy sequence $(x_{m,n})_n$ in X representing ξ_m. Replacing $(x_{m,n})_n$ by a tail of itself if necessary, which is harmless because a Cauchy sequence and any of its tails represent the same class, differing in only finitely many terms, we may assume each representative is already "*regularized*":

$$d(x_{m,n}, x_{m,n'}) < \tfrac{1}{m+1} \qquad \text{for all } n, n' \ge 0.$$

Set $y_m := x_{m,m}$, the diagonal sequence the exercise asks about. Letting $n' \to \infty$ in the regularization bound (with $n = m$) gives

$$\overline{d}\big(\iota(y_m),\, \xi_m\big) = \lim_{n\to\infty} d(x_{m,m}, x_{m,n}) \;\le\; \tfrac{1}{m+1}. \tag{6.1}$$

$(y_m)_m$ is Cauchy in X. By the triangle inequality for $\overline{d}$ (already established in Exercise 6.b.14) and (6.1),

$$\begin{aligned} d(y_m, y_{m'}) &= \overline{d}\big(\iota(y_m), \iota(y_{m'})\big) \\ &\le\; \overline{d}(\iota(y_m), \xi_m) + \overline{d}(\xi_m, \xi_{m'}) + \overline{d}(\xi_{m'}, \iota(y_{m'})) \\ &\le\; \tfrac{1}{m+1} + \overline{d}(\xi_m, \xi_{m'}) + \tfrac{1}{m'+1}. \end{aligned}$$

Given $\varepsilon > 0$, Cauchyness of $(\xi_m)_m$ gives M_1 with $\overline{d}(\xi_m, \xi_{m'}) < \varepsilon/3$ for $m, m' \ge M_1$; taking also $M \ge \max\{M_1, 3/\varepsilon\}$ makes all three terms $< \varepsilon/3$ for $m, m' \ge M$, so $(y_m)_m$ is Cauchy in X and defines a class $\overline{x} := [(y_m)_m] \in \overline{X}$.

$\xi_m \to \overline{x}$. Using (6.1) again and the same triangle inequality,

$$\overline{d}(\xi_m, \overline{x}) \;\le\; \overline{d}(\xi_m, \iota(y_m)) + \overline{d}(\iota(y_m), \overline{x}) \;\le\; \tfrac{1}{m+1} + \lim_{k\to\infty} d(y_m, y_k).$$

The first term $\to 0$ as $m \to \infty$ by construction, and the second does too, being (for each fixed m) bounded by $\sup_{k \geq m} d(y_m, y_k)$, which $\to 0$ as $m \to \infty$ since $(y_m)_m$ is Cauchy. Hence $\overline{d}(\xi_m, \overline{x}) \to 0$, i.e. $\xi_m \to \overline{x}$ in $\overline{X}$, as required. Q.E.D.

Proof of Exercise 6.b.16:

Construction. For $\overline{x} = [(x_n)_n] \in \overline{X}$, the sequence $(f(x_n))_n$ is Cauchy in Y, since $d_Y(f(x_n), f(x_m)) = d(x_n, x_m) \to 0$ as $(x_n)_n$ is Cauchy in X. As Y is complete, $f(x_n) \to y$ for a unique $y \in Y$ (Proposition 4.a.4); define $\overline{f}(\overline{x}) := y$.

Well-defined. If $(x_n) \sim (x'_n)$, then $d_Y(f(x_n), f(x'_n)) = d(x_n, x'_n) \to 0$, so $(f(x_n))_n$ and $(f(x'_n))_n$ have the same limit in Y by uniqueness of limits; hence $\overline{f}(\overline{x})$ does not depend on the representative chosen.

$f = \overline{f} \circ \iota_X$. For $x \in X$, $\iota_X(x)$ is represented by the constant sequence $(x, x, \dots)$, so $\overline{f}(\iota_X(x)) = \lim_n f(x) = f(x)$.

$\overline{f}$ satisfies the displayed identity. For $\overline{x} = [(x_n)_n]$, $\overline{y} = [(y_n)_n]$, continuity of d_Y in both arguments (itself 1-Lipschitz, by the same reverse-triangle-inequality argument as in Exercise 6.b.14) gives

$$d_Y\big(\overline{f}(\overline{x}), \overline{f}(\overline{y})\big) = d_Y\Big(\lim_n f(x_n),\ \lim_n f(y_n)\Big) = \lim_n d_Y\big(f(x_n), f(y_n)\big) = \lim_n d(x_n, y_n) = \overline{d}(\overline{x}, \overline{y}).$$

Uniqueness. Let $g : \overline{X} \to Y$ satisfy the same two conditions. The displayed identity makes g (like $\overline{f}$) 1-Lipschitz, hence continuous. For $\overline{x} = [(x_n)_n]$, the sequence $\iota_X(x_n)$ converges to $\overline{x}$ in $\overline{X}$: indeed $\overline{d}(\iota_X(x_n), \overline{x}) = \lim_k d(x_n, x_k) \to 0$ as $n \to \infty$, since $(x_n)_n$ is Cauchy. By continuity of g and $f = g \circ \iota_X$,

$$g(\overline{x}) = \lim_n g(\iota_X(x_n)) = \lim_n f(x_n) = \overline{f}(\overline{x}),$$

using uniqueness of limits in Y once more for the last equality. As $\overline{x}$ was arbitrary, $g = \overline{f}$. Q.E.D.

Proof of Exercise 6.c.21:

(a) This is Remark 6.c.20's first bullet: $X := (0, 1]$ is not complete, and $T(x) := x/2$ is a $\frac{1}{2}$-contraction on X with iterates marching towards the missing point 0, hence no fixed point in X.

(b) Take $X := \mathbb{R}$ (complete, Proposition 6.a.5) with the translation $T(x) := x + 1$. Then $d(T(x), T(y)) = |(x+1) - (y+1)| = |x - y| = d(x, y)$, so T is an isometry, and in particular Lipschitz with constant $L = 1$ rather than $L < 1$. Consequently, this does not contradict Theorem 6.c.19. A fixed point would need $x + 1 = x$, which is impossible, so T has none.

Q.E.D.

Proof of Exercise 6.c.23:

Write $K := \overline{B}_r(x_0)$. The whole content of the exercise is that f restricts to a self-map of K, after which Theorem 6.c.19 finishes the job.

K is complete. A closed ball is a closed subset of X, and a closed subset of a complete metric space is complete: a Cauchy sequence in K is Cauchy in X, hence converges in X, and its limit lies in K because K is closed. It is also non-empty, since $x_0 \in K$.

$f(K) \subseteq K$. Let $x \in K$, so $d(x, x_0) \leq r$. Since $K \subseteq E$, both x and x_0 lie in the domain of f, and the triangle inequality together with the contraction estimate gives

$$d\big(f(x), x_0\big) \ \leq\ d\big(f(x), f(x_0)\big) + d\big(f(x_0), x_0\big) \ \leq\ L\, d(x, x_0) + (1 - L)r \ \leq\ Lr + (1 - L)r \ =\ r,$$

so $f(x) \in K$. The hypothesis $d(f(x_0), x_0) \leq (1 - L)r$ is used exactly once, here, and it is what makes the two terms add up to r rather than to something larger.

Conclusion. The restriction $f|_K : K \to K$ is a contraction with the same constant L on the non-empty complete metric space K, so Theorem 6.c.19 supplies a unique $\bar{x} \in K$ with $f(\bar{x}) = \bar{x}$. Since $K \subseteq E$, this $\bar{x}$ is a fixed point in E, as required.

Uniqueness comes for free, and in all of E. Theorem 6.c.19 gives uniqueness only within K, but the two-line argument used for the theorem itself needs nothing beyond the contraction estimate, which holds on all of E: if $f(x) = x$ and $f(y) = y$ with $x, y \in E$, then $d(x, y) = d(f(x), f(y)) \leq L\, d(x, y)$, and $L < 1$ forces $d(x, y) = 0$. So $\bar{x}$ is the only fixed point of f in E, even though the existence argument saw only the ball. **Q.E.D.**

Proof of AI-Exercise 6.c.25:

(a) By the Mean Value Theorem, for any $x, y \in [0, 1]$, there exists $\xi \in (0, 1)$ such that $|f(x) - f(y)| = |f'(\xi)||x - y| = \frac{1}{2}\sin(\xi)|x - y|$. Since $\sin(x)$ is strictly increasing on $[0, 1]$, $\sup_{\xi \in [0,1]} \frac{1}{2}\sin(\xi) = \frac{1}{2}\sin(1) \approx 0.4207 < 1$. Thus f is a contraction (Definition 6.c.18) with $L = \frac{1}{2}\sin(1) < 1$.

(b) Since $[0, 1]$ equipped with the Euclidean metric is a complete metric space (Definition 6.a.1) and $f([0, 1]) \subseteq [0, 1]$, Theorem 6.c.19 guarantees the existence and uniqueness of a fixed point $x^* \in [0, 1]$ satisfying $f(x^*) = x^*$.

Q.E.D.

Solution to Exercise 6.c.27:

(a), (c) and (e) are true; (b) and (d) are false.

(a) **True,** and this is the classic illustration of Theorem 6.c.19. The map of Zurich lying on the desk defines a function from the (compact, hence complete) region of the city depicted onto a smaller region of the desk, and that function is a similarity: a rotation and translation composed with a scaling by the map's ratio $L < 1$. It is therefore a contraction of a complete metric space into itself, and so has exactly one fixed point: one spot on the map lying directly above the point of the city it represents.

(b) **False.** The hypothesis says nothing about f mapping its domain into itself, and the codomain $[0, 2]$ is strictly larger than the domain $[0, 1]$. Take

$$f(x) := \tfrac{x}{2} + \tfrac{3}{2}, \qquad f : [0, 1] \to [\tfrac{3}{2}, 2] \subseteq [0, 2],$$

which is $\frac{1}{2}$-Lipschitz. A fixed point would need $x = \frac{x}{2} + \frac{3}{2}$, i.e. $x = 3$, which is not in $[0, 1]$.

(c) **True.** Now the inclusion runs the right way: f maps $[0, 2]$ into $[0, 1] \subseteq [0, 2]$, so f is a self-map of the complete space $[0, 2]$ (closed and bounded in $\mathbb{R}$, hence compact, hence complete by Exercise 7.a.24), and it is a contraction with $L = \frac{1}{2} < 1$. Theorem 6.c.19 applies directly.

(d) **False,** and this is the instructive one: the domain is **disconnected**, and the condition $|f'| < 1$ is purely local, so it cannot see across the gap. Define

$$f(x) := \begin{cases} \frac{5}{2} & x \in [0, 1], \\ \frac{1}{2} & x \in [2, 3]. \end{cases}$$

Then f is continuously differentiable with $f' \equiv 0$, it maps $[0, 1] \cup [2, 3]$ into itself, and it has no fixed point, because each of the two pieces is sent into the **other** one. Compare the local-to-global reading of Remark 7.c.38: a pointwise derivative bound upgrades to a global contraction only along paths that stay inside the domain, and here there are none.

(e) **True,** but **not** by the fixed point theorem, and that is the point of the star. The hypothesis $|f'(x)| < 1$ gives no uniform $L < 1$, since the supremum of $|f'|$ may well equal 1, so Theorem 6.c.19 is unavailable. Existence follows from the intermediate value theorem instead. Put $g(x) := f(x) - x$, which is continuous on $[0, 1]$. Since f takes values in $[0, 1]$,

$$g(0) = f(0) \geq 0 \qquad \text{and} \qquad g(1) = f(1) - 1 \leq 0,$$

so g vanishes somewhere in $[0,1]$ by Corollary 8.a.8, and that zero is a fixed point of f. Note that the derivative hypothesis was never used: **every** continuous self-map of $[0,1]$ has a fixed point. What $|f'|<1$ buys is **uniqueness**, by the same two-line argument as in the proof of uniqueness for Theorem 6.c.19 combined with the mean value theorem.

Q.E.D.

Solution to Exercise 6.c.26:

Recall that Theorem 6.c.19 needs three things at once: X complete, T mapping X into X, and a Lipschitz constant $\lambda<1$ that is **actually attained as a bound**, not merely approached.

- $T(x)=\sin(x)$ **on** $\mathbb{R}$**. Does not apply; fixed points** $\{0\}$**.** $\mathbb{R}$ is complete and T maps it into itself, but $\lambda=\sup_x|\cos x|=1$, attained at $x=0$. So the theorem is unavailable. A fixed point exists all the same: $\sin x=x$ holds at $x=0$ and nowhere else, since $|\sin x|<|x|$ for $x\neq 0$. This is the first lesson of the table: the theorem is **sufficient**, never necessary.
- $T(x)=\frac{9}{10}\cos(x)$ **on** $\mathbb{R}$**. Applies; one fixed point.** Here $|T'(x)|=\frac{9}{10}|\sin x|\le\frac{9}{10}<1$, so by the mean value theorem T is a $\frac{9}{10}$-contraction; $\mathbb{R}$ is complete and $T(\mathbb{R})\subseteq[-\frac{9}{10},\frac{9}{10}]\subseteq\mathbb{R}$. All three hypotheses hold, so there is exactly one fixed point. It has no closed form; iterating from any starting value gives $x^*\approx 0.693$.
- $T(x)=x^2$ **on** $\mathbb{R}$**. Does not apply; fixed points** $\{0,1\}$**.** T is not Lipschitz at all on $\mathbb{R}$, since $T'(x)=2x$ is unbounded. That it cannot apply is visible without any computation: $x^2=x$ has the two solutions 0 and 1, and the theorem asserts **uniqueness**, so any set of hypotheses implying it must fail here.
- $T(x)=\frac{1}{2}x^2$ **on** $(-1,1)$**. Does not apply; fixed points** $\{0\}$**.** Two hypotheses fail together. The domain $(-1,1)$ is not complete, and $\lambda=\sup_{|x|<1}|T'(x)|=\sup|x|=1$. This supremum is not attained, but a supremum of 1 is still not <1, which is exactly the trap of Remark 6.c.20. The map does send $(-1,1)$ into $[0,\frac{1}{2})\subseteq(-1,1)$. Solving $\frac{1}{2}x^2=x$ gives $x\in\{0,2\}$, of which only 0 lies in the domain, so a unique fixed point exists regardless.
- $T(x)=\frac{1}{2}\big(x+\frac{2}{x}\big)$ **on** $\mathbb{Q}\cap(1,2)$**. Does not apply; no fixed points.** This row is the interesting one, and it is the Babylonian iteration for $\sqrt{2}$. Everything works except completeness:
  - $T'(x)=\frac{1}{2}-\frac{1}{x^2}$, so on $(1,2)$ we have $-\frac{1}{2}\le T'(x)\le\frac{1}{4}$ and hence $\lambda=\frac{1}{2}<1$: a genuine contraction.
  - T maps X into X: it sends rationals to rationals, and on $(1,2)$ its minimum is $T(\sqrt{2})=\sqrt{2}\approx 1.414$ while $T(1)=T(2)=\frac{3}{2}$, so $T\big((1,2)\big)\subseteq[\sqrt{2},\frac{3}{2}]\subseteq(1,2)$.
  - But $\mathbb{Q}\cap(1,2)$ is **not complete**.

  And there is indeed no fixed point: $x=\frac{1}{2}(x+\frac{2}{x})$ forces $x^2=2$, so $x=\sqrt{2}\notin\mathbb{Q}$. The iterates are the familiar rational approximations $\frac{3}{2},\frac{17}{12},\frac{577}{408},\dots$, which form a Cauchy sequence in X converging to a point that X does not contain.

  This is worth contrasting with Remark 6.c.20, where completeness failed on $(0,1]$ with the missing point sitting at an endpoint one is already suspicious of. Here the domain is an interval of rationals, the hole is invisible, and the contraction is a thoroughly practical algorithm. Completeness is not a technicality.

Q.E.D.

Solution of Exercise 6.a.11:

Take $X:=(0,1]$ with the Euclidean distance $d(x,y):=|x-y|$ inherited from $\mathbb{R}$. The sequence $x_k:=1/k$ lies in X and is Cauchy, since $|x_k-x_l|\le\max\{1/k,1/l\}\to 0$. But it does not converge in X: its only candidate limit in $\mathbb{R}$ is 0, and $0\notin X$, so no point of X can be its limit. Hence (X,d) is not complete.

The same mechanism gives the other standard answers. Taking $X:=\mathbb{Q}$ with $d(x,y):=|x-y|$,

any rational sequence converging to $\sqrt{2}$ is Cauchy without a limit in X; taking $X := \mathbb{R}\setminus\{0\}$ works for the same reason as $(0,1]$.

AI-Note 6.1: What is being exploited each time is that completeness is **not** a property of the distance alone: $(0,1]$ and $[0,1]$ carry the same formula for d, and one is complete while the other is not. A Cauchy sequence is a sequence that is trying to converge, and the question is only whether the space has kept the point it is aiming at. This is exactly the gap that the completion of a metric space (Section 6.b) is built to fill.

Q.E.D.

Solution of Exercise 6.a.12:

Only **(d)** is correct.

- **(a)** **False.** A Cauchy sequence converges precisely when the ambient space has kept the point it is aiming at, and that is the definition of completeness (Definition 6.a.1), not a theorem about Cauchy sequences. Exercise 6.a.11 is a counterexample in one line.
- **(b)** **False**, and this is the option the question is built around. Closedness is not a property a space has, only a property a subset has **relative to** something larger: every metric space is closed in itself, so reading (b) as a hypothesis on X alone makes it say nothing at all, and $X := (0,1]$ is again a counterexample. What is true is the relative statement: a closed subset of a **complete** space is complete, and the ambient completeness is the part (b) omits.
- **(c)** **False.** An interval need not be complete either. On $X := (0,1)$ the sequence $a_i := 1/i$ is Cauchy with no limit in X. Only the closed bounded intervals, and the closed unbounded ones, are complete.
- **(d)** **True.** A compact metric space is sequentially compact (Theorem 7.a.3), so (a_i) has a convergent subsequence, and a Cauchy sequence with a convergent subsequence converges (Exercise 6.a.2**(c)**). Hence every compact metric space is complete, which is Exercise 7.a.24.

Q.E.D.

Solution of Exercise 6.c.29:

- **(a)** **True.** For $x,y\in[0,1]$, $|x^2-y^2| = |x+y|\,|x-y| \le 2|x-y|$, since $x+y\le 2$ on $[0,1]$. So 2 is a Lipschitz constant. It is even the optimal one, as $|f'(x)| = 2x$ approaches 2 at $x=1$.
- **(b)** **False.** The space $X=[0,1]$ is complete, and f does map X into X; what fails is the contraction hypothesis. A contraction needs a Lipschitz constant **strictly less than** 1, and by **(a)** the best available constant here is 2. Concretely, taking $x=1$ and $y=1-\varepsilon$ gives $|f(x)-f(y)| = (2-\varepsilon)\varepsilon > |x-y|$ for small $\varepsilon>0$: near $x=1$ the map **expands** distances.
- **(c)** **True**, and note the plural. Solving $x^2=x$ on $[0,1]$ gives $x=0$ and $x=1$, so f has exactly two fixed points.

Remark 6.d.32: Parts **(b)** and **(c)** together are the point of the exercise. The Banach fixed point theorem (Theorem 6.c.19) is a sufficient condition, not a necessary one: its hypotheses fail here, and fixed points exist anyway. What the failure does cost is the **uniqueness** half of the conclusion, and sure enough there are two of them. Reading *"the hypotheses fail"* as *"there is no fixed point"* is the standard mistake.

Q.E.D.

Proof of Exercise 6.c.28:

(a) Fix $x, y \in \mathbb{R}^n$. Since the Euclidean norm is homogeneous and f is L-Lipschitz,

$$|g(x) - g(y)| = \left| \frac{f(x) - f(y)}{\lambda} \right| = \frac{1}{\lambda} |f(x) - f(y)| \leq \frac{L}{\lambda} |x - y|.$$

Set $k := L/\lambda$. Since $\lambda > L > 0$ we have $k < 1$, and therefore g is a contraction with contraction constant k.

(b) The space $\mathbb{R}^n$ is complete, so the Banach Fixed Point Theorem (Theorem 6.c.19) applies to g and yields a unique $x \in \mathbb{R}^n$ with $g(x) = x$. Multiplying $\lambda^{-1} f(x) = x$ by λ turns this into $f(x) = \lambda x$, and the two equations have exactly the same solutions. Hence $f(x) = \lambda x$ has a unique solution.

AI-Note 6.2: The hypothesis is what makes this work, and it is worth naming: a Lipschitz f need not be a contraction, and need have no fixed point at all (translation $x \mapsto x + v$ is 1-Lipschitz). Dividing by $\lambda > L$ buys the missing factor. This is also why the original exam question says *"for all sufficiently large λ"* rather than naming a threshold: any $\lambda > L$ works, but L is not part of the data the question hands you.

Q.E.D.

Proof of Exercise 6.c.30:

(a) **True:** The derivative is $f'(x) = -\sin(x)$. Since $|f'(x)| = |\sin x| \leq 1$ for all $x \in \mathbb{R}$, the mean value theorem gives $|f(x) - f(y)| \leq 1 \cdot |x - y|$, so f has Lipschitz constant 1.

(b) **False:** The Banach fixed point theorem requires a **strict** contraction, that is, a Lipschitz constant $\lambda < 1$. Here, the tightest possible constant is $\sup_{x \in [-\pi, \pi]} |f'(x)| = 1$ (attained at $x = \pm\pi/2$), so the map is non-expansive but not strictly contractive.

(c) **True:** The function $f(x) = \cos x$ is continuous and maps $[-\pi, \pi]$ into $[-1, 1] \subseteq [-\pi, \pi]$. Since $g(x) := \cos x - x$ has $g(-\pi) = \pi - 1 > 0$ and $g(\pi) = -1 - \pi < 0$, the intermediate value theorem supplies a root, which is a fixed point of f.

Q.E.D.

Compactness (Chapter 7)

Compactness is the least self-explanatory definition in this part, so it is worth saying in advance what it is **for**. Finite sets are easy to handle. One can take a maximum over them, choose a smallest radius, or check every case in turn. Infinite sets allow none of that. Compactness is the property that lets an infinite space be treated as though it were finite, not because it is small, but because every attempt to spread it out over infinitely many open pieces collapses back to finitely many.

The chapter defines compactness by open covers, proves it equivalent to sequential compactness, records the Heine–Borel theorem for $\mathbb{R}^n$ together with how badly it fails outside $\mathbb{R}^n$, and then collects the payoffs that make the definition worth having. The last section moves the whole question up one level. It asks which **families of functions** are compact in the supremum metric, and the answer, Ascoli–Arzelà, needs one hypothesis (equicontinuity) that has no counterpart for sets of points.

Compactness (Section 7.a)

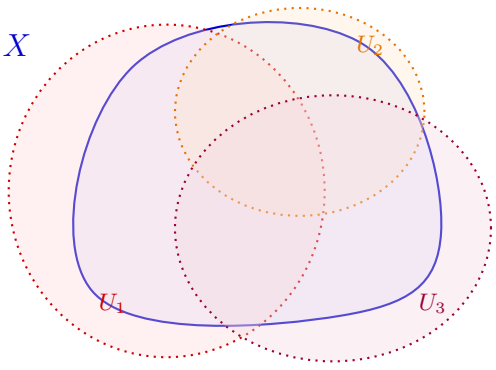
The definition itself is one line long, and the difficulty is entirely in the order of its quantifiers. It is worth reading it twice before reading anything else.

Definition 7.a.1 (*Compact metric space*): A metric space (X, d) is "*compact*" (German: „*kompakt*“) if every open cover (by sets open in the sense of Definition 3.a.6) has a finite subcover.

Notation 7.a.2 (*Open covers*): An "*open cover*" (German: „*offene Überdeckung*“) of X is a family $\{U_i\}_{i\in I}$ of open sets with $X = \bigcup_{i\in I} U_i$; the index set I is allowed to be arbitrary, in particular uncountable. A "*subcover*" is a subfamily $\{U_i\}_{i\in J}$, $J \subseteq I$, that still covers X, and it is **finite** when J is. The order of quantifiers is what carries the content: **every** cover must admit **some** finite subcover. Note the asymmetry this produces. Exhibiting one cover with a finite subcover proves nothing at all, since every space has the one-element cover $\{X\}$. Exhibiting one cover **without** a finite subcover, on the other hand, disproves compactness outright, and that is how Exercise 7.b.34 and the open-interval example below both work.

Theorem 7.a.3 (*Sequential compactness*): A metric space (X, d) is compact if and only if every sequence in X has a convergent (Definition 4.a.1) subsequence in X.

AI-Note 7.1: The proof is given below as Theorem 7.a.13, once the tools it needs (total boundedness) are available.



$X = U_1 \cup U_2 \cup U_3$

Lemma 7.a.4 (*Nesting principle*): Let X be a metric space. A subset $K \subseteq X$ is topologically compact (Item **(b)**) if and only if it has the following property: for every collection $\mathcal{A} = \{A_i\}_{i\in I}$

of closed subsets of X, if every intersection of finitely many sets in $\mathcal{A}$ meets K, then $K \cap \bigcap_{i\in I} A_i$ is non-empty.

Proof:

(⇒) Let K be compact and let $\mathcal{A}$ be a collection of closed subsets of X with $K \cap \bigcap_{i\in I} A_i = \emptyset$. The complements $\mathcal{U} := \{X \setminus A_i : i \in I\}$ then form an open cover of K, so compactness gives a finite subcover $K \subseteq (X \setminus A_{i_1}) \cup \cdots \cup (X \setminus A_{i_n})$, i.e. $K \cap A_{i_1} \cap \cdots \cap A_{i_n} = \emptyset$: some finite subcollection already fails to meet K.

(⇐) Let K have the nesting property and let $\mathcal{U}$ be an open cover of K. The complements $\mathcal{A} := \{X \setminus U : U \in \mathcal{U}\}$ form a collection of closed sets with $K \cap \bigcap_{U\in\mathcal{U}} (X \setminus U) = K \setminus \bigcup_{U\in\mathcal{U}} U = \emptyset$, since $\mathcal{U}$ covers K. By the contrapositive of the nesting property, some finite subcollection $X \setminus U_1, \ldots, X \setminus U_n$ already has $K \cap (X \setminus U_1) \cap \cdots \cap (X \setminus U_n) = \emptyset$, i.e. $K \subseteq U_1 \cup \cdots \cup U_n$: a finite subcover. **Q.E.D.**

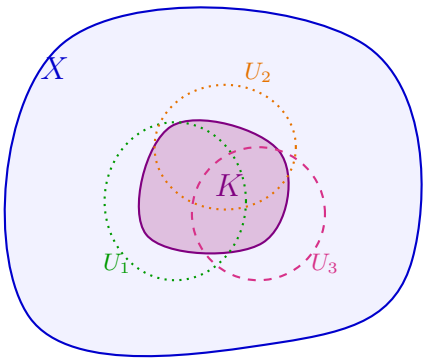
Remark 7.a.5: The nesting principle is Definition 7.a.1 read through De Morgan's laws. Open covers become closed families, unions become intersections, and *"some finite piece already covers"* becomes *"some finite piece already misses"*. It earns its own name because closed sets, not open ones, are the natural language for several later arguments (nested closed sets of shrinking size, in particular), where phrasing compactness this way saves a complementation step every time.

Compact subsets (Subsection 7.a.1)

A subset K of a metric space (X, d) is compact if the metric space $(K, d|_{K\times K})$ is compact (Definition 7.a.1). This is equivalent to the definition in the script:

Definition 7.a.6 (*Definition 9.62, quoted from the official lecture notes*)**:** Let (X, d) be a metric space. A subset $K \subset X$ is called **compact** if one of the following equivalent conditions hold:

- **(a)** K is **sequentially compact**: every sequence $(x_n)_{n\in\mathbb{N}}$ in K has a subsequence that is convergent (Definition 4.a.1) in K.
- **(b)** K is **topologically compact**: every family of open sets (Definition 3.a.6) $\{U_i\}_{i\in I}$ that cover K, has a finite subcover.
- **(c)** K is **complete** (every Cauchy sequence contained in K has a limit in K) and **totally bounded** (Definition 7.a.9): for every $r > 0$ there exist finitely many $x_1, \ldots, x_n \in K$ such that the balls $B_r(x_1), \ldots, B_r(x_n)$ cover K.



$K \subseteq U_1 \cup U_2 \cup U_3 \subseteq X$

Sequential and topological compactness agree (Subsection 7.a.2)

This subsection proves Theorem 7.a.3 and the equivalence of sequential and topological compactness (Theorem 7.a.13). By the reduction noted above, compactness of $K \subseteq X$ being compactness of the metric space $(K, d|_{K\times K})$, it suffices to treat a whole metric space X rather than a subset K.

Lemma 7.a.7 (*Continuous functions attain extrema on sequentially compact spaces*): Let (X, d) be sequentially compact (Item **(a)**) and $f : X \to \mathbb{R}$ continuous. Then f attains a maximum and a minimum on X.

Proof:

It suffices to find a minimum (apply the result to $-f$ for the maximum). Let $m := \inf_{x\in X} f(x) \in [-\infty, \infty)$ and choose a sequence $(x_n)_n$ with $f(x_n) \to m$. By sequential compactness, some subsequence $x_{n_k} \to x \in X$; by continuity, $f(x) = \lim_k f(x_{n_k}) = m$. In particular $m \neq -\infty$, so f attains its minimum m at x. Q.E.D.

Remark 7.a.8: This is exactly the extreme value theorem (Item **(b)**) proved from first principles, without presupposing Theorem 7.a.3. Avoiding that appeal is essential here, since that theorem is precisely what we are about to prove.

Definition 7.a.9 (*Totally bounded*): A metric space (X, d) is "*totally bounded*" (German: „*total beschränkt*“) if for every $\varepsilon > 0$ there exist finitely many points $x_1, \dots, x_n \in X$ such that

$$X = \bigcup_{k=1}^{n} B_\varepsilon(x_k).$$

Exercise 7.a.10 (*Totally bounded is strictly stronger than bounded*): Let X be a metric space. Show that if X is totally bounded, then it is bounded, i.e. $\sup_{x,x'\in X} d(x, x') < \infty$. Find an example of a bounded metric space that is **not** totally bounded.

Proof of Exercise 7.a.10:

Totally bounded ⇒ bounded. Apply Definition 7.a.9 with $\varepsilon := 1$: there are $x_1, \dots, x_n$ with $X = \bigcup_{k=1}^{n} B_1(x_k)$. Let $M := \max_{i,j} d(x_i, x_j)$ (finite, a maximum over finitely many pairs). For any $x, x' \in X$, choose i, j with $x \in B_1(x_i)$, $x' \in B_1(x_j)$; the triangle inequality gives $d(x, x') \leq d(x, x_i) + d(x_i, x_j) + d(x_j, x') < 1 + M + 1$, so X is bounded.

Not conversely. Let X be any infinite set with the discrete metric. Then $\sup_{x,x'} d(x, x') = 1 < \infty$, so X is bounded. But X is not totally bounded: for $\varepsilon := \frac{1}{2}$, every ball $B_{1/2}(x) = \{x\}$ (Definition 3.a.1), so no **finite** union of such balls can cover the infinite set X. Q.E.D.

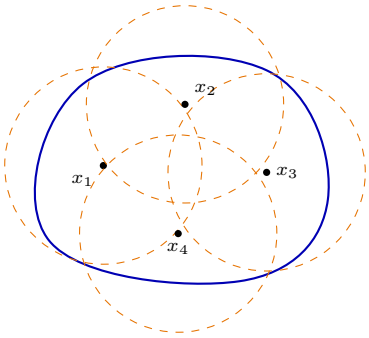
Remark 7.a.11: This is the same asymmetry Lemma 7.a.12 proves in one direction only: sequential compactness forces **both** boundedness and total boundedness, and this exercise shows the two are genuinely different hypotheses. Total boundedness is the strictly stronger of the two, and the gap opens up wherever a bounded space has room for infinitely many points at a fixed distance from one another: the infinite discrete space above, or the unit ball of an infinite-dimensional normed space.

Lemma 7.a.12 (*Sequential compactness gives (total) boundedness*): Let (X, d) be sequentially compact. Then X is bounded and totally bounded (Definition 7.a.9).

Proof:

Bounded. Fix $x_0 \in X$; the function $f(y) := d(x_0, y)$ is continuous, so by Lemma 7.a.7 it attains a maximum $M \geq 0$ on X. Then $X = B_{M+1}(x_0)$.

Totally bounded. Fix $\varepsilon > 0$. Construct a sequence inductively: let $x_1 \in X$ be arbitrary, and, having chosen $x_1, \dots, x_n$, let $U_n := \bigcup_{k=1}^{n} B_\varepsilon(x_k)$. If $U_n = X$, stop, since we then have the desired finite cover. Otherwise pick $x_{n+1} \in X \setminus U_n$, so that $d(x_{n+1}, x_k) \geq \varepsilon$ for all $k \leq n$. If this process never terminates, sequential compactness gives a convergent, hence Cauchy, subsequence $(x_{n_k})_k$; but $d(x_{n_k}, x_{n_{k+1}}) \geq \varepsilon$ for all k contradicts the Cauchy property. So, the process must terminate after finitely many steps, giving the desired finite ε-net.



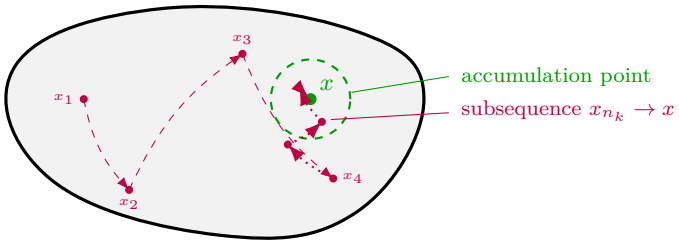
$X = B_\varepsilon(x_1) \cup \cdots \cup B_\varepsilon(x_4)$

Q.E.D.

Theorem 7.a.13 (*Sequential and topological compactness agree*)**:** A metric space (X, d) is sequentially compact (Item **(a)**) if and only if it is topologically compact (Definition 7.a.1). In particular, Theorem 7.a.3 holds.

Proof:

(⇐) Suppose X is not sequentially compact, so some sequence $(x_n)_{n\in\mathbb{N}}$ has no convergent subsequence. Then no $x \in X$ is the limit of any subsequence, so there exist $\varepsilon_x > 0$ and $N_x \in \mathbb{N}$ such that $B_{\varepsilon_x}(x) \cap \{x_n : n \geq N_x\} = \emptyset$. The balls $\{B_{\varepsilon_x}(x)\}_{x\in X}$ cover X, so topological compactness gives finitely many $x_1, \dots, x_m$ with $X = \bigcup_{k=1}^{m} B_{\varepsilon_{x_k}}(x_k)$. With $N := \max\{N_{x_1}, \dots, N_{x_m}\}$, every x_n with $n \geq N$ then avoids $B_{\varepsilon_{x_1}}(x_1) \cup \cdots \cup B_{\varepsilon_{x_m}}(x_m) = X$, which is absurd. Therefore, X is sequentially compact.



(⇒) Suppose X is sequentially compact and let $\{U_i\}_{i\in I}$ be an open cover; we may assume no single U_i equals X (else $\{U_i\}$ is itself a finite subcover). Define

$$r(x) := \tfrac{1}{2} \sup\{r \geq 0 : B_r(x) \subseteq U_i \text{ for some } i \in I\}, \qquad x \in X.$$

r is finite and positive. Since $\{U_i\}$ covers X and each U_i is open, some $U_{i_0} \ni x$ contains a ball around x, so $r(x) > 0$; and since X is bounded (Lemma 7.a.12) but no U_i equals X, $r(x) < \infty$.

r is Lipschitz, hence continuous. (That is, $|r(x) - r(y)| \leq L\, d(x, y)$ for a constant L; the notion is named and placed in context as Definition 5.b.20, but only the displayed estimate is used here.)

Write $R(x) := \sup\{\rho \geq 0 : B_\rho(x) \subseteq U_i \text{ for some } i\}$, so that $r = \frac{1}{2}R$; it is R that the argument bounds, and the factor $\frac{1}{2}$ is carried through at the end.

Fix $x, y \in X$ and let $\rho < R(x)$ be any radius with $B_\rho(x) \subseteq U_i$ for some i. The triangle inequality gives

$$B_{\rho - d(x,y)}(y) \subseteq B_\rho(x) \subseteq U_i \qquad \Longrightarrow \qquad R(y) \geq \rho - d(x,y),$$

trivially so when $\rho \leq d(x,y)$, the ball on the left being empty. Letting $\rho \uparrow R(x)$, which is legitimate because $R(x)$ is the supremum of such radii and is finite, gives $R(y) \geq R(x) - d(x,y)$; exchanging x and y gives the reverse inequality, so $|R(x) - R(y)| \leq d(x,y)$. Halving,

$$|r(x) - r(y)| \leq \tfrac{1}{2}\, d(x,y),$$

so r is $\frac{1}{2}$-Lipschitz, and in particular continuous, which is all that is needed below.

By Lemma 7.a.7, r attains a minimum $\varepsilon := \min_{x \in X} r(x) > 0$ on X. Then $B_\varepsilon(x) \subseteq U_i$ for some i, for every $x \in X$. By Lemma 7.a.12, X is totally bounded, so there are $x_1, \ldots, x_n \in X$ with $X = \bigcup_{k=1}^{n} B_\varepsilon(x_k)$; choosing, for each k, an index i_k with $B_\varepsilon(x_k) \subseteq U_{i_k}$, the sets $U_{i_1}, \ldots, U_{i_n}$ form a finite subcover of X. Q.E.D.

Remark 7.a.14 (*Total boundedness, more generally*): A sharper characterization holds: (X,d) is compact if and only if it is complete and totally bounded. Lemma 7.a.12 gives one direction (compact $\Longrightarrow$ complete, by Exercise 7.a.24 below, and totally bounded); for the converse, a Cauchy subsequence can be extracted from any sequence using total boundedness (cover by finitely many $\frac{1}{k}$-balls for $k = 1, 2, \ldots$ and pass to nested subsequences), and it converges by completeness.

Definition 7.a.15 (*Lebesgue number*): Let (X,d) be a metric space and $\{U_i\}_{i \in I}$ an open cover of X. A number $\varepsilon > 0$ is a "*Lebesgue number*" (German: „*Lebesgue-Zahl*“) for the cover if

$$\forall x \in X \ \exists i \in I : B_\varepsilon(x) \subseteq U_i.$$

Remark 7.a.16 (*Lebesgue's covering lemma*): The proof of Theorem 7.a.13 ($\Rightarrow$) shows more than stated: on a compact metric space, **every** open cover has a Lebesgue number (take $\varepsilon := \min_{x \in X} r(x)$ as constructed there). This is known as "*Lebesgue's covering lemma*" (German: „*Lebesguesches Überdeckungslemma*“).

Proposition 7.a.17 (*Compactness implies uniform continuity*): Let (X, d_X), (Y, d_Y) be metric spaces with X compact, and let $f : X \to Y$ be continuous. Then f is uniformly continuous (Definition 5.a.4).

Proof:

Let $\varepsilon > 0$. The sets $f^{-1}(B_{\varepsilon/2}(y))$, $y \in Y$, are open (Definition 5.a.1) and cover X, so they admit a Lebesgue number $\delta > 0$ (Definition 7.a.15). For any $x, x' \in X$ with $d_X(x,x') < \delta$, the ball $B_\delta(x)$ lies inside some $f^{-1}(B_{\varepsilon/2}(y))$, so $x' \in B_\delta(x)$ gives $f(x), f(x') \in B_{\varepsilon/2}(y)$, hence $d_Y(f(x), f(x')) < \varepsilon$ by the triangle inequality. Since δ did not depend on x, f is uniformly continuous. Q.E.D.

Example 7.a.18 (*Locally Lipschitz on a compact space means Lipschitz*): Call $f : X \to \mathbb{R}$ "*locally Lipschitz*" (German: „*lokal Lipschitz-stetig*“) if every $x_0 \in X$ admits an open neighbourhood U_{x_0} and a constant $L_{x_0} > 0$ with

$$|f(x) - f(y)| \leq L_{x_0}\, d(x,y) \qquad \text{for all } x, y \in U_{x_0}.$$

The constant is allowed to depend on the point, and on a general space it must: on $(0,1]$ the function $f(x) := 1/x$ is locally Lipschitz and not Lipschitz, the local constants blowing up as $x_0 \to 0$. If X is compact, that cannot happen.

Claim. If (X,d) is compact and $f : X \to \mathbb{R}$ is locally Lipschitz, then f is Lipschitz (Definition 5.b.20).

The neighbourhoods $\{U_{x_0}\}_{x_0\in X}$ cover X, so compactness extracts a finite subcover $U_{x_1},\dots,U_{x_N}$. Put $L := \max_{1\le i\le N} L_{x_i}$, which is finite because the maximum is now over finitely many indices, and let $\delta > 0$ be a Lebesgue number for that finite subcover (Definition 7.a.15). Fix $x, y \in X$ and split on how far apart they are.

(a) Nearby points, $d(x,y) < \delta$. The ball $B_\delta(x)$ lies inside some U_{x_i}, and both x and y belong to it. The local estimate on U_{x_i} therefore applies to this pair, giving $|f(x)-f(y)| \le L_{x_i} d(x,y) \le L\, d(x,y)$.

(b) Distant points, $d(x,y) \ge \delta$. Here no local estimate is available, and none is needed. A locally Lipschitz function is continuous, so f is bounded on the compact X, say $|f| \le M$. Then

$$|f(x) - f(y)| \le 2M = \frac{2M}{\delta}\,\delta \le \frac{2M}{\delta}\, d(x,y).$$

Consequently f is Lipschitz with constant $\max\{L,\, 2M/\delta\}$.

Remark 7.a.19 (*Why two regimes, and why neither alone suffices*): The finite subcover alone does **not** finish the argument, and this is the step most solutions skip. It bounds the local constants, but a pair x, y lying in **no** common U_{x_i} is never reached by any of them. The Lebesgue number is exactly what repairs that: it converts *"some member of the cover"* into *"a uniform radius around every point"*, so that every sufficiently close pair is covered by one estimate. The far regime is then free, because over a fixed distance a bounded function cannot vary fast.

> **Info:** *This example is taken from the Analysis I & II examination of winter 2018/2019 (Prof. Manfred Einsiedler), Problem 9, where it is set as a proof exercise. The counterexample on $(0,1]$ and the two-regime commentary are added here.*

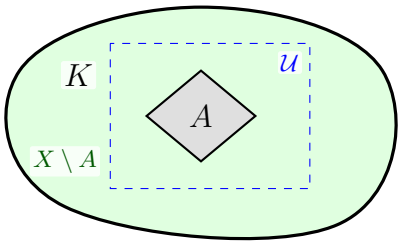
Proposition 7.a.20 (*A closed subset of a compact set is compact*): Let (X,d) be a metric space, $K \subseteq X$ compact, and $A \subseteq K$ closed in X. Then A is compact.

Proof:

Let $\mathcal{U}$ be an open cover of A. Since A is closed, $X \setminus A$ is open, so

$$\mathcal{U}' := \mathcal{U} \cup \{X \setminus A\}$$

is an open cover of K, covering A by assumption and everything outside A by the added set. Compactness of K gives a finite subcover $U_1,\dots,U_m \in \mathcal{U}'$ of K, hence of A.



$\mathcal{U}$ covers A; adding $X \setminus A$ covers K

Now simply discard $X \setminus A$ from that list if it occurs: it contains no point of A, so the remaining sets still cover A. They all belong to $\mathcal{U}$, and there are finitely many. As $\mathcal{U}$ was arbitrary, A is compact. Q.E.D.

Remark 7.a.21 (*Enlarge the cover, then discard the extra set*): The trick just used is the standard way to transfer compactness downward, and it is worth remembering in the form **enlarge the cover by the complement of the set you care about, then throw that complement away again**. It reappears whenever one needs a compact subset of a compact set. Note that the hypothesis *"closed"* is essential, since $(0,1) \subseteq [0,1]$ sits inside a compact set without being compact.

Proposition 7.a.22 (*A compact subset is always closed*): Let (X, d) be a metric space and $K \subseteq X$ compact. Then K is closed in X.

Proof:

Let $(x_n) \subseteq K$ converge to some $x \in X$; by Proposition 4.a.7 it suffices to show $x \in K$. Compactness gives a subsequence $x_{n_k} \to y$ with $y \in K$ (Theorem 7.a.13). But a subsequence of a convergent sequence converges to the same limit, so $y = x$ by uniqueness of limits (Proposition 4.a.4), and therefore $x \in K$. Q.E.D.

Remark 7.a.23 (*This is the cheap half of Heine–Borel*): Together with Lemma 7.a.12, which gives boundedness, this proposition says that in **every** metric space a compact set is closed and bounded. That direction is short and completely general, and it is the one Theorem 7.b.28 gets for free. What Heine–Borel adds, and what is special to $\mathbb{R}^n$, is the converse, and Exercise 7.b.34 shows how badly the converse can fail elsewhere: there is a metric space with a closed bounded set that is not compact.

The pairing with Proposition 7.a.20 is worth noticing too. Compactness passes **down** to closed subsets and **up** to the property of being closed, so within a compact ambient space *"compact"* and *"closed"* are the same word.

Exercise 7.a.24 (*Compact sets are complete*): Let (X, d) be a metric space.

(a) Show that every finite subset $\{x_1, \dots, x_n\} \subseteq X$ is compact.

(b) If X is compact, is it complete (Definition 6.a.1)?

Proof of Exercise 7.a.24:

(a) Let $\{U_\alpha\}_{\alpha \in I}$ be an open cover of $\{x_1, \dots, x_n\}$. For each $i = 1, \dots, n$, pick U_{α_i} such that $x_i \in U_{\alpha_i}$. Then $\{U_{\alpha_i}\}_{i=1,\dots,n}$ is a finite subcover.

(b) **Yes.** Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence in X. Then, because X is compact, we can extract a convergent subsequence such that $\lim_{i \to \infty} x_{n_i} = x \in X$.

Fix $\varepsilon > 0$. We find $N_1 \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \quad \forall n, m > N_1$$

and $N_2 \in \mathbb{N}$ such that

$$d(x, x_{n_i}) < \varepsilon \quad \forall i > N_2.$$

Let $N := \max\{N_1, N_2\}$. For any $k > N$, we have:

$$d(x, x_k) \leq d(x, x_{n_k}) + d(x_{n_k}, x_k) < 2\varepsilon$$

where we used the fact that $n_k \geq k > N$. Thus, any Cauchy sequence converges.

Q.E.D.

Exercise 7.a.25 (*Compactness of a convergent sequence*): Let (X, d) be a metric space. Let $(x_n)_{n=0}^{\infty}$ be a sequence such that $x_n \to \bar{x}$. Prove that the set

$$E := \{x_n \mid n \in \mathbb{N}\} \cup \{\bar{x}\}$$

is compact.

Proof of Exercise 7.a.25:

Using topological compactness: Let $\{U_i\}_{i \in I}$ be an open cover of E. Since $\bar{x} \in E$, there exists some $i_* \in I$ such that $\bar{x} \in U_{i_*}$. Because U_{i_*} is open and $x_n \to \bar{x}$, there exists $N \in \mathbb{N}$ such that $x_n \in U_{i_*}$ for all $n \geq N$. For the remaining finitely many points $x_0, \dots, x_{N-1}$, we can pick

sets $U_{i_0}, \dots, U_{i_{N-1}}$ from the cover containing each of them. Then $\{U_{i_*}, U_{i_0}, \dots, U_{i_{N-1}}\}$ is a finite subcover of E.

Using sequential compactness: Let $(y_k)_{k=0}^{\infty}$ be a sequence in E. If (y_k) takes values in a finite subset of E, it has a constant (hence convergent) subsequence. Otherwise, it takes infinitely many distinct values from E, which means it must contain a subsequence of (x_n) progressing towards $\bar{x}$. In either case, it contains a convergent subsequence with limit in E. Thus E is sequentially compact, hence compact. **Q.E.D.**

Exercise 7.a.26 (*Compactness of a finite set grid*): Let

$$X := \{(x, y) \in \mathbb{R}^2 \mid x, y \in \mathbb{Z} \wedge |x|, |y| \leq 3\}.$$

Show that X is compact.

Proof of Exercise 7.a.26:

The set X contains exactly $7 \times 7 = 49$ points, so it is finite. By Exercise 7.a.24, every finite subset of a metric space is compact. Alternatively, using sequential compactness (as hinted in the notes), any sequence in X must take at least one value infinitely often (by the pigeonhole principle), thus it contains a constant, hence convergent, subsequence. **Q.E.D.**

Exercise 7.a.27 (*Infinite discrete versus closed discrete subsets of a compact space*): Let (X, d) be a metric space. A subset $S \subseteq X$ is called "*discrete*" if every point $p \in S$ is an isolated point of S, that is, there exists $r > 0$ such that $B_r(p) \cap S = \{p\}$.

(a) Give an example of a compact metric space K containing an infinite discrete subset $S \subset K$.

(b) Prove that if K is a compact metric space, then every **closed** discrete subset $S \subseteq K$ must be finite.

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *of 31 January 2023, Problem 5(a)–(b). That paper names no lecturer.*

Solution: *See solution on Page 110*

Heine–Borel (Section 7.b)

Which subsets of $\mathbb{R}^n$ are compact? Neither definition is a practical test, since both quantify over all covers or all sequences. In $\mathbb{R}^n$ there is a test one can check by inspection, and it is the familiar one from **Analysis I**.

Theorem 7.b.28 (*Heine–Borel*): A subset $K \subseteq \mathbb{R}^n$ equipped with the **standard metric** is **compact** if and only if it is "*closed*" and "*bounded*" (German: „*beschränkt*").

Proof:

Every ingredient has already appeared; the proof only assembles them.

- **(⇒)** holds in every metric space. A compact set is bounded (Lemma 7.a.12) and closed (Proposition 7.a.22).
- **(⇐)** is where the real content sits, and it is the **Bolzano–Weierstrass theorem** in $\mathbb{R}^n$. Let K be closed and bounded and let (x_m) be a sequence in K. Each coordinate sequence $(x_{m,j})_m$ is a bounded sequence of real numbers. By Bolzano–Weierstrass in $\mathbb{R}$ (**Analysis I**), the first coordinate converges along some subsequence; along a further subsequence of that one, so does the second; after n such extractions every coordinate converges along one common subsequence $(x_{m_k})_k$. By Proposition 6.a.5**(a)** that subsequence converges in $\mathbb{R}^n$, and since

K is closed its limit lies in K (Proposition 4.a.7**(b)**). So K is sequentially compact, which is compactness by Theorem 7.a.13. **Q.E.D.**

The theorem is special to $\mathbb{R}^n$ with the standard metric because the second step uses both the completeness of $\mathbb{R}$, hidden inside Bolzano–Weierstrass, and the finiteness of the dimension, which is what lets the extraction stop after n steps. Exercise 7.b.34 below shows how thoroughly it fails otherwise.

Proposition 7.b.29 (*Topologically compact subsets of $\mathbb{R}^n$ are bounded*): If a subset $S \subseteq \mathbb{R}^n$ is topologically compact, then S is bounded.

Proof:

Consider the collection of open balls $\mathcal{U} := \{B_k(0) \mid k \in \mathbb{N}_{\geq 1}\}$ centered at the origin. Since every point $x \in \mathbb{R}^n$ has $|x| < k$ for some large integer k, $\mathcal{U}$ forms an open cover of $\mathbb{R}^n$, and hence an open cover of S:

$$S \subseteq \mathbb{R}^n = \bigcup_{k=1}^{\infty} B_k(0).$$

By topological compactness of S, there exists a finite subcover $\{B_{k_1}(0), B_{k_2}(0), \ldots, B_{k_m}(0)\}$. Setting $K := \max\{k_1, k_2, \ldots, k_m\}$, we obtain

$$S \subseteq \bigcup_{j=1}^{m} B_{k_j}(0) = B_K(0).$$

Therefore, S is contained in the ball of radius K centered at the origin, which means S is bounded. **Q.E.D.**

Example 7.b.30 (*Compactness via Heine-Borel and continuity*): Consider the following subsets and determine whether they are compact under the standard Euclidean metric:

(a) **Is $\mathbb{N} \subset \mathbb{R}$ compact?** No. While it is closed in $\mathbb{R}$, it is not bounded. By Heine-Borel, it cannot be compact.

(b) **Is $[0,1] \times [0,1] \subset \mathbb{R}^2$ complete?** Yes. The square $[0,1]^2$ is closed and bounded in $\mathbb{R}^2$, hence compact by Heine-Borel. As every compact metric space is complete (Exercise 7.a.24), it is complete.

(c) Let $f : \mathbb{R} \to \mathbb{R}^3, x \mapsto (\sin x, \cos x, 1)$. **Is $f([0, \pi/2])$ compact in $\mathbb{R}^3$?** Yes. We do not need to show this manually using sequences or open covers. Since the domain $[0, \pi/2]$ is compact in $\mathbb{R}$ (closed and bounded) and the function f is continuous, its image $f([0, \pi/2])$ is compact in $\mathbb{R}^3$ (by Item **(a)**).

AI-Example 7.b.31 (*Open Cover of $(0,1)$ with No Finite Subcover*): Consider the open interval $X = (0,1)$ under the standard Euclidean metric. The family of open sets

$$U_n := \left(\frac{1}{n}, 1\right) \quad \text{for } n \in \mathbb{N}_{\geq 2}$$

forms an open cover of $(0,1)$ because for any $x \in (0,1)$, there exists $n \in \mathbb{N}$ with $1/n < x$. However, any finite subcollection $\{U_{n_1}, \ldots, U_{n_k}\}$ has a maximum index $N := \max\{n_1, \ldots, n_k\}$, so its union is $U_N = (1/N, 1)$, which fails to cover points in $(0, 1/N]$. Thus $(0,1)$ is bounded but not compact.

Example 7.b.32 (*Heine–Borel needs $\mathbb{R}^n$ itself, not just the standard-looking metric*): $\mathbb{Q} \cap [0,2]$, with the metric inherited from the standard metric on $\mathbb{R}$, is closed and bounded **as a subset of $\mathbb{Q}$**, and yet it is not compact. The family

$$\mathbb{Q} \cap [0,2] = \left(\mathbb{Q} \cap [0, \sqrt{2})\right) \cup \bigcup_{p \in \mathbb{Q},\, p > \sqrt{2}} \left(\mathbb{Q} \cap (p, 2]\right)$$

is an open cover of $\mathbb{Q} \cap [0,2]$ (each set is the trace of an open subset of $\mathbb{R}$, hence open in the subspace $\mathbb{Q} \cap [0,2]$): every rational in $[0,2]$ is either $< \sqrt{2}$ or $> \sqrt{2}$, since $\sqrt{2} \notin \mathbb{Q}$. Any finite subcover uses only finitely many of the sets $\mathbb{Q} \cap (p,2]$, hence has a largest such p, call it $p_{\max}$, and then misses every rational in $(\sqrt{2}, p_{\max}]$, of which there are infinitely many. This is sharper than Example 7.b.30**(a)**'s $\mathbb{N}$. There, compactness fails because the set is unbounded. Here it fails on a closed **and** bounded set, purely because the ambient space $\mathbb{Q}$ is not $\mathbb{R}^n$. The irrationality of $\sqrt{2}$ is doing all the work, leaving a *"hole"* exactly where the cover would need to close up.

Exercise 7.b.33 (*Topological properties of various subsets*): Decide whether the following subsets are open, closed, compact, or complete (under the standard Euclidean metric):

(a) $[0,1] \subseteq \mathbb{R}$
(b) $\mathbb{Q} \subseteq \mathbb{R}$
(c) $B_1(0) \subseteq \mathbb{R}^3$
(d) $B_1(0) \cap B_1(1) \subseteq \mathbb{R}^2$
(e) $\{0\} \cup \{1/n \mid n \in \mathbb{N}_{\geq 1}\} \subseteq \mathbb{R}$
(f) $\bigcup_{n=1}^{\infty} B_{1/n}(n) \subseteq \mathbb{R}$
(g) $(-\infty, 1] \subseteq \mathbb{R}$
(h) $f([0,1])$ where $f : \mathbb{R} \to \mathbb{R}^n$ is continuous
(i) $f^{-1}([0,1])$ where $f : \mathbb{R}^n \to \mathbb{R}$ is continuous
(j) $f^{-1}((0,1))$ where $f : \mathbb{R}^n \to \mathbb{R}$ is continuous
(k) $S := \operatorname{span}\{(1,0,0)^\mathsf{T}, (3,1,0)^\mathsf{T}\} \subseteq \mathbb{R}^3$
(l) $\{x \in \mathbb{R}^3 \mid \exists s \in S \text{ such that } d(x,s) < 1\}$ for a subset $S \subseteq \mathbb{R}^3$

Solution to Exercise 7.b.33:

(a) $[0,1] \subseteq \mathbb{R}$: **Closed**, **compact**, **complete**. (Not open).
(b) $\mathbb{Q} \subseteq \mathbb{R}$: **None**. It is neither open nor closed, not bounded (so not compact), and not complete.
(c) $B_1(0) \subseteq \mathbb{R}^3$: **Open**.
(d) $B_1(0) \cap B_1(1) \subseteq \mathbb{R}^2$: **Open** (finite intersection of open sets).
(e) $\{0\} \cup \{1/n \mid n \in \mathbb{N}_{\geq 1}\} \subseteq \mathbb{R}$: **Closed**, **compact**, **complete** (bounded and contains its only limit point 0).
(f) $\bigcup_{n=1}^{\infty} B_{1/n}(n) \subseteq \mathbb{R}$: **Open** (arbitrary union of open sets).
(g) $(-\infty, 1] \subseteq \mathbb{R}$: **Closed**, **complete**.
(h) $f([0,1])$: **Closed**, **compact**, **complete** (continuous image of a compact set).
(i) $f^{-1}([0,1])$: **Closed**, **complete** (continuous preimage of a closed set).
(j) $f^{-1}((0,1))$: **Open** (continuous preimage of an open set).
(k) $S := \operatorname{span}\{(1,0,0)^\mathsf{T}, (3,1,0)^\mathsf{T}\} \subseteq \mathbb{R}^3$: **Closed**, **complete** (it is a 2-dimensional subspace, hence closed in $\mathbb{R}^3$).
(l) $\{x \in \mathbb{R}^3 \mid \exists s \in S \text{ such that } d(x,s) < 1\}$: **Open**. This set is equivalently written as $\{x \in \mathbb{R}^3 \mid d(x,S) < 1\}$ or $\bigcup_{s \in S} B_1(s)$, which is an arbitrary union of open balls.

✔ Solved

This captures our intuition of a *"compact"* set. We will briefly show that Heine–Borel is **false** for $\mathbb{R}^n$ with an **arbitrary metric**.

Exercise 7.b.34 (*Heine–Borel fails for arbitrary metrics*): Let (X,d) be a metric space.

(a) *(Optional)* Show that $\tilde{d}(x,y) := \dfrac{d(x,y)}{1+d(x,y)}$ is a metric on X.
(b) Show that, if $U \subseteq X$ is open with respect to d, then U is open with respect to $\tilde{d}$.

(c) Show that, for $X = \mathbb{R}$, $d(x,y) = \dfrac{|x-y|}{1+|x-y|}$, $\mathbb{N} \subseteq X$ is closed and bounded but not compact.

AI-Note 7.2: Part **(c)** is stated in the source for $X = \mathbb{R}^n$, but it then works with $\mathbb{N} \subseteq X$ and the absolute value $|x-y|$, which only parses for $n = 1$. It is corrected to $X = \mathbb{R}$ above. Nothing is lost: the same argument runs verbatim in $\mathbb{R}^n$ once $|x-y|$ is read as the Euclidean norm and $\mathbb{N}$ as $\mathbb{N} \times \{0\}^{n-1}$.

Proof of Exercise 7.b.34:

(a) Non-negativity, definiteness and symmetry are inherited directly from the corresponding properties of d, since $t \mapsto \frac{t}{1+t}$ is non-negative on $[0,\infty)$ and vanishes only at $t = 0$.

Triangle inequality. We want to show

$$\frac{d(x,y)}{1+d(x,y)} \leq \frac{d(x,z)}{1+d(x,z)} + \frac{d(z,y)}{1+d(z,y)}.$$

Denote $d(x,y) = d_1$, $d(x,z) = d_2$, $d(z,y) = d_3$ and multiply out:

$$\begin{aligned}
& d_1(1+d_2)(1+d_3) \leq d_2(1+d_1)(1+d_3) + d_3(1+d_1)(1+d_2) \\
\iff\ & (d_1 + d_1d_2)(1+d_3) \leq (d_2 + d_2d_1)(1+d_3) + (d_3 + d_1d_3)(1+d_2) \\
\iff\ & d_1 + d_1d_3 + d_1d_2d_3 + d_1d_2 \leq d_2 + d_3d_2 + d_2d_1d_3 + d_2d_1 \\
& \qquad\qquad + d_3 + d_3d_2 + d_1d_3 + d_1d_3d_2 \\
\iff\ & d_1 \leq d_2 + 2d_2d_3 + d_3 + d_1d_2d_3 \\
\impliedby\ & 0 \leq 2d_2d_3 + d_1d_2d_3 \quad \text{together with } d_1 \leq d_2 + d_3 \text{ (the triangle inequality for } d)
\end{aligned}$$

which is true by non-negativity.

(b) Let $U \subseteq X$ be open. Then, given $x \in U$, we find $r > 0$ such that $B_r(x) := \{y \in X : d(x,y) < r\} \subseteq U$. Let $\varepsilon := \dfrac{r}{1+r}$. Then $\tilde{B}_\varepsilon(x) \subseteq B_r(x) \subseteq U$, where $\tilde{B}_\varepsilon(x) := \{y \in X : \tilde{d}(x,y) < \varepsilon\}$.

(c) Note that for all $x, y \in \mathbb{R}$, $\dfrac{|x-y|}{1+|x-y|} < 1$. So $\mathbb{N} \subseteq B_1(0)$, thus it is **bounded** with respect to this metric. Also, $\mathbb{N}$ is **closed**, as follows from **(b)**. But $x_n = n$ defines a sequence in $\mathbb{N}$ with no convergent subsequence, thus $\mathbb{N}$ is **not compact**.

Q.E.D.

AI-Note 7.3: The cancellation in the derivation above originally contained a stray $d_2d_1d_2$ where the expansion gives $d_2d_1d_3$; the cancelled terms and the conclusion $0 \leq 2d_2d_3 + d_1d_2d_3$ are unaffected. Re-derived cleanly above.

Remark 7.b.35 (*The moral: boundedness is not a topological property*): It is worth saying out loud what Exercise 7.b.34 has just demonstrated, because the three parts are more striking together than separately.

Part **(b)** shows that d and $\tilde{d}$ have the **same open sets**. (Only one direction is asked for, but the converse follows the same way: given $\tilde{B}_\varepsilon(x)$, the choice $r := \frac{\varepsilon}{1-\varepsilon}$ gives $B_r(x) \subseteq \tilde{B}_\varepsilon(x)$.) Two metrics with the same open sets have the same convergent sequences, the same continuous functions, and, crucially, the same **compact** sets, since compactness is defined purely by open covers (Definition 7.a.1).

Part **(c)** shows that they do **not** have the same bounded sets. Indeed $\tilde{d} < 1$ always, so under $\tilde{d}$ **every** subset of **every** metric space is bounded. The notion collapses entirely.

Putting those together: **compactness is topological, boundedness is not.** Boundedness is a property of the chosen metric, not of the open sets it generates, so no theorem of the form *"compact if and only if closed and bounded"* can hold for a general metric. That is precisely why Theorem 7.b.28

carries the hypothesis **standard metric**. That hypothesis is not a technicality to be skimmed; it is the entire content of the theorem.

Remark 7.b.36 (*What boundedness should have been*): The right general replacement is already available: compact if and only if complete **and** "*totally bounded*" (Definition 7.a.9, Remark 7.a.14). Total boundedness is the notion that boundedness was trying and failing to be, and $\mathbb{N}$ under $\tilde{d}$ separates the two cleanly. For $\varepsilon < \frac{1}{2}$ the ball $\tilde{B}_\varepsilon(n)$ contains no integer other than n itself, so no finite family of ε-balls covers $\mathbb{N}$. The set is therefore bounded but not totally bounded, and that is exactly the gap through which compactness escapes.

Exercise 7.b.37 (*True or False* (what actually characterises compactness)): Let $E \subseteq X$ be a subset of a metric space. For each of the following completions of the statement "*E is compact if ...*", decide whether it is true or false. More than one may be correct, and so may none.

(a) E is closed and bounded.

(b) E is simultaneously open and closed.

(c) E is complete and totally bounded.

(d) E is open and totally bounded.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 10, the opening question of the multiple-choice section.*

Solution: *See solution on Page 109*

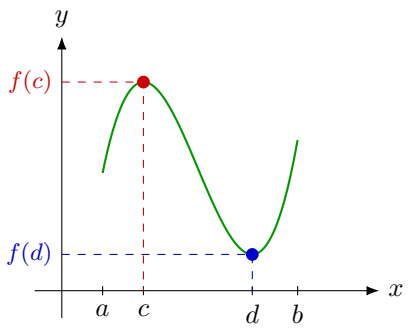
Why compactness matters (Section 7.c)

(a) Preserved by continuous functions: X, Y metric spaces, $K \subseteq X$ compact, $f : X \to Y$ continuous. Then $f(K) \subseteq Y$ is compact.

(b) Weierstrass theorem / Extreme value theorem: X metric space, $K \subseteq X$ compact, $f : K \to \mathbb{R}$ continuous. Then f attains its maximum and minimum in K. In particular, f is bounded. This is proved as Lemma 7.a.7.

AI-Note 7.4: The official script's statement of this theorem (Cor. 9.79) writes the conclusion as "*there exist $\bar{x} \in X$ such that $f(\bar{x}) = \sup_K f$*", with $\bar{x}$ ranging over the whole ambient space X rather than over K; the proof given there repeats the slip, writing $f^{-1}(\sup f(X))$ where it must mean $f^{-1}(\sup f(K))$. As stated, the claim is false as soon as f is unbounded off K: e.g. $K := \{0\} \subset X := \mathbb{R}$, $f(x) := x$. The maximiser $\bar{x}$ must be sought in K, as in the statement above.

(c) Uniform continuity: X, Y metric spaces, $f : X \to Y$ continuous. If X is compact, then f is uniformly continuous (Definition 5.a.4). This is proved as Proposition 7.a.17, via the Lebesgue number of the cover $\{f^{-1}(B_{\varepsilon/2}(y))\}_y$.



(b) on the compact $K = [a, b]$, f attains its maximum at c and its minimum at d

Both extrema are attained **inside** K. That is the content of **(b)**, and it is exactly what fails on a non-compact domain: on the open interval (a, b) the same f may approach a supremum it never reaches.

Remark 7.c.38 (*What compactness and connectedness are for: local to global*): The three facts above look like a list of unrelated services compactness happens to provide. One idea unifies them, and it is worth having early because it also explains why connectedness sits next to compactness.

Local-to-global arguments are a recurring theme across mathematics. It is often easy to check a property **locally**, on some ball $B_\varepsilon(x)$ around each point, while what one actually wants is the property **globally**, on all of X. Compactness and connectedness are the two hypotheses that license that jump. Schematically, a local-to-global statement reads:

> *If for every $x \in X$ there is $\varepsilon > 0$ such that $B_\varepsilon(x)$ has property P, then X has property P.*

Each of the two hypotheses powers its own version, and the proofs are three lines each.

Connectedness turns *"locally constant"* into *"constant"*. Let (X, d) be connected and $f : X \to \mathbb{R}$ locally constant. Every level set $f^{-1}(a)$ is then open. Fix $a \in f(X)$ and split

$$X = \underbrace{f^{-1}(a)}_{=:U} \sqcup \underbrace{\bigcup_{b \neq a} f^{-1}(b)}_{=:V},$$

a disjoint union of two open sets. Since $U \neq \emptyset$, connectedness forces $V = \emptyset$, i.e. $f \equiv a$.

Compactness turns *"locally bounded"* into *"bounded"*. Let X be compact and suppose each x has an open $U_x \ni x$ on which f is bounded. The $\{U_x\}_{x \in X}$ cover X, so finitely many suffice, say $U_1, \dots, U_n$, and then

$$\sup_{x \in X} |f(x)| \;=\; \max_{1 \le i \le n} \sup_{x \in U_i} |f(x)| \;<\; \infty,$$

because a **maximum of finitely many** finite numbers is finite. A continuous f is locally bounded, since $|f| < |f(x)| + 1$ near each x, so this recovers the boundedness half of Item **(b)**.

Read this way the two proofs are the same proof twice, with different notions of *"finitely many"*: connectedness says a partition into two open pieces must be trivial, compactness says a cover can be taken finite. Both convert local information into global. Every application in Items **(a)** to **(c)** is an instance. Uniform continuity is the most visible of them, since there the whole difficulty is that the δ supplied by continuity is local and one needs a single global one (Important remark 5.a.5).

The exercise below covers the same ground as Exercise 7.b.33, and is worth doing anyway. It adds the explicit **?** answers for the cases where the data is insufficient, a trap in the final item involving two **parallel** spanning vectors, and a four-column solution table which is where the lesson actually lands. Read Exercise 7.b.33 as the warm-up and this one as the real test.

Exercise 7.c.39 (*Classify these sets*): This self-test is the fastest way to find out whether the four notions have actually separated in your head. For each subset of $\mathbb{R}^n$ (standard metric), decide whether it is **open**, **closed**, **compact**, **complete**. Throughout, f denotes a continuous function and **?** means *"not decidable from the information given"*.

(a) $[0,1] \subseteq \mathbb{R}$
(b) $\mathbb{Q} \subseteq \mathbb{R}$
(c) $B_1(0) \subseteq \mathbb{R}^3$
(d) $\left\{\frac{1}{n} : n \geq 1\right\} \subseteq \mathbb{R}$, and separately $\{0\} \cup \left\{\frac{1}{n} : n \geq 1\right\}$
(e) $\bigcup_{n\geq 1} B_{1/n}(n) \subseteq \mathbb{R}$
(f) $(-\infty, 1] \subseteq \mathbb{R}$
(g) $f([0,1]), \quad f^{-1}([0,1]), \quad f^{-1}\big((0,1)\big)$
(h) $S := \operatorname{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}3\\0\end{pmatrix}\right\} \subseteq \mathbb{R}^2$

Info: *This exercise is taken from Sascha Brack's class notes (Week 2, Friday, p. 8).*

Solution: *See solution on Page 103*

Remark 7.c.40 (*What continuous maps do and do not preserve*): The table below collects Item **(a)** and the analogous fact for connectedness (Theorem 8.a.7), together with the dual statement for preimages, and a caution worth stating explicitly:

	Preserved by f^{-1}	Preserved by f
open	yes (Definition 5.a.1)	not in general
closed	yes	not in general
compact	not in general	yes (Item **(a)**)
connected	not in general	yes (Theorem 8.a.7)

Caution: completeness and boundedness are **not** topological properties, so in general they are preserved neither by continuous maps nor by their inverses. For instance, $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is a continuous bijection carrying a bounded set onto an unbounded one.

Exercise 7.c.41 (*Open, closed, complete and compact*): For each of the following subsets of $\mathbb{R}^N$ say whether they are open / closed / none / compact (with respect to the standard Euclidean structure). Try to prove your assertions "efficiently".

(a) $E_1 := \{x \in \mathbb{R}^3 : 0 < x_2 \leq 2\}$
(b) $E_2 := \{x \in \mathbb{R}^n : \sin(|x|) \geq \frac{1}{4}\}$
(c) $E_3 := \bigcup_{n\geq 1}\{x \in \mathbb{R}^3 : x_1^4 - \frac{1}{n}x_3^2 - x_2^2 > \frac{1}{n},\ x_1 + x_2 < 6\}$
(d) $E_4 := \{(x,y) \in (\mathbb{R}^n \times \mathbb{R}^n) : x \cdot y > \frac{1}{2}|x||y|\}$
(e) $E_5 := \{X \in \mathbb{R}^{n\times n} : \text{the matrix } X \text{ is invertible}\}$
(f) $E_6 := \{X \in \mathbb{R}^{n\times n} : \text{the matrix } X \text{ is symmetric}\}$
(g) $E_7 := \{X \in \mathbb{R}^{n\times n} : \text{the entries of } X \text{ are either } 0 \text{ or } 1\}$
(h) $E_8 := \{x \in \mathbb{R}^n : |x_i| \leq 6 \text{ for all } 1 \leq i \leq n\}$
(i) $E_9 := E_8 \times E_3 \subset \mathbb{R}^{n+3}$
(j) $E_{10} := E_2 \cap E_8 \subset \mathbb{R}^n$

Info: *This exercise is Problem 3.3 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 103*

Exercise 7.c.42 (*Multiple choice (open, complete, compact)*): Select all the statements below that are true.

(a) A nonempty open strict subset of $\mathbb{R}^n$ cannot be compact.
(b) A nonempty open strict subset of $\mathbb{R}^n$ cannot be complete.
(c) A complete subset of $\mathbb{R}^n$ contains all its accumulation points.
(d) Countable union of complete subsets of $\mathbb{R}^n$ is complete.
(e) A subset is closed in $\mathbb{R}^n$ if and only if it is complete as a metric space itself (with the distance inherited from $\mathbb{R}^n$).

Info: *This exercise is Problem 3.4 of Problem Sheet 3, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 105*

Exercise 7.c.43 (*Continuous image of an annulus*): Consider the closed annulus $U := \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq 4\}$ and the continuous function $f : U \to \mathbb{R}$, $f(x, y) := \sin(xy) - y^4$.

(a) Determine whether U is compact and whether U is connected.
(b) Prove that the image $f(U) \subseteq \mathbb{R}$ is a compact, connected interval $[a, b]$ for some real numbers $a < b$.
(c) Is U simply connected?

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the first block of true/false questions. It is reformulated here as a proof exercise.*

Solution: *See solution on Page 106*

Exercise 7.c.44 (*Nearest point to the origin*): Let $A \subseteq \mathbb{R}^n$ be non-empty and let $f : A \to \mathbb{R}_{\geq 0}$, $f(x) := |x|$, be the restriction of the Euclidean norm to A. Prove the following.

(a) If A is closed, then f attains a minimum on A: there is $P_0 \in A$ with $f(P_0) \leq f(P)$ for all $P \in A$.
(b) Statement **(a)** is false in general when A is not assumed closed.
(c) If A is convex, then the minimum is attained at no more than one point of A.
(d) If $0 \notin A$, then every minimiser $P_0 \in A$ lies in ∂A.

Info: *This exercise is taken from the Analysis I & II exam of 9 August 2021 (Prof. Giovanni Felder), Part B, Problem 3.*

Solution: *See solution on Page 106*

Exercise 7.c.45 (*What boundedness of every continuous function buys*): Let (X, d) be a non-empty metric space and consider the following four statements about it.

- **(i)** Every continuous function $X \to \mathbb{R}$ is bounded.
- **(ii)** Every continuous function $X \to \mathbb{R}$ attains a maximum and a minimum.
- **(iii)** X is compact.
- **(iv)** Every sequence in X has an accumulation point.

(a) Prove the equivalence **(i)** $\iff$ **(ii)**.
(b) Prove the implication **(iii)** $\implies$ **(i)**, working directly from the open-cover definition of compactness.
(c) Prove the implication **(iv)** $\implies$ **(i)**.

Info: *This exercise is taken from the Analysis I & II examination of 12 August 2020 (Prof. Peter S. Jossen), Part B, Problem 2. Its first two parts, on the definition of an open cover and on the non-compactness of $(0, 1]$, are omitted here.*

Solution: *See solution on Page 107*

Exercise 7.c.46 (*Sums of sets, and which property survives*): For subsets $A, B \subseteq \mathbb{R}^2$ write $A+B := \{a+b \mid a \in A,\ b \in B\}$. Prove the following, using in each case whichever characterisation of compactness is most convenient.

(a) If A is open or B is open, then $A+B$ is open.

(b) If A is closed and B is compact, then $A+B$ is closed.

(c) If A and B are compact, then $A+B$ is compact.

Info: *This exercise is taken from the Analysis I & II examination of 15 August 2019 (Prof. Peter S. Jossen), Part B, Problem 2, whose first three parts ask for the definitions of open, closed and compact.*

Solution: *See solution on Page 108*

Exercise 7.c.47 (*Box question: a compact set with a non-compact preimage*): Give an explicit example of a continuous, nonconstant map $g : \mathbb{R}^2 \to \mathbb{R}^2$ and a compact set $K \subseteq \mathbb{R}^2$ such that $g^{-1}(K)$ is **not** compact.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), Box Problem 16, where only the final answer was graded.*

Solution: *See solution on Page 109*

Compactness in spaces of functions: Ascoli–Arzelà (Section 7.d)

Theorem 7.b.28 settles compactness in $\mathbb{R}^n$ and says nothing about spaces of **functions**. The theorem of this section (Theorem 15.47 of the lecture notes) is the Bolzano–Weierstrass theorem one level up, and it is worth having for a concrete reason: it is the compactness input that Remark A.d.23 appeals to without naming, when it says Peano's theorem extracts a convergent subsequence from a family of approximate solutions.

The space of continuous functions on a compact set (Subsection 7.d.1)

Definition 7.d.48 (*Supremum norm on $C(K,\mathbb{R}^m)$*): Let $K \subseteq \mathbb{R}^n$ be compact. Write $C(K,\mathbb{R}^m)$ for the set of continuous maps $f : K \to \mathbb{R}^m$, a real vector space under pointwise operations, and equip it with the "*supremum norm*" (German: „*Supremumsnorm*“)

$$\|f\|_\infty := \sup_{x \in K} |f(x)|.$$

The supremum is finite because $|f|$ is continuous on a compact set, hence bounded (Section 7.c), which is the same point already made for the supremum metric in Exercise 2.b.24. Convergence in this norm is exactly **uniform** convergence.

Proposition 7.d.49 (*$C(K,\mathbb{R}^m)$ is complete*): For $K \subseteq \mathbb{R}^n$ compact, $\big(C(K,\mathbb{R}^m), \|\cdot\|_\infty\big)$ is a complete normed vector space.

Proof:

Let $(f_k)_{k\in\mathbb{N}}$ be Cauchy for $\|\cdot\|_\infty$. For each fixed $x \in K$ we have $|f_k(x) - f_\ell(x)| \le \|f_k - f_\ell\|_\infty$, so $(f_k(x))_k$ is Cauchy in $\mathbb{R}^m$ and converges, by completeness of $\mathbb{R}^m$, to a limit we call $f(x)$. This defines a function $f : K \to \mathbb{R}^m$.

The convergence is uniform. Given $\varepsilon > 0$ choose N with $\|f_k - f_\ell\|_\infty < \varepsilon$ for $k, \ell \ge N$; fixing $k \ge N$ and letting $\ell \to \infty$ in $|f_k(x) - f_\ell(x)| < \varepsilon$ gives $|f_k(x) - f(x)| \le \varepsilon$ for **every** x, i.e. $\|f_k - f\|_\infty \le \varepsilon$. Finally f is continuous, being a uniform limit of continuous functions, by the standard $\varepsilon/3$ argument from **Analysis I**. Hence $f \in C(K,\mathbb{R}^m)$ and $f_k \to f$ in the norm. **Q.E.D.**

Why Bolzano–Weierstrass fails one level up (Subsection 7.d.2)

In $\mathbb{R}^n$, bounded sequences have convergent subsequences. The naive transfer of that statement to $C(K,\mathbb{R}^m)$ is **false**, and it is worth seeing exactly how.

AI-Example 7.d.50 (*A bounded sequence of functions with no convergent subsequence*): Take $K := [0,1]$, $m := 1$ and $f_k(x) := x^k$. Every f_k is continuous with $\|f_k\|_\infty = 1$, so the sequence is bounded, and indeed lies on the unit sphere.

Yet no subsequence converges in $C([0,1])$. Pointwise,

$$f_k(x) \longrightarrow g(x) := \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1, \end{cases}$$

and the same holds for every subsequence. If some f_{k_j} converged uniformly to a limit $f \in C([0,1])$, then it would converge pointwise to f as well, forcing $f = g$; but g is not continuous. So no subsequence can converge.

Remark 7.d.51 (*The diagnosis*): Boundedness was never the issue. What goes wrong is that the f_k become arbitrarily **steep** near $x = 1$: to keep $|f_k(x) - f_k(y)|$ below a fixed ε there, one needs $|x-y|$ smaller and smaller as k grows. Each individual f_k is uniformly continuous (Definition 5.a.4), being continuous on a compact interval. What fails is that no single δ works for all of them at once. Ruling that out is precisely what the next definition does.

Equicontinuity, and the theorem (Subsection 7.d.3)

Definition 7.d.52 (*Equicontinuity*): A family $(f_k)_{k\in\mathbb{N}}$ in $C(K,\mathbb{R}^m)$ is "*equicontinuous*" (German: „*gleichgradig stetig*“) if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\; \forall k \in \mathbb{N} \;\; \forall x, y \in K : \quad |x - y| < \delta \implies |f_k(x) - f_k(y)| < \varepsilon.$$

Remark 7.d.53 (*Reading the quantifiers*): Compare the three conditions, which differ only in what δ is allowed to depend on:

- **Continuity** of one f: δ may depend on ε **and on the point** x.
- **Uniform continuity** of one f: δ may depend on ε only, so there is one δ for all points.
- **Equicontinuity** of a family: δ may depend on ε only, so there is one δ for all points **and all members of the family**.

So equicontinuity is uniform continuity made uniform in one further variable, the index. Each x^k of AI-Example 7.d.50 is uniformly continuous, and the family is not equicontinuous; that is the whole gap.

Theorem 7.d.54 (*Ascoli–Arzelà*): Let $K \subseteq \mathbb{R}^n$ be compact and let $(f_k)_{k\in\mathbb{N}}$ be a sequence in $C(K,\mathbb{R}^m)$ that is

(a) **bounded**: $\sup_k \|f_k\|_\infty < \infty$; and
(b) **equicontinuous** in the sense of Definition 7.d.52.

Then (f_k) has a subsequence converging uniformly to some $f \in C(K,\mathbb{R}^m)$.

Proof:

The mechanism is a diagonal argument, and it shows why **both** hypotheses are needed. Fix a countable dense subset $D = \{q_1, q_2, \dots\} \subseteq K$, which exists because K is compact and therefore totally bounded: take a finite $\frac{1}{j}$-net for each j and let D be the union of them all.

Step 1: converge on D, using boundedness. The values $(f_k(q_1))_k$ form a bounded sequence in $\mathbb{R}^m$, so by Bolzano–Weierstrass some subsequence converges at q_1. Pass to a further subsequence converging also at q_2, then at q_3, and so on. Each refinement is legitimate, but no single one

of them handles all of D. Taking the **diagonal** sequence $(g_j)_j$, whose j-th member is the j-th member of the j-th subsequence, produces one subsequence of (f_k) that converges at every point of D, since from the i-th term on it is a subsequence of the i-th refinement.

Step 2: upgrade to uniform convergence, using equicontinuity. Convergence on a dense set says nothing on its own about the rest of K; a sequence can converge at every rational and oscillate wildly in between. Equicontinuity forbids that. Let $\varepsilon > 0$ and let δ be supplied by Definition 7.d.52 for $\varepsilon/3$. The balls $B_\delta(q)$, $q \in D$, cover K because D is dense, so finitely many of them do, say those centred at $q_{i_1}, \dots, q_{i_r}$. Since $(g_j(q_{i_s}))_j$ converges for each of these finitely many s, there is N with $|g_j(q_{i_s}) - g_l(q_{i_s})| < \varepsilon/3$ for all $j, l \geq N$ and all s. Now let $x \in K$ and pick s with $|x - q_{i_s}| < \delta$. For $j, l \geq N$,

$$|g_j(x) - g_l(x)| \leq |g_j(x) - g_j(q_{i_s})| + |g_j(q_{i_s}) - g_l(q_{i_s})| + |g_l(q_{i_s}) - g_l(x)| < \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3},$$

the outer two terms by equicontinuity and the middle one by the choice of N. As x was arbitrary, $\|g_j - g_l\|_\infty \leq \varepsilon$ for $j, l \geq N$: the diagonal subsequence is Cauchy in $C(K, \mathbb{R}^m)$, hence convergent by Proposition 7.d.49. **Q.E.D.**

Remark 7.d.55 (*Where each hypothesis does its work*): The two hypotheses are not interchangeable, and the proof shows which does what: **boundedness** is what lets Bolzano–Weierstrass run at each individual point, and **equicontinuity** is what converts pointwise control on a dense set into uniform control everywhere. Delete the second and AI-Example 7.d.50 is a counterexample; delete the first and the constant sequences $f_k \equiv k$ are equicontinuous (they are all constant, so any δ works) with no convergent subsequence at all.

AI-Exercise 7.d.56 (*A uniform derivative bound gives compactness*): Let $(f_k)_{k \in \mathbb{N}}$ be a sequence in $C^1([0,1], \mathbb{R})$ with

$$|f_k'(x)| \leq 1 \quad \text{for all } x \in [0,1] \text{ and all } k, \qquad \text{and} \qquad f_k(0) = 0 \text{ for all } k.$$

(a) Show that the family is equicontinuous.

(b) Show that it is bounded in $\|\cdot\|_\infty$.

(c) Conclude that some subsequence converges uniformly, and explain why the argument would break down if the bound $|f_k'| \leq 1$ were replaced by *"each f_k is differentiable"*.

Solution: *See solution on Page 105*

That closes the account of compactness. One topological property remains before the part ends, connectedness, which Remark 7.c.38 has already shown to be compactness's partner in turning local information into global. After it the course returns to $\mathbb{R}^n$ and to differentiation, where compactness reappears as a hypothesis rather than a subject.

Solutions (Section 7.e)

Solution to Exercise 7.c.39:

Set	open	closed	compact	complete
$[0,1]$	×	✓	✓	✓
$\mathbb{Q}$	×	×	×	×
$B_1(0) \subseteq \mathbb{R}^3$	✓	×	×	×
$\{\frac{1}{n}\}$	×	×	×	×
$\{0\} \cup \{\frac{1}{n}\}$	×	✓	✓	✓
$\bigcup_{n\geq 1} B_{1/n}(n)$	✓	×	×	×
$(-\infty, 1]$	×	✓	×	✓
$f([0,1])$	×	✓	✓	✓
$f^{-1}([0,1])$	**?**	✓	**?**	✓
$f^{-1}\big((0,1)\big)$	✓	**?**	**?**	**?**
$S = \operatorname{span}\{(1,0),(3,0)\}$	×	✓	×	✓

The rows that carry the lesson:

$\{\frac{1}{n}\}$ versus $\{0\} \cup \{\frac{1}{n}\}$. Adding the single point 0 changes all four answers. Without it the set misses its own limit point, so it is not closed, and the Cauchy sequence $\frac{1}{n}$ has nowhere to converge inside the set. With it, the set is closed and bounded, hence compact by Theorem 7.b.28.

$\bigcup_{n\geq 1} B_{1/n}(n)$. Open as a union of open sets (Item **(a)**); not closed, since 0 is a limit point of the union that does not belong to it; unbounded, so certainly not compact.

S is a trap. The two spanning vectors are parallel, so S is not $\mathbb{R}^2$ but the x-axis, a line, which is closed, unbounded and not compact.

The f rows show exactly how far continuity gets you. Preimages behave: f^{-1} of a closed set is closed, of an open set is open (Definition 5.a.1). Images do not, and only compactness transfers forward (Item **(a)**), which is why $f([0,1])$ is compact but $f^{-1}([0,1])$ need not be: take $f \equiv 0$, giving $f^{-1}([0,1]) = \mathbb{R}^n$. That same f makes $f^{-1}([0,1])$ open, while $f = \mathrm{id}$ makes it $[0,1]$, which is not, hence the **?**. And $f([0,1])$ is never open: it is compact, hence closed and bounded, and a non-empty proper subset of $\mathbb{R}^m$ cannot be both open and closed, since $\mathbb{R}^m$ is connected (Definition 8.a.1).

Remark 7.e.57 (*Why the closed and complete columns agree*): The punchline of the table is that its **closed** and **complete** columns are **identical**. That is not a coincidence of this particular list. Inside a **complete** ambient space a subset is complete exactly when it is closed (Remark 6.a.8), and $\mathbb{R}^n$ is complete. Build the same table inside $\mathbb{Q}$ instead and the two columns come apart immediately.

Q.E.D.

Not every solution in this chapter is collected here. Exercise 7.a.10 (Page 87), Exercise 7.a.24 (Page 91), Exercise 7.a.25 (Page 91), Exercise 7.a.26 (Page 92), Exercise 7.b.33 (Page 94) and Exercise 7.b.34 (Page 94) are solved where they appear.

Solution to Exercise 7.c.41:

The efficient route is to write each set as a preimage of an open or closed set under a continuous map, and then read openness or closedness off Definition 5.a.1**(c)** rather than chasing balls. Compactness is then Heine–Borel (Theorem 7.b.28): closed **and** bounded.

	Set	open	closed	compact
1	$E_1 = \{0 < x_2 \leq 2\} \subseteq \mathbb{R}^3$	×	×	×
2	$E_2 = \{\sin\lvert x\rvert \geq \frac{1}{4}\}$	×	✓	×
3	E_3 (union of open sets)	✓	×	×
4	$E_4 = \{x \cdot y > \frac{1}{2}\lvert x\rvert\lvert y\rvert\}$	✓	×	×
5	$E_5 = \{X$ invertible$\}$	✓	×	×
6	$E_6 = \{X$ symmetric$\}$	×	✓	×
7	$E_7 = \{X$ with entries $0, 1\}$	×	✓	✓
8	$E_8 = [-6, 6]^n$	×	✓	✓
9	$E_9 = E_8 \times E_3$	×	×	×
10	$E_{10} = E_2 \cap E_8$	×	✓	✓

(a) Half-open in the x_2 direction, so **neither**: no ball around a point with $x_2 = 2$ stays inside, and the sequence $(0, \frac{1}{k}, 0) \to 0 \notin E_1$ shows it is not closed. Unbounded, hence not compact.

(b) **Closed:** $E_2 = h^{-1}\big([\frac{1}{4}, \infty)\big)$ for the continuous $h(x) := \sin(\lvert x\rvert)$. Not open, since a point with $\sin\lvert x\rvert = \frac{1}{4}$ has points of E_2^c arbitrarily close. Not compact: sin is periodic, so E_2 contains infinitely many concentric annuli and is unbounded.

(c) **Open:** each set in the union is defined by two **strict** inequalities between continuous functions, hence open, and an arbitrary union of open sets is open (Item **(a)**). It is non-empty (take $x = (2, 0, 0)$ and $n = 1$) and unbounded (take $x = (-M, 0, 0)$ for large M), so it is not compact; and a non-empty proper open subset of the connected space $\mathbb{R}^3$ cannot also be closed (Definition 8.a.1).

(d) **Open:** $E_4 = F^{-1}\big((0, \infty)\big)$ for the continuous $F(x, y) := x \cdot y - \frac{1}{2}\lvert x\rvert\lvert y\rvert$. Unbounded (scale any element), hence not compact, and not closed by the connectedness argument of part **(c)**. Geometrically E_4 is the set of pairs enclosing an angle strictly less than $60°$ (Remark 2.a.6).

(e) **Open:** $E_5 = \det^{-1}\left(\mathbb{R} \setminus \{0\}\right)$ and det is a polynomial in the entries, hence continuous. Unbounded, so not compact.

(f) **Closed:** E_6 is the solution set of the linear equations $X_{ij} - X_{ji} = 0$, i.e. the preimage of the closed set $\{0\}$ under a continuous (indeed linear) map. Being a linear subspace it is unbounded, hence not compact, and for $n \geq 2$ it is a proper subspace, hence has empty interior and is not open.

(g) **Closed and compact:** E_7 is **finite**, with 2^{n^2} elements. Finite sets are compact (Exercise 7.a.24**(a)**) and closed. Not open, since no ball around a matrix of 0s and 1s consists only of such matrices.

(h) **Closed and compact:** $E_8 = [-6, 6]^n$ is closed (a finite intersection of the closed sets $\{\lvert x_i\rvert \leq 6\}$) and bounded, so Theorem 7.b.28 applies. Not open: no ball around a point with some $\lvert x_i\rvert = 6$ fits inside.

(i) **Neither, and not compact.** For non-empty A, B, the product $A \times B$ is open exactly when both factors are, closed exactly when both are, and compact exactly when both are. Here E_8 is not open and E_3 is neither closed nor compact, so E_9 fails all three.

AI-Note 7.5: The official solution opens this part with *"Since E_8 is unbounded so is E_9"*. That is a slip, since $E_8 = [-6, 6]^n$ is bounded. It is E_3 that is unbounded, as the official solution itself established two parts earlier. The conclusion (that E_9 is not compact) is correct; only the reason is misattributed.

(j) **Closed and compact:** an intersection of the closed set E_2 with the closed set E_8 is closed (Proposition 3.a.8**(c)**), and it is bounded because E_8 is. Heine–Borel again. Alternatively, quote Proposition 7.a.20: a closed subset of the compact E_8 is compact.

Remark 7.e.58 (*The efficient method, extracted*): Nine of the ten sets were settled without producing a single ε. The recipe is: **strict** inequalities between continuous functions give open sets, **weak** ones give closed sets, and mixing the two gives neither. Compactness then reduces to checking boundedness, by Heine–Borel. This is what the problem means by proving assertions *"efficiently"*, and it is the practical payoff of Definition 5.a.1**(c)** over the ε–δ formulation.

Q.E.D.

Solution to Exercise 7.c.42:

(a), (b), (c) and (e) are true; (d) is false.

The engine behind **(a)**, **(b)**, **(c)** and **(e)** is a single fact, Remark 6.a.8: inside a **complete** ambient space, a subset is complete if and only if it is closed. Here the ambient space is $\mathbb{R}^n$, which is complete by Proposition 6.a.5.

(a) True. A compact subset of $\mathbb{R}^n$ is closed and bounded (Theorem 7.b.28). If it were also open and non-empty, it would be a non-empty **clopen** proper subset of $\mathbb{R}^n$, which is impossible because $\mathbb{R}^n$ is connected. Indeed $\mathbb{R}^n$ is path-connected, any two points being joined by a segment, and hence connected by Lemma 8.b.22.

(b) True, by the same argument: complete $\Rightarrow$ closed, and non-empty, open, closed and proper is again impossible in the connected space $\mathbb{R}^n$.

(c) True. Complete $\Rightarrow$ closed, and a closed set contains all its accumulation points. Indeed, if x is an accumulation point of A then some sequence in A converges to x, and Proposition 4.a.7**(b)** puts x in A.

(d) False. Every singleton $\{q\}$ is complete (its only Cauchy sequences are eventually constant), yet

$$\mathbb{Q} = \bigcup_{q\in\mathbb{Q}} \{q\}$$

is a countable union of complete sets which is famously **not** complete (Remark 6.a.8). Equivalently, in terms of closedness: a countable union of closed sets need not be closed, exactly as in Important remark 3.a.9.

(e) True, this being Remark 6.a.8 stated for the ambient space $\mathbb{R}^n$, and it is worth stressing that the *"if"* direction is where completeness of the ambient space is used. Inside $\mathbb{Q}$ instead of $\mathbb{R}$ the statement collapses: $\mathbb{Q}$ is closed in itself but not complete. Q.E.D.

Remark 7.e.59 (*A second route to (b), avoiding connectedness*): Part **(b)** is also script Exercise 9.76, and the script proves it without invoking connectedness at all. The argument is more elementary and worth recording. Fix $x \in U$ and $y \notin U$, and let $\gamma(t) := x + t(y - x)$, $t \in [0, 1]$, be the segment between them. Put $t_* := \sup\{t \in [0, 1] : \gamma(t) \in U\}$. The point $\gamma(t_*)$ cannot lie in U, since openness of U would let t_* be pushed further out, and yet it is a limit of points $\gamma(t_k) \in U$ with $t_k \nearrow t_*$. That is a convergent, hence Cauchy, sequence in U whose limit escapes U.

This avoids Lemma 8.b.22 entirely, at the cost of only working in $\mathbb{R}^n$, since the segment needs the vector space structure. The connectedness proof above generalises to any connected ambient metric space, since *"complete $\Rightarrow$ closed"* holds in every metric space.

Solution to AI-Exercise 7.d.56:

(a) By the mean value theorem, for any $x, y \in [0, 1]$ there is ξ between them with

$$|f_k(x) - f_k(y)| = |f_k'(\xi)|\,|x - y| \le |x - y|,$$

so every f_k is 1-Lipschitz **with the same constant**. Given $\varepsilon > 0$, the choice $\delta := \varepsilon$ then satisfies Definition 7.d.52 for every k and every pair x, y at once. A uniformly Lipschitz

family is always equicontinuous, and this is by far the most common way the hypothesis is verified in practice.

(b) Apply the same bound with $y := 0$ and use $f_k(0) = 0$:

$$|f_k(x)| = |f_k(x) - f_k(0)| \le |x| \le 1 \quad \text{for all } x \in [0,1],$$

so $\|f_k\|_\infty \le 1$ for every k. Note that the normalisation $f_k(0) = 0$ is doing real work: without it the family $f_k := k$ satisfies the derivative bound and is unbounded.

(c) Both hypotheses of Theorem 7.d.54 hold on the compact set $[0,1]$, so some subsequence converges uniformly to a continuous limit.

Replacing the uniform bound by mere differentiability destroys this, and AI-Example 7.d.50 is already the counterexample: $f_k(x) := x^k$ is C^1 on $[0,1]$ with $f_k(0) = 0$, so it satisfies every hypothesis **except** the uniform one. Its derivative $f_k'(x) = kx^{k-1}$ has $\|f_k'\|_\infty = k$, which grows without bound. And we saw that no subsequence of (x^k) converges. What matters is therefore not that each f_k is well behaved, but that they are well behaved **by a common margin**.

Q.E.D.

Proof of Exercise 7.c.43:

(a) The map $(x,y) \mapsto x^2 + y^2$ is continuous, and U is the preimage of the closed set $[1,4]$ under it, so U is closed. It is also bounded, being contained in $\overline{B_2(0)}$. By Heine–Borel (Theorem 7.b.28), U is compact. It is moreover path-connected, since any two of its points can be joined by moving along a circle and then radially, and therefore connected.

(b) Since f is continuous and U is compact, $f(U)$ is compact (Item **(a)**); since U is connected, $f(U)$ is connected (Theorem 8.a.7). The connected subsets of $\mathbb{R}$ are exactly the intervals, and a compact interval is one of the form $[a,b]$. It remains to rule out $a = b$, that is, to check f is non-constant: $(1,0)$ and $(0,2)$ both lie in U, and $f(1,0) = \sin 0 - 0 = 0$ while $f(0,2) = \sin 0 - 16 = -16$. Hence $f(U) = [a,b]$ with $a < b$.

(c) **No.** A loop running once around the inner hole cannot be contracted to a point within U without crossing the removed disc $\{x^2 + y^2 < 1\}$. The annulus is the standard example separating *"connected"* from *"simply connected"*, and it is worth keeping the two apart: part **(a)** is what **(b)** needs, and simple connectedness plays no role there at all.

Q.E.D.

Proof of Exercise 7.c.44:

(a) The set A need not be bounded, so Weierstrass does not apply to it directly. The remedy is to cut it down to a compact piece without losing the minimum. Fix any $a_0 \in A$ and set $R := |a_0|$. The set

$$K := A \cap \overline{B_R(0)}$$

is closed (an intersection of two closed sets) and bounded, hence compact by Heine–Borel (Theorem 7.b.28), and it is non-empty because $a_0 \in K$. Since f is continuous, it attains a minimum on K (Item **(b)**), say at $P_0 \in K \subseteq A$. This P_0 minimises over all of A: any $P \in A \setminus K$ has $|P| > R = |a_0| \ge f(P_0)$, because $a_0 \in K$.

(b) Take $A := \{x \in \mathbb{R}^n \mid 0 < |x| \le 1\}$, the closed unit ball with its centre removed. Then $\inf_A f = 0$, since $|x|$ can be made as small as we like, but $f(x) = 0$ would force $x = 0 \notin A$. So the infimum is not attained. The set is bounded, which shows that closedness, not boundedness, is the hypothesis doing the work in **(a)**.

(c) Let A be convex and suppose $P_0, P_1 \in A$ both minimise, with common value $m := |P_0| = |P_1|$. If $m = 0$ then $P_0 = P_1 = 0$ and there is nothing to prove, so assume $m > 0$. By

convexity the midpoint $M := \frac{1}{2}(P_0 + P_1)$ lies in A, and the triangle inequality gives

$$|M| \le \tfrac{1}{2}\big(|P_0| + |P_1|\big) = m.$$

Equality in the triangle inequality for the Euclidean norm forces P_0 and P_1 to be non-negative multiples of one another; having equal non-zero norms, they would then be equal. So if $P_0 \neq P_1$ the inequality is strict, $|M| < m$, contradicting minimality of m. Hence $P_0 = P_1$.

(d) Let $P_0 \in A$ be a minimiser and put $m := |P_0|$, which is positive because $0 \notin A$. Suppose, for contradiction, that P_0 were an interior point of A, so that $B_\varepsilon(P_0) \subseteq A$ for some $\varepsilon > 0$. Note first that $\varepsilon \le m$: otherwise $|0 - P_0| = m < \varepsilon$ would put the origin inside $B_\varepsilon(P_0) \subseteq A$, contrary to $0 \notin A$. Now move from P_0 straight towards the origin: the point

$$P_1 := \Big(1 - \frac{\varepsilon}{2m}\Big)\, P_0$$

satisfies $|P_1 - P_0| = \frac{\varepsilon}{2m}|P_0| = \frac{\varepsilon}{2} < \varepsilon$, so $P_1 \in A$. Since $\varepsilon \le m$, the scalar $1 - \frac{\varepsilon}{2m}$ lies in $[\frac{1}{2}, 1)$ and in particular is positive, so $|P_1| = \big(1 - \frac{\varepsilon}{2m}\big)\, m = m - \frac{\varepsilon}{2} < m$. This contradicts the minimality of $|P_0|$. Therefore P_0 is not interior, and since $P_0 \in A \subseteq \overline{A}$ it must lie in ∂A.

Remark 7.e.60: The four parts isolate four different hypotheses and show what each one buys: closedness gives **existence**, convexity gives **uniqueness**, and $0 \notin A$ gives **location**. Part **(b)** is what stops the list reading as decoration. It is worth noticing that boundedness is never assumed: the trick in **(a)** of intersecting with a ball large enough to catch one point of A is the standard way to run a Weierstrass argument on an unbounded closed set, and it recurs whenever a coercive function is minimised.

Q.E.D.

Solution to Exercise 7.c.45:

(a) Assume **(ii)** and let $f : X \to \mathbb{R}$ be continuous. It attains a maximum at some p and a minimum at some q, so $f(X) \subseteq [f(q), f(p)]$ and f is bounded.

Conversely, assume **(i)** and let $f : X \to \mathbb{R}$ be continuous. Then $M := \sup f(X)$ is a real number, since $f(X)$ is non-empty and bounded. Suppose f never took the value M. Then $M - f(x) > 0$ for every x, so

$$g : X \to \mathbb{R}, \qquad g(x) := \frac{1}{M - f(x)}$$

is well defined and continuous. By definition of the supremum there is a sequence (x_k) in X with $f(x_k) \to M$, whence $g(x_k) \to +\infty$. Therefore g is an unbounded continuous function, contradicting **(i)**. Consequently f attains its maximum, and applying this to $-f$ gives the minimum.

(b) Assume **(iii)** and let $f : X \to \mathbb{R}$ be continuous. The preimages $f^{-1}\big((-k, k)\big)$, $k \in \mathbb{N}$, are open (Definition 5.a.1) and cover X, because every $f(x)$ is a real number and so lies in $(-k, k)$ for k large. Compactness yields finitely many indices $k_1 < \cdots < k_r$ whose preimages already cover X, and these are nested, so $X = f^{-1}\big((-k_r, k_r)\big)$. Hence $|f| < k_r$ throughout.

(c) Assume **(iv)** and suppose some continuous $f : X \to \mathbb{R}$ were unbounded. Choose $x_k \in X$ with $|f(x_k)| \ge k$ for every k. By **(iv)** the sequence (x_k) has an accumulation point $x \in X$, so some subsequence satisfies $x_{k_j} \to x$. Continuity gives $f(x_{k_j}) \to f(x)$, so the numbers $f(x_{k_j})$ are bounded. But $|f(x_{k_j})| \ge k_j \to \infty$, a contradiction.

Remark 7.e.61 (*Weierstrass, split into its two halves*): Part **(a)** is the striking one, and it is worth seeing what it says: the two conclusions of the extreme value theorem, **bounded** and **attained**, are not independent facts about the space. Either one forces the other, with no compactness assumed anywhere. What compactness (or **(iv)**) supplies is only the entry ticket **(i)**.

The trick in **(a)** is worth keeping: to force a supremum to be attained, divide by the gap $M - f$ and let the failure to attain it manufacture an unbounded function. It reappears whenever a statement about suprema has to be converted into a statement about boundedness.

A space satisfying **(i)** is called "*pseudocompact*". The chain proved above is **(iii)** $\Rightarrow$ **(i)** $\Leftrightarrow$ **(ii)** and **(iv)** $\Rightarrow$ **(i)**; that **(iii)** and **(iv)** are themselves equivalent for metric spaces is Theorem 7.a.13, which is a much deeper statement than either implication above.

What is worth adding is that the reverse implications hold as well, so for a metric space all four statements are equivalent and pseudocompactness is not a weaker notion at all. Given Theorem 7.a.13 it is enough to prove **(i)** $\Rightarrow$ **(iv)**, and the contrapositive is a direct construction. Suppose some sequence has no accumulation point. Passing to distinct terms, the set $S := \{x_1, x_2, \dots\}$ is then closed, and every point of X has a neighbourhood meeting S finitely often. Put

$$r_k := \min\Big\{\tfrac{1}{3}\, d\big(x_k,\, S \setminus \{x_k\}\big),\ \tfrac{1}{k}\Big\} > 0, \qquad f(x) := \sum_{k\geq 1} k \max\Big\{0,\ 1 - \tfrac{1}{r_k} d(x, x_k)\Big\}.$$

The factor $\frac{1}{3}$ makes the balls $B_{r_k}(x_k)$ pairwise disjoint, and the cap $\frac{1}{k}$ makes every point of X have a neighbourhood on which all but finitely many summands vanish; so f is locally a finite sum of continuous functions, hence continuous. But $f(x_k) = k$, so f is unbounded and **(i)** fails. (The slicker route is the same idea packaged: S is closed and discrete, $x_k \mapsto k$ is continuous on it, and Tietze extends that to X.)

Q.E.D.

Solution to Exercise 7.c.46:

(a) Suppose A is open. For fixed $b \in \mathbb{R}^2$ the translate $A + b := \{a + b \mid a \in A\}$ is open, since translation is a homeomorphism of $\mathbb{R}^2$; concretely, $B_r(a) \subseteq A$ gives $B_r(a+b) \subseteq A + b$. Now

$$A + B = \bigcup_{b \in B} (A + b)$$

is a union of open sets and therefore open. The case of B open is symmetric, since $A + B = B + A$.

(b) Let $c \in \overline{A + B}$ and pick $c_k = a_k + b_k \in A + B$ with $c_k \to c$, where $a_k \in A$ and $b_k \in B$. Since B is compact, it is sequentially compact (Theorem 7.a.13), so a subsequence $b_{k_j} \to b \in B$ converges. Then

$$a_{k_j} = c_{k_j} - b_{k_j} \longrightarrow c - b,$$

and A is closed, so $c - b =: a$ lies in A. Hence $c = a + b \in A + B$, and $A + B$ is closed.

(c) The product $A \times B \subseteq \mathbb{R}^4$ is compact: it is closed and bounded, because A and B are, so Theorem 7.b.28 applies. Addition

$$s : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2, \qquad s(a, b) := a + b$$

is continuous, being linear, and $A + B = s(A \times B)$ is the continuous image of a compact set, hence compact.

Remark 7.e.62 (*Compactness of B is not decoration*): Two closed sets can have a non-closed sum, so **(b)** genuinely needs more than closedness. Take

$$A := \{(n, 0) \mid n \geq 1\}, \qquad B := \{(-n, \tfrac{1}{n}) \mid n \geq 1\}.$$

Both are closed, since each is a set of isolated points running off to infinity and so has no accumulation point at all. Their sum contains $(0, \frac{1}{n})$ for every n, and those converge to the origin. But $(0, 0) \in A + B$ would need $n = m$ and $\frac{1}{m} = 0$ at once, so the origin is missing and $A + B$ is not closed.

What compactness of B supplies in the proof is precisely the convergent subsequence of (b_k): closedness of B alone would let (b_k) escape to infinity, which is exactly what the counterexample

arranges. Note also that nothing in any of the three arguments used the dimension, so all three hold verbatim in $\mathbb{R}^n$, and **(a)** and **(b)** hold in any normed space.

Q.E.D.

Solution to Exercise 7.b.37:

Only **(c)** is true.

- **(a)** **False.** This is Heine–Borel (Theorem 7.b.28), and it is a theorem about $\mathbb{R}^n$ with the standard metric, not about metric spaces. Exercise 7.b.34 breaks it with the bounded metric $\tilde{d}$, under which $\mathbb{N}$ is closed and bounded and not compact; an infinite set with the discrete metric does the same.
- **(b)** **False.** Being open and closed is a statement about how E sits in X, and it has no bearing on compactness. Any space is open and closed in itself, so $E = X = \mathbb{R}$ is a counterexample.
- **(c)** **True**, and it is the honest general replacement for **(a)**: a metric space is compact if and only if it is complete and totally bounded (Remark 7.a.14, built on Lemma 7.a.12).
- **(d)** **False.** Total boundedness alone is not enough, since completeness is the other half of **(c)**, and openness does not supply it. The interval $(0, 1) \subseteq \mathbb{R}$ is open and totally bounded and not compact.

Remark 7.e.63 (*Why **(a) is the tempting one*):** Remark 7.b.35 already contains the reason **(a)** fails and **(c)** succeeds, and it is worth restating in one line, because the question is designed to catch a reader who remembers Heine–Borel without its hypothesis. Compactness is a topological property; boundedness is not, since two metrics with the same open sets can disagree about which sets are bounded. Total boundedness, unlike boundedness, is not so easily destroyed by rescaling the metric, and that is exactly why it is the notion that survives into the general statement.

Q.E.D.

Solution to Exercise 7.c.47:

Take the projection onto the first coordinate,

$$g(x, y) := (x,\, 0), \qquad K := \{(0, 0)\}.$$

Then g is linear and therefore continuous, and it is nonconstant since $g(1, 0) = (1, 0)$ while $g(0, 0) = (0, 0)$. The set K is a single point, hence compact. But

$$g^{-1}(K) = \{(x, y) \mid x = 0\} = \{0\} \times \mathbb{R}$$

is the whole y-axis, which is closed but unbounded and so not compact by Theorem 7.b.28.

Remark 7.e.64 (*Continuity preserves compactness in one direction only*): Item **(a)** says that $g(C)$ is compact whenever C is, and it is easy to carry that away as a statement about g and compactness rather than about images. It is not: the preimage of a compact set under a continuous map can be as large as one likes, and the example above makes it a whole line by collapsing one direction. A map that **does** have the property is called "*proper*", and properness is a genuine extra hypothesis, not a consequence of continuity. Exercise 15.a.15 constructs one, and shows that properness in turn buys much less than one might hope.

The official key gives a different answer with the same moral, $g(x, y) := (\arctan x, \arctan y)$ together with any compact K containing the open square $(-\frac{\pi}{2}, \frac{\pi}{2})^2$, for which $g^{-1}(K) = \mathbb{R}^2$. There the map is injective; it is the **boundedness** of its image that does the work, rather than a collapsed direction.

Q.E.D.

Solution to Exercise 7.a.27:

(a) Consider the closed unit interval $K := [0,1] \subset \mathbb{R}$ with the Euclidean metric, which is compact by the Heine–Borel theorem (Theorem 7.b.28). Define the infinite subset

$$S := \left\{ \frac{1}{n} \;\middle|\; n \in \mathbb{N}_{\geq 1} \right\} \subset [0,1].$$

For each $n \geq 1$, choose $r_n := \frac{1}{2}\left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{1}{2n(n+1)} > 0$. Then the open interval $\left(\frac{1}{n} - r_n, \frac{1}{n} + r_n\right)$ contains no other point of S. Hence every point of S is isolated in S, so S is a discrete subset. Yet S contains infinitely many distinct points. (Notice that S is not closed in $[0,1]$, since the limit point $0 = \lim_{n\to\infty} 1/n$ belongs to $K \setminus S$.)

(b) Let K be a compact metric space, and let $S \subseteq K$ be a closed discrete subset. We give two proofs:

Method 1: via open covers. Since S is discrete, for each point $x \in S$ there exists an open ball $B_{r_x}(x)$ such that $B_{r_x}(x) \cap S = \{x\}$. Because S is closed in K, the complement $K \setminus S$ is open. The collection of open sets

$$\mathcal{U} := \left\{ B_{r_x}(x) \;\middle|\; x \in S \right\} \cup \{K \setminus S\}$$

covers all of K, because every point of S is covered by its own ball $B_{r_x}(x)$, and every point not in S is covered by $K \setminus S$. By compactness of K, $\mathcal{U}$ has a finite subcover:

$$K \subseteq B_{r_{x_1}}(x_1) \cup \cdots \cup B_{r_{x_m}}(x_m) \cup (K \setminus S)$$

for finitely many points $x_1, \ldots, x_m \in S$. Intersecting both sides with S:

$$S = S \cap K \subseteq \bigcup_{i=1}^{m} \left(B_{r_{x_i}}(x_i) \cap S\right) \cup \left((K \setminus S) \cap S\right) = \bigcup_{i=1}^{m} \{x_i\} \cup \emptyset = \{x_1, \ldots, x_m\}.$$

Therefore S is a finite set.

Method 2: via sequential compactness. Suppose for contradiction that S were infinite. Then we could choose a sequence $(x_n)_{n\in\mathbb{N}}$ of pairwise distinct points in S. By sequential compactness of K (Theorem 7.a.13), there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ converging to some limit $x^* \in K$. Since S is closed in K, the limit must lie in S, so $x^* \in S$. Because S is discrete, x^* is isolated in S, so there exists a ball $B_\rho(x^*)$ with $B_\rho(x^*) \cap S = \{x^*\}$. However, by definition of convergence, $x_{n_k} \in B_\rho(x^*)$ for all sufficiently large k, which forces $x_{n_k} = x^*$ for all large k. This contradicts that all terms in the original sequence (x_n) were chosen to be pairwise distinct. Thus S must be finite.

Q.E.D.

Connectedness (Chapter 8)

Compactness told us when a space has *"enough finiteness"* to guarantee extrema and uniform behaviour. This chapter turns to a completely different, qualitative kind of structure: whether a space can be split into separate pieces at all. Connectedness and path connectedness are the two answers, they are not equivalent, and the topologist's sine curve is the standard witness. The chapter closes with a finer question about paths: whether any two with the same endpoints can be deformed into each other. That is simple connectedness. Nothing needs it for a long time, but it returns in the last chapter as the hypothesis that guarantees a curl-free field a potential.

Connectedness (Section 8.a)

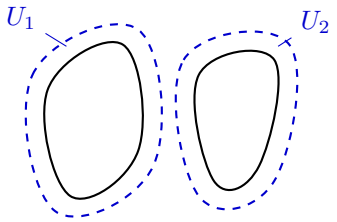
Definition 8.a.1 (*Connected and disconnected metric spaces*): A metric space (X, d) is "*disconnected*" (German: „*unzusammenhängend*“) if there exist subsets $U_1, U_2 \subseteq X$ which are

(a) **open**,
(b) **non-empty**, $U_1 \neq \emptyset$ and $U_2 \neq \emptyset$,
(c) **disjoint**, $U_1 \cap U_2 = \emptyset$,

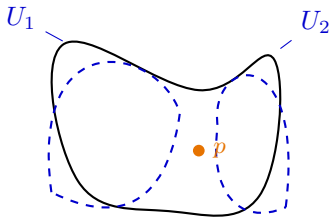
such that $X = U_1 \cup U_2$. Similarly, for a subset $A \subseteq X$, it is disconnected if it is disconnected as a metric space with the induced metric $d|_{A\times A}$.

If no such pair U_1, U_2 exists, X is "*connected*" (German: „*zusammenhängend*“). Connected is thus defined as the negation of disconnected.

Typically, proving that a set or metric space is connected is done **by contradiction**, which is why one should remember the definition of *disconnected*. Intuitively, connected means: *"the space cannot be separated by open sets."*



Disconnected
$X \subseteq U_1 \cup U_2$, and $U_1 \cap U_2 = \emptyset$



Connected
$p \in X$ lies in neither: no separation

The figure draws U_1 and U_2 as open sets of the surrounding plane, while Definition 8.a.1 asks, for a subset, for sets that are open **in the subset**. The two pictures agree, and several proofs below use the first one.

Lemma 8.a.2 (*Separating a subset by open sets of the whole space*): Let (X, d) be a metric space and $A \subseteq X$. Then A is disconnected if and only if there are open sets $U_1, U_2 \subseteq X$ with

$$A \subseteq U_1 \cup U_2, \qquad U_1 \cap U_2 = \emptyset, \qquad A \cap U_1 \neq \emptyset, \qquad A \cap U_2 \neq \emptyset.$$

Proof:

($\Leftarrow$) Put $W_i := A \cap U_i$. If $a \in W_i$, some ball $B_\varepsilon(a)$ lies in U_i, so the ball of radius ε about a in A, which is $B_\varepsilon(a) \cap A$, lies in W_i. So W_1 and W_2 are open in A, and they are non-empty, disjoint and cover A. They disconnect A.

(⇒) Let W_1, W_2 be open in A, non-empty and disjoint, with $A = W_1 \cup W_2$, and set

$$U_1 := \{x \in X : d(x, W_1) < d(x, W_2)\}, \qquad U_2 := \{x \in X : d(x, W_2) < d(x, W_1)\}.$$

Both are open, since $x \mapsto d(x, W_i)$ is continuous by Corollary 5.a.17, and they are disjoint by construction. Let $a \in W_1$. Then $d(a, W_1) = 0$, and since W_1 is open in A there is $\varepsilon > 0$ with $B_\varepsilon(a) \cap A \subseteq W_1$, so no point of W_2 lies within ε of a and $d(a, W_2) \geq \varepsilon > 0$. Hence $W_1 \subseteq U_1$, and likewise $W_2 \subseteq U_2$. This gives $A \subseteq U_1 \cup U_2$, with both U_i meeting A. **Q.E.D.**

The work is all in the forward direction. Open sets of X that merely cut W_1 and W_2 out of A always exist, but they may overlap away from A, and the distance functions are what push them apart.

Proposition 8.a.3 (*Intervals are connected*): Every interval $I \subseteq \mathbb{R}$, whether open, closed, half-open, bounded or unbounded, is connected (Definition 8.a.1). Conversely, a connected subset of $\mathbb{R}$ is an interval.

Proof:

Suppose $I = U_1 \sqcup U_2$ with U_1, U_2 open in I, disjoint and both non-empty. Pick $a \in U_1$ and $b \in U_2$; without loss of generality $a < b$, and $[a, b] \subseteq I$ because I is an interval. Set

$$c := \sup\{\, t \in [a, b] \;:\; [a, t] \subseteq U_1 \,\},$$

which is well defined since a belongs to the set. Now $c \in [a, b] \subseteq I$, so c lies in U_1 or in U_2, and both lead to a contradiction:

- If $c \in U_1$, openness gives $\varepsilon > 0$ with $(c - \varepsilon, c + \varepsilon) \cap I \subseteq U_1$; together with $[a, c) \subseteq U_1$ this yields $[a, t] \subseteq U_1$ for some $t > c$ unless $c = b$, contradicting either the supremum or $b \in U_2$.
- If $c \in U_2$, openness gives $\varepsilon > 0$ with $(c - \varepsilon, c] \cap I \subseteq U_2$. But $c > a$ (as $a \in U_1$ is open in I), so by definition of the supremum there is $t \in (c - \varepsilon, c]$ with $[a, t] \subseteq U_1$, contradicting disjointness.

So no such splitting exists, and I is connected. For the converse, a set $A \subseteq \mathbb{R}$ that is not an interval admits $a < c < b$ with $a, b \in A$ and $c \notin A$; then $A \cap (-\infty, c)$ and $A \cap (c, \infty)$ disconnect it. **Q.E.D.**

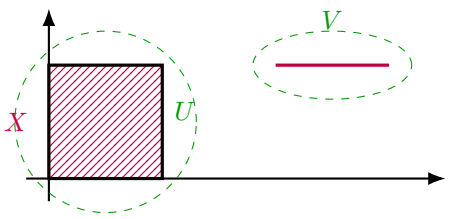
Remark 8.a.4 (*Why this is the only example one ever needs*): Every connectedness proof in this chapter reduces to Proposition 8.a.3: a path is a continuous image of $[0, 1]$ (Definition 8.b.20), so path-connected sets are connected (Lemma 8.b.22) purely because the **interval** is; and the intermediate value theorem, invoked in the topologist's sine curve below, is Theorem 8.a.7 applied to an interval, since the connected subsets of $\mathbb{R}$ are exactly the intervals. In other words, all of it flows from the single fact that $\mathbb{R}$ has no gaps. That fact is the completeness axiom, and it is what powers the supremum in the proof above.

AI-Example 8.a.5 (*Disconnectedness of Rational Numbers*): The set of rational numbers $\mathbb{Q} \subset \mathbb{R}$ equipped with the subspace topology is disconnected. To see this, define

$$U := \mathbb{Q} \cap (-\infty, \sqrt{2}) \quad \text{and} \quad V := \mathbb{Q} \cap (\sqrt{2}, \infty).$$

Since $\sqrt{2} \notin \mathbb{Q}$, we have $U \cup V = \mathbb{Q}$ and $U \cap V = \emptyset$. Both U and V are non-empty and open in the subspace topology of $\mathbb{Q}$ (as intersections of $\mathbb{Q}$ with open intervals in $\mathbb{R}$). Hence U and V form a separation of $\mathbb{Q}$.

Example 8.a.6 (*A disconnected set in $\mathbb{R}^2$*): Let $X := ([0, 1] \times [0, 1]) \cup ([2, 3] \times \{1\}) \subseteq \mathbb{R}^2$. Is X connected? No. Let $U := \{(x, y) \in X \mid x < 1.5\}$ and $V := \{(x, y) \in X \mid x > 1.5\}$. We can write these as $U = X \cap \{(x, y) \in \mathbb{R}^2 \mid x < 1.5\}$ and $V = X \cap \{(x, y) \in \mathbb{R}^2 \mid x > 1.5\}$. Both U and V are open in the subspace topology of X because they are intersections of X with open half-planes in $\mathbb{R}^2$. Furthermore, $X = U \cup V$, and U, V are disjoint and non-empty. Thus, X is disconnected.



Theorem 8.a.7 (*Continuous functions preserve connectedness*): Let X, Y be metric spaces, $f : X \to Y$ continuous. If $A \subseteq X$ is connected, then $f(A)$ is connected.

Proof:

We may assume $A = X$, replacing X by A with the induced metric and Y by $f(A)$. Suppose $f(X)$ were disconnected, say $f(X) = V_1 \sqcup V_2$ with V_1, V_2 open in $f(X)$, disjoint and non-empty. Then $U_i := f^{-1}(V_i)$ is open by Definition 5.a.1**(c)**, non-empty because V_i meets the image, and $U_1 \cap U_2 = \emptyset$ with $U_1 \cup U_2 = X$ because the V_i partition $f(X)$. So U_1, U_2 disconnect X, contrary to hypothesis. **Q.E.D.**

Corollary 8.a.8 (*Intermediate value theorem*): Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f attains every value between $f(a)$ and $f(b)$.

Proof:

By Proposition 8.a.3, $[a, b]$ is connected, so $f([a, b])$ is connected by Theorem 8.a.7, hence is an interval (Proposition 8.a.3 again). An interval containing $f(a)$ and $f(b)$ contains everything between them. **Q.E.D.**

The intermediate value theorem is a result from **Analysis I**, and re-deriving it here costs one line. It is worth the line twice over. This is the first place where connectedness earns its keep, and the derivation makes visible that the theorem is really a statement about $\mathbb{R}$ having no gaps rather than a statement about f. It is used again in the topologist's sine curve (Page 117) below.

Corollary 8.a.9 (*Intermediate value theorem on a connected space*): Let X be a connected metric space and $f : X \to \mathbb{R}$ continuous. For any $a, b \in X$ and any $y \in \mathbb{R}$ between $f(a)$ and $f(b)$, there exists $x \in X$ with $f(x) = y$.

Proof:

Word for word Corollary 8.a.8's proof, with X in place of $[a, b]$: X is connected by hypothesis (rather than by Proposition 8.a.3), so $f(X)$ is connected by Theorem 8.a.7, hence an interval by Proposition 8.a.3; an interval containing $f(a)$ and $f(b)$ contains everything between them. **Q.E.D.**

Remark 8.a.10: Corollary 8.a.8 is the special case $X = [a, b]$. Stating the general version separately is worth doing once: it makes clear that connectedness, not any special property of an interval, is what the intermediate value theorem actually needs. That is exactly the moral of Remark 8.a.4.

Corollary 8.a.11 (*Linear images of the unit cube are connected and compact*): Given the n-dimensional unit cube $[0, 1]^n \subset \mathbb{R}^n$, any linear map $L : \mathbb{R}^n \to \mathbb{R}^n$ maps $[0, 1]^n$ to a parallelepiped, possibly degenerate: if L is singular the image collapses to a parallelepiped of lower dimension, and to the single point $\{0\}$ when $L = 0$. Since linear maps in Euclidean space are continuous, the image $L([0, 1]^n)$ is connected. Furthermore, since $[0, 1]^n$ is compact, $L([0, 1]^n)$ is also compact.

Exercise 8.a.12 (*Properties of connectedness*): Are the following true or false?

(a) The union of connected subsets with a common point is connected.

(b) The intersection of connected subsets is connected.

(c) The closure (Definition 3.b.13) of a connected subset is connected.

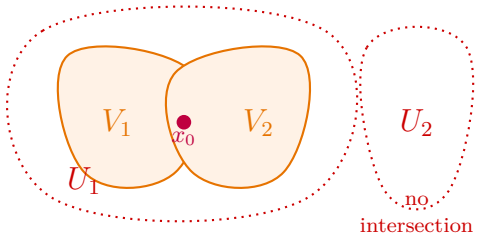
Proof of Exercise 8.a.12:

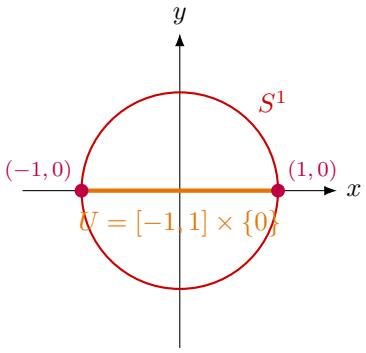
(a) True. *Proof.* Let X be a metric space, $V_1, V_2 \subseteq X$ connected, and $x_0 \in V_1 \cap V_2$. Suppose $V_1 \cup V_2 \subseteq U_1 \cup U_2$ with $U_1, U_2 \subseteq X$ open and disjoint. By Lemma 8.a.2 it is enough to show that one of them misses $V_1 \cup V_2$. Assume without loss of generality $x_0 \in U_1$. Then $U_1 \cap V_1 \neq \emptyset$ and $U_1 \cap V_2 \neq \emptyset$. So, since V_1, V_2 are connected, Lemma 8.a.2 gives $V_1 \cap U_2 = V_2 \cap U_2 = \emptyset$. Therefore $(V_1 \cup V_2) \cap U_2 = \emptyset$ and $V_1 \cup V_2$ is connected.

(b) False. *Counterexample:*

$$X = \mathbb{R}^2, \quad U = [-1,1] \times \{0\}, \quad V = S^1 = \{x \in \mathbb{R}^2 : |x| = 1\},$$
$$U \cap V = \{(-1,0), (1,0)\}.$$

(c) True. Suppose X is a metric space, $A \subseteq X$ a subset with closure $\overline{A}$. Suppose $U_1, U_2 \subseteq X$ are disjoint open sets covering $\overline{A}$, and assume without loss of generality $\overline{A} \cap U_1 \neq \emptyset$. Then, since U_1 is open, $A \cap U_1 \neq \emptyset$. By connectedness of A and Lemma 8.a.2, $U_2 \cap A = \emptyset$, and since U_2 is open, also $U_2 \cap \overline{A} = \emptyset$. So no such pair separates $\overline{A}$, which is therefore connected by Lemma 8.a.2 once more. **Q.E.D.**





Corollary 8.a.13 (*Arbitrary unions of connected sets through a common point*): Let (X, d) be a metric space, $p \in X$, and $\{V_i\}_{i \in I}$ a family of connected subsets of X with $p \in V_i$ for every $i \in I$. Then $\bigcup_{i \in I} V_i$ is connected.

Proof:

By Lemma 8.a.2 it suffices to rule out disjoint open sets $U_1, U_2 \subseteq X$ that cover the union and both meet it. So suppose $\bigcup_i V_i \subseteq U_1 \cup U_2$ with U_1, U_2 open and disjoint, and say $p \in U_1$. For each i, the argument of Exercise 8.a.12**(a)** applies to V_i alone: it meets U_1 (at p) and is connected, so it

cannot also meet U_2, i.e. $V_i \subseteq U_1$. This holds for every $i \in I$, so $\bigcup_i V_i \subseteq U_1$, and U_2 cannot meet $\bigcup_i V_i$: no separation of the union exists. Q.E.D.

Definition 8.a.14 (*Connected component*): Let (X, d) be a metric space and $x \in X$. The "*connected component*" (German: „*Zusammenhangskomponente*") of x is

$$C_x := \bigcup \{V \subseteq X : V \text{ connected and } x \in V\},$$

the union of all connected subsets of X containing x.

Remark 8.a.15: By Corollary 8.a.13 (with I indexing **all** connected sets containing x, and common point $p := x$), C_x is itself connected. It is therefore the **largest** connected subset of X containing x, since it contains every connected set that does. Two connected components are either equal or disjoint: if $C_x \cap C_y \neq \emptyset$, pick p in the intersection; then $C_x \cup C_y$ is connected (Corollary 8.a.13 again, common point p) and contains both x and y, so maximality of C_x and C_y forces $C_x \cup C_y \subseteq C_x$ and $C_x \cup C_y \subseteq C_y$, that is, $C_x = C_y$. Hence the connected components partition X into disjoint pieces, which is the connectedness analogue of the way an equivalence relation partitions a set into equivalence classes.

Remark 8.a.16 (*Summary of topological properties under continuous maps*): Let X, Y be metric spaces and $f : X \to Y$ a continuous function. The following table summarizes which topological properties of a subset $A \subseteq X$ or $B \subseteq Y$ are necessarily preserved under images $f(A)$ and preimages $f^{-1}(B)$.

If A or B is ...	Then $f^{-1}(B)$ is ...	Then $f(A)$ is ...
open	**open**	(not necessarily)
closed	**closed**	(not necessarily)
compact	(not necessarily)	**compact**
connected	(not necessarily)	**connected**

Exercise 8.a.17 (*Connected but not simply connected*): Give an example of a subset $U \subseteq \mathbb{R}^2$, with the Euclidean distance, that is closed and unbounded, connected, but **not** simply connected. Justify each of the four properties.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Box Problem 1, where only the final answer was graded.*

Solution: *See solution on Page 121*

Exercise 8.a.18 (*True or False* (topology and continuous images)): Let $U = \{(x, y) \in \mathbb{R}^2 : -x^2 \leq y \leq x^2\}$ and $f(x, y) = \frac{x^2}{1+y^2}$. Decide whether each of the following statements is true or false.

(a) U is closed.
(b) U is simply connected.
(c) U is compact.
(d) $f(U) \subseteq \mathbb{R}$ is connected.
(e) f is bounded.
(f) There exists a point $p \in U$ such that $f(p) = \inf\{f(q) \mid q \in U\}$.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the first block of true/false questions.*

Solution: *See solution on Page 122*

Exercise 8.a.19 (*Box question: connected union, disconnected intersection*): Sketch two open connected sets $U_1, U_2 \subseteq \mathbb{R}^2$ such that $U_1 \cap U_2$ is **not** connected while $U_1 \cup U_2$ is connected, and justify all four claims.

> **Info:** *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), Box Problem 19, where only the final answer was graded. The request for justifications is added here.*

Solution: *See solution on Page 122*

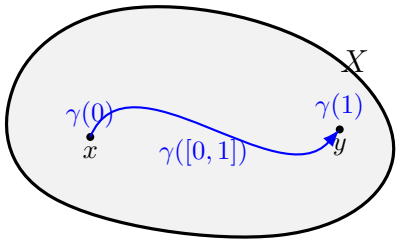
Path connectedness (Section 8.b)

Definition 8.a.1 defines connectedness negatively. A space is connected when it **cannot** be split, which is why proofs about it run by contradiction. That is mathematically convenient and intuitively unsatisfying, since it says nothing whatever about what holds a connected space together.

Path connectedness is the positive version, and it says the obvious thing: any two points can be joined by a curve running inside the space. It implies connectedness (Lemma 8.b.22), and on open subsets of $\mathbb{R}^n$ the two notions agree (Proposition 8.b.24). In general they come apart, and the topologist's sine curve at the end of this section is the standard witness.

Definition 8.b.20 (*Path*): Let (X, d) be a metric space. A continuous function $\gamma : [0,1] \to X$, where $[0,1]$ carries the standard metric, is called a "*path*" (German: „*Weg*"). More generally, we can replace $[0,1]$ by $[a,b]$ for $a < b$.

Definition 8.b.21 (*Path-connected*): A metric space X (or subset thereof) is "*path-connected*" (German: „*wegzusammenhängend*") if for all $x, y \in X$ there exists a path (Definition 8.b.20) $\gamma : [0,1] \to X$ with $\gamma(0) = x$, $\gamma(1) = y$.



Lemma 8.b.22 (*Path-connected implies connected*): Let X be a metric space. Then X path-connected (Definition 8.b.21) implies X connected (Definition 8.a.1).

Proof:

Suppose X were path-connected but disconnected, say $X = U_1 \sqcup U_2$ with U_1, U_2 open, disjoint and non-empty. Pick $x \in U_1$, $y \in U_2$ and a path $\gamma : [0,1] \to X$ with $\gamma(0) = x$, $\gamma(1) = y$. Then $\gamma^{-1}(U_1)$ and $\gamma^{-1}(U_2)$ are open in $[0,1]$ by Definition 5.a.1**(c)**, disjoint, cover $[0,1]$, and are non-empty (they contain 0 and 1 respectively). This disconnects $[0,1]$, contradicting Proposition 8.a.3. Q.E.D.

Exercise 8.b.23 (*The unit disk is connected*): Prove that the set $S^1 := \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ is connected.

Proof of Exercise 8.b.23:

Instead of proving connectedness directly via open sets, we can show that S^1 is path-connected, which implies connectedness (Lemma 8.b.22). Given any two points $a, b \in S^1$, we can connect them

via a straight line segment. The segment can be parameterized as $\gamma(t) := (1-t)a+tb$ for $t \in [0,1]$. Since S^1 (the unit disk in $\mathbb{R}^2$) is convex, for any $t \in [0,1]$ we have $|\gamma(t)| \leq (1-t)|a| + t|b| \leq 1-t+t = 1$, so the entire path lies in S^1. Because any two points can be connected by a continuous path within S^1, the set is path-connected, and therefore connected.

(Note: The standard notation for the unit disk is D^2 or $B_1(0)$, whereas S^1 usually denotes the boundary circle $\{x^2+y^2=1\}$, but we reproduce the notation S^1 here as it appears in the source notes.) **Q.E.D.**

Proposition 8.b.24 (*Connectedness and path-connectedness in open Euclidean subsets*): Let $U \subseteq \mathbb{R}^n$ be open (Definition 3.a.6). Then

$$U \text{ path-connected} \iff U \text{ connected}.$$

Proof:

($\Rightarrow$) is Lemma 8.b.22, which needs neither openness nor $\mathbb{R}^n$.

($\Leftarrow$) Assume U is connected and non-empty (the empty set is vacuously path-connected), and fix $x_0 \in U$. Let

$$A := \{x \in U \ : \ \text{some path in } U \text{ joins } x_0 \text{ to } x\}.$$

We show A and $U \setminus A$ are both open; since $x_0 \in A$ (take the constant path), connectedness then forces $U \setminus A = \emptyset$, i.e. $A = U$, which is the claim.

Let $x \in U$ and choose $r > 0$ with $B_r(x) \subseteq U$, using that U is open. The ball is convex, so every $y \in B_r(x)$ is joined to x by the straight segment $t \mapsto (1-t)x + ty$, which stays inside $B_r(x) \subseteq U$. Consequently:

- If $x \in A$, concatenating a path from x_0 to x with that segment joins x_0 to any $y \in B_r(x)$, so $B_r(x) \subseteq A$ and A is open.
- If $x \notin A$, then no $y \in B_r(x)$ can lie in A, since otherwise running the segment backwards from y to x would join x_0 to x. So $B_r(x) \subseteq U \setminus A$ and $U \setminus A$ is open.

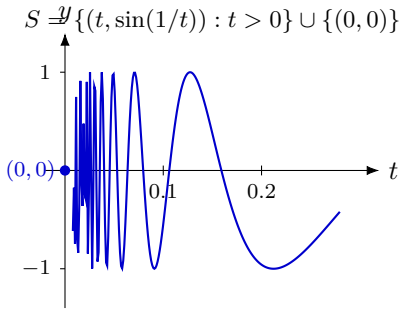
(Concatenation of two paths is again a path: traverse the first on $[0, \frac{1}{2}]$ and the second on $[\frac{1}{2}, 1]$, each at double speed; the result is continuous because the two pieces agree at $t = \frac{1}{2}$.) **Q.E.D.**

Important remark 8.b.25 (*Only for open subsets*): The implication *"connected $\Rightarrow$ path-connected"* is **false** without openness, and the proof shows exactly why: the ball $B_r(x)$ around a point of U was needed to supply the straight segment. For a set that is not open there may be points with no room around them at all, and the topologist's sine curve (Example 8.b.26) below is precisely such a case. There the origin has no neighbourhood in S that is a nice piece of curve.

Example 8.b.26 (*The topologist's sine curve*):

$$S := \left\{\left(t, \sin\left(\tfrac{1}{t}\right)\right) : t > 0\right\} \cup \{(0,0)\} \subseteq \mathbb{R}^2$$

S is **connected** but **not path-connected**!



S is connected:

Write $G := S \setminus \{(0,0)\} = \{(t, \sin(1/t)) : t > 0\}$ for the graph part, and suppose for contradiction that S is disconnected, say $S = V_1 \sqcup V_2$ with V_1, V_2 non-empty, disjoint and open in S. Say $(0,0) \in V_1$.

G is the continuous image of the interval $(0, \infty)$ under $t \mapsto (t, \sin(1/t))$, hence path-connected, hence connected (Lemma 8.b.22). A connected set meeting both parts of a separation would itself be separated, so G lies entirely in V_1 or entirely in V_2.

It cannot lie in V_1: then $V_2 \subseteq S \setminus (G \cup \{(0,0)\}) = \emptyset$, contradicting $V_2 \neq \emptyset$. So $G \subseteq V_2$, and therefore $V_1 = \{(0,0)\}$.

But V_1 is open in S, so some ball $B_\varepsilon\big((0,0)\big) \cap S$ equals $\{(0,0)\}$, and that is false. For $t := \frac{1}{k\pi}$ with k large, the point $(t, \sin(1/t)) = (t, 0) \in G$ lies within ε of the origin. This is the required contradiction, so S is connected. **Q.E.D.**

S is not path-connected:

Assume by contradiction there is a path from $(0,0)$ to $(\frac{1}{\pi}, 0)$,

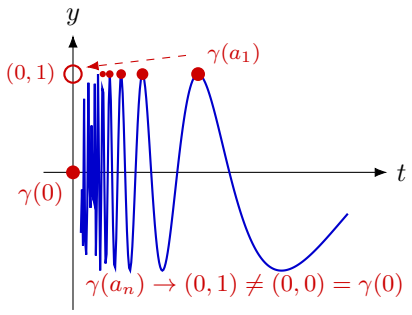
$$\gamma : [0,1] \to S, \qquad t \mapsto (x(t), y(t)).$$

Since γ is continuous, x and y must also be continuous. By the "*intermediate value theorem*" (Corollary 8.a.8), $x : [0,1] \to [0, \infty)$ attains every value in $[0, \frac{1}{\pi}]$, since $x(0) = 0$ and $x(1) = \frac{1}{\pi}$. Therefore, we find a sequence $(a_n)_{n=1}^{\infty}$ such that

$$x(a_n) = \frac{1}{2\pi n + \frac{\pi}{2}}.$$

We may moreover choose the a_n **decreasing**: having found a_{n-1}, apply the intermediate value theorem on $[0, a_{n-1}]$ rather than on $[0,1]$, which is legitimate because $x(0) = 0$ and $x(a_{n-1}) = \frac{1}{2\pi(n-1)+\pi/2}$ already straddle the next target value. This yields $a_n \in (0, a_{n-1})$, so (a_n) is decreasing and bounded below, hence convergent, say $a_n \to \ell \in [0,1]$.

Now evaluate γ along this sequence. By construction $x(a_n) \to 0$, so continuity gives $x(\ell) = 0$; and $y(a_n) = \sin\big(2\pi n + \frac{\pi}{2}\big) = 1$ for every n, so $y(\ell) = 1$. Hence $\gamma(\ell) = (0,1)$. But $(0,1) \notin S$: the only point of S whose first coordinate vanishes is the origin. This contradicts $\gamma([0,1]) \subseteq S$, so no such path exists. **Q.E.D.**



Simple connectedness (Subsection 8.b.1)

Connectedness asks whether a set falls into separate pieces, and path-connectedness asks whether any two points can be joined. Neither notices a **hole**: an annulus is connected and path-connected, and yet a loop running around its inner boundary cannot be shrunk to a point without leaving the set. The third notion in the family is the one that sees this, and it is needed as early as Exercise 7.c.43, well before the chapter on the Poincaré lemma that puts it to work.

Definition 8.b.27 (*Homotopy, simply connected set*): Let $U \subseteq \mathbb{R}^n$, and let $\gamma_0, \gamma_1 : [0,1] \to U$ be continuous paths with the same initial point $x_0 = \gamma_0(0) = \gamma_1(0)$ and the same endpoint $x_1 = \gamma_0(1) = \gamma_1(1)$. A "*homotopy*" (German: „*Homotopie*“) from γ_0 to γ_1 within U is a continuous map $H : [0,1] \times [0,1] \to U$ with

$$H(0,t) = \gamma_0(t), \qquad H(1,t) = \gamma_1(t), \qquad H(s,0) = x_0, \qquad H(s,1) = x_1$$

for all $s, t \in [0,1]$. If such an H exists we say γ_1 is "*homotopic*" (German: „*homotop*“) to γ_0.

The set U is "*simply connected*" (German: „*einfach zusammenhängend*“) if it is path-connected and any two paths in U with the same endpoints are homotopic.

Remark 8.b.28 (*Reading the definition as a deformation*): Fix s and the map $t \mapsto H(s,t)$ is itself a path in U from x_0 to x_1. So H is a family of paths depending continuously on the parameter s, starting at γ_0 when $s = 0$ and ending at γ_1 when $s = 1$. The last two conditions pin the two endpoints down throughout the deformation, which is what makes the notion non-trivial: without them, every path could be contracted to a point in any path-connected set and the definition would say nothing.

Taking γ_1 constant gives the usual informal phrasing, that every loop in U can be continuously shrunk to a point inside U.

Important remark 8.b.29 (*Three notions, and how they are ordered*): For subsets of $\mathbb{R}^n$ the three properties are ordered, each implying the one after it:

$$\text{simply connected} \implies \text{path-connected} \implies \text{connected}.$$

The second implication is Lemma 8.b.22. The first holds because path-connectedness is part of Definition 8.b.27 itself, and it has to be stated there rather than deduced, since the homotopy condition alone is vacuous on a set whose components cannot be joined at all. Neither converse holds in general, and each failure has a standard witness:

- **Connected but not path-connected:** the topologist's sine curve, Example 8.b.26. This one needs the set to be non-open: Proposition 8.b.24 says the two notions coincide for open subsets of $\mathbb{R}^n$, so for an open set the chain has only two distinct steps.
- **Path-connected but not simply connected:** the punctured plane $\mathbb{R}^2 \setminus \{0\}$, or any open annulus, so this failure survives for open sets. The loop around the hole cannot be contracted, and Example 26.c.28 turns exactly that failure into a vector field with no potential.

Two warnings that the annulus already illustrates. Simple connectedness is **not** about convexity or about having no dents: $\mathbb{R}^n$ and any convex set are simply connected, but so is a spiral-shaped region. And it depends on the dimension in a way the other two do not: $\mathbb{R}^3 \setminus \{0\}$ **is** simply connected, because a loop in space can be slid off a single missing point, while $\mathbb{R}^2 \setminus \{0\}$ is not. Removing a whole line from $\mathbb{R}^3$ destroys it again.

Of the three notions, only connectedness is used heavily in what follows, and it is used in one particular way: continuous images of connected sets are connected (Theorem 8.a.7), which is what turns a statement about a domain into a statement about the values a function takes on it. Simple connectedness returns in Chapter 26, where it becomes the exact condition for a curl-free field to have a potential.

AI-Exercise 8.b.30 (*Continuous image of a path-connected set*): Let $f : X \to Y$ be a continuous map between metric spaces.

(a) Prove that if X is path-connected (Definition 8.b.21), then so is $f(X)$.

(b) Conclude that the graph $\Gamma_g := \{(x, g(x)) : x \in U\}$ of a continuous $g : U \to \mathbb{R}^m$ is path-connected whenever $U \subseteq \mathbb{R}^n$ is convex. (Compare Exercise 8.b.31, which asks the same question with *"connected"* in place of *"path-connected"*.)

Solution: *See solution on Page 120*

Exercise 8.b.31 (*Connected graphs* (important)): Let $U \subset \mathbb{R}^n$ be open and connected and let $f \in C^1(U, \mathbb{R}^m)$. Show that its graph

$$\Gamma_f := \{(x, f(x)) : x \in U\}$$

is a connected subset of $\mathbb{R}^n \times \mathbb{R}^m$.

Info: *This exercise is Problem 4.2 of Problem Sheet 4, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 120*

AI-Note 8.1: The hypothesis $f \in C^1$ is the sheet's, and it is far stronger than the statement needs: continuity of f is enough, and differentiability plays no part in the argument. The graph map $x \mapsto (x, f(x))$ is continuous precisely when f is, and connectedness is preserved by continuous maps (Theorem 8.a.7). AI-Exercise 8.b.30**(b)** is the same statement under the weaker hypothesis. Do not go looking for the place where C^1 is used; there is none.

Exercise 8.b.32 (*When is the sphere path-connected?*): Let $n \in \{1, 2, 3, \dots\}$ and let $S^{n-1} := \{x \in \mathbb{R}^n \mid \sum_{i=1}^n x_i^2 = 1\}$ be the unit sphere. Show that S^{n-1} is path-connected if and only if $n \geq 2$.

Info: *This exercise is taken from the Analysis I & II exam of 24 January 2022 (Prof. Giovanni Felder), Part A, Problem 7, which opens by asking for the definition of path-connectedness (Definition 8.b.21).*

Solution: *See solution on Page 121*

Solutions (Section 8.c)

Not every solution in this chapter is collected here. Exercise 8.a.12 (Page 114) and Exercise 8.b.23 (Page 116) are solved where they appear.

Proof of AI-Exercise 8.b.30:

(a) Let $p, q \in f(X)$, say $p = f(x)$ and $q = f(y)$. Since X is path-connected there is a path $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. Then $f \circ \gamma : [0,1] \to f(X)$ is continuous as a composition of continuous maps, and it satisfies $(f \circ \gamma)(0) = p$, $(f \circ \gamma)(1) = q$. So it is a path in $f(X)$ from p to q, and $f(X)$ is path-connected.

(b) A convex set U is path-connected: the straight segment $t \mapsto (1-t)x + ty$ lies in U and is continuous. The map $\varphi : U \to \mathbb{R}^n \times \mathbb{R}^m$, $\varphi(x) := (x, g(x))$, is continuous because each component is, and $\Gamma_g = \varphi(U)$. Apply **(a)**.

Remark 8.c.33: Part **(a)** is strictly easier than its connectedness counterpart (Theorem 8.a.7), because path-connectedness is a statement one verifies by **exhibiting** something, namely a path, and continuous maps push paths forward directly. Connectedness has no such witness to push forward, which is why its proof has to run by contradiction through preimages instead. This asymmetry is worth noticing: it is the reason path-connectedness is usually the easier of the two to prove, and connectedness the easier of the two to **use**.

Q.E.D.

Proof of Exercise 8.b.31:

Consider the map

$$\varphi : U \to \mathbb{R}^n \times \mathbb{R}^m, \qquad x \mapsto (x, f(x)).$$

Since $f \in C^1(U, \mathbb{R}^m)$ is in particular continuous, and the identity on U is continuous, φ is continuous. By definition $\Gamma_f = \varphi(U)$. Since U is connected and φ is continuous, Theorem 8.a.7 gives that $\varphi(U) = \Gamma_f$ is connected. Q.E.D.

Solution of Exercise 8.a.17:

Take the closed exterior of the unit disc,

$$U := \{x \in \mathbb{R}^2 \mid |x| \geq 1\}.$$

- **Closed:** U is the preimage of the closed set $[1, \infty)$ under the continuous map $x \mapsto |x|$.
- **Unbounded:** it contains $(k, 0)$ for every $k \in \mathbb{N}$.
- **Connected:** it is in fact path-connected. Given $p, q \in U$, travel radially outwards from p to the circle of radius $R := \max\{|p|, |q|\}$, then along that circle, then radially inwards to q. Every point on this path has norm between 1 and R, so the path stays in U.
- **Not simply connected:** the loop $\gamma(t) := (2\cos 2\pi t,\, 2\sin 2\pi t)$ winds once around the origin, and no loop with winding number 1 about a point outside U can be contracted to a point within U. Contracting it would force the loop to sweep across the removed open disc $\{|x| < 1\}$.

AI-Note 8.2: The exercise is really a test of whether the three notions have been kept apart. Closed-and-unbounded rules out the reflex answer of an annulus $\{1 \leq |x| \leq 2\}$, and *"connected but not simply connected"* rules out a half-plane or all of $\mathbb{R}^2$. Deleting an **open** disc rather than a point is what keeps the result closed while still puncturing it.

Q.E.D.

Proof of Exercise 8.b.32:

Both directions are proved by looking at what S^{n-1} actually is in the two cases.

The case $n = 1$. Here $S^0 = \{x \in \mathbb{R} \mid x^2 = 1\} = \{-1, +1\}$, a two-point space. Suppose $\gamma : [0, 1] \to S^0$ were a path with $\gamma(0) = -1$ and $\gamma(1) = 1$. Regarded as a map into $\mathbb{R}$, it is continuous on an interval, so by the intermediate value theorem it takes the value 0 somewhere. But $0 \notin S^0$, a contradiction. Hence S^0 is not path-connected.

The case $n \geq 2$. Fix $p, q \in S^{n-1}$, and suppose first that $q \neq -p$. Then the straight segment between them misses the origin: if $(1-t)p+tq = 0$ for some $t \in [0, 1]$, taking norms gives $(1-t) = t$, so $t = \frac{1}{2}$ and $p = -q$, which we excluded. We may therefore normalise the segment and set

$$\gamma(t) := \frac{(1-t)p + tq}{|(1-t)p + tq|}, \qquad t \in [0, 1].$$

This is continuous, takes values in S^{n-1}, and satisfies $\gamma(0) = p$, $\gamma(1) = q$.

It remains to handle $q = -p$, and this is exactly where $n \geq 2$ enters: the sphere S^{n-1} then contains a point r with $r \neq \pm p$. Take for r any unit vector orthogonal to p, which exists because $p^\perp$ has dimension $n - 1 \geq 1$. Since $r \neq -p$ and $r \neq p = -q$, the previous construction gives a path from p to r and a path from r to q; concatenating them joins p to q. Hence S^{n-1} is path-connected.

Remark 8.c.34: The construction in the main case is the radial projection of the segment, and it is the standard way to move around a sphere: it works verbatim on any S^{n-1} and explains why the antipodal pair is the only obstruction. Note also that $n \geq 2$ is used only once, to produce a third point, which is precisely what S^0 does not have.

AI-Note 8.3: The official solution introduces the same construction with the words *"now let $n \geq 1$"*, and then disposes of the antipodal case $x = -y$ by composing two paths through *"an*

arbitrary other point $w \in S^{n-1}$ ". For $n = 1$ there is no such w: $S^0 = \{\pm 1\}$ consists of x and $-x$ and nothing else. The bound is therefore $n \geq 2$, exactly as the statement being proved requires, and as the official solution's own preceding paragraph, which shows S^0 is **not** path-connected, confirms. The step is written out explicitly above rather than left to the reader, since it is the only place the hypothesis is used.

Q.E.D.

Proof of Exercise 8.a.18:

(a) **True:** U is defined by non-strict inequalities involving continuous functions (specifically, the pre-image of $[0, \infty)$ under $g_1(x, y) = y + x^2$ and $g_2(x, y) = x^2 - y$). Consequently, U is closed.

(b) **True:** We can contract U to the origin via the homotopy $H((x, y), t) = (tx, t^2 y)$ for $t \in [0, 1]$. Note that $(tx)^2 = t^2 x^2$, and since $-x^2 \leq y \leq x^2$, multiplying by $t^2 \geq 0$ gives $-(tx)^2 \leq t^2 y \leq (tx)^2$, which ensures the contraction stays within U.

(c) **False:** U is not bounded (for example, the sequence $(n, 0)$ is in U for all $n \in \mathbb{N}$), and therefore not compact.

(d) **True:** Since U is path-connected (from the homotopy to the origin) and therefore connected, its continuous image $f(U)$ is connected in $\mathbb{R}$.

(e) **False:** The function is not bounded on U. Consider the path $(x, 0)$ inside U; here $f(x, 0) = x^2$, which grows without bound as $x \to \infty$.

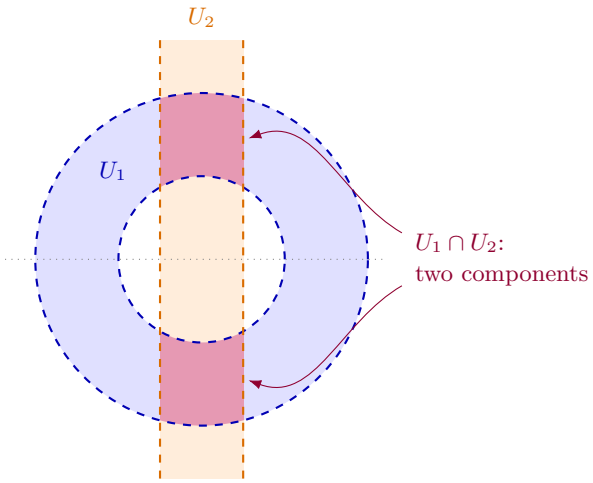
(f) **True:** Since $f(x, y) \geq 0$ everywhere on U and $f(0, 0) = 0$ with $(0, 0) \in U$, the infimum is 0 and it is attained at the origin.

Q.E.D.

Solution to Exercise 8.a.19:

Take an open annulus and an open vertical strip through its hole:

$$U_1 := B_1(0) \setminus \overline{B_{1/2}(0)} = \left\{p \in \mathbb{R}^2 \;\middle|\; \tfrac{1}{2} < |p| < 1\right\}, \qquad U_2 := \left(-\tfrac{1}{4}, \tfrac{1}{4}\right) \times \mathbb{R}.$$



Four things have to be checked, and the figure does none of them by itself.

(a) U_1 **is open and connected.** It is the intersection of the open set $B_1(0)$ with the complement of the closed set $\overline{B_{1/2}(0)}$, hence open. It is path-connected, since any two of its points can

be joined by moving radially to the circle of radius $\frac{3}{4}$ and then along that circle, and a path-connected set is connected.

(b) U_2 **is open and connected.** It is a product of an open interval with $\mathbb{R}$, hence open, and it is convex, hence path-connected and so connected.

(c) $U_1 \cap U_2$ **is not connected.** If (x, y) lies in it then $|x| < \frac{1}{4}$ and $x^2 + y^2 > \frac{1}{4}$, so

$$y^2 > \tfrac{1}{4} - x^2 > \tfrac{1}{4} - \tfrac{1}{16} = \tfrac{3}{16} > 0.$$

In particular $y \neq 0$ throughout, so the two open half-planes $\{y > 0\}$ and $\{y < 0\}$ separate $U_1 \cap U_2$. Both pieces are non-empty, since $(0, \frac{3}{4})$ and $(0, -\frac{3}{4})$ both lie in $U_1 \cap U_2$. That is a separation, so the intersection is disconnected.

(d) $U_1 \cup U_2$ **is connected.** Both sets are connected and they share the point $(0, \frac{3}{4})$, so Corollary 8.a.13 applies to the two-element family $\{U_1, U_2\}$.

Remark 8.c.35 (*Why the intersection is the odd one out*): Corollary 8.a.13 makes unions easy, and nothing in this chapter says anything at all about intersections. That silence is not an oversight. A union of connected sets sharing a point cannot be split, because any separation would have to split one of the pieces; an intersection carries no such argument, and the picture above shows why: the strip enters the annulus twice, and what it takes away in between is exactly what would have joined the two pieces.

Note also that the construction needs the hole. If U_1 were a disc, then $U_1 \cap U_2$ would be convex and the example would collapse; the failure of connectedness in **(c)** is the annulus's non-simple-connectedness showing up in a second guise.

Q.E.D.

PART III

Differential Calculus

In one variable the derivative is a number, which makes it tempting to expect the several-variable derivative to be a list of numbers. It is not. The right object is a linear map, the one that best approximates f near a point, and the partial derivatives are merely its matrix in the standard basis. Getting that order of ideas right is what turns the chain rule into a statement about composing linear maps rather than a formula to be memorised, and it is also what accounts for the counterexamples running through these chapters, in which a function possesses every directional derivative at a point and is not even continuous there. The rest of the part asks how good the linear approximation actually is. The mean value theorem measures its error and is the first result in the course to degrade on the way up, surviving for vector-valued maps only as an inequality, and Taylor's theorem then sharpens the approximation to any order and supplies every quantitative estimate the optimization theory of the next part runs on.

The Differential and the Jacobi Matrix (Chapter 9)

The previous part treated continuous functions as structure-preserving maps between metric spaces. This one asks for considerably more: that a map be well approximated, near each point, by a **linear** one. That approximating linear map is the differential.

Three notions compete for the name *"derivative"* in several variables, and they are not equivalent. Partial derivatives ask only what happens along the coordinate axes, directional derivatives along an arbitrary straight line, and the differential asks for a linear map approximating f from every direction at once. Only the last deserves to be called differentiability, and the gap between them is real: a function can have every partial derivative at a point and still fail to be continuous there. This chapter sets out that hierarchy and identifies the matrix representing the differential.

The differential (Section 9.a)

For $f:\mathbb{R}\to\mathbb{R}$ differentiable at $x_0\in\mathbb{R}$:

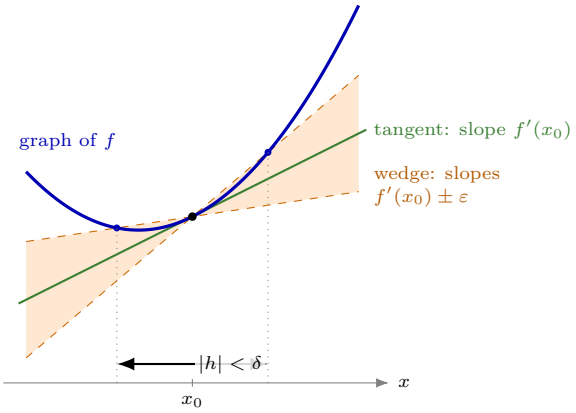
$$f(x_0+h)=f(x_0)+f'(x_0)h+o(h)$$
$$\Updownarrow$$
$$\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}=f'(x_0)$$
$$\Updownarrow$$
$$\lim_{h\to 0}\left|\frac{f(x_0+h)-f(x_0)-f'(x_0)h}{h}\right|=0$$

For $F:\mathbb{R}^n\to\mathbb{R}^m$ differentiable at $x_0\in\mathbb{R}^n$:

$$F(x_0+h)=F(x_0)+\underbrace{DF_{x_0}(h)}_{\text{linear map}}+o(|h|)$$
$$\Updownarrow$$
$$\lim_{h\to 0}\frac{|F(x_0+h)-F(x_0)-DF_{x_0}(h)|}{|h|}=0$$
$$\Downarrow$$
$$\lim_{s\to 0}\frac{F(x_0+sv)-F(x_0)}{s}=DF_{x_0}(v)=\partial_v F(x_0)\quad\text{for each fixed } v\in\mathbb{R}^n$$
(the "*directional derivative*" of F at x_0 in direction v)

Note that the last arrow points **one way only**. The first two lines say the same thing; the third is a strictly weaker consequence, and recovering differentiability from it is exactly what fails in several variables; see AI-Example 9.b.8 below.

It is worth pausing on why the definition had to be restated rather than merely copied. The one-variable derivative is defined as a **quotient**, and that is precisely what does not survive: for $h\in\mathbb{R}^n$ with $n\geq 2$ the expression $\big(F(x_0+h)-F(x_0)\big)/h$ is meaningless, because one cannot divide by a vector. The first line above is the version that does survive, and it is the one to remember: **F is differentiable at x_0 if it is approximated near x_0 by an affine map, to an error smaller than $|h|$.** Everything that follows, the Jacobian, the chain rule and Taylor expansions alike, is bookkeeping for that single idea. The reason the derivative becomes a **linear map** rather than a number is simply that a linear map is what an approximation in n variables looks like.



For every $\varepsilon > 0$ there is a $\delta > 0$ such that, over $|h| < \delta$, the graph lies inside the shaded wedge around the tangent line. That is the statement $|f(x_0 + h) - f(x_0) - f'(x_0)h| \le \varepsilon|h|$, and it is the form that generalises: in several variables the line becomes the graph of the affine map $h \mapsto F(x_0) + DF_{x_0}(h)$, and the wedge becomes a cone around it.

Definition 9.a.1 (*Differentiability*): We say that $F : \mathbb{R}^n \to \mathbb{R}^m$ is "*differentiable*" (German: „*differenzierbar*“) at x_0 if a linear map DF_{x_0} as above exists, where $x_0, h \in \mathbb{R}^n$. The same definition applies verbatim to $F : U \to \mathbb{R}^m$ for any open subset $U \subseteq \mathbb{R}^n$, and we use it in that generality throughout.

Remark 9.a.2: The linear map DF_{x_0} in Definition 9.a.1, when it exists, is automatically unique: if L_1, L_2 both satisfy the defining limit, then $(L_1 - L_2)(h)/|h| \to 0$ as $h \to 0$ along every direction, which forces the linear map $L_1 - L_2$ to vanish identically. Concretely: write $L := L_1 - L_2$ and suppose $L(h) \neq 0$ for some h. Along the ray th, $t > 0$, linearity gives

$$\frac{|L(th)|}{|th|} = \frac{t\,|L(h)|}{t\,|h|} = \frac{|L(h)|}{|h|} \neq 0,$$

a nonzero **constant**. Consequently, the quotient cannot tend to 0 as $t \to 0$, and therefore $L(h) = 0$ for every h. This is the multivariable analogue of the uniqueness-of-limits argument from Proposition 4.a.4.

Lemma 9.a.3 (*Differentiability is componentwise*): Let $U \subseteq \mathbb{R}^n$ be open, $F : U \to \mathbb{R}^m$, and write $F = (F_1, \dots, F_m)$ for its components $F_j : U \to \mathbb{R}$. Then F is differentiable at $x \in U$ if and only if every F_j is, and in that case

$$\big(DF_x(v)\big)_j = D(F_j)_x(v) \qquad \text{for all } v \in \mathbb{R}^n \text{ and } 1 \le j \le m.$$

Proof:

Both directions come from the same observation: a vector in $\mathbb{R}^m$ is $o(|h|)$ exactly when each of its m components is, since $\max_j |w_j| \le |w| \le \sqrt{m}\max_j |w_j|$.

If each F_j is differentiable at x with differential L_j, assemble $L(v) := (L_1(v), \dots, L_m(v))$, which is linear. The remainder of F at x has j-th component the remainder of F_j, each of which is $o(|h|)$, hence so is the vector. Conversely, if F is differentiable with differential L, then composing with the j-th coordinate projection, which is linear and 1-Lipschitz, shows F_j is differentiable with differential $v \mapsto L(v)_j$. Q.E.D.

Remark 9.a.4 (*What this buys, and what it does not*): This is the licence to reduce to $m = 1$, and it is used in exactly that way twice below: the rows of the Jacobi matrix are the gradients of the components (Section 9.b), and the proof that continuous partial derivatives suffice (Theorem 10.c.14) treats a scalar function only.

It does **not** say anything about the other index. Differentiability in n variables does not reduce to differentiability in each variable separately: that is precisely the failure AI-Example 9.b.8 exhibits. The target $\mathbb{R}^m$ splits; the source $\mathbb{R}^n$ does not.

Proposition 9.a.5 (*Differentiability implies continuity*): If $F : U \to \mathbb{R}^m$, $U \subseteq \mathbb{R}^n$ open, is differentiable (Definition 9.a.1) at $x_0 \in U$, then F is continuous at x_0.

Proof:

By Definition 9.a.1, $F(x_0+h) - F(x_0) = DF_{x_0}(h) + o(|h|)$ as $h \to 0$. Since DF_{x_0} is linear on the finite-dimensional space $\mathbb{R}^n$, it is bounded, i.e. there exists $C \geq 0$ with $|DF_{x_0}(h)| \leq C|h|$ for all h. Hence

$$|F(x_0+h) - F(x_0)| \leq |DF_{x_0}(h)| + o(|h|) \leq C|h| + o(|h|) \xrightarrow{h\to 0} 0,$$

so $F(x_0+h) \to F(x_0)$ as $h \to 0$, i.e. F is continuous at x_0. Q.E.D.

Remark 9.a.6: The converse fails: continuity does not imply differentiability, exactly as in one variable (e.g. $x \mapsto |x|$ at 0). AI-Example 9.b.8 below shows the gap is even wider in several variables, where **all** partial derivatives can exist at a point without F even being continuous there, let alone differentiable.

The Jacobi matrix (Section 9.b)

Since $DF_{x_0} : \mathbb{R}^n \to \mathbb{R}^m$ (Definition 9.a.1) is a linear map, there is a **1–1 correspondence with a matrix once a basis is fixed**! In the standard basis of $\mathbb{R}^n$, $(e_1, \ldots, e_n)$, this is the "*Jacobi matrix*" (German: „*Jacobi-Matrix*"):

$$DF_x \;\cong\; \mathrm{J}\,F(x) = \begin{pmatrix} \dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_m}{\partial x_1} & \cdots & \dfrac{\partial F_m}{\partial x_n} \end{pmatrix}$$

(m rows, n columns), where we wrote $F(x) = (F_1, \ldots, F_m)^{\mathsf{T}}$ and

$$\frac{\partial F}{\partial x_i} =: \partial_i F(x)$$

is the "*i-th partial derivative*" (German: „*partielle Ableitung*") at x:

$$\partial_i F(x) = \lim_{s\to 0} \frac{F(x + s e_i) - F(x)}{s}.$$

This defines a function $\partial_i F : U \to \mathbb{R}^m$. If F is differentiable, then

$$\partial_i F(x) = DF_x(e_i).$$

Important remark 9.b.7: The partial derivatives can exist even if $F : U \to \mathbb{R}^m$ is not differentiable (recall AI-Example 9.b.8)!

AI-Example 9.b.8 (*All partial derivatives exist, yet f is not even continuous*): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) := \begin{cases} \dfrac{xy}{x^2+y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0). \end{cases}$$

Along each axis, f vanishes identically ($f(x,0) = f(0,y) = 0$), so both partial derivatives at the origin exist and equal 0:

$$\partial_x f(0,0) = \lim_{s\to 0} \frac{f(s,0) - f(0,0)}{s} = 0, \qquad \partial_y f(0,0) = 0.$$

Yet f is not even continuous at $(0,0)$: along the diagonal $y=x$, $f(x,x)=\frac{x^2}{2x^2}=\frac{1}{2}$ for all $x\neq 0$, so $f(x,x)\to\frac{1}{2}\neq 0=f(0,0)$ as $x\to 0$. By Proposition 9.a.5, f is therefore not differentiable at $(0,0)$, even though both partial derivatives exist there.

Remark 9.b.9 (*The same expression, squared, behaves completely differently*): The function above has a near twin, and telling the two apart is a good test:

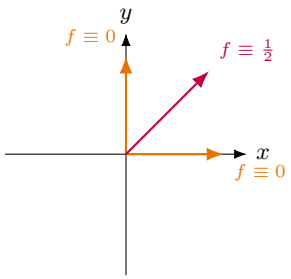
$$f_1(x,y):=\frac{xy}{x^2+y^2},\qquad f_2(x,y):=\frac{x^2y^2}{x^2+y^2},$$

both extended by 0 at the origin. Substituting $x=r\cos\theta$, $y=r\sin\theta$ settles both at once:

$$f_1=\frac{r^2\cos\theta\sin\theta}{r^2}=\cos\theta\sin\theta,\qquad f_2=\frac{r^4\cos^2\theta\sin^2\theta}{r^2}=r^2\cos^2\theta\sin^2\theta.$$

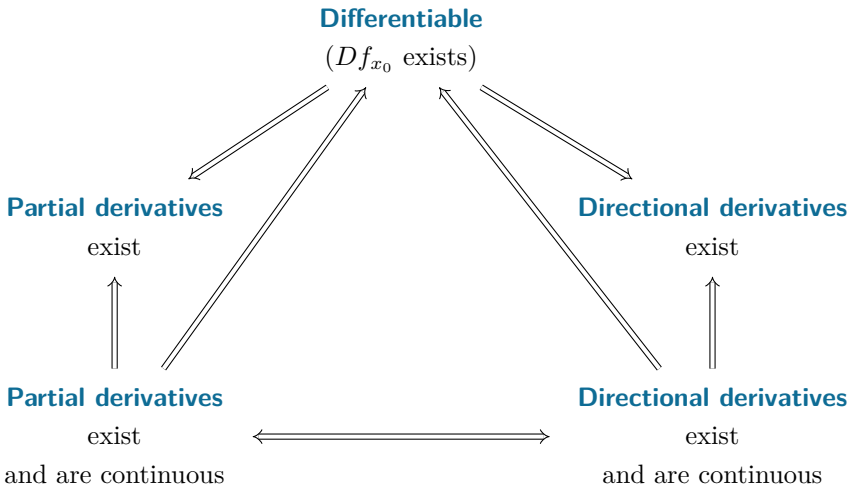
In f_1 every r cancels: the function is constant along each ray and takes a different constant on different rays, so no limit exists and f_1 is discontinuous at the origin. In f_2 a factor r^2 survives, and since $|\cos^2\theta\sin^2\theta|\leq 1$ we get $|f_2|\leq r^2\to 0$ **uniformly in** θ, so f_2 is continuous at the origin.

The pair is worth doing once because it shows what the polar substitution is actually testing. The question is never *"does the expression go to zero along each fixed ray?"*, which f_1 does not even manage. The question is **does the θ-dependence get killed by a positive power of r?** If the r's cancel completely, the direction survives into the limit and there is none. This is the same test used in Question C.a.7 on the repetition quiz, and the reason step 3 of Remark 10.c.15 insists on *"uniformly in θ"*.



$f=\frac{xy}{x^2+y^2}$ near the origin: constant along every ray,
but a **different** constant on the diagonal than on the axes

The picture below collects these relationships. Read it as a hierarchy: the strongest condition is at the bottom, and every arrow points at something weaker.



Important remark 9.b.10 (*Not one of these arrows reverses*): The diagram is easy to misread as a set of equivalences, and it is nothing of the kind: every $\Rightarrow$ in it is strict. AI-Example 9.b.8 already shows the bottom-left arrow failing backwards, since there both partial derivatives exist at the origin while f is not even continuous. The other two failures, with their counterexamples, are collected in Section 10.c, which redraws this same landscape as a single vertical chain and attaches a witness to each broken link.

The one genuine equivalence is the horizontal arrow along the bottom, and it is worth seeing why. Left to right: continuous partials make f continuously differentiable (Theorem 10.c.14), so every directional derivative exists and equals $Df_x(v)$, which depends continuously on x. Right to left is immediate: take $v = e_i$.

Examples (Subsection 9.b.1)

The three examples that follow are the three shapes the Jacobian takes, and it is worth naming them, because the layout of the matrix changes completely between them while the underlying object, the linear map DF_{x_0}, does not.

- **A curve** ($n = 1$, m arbitrary): one column, so $\mathrm{J}\,\gamma(t)$ is a **column vector**, the velocity.
- **A scalar field** (n arbitrary, $m = 1$): one row, so $\mathrm{J}\,f(x)$ is a **row vector**, namely the transposed gradient $(\nabla f)^{\mathsf{T}}$.
- **A vector field** (both > 1): the full $m \times n$ matrix, whose columns are the partial derivatives and whose rows are the gradients of the components.

That the Jacobian of a scalar field is a row while the Jacobian of a curve is a column is not a convention to memorise: it follows from *"m rows, n columns"* and nothing else. Note the transpose in the second bullet, which is the one place a real convention enters: the **gradient** ∇f is a column vector, so it is $\mathrm{J}\,f = (\nabla f)^{\mathsf{T}}$ that is the row. Confusing the two is the commonest source of shape errors when the chain rule is applied in the next chapter.

Example 9.b.11 (*Derivative of a curve*):

$$\gamma : \mathbb{R} \to \mathbb{R}^2, \qquad t \mapsto \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

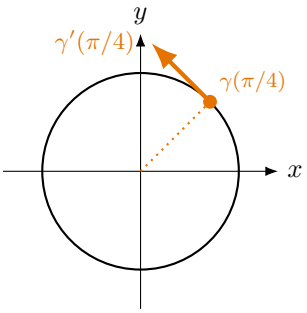
What is $\gamma'(\frac{\pi}{4}) := D\gamma_{\pi/4}$?

$$\gamma'\left(\tfrac{\pi}{4}\right) = \begin{pmatrix} \partial_t \gamma_1(\pi/4) \\ \partial_t \gamma_2(\pi/4) \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

AI-Example 9.b.12 (*Jacobian Matrix of Polar Coordinates*): Let $f\colon (0,\infty) \times (0, 2\pi) \to \mathbb{R}^2$ be the polar coordinate map $f(r,\theta) := (r\cos\theta, r\sin\theta)^{\mathsf{T}}$. The Jacobian matrix $\mathrm{J}\,f(r,\theta)$ is:

$$\mathrm{J}\,f(r,\theta) = \begin{pmatrix} \dfrac{\partial f_1}{\partial r} & \dfrac{\partial f_1}{\partial \theta} \\ \dfrac{\partial f_2}{\partial r} & \dfrac{\partial f_2}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}.$$

Its determinant is $\det \mathrm{J}\,f(r,\theta) = r\cos^2\theta + r\sin^2\theta = r > 0$.



This vector is the **velocity of the curve** (German: „*Geschwindigkeit*“) at $t = \frac{\pi}{4}$; its length $|\gamma'(t)|$ is the **speed**. We get that

$$\gamma\left(\tfrac{\pi}{4} + h\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} + \frac{h}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix} + o(|h|).$$

From this example, we see that the **column** vectors of the Jacobian are the partial derivatives:

$$\mathrm{J}\,F(x) = \left(\frac{\partial F}{\partial x_1} \;\middle|\; \cdots \;\middle|\; \frac{\partial F}{\partial x_n}\right)$$

and therefore, in the canonical basis, with $v := (v_1, \ldots, v_n)^\mathsf{T}$:

$$\partial_v F(x) = DF_x(v) = \mathrm{J}\,F(x)\,v = \sum_{i=1}^{n} v_i \frac{\partial F}{\partial x_i}.$$

Example 9.b.13 (*Scalar field gradient*):

$$f : \mathbb{R}^3 \to \mathbb{R}, \qquad (x, y, z) \mapsto 2x^2 + y^2 + 3z^2 - 2xyz$$

What is $\mathrm{J}\,f(x, y, z)$?

$$\mathrm{J}\,f(x, y, z) = (4x - 2yz,\ 2y - 2xz,\ 6z - 2xy) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (\nabla f)^\mathsf{T}$$

Example 9.b.14 (*Jacobian matrix of a vector field*): Consider $f : \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$f(x, y) := \begin{pmatrix} x^2 + y^2 \\ 2xy \\ e^x \sin(y) \end{pmatrix}.$$

The partial derivatives with respect to x and y are the column vectors:

$$\partial_1 f(x, y) = \begin{pmatrix} 2x \\ 2y \\ e^x \sin(y) \end{pmatrix}, \qquad \partial_2 f(x, y) = \begin{pmatrix} 2y \\ 2x \\ e^x \cos(y) \end{pmatrix}.$$

Because these partial derivatives exist everywhere and are continuous, f is differentiable on all of $\mathbb{R}^2$. The differential $Df_{(x_0,y_0)}$ is represented by the 3×2 Jacobian matrix:

$$\mathrm{J}\,f(x_0, y_0) = \begin{pmatrix} 2x_0 & 2y_0 \\ 2y_0 & 2x_0 \\ e^{x_0}\sin(y_0) & e^{x_0}\cos(y_0) \end{pmatrix}.$$

From this, we see that the **rows** of the Jacobian are the “*gradients*” (German: „*Gradient*“):

$$\mathrm{J}\,F(x) = \begin{pmatrix} \nabla F_1 \\ \hline \vdots \\ \hline \nabla F_m \end{pmatrix}$$

Example 9.b.15 (*A function that is differentiable but not C^1*): Let $f:\mathbb{R}^2\to\mathbb{R}$ be defined by

$$f(x,y):=\begin{cases}(x^2+y^2)\sin\left(\dfrac{1}{\sqrt{x^2+y^2}}\right), & \text{if } (x,y)\neq(0,0),\\ 0, & \text{if } (x,y)=(0,0).\end{cases}$$

We claim that f is differentiable at $(0,0)$ with $Df_{(0,0)}=(0,0)$, but the partial derivatives are not continuous at $(0,0)$.

Step 1: Differentiability at $(0,0)$. We use the definition of the differential. Let $L=(0,0)$. We need to show that

$$\lim_{h\to(0,0)}\frac{f(h_1,h_2)-f(0,0)-L\begin{pmatrix}h_1\\h_2\end{pmatrix}}{|h|}=0.$$

Plugging in the definitions and changing to polar coordinates $(h_1,h_2)=(r\cos\theta,r\sin\theta)$ with $r=|h|=\sqrt{h_1^2+h_2^2}$:

$$\lim_{r\to0}\frac{r^2\sin(1/r)-0-0}{r}=\lim_{r\to0}r\sin(1/r)=0.$$

The sine term is bounded between -1 and 1, so the limit evaluates to 0. Thus, f is differentiable at $(0,0)$ and $Df_{(0,0)}=(0,0)$, which implies $\partial_1 f(0,0)=0$ and $\partial_2 f(0,0)=0$.

Step 2: Discontinuity of partial derivatives. Let us compute $\partial_1 f(x,y)$ for $(x,y)\neq(0,0)$. Using the product and chain rules:

$$\partial_1 f(x,y)=2x\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right)-\frac{x}{\sqrt{x^2+y^2}}\cos\left(\frac{1}{\sqrt{x^2+y^2}}\right).$$

Let us choose a sequence $(x_n,y_n):=(\frac{1}{n},0)$ which converges to $(0,0)$ as $n\to\infty$. Then

$$\lim_{n\to\infty}\partial_1 f\left(\frac{1}{n},0\right)=\lim_{n\to\infty}\left[\frac{2}{n}\sin(n)-\cos(n)\right].$$

The first term vanishes, but $\lim_{n\to\infty}\cos(n)$ does not exist. Thus, $\partial_1 f$ is not continuous at $(0,0)$, and so f is not continuously differentiable (C^1) at $(0,0)$. (By symmetry, the same holds for $\partial_2 f$).

AI-Exercise 9.b.16 (*Differential and Linearization at $(1,0)$*): Let $f(x,y):=x^2y+\sin(xy)$ defined on $\mathbb{R}^2$.

(a) Compute the gradient $\nabla f(x,y)$ and the total differential $Df(1,0)$.

(b) Find the linear approximation $L(x,y)$ of f around the point $(1,0)$.

Solution: *See solution on Page 133*

Differentiating with respect to a matrix (Subsection 9.b.2)

Nothing in Definition 9.a.1 requires the domain to be $\mathbb{R}^n$ with n small, or even to be written as a column of numbers. A function of an $n\times m$ matrix is a function of nm real variables, and its partial derivatives $\partial f/\partial A_{ij}$ are defined exactly as before, by varying one entry and freezing the rest.

Remark 9.b.17 (*Why this is worth a subsection*): Taking derivatives with respect to a matrix argument can look like a formal exercise with no purpose. It is not, and the reason is worth recording: **it is how neural networks are trained.** The parameters of such a network are matrices. The function being minimised sends those matrices to a single number measuring how badly the network fits the data, and training consists of computing the gradient with respect to those matrices and then stepping a little way against it, because $-\nabla f$ points in the direction of steepest decrease. Every step of that loop is the material of this chapter.

The simplest instance is already worth doing by hand, and it is exactly linear regression.

AI-Example 9.b.18 (*The gradient of a least-squares fit*): Fix data matrices $X \in \mathbb{R}^{m\times k}$ (the k inputs, each a column in $\mathbb{R}^m$) and $Y \in \mathbb{R}^{n\times k}$ (the k desired outputs). The model applies an unknown matrix $A \in \mathbb{R}^{n\times m}$ and adds an unknown offset $b \in \mathbb{R}^n$ to every column, so its total error is

$$f(A,b) := \big\| \underbrace{AX + b\mathbf{1}^{\mathsf{T}} - Y}_{=:\,R} \big\|^2, \qquad \mathbf{1} := (1,\dots,1)^{\mathsf{T}} \in \mathbb{R}^k,$$

where $\|\cdot\|$ is the Hilbert–Schmidt norm of Section 2.a, so that $f(A,b) = \sum_{p,q} R_{pq}^2$ is the total squared error over all k data points. Writing out one entry of the residual,

$$R_{pq} = \sum_{s=1}^{m} A_{ps}X_{sq} + b_p - Y_{pq},$$

so that $\partial R_{pq}/\partial A_{ij} = \delta_{pi}X_{jq}$ and $\partial R_{pq}/\partial b_i = \delta_{pi}$. The chain rule then gives

$$\frac{\partial f}{\partial A_{ij}} = 2\sum_{p,q} R_{pq}\,\delta_{pi}X_{jq} = 2\sum_{q} R_{iq}X_{jq} = 2\big(RX^{\mathsf{T}}\big)_{ij},$$

$$\frac{\partial f}{\partial b_i} = 2\sum_{p,q} R_{pq}\,\delta_{pi} = 2\sum_{q} R_{iq} = 2\big(R\mathbf{1}\big)_i.$$

Collecting the entries back into arrays of the same shape as the variables,

$$\nabla_A f = 2\big(AX + b\mathbf{1}^{\mathsf{T}} - Y\big)X^{\mathsf{T}}, \qquad \nabla_b f = 2\big(AX + b\mathbf{1}^{\mathsf{T}} - Y\big)\mathbf{1}.$$

The gradient with respect to b is twice the vector of row sums of the residual, and the gradient with respect to A is the residual correlated against the inputs. Both are entirely explicit, and both were computed with nothing beyond partial differentiation and the chain rule.

Remark 9.b.19 (*A notational caution*): One point trips people up here. Since f above has domain $\mathbb{R}^{n\times m}\times\mathbb{R}^n$ of dimension $nm+n$, its Jacobian is formally a $1\times(nm+n)$ row, and $\partial_v f(A,b) = \mathrm{J}\,f(A,b)\,v$ would require flattening the matrix direction v into a column of length nm. That flattening is pure bookkeeping and is almost never performed: one keeps $\nabla_A f$ shaped like A, as above. The identity $\partial_v f = \mathrm{J}\,f\cdot v$ of Section 9.b still holds, and is simply being read in a reshaped coordinate system.

Exercise 9.b.20 (*Derivative computation* (important)): Consider the function $u : (x,y) \mapsto x^{\sin(y)}$, defined for $(x,y) \in (0,\infty)\times\mathbb{R} \subset \mathbb{R}^2$. Compute $\partial_x u$ and $\partial_y u$.

Info: *This exercise is Problem 4.1 of Problem Sheet 4, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 133*

Exercise 9.b.21 (*True or False* (piecewise differentiability)): Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x,y) = \begin{cases} y^2\cos x & \text{if } y \geq 0,\\ 0 & \text{if } y < 0. \end{cases}$$

Decide whether each of the following statements is true or false.

(a) $f \in C^0(\mathbb{R}^2)$.
(b) $\partial_y f \in C^0(\mathbb{R}^2)$.
(c) $\partial_y f \in C^1(\mathbb{R}^2)$.
(d) $f \in C^2(\mathbb{R}^2)$.
(e) $\frac{\partial^2 f}{\partial x^2}(0,0)$ exists.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the fourth block of true/false questions.*

Solution: *See solution on Page 133*

Exercise 9.b.22 (*Single choice: which condition gives differentiability at the origin?*)**:** Let f be a function defined on the unit ball $B_1(0) \subseteq \mathbb{R}^n$. Exactly one of the following guarantees that f is differentiable at the origin. Which?

(a) All partial derivatives of f exist on the unit ball.

(b) There exists an **arbitrary** map L such that

$$\lim_{h\to 0} \frac{\|f(h) - f(0) - L(h)\|}{\|h\|} = 0.$$

(c) There exists a **linear** map L such that $f(h) - f(0) - L(h) = o\big(\|h\|^2\big)$.

(d) All partial derivatives of f are bounded on the unit ball.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 3.*

Solution: *See solution on Page 134*

Solutions (Section 9.c)

Solution to Exercise 9.b.20:

Write $u(x,y) = x^{\sin(y)} = e^{\sin(y)\ln x}$ for $x > 0$. Differentiating with the chain rule:

$$\partial_x u = \sin(y)\, x^{\sin(y)-1}, \qquad \partial_y u = \cos(y)\ln(x)\, x^{\sin(y)} = \cos(y)\ln(x)\, e^{\sin(y)\ln x}.$$

Q.E.D.

Proof of AI-Exercise 9.b.16:

(a) Partial derivatives are $\frac{\partial f}{\partial x} = 2xy + y\cos(xy)$ and $\frac{\partial f}{\partial y} = x^2 + x\cos(xy)$. Evaluating at $(1,0)$ yields $\frac{\partial f}{\partial x}(1,0) = 0$ and $\frac{\partial f}{\partial y}(1,0) = 1 + 1 = 2$. Thus $\nabla f(1,0) = (0,2)^\mathsf{T}$, and the total differential $Df(1,0)\colon \mathbb{R}^2 \to \mathbb{R}$ is $Df(1,0)(h_1,h_2) = 2h_2$.

(b) Since $f(1,0) = 0$, the linearization at $(1,0)$ is $L(x,y) = f(1,0) + Df(1,0)(x-1, y-0) = 0 + 0(x-1) + 2y = 2y$.

Q.E.D.

Proof of Exercise 9.b.21:

(a) **True:** For $y > 0$ and $y < 0$ the function is given by continuous formulas. On the interface $y = 0$, we have $\lim_{y\to 0^+} y^2\cos x = 0 = \lim_{y\to 0^-} 0$, which matches $f(x,0) = 0$.

(b) **True:** For $y > 0$, $\partial_y f = 2y\cos x$; for $y < 0$, $\partial_y f = 0$. At $y = 0$, the one-sided derivatives are $\lim_{h\to 0^+} \frac{h^2\cos x - 0}{h} = 0$ and $\lim_{h\to 0^-} \frac{0-0}{h} = 0$. So $\partial_y f$ exists everywhere and evaluates to $2y\cos x$ for $y \geq 0$ and 0 for $y < 0$. These two pieces patch continuously at $y = 0$.

(c) **False:** For $\partial_y f$ to be C^1, its own partial derivatives must be continuous. However, for $y > 0$, $\partial_y(\partial_y f) = 2\cos x$, while for $y < 0$, it is 0. As $y \to 0$, the upper half approaches $2\cos x$, which does not match the lower half limit of 0 unless $\cos x = 0$. Thus $\partial_{yy} f$ has jump discontinuities along the x-axis.

(d) **False:** Since $\partial_{yy} f$ is not continuous (from the previous part), f is not twice continuously differentiable on $\mathbb{R}^2$.

(e) **True:** Along the x-axis ($y = 0$), the function is $f(x,0) = 0$ everywhere. Therefore, the function restricted to the x-axis is identically zero, meaning $\partial_x f(x,0) = 0$ everywhere, and consequently its derivative with respect to x again, $\partial_{xx} f(0,0)$, is also 0.

Q.E.D.

Solution of Exercise 9.b.22:

The correct answer is **(c)**.

(a) **False**, and this is the misconception the chapter opens with. Existence of every partial derivative does not even force continuity, let alone differentiability: AI-Example 9.b.8 exhibits a function whose two partials exist everywhere and which is discontinuous at the origin. Partial derivatives see only the coordinate directions, and differentiability is a statement about all directions at once.

(b) **False**, and the word doing the damage is **arbitrary**. Compare the displayed limit with Definition 9.a.1: they are the same limit, except that the definition requires L to be **linear**. Drop that and the condition becomes vacuous: take $L(h) := f(h) - f(0)$, which makes the numerator identically zero for every f whatsoever. The linearity of L is not a technical convenience; it is the entire content of the definition, since differentiability means precisely that f is approximated by a linear map.

(c) **True**, and it is true for the unexciting reason that it asks for too much. Differentiability at 0 is the existence of a linear L with $f(h) - f(0) - L(h) = o(\|h\|)$. Here the error is required to be $o(\|h\|^2)$, which for small h is a strictly stronger demand: if $\|f(h) - f(0) - L(h)\|/\|h\|^2 \to 0$, then

$$\frac{\|f(h) - f(0) - L(h)\|}{\|h\|} = \|h\| \cdot \frac{\|f(h) - f(0) - L(h)\|}{\|h\|^2} \to 0$$

as well. So f is differentiable, with $Df_0 = L$.

(d) **False.** Bounded partial derivatives are no better than existing ones for this purpose. A witness is

$$f(x,y) := \frac{xy^2}{x^2+y^2} \quad \text{for } (x,y) \neq (0,0), \qquad f(0,0) := 0,$$

the function of Exercise 10.c.18. Away from the origin

$$\partial_x f = \frac{y^2(y^2-x^2)}{(x^2+y^2)^2}, \qquad \partial_y f = \frac{2x^3 y}{(x^2+y^2)^2},$$

both homogeneous of degree 0 and continuous off the origin, hence bounded on the whole plane; at the origin both partials exist and vanish, since f vanishes on each axis. So the only candidate derivative is $L = 0$. But along $h = (s,s)$ with $s > 0$,

$$\frac{f(h) - f(0) - L(h)}{\|h\|} = \frac{s^3/(2s^2)}{s\sqrt{2}} = \frac{1}{2\sqrt{2}} \not\to 0,$$

so f is not differentiable at 0. Boundedness constrains the partials only in the coordinate directions and says nothing about the behaviour in between. What **does** suffice is **continuity** of the partials, which is Theorem 10.c.14; note that this is a genuinely stronger hypothesis than the one offered here, and that Example 9.b.15 shows it is not necessary either.

Q.E.D.

The Chain Rule and Directional Derivatives (Chapter 10)

Having encoded the differential as a matrix, we now ask how it interacts with composition. The answer is the chain rule, and in this setting it says something cleaner than its one-variable version does: the differential of a composition is the **composition of the differentials**, which is to say the product of their matrices. Reading it that way makes it a statement of linear algebra rather than a formula to be memorised, and explains where the sums of products in the coordinate form come from.

The same machinery yields directional derivatives, and with them the standard trap of several-variable calculus. A function can possess every directional derivative at a point and still fail to be differentiable there, or even continuous. Section 10.c says what does suffice, and it is the criterion used in practice for the rest of the course.

Chain rule (Section 10.a)

Theorem 10.a.1 (*Chain rule*): The "*chain rule*" (German: „*Kettenregel*") states: for $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $g : V \subseteq \mathbb{R}^m \to \mathbb{R}^d$ differentiable, it holds:

$$D(g \circ f)(x) = Dg(f(x))\, Df(x).$$

The same identity holds for the corresponding Jacobi matrices, where the composition on the right becomes an ordinary matrix product.

Proof:

Read the statement at a point: f differentiable at x, g differentiable at $y := f(x)$, and $f(U) \subseteq V$ with U, V open. Write $A := Df_x$ and $B := Dg_y$, so that

$$f(x+h) = y + Ah + r(h), \qquad g(y+k) = g(y) + Bk + s(k),$$

with $r(h) = o(|h|)$ and $s(k) = o(|k|)$. The idea is simply to substitute the first expansion into the second. Put $k(h) := f(x+h) - y = Ah + r(h)$; then

$$g(f(x+h)) = g(y) + BAh + \underbrace{Br(h) + s\big(k(h)\big)}_{=:\,E(h)},$$

and it remains to show $E(h) = o(|h|)$.

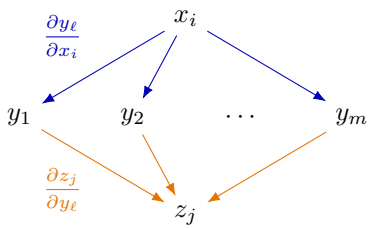
- $|Br(h)| \le \|B\|_2\, |r(h)|$ by Lemma 2.a.4, and this is $o(|h|)$.
- For the second term, define $\sigma(k) := |s(k)|/|k|$ for $k \neq 0$ and $\sigma(0) := 0$, so that $|s(k)| = \sigma(k)|k|$ for **every** k and $\sigma(k) \to 0$ as $k \to 0$. Since $|k(h)| \le \|A\|_2|h| + |r(h)| \le (\|A\|_2 + 1)|h|$ for all small h, we get $|s(k(h))| \le (\|A\|_2 + 1)\, \sigma\big(k(h)\big)\, |h|$, and $\sigma(k(h)) \to 0$ because $k(h) \to 0$.

Hence $g \circ f$ is differentiable at x with $D(g\circ f)_x = BA = Dg_{f(x)}\, Df_x$. The matrix statement follows because the matrix of a composition of linear maps is the product of their matrices. **Q.E.D.**

The proof is the one-variable proof with $|\cdot|$ in place of absolute values, and its one delicate point is the same too: $k(h)$ may vanish for $h \neq 0$, so one cannot simply divide by $|k(h)|$. The function σ is what makes that harmless.

Remark 10.a.2 (*The chain rule as a sum over paths*): Stated as $D(g \circ f) = Dg \cdot Df$ the rule is compact but rather abstract. It says nothing about **why** a sum appears when the rule is written out in coordinates. A useful device is to draw the variables as a graph and read the formula off it.

Write $x \in \mathbb{R}^n \xrightarrow{\ f\ } y \in \mathbb{R}^m \xrightarrow{\ g\ } z \in \mathbb{R}^k$ and join each variable to those it feeds into. Then x_i influences z_j only by first moving some intermediate y_ℓ, and the contribution of that route is the product of the two rates along it. Summing over the routes gives the chain rule:



$$\frac{\partial z_j}{\partial x_i} = \sum_{\ell=1}^{m} \underbrace{\frac{\partial z_j}{\partial y_\ell}}_{(\mathrm{J}\,g)_{j\ell}} \underbrace{\frac{\partial y_\ell}{\partial x_i}}_{(\mathrm{J}\,f)_{\ell i}} = \big(\mathrm{J}\,g(f(x))\ \mathrm{J}\,f(x)\big)_{ji}.$$

The right-hand equality is just the definition of matrix multiplication, and that is precisely the point: **matrix multiplication is a sum over paths**, and the chain rule looks like a product of matrices precisely because composing functions composes those graphs. Computing $\partial z_j/\partial x_i$ this way is nothing more than computing single components of the chain rule.

The reading is worth having because it makes the coordinate form reconstructible instead of memorised: count the intermediate variables, one term each, each term a product of two factors, and the index that is summed over is the one belonging to the middle layer. It is also the form in which the rule is actually used in physics. Compare Section 10.d, which is this picture with a single intermediate layer $\gamma_1, \dots, \gamma_n$ and a single source variable t.

Theorem 10.a.3 (*Euler's homogeneous function theorem*): Let $f \in C^1(\mathbb{R}^n \setminus \{0\}, \mathbb{R})$ be "*positively homogeneous of degree k*" for some $k \in \mathbb{R}$, meaning that

$$f(tx) = t^k f(x) \qquad \text{for all } x \in \mathbb{R}^n \setminus \{0\} \text{ and all } t > 0.$$

Then $\langle x, \nabla f(x)\rangle = k\, f(x)$ for all $x \in \mathbb{R}^n \setminus \{0\}$.

Proof:

Fix $x \neq 0$ and read both sides of the homogeneity identity as functions of $t > 0$ alone. The left-hand side is f composed with the path $\gamma(t) := tx$, whose velocity is $\gamma'(t) = x$, so the chain rule along a path (Section 10.d) gives

$$\frac{d}{dt} f(tx) = \big\langle \nabla f(tx),\ x \big\rangle.$$

On the right-hand side $f(x)$ is a constant in t, and therefore $\frac{d}{dt}\big(t^k f(x)\big) = k t^{k-1} f(x)$. The two derivatives agree for every $t > 0$; evaluating at $t = 1$ leaves $\langle \nabla f(x), x\rangle = k f(x)$, which is the claim by symmetry of the inner product. Q.E.D.

AI-Example 10.a.4 (*Chain Rule for Vector Fields*): Let $f(u, v) := u^2 + v^2$ and let $g(r, \theta) := (r\cos\theta, r\sin\theta)^{\mathsf{T}}$. The composite function is $(f \circ g)(r, \theta) = r^2\cos^2\theta + r^2\sin^2\theta = r^2$. By the multi-variable Chain Rule, the Jacobian matrix of $f \circ g$ satisfies:

$$D(f \circ g)(r, \theta) = Df(g(r, \theta)) \cdot Dg(r, \theta) = \begin{pmatrix} 2r\cos\theta & 2r\sin\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = \begin{pmatrix} 2r & 0 \end{pmatrix},$$

which matches the direct calculation $\frac{\partial}{\partial r}(r^2) = 2r$ and $\frac{\partial}{\partial \theta}(r^2) = 0$.

Exercise 10.a.5 (*Jacobian of a composite map*): Compute $\mathrm{J}(g \circ f)$ for

$$f : \mathbb{R}^3 \to \mathbb{R}^2,\ x \mapsto (x_1 x_2,\ x_2 + x_3), \qquad g : \mathbb{R}^2 \to \mathbb{R}^2,\ x \mapsto (e^{x_1},\ x_1 x_2).$$

Solution to Exercise 10.a.5:

We have

$$\mathrm{J}\, f(x) = \begin{pmatrix} x_2 & x_1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \mathrm{J}\, g(x) = \begin{pmatrix} e^{x_1} & 0 \\ x_2 & x_1 \end{pmatrix}.$$

So:

$$\begin{aligned} \mathrm{J}(g \circ f) &= \begin{pmatrix} e^{x_1 x_2} & 0 \\ x_2 + x_3 & x_1 x_2 \end{pmatrix} \begin{pmatrix} x_2 & x_1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} x_2 e^{x_1 x_2} & x_1 e^{x_1 x_2} & 0 \\ x_2^2 + x_2 x_3 & 2x_1 x_2 + x_1 x_3 & x_1 x_2 \end{pmatrix} \end{aligned}$$

Q.E.D.

Exercise 10.a.6 (*Computing differentials with and without the chain rule*)**:** Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ and $g : \mathbb{R}^3 \to \mathbb{R}$ be defined by

$$f(x, y) := \begin{pmatrix} x + y \\ x - y \\ x^2 + y^2 - 1 \end{pmatrix}, \qquad g(x, y, z) := x^2 + y^2 + z^2.$$

Compute the differential $D(g \circ f)$ in two ways:

(a) By directly computing the composition $g \circ f$ and then differentiating.

(b) By computing Df and Dg and using the chain rule.

Solution to Exercise 10.a.6:

(a) Directly: First compute $g(f(x, y))$:

$$\begin{aligned} (g \circ f)(x, y) &= (x + y)^2 + (x - y)^2 + (x^2 + y^2 - 1)^2 \\ &= (x^2 + 2xy + y^2) + (x^2 - 2xy + y^2) + (x^4 + y^4 + 1 + 2x^2y^2 - 2x^2 - 2y^2) \\ &= 2x^2 + 2y^2 + x^4 + y^4 + 1 + 2x^2y^2 - 2x^2 - 2y^2 \\ &= x^4 + y^4 + 2x^2y^2 + 1 \\ &= (x^2 + y^2)^2 + 1. \end{aligned}$$

Now differentiate this scalar function with respect to x and y:

$$D(g \circ f)(x, y) = \begin{pmatrix} 2(x^2 + y^2) \cdot 2x & 2(x^2 + y^2) \cdot 2y \end{pmatrix} = \begin{pmatrix} 4x(x^2 + y^2) & 4y(x^2 + y^2) \end{pmatrix}.$$

(b) Via chain rule: Compute the differentials of f and g individually.

$$Df(x, y) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2x & 2y \end{pmatrix}, \qquad Dg(u, v, w) = \begin{pmatrix} 2u & 2v & 2w \end{pmatrix}.$$

Evaluate Dg at $f(x, y)$:

$$Dg(f(x, y)) = \begin{pmatrix} 2(x + y) & 2(x - y) & 2(x^2 + y^2 - 1) \end{pmatrix}.$$

Multiply the matrices:

$$\begin{aligned} D(g\circ f)(x,y) &= Dg(f(x,y))Df(x,y) \\ &= \begin{pmatrix} 2(x+y) & 2(x-y) & 2(x^2+y^2-1) \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2x & 2y \end{pmatrix} \\ &= \begin{pmatrix} 4x+4x(x^2+y^2-1) & 4y+4y(x^2+y^2-1) \end{pmatrix} \\ &= \begin{pmatrix} 4x(x^2+y^2) & 4y(x^2+y^2) \end{pmatrix}. \end{aligned}$$

Both methods yield the same result.

Q.E.D.

Exercise 10.a.7 (*True or False* (the higher-order chain rule)): Let $u \in C^2(\mathbb{R}^n)$ and $F = (F_1, \dots, F_n) \in C^2(\mathbb{R}^n, \mathbb{R}^n)$, with $n \geq 2$. Decide whether each of the following identities is true or false.

(a) The identity $\partial_i(u \circ F) = \sum_{j=1}^n \partial_j u \circ F \, \partial_i F_j$ holds for all $i \in \{1, \dots, n\}$.

(b) The identity $\partial_{ii}(u\circ F) = \sum_{j=1}^n \partial_{jj} u\circ F \, (\partial_i F_j)^2 + \sum_{j=1}^n \partial_j u\circ F \, \partial_{ii} F_j$ holds for all $i \in \{1, \dots, n\}$.

(c) The identity $\partial_i \partial_j (u \circ F) = \partial_j \partial_i (u \circ F)$ holds for all $i, j \in \{1, \dots, n\}$.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the third block of true/false questions.*

Solution: *See solution on Page 149*

Example 10.a.8 (*Which identities are forced, and by what*): Let $u \in C^2(\mathbb{R}^n)$ and let $F = (F_1, \dots, F_n) \in C^1(\mathbb{R}^n, \mathbb{R}^n)$, with $n \geq 2$. Exactly one of the following three identities is forced by those hypotheses, and the two that fail do so for quite different reasons.

(a) $\partial_i F_j = \partial_j F_i$ **for all** i, j**: not forced.** This asks the Jacobi matrix $\mathrm{J}F$ to be symmetric at every point, which is the integrability condition of Lemma 26.b.12: it holds precisely when F is a candidate for a gradient, and nothing in the hypotheses makes it one. The rotation field $F(x) := (-x_2, x_1, 0, \dots, 0)$ is as smooth as one likes and has $\partial_1 F_2 = 1$ while $\partial_2 F_1 = -1$.

(b) $\partial_j \partial_i u = \partial_i \partial_j u$ **for all** i, j**: forced.** This is Schwarz's theorem (Theorem 11.b.13), and $u \in C^2$ is exactly its hypothesis. Note how differently the two statements read once they are set side by side: **(a)** constrains an arbitrary vector field, **(b)** constrains the gradient of a function. Only the second is automatic.

(c) $u \circ F \in C^2$**: not forced.** The chain rule gives $\partial_i(u \circ F) = \sum_j (\partial_j u \circ F)\, \partial_i F_j$, and the factors $\partial_i F_j$ are merely continuous, so this says only that $u \circ F \in C^1$. To differentiate once more one would have to differentiate $\partial_i F_j$, which the hypotheses do not permit. Take $n = 2$, $u(y_1, y_2) := y_1$ and $F(x_1, x_2) := (x_1|x_1|, x_2)$. Then $F \in C^1$, since $\partial_1 F_1 = 2|x_1|$ is continuous, and $(u\circ F)(x) = x_1|x_1|$ has the same non-differentiable first derivative at $x_1 = 0$. Consequently $u \circ F \notin C^2$.

The general rule that **(c)** illustrates is that a composition inherits the **weaker** of the two regularities. Raise F to C^2 and $u \circ F$ becomes C^2, which is what Exercise 10.a.7**(c)** is then free to assume.

Info: *This example is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the third block of true/false questions.*

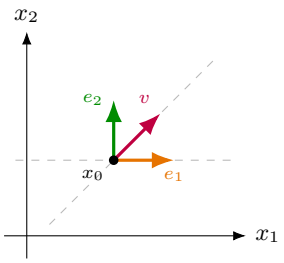
Directional derivatives (Section 10.b)

Sometimes the "*differential*" (German: „*das Differential*“) is not easily expressed as a Jacobian. Then it can be convenient to use the formula for "*directional derivatives*" (German: „*Rich-*

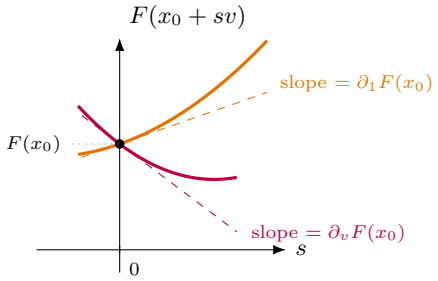
tungsableitung“):

$$DF_{x_0}(v) = \partial_v F(x_0) = \frac{d}{ds}\Big|_{s=0} F(x_0 + sv) \qquad (*)$$

Formula $(*)$ describes a two-step recipe, and the picture below is worth carrying around. **Step one:** pick a direction v and walk away from x_0 along the straight ray $s \mapsto x_0 + sv$, ignoring the rest of the domain. **Step two:** record the values of F along that ray. What you get is an ordinary function of **one** variable s, and $\partial_v F(x_0)$ is simply its derivative at $s = 0$, an **Analysis I** object. The partial derivative $\partial_i F$ is the special case $v = e_i$: walk parallel to the i-th axis.



pick a direction and walk along that ray



each ray gives a one-variable function; its slope at 0 is $\partial_v F(x_0)$

The right-hand panel is also the cleanest way to see what differentiability adds. Every direction produces **some** slope; nothing so far forces those slopes to be related to one another. Being differentiable is precisely the demand that all of them are read off a **single** linear map via $\partial_v F(x_0) = DF_{x_0}(v)$, so that knowing the n partial derivatives already determines every other direction. AI-Example 10.b.10 below is the cautionary case where all the slopes exist but no such linear map governs them.

Important remark 10.b.9: The directional derivative $\partial_v F(x_0)$ can exist (depending on v and x_0) even if F is not differentiable at x_0, i.e. DF_{x_0} does not exist. However, if DF_{x_0} exists, the formula $(*)$ holds.

AI-Example 10.b.10 (*Every direction, and still not continuous*): AI-Example 9.b.8 showed that the two **partial** derivatives are too little. One might hope that asking for **all** directions is enough. It is not. Let

$$f(x,y) := \begin{cases} \dfrac{x^2 y}{x^4 + y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0). \end{cases}$$

Fix a direction $v = (a,b) \neq 0$ and compute along the ray $t \mapsto tv$:

$$\frac{f(ta,tb)}{t} = \frac{1}{t} \cdot \frac{t^2 a^2 \cdot tb}{t^4 a^4 + t^2 b^2} = \frac{a^2 b}{t^2 a^4 + b^2} \xrightarrow{t \to 0} \begin{cases} \dfrac{a^2}{b} & b \neq 0, \\ 0 & b = 0. \end{cases}$$

So $\partial_v f(0,0)$ exists for **every** direction v. Yet f is not even continuous at the origin: approaching along the **parabola** $y = x^2$ instead of along a straight line,

$$f(x, x^2) = \frac{x^2 \cdot x^2}{x^4 + x^4} = \frac{1}{2} \quad \text{for all } x \neq 0.$$

By Proposition 9.a.5, f is therefore not differentiable at the origin.

Two lessons worth keeping.

- **Straight lines cannot see everything.** Every directional derivative is computed along a line, and this f is well behaved along every line while misbehaving along a parabola. Approaching a point in $\mathbb{R}^n$ is genuinely harder than in $\mathbb{R}$, where there are only two directions.
- **Watch whether $v \mapsto \partial_v f(x_0)$ is linear.** If f were differentiable we would need $\partial_v f(0,0) = DF_0(v)$, a **linear** function of v; but a^2/b is not linear in (a,b). That failure alone rules out differentiability, without any mention of continuity, and it is precisely the test that settles Exercise 10.b.11**(b)** below.

Exercise 10.b.11 (*Derivatives* (important)):

(a) Consider $f(x,y) := xy^2 e^{-x^2-y^2}$ for $(x,y) \in \mathbb{R}^2$. Find all the critical points of f, that is all the points where the gradient of f vanishes. (The point will be given if and only if all the numerical values you find are correct… so check your computations twice.)

(b) We want to find a function $g \in C^1(\mathbb{R}^2)$ with the following directional derivative:

$$\partial_v g(x,y) = 2\cos(x^2y)xv_2 + \cos(x^2y)v_1^2 y \quad \text{for all } (x,y) \text{ and } (v_1,v_2) \in \mathbb{R}^2.$$

Give an explicit example of such a function g or prove that a function with these properties cannot exist.

Info: *This exercise is Problem 5.1 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 147*

Remark 10.b.12: The linearity of the differential is the whole of Exercise 10.b.11**(b)**: a directional derivative must be **linear** in v, and v_1^2 is not.

Exercise 10.b.13 (*Computation of derivatives* (semi-important)): For each of the following functions compute the directional derivative in the general direction v at a general point x, that is $\partial_v f(x)$, where applicable.

(a) x_1/x_2, e^{-x_1/x_2} for $x \in \mathbb{R}^2 \setminus \{x_2 = 0\}$. (Also: can they be extended continuously or differentiably across $\{x_2 = 0\}$?)

(b) $e^{x_1x_2}\sin(x_1+x_2^2)$, $\dfrac{x_1^2}{x_1^2+x_2^2}$, $\dfrac{x_1^2x_2}{x_1^2+x_2^2}$, $\dfrac{x_1^2x_2}{x_1^2+x_2^4}$ for $x \in \mathbb{R}^2 \setminus \{0\}$. (Also: can they be extended continuously or differentiably across the origin?)

(c) $|x|$, $|x|^\alpha$, $x/|x|$, $g(|x|)$, $x \cdot e$, $g(x \cdot e)$, for $\alpha \in \mathbb{R}$, $e \in \mathbb{R}^n$ and $g \in C^1(\mathbb{R})$.

(d) ax, x^T, $\mathrm{Tr}(x)$, x^2, x^3, $\mathrm{Tr}(x^2)$ for $x, a \in \mathbb{R}^{n\times n}$.

(e) **(*)** x^{-1}, x^{-2} for $x \in \mathbb{R}^{n\times n}$ with $\det x \neq 0$.

Official hints: *To prove that a function **is** differentiable at 0 you have two tools, namely the definition of differentiability and the sufficient condition of Theorem 10.16. To prove that a function **is not** differentiable at 0 you again have two: show that the necessary condition of Proposition 10.14 does not hold, or show that for some C^1 curve $\gamma : (-\delta,\delta) \to \mathbb{R}^n$ the function $(f \circ \gamma)(t)$ is **not** differentiable at $t = 0$ (simpler, as you study a function of one variable). **5.4.3:** these functions are C^∞, so compute them on $x + tv$ and find an expansion in powers of t; by Corollary 10.45 it must be the Taylor polynomial, from which the derivatives can be read off. For example $(x+tv)^2 = x^2 + (xv+vx)t + O(t^2)$.*

Info: *This exercise is Problem 5.4 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 147*

What does imply differentiability (Section 10.c)

AI-Note 10.1: The course gives the definition of differentiability and the warnings that go with it, but never states a **sufficient** condition, so up to this point we have two counterexamples and no

positive test at all. The theorem below supplies one. Without it the practical question *"how do I actually check differentiability?"* has no answer.

Theorem 10.c.14 (*Continuous partial derivatives suffice*): Let $U \subseteq \mathbb{R}^n$ be open and $F : U \to \mathbb{R}^m$. If all partial derivatives $\partial_i F$ exist on U **and are continuous** at every $x \in U$, then F is differentiable at every $x \in U$, and

$$DF_x = \big(\partial_1 F(x),\ \ldots,\ \partial_n F(x)\big).$$

Such an F is called "*continuously differentiable*", written $F \in C^1(U)$.

Proof:

By Lemma 9.a.3 it suffices to treat each component on its own, so assume $m = 1$ and write $f := F$. Fix $x \in U$ and $r > 0$ with $B_r(x) \subseteq U$, and let $|h| < r$.

The idea is to walk from x to $x + h$ not along the segment but along a **staircase**, changing one coordinate at a time, because along each step only one partial derivative is involved. Put

$$p_0 := x, \qquad p_k := x + h_1 e_1 + \cdots + h_k e_k \quad (k = 1, \ldots, n),$$

so $p_n = x + h$, and every point of every step lies within distance $|h|$ of x, hence in U. On the k-th step only the k-th coordinate moves, so the one-variable function $t \mapsto f(p_{k-1} + t e_k)$ is differentiable with derivative $\partial_k f(p_{k-1} + t e_k)$. The mean value theorem of Analysis I gives a point ξ_k on the k-th step with

$$f(p_k) - f(p_{k-1}) = h_k\, \partial_k f(\xi_k).$$

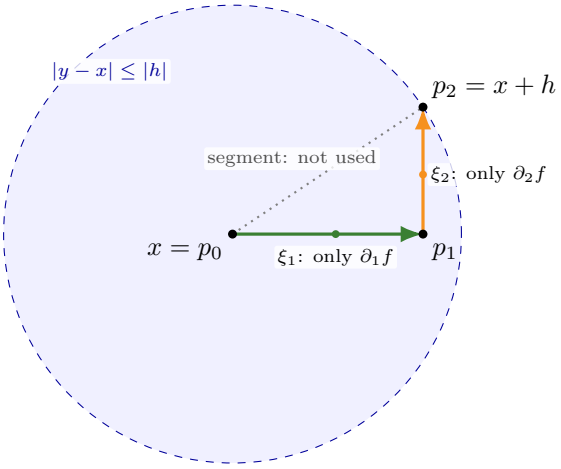
Summing the telescoping differences,

$$f(x+h) - f(x) - \sum_{k=1}^{n} h_k\, \partial_k f(x) = \sum_{k=1}^{n} h_k \big(\partial_k f(\xi_k) - \partial_k f(x)\big),$$

and therefore, since $|h_k| \le |h|$,

$$\frac{\big|f(x+h) - f(x) - \sum_k h_k\, \partial_k f(x)\big|}{|h|} \ \le\ \sum_{k=1}^{n} \big|\partial_k f(\xi_k) - \partial_k f(x)\big|.$$

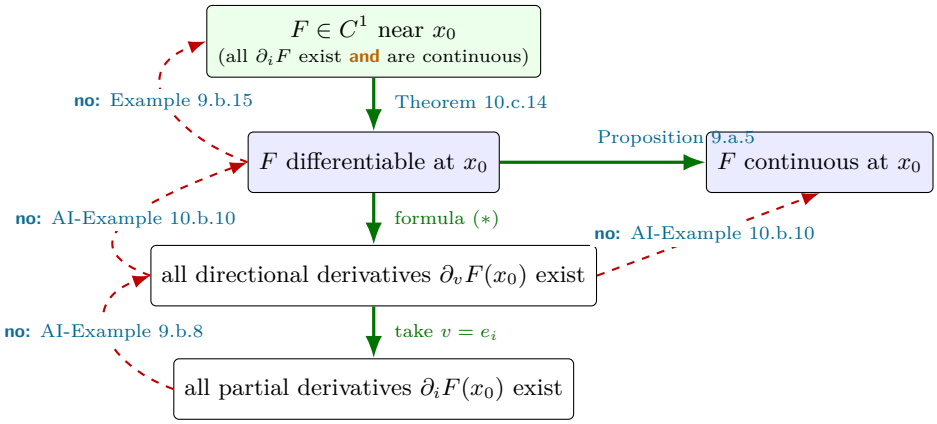
Each ξ_k satisfies $|\xi_k - x| \le |h|$, so $\xi_k \to x$ as $h \to 0$, and continuity of $\partial_k f$ at x sends every summand to 0. This is exactly the definition of differentiability at x with $Df_x(h) = \sum_k h_k\, \partial_k f(x)$. **Q.E.D.**



This is the theorem that makes the subject workable: in practice one almost never checks the definition directly. Compute the partials, observe that they are built from continuous pieces, and

differentiability follows. That is exactly why the course can write $f \in C^1$, C^2 and C^k so freely from here onward.

Together with the two counterexamples, the whole landscape is a single chain, each link one-directional:



Every downward arrow holds, and **not one of them reverses**. Each failure is witnessed by an example already in this chapter, so the diagram doubles as a map of where those counterexamples belong:

- **Differentiable $\not\Rightarrow$ C^1:** Example 9.b.15, whose partial derivatives oscillate without limit at the origin even though the differential exists there.
- **All directional derivatives $\not\Rightarrow$ differentiable:** AI-Example 10.b.10, where every ray gives a slope but $v \mapsto \partial_v f(0,0)$ is not linear. The same example defeats the branch to the right as well: it is not even **continuous** at the origin, so having every directional derivative buys strictly less than continuity does.
- **Partial derivatives $\not\Rightarrow$ all directional derivatives:** AI-Example 9.b.8 again. For $f = \frac{xy}{x^2+y^2}$ and a direction $v = (a, b)$ with $ab \neq 0$,

$$\frac{f(ta, tb)}{t} = \frac{1}{t} \cdot \frac{t^2 ab}{t^2(a^2+b^2)} = \frac{ab}{t\,(a^2+b^2)} \xrightarrow{t \to 0} \pm\infty,$$

  so $\partial_v f(0,0)$ exists only along the two axes, which are precisely the two directions the partial derivatives happen to look at.

The three counterexamples are not variations on one theme; they fail at three different heights, which is precisely why the chain has three separate links.

Remark 10.c.15 (*A recipe for checking differentiability at a point*): The above distils into three steps, worth following in this order.

(a) Compute the partial derivatives $\partial_i F(x_0)$. If one of them fails to exist, F is not differentiable and you are done. If they all exist **and are continuous near x_0**, F **is** differentiable by Theorem 10.c.14, and you are done again. Most exercises end here.

(b) Otherwise, guess the candidate. If F is differentiable at all, the differential can only be $L := \big(\partial_1 F(x_0), \ldots, \partial_n F(x_0)\big)$, since the partials are forced (Section 9.b).

(c) Verify the candidate against the definition:

$$\lim_{h \to 0} \frac{\big|F(x_0 + h) - F(x_0) - L(h)\big|}{|h|} \overset{?}{=} 0.$$

In $\mathbb{R}^2$, substituting $h = (r\cos\theta,\ r\sin\theta)$ and checking that the result tends to 0 **uniformly in θ** is usually the quickest route. This is the same polar trick used in AI-Example 10.b.10 and, later, in Question C.a.7.

Step 1 is the one students skip; it settles the overwhelming majority of cases without any ε–δ work at all.

Example 10.c.16 (*Checking differentiability at the origin*): Let $f(x,y) := \frac{x^2 y}{x^2+y^2}$ for $(x,y) \neq (0,0)$ and $f(0,0) := 0$. We check if f is differentiable at the origin.

(a) Partial derivatives:

$$\partial_1 f(0,0) = \lim_{t\to 0} \frac{f(t,0)-f(0,0)}{t} = \lim_{t\to 0} \frac{0}{t} = 0, \qquad \partial_2 f(0,0) = \lim_{t\to 0} \frac{f(0,t)-f(0,0)}{t} = 0.$$

They exist, but to use Theorem 10.c.14 we would have to compute them everywhere and show they are continuous at $(0,0)$, which is tedious and (as we will see) false. We proceed to step 2.

(b) Guess the candidate: The only possible differential is $L(h_1,h_2) = \partial_1 f(0,0)h_1 + \partial_2 f(0,0)h_2 = 0$.

(c) Verify against the definition: We check whether the following limit is zero:

$$\lim_{h\to 0} \frac{|f(h_1,h_2)-f(0,0)-L(h)|}{|h|} = \lim_{h\to 0} \frac{\frac{h_1^2 h_2}{h_1^2+h_2^2}}{\sqrt{h_1^2+h_2^2}} = \lim_{h\to 0} \frac{h_1^2 h_2}{(h_1^2+h_2^2)^{3/2}}.$$

Passing to polar coordinates $h = (r\cos\theta, r\sin\theta)$, the expression becomes

$$\frac{r^3 \cos^2\theta \sin\theta}{r^3} = \cos^2\theta \sin\theta.$$

This depends on the angle of approach θ (for instance, it is 0 along the axes but $\frac{1}{2\sqrt{2}} \neq 0$ along $h_1 = h_2$), so the limit as $r \to 0$ is not uniformly 0. Hence f is **not differentiable** at $(0,0)$.

Example 10.c.17 (*A function that is differentiable at the origin*): Let $f(x,y) := \frac{x^2 y^2}{\sqrt{x^2+y^2}}$ for $(x,y) \neq (0,0)$ and $f(0,0) := 0$. We check differentiability at the origin following the recipe.

(a) Partial derivatives: For $(x,y) \neq (0,0)$, the quotient and chain rules give

$$\partial_1 f(x,y) = xy^2 \frac{x^2+2y^2}{(x^2+y^2)^{3/2}}, \qquad \partial_2 f(x,y) = x^2 y \frac{2x^2+y^2}{(x^2+y^2)^{3/2}}.$$

At the origin, computing directly from the definition yields $\partial_1 f(0,0) = 0$ and $\partial_2 f(0,0) = 0$. To see if they are continuous at $(0,0)$, we substitute $x = r\cos\theta$ and $y = r\sin\theta$:

$$\partial_1 f(r\cos\theta, r\sin\theta) = \frac{r\cos\theta\, r^2 \sin^2\theta \cdot r^2(1+\sin^2\theta)}{r^3} = r^2 \cos\theta \sin^2\theta (1+\sin^2\theta).$$

As $r \to 0$, this tends to 0 uniformly in θ, so $\partial_1 f$ is continuous at $(0,0)$. The same holds for $\partial_2 f$. By Theorem 10.c.14, f is differentiable at the origin.

(b) Guess the candidate: We already know f is differentiable and $L = (0,0)$. But for practice, let's verify it with the definition directly.

(c) Verify against the definition: We check the limit

$$\lim_{h\to 0} \frac{|f(h_1,h_2)-f(0,0)-L(h)|}{|h|} = \lim_{h\to 0} \frac{\frac{h_1^2 h_2^2}{\sqrt{h_1^2+h_2^2}}}{\sqrt{h_1^2+h_2^2}} = \lim_{h\to 0} \frac{h_1^2 h_2^2}{h_1^2+h_2^2}.$$

In polar coordinates, this is $\frac{r^4\cos^2\theta\sin^2\theta}{r^2} = r^2\cos^2\theta\sin^2\theta$. As $r \to 0$, the limit is clearly 0 uniformly in θ, confirming f is differentiable with $Df_0 = 0$.

Exercise 10.c.18 (*True or False* (how much regularity does this function have?)): Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x,y) := \begin{cases} \dfrac{xy^2}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Decide whether each of the following is true or false, with justification.

(a) $f \in C^\infty\big(\mathbb{R}^2 \setminus \{(0,0)\}\big)$.
(b) $\partial_x f(0,0)$ exists.
(c) $f \in C^0(\mathbb{R}^2)$.
(d) $f \in C^1(\mathbb{R}^2)$.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the fourth block of true/false questions.*

Solution: *See solution on Page 149*

The chain rule along a path (Section 10.d)

An important special case of the chain rule is that of a composition of a scalar field with a path. Suppose $f : \mathbb{R}^n \supseteq U \to \mathbb{R}$ and $\gamma : [0,1] \to U$ are a C^1 "*scalar field*" (German: „*Skalarfeld*“) and path, respectively. Then

$$\begin{aligned} D(f \circ \gamma)(t) &= Df(\gamma(t)) \cdot D\gamma(t) \\ &= \langle \nabla f(\gamma(t)),\ \gamma'(t) \rangle \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial \gamma_i} \frac{\partial \gamma_i}{\partial t} \end{aligned}$$

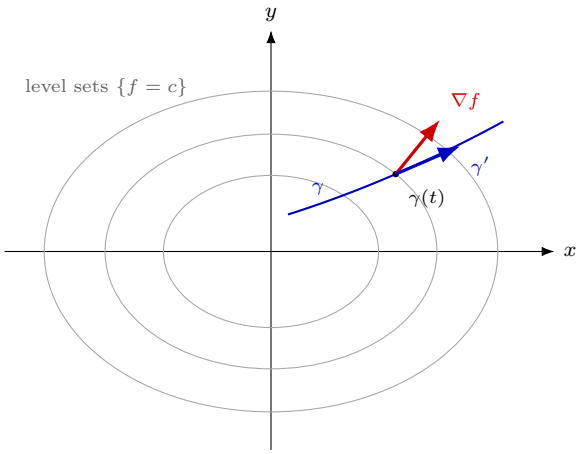
where $\gamma = (\gamma_1, \dots, \gamma_n)$ and $\dfrac{\partial f}{\partial \gamma_i} := \partial_i f(\gamma(t))$.

This operation is so common in physics that it has its own notation: for f as above, we look at each coordinate as a function of **time** and differentiate:

$$\frac{d}{dt} f\big(x_1(t), \dots, x_n(t)\big) = Df\big(x(t)\big)\, x'(t) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{dx_i}{dt}.$$

The middle line, $D(f \circ \gamma)(t) = \langle \nabla f(\gamma(t)), \gamma'(t) \rangle$, is worth dwelling on, because it is the formula that gives the gradient its meaning. It says the rate at which f changes along a path depends on the velocity only through its inner product with ∇f. Consequently, by Cauchy–Schwarz (Theorem 2.c.32), the change is fastest when γ' points along ∇f, and zero when γ' is perpendicular to it. Two standard facts fall straight out:

- ∇f **points in the direction of steepest ascent**, with $|\nabla f|$ the rate of ascent in that direction.
- ∇f **is perpendicular to the level sets of** f**.** If γ stays inside a level set $\{f = c\}$, then $f \circ \gamma$ is constant, so $\langle \nabla f(\gamma(t)), \gamma'(t) \rangle = 0$ for every t, and γ' runs through the directions tangent to the level set.



$\frac{d}{dt} f(\gamma(t)) = \langle \nabla f, \gamma' \rangle$: only the component of γ' along ∇f contributes, so f is constant along a level set and grows fastest along ∇f

AI-Exercise 10.d.19 (*Directional Derivative along a Unit Vector*): Let $f(x,y) := x^2 y$ defined on $\mathbb{R}^2$. Compute the directional derivative $D_v f(1,2)$ of f at the point $(1,2)$ along the unit vector $v := \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$.

Solution: *See solution on Page 147*

Exercise 10.d.20 (*A directional derivative vanishes* (important)): Let $u \in C^1(\mathbb{R}^n)$ and $\nu \in \mathbb{R}^n$. Show that

(a) If $\partial_1 u \equiv 0$ then "u does not depend on x_1"; more rigorously: there exists a unique function $v \in C^1(\mathbb{R}^{n-1})$ such that

$$u(x_1, \dots, x_n) = v(x_2, \dots, x_n) \quad \text{for all } x \in \mathbb{R}^n. \tag{2}$$

(b) If $\partial_\nu u \equiv 0$ and $\nu \cdot e_1 \neq 0$ then "u is a function of $n-1$ variables"; more rigorously: there exists a unique function $w \in C^1(\mathbb{R}^{n-1})$ such that

$$u(x_1, \dots, x_n) = w\left(x_2 - \frac{x_1 \nu_2}{\nu_1}, \dots, x_n - \frac{x_1 \nu_n}{\nu_1}\right) \quad \text{for all } x \in \mathbb{R}^n.$$

(c) **(*)** What can we conclude if we assume only that $\partial_1 u = 0$ in an open connected subset $U \subset \mathbb{R}^n$?

Official hints: ***4.6.2:*** *apply the mean value theorem to* $u(x + t\nu)$. ***4.6.3:*** *consider the domain* $U := \{(x,y) : x^2 < y < x^2 + 1\}$ *and the function*

$$u(x,y) := \begin{cases} \max\{0, y-2\}^2 & \text{if } x \geq 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

Info: *This exercise is Problem 4.6 of Problem Sheet 4, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 146*

Exercise 10.d.21 (*Chain rule for a curve, or: a car on Earth* (optional — "interesting for physicists")): Let $R > 0$ and $\omega \in \mathbb{R}$ be constants. Let $\sigma : I \subset \mathbb{R} \to \mathbb{R}^2$, $\sigma(t) = (a(t), b(t))$, which we interpret as a curve in a "map" representing the longitude and latitude of a car at time t.

Define the "rotating" spherical coordinates $F : \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}^3$,

$$F((x,y),t) = \big(R\cos y \cos(\omega t + x),\ R\cos y \sin(\omega t + x),\ R \sin y\big),$$

representing what a point on the map looks like on Earth from a satellite (thus we consider the Earth as a rotating sphere with speed given by ω, and we forget about translation around the Sun) at time t.

Consider $r : \mathbb{R} \to \mathbb{R}^3$, $r(t) = F(\sigma(t), t)$, which is the position of the car at time t.

(a) Compute the acceleration $r''(t)$ using the chain rule.

(b) Assume that the acceleration is always proportional to the radial vector $r(t)$, i.e. $r''(t) = \lambda(t)\, r(t)$ for some scalar function $\lambda(t)$. Show that a and b satisfy

$$b'' + \sin b \cos b\, (\omega + a')^2 = 0, \qquad \cos b\, a'' - 2 \sin b\, (\omega + a') b' = 0.$$

Equivalently, away from the poles ($\cos b \neq 0$),

$$a'' - 2 \tan b\, (\omega + a')\, b' = 0.$$

Info: *This exercise is Problem 5.7 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 148*

Solutions (Section 10.e)

Not every solution in this chapter is collected here. Exercise 10.a.5 (Page 136) and Exercise 10.a.6 (Page 137) are solved where they appear.

Solution to Exercise 10.d.20:

(a) Define $v(x_2, \dots, x_n) := u(0, x_2, \dots, x_n)$, which is C^1 as the restriction of u to the hyperplane $x_1 = 0$. For fixed $(x_2, \dots, x_n)$, the function $t \mapsto u(t, x_2, \dots, x_n)$ has derivative $\partial_1 u \equiv 0$ everywhere, hence is constant in t; evaluating at $t = 0$ gives $u(x_1, \dots, x_n) = u(0, x_2, \dots, x_n) = v(x_2, \dots, x_n)$ for all x_1. Uniqueness: if some $\tilde{v}$ also satisfies $u(x) = \tilde{v}(x_2, \dots, x_n)$, then setting $x_1 = 0$ gives $\tilde{v} = v$.

(b) Since $\nu \cdot e_1 = \nu_1 \neq 0$, consider the invertible linear change of coordinates $y_i := x_i - \dfrac{x_1 \nu_i}{\nu_1}$ for $i = 2, \dots, n$, together with x_1 itself. Moving x_1 by t while holding $y_2, \dots, y_n$ fixed forces x_i to move by $t\nu_i/\nu_1$, i.e. moves x in the direction $\frac{1}{\nu_1}\nu$. By the hint, applying the mean value theorem to $u(x + t\nu)$ shows $t \mapsto u(x + t\nu)$ is constant since $\partial_\nu u \equiv 0$; consequently u, expressed in the coordinates $(x_1, y_2, \dots, y_n)$, has vanishing x_1-derivative and is therefore (by part **(a)**) independent of x_1 in these coordinates. Hence, there is a (unique, by the same argument as in part **(a)**) function $w \in C^1(\mathbb{R}^{n-1})$ with

$$u(x_1, \dots, x_n) = w\left(x_2 - \frac{x_1 \nu_2}{\nu_1}, \dots, x_n - \frac{x_1 \nu_n}{\nu_1}\right).$$

(c) In general, only **locally**: on any open subset where every fiber $\{x_1 : (x_1, x_2, \dots, x_n) \in U\}$ (for fixed $(x_2, \dots, x_n)$) is connected, the same argument as in part **(a)** shows u is locally of the form $v(x_2, \dots, x_n)$. But this need not hold **globally** on all of U, even though U itself is connected: with $U := \{(x, y) : x^2 < y < x^2 + 1\}$ and $u(x, y) := \max\{0, y - 2\}^2$ for $x \geq 0$, $u := 0$ for $x \leq 0$, one checks $\partial_1 u \equiv 0$ throughout U (near $x = 0$ both branches agree, since $y < 2$ there), yet u is not globally independent of x: for $y > 2$, the fiber of U over y splits into two disconnected intervals (one with $x > 0$, one with $x < 0$), and u takes different values on each. So, connectedness of U alone does not suffice; what is needed is connectedness of each vertical fiber of U.

Q.E.D.

Proof of AI-Exercise 10.d.19:

The gradient of $f(x,y)$ is $\nabla f(x,y) = (2xy, x^2)^\mathsf{T}$. Evaluating at $(1,2)$ gives $\nabla f(1,2) = (4,1)^\mathsf{T}$. Since f is continuously differentiable (C^∞), the directional derivative along the unit vector v is given by the inner product:

$$D_v f(1,2) = \nabla f(1,2) \cdot v = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{4+1}{\sqrt{2}} = \frac{5}{\sqrt{2}}.$$

Q.E.D.

Solution to Exercise 10.b.11:

(a) Since $e^{-x^2-y^2} > 0$ everywhere, the gradient of $f(x,y) = xy^2 e^{-x^2-y^2}$ vanishes exactly where its two factors below vanish:

$$\partial_x f = y^2(1-2x^2)e^{-x^2-y^2}, \qquad \partial_y f = 2xy(1-y^2)e^{-x^2-y^2}.$$

If $y = 0$: both partials vanish for every x (indeed $f(x,0) \equiv 0$), so the entire line $\{(x,0) : x \in \mathbb{R}\}$ consists of critical points. **If $y \neq 0$:** $\partial_x f = 0$ forces $x = \pm\frac{1}{\sqrt{2}}$, and since then $x \neq 0$, $\partial_y f = 0$ forces $y = \pm 1$. This gives the four additional critical points $\left(\pm\frac{1}{\sqrt{2}}, \pm 1\right)$.

(b) A directional derivative $\partial_v g(x,y) = \nabla g(x,y) \cdot v$ is always **linear** in v. The prescribed expression $2\cos(x^2y)xv_2 + \cos(x^2y)v_1^2 y$ contains a v_1^2 term with coefficient $y\cos(x^2y)$, which is not identically zero (e.g. at $(x,y) = (0,1)$ it equals 1). Hence the expression is not linear in v at every point, so **no such $g \in C^1(\mathbb{R}^2)$ can exist**.

Q.E.D.

Solution to Exercise 10.b.13:

Throughout, v is the direction and x the base point. Where f is differentiable the answer is just $\partial_v f(x) = \langle \nabla f(x), v \rangle$; the interest is in the points where it is not, which is what the second half of each part is asking about.

(a) On $\mathbb{R}^2 \setminus \{x_2 = 0\}$ both functions are smooth, so

$$\partial_v\left(\frac{x_1}{x_2}\right) = \frac{v_1}{x_2} - \frac{x_1 v_2}{x_2^2}, \qquad \partial_v\left(e^{-x_1/x_2}\right) = -e^{-x_1/x_2}\left(\frac{v_1}{x_2} - \frac{x_1 v_2}{x_2^2}\right).$$

Neither extends continuously across $\{x_2 = 0\}$. For x_1/x_2, fix $x_1 = 1$ and let $x_2 \to 0^\pm$: the values run off to $\pm\infty$. For e^{-x_1/x_2} the failure is more interesting, since the function is **bounded** on one side: with $x_1 = 1$ fixed, $x_2 \to 0^+$ sends $-1/x_2 \to -\infty$ and the value to 0, while $x_2 \to 0^-$ sends it to $+\infty$. A limit existing from one side is not enough.

(b) The first function is smooth on all of $\mathbb{R}^2$, with

$$\begin{aligned}\partial_v\left(e^{x_1x_2}\sin(x_1+x_2^2)\right) = {} & e^{x_1x_2}\left(x_2\sin(x_1+x_2^2) + \cos(x_1+x_2^2)\right)v_1 \\ & + e^{x_1x_2}\left(x_1\sin(x_1+x_2^2) + 2x_2\cos(x_1+x_2^2)\right)v_2,\end{aligned}$$

and nothing needs extending. The other three are the point of the exercise, and they separate three different behaviours at the origin.

- $\dfrac{x_1^2}{x_1^2+x_2^2}$ is homogeneous of degree 0, with

$$\partial_v f(x) = \frac{2x_1x_2\,(x_2v_1 - x_1v_2)}{(x_1^2+x_2^2)^2}.$$

It has **no continuous extension**: it equals 1 on $\{x_2 = 0\}$ and 0 on $\{x_1 = 0\}$.

- $\dfrac{x_1^2 x_2}{x_1^2+x_2^2}$ **does** extend continuously by $f(0):=0$, since $x_1^2 \le x_1^2+x_2^2$ gives $|f| \le |x_2|$. It is not differentiable there:
$$\partial_v f(0) = \lim_{t\to 0} \frac{f(tv)}{t} = \lim_{t\to 0} \frac{t^3 v_1^2 v_2}{t^3(v_1^2+v_2^2)} = \frac{v_1^2 v_2}{v_1^2+v_2^2},$$
which exists in every direction and is **not linear** in v.
- $\dfrac{x_1^2 x_2}{x_1^2+x_2^4}$ also satisfies $|f| \le |x_2|$, so it too extends continuously by 0. Here
$$\partial_v f(0) = \lim_{t\to 0} \frac{t^3 v_1^2 v_2}{t^3\,(v_1^2+t^2 v_2^4)} = \begin{cases} v_2 & v_1 \neq 0, \\ 0 & v_1 = 0, \end{cases}$$
which is not even continuous in v, let alone linear. Again continuous, not differentiable.

All three are instances of AI-Example 10.b.10: every directional derivative can exist without the map $v \mapsto \partial_v f(0)$ being linear, and linearity is exactly what differentiability demands.

(c) Here $x \in \mathbb{R}^n \setminus \{0\}$ for the first four, $e \in \mathbb{R}^n$ and $g \in C^1(\mathbb{R})$:
$$\partial_v |x| = \frac{\langle x, v\rangle}{|x|}, \qquad \partial_v |x|^\alpha = \alpha\,|x|^{\alpha-2}\langle x, v\rangle, \qquad \partial_v\, g(|x|) = g'(|x|)\,\frac{\langle x, v\rangle}{|x|},$$
$$\partial_v\left(\frac{x}{|x|}\right) = \frac{1}{|x|}\left(v - \frac{\langle x, v\rangle}{|x|^2}\,x\right), \qquad \partial_v\,(x\cdot e) = \langle e, v\rangle, \qquad \partial_v\, g(x\cdot e) = g'(\langle x, e\rangle)\,\langle e, v\rangle.$$
The fourth is worth reading geometrically: $x/|x|$ records only a direction, so moving x along itself changes nothing, and indeed the bracket is exactly v with its component along x removed.

(d) For matrix arguments the cleanest route is to expand $f(x+tv)$ in powers of t and read off the coefficient of t. The first three maps are linear, hence their own derivatives:
$$\partial_v(ax) = av, \qquad \partial_v(x^{\mathsf T}) = v^{\mathsf T}, \qquad \partial_v \operatorname{Tr}(x) = \operatorname{Tr}(v).$$
For the rest, $(x+tv)^2 = x^2 + t(xv+vx) + t^2 v^2$ and $(x+tv)^3 = x^3 + t(x^2 v + xvx + vx^2) + O(t^2)$, so
$$\partial_v(x^2) = xv + vx, \qquad \partial_v(x^3) = x^2 v + xvx + vx^2, \qquad \partial_v \operatorname{Tr}(x^2) = \operatorname{Tr}(xv+vx) = 2\operatorname{Tr}(xv),$$
the last using cyclicity of the trace. Note that $\partial_v(x^2) \neq 2xv$: matrices do not commute, and the familiar one-variable answer is precisely what fails.

(e) **(*)** Differentiate the identity $(x+tv)(x+tv)^{-1} = \mathrm{id}$ at $t=0$. The product rule gives $v\,x^{-1} + x\,\partial_v(x^{-1}) = 0$, hence
$$\partial_v(x^{-1}) = -\,x^{-1} v\, x^{-1}.$$
Writing $x^{-2} = x^{-1}x^{-1}$ and applying the product rule again,
$$\partial_v(x^{-2}) = -\,x^{-1} v\, x^{-2} \;-\; x^{-2} v\, x^{-1}.$$
Both reduce to the scalar formulas $-v/x^2$ and $-2v/x^3$ when $n=1$; for $n>1$ the two terms genuinely differ and cannot be combined.

Q.E.D.

Solution to Exercise 10.d.21:

Write $\theta(t) := \omega t + a(t)$ for the longitude actually occupied in space and $\varphi(t) := b(t)$ for the latitude, so that
$$r(t) = R\big(\cos\varphi\cos\theta,\ \cos\varphi\sin\theta,\ \sin\varphi\big).$$

(a) The computation is far shorter in the moving orthonormal frame
$$e_r := (\cos\varphi\cos\theta,\ \cos\varphi\sin\theta,\ \sin\varphi),$$
$$e_\varphi := (-\sin\varphi\cos\theta,\ -\sin\varphi\sin\theta,\ \cos\varphi),$$
$$e_\theta := (-\sin\theta,\ \cos\theta,\ 0),$$

in which $r = R\,e_r$. Differentiating the three frame vectors by the chain rule gives

$$\dot{e}_r = \varphi'\,e_\varphi + \theta'\cos\varphi\,e_\theta, \qquad \dot{e}_\varphi = -\varphi'\,e_r - \theta'\sin\varphi\,e_\theta, \qquad \dot{e}_\theta = \theta'\big(-\cos\varphi\,e_r + \sin\varphi\,e_\varphi\big).$$

Hence $r' = R\big(\varphi'\,e_\varphi + \theta'\cos\varphi\,e_\theta\big)$, and differentiating once more and collecting the three components,

$$\begin{aligned} r''(t) = R\Big[&-\big(\varphi'^2 + \theta'^2\cos^2\varphi\big)\,e_r \\ &+\big(\varphi'' + \theta'^2\sin\varphi\cos\varphi\big)\,e_\varphi \\ &+\big(\theta''\cos\varphi - 2\,\theta'\varphi'\sin\varphi\big)\,e_\theta\Big]. \end{aligned}$$

The first component is the centripetal term; the other two are tangential.

(b) Substituting $\varphi = b$, $\theta' = \omega + a'$ and $\theta'' = a''$, the assumption $r''(t) = \lambda(t)\,r(t) = \lambda(t)R\,e_r$ says precisely that the e_φ and e_θ components vanish. The frame is orthonormal, so there is no cancellation between them to hope for. That is

$$b'' + \sin b\cos b\,(\omega + a')^2 = 0, \qquad \cos b\;a'' - 2\sin b\,(\omega + a')\,b' = 0,$$

which is the claim. Away from the poles $\cos b \neq 0$, so dividing the second equation by $\cos b$ gives

$$a'' - 2\tan b\,(\omega + a')\,b' = 0.$$

Remark 10.e.22 (*Reading the two equations physically*): Two features are worth noticing. First, ω enters only through the combination $\omega + a'$, which is the **absolute** angular velocity. That is what one would hope for, since the equations ought not to distinguish a car driving east on a still Earth from a parked car on a rotating one. Second, the factor 2 in the e_θ equation is the Coriolis term, and it is the same $2\theta'\varphi'$ that appears in the acceleration in polar coordinates one dimension lower.

Q.E.D.

Proof of Exercise 10.a.7:

(a) **True:** This is the standard chain rule $\nabla(u \circ F) = \nabla u(F) \cdot \mathrm{J}\,F$, where the i-th component (derivative with respect to x_i) is the product of $\nabla u(F)$ with the i-th column of the Jacobian $\mathrm{J}\,F$.

(b) **False:** Applying the product rule to the expression from part **(a)** yields $\partial_{ii}(u \circ F) = \partial_i\big(\sum_{j=1}^n (\partial_j u \circ F)\,\partial_i F_j\big) = \sum_{j=1}^n \big(\partial_i(\partial_j u \circ F)\,\partial_i F_j + (\partial_j u \circ F)\,\partial_{ii} F_j\big)$. Expanding the first term with the chain rule again gives $\sum_{j=1}^n \big(\sum_{k=1}^n (\partial_{jk} u \circ F)\,\partial_i F_k\big)\partial_i F_j$. The given formula erroneously collapses the double sum $\sum_{j,k}$ into a single sum $\sum_j \partial_{jj} u(\partial_i F_j)^2$, missing all the cross terms where $j \neq k$.

(c) **True:** Since F and u are both C^2, their composition $u \circ F$ is also C^2. By Schwarz's theorem, the mixed partial derivatives commute.

Q.E.D.

Solution of Exercise 10.c.18:

One substitution settles everything. In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$,

$$f(r\cos\theta,\, r\sin\theta) = \frac{r\cos\theta \cdot r^2\sin^2\theta}{r^2} = r\,\cos\theta\sin^2\theta \qquad (r > 0),$$

so f is positively homogeneous of degree 1: it grows linearly along every ray, with a ray-dependent slope. That single formula answers **(c)** and **(d)** at once.

(a) **True.** Away from the origin f is a quotient of polynomials whose denominator does not vanish, and such a quotient is C^∞.

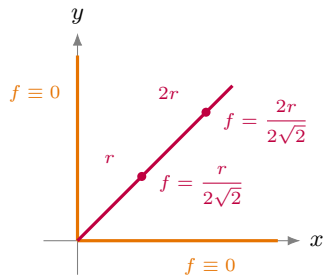
(b) True. Along the x-axis $f(x,0)=0$ for every x, so $\partial_x f(0,0)=\lim_{s\to 0}\big(f(s,0)-f(0,0)\big)/s=0$. The same computation on the y-axis gives $\partial_y f(0,0)=0$, so both partial derivatives exist and vanish.

(c) True. From the polar formula, $|f| = r\,|\cos\theta|\sin^2\theta \le r$, and the bound is uniform in θ. Consequently $f(x,y)\to 0=f(0,0)$ as $(x,y)\to(0,0)$, and f is continuous there. Continuity away from the origin is **(a)**.

(d) False, and f is not even differentiable at the origin. By **(b)** the only candidate for the differential is $L=0$, so Remark 10.c.15 asks whether $f(h)/|h|\to 0$. In polar coordinates that quotient is

$$\frac{f(r\cos\theta, r\sin\theta)}{r} = \cos\theta\sin^2\theta,$$

which does not depend on r at all. It is 0 along the axes but $\frac{1}{2\sqrt{2}}\neq 0$ along $\theta=\pi/4$, so the limit is not 0 and f is not differentiable at $(0,0)$. Since differentiability is weaker than C^1 (Theorem 10.c.14 and the chain of implications above it), $f\notin C^1(\mathbb{R}^2)$.



Along every ray f is **proportional** to r, so $f\to 0$ and f is continuous;
but the ratio f/r is a different constant on different rays, so f is not differentiable

Remark 10.e.23 (*Where this function sits in the chain*): The four verdicts place f at a definite height in the diagram of Section 10.c: strictly above $\frac{xy}{x^2+y^2}$, which fails already at continuity, and strictly below $\frac{x^2y^2}{\sqrt{x^2+y^2}}$, which is differentiable. Degree-1 homogeneity is what pins it there, and it does so in general. Such an f is automatically continuous at the origin, since $|f|\le Cr$; and it is differentiable there only if it is already **linear**, because $f(h)/|h|$ then depends on the direction alone, so it can converge only by agreeing with $L(h)/|h|$ exactly, in every direction at once.

Q.E.D.

The Mean Value Theorem and Taylor Expansions (Chapter 11)

Having linearized a map once, we ask two further questions. How good is that linear approximation, and how is it improved order by order? The answer is the several-variable analogue of the Taylor series from **Analysis I**, and it is what supplies every quantitative estimate in the optimization theory of the next part.

The mean value theorem comes first, for two reasons. It is the instrument the error terms are estimated with, and it is the first result in this course that survives the move to $\mathbb{R}^m$ only as an **inequality** rather than an equality once $m > 1$ (Exercise 11.a.11). That degradation is not a defect of the proof. There is simply no single intermediate point at which all m components can be caught at once. A short closing section asks when the expansion can be carried to infinite order, that is, when a smooth function equals its Taylor series.

The mean value theorem (Section 11.a)

AI-Note 11.1: The mean value theorem is used twice in the previous chapter, in the official hint to Exercise 10.d.20 and again in its solution. It is stated here, together with the warning that follows, which is what Exercise 10.d.20 actually turns on. In the lecture notes it is Theorem 10.28 (*"Higher dimensional Mean Value Theorem"*, p. 44), proved exactly as below.

Theorem 11.a.1 (*Mean value theorem*): Let $f : U \to \mathbb{R}$ be differentiable, $x \in U$ and $h \in \mathbb{R}^n$ with $x + th \in U$ for all $t \in [0,1]$. Then there is $t_0 \in (0,1)$ such that, writing $\xi := x + t_0 h$,

$$f(x+h) - f(x) = \partial_h f(\xi).$$

Proof:

Apply the one-variable mean value theorem to $\varphi(t) := f(x+th)$ on $[0,1]$, which is differentiable with $\varphi'(t) = \partial_h f(x+th)$ by the chain rule along a path (Section 10.d). It gives $t_0 \in (0,1)$ with $\varphi(1) - \varphi(0) = \varphi'(t_0)$, which is the claim. Q.E.D.

Important remark 11.a.2 (*It fails for vector-valued f*): The hypothesis "$f : U \to \mathbb{R}$" is essential. **The theorem is false the moment the target is $\mathbb{R}^m$ with $m \geq 2$**, and this is exactly what Exercise 11.a.11 below probes.

Applying the scalar result to each component $f_1, \dots, f_m$ separately does produce points $\xi_1, \dots, \xi_m$ with

$$f(x+h) - f(x) = \begin{pmatrix} \partial_h f_1(\xi_1) \\ \vdots \\ \partial_h f_m(\xi_m) \end{pmatrix},$$

but the ξ_i are **different points** in general, and no single ξ need work for all components at once. The standard witness is the **circle** $\gamma(t) := (\cos t, \sin t)$ on $[0, 2\pi]$: here $\gamma(2\pi) - \gamma(0) = 0$, while $\gamma'(t) = (-\sin t, \cos t)$ has $|\gamma'(t)| = 1$ and so never vanishes. There is no such ξ at all. Geometrically this is entirely reasonable: a walker who returns to the starting point must have had zero **average** velocity, but need never have stood still. In one dimension *"returning"* forces a turning point; in two it does not.

What survives for vector-valued f is the "*mean value inequality*" (German: „*Mittelwertsatz-Ungleichung*"), which replaces exact equality by a sharp upper bound controlling the displacement in terms of the maximum derivative along the line segment.

Theorem 11.a.3 (*Mean value inequality*): Let $U \subseteq \mathbb{R}^n$ be open, $x \in U$, and $h \in \mathbb{R}^n$ such that the line segment $[x, x+h] := \{x+th \mid t \in [0,1]\}$ lies entirely in U. If $F : U \to \mathbb{R}^m$ is differentiable at every point of $[x, x+h]$, then

$$|F(x+h) - F(x)| \leq \left(\sup_{t \in [0,1]} \|DF_{x+th}\|_{\mathrm{op}} \right) |h|,$$

where $\|DF_z\|_{\mathrm{op}} := \sup_{|v| \leq 1} |DF_z(v)|$ is the operator norm of the linear map DF_z.

Proof:

Set $y := F(x+h) - F(x) \in \mathbb{R}^m$. If $y = 0$, the inequality holds trivially. Assuming $y \neq 0$, let $u := \frac{y}{|y|}$, so that $|u| = 1$ and $\langle u, y \rangle = |y|$.

Define the auxiliary scalar function $\varphi : [0,1] \to \mathbb{R}$ by

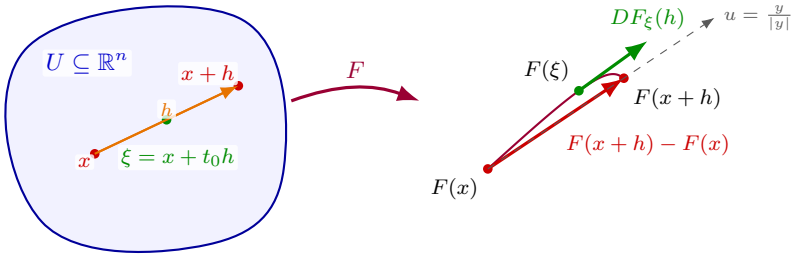
$$\varphi(t) := \langle u, F(x+th) \rangle.$$

By the chain rule along a path (Section 10.d), φ is continuous on $[0,1]$ and differentiable on $(0,1)$ with derivative $\varphi'(t) = \langle u, DF_{x+th}(h) \rangle$. Applying the scalar Mean Value Theorem (Theorem 11.a.1) to φ on $[0,1]$, there exists some $t_0 \in (0,1)$ such that $\varphi(1) - \varphi(0) = \varphi'(t_0)$, which expands to

$$|F(x+h) - F(x)| = \langle u, F(x+h) - F(x) \rangle = \langle u, DF_{x+t_0 h}(h) \rangle.$$

Applying the Cauchy–Schwarz inequality and the definition of the operator norm, we obtain

$$|F(x+h) - F(x)| \leq |u|\,|DF_{x+t_0h}(h)| \leq 1 \cdot \|DF_{x+t_0h}\|_{\mathrm{op}}\,|h| \leq \left(\sup_{t \in [0,1]} \|DF_{x+th}\|_{\mathrm{op}} \right) |h|. \quad \text{Q.E.D.}$$



Remark 11.a.4 (*Global Lipschitz bound on convex domains*): Theorem 11.a.3 is the principal multivariable tool for establishing Lipschitz continuity. If $U \subseteq \mathbb{R}^n$ is **convex** and $F : U \to \mathbb{R}^m$ is differentiable with $\|DF_z\|_{\mathrm{op}} \leq M$ for all $z \in U$, then for any pair $x, y \in U$, the line segment $[x, y]$ lies in U, so

$$|F(y) - F(x)| \leq M|y - x|.$$

Thus F is M-Lipschitz continuous on U.

Corollary 11.a.5 (*Every C^1 function is locally Lipschitz*): Let $U \subseteq \mathbb{R}^n$ be open, not necessarily convex, and let $F \in C^1(U, \mathbb{R}^m)$. Then every point of U has a ball around it on which F is Lipschitz, with a constant that may depend on the ball.

Proof:

Given $z \in U$, choose $r > 0$ with $\overline{B_r(z)} \subseteq U$. This closed ball is closed and bounded, hence compact (Theorem 7.b.28), and $x \mapsto \|\,\mathrm{J}\,F(x)\|_2$ is continuous because F is C^1 and the Hilbert–Schmidt norm is a polynomial in the entries. So it attains a finite maximum M there (Lemma 7.a.7). By Lemma 2.a.4 that bounds the operator norm too, $\|DF_x\|_{\mathrm{op}} \leq \|\,\mathrm{J}\,F(x)\|_2 \leq M$ on $\overline{B_r(z)}$. The open ball $B_r(z)$ is convex, so Remark 11.a.4 applies on it with that M. Q.E.D.

Important remark 11.a.6 (*Local, and not more than local*): The word **locally** is not decoration. On $U = \mathbb{R}$ the function $F(x) := x^2$ is C^1, hence Lipschitz on every bounded interval, and Lipschitz on no unbounded one: the constant $M = \max |F'|$ produced by the proof grows with the ball and has no finite supremum. What Remark 11.a.4 needs, and what fails here, is a bound on the derivative that is uniform over the whole domain.

Locally Lipschitz is nevertheless the right hypothesis surprisingly often, because it is what Picard–Lindelöf asks of the right-hand side of an ODE (Theorem A.d.13), and Exercise A.d.16 is this corollary again, proved from scratch in the form that theorem wants.

Theorem 11.a.7 (*Vanishing differential on a connected set*): Let $U \subseteq \mathbb{R}^n$ be open and **connected**, and let $F : U \to \mathbb{R}^m$ be differentiable with $DF_x = 0$ for every $x \in U$. Then F is constant.

Without connectedness the conclusion weakens to: F is constant on each connected component of U.

Proof:

Assume $U \neq \emptyset$, fix $x_0 \in U$ and put

$$G := \{x \in U : F(x) = F(x_0)\}.$$

G is non-empty, since $x_0 \in G$, and closed in U, being the preimage of the closed set $\{F(x_0)\}$ under the continuous map F (Proposition 9.a.5).

G is also **open**. Given $x \in G$, choose $r > 0$ with $B_r(x) \subseteq U$. A ball is convex, so Remark 11.a.4 applies on it with $M = 0$, giving $|F(y) - F(x)| \leq 0$, that is $F(y) = F(x) = F(x_0)$, for every $y \in B_r(x)$. Hence $B_r(x) \subseteq G$.

So G is a non-empty subset of U that is both open and closed in U. Since U is connected, $G = U$ (Definition 8.a.1), which is the claim. If U is not connected, run the argument inside a single component, which is itself open and connected. Q.E.D.

Remark 11.a.8 (*Why connectedness cannot be dropped*): On $U := (0,1) \cup (2,3) \subseteq \mathbb{R}$ the function that is 0 on the first interval and 1 on the second is differentiable with vanishing derivative everywhere, and is not constant. Nothing in the local argument can see this: the derivative is a purely local object, and *"constant"* is not. Connectedness is precisely the hypothesis that lets a local statement be propagated to a global one, which is why the proof above is a connectedness argument (*"non-empty, open and closed, hence everything"*) with a one-line differential calculus input rather than the other way round.

Example 11.a.9 (*Computing a Lipschitz constant via the Mean Value Inequality*): Consider the vector field $F : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $F(x,y) := (\sin(x+y), \cos(x-y))^{\mathsf{T}}$. We use the Mean Value Inequality to find an explicit Lipschitz constant for F on $\mathbb{R}^2$.

The Jacobi matrix of F at (x,y) is

$$\mathrm{J}\,F(x,y) = \begin{pmatrix} \cos(x+y) & \cos(x+y) \\ -\sin(x-y) & \sin(x-y) \end{pmatrix}.$$

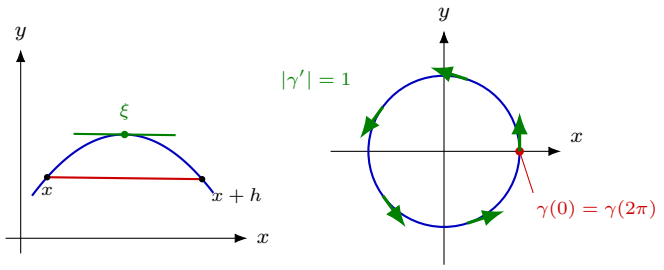
For any vector $v = (v_1, v_2)^{\mathsf{T}} \in \mathbb{R}^2$, the squared norm of $\mathrm{J}\,F(x,y)v$ is

$$|\,\mathrm{J}\,F(x,y)v|^2 = \cos^2(x+y)(v_1+v_2)^2 + \sin^2(x-y)(v_1-v_2)^2.$$

Since $\cos^2(x+y) \leq 1$ and $\sin^2(x-y) \leq 1$, we estimate:

$$|\,\mathrm{J}\,F(x,y)v|^2 \leq (v_1+v_2)^2 + (v_1-v_2)^2 = (v_1^2 + 2v_1v_2 + v_2^2) + (v_1^2 - 2v_1v_2 + v_2^2) = 2(v_1^2+v_2^2) = 2|v|^2.$$

Taking square roots gives $|\,\mathrm{J}\,F(x,y)v| \leq \sqrt{2}|v|$, so $\|DF_{(x,y)}\|_{\mathrm{op}} \leq \sqrt{2}$ for all $(x,y) \in \mathbb{R}^2$. Since $\mathbb{R}^2$ is convex, Remark 11.a.4 implies that F is $\sqrt{2}$-Lipschitz continuous on $\mathbb{R}^2$.



scalar f: some ξ has tangent parallel to the chord

vector-valued γ: $\gamma(2\pi) - \gamma(0) = 0$, yet γ' never vanishes, so no ξ exists

Exercise 11.a.10 (*Differential of $A \mapsto A^\mathsf{T}A$*): We identify the inner product space of real-valued matrices $\mathbb{R}^{n\times n}$ with inner product

$$\langle A, B\rangle = \sum_{i,j=1}^{n} A_{ij}B_{ij} = \mathrm{Tr}(A^\mathsf{T}B)$$

for $A, B \in \mathbb{R}^{n\times n}$, with the euclidean space $\mathbb{R}^{n^2} \cong \mathbb{R}^{n\times n}$, with the standard inner product. Then, consider the function

$$F : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}, \qquad A \mapsto A^\mathsf{T}A.$$

Compute the differential at the identity matrix $\mathrm{id} \in \mathbb{R}^{n\times n}$ applied to $X \in \mathbb{R}^{n\times n}$:

$$DF_{\mathrm{id}}(X) = \partial_X F(\mathrm{id}).$$

Solution to Exercise 11.a.10:

We use $(*)$:

$$\begin{aligned} DF_{\mathrm{id}}(X) &= \frac{d}{ds}\Big|_{s=0} F(\mathrm{id}+sX) \\ &= \frac{d}{ds}\,(\mathrm{id}+sX)^\mathsf{T}(\mathrm{id}+sX)\Big|_{s=0} \\ &= \frac{d}{ds}\left[\mathrm{id}+s(X^\mathsf{T}+X)+s^2X^\mathsf{T}X\right]\Big|_{s=0} \\ &= X^\mathsf{T}+X \end{aligned}$$

Q.E.D.

Exercise 11.a.11 (*Mean value for vector-valued functions* (optional)): Let $f \in C^1(\mathbb{R}, \mathbb{R}^m)$ for $m > 1$. Is it true that there is $t \in [0, 1]$ such that

$$f(1) - f(0) = Df_t(1) = \begin{pmatrix} f_1'(t) \\ \vdots \\ f_m'(t) \end{pmatrix}?$$

Prove it or provide a counterexample.

Official hint: *Try* $f(t) = (\sin(2\pi t), \cos(2\pi t))$.

Info: *This exercise is Problem 4.5 of Problem Sheet 4, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 166*

Exercise 11.a.12 (*Bounded gradient on a bounded domain does not force boundedness*): Let $U \subset \mathbb{R}^2$ be an open and bounded set, and let $f \in C^1(U, \mathbb{R})$ have bounded gradient, meaning there exists a constant $M < \infty$ such that

$$\|\nabla f(x)\| \le M \qquad \text{for all } x \in U.$$

Is f necessarily bounded on U? Prove the statement or provide a counterexample.

Info: *This exercise is taken from the examination of 18 August 2023 (Prof. Urs Lang), Problem 2.MC2 (D).*

Solution: *See solution on Page 171*

Taylor expansions (Section 11.b)

The general formula (Subsection 11.b.1)

In the lecture, you have seen the following formula. For $f \in C^k(U \subseteq \mathbb{R}^n, \mathbb{R})$, $\bar{x} \in U$:

$$f(\bar{x}+h) = \sum_{|\alpha|=0}^{k} \partial^\alpha f(\bar{x}) \, \frac{h^\alpha}{\alpha!} + o(|h|^k)$$

where α is a "*multi-index*" (German: „*Multiindex*“) in $\mathbb{N}^n$.

In practice, we never use this formula directly, but it is worth unpacking once, which is what Example 11.b.16 does.

Before unpacking the formula, one hypothesis hidden in the notation deserves to be made explicit. Writing ∂^α for a multi-index α presumes that it does not matter **in which order** the derivatives are taken, since otherwise $\partial^{(1,1)}$ would be ambiguous between $\partial_1\partial_2$ and $\partial_2\partial_1$. That the presumption is legitimate is itself a theorem, "*Schwarz's theorem*" (German: „*Satz von Schwarz*“), and it is stated next.

Theorem 11.b.13 (*Schwarz / Clairaut — mixed partial derivatives commute*): Let $U \subseteq \mathbb{R}^n$ be open and $f \in C^2(U, \mathbb{R})$. Then for all $i, j \in \{1, \dots, n\}$ and all $x \in U$,

$$\partial_i \partial_j f(x) = \partial_j \partial_i f(x).$$

More generally, for $f \in C^k(U, \mathbb{R})$ any k partial derivatives may be permuted freely, so $\partial^\alpha f$ depends only on the multi-index α and not on the order of differentiation. In particular the Hessian $\operatorname{Hess} f(x)$ of Remark 11.b.30 is a **symmetric** matrix.

The proof, by two applications of the mean value theorem to a second difference, is given at the start of Section 12.b, where the symmetry of the Hessian is first put to work.

Remark 11.b.14 (*The hypothesis is not decoration*): Continuity of the second derivatives is genuinely needed. The standard counterexample is

$$f(x,y) := \begin{cases} xy\,\dfrac{x^2-y^2}{x^2+y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0), \end{cases}$$

for which a direct computation from the definition gives $\partial_1\partial_2 f(0,0) = 1$ but $\partial_2\partial_1 f(0,0) = -1$. This f has both mixed partials everywhere, but they are not continuous at the origin, so Theorem 11.b.13 does not apply, and indeed fails. Since everything in Section 11.b assumes $f \in C^k$, the issue never arises here. It is worth knowing that it **could**.

Exercise 11.b.15 (*Applying Schwarz's theorem*): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) := x^2 e^{y^2} + e^{-x^2} \sin\left(e^{x^2}\left(x^5 + 3x^2 + 100x + \frac{1}{2+\sin(x)}\right)\right).$$

(a) Show that $f \in C^2(\mathbb{R}^2, \mathbb{R})$.

(b) Compute $\partial_2\partial_1 f(x,y)$. (Make it as easy as possible for yourself.)

Solution: *See solution on Page 165*

Example 11.b.16 (*Maximal Taylor expansion in $\mathbb{R}^2$*): Let $f \in C^3(\mathbb{R}^2, \mathbb{R})$, and write out the maximal Taylor approximation. The multi-indices are:

- $|\alpha| = 0$: $\alpha = (0,0)$
- $|\beta| = 1$: $\beta_1 = (1,0)$, $\beta_2 = (0,1)$
- $|\gamma| = 2$: $\gamma_1 = (1,1)$, $\gamma_2 = (2,0)$, $\gamma_3 = (0,2)$
- $|\delta| = 3$: $\delta_1 = (3,0)$, $\delta_2 = (2,1)$, $\delta_3 = (1,2)$, $\delta_4 = (0,3)$

Recall that

$$\partial^\alpha = \partial_1^{\alpha_1}\partial_2^{\alpha_2}\cdots\partial_n^{\alpha_n}, \qquad \text{e.g. } \partial^{\delta_2} = \partial^{(2,1)} = \partial_1^2\partial_2,$$

and

$$\alpha! = \alpha_1!\cdots\alpha_n!, \qquad \text{e.g. } \gamma_2! = (2,0)! = 2\cdot 1 = 2.$$

So:

$$\begin{aligned} f(\bar{x}+h) = {} & f(\bar{x}) && (\alpha)\\ & + \partial_1 f(\bar{x})h_1 + \partial_2 f(\bar{x})h_2 && (\beta_1,\beta_2)\\ & + \partial_1\partial_2 f(\bar{x})h_1h_2 + \partial_1^2 f(\bar{x})\frac{h_1^2}{2} + \partial_2^2 f(\bar{x})\frac{h_2^2}{2} && (\gamma_1,\gamma_2,\gamma_3)\\ & + \partial_1^3 f(\bar{x})\frac{h_1^3}{6} + \partial_1^2\partial_2 f(\bar{x})\frac{h_1^2h_2}{2} && (\delta_1,\delta_2)\\ & + \partial_1\partial_2^2 f(\bar{x})\frac{h_1h_2^2}{2} + \partial_2^3 f(\bar{x})\frac{h_2^3}{6} && (\delta_3,\delta_4)\\ & + o(|h|^3) \end{aligned}$$

Example 11.b.17 (*Reading a multi-index: evaluating $\partial^\alpha f$*): Let $f(x,y,z) := x^3 + 2y + e^{2x}\cos(z)$, and take the multi-index $\alpha = (2,0,1)$. It calls for differentiating twice with respect to x, not at all with respect to y, and once with respect to z. Since f is smooth, Theorem 11.b.13 says the order does not matter, so we may take the cheapest one:

$$\begin{aligned} \partial_x f &= 3x^2 + 2e^{2x}\cos(z)\\ \partial_{xx} f &= 6x + 4e^{2x}\cos(z)\\ \partial_{xxz} f &= -4e^{2x}\sin(z). \end{aligned}$$

Hence $\partial^{(2,0,1)} f(x,y,z) = -4e^{2x}\sin(z)$.

The working formula (Subsection 11.b.2)

We can rewrite this as

$$f(\bar{x}+h) = \sum_{\ell=0}^{k} \frac{1}{\ell!}\big(h_1\partial_1 + h_2\partial_2\big)^\ell f(\bar{x}).$$

Theorem 11.b.18 (*Taylor working formula*): For $f \in C^k(U \subseteq \mathbb{R}^n, \mathbb{R})$ with $\bar{x} + th \in U$ for all $t \in [0,1]$:

$$f(\bar{x}+h) = \sum_{\ell=0}^{k} \frac{1}{\ell!}\left(\sum_{m=1}^{n} h_m\partial_m\right)^\ell f(\bar{x}) + o(|h|^k)$$

AI-Note 11.2: Both sums in Theorem 11.b.18 are written with lower limit $\ell = 1$ (resp. starting index) in the source. The $\ell = 0$ term is $f(\bar{x})$ itself, which the explicit expansion above does include, so the lower limit must be 0. Corrected above; compare the $|\alpha| = 0$ line of Example 11.b.16, which is the same term written in multi-index form.

Theorem 11.b.19 (*Taylor's theorem with Lagrange remainder*): Let $U \subseteq \mathbb{R}^n$ be open, $f \in C^{k+1}(U, \mathbb{R})$, $\bar{x} \in U$, and $h \in \mathbb{R}^n$ with $\bar{x} + th \in U$ for all $t \in [0,1]$. Then there exists $\theta \in (0,1)$

such that

$$f(\bar{x}+h)=\sum_{\ell=0}^{k}\frac{1}{\ell!}\Big(\sum_{m=1}^{n}h_m\partial_m\Big)^{\ell}f(\bar{x})\;+\;\frac{1}{(k+1)!}\Big(\sum_{m=1}^{n}h_m\partial_m\Big)^{k+1}f(\bar{x}+\theta h).$$

Equivalently, in multi-index notation,

$$f(\bar{x}+h)=\sum_{|\alpha|\le k}\partial^{\alpha}f(\bar{x})\,\frac{h^{\alpha}}{\alpha!}\;+\;\sum_{|\alpha|=k+1}\partial^{\alpha}f(\bar{x}+\theta h)\,\frac{h^{\alpha}}{\alpha!}.$$

Proof:

Let $\varphi(t):=f(\bar{x}+th)$ for $t\in[0,1]$, well defined and C^{k+1} since $f\in C^{k+1}$ and the segment $\bar{x}+th$ stays in U. By induction on ℓ via the chain rule, which is exactly the computation underlying Theorem 11.b.18 itself,

$$\varphi^{(\ell)}(t)=\Big(\sum_{m=1}^{n}h_m\partial_m\Big)^{\ell}f(\bar{x}+th)\qquad\text{for } 0\le\ell\le k+1.$$

Apply the one-variable Taylor theorem with Lagrange remainder (**Analysis I**) to φ on $[0,1]$: there is $\theta\in(0,1)$ with

$$\varphi(1)=\sum_{\ell=0}^{k}\frac{\varphi^{(\ell)}(0)}{\ell!}+\frac{\varphi^{(k+1)}(\theta)}{(k+1)!}.$$

Substituting $\varphi(1)=f(\bar{x}+h)$, $\varphi^{(\ell)}(0)=\big(\sum_m h_m\partial_m\big)^{\ell}f(\bar{x})$, and $\varphi^{(k+1)}(\theta)=\big(\sum_m h_m\partial_m\big)^{k+1}f(\bar{x}+\theta h)$ gives the working-formula version. The multi-index version follows by expanding $\big(\sum_m h_m\partial_m\big)^{\ell}$ via the multinomial theorem, exactly as in Example 11.b.16. Q.E.D.

Proof of Theorem 11.b.18:

For $k=0$ the claim is continuity of f at $\bar{x}$, so let $k\ge1$ and h be small enough that $\bar{x}+th\in U$ for $t\in[0,1]$. Apply Theorem 11.b.19 with $k-1$ in place of k, which needs only $f\in C^{k}$: for some $\theta\in(0,1)$,

$$f(\bar{x}+h)=\sum_{|\alpha|\le k-1}\partial^{\alpha}f(\bar{x})\,\frac{h^{\alpha}}{\alpha!}+\sum_{|\alpha|=k}\partial^{\alpha}f(\bar{x}+\theta h)\,\frac{h^{\alpha}}{\alpha!}.$$

Replacing $\bar{x}+\theta h$ by $\bar{x}$ in the last sum produces exactly the order-k Taylor polynomial, at the cost of the error

$$E(h):=\sum_{|\alpha|=k}\Big(\partial^{\alpha}f(\bar{x}+\theta h)-\partial^{\alpha}f(\bar{x})\Big)\frac{h^{\alpha}}{\alpha!}.$$

Since $|h^{\alpha}|\le|h|^{k}$ for $|\alpha|=k$, we get $|E(h)|\le|h|^{k}\sum_{|\alpha|=k}\big|\partial^{\alpha}f(\bar{x}+\theta h)-\partial^{\alpha}f(\bar{x})\big|$, and every summand tends to 0 as $h\to0$ because the k-th partial derivatives are continuous at $\bar{x}$. So $E(h)=o(|h|^{k})$. Q.E.D.

Remark 11.b.20: Unlike Theorem 11.b.18's $o(|h|^{k})$ remainder, this is an **exact** equality for the specific h in question, not merely a statement about what happens as $h\to0$. That is genuinely different information, and it is what is needed whenever a hard, uniform error bound is wanted rather than an asymptotic one (e.g. bounding $|f(\bar{x}+h)-T_k(h)|$ over an entire region at once, given a bound on the $(k+1)$-th derivatives there). The price is that θ depends on $\bar{x}$, h **and** k together, and is almost never computable in practice. That is exactly why the rest of this chapter uses the asymptotic form for actually computing expansions, and reserves this one for error estimates.

Corollary 11.b.21 (*A polynomial that fits to order k **is** the Taylor polynomial*)**:** Let $U\subseteq\mathbb{R}^{n}$ be open, $x_0\in U$, $f\in C^{k+1}(U)$, and let P be a polynomial of degree at most k with

$$|f(x_0+h)-P(h)|=o\big(|h|^{k}\big)\qquad\text{as } h\to0.$$

Then $\partial^\alpha f(x_0) = \partial^\alpha P(0)$ for every multi-index with $|\alpha| \le k$. Equivalently, P is **the** Taylor polynomial of f of degree k at x_0.

Proof:

By Theorem 11.b.18, the Taylor polynomial T of degree k also satisfies $|f(x_0+h)-T(h)| = o(|h|^k)$. Subtracting, the polynomial $Q := P - T$ has degree at most k and satisfies $|Q(h)| = o(|h|^k)$. Suppose $Q \not\equiv 0$ and let $d \le k$ be the degree of its lowest non-vanishing homogeneous part Q_d. Choosing v with $Q_d(v) \neq 0$ and restricting to the ray $h := tv$,

$$Q(tv) = t^d Q_d(v) + O(t^{d+1}), \qquad \text{so} \qquad \frac{|Q(tv)|}{t^k} \sim |Q_d(v)|\, t^{d-k} \not\to 0$$

as $t \downarrow 0$, since $d \le k$. That contradicts $|Q(h)| = o(|h|^k)$, so $Q \equiv 0$ and $P = T$. **Q.E.D.**

Remark 11.b.22 (*Reading derivatives off an asymptotic expansion*): This is more useful than it looks, because it runs **backwards**. Normally one computes derivatives in order to produce an expansion; the corollary says that an expansion, however it was obtained, hands back the derivatives. Given

$$f(x_0 + h) = c + a^{\mathsf T}h + \tfrac{1}{2}h^{\mathsf T}Bh + o(|h|^2)$$

one may read off $f(x_0) = c$, $\nabla f(x_0) = a$ and $\operatorname{Hess} f(x_0) = B$ without differentiating anything. This is the same move by which Definition 9.a.1 recognises a differential from an $o(|h|)$ estimate, now carried to higher order. It is the standard route to a Hessian for functions built out of known series.

Two cautions on the bookkeeping, both of which the exercise below turns on. The estimate must be $o(|h|^k)$, not $O(|h|^k)$: an $O(|h|^k)$ error can contain a genuine degree-k term and so corrupts the order-k coefficients, leaving only the derivatives of order $< k$ determined. And the expansion determines derivatives only up to order k, and nothing beyond it.

Exercise 11.b.23 (*What does an expansion actually tell you?*): For each of the following, state precisely which values of f and of its derivatives at the indicated point are determined by the given information, and which are not.

(a) $f : \mathbb{R}^3 \to \mathbb{R}$ with $f(x_1, x_2, x_3) - x_1x_2 \in O\big(|x|\big)$, at $x = 0$.

(b) $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(1 + x_1,\, -1 + x_2) - 2 - x_1 \in o\big(|x|\big)$, at $(1, -1)$.

(c) $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(-1 + x_1,\, x_2) + 2 - 2x_1x_2 + 2x_2^2 - 3x_2^3 \in o\big(|x|^3\big)$, at $(-1, 0)$.

Assume in each case that f is as smooth as the relevant statement requires.

Solution: *See solution on Page 166*

Example 11.b.24 (*One expansion, five verdicts*): Suppose $f \in C^\infty(\mathbb{R}^2)$ satisfies

$$f(x_1, x_2) = 10x_2 + x_1^2 + x_2^2 + o\big(|x|^2\big) \qquad \text{as } |x| \to 0.$$

Corollary 11.b.21 applies with $k = 2$ and $x_0 = 0$, so the displayed polynomial **is** the second-order Taylor polynomial of f at the origin. Matching it term by term against

$$f(h) = f(0) + \nabla f(0)^{\mathsf T}h + \tfrac{1}{2}h^{\mathsf T} \operatorname{Hess} f(0)\, h + o\big(|h|^2\big)$$

gives everything at once:

$$f(0) = 0, \qquad \nabla f(0) = \begin{pmatrix} 0 \\ 10 \end{pmatrix}, \qquad \operatorname{Hess} f(0) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The Hessian entries are 2 rather than 1 because it is $\frac{1}{2}h^{\mathsf T} \operatorname{Hess} f(0) h$, not $h^{\mathsf T} \operatorname{Hess} f(0) h$, that must reproduce $h_1^2 + h_2^2$. Five statements now settle themselves.

(a) $(0, 0)$ **must be a critical point of** f**: false.** The expansion carries a linear term $10x_2$, so $\nabla f(0) \neq 0$.

(b) $f(0,0)=0$**: true.** It is the constant term, and there is none.

(c) $\partial^2 f/\partial x_1^2(0,0)=1$**: false.** That entry is 2.

(d) $\operatorname{Hess} f(0,0)=\left(\begin{smallmatrix}2&0\\0&2\end{smallmatrix}\right)$**: true**, as computed above.

(e) f **must have a strict local minimum at** $(0,0)$**: false.** Along the x_2-axis, $f(0,t)=10t+t^2+o(t^2)$, which is negative for small $t<0$ and positive for small $t>0$. The origin is not even a local extremum.

Items **(d)** and **(e)** are the pair the block is built around. The Hessian really is positive definite, and that really does not make the origin a minimum: the second-derivative test of Theorem 12.b.11 is a statement about **critical** points, and **(a)** has already shown this one is not. A definite Hessian at a non-critical point says nothing whatever about extrema, because the linear term dominates the quadratic one near the origin.

Info: *This example is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the fifth block of true/false questions.*

Remark 11.b.25 (*Two ways to weight the same expansion***):** Example 11.b.16 above and Theorem 11.b.18 appear to disagree on how heavily mixed partial derivatives are weighted. Take the coefficient of $h_1^2h_2$, so $\alpha=(2,1)$ and $\ell=|\alpha|=3$. The explicit multi-index expansion gives $\frac{1}{2}$ directly. The working formula instead offers $\frac{1}{\ell!}(h_1\partial_1+h_2\partial_2)^\ell$, and expanding that by the multinomial theorem produces

$$\frac{\ell!}{\alpha!}\cdot\frac{1}{\ell!}=\frac{1}{\alpha!}=\frac{1}{2!\,1!}=\frac{1}{2}$$

for the term $h^\alpha\partial^\alpha f$. The two agree, but they arrive by different routes, and seeing why is worth the paragraph.

The working formula's $\frac{1}{\ell!}$ weights **all** ℓ-fold products of partials globally, including repeated ones such as $\partial_1\partial_2$, which appears in more than one order in the multinomial expansion of $(h_1\partial_1+h_2\partial_2)^\ell$. By Schwarz's theorem (equality of mixed partial derivatives, valid here because $f\in C^k$) all those repeats collapse onto the same ∂^α, and there are exactly $\ell!/\alpha!$ of them. That collapsing is what turns the global $1/\ell!$ into the per-term $1/\alpha!$ seen in Example 11.b.16.

In short, $1/\ell!$ weights **paths** through the mixed partial derivatives, whereas $1/\alpha!$ weights the **distinct** derivatives themselves, once those paths have been identified with one another.

Worked examples (Subsection 11.b.3)

Exercise 11.b.26 (*Taylor expansion of* $(x+y^2)e^{-x^2-y^2}$**):** Compute the Taylor expansion of

$$f:\mathbb{R}^2\to\mathbb{R},\qquad (x,y)\mapsto(x+y^2)e^{-x^2-y^2}$$

around $(0,0)$ up to third order.

The cumbersome solution to Exercise 11.b.26:

Calculate all partial derivatives up to third order and use the definition. (Please never do this.)

Q.E.D.

The easy solution to Exercise 11.b.26:

The substitution and multiplication rules we used in Analysis 1 still apply:

$$\begin{aligned}(x+y^2)e^{-x^2-y^2}&=(x+y^2)\left(1-x^2-y^2+o(|x^2+y^2|)\right)\\&=x+y^2-x^3-xy^2+o\big(|(x,y)|^3\big)\end{aligned}$$

Q.E.D.

Example 11.b.27 (*More Taylor expansions via Analysis 1*): The substitution and multiplication rules from Analysis 1 often make expansions much faster than using the formula directly, provided the expansion is around $(0,0)$.

(a) Expand $f(x,y) := \cos(x)e^y$ up to second order around $(0,0)$:

$$f(x,y) = \left(1 - \frac{x^2}{2} + o(|(x,y)|^2)\right)\left(1 + y + \frac{y^2}{2} + o(|(x,y)|^2)\right) = 1+y-\frac{1}{2}x^2+\frac{1}{2}y^2+o(|(x,y)|^2).$$

(b) Expand $f(x,y) := \cos(x+y^2)$ up to second order around $(0,0)$. Setting $z := x+y^2$:

$$\cos(z) = 1-\frac{z^2}{2}+o(|z|^2) \implies \cos(x+y^2) = 1-\frac{(x+y^2)^2}{2}+o(|(x,y)|^2) = 1-\frac{1}{2}x^2+o(|(x,y)|^2),$$

where terms like xy^2 and y^4 are absorbed into $o(|(x,y)|^2)$ since their total degree is strictly greater than 2.

(c) Expand $f(x,y) := \sqrt{y}e^{\sqrt{x}}$ to first order around $(1,1)$. Here the formula is better since the expansion point is not the origin. We have $f(1,1) = e$. $\partial_1 f(x,y) = \sqrt{y}e^{\sqrt{x}}\frac{1}{2\sqrt{x}} \implies \partial_1 f(1,1) = \frac{e}{2}$. $\partial_2 f(x,y) = \frac{1}{2\sqrt{y}}e^{\sqrt{x}} \implies \partial_2 f(1,1) = \frac{e}{2}$. Writing $h = (h_1,h_2) = (x-1,y-1)$:

$$f(1+h_1,1+h_2) = f(1,1) + \partial_1 f(1,1)h_1 + \partial_2 f(1,1)h_2 + o(|h|) = e + \frac{e}{2}h_1 + \frac{e}{2}h_2 + o(|h|).$$

The claim that the substitution route is cheaper is easier to believe when both routes are run on the same function. They are, next, on a function chosen so that the general formula has to inspect ten multi-indices in order to discover that nine of them vanish.

Example 11.b.28 (*Taylor expansion of $\sin(xy)$, the long way and the short way*): Compute the Taylor expansion of $f(x,y) := \sin(xy)$ to order 3 around $(0,0)$.

(a) Using the general formula: We list the multi-indices α with $|\alpha| \le 3$ and compute the partial derivatives at $(0,0)$:

- $|\alpha| = 0$: $f(0,0) = \sin(0) = 0$.
- $|\alpha| = 1$: $\partial_x f = y\cos(xy) \implies 0$. $\partial_y f = x\cos(xy) \implies 0$.
- $|\alpha| = 2$: $\partial_{xx} f = -y^2\sin(xy) \implies 0$. $\partial_{yy} f = -x^2\sin(xy) \implies 0$. $\partial_{xy} f = \cos(xy) - xy\sin(xy) \implies 1$.
- $|\alpha| = 3$: $\partial_{xxx} f = -y^3\cos(xy) \implies 0$. $\partial_{yyy} f = -x^3\cos(xy) \implies 0$. $\partial_{xxy} f = -2y\sin(xy) - xy^2\cos(xy) \implies 0$. $\partial_{xyy} f = -2x\sin(xy) - x^2 y\cos(xy) \implies 0$.

The only non-zero term is $\partial^{(1,1)} f(0,0) = 1$. The corresponding factor in the formula is $\frac{1}{\alpha!}h^\alpha = \frac{1}{(1!)(1!)}xy = xy$. Thus,

$$f(x,y) = xy + o(|(x,y)|^3).$$

(b) Using 1D expansions (the clever way): We know that $\sin(t) = t - \frac{t^3}{6} + O(t^5)$ as $t \to 0$. Substituting $t = xy$ yields

$$\sin(xy) = xy - \frac{(xy)^3}{6} + O((xy)^5) = xy + o(|(x,y)|^3).$$

Example 11.b.29 (*Taylor expansion of $\frac{\cos x}{1-y^2}$, where two series meet*): Compute the Taylor expansion of $f(x,y) := \frac{\cos x}{1-y^2}$ to order 4 around $(0,0)$. Here nothing is substituted into anything. The two factors are expanded separately, in their own variables, and then multiplied:

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6) \qquad (x \to 0),$$

$$\frac{1}{1-y^2} = 1 + y^2 + y^4 + O(y^6) \qquad (y \to 0).$$

Multiplying and discarding everything of total degree above 4,

$$\begin{aligned}\frac{\cos x}{1-y^2} &= \left(1-\frac{x^2}{2}+\frac{x^4}{24}+O(x^6)\right)\left(1+y^2+y^4+O(y^6)\right)\\ &= 1+y^2+y^4-\frac{x^2}{2}-\frac{x^2y^2}{2}+\frac{x^4}{24}+o(|(x,y)|^4)\\ &= 1-\frac{1}{2}x^2+y^2+\frac{1}{24}x^4-\frac{1}{2}x^2y^2+y^4+o(|(x,y)|^4).\end{aligned}$$

The one term needing care is $-\frac{1}{2}x^2y^2$. It is the product of two second-order terms, so it is of total degree 4 and has to be kept, whereas the analogous product one order further out is of degree 6 and does not.

Second-order form and the Hessian (Subsection 11.b.4)

Remark 11.b.30 (*Second-order Taylor formula with Hessian*): The second-order Taylor approximation of $f \in C^2(\mathbb{R}^n, \mathbb{R})$ around $x_0 \in \mathbb{R}^n$ can be expressed compactly as

$$f(x_0+x) = f(x_0) + \nabla f(x_0)\cdot x + \tfrac{1}{2}x^{\mathsf{T}}\cdot \operatorname{Hess} f(x_0)\cdot x + o(|x|^2)$$

where

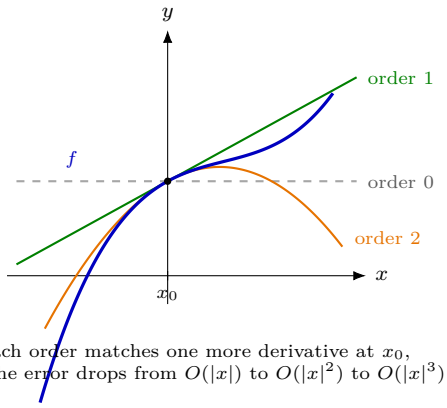
$$\operatorname{Hess} f(x_0) = \big(\partial_i\partial_j f(x_0)\big)_{i,j=1,\dots,n}$$

is the "*Hessian matrix*" (German: „*Hesse-Matrix*“) of f.

This is the form the rest of the course actually uses, and it is worth reading term by term as a sequence of ever-better approximations to f near x_0:

- the **constant** term $f(x_0)$, the crudest guess, exact only at x_0;
- the **linear** term $\nabla f(x_0)\cdot x$, which is the tangent plane and hence exactly the differential of Definition 9.a.1;
- the **quadratic** term $\frac{1}{2}x^{\mathsf{T}}\operatorname{Hess} f(x_0)\,x$, the first term that sees curvature, and the one that decides in Section 12.b whether a critical point is a minimum, a maximum, or a saddle.

By Theorem 11.b.13 the Hessian is symmetric, which is what makes that last decision a question about the **signs of its eigenvalues** rather than anything harder.



each order matches one more derivative at x_0,
and the error drops from $O(|x|)$ to $O(|x|^2)$ to $O(|x|^3)$

In several variables the same picture holds with *"tangent line"* replaced by **tangent plane** and *"parabola"* by a paraboloid, bowl-shaped or saddle-shaped according to the signs of the eigenvalues of $\operatorname{Hess} f(x_0)$. That is exactly the classification carried out in Section 12.b.

Question 11.b.31: For $f \in C^k(U, \mathbb{R})$, do the Taylor expansions of $t \mapsto f(x_0+th)$ in t and of $(th) \mapsto f(x_0+th)$ in (th) around 0 coincide?

Answer 11.b.32: Yes. This is, in fact, how we prove the multi-dimensional Taylor approximation, and why we need the assumption $x_0 + th \in U$ for all $t \in [0,1]$.

Exercise 11.b.33 (*Proof of second-order Taylor formula*): Prove the formula from above:

$$f(x_0 + h) = f(x_0) + \nabla f(x_0)h + \tfrac{1}{2}h^{\mathsf{T}} \operatorname{Hess} f(x_0)h + \dots$$

for $f \in C^k(U, \mathbb{R})$ with $x_0 + th \in U$ for all $t \in [0,1]$.

Proof of Exercise 11.b.33:

We expand $t \mapsto f(x_0 + th)$ in t:

$$\begin{aligned} f(x_0 + th) &= f(x_0) + t \left.\frac{d}{dt}\right|_{t=0} f(x_0 + th) + \frac{t^2}{2} \left.\frac{d^2}{dt^2}\right|_{t=0} f(x_0 + th) + o(|th|^2) \\ &= f(x_0) + t \sum_{i=1}^{n} \partial_i f(x_0) h_i + \frac{t^2}{2} \frac{d}{dt} \sum_{i=1}^{n} \partial_i f(x_0 + th) h_i \Bigg|_{t=0} + o(|h|^2) \\ &= f(x_0) + t\nabla f(x_0)h + \frac{t^2}{2} \sum_{i,j=1}^{n} \partial_i \partial_j f(x_0) h_i h_j + o(|th|^2) \\ &= f(x_0) + t\nabla f(x_0)h + \frac{t^2}{2} h^{\mathsf{T}} \operatorname{Hess} f(x_0)h + o(|th|^2) \end{aligned}$$

Setting $t = 1$ gives the result. **Q.E.D.**

AI-Note 11.3: The last step of the argument above needs care. The remainder $o(|th|^2)$ is a statement about $t \to 0$ with h fixed, so it says nothing at the single value $t = 1$; what the formula needs is a remainder that is small as $h \to 0$. The computation above is the right one, but the limit has to be taken in h, which is what the proof of Theorem 11.b.18 does: it uses the exact Lagrange form for the fixed h, and only then lets $h \to 0$.

Exercise 11.b.34 (*Multiple choice (Landau notation)* (important)): Mark all and only the statements which are true.

- **(a)** As $|x| \to 0$, if $f(x) = x_2 x_1^2 + O(x_1^2)$, then $f(x) = O(|x|^2)$.
- **(b)** As $x \to 0$, if $f(x) = x_2 x_1^2 + O(x_1^3)$, then $f(x) = O(x_1^3)$.
- **(c)** As $|x| \to 0$, if $f(x) = x_2 x_1 + O(|x|^2)$, then f is differentiable at $x = 0$.
- **(d)** As $x \to 0$, if $f(x) = x_2 x_1 + O(|x|^2)$, then f is twice differentiable around $x = 0$.
- **(e)** As $|x| \to 0$, if $f(x) = x_2 x_1 + O(|x|^3)$, then f is twice differentiable around $x = 0$ and $\partial_{11} f(0) = 0$.

Official hint: *Revise Corollary 10.45.*

Info: *This exercise is Problem 5.3 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 167*

Exercise 11.b.35 (*Multiple choice (cost of a Taylor expansion)* (optional)): Assume $g \in C^\infty(\mathbb{R})$ and $f \in C^\infty(\mathbb{R}^n)$ is such that

$$f(x) = x_1 + x_2 x_1 + O(|x|^4) \quad \text{as } x \to 0.$$

We want to compute the Taylor expansion of $(g \circ f)$ at $x = 0$ of the highest possible order with the information we have. We can ask for the value $g^{(k)}(t)$ for arbitrary $k \in \mathbb{N}$, $t \in \mathbb{R}$, but we have to pay 5 CHF each time.

For a general g, which is the degree of the best (i.e. highest-order) Taylor polynomial we can compute? How expensive will it be to compute it?

- **(a)** degree 3 and 20 CHF

(b) degree 2 and 15 CHF
(c) degree 2 and 10 CHF
(d) degree 3 and 15 CHF

Info: *This exercise is Problem 5.5 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 168*

Exercise 11.b.36 (*Polynomials* (optional — "interesting, but irrelevant for the course")): Let $P, Q \in \mathbb{R}[x_1, \dots, x_n]$ be polynomials of n variables; assume that there are positive numbers M, σ such that

$$|P(x) - Q(x)| \leq M|x|^\sigma \quad \text{for all } |x| \leq 1.$$

Show that P and Q have the same coefficients of order smaller than σ. That is, if we write $P(X) = \sum_{\alpha \in \mathbb{N}^n} a_\alpha X^\alpha$, $Q(X) = \sum_{\alpha \in \mathbb{N}^n} b_\alpha X^\alpha$, then $a_\alpha = b_\alpha$ for all $|\alpha| < \sigma$.

Info: *This exercise is Problem 5.6 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 168*

The three expansions below are the same technique again, unassisted. Each yields to a single one-variable series, and the only judgement required is which substitution to make and how many terms of it survive the truncation.

Exercise 11.b.37 (*Taylor expansion of* $\sqrt{1+x+y^2}$): Compute the Taylor expansion of $f(x,y) := \sqrt{1+x+y^2}$ to order 2 around $(0,0)$.

Solution: *See solution on Page 169*

Exercise 11.b.38 (*Taylor expansion of* $\exp(\arctan(x-y))$): Compute the Taylor expansion of $f(x,y) := \exp(\arctan(x-y))$ to order 2 around $(0,0)$.

Solution: *See solution on Page 169*

Exercise 11.b.39 (*Taylor expansion of* $1/(1-x^2-y^2)$): Compute the Taylor expansion of $f(x,y) := \dfrac{1}{1-x^2-y^2}$ to order 2 around $(0,0)$.

Solution: *See solution on Page 169*

Remark 11.b.40: Together with Example 11.b.28, these three are exactly script Exercise 10.36: *"compute the Taylor polynomials up to the quadratic order of* $\sin(xy)$, $\sqrt{1+x+y^2}$, $\exp(\arctan(x-y))$, $1/(1-x^2-y^2)$ *— don't use the general formula!"* The last clause is the instruction that matters. Each of the four collapses to a line by substitution and to most of a page by the general formula, and Example 11.b.28 runs both routes so that the difference can be seen rather than asserted.

Exercise 11.b.41 (*Taylor expansion of a parametric integral*): Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by the parametric integral

$$f(x) := \int_{x-1}^{x+1} \cos(\pi t^2)\, dt.$$

The integral need not be evaluated explicitly.

(a) Compute $f'(x)$, and simplify it to a product of two sines.
(b) Find the coefficients of the fourth-order Taylor polynomial of f around $x = 0$,

$$f(x) = f(0) + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + O(x^5).$$

(c) Determine **all** critical points of f.

(d) Decide, for each of the critical points $x = 0$ and $x = 1$, whether it is a local maximum, a local minimum, or neither.

Info: *This exercise is taken from the Analysis I & II exam of 9 August 2021 (Prof. Giovanni Felder), Part B, Problem 1. The original opens with a part asking whether f is continuously differentiable, bounded, and even.*

Solution: *See solution on Page 169*

Analytic functions (Section 11.c)

Just as in one variable, we can ask when a smooth function is exactly equal to its Taylor series. This requires a bound on the growth of its derivatives.

Definition 11.c.42 (*Analytic estimates*): Let $U \subseteq \mathbb{R}^n$ be open. A smooth function $f \in C^\infty(U, \mathbb{R})$ satisfies "*analytic estimates*" if for every $x_0 \in U$, there exist $\rho > 0$ and $C > 0$ such that for all multi-indices $\alpha \in \mathbb{N}^n$,

$$\sup_{x \in B_\rho(x_0)} |\partial^\alpha f(x)| \leq C|\alpha|! \, (n\rho)^{-|\alpha|} .$$

Such functions are called "*real analytic*".

Theorem 11.c.43 (*Convergence of the Taylor series*): If $f \in C^\infty(U, \mathbb{R})$ satisfies analytic estimates, then its Taylor series converges to $f(x)$ near x_0. That is, for all $h \in B_\rho(0)$,

$$f(x_0 + h) = \sum_{k=0}^{\infty} \sum_{|\alpha|=k} \frac{\partial^\alpha f(x_0)}{\alpha!} h^\alpha .$$

Proof:

Let $|h| < \rho$, so the segment from x_0 to $x_0 + h$ lies in $B_\rho(x_0)$. By Theorem 11.b.19, the difference between $f(x_0 + h)$ and the k-th partial sum is $R_k(h) = \sum_{|\alpha|=k+1} \partial^\alpha f(x_0 + \theta h)\, h^\alpha/\alpha!$ for some $\theta \in (0, 1)$. The analytic estimate and the multinomial theorem $\sum_{|\alpha|=m} \frac{m!}{\alpha!} |h^\alpha| = \big(|h_1| + \cdots + |h_n|\big)^m$ give

$$|R_k(h)| \leq C\,(k+1)!\,(n\rho)^{-(k+1)} \sum_{|\alpha|=k+1} \frac{|h^\alpha|}{\alpha!} = C \left(\frac{|h_1| + \cdots + |h_n|}{n\rho} \right)^{k+1} .$$

By Cauchy–Schwarz $|h_1| + \cdots + |h_n| \leq \sqrt{n}\,|h| < \sqrt{n}\,\rho \leq n\rho$, so the base of the power is a number $q < 1$, and $|R_k(h)| \leq C q^{k+1} \to 0$ as $k \to \infty$. The partial sums therefore converge to $f(x_0 + h)$. Q.E.D.

The factor n in the estimate is what the multinomial theorem consumes: it turns the sum over all $\binom{k+n}{n-1}$ multi-indices of order $k+1$ into a single power.

Example 11.c.44 (*An analytic function*): The function $f(x, y) = \sin(xy)$ satisfies the analytic estimates everywhere, and is therefore equal to its Taylor series.

Corollary 11.c.45 (*Unique continuation principle*): Let $U \subseteq \mathbb{R}^n$ be connected and open, and let f, g be real analytic on U (Definition 11.c.42). If $\partial^\alpha f(x_0) = \partial^\alpha g(x_0)$ for every multi-index α at some single point $x_0 \in U$, then $f = g$ on all of U. In particular, if f and g agree on a non-empty open subset $V \subseteq U$, they must agree on all of U.

Proof:

Let $U' := \{x \in U : \partial^\alpha f(x) = \partial^\alpha g(x) \text{ for all } \alpha \in \mathbb{N}^n\}$; by hypothesis $x_0 \in U'$, so U' is non-empty. Each $\partial^\alpha(f-g)$ is continuous, so

$$U \setminus U' = \bigcup_{\alpha\in\mathbb{N}^n} \big(\partial^\alpha(f-g)\big)^{-1}\big((-\infty,0)\cup(0,\infty)\big)$$

is a union of open sets, hence open. To see U' is also open, let $x \in U'$; since f and g are analytic at x, Theorem 11.c.43 gives $\rho > 0$ (the smaller of the two radii for f and g at x) such that both are equal to their Taylor series on $B_\rho(x)$; but those Taylor series agree term by term, since $x \in U'$, so $f = g$ on all of $B_\rho(x)$, giving $\partial^\alpha f = \partial^\alpha g$ throughout $B_\rho(x)$ (differentiate the identity $f \equiv g$ there) and hence $B_\rho(x) \subseteq U'$.

So, U' and $U \setminus U'$ are both open, U' is non-empty, and U is connected. Connectedness therefore forces $U \setminus U' = \emptyset$, that is, $U' = U$. In particular $f = g$ on all of U (the case $\alpha = 0$).

For the last statement, if $f = g$ on a non-empty open $V \subseteq U$, pick any $x_0 \in V$: then $\partial^\alpha f(x_0) = \partial^\alpha g(x_0)$ for every α (differentiating the identity $f \equiv g$ near x_0), so the first statement applies.

Q.E.D.

Remark 11.c.46: This is the payoff that makes real-analyticity worth having beyond C^∞: an analytic function is determined, on its **entire** connected domain, by its behaviour arbitrarily close to a single point. Knowing every derivative at one x_0, or merely agreeing with another analytic function on a tiny open patch, pins the function down everywhere. This is dramatically false for merely C^∞ functions: the standard bump function $\varphi(x) := e^{-1/x^2}$ for $x > 0$, $\varphi(x) := 0$ for $x \le 0$, has $\partial^\alpha\varphi(0) = 0$ for every α, agreeing to all orders with the zero function at 0, yet $\varphi \not\equiv 0$ on any neighbourhood of 0. That is exactly the failure Corollary 11.c.45 rules out for analytic functions.

Solutions (Section 11.d)

Three of this chapter's solutions are not collected here but sit beside their exercises, namely those to Exercise 11.a.10 (Page 154), Exercise 11.b.26 (Page 159) and Exercise 11.b.33 (Page 162). That is four solutions rather than three, because Exercise 11.b.26 carries two, a cumbersome one and an easy one.

Solution to Exercise 11.b.15:

(a) Every term in f is a composition of C^2 functions (polynomials, exponentials, sine), and the denominator $2 + \sin(x)$ is bounded away from zero. Thus f is C^2.

(b) Since $f \in C^2$, by Schwarz's theorem (Theorem 11.b.13) we have $\partial_2\partial_1 f = \partial_1\partial_2 f$. Computing ∂_1 of the second term would be tedious. However, notice that the second term depends only on x, so its derivative with respect to y is 0. Thus, we first compute $\partial_2 f$:

$$\partial_2 f(x,y) = x^2(2ye^{y^2}) + 0 = 2x^2ye^{y^2}.$$

Then we compute ∂_1 of this result:

$$\partial_1\partial_2 f(x,y) = \partial_1(2x^2ye^{y^2}) = 4xye^{y^2}.$$

By Schwarz's theorem, $\partial_2\partial_1 f(x,y) = 4xye^{y^2}$.

Q.E.D.

Remark 11.d.47 (*The monstrous term is the exercise*): The second term of f was not chosen to be differentiated. It was chosen so that differentiating it is out of the question, which leaves the reader with only two options: give up, or notice that it does not depend on y and let Theorem 11.b.13 take the derivatives in the other order. An exercise of this shape has a single decision in it, and taking the decision is the entire content.

Solution to Exercise 11.b.23:

Throughout, the tool is Corollary 11.b.21: an $o(|h|^k)$ fit by a polynomial of degree $\le k$ pins down every derivative of order $\le k$, and nothing beyond.

(a) Only $f(0) = 0$. The hypothesis is an O, not an o, so the corollary does not apply at any order, and that is the entire point of the item. Setting $x = 0$ in $|f(x) - x_1x_2| \le C|x|$ gives $|f(0)| \le 0$, so $f(0) = 0$. Nothing else is determined: both

$$f(x) := x_1x_2 \qquad \text{and} \qquad g(x) := x_1x_2 + 5x_1$$

satisfy the hypothesis, yet $\nabla f(0) = (0,0)$ while $\nabla g(0) = (5,0)$. An $O(|x|)$ remainder is free to hide a whole linear term, so not even the first derivatives survive.

Note also that the visible x_1x_2 is a red herring: it is $O(|x|^2)$, hence already absorbed by the stated error, and contributes nothing.

(b) Value and first derivatives; nothing further. Put $x_0 := (1,-1)$ and $h := (x_1, x_2)$. The hypothesis reads $f(x_0 + h) = 2 + h_1 + o(|h|)$, and $P(h) := 2 + h_1$ has degree 1, so by Corollary 11.b.21 with $k = 1$ it **is** the first-order Taylor polynomial:

$$f(1,-1) = 2, \qquad \partial_1 f(1,-1) = 1, \qquad \partial_2 f(1,-1) = 0.$$

Equivalently $\nabla f(1,-1) = (1,0)^{\mathsf{T}}$. Second derivatives are not determined: adding any quadratic to f preserves the hypothesis.

(c) Everything up to third order. Put $x_0 := (-1,0)$. Rearranging,

$$f(x_0 + h) = \underbrace{-2 + 2h_1h_2 - 2h_2^2 + 3h_2^3}_{=:\,P(h)} + o\big(|h|^3\big),$$

with $\deg P = 3$, so P is the third-order Taylor polynomial and every derivative of order ≤ 3 can be read off.

Order 0 and 1. $f(-1,0) = -2$, and P has no linear part, so $\nabla f(-1,0) = (0,0)^{\mathsf{T}}$, so the point is critical.

Order 2. The quadratic part of a Taylor expansion is $\frac{1}{2}h^{\mathsf{T}} \operatorname{Hess} f(x_0)h$, so $\frac{1}{2}h^{\mathsf{T}} \operatorname{Hess} f(-1,0)\, h = 2h_1h_2 - 2h_2^2$. Comparing coefficients in $h^{\mathsf{T}}Hh = H_{11}h_1^2 + 2H_{12}h_1h_2 + H_{22}h_2^2 = 4h_1h_2 - 4h_2^2$ gives

$$\operatorname{Hess} f(-1,0) = \begin{pmatrix} 0 & 2 \\ 2 & -4 \end{pmatrix}.$$

Its determinant is $-4 < 0$, so the Hessian is indefinite and, by Theorem 12.b.11, the critical point $(-1,0)$ is a **saddle**. Note that this was settled without ever differentiating f.

Order 3. A third-order term is $\partial^\alpha f(x_0)\, h^\alpha/\alpha!$. The only cubic monomial present is h_2^3, with $\alpha = (0,3)$ and $\alpha! = 6$, so $\partial_2^3 f(-1,0)/6 = 3$, giving $\partial_2^3 f(-1,0) = 18$; every other third derivative vanishes. Fourth-order derivatives are not determined.

Q.E.D.

Solution to Exercise 11.a.11:

No, and the official hint's function is the cleanest witness. Take

$$f(t) := \big(\sin(2\pi t),\ \cos(2\pi t)\big), \qquad f \in C^1(\mathbb{R}, \mathbb{R}^2).$$

Then $f(1) = (0,1) = f(0)$, so the left-hand side is

$$f(1) - f(0) = (0,0).$$

On the other hand $f'(t) = 2\pi\big(\cos(2\pi t),\ -\sin(2\pi t)\big)$, whose norm is

$$|f'(t)| = 2\pi\sqrt{\cos^2(2\pi t) + \sin^2(2\pi t)} = 2\pi \neq 0 \qquad \text{for every } t.$$

So $f'(t)$ never equals $(0,0)$, and no $t \in [0,1]$ can satisfy the claimed identity.

The reason is the one given in Important remark 11.a.2: applying the scalar mean value theorem componentwise produces one intermediate point **per component**,

$$f_1(1) - f_1(0) = f_1'(\xi_1), \qquad f_2(1) - f_2(0) = f_2'(\xi_2),$$

and nothing forces $\xi_1 = \xi_2$. Here $\xi_1 = \frac{1}{4}$ and $\xi_2 = \frac{1}{2}$ work componentwise. Both components do vanish, just not at the same time.

AI-Note 11.4: Note what does **not** fail. The **mean value inequality** $|f(1) - f(0)| \leq \sup_{t\in[0,1]} |f'(t)|$ holds here as $0 \leq 2\pi$, comfortably. That inequality is what survives into $\mathbb{R}^m$ and what every later estimate in this document actually uses; the **equality** is a one-dimensional accident. Geometrically the counterexample is a walker going once round a circle: they return to the start, so the total displacement is zero, but they never once stand still.

Q.E.D.

Solution to Exercise 11.b.34:

Throughout, recall the hint: *as* $x \to 0$ and *as* $|x| \to 0$ describe the same limit, so both phrasings are treated identically below. The recurring subtlety is instead between a statement holding **at** the single point $x = 0$ (which a remainder estimate as $x \to 0$ can establish) versus **around**/in a whole neighbourhood of $x = 0$ (which it cannot).

(a) **True.** Since $|x_2x_1^2| \leq |x| \cdot |x_1|^2 \leq |x|^3 = O(|x|^2)$ (as $|x|^3/|x|^2 = |x| \to 0$) and $O(x_1^2) = O(|x|^2)$ (since $x_1^2 \leq |x|^2$), the sum $f(x) = x_2x_1^2 + O(x_1^2)$ is a sum of two $O(|x|^2)$ terms, hence $f(x) = O(|x|^2)$.

(b) **False.** It suffices to show $x_2x_1^2$ itself is not $O(x_1^3)$. Along the path $x_1 = t^2$, $x_2 = t$ (so $x \to 0$ as $t \to 0$), $\dfrac{|x_2x_1^2|}{|x_1|^3} = \dfrac{t \cdot t^4}{t^6} = \dfrac{1}{t} \to \infty$, so no bound $|x_2x_1^2| \leq C|x_1|^3$ can hold near 0. Adding an $O(x_1^3)$ term cannot repair this, so $f(x) \neq O(x_1^3)$ in general.

(c) **True.** Since $x_2x_1 = O(|x|^2)$ and the hypothesis adds a further $O(|x|^2)$ term, $f(x) = O(|x|^2) = o(|x|)$ as $|x| \to 0$. Hence $f(x) = 0 \cdot x + o(|x|)$, i.e. f is differentiable **at** $x = 0$ with $Df(0) = 0$.

(d) **False.** The hypothesis only constrains the behaviour of f as $x \to 0$, i.e. at the single point 0; it says nothing about f at nearby points $x_0 \neq 0$, so twice differentiability **around** (in a neighbourhood of) $x = 0$ does not follow. Concretely, take $f(x) := \max\{0, x_1\}^2$: this satisfies $f(x) = x_2x_1 + O(|x|^2)$ as $x \to 0$ (indeed even $f(x) = O(|x|^2)$, since x_2x_1 itself is $O(|x|^2)$ and so contributes nothing extra to the hypothesis), yet along x_1 alone, $t \mapsto \max\{0, t\}^2$ has derivative $2\max\{0, t\}$, which is not differentiable at $t = 0$ (its one-sided difference quotients at 0 are 2 from the right and 0 from the left). So f is not twice differentiable at $x = 0$, let alone around it.

(e) **False**, for the same reason as **(d)**: even the sharper estimate $f(x) = x_2x_1 + O(|x|^3)$ only pins down the second-order Taylor data of f **at** the point $x = 0$, in the sense that the second-order Taylor **polynomial** is determined. It does not force twice differentiability in a whole neighbourhood **around** $x = 0$, and it does not even produce a second partial derivative **at** 0. Without an a priori assumption such as $f \in C^2$ near 0, one can cook up genuinely pathological examples, e.g. $f(x) := x_1x_2 + |x|^4 \sin(|x|^{-100})$, which does satisfy $f(x) = x_1x_2 + O(|x|^3)$ and yet has, along the x_1-axis and for $t > 0$,

$$\partial_1 f(t, 0) = 4t^3 \sin\big(t^{-100}\big) - 100\, t^{-97} \cos\big(t^{-100}\big).$$

So $\partial_1 f$ is not even continuous at 0; and since $\partial_1 f(0) = 0$, the difference quotient $\partial_1 f(t, 0)/t$ carries the unbounded term $-100\, t^{-98} \cos(t^{-100})$ and has no limit, so $\partial_{11} f(0)$ does not exist.

Q.E.D.

Solution to Exercise 11.b.35:

The answer is (a): degree 3, at a cost of 20 CHF.

The hypothesis $f(x) = x_1 + x_2x_1 + O(|x|^4)$ says two things at once: $f(0) = 0$, and f is pinned down exactly up to order 3 (its degree-3 part happens to be zero). Expanding g about $f(0) = 0$,

$$(g \circ f)(x) = \sum_{k\geq 0} \frac{g^{(k)}(0)}{k!}\, f(x)^k.$$

Since $f = O(|x|)$, we have $f^k = O(|x|^k)$, so every term with $k \geq 4$ is already $O(|x|^4)$ and invisible at order 3. The unknown $O(|x|^4)$ inside f likewise only perturbs the answer at order 4. So the degree-3 Taylor polynomial of $g \circ f$ is determined, and no more than that is: nothing in the data constrains the degree-4 coefficients of f.

Concretely, truncating $f = x_1 + x_1x_2$ at order 3 gives $f^2 = x_1^2 + 2x_1^2x_2 + O(|x|^4)$ and $f^3 = x_1^3 + O(|x|^4)$, so

$$(g \circ f)(x) = g(0) + g'(0)\big(x_1 + x_1x_2\big) + \frac{g''(0)}{2}\big(x_1^2 + 2x_1^2x_2\big) + \frac{g'''(0)}{6}\, x_1^3 + O(|x|^4).$$

That uses the four values $g(0), g'(0), g''(0), g'''(0)$, so four queries at 5 CHF each, giving **20 CHF**. Options (b) and (c) under-pay and cannot reach degree 3; (d) claims degree 3 for three queries, which is one derivative short. Q.E.D.

Solution to Exercise 11.b.36:

Put $R := P - Q = \sum_\alpha c_\alpha X^\alpha$ with $c_\alpha := a_\alpha - b_\alpha$, so the hypothesis reads $|R(x)| \leq M|x|^\sigma$ for $|x| \leq 1$ and the claim is that $c_\alpha = 0$ whenever $|\alpha| < \sigma$.

Suppose not, and let

$$d := \min\{\, |\alpha| \;:\; c_\alpha \neq 0 \,\} < \sigma.$$

Split $R = R_d + \widetilde{R}$, where $R_d := \sum_{|\alpha|=d} c_\alpha X^\alpha$ is the homogeneous part of degree d, non-zero by the choice of d, and $\widetilde{R}$ collects the terms of degree $> d$, so that $\widetilde{R}(tu) = O(t^{d+1})$ uniformly over unit vectors u.

Fix a unit vector u and evaluate at $x := tu$ for $t \in (0, 1]$. Homogeneity gives $R_d(tu) = t^d R_d(u)$, so

$$\big|t^d R_d(u) + O(t^{d+1})\big| \;=\; |R(tu)| \;\leq\; M\, t^\sigma,$$

and dividing by $t^d > 0$,

$$\big|R_d(u) + O(t)\big| \;\leq\; M\, t^{\sigma - d}.$$

Here is where the hypothesis $d < \sigma$ is spent: the exponent $\sigma - d$ is **strictly positive**, so the right-hand side tends to 0 as $t \to 0^+$, and the left-hand side tends to $|R_d(u)|$. Hence $R_d(u) = 0$ for every unit vector u, and by homogeneity $R_d \equiv 0$ on all of $\mathbb{R}^n$.

A polynomial vanishing identically on $\mathbb{R}^n$ has all coefficients zero, so $c_\alpha = 0$ for every $|\alpha| = d$, contradicting the choice of d. Therefore no such d exists, i.e. $a_\alpha = b_\alpha$ for all $|\alpha| < \sigma$.

AI-Note 11.5: The statement is sharp in the obvious way: coefficients of order exactly σ need not agree. With $n = 1$, $\sigma = 2$, $P(x) := x^2$ and $Q := 0$ we get $|P - Q| = |x|^2 \leq 1 \cdot |x|^2$, and the order-2 coefficients differ. The argument above breaks at precisely the right place: for $d = \sigma$ the bound becomes $|R_d(u) + O(t)| \leq M$, which says nothing.

Q.E.D.

Solution to Exercise 11.b.37:

Using the 1D expansion $\sqrt{1+t} = 1 + \frac{1}{2}t - \frac{1}{8}t^2 + O(t^3)$ with $t = x + y^2$, we obtain:

$$\begin{aligned}\sqrt{1+x+y^2} &= 1 + \frac{1}{2}(x+y^2) - \frac{1}{8}(x+y^2)^2 + O(|(x,y)|^3) \\ &= 1 + \frac{1}{2}x + \frac{1}{2}y^2 - \frac{1}{8}(x^2 + 2xy^2 + y^4) + o(|(x,y)|^2) \\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{2}y^2 + o(|(x,y)|^2).\end{aligned}$$

Note that $t = x + y^2$ is of order $|h|$ and not of order $|h|^2$, because of its x. That is why the t^2 term has to be carried and expanded, and why $2xy^2$ and y^4 are then dropped from it. **Q.E.D.**

Solution to Exercise 11.b.38:

Write $s := x - y$, itself of order $|h|$. Since arctan is odd, $\arctan(t) = t + O(t^3)$, so

$$\arctan(s) = s + O(|h|^3) = s + o(|h|^2).$$

Since exp is smooth and bounded near 0, an $o(|h|^2)$ correction inside exp contributes only an $o(|h|^2)$ correction to the output. Consequently, to order 2, $\exp(\arctan(s))$ and $\exp(s)$ agree:

$$\exp(\arctan(x-y)) = \exp(x-y) + o(|(x,y)|^2).$$

Now $\exp(s) = 1 + s + \frac{1}{2}s^2 + O(s^3)$ with $s^2 = x^2 - 2xy + y^2$, so

$$\exp(\arctan(x-y)) = 1 + x - y + \tfrac{1}{2}x^2 - xy + \tfrac{1}{2}y^2 + o(|(x,y)|^2).$$

Q.E.D.

Solution to Exercise 11.b.39:

Using $\frac{1}{1-t} = 1 + t + O(t^2)$ with $t := x^2 + y^2$, itself already $O(|h|^2)$, so $t^2 = O(|h|^4) = o(|h|^2)$:

$$\frac{1}{1-x^2-y^2} = 1 + (x^2+y^2) + o(|(x,y)|^2).$$

Q.E.D.

Remark 11.d.48 (*The substitution's own order decides how far to expand*): Read the three together and one number governs all of them, namely the order of the substituted quantity t itself. For $1/(1-x^2-y^2)$ we have $t = x^2 + y^2 = O(|h|^2)$, so a single term of the geometric series already exhausts order 2 and t^2 is invisible. For $\sqrt{1+x+y^2}$ the same t carries an x, drops to order $|h|$, and consequently the quadratic term of the series has to be expanded in full. The habit worth taking away is to ask what order t is **before** deciding how many terms of the one-variable series to write down.

Proof of Exercise 11.b.41:

(a) The integrand is continuous and both limits are differentiable in x, so by the Leibniz rule together with the fundamental theorem of calculus,

$$f'(x) = \frac{d}{dx}\int_{x-1}^{x+1} \cos(\pi t^2)\,dt = \cos\left(\pi(x+1)^2\right) - \cos\left(\pi(x-1)^2\right).$$

Since $(x \pm 1)^2 = x^2 \pm 2x + 1$ and $\cos(\theta + \pi) = -\cos\theta$, each term simplifies:

$$\cos\left(\pi(x^2+1) \pm 2\pi x\right) = -\cos(\pi x^2 \pm 2\pi x).$$

Therefore $f'(x) = \cos(\pi x^2 - 2\pi x) - \cos(\pi x^2 + 2\pi x)$, and the identity $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ collapses this to

$$f'(x) = 2\sin(\pi x^2)\sin(2\pi x).$$

(b) Expand each factor as $x \to 0$: $\sin(2\pi x) = 2\pi x - \frac{4\pi^3}{3}x^3 + O(x^5)$ and $\sin(\pi x^2) = \pi x^2 + O(x^6)$. Multiplying,

$$f'(x) = 2\big(\pi x^2 + O(x^6)\big)\left(2\pi x - \frac{4\pi^3}{3}x^3 + O(x^5)\right) = 4\pi^2 x^3 + O(x^5).$$

Integrating from 0,

$$f(x) = f(0) + \int_0^x \big(4\pi^2 t^3 + O(t^5)\big)\,dt = f(0) + \pi^2 x^4 + O(x^6),$$

where $f(0) = \int_{-1}^{1} \cos(\pi t^2)\,dt = 2\int_0^1 \cos(\pi t^2)\,dt$. Hence $a_1 = a_2 = a_3 = 0$ and $a_4 = \pi^2$.

(c) By part **(a)**, $f'(x) = 0$ if and only if one of the two sine factors vanishes. Now $\sin(\pi x^2) = 0$ exactly when x^2 is a non-negative integer, and $\sin(2\pi x) = 0$ exactly when $2x$ is an integer. The set of critical points is therefore

$$\big\{\pm\sqrt{n} \;\big|\; n \in \mathbb{Z}_{\geq 0}\big\} \cup \tfrac{1}{2}\mathbb{Z}.$$

The two families overlap exactly at the integers: an integer m is $\pm\sqrt{m^2}$, and a half-integer that is not an integer has a non-integer square. So the critical points $x = \pm 1, \pm 2, \ldots$ are "*double*" zeros of f' (and $x = 0$ a triple one), and those are the ones where the classification changes character.

(d) **At $x = 0$:** by part **(b)**, $f(x) - f(0) = \pi^2 x^4 + O(x^6) > 0$ for all small $x \neq 0$. Equivalently, $f'(x) = 4\pi^2 x^3 + O(x^5)$ changes sign from negative to positive. Either way, $x = 0$ is a strict local **minimum** (and $f^{(4)}(0) = 4!\,a_4 = 24\pi^2 > 0$).

At $x = 1$: write $x = 1 + h$. Then $\sin(\pi x^2) = \sin\big(\pi + \pi(2h + h^2)\big) = -\sin\big(\pi(2h + h^2)\big) = -2\pi h + O(h^2)$ and $\sin(2\pi x) = \sin(2\pi h) = 2\pi h + O(h^3)$, so

$$f'(1 + h) = 2(-2\pi h + O(h^2))(2\pi h + O(h^3)) = -8\pi^2 h^2 + O(h^3).$$

This is strictly negative for all small $h \neq 0$, on **both** sides. So f is decreasing through $x = 1$ with a horizontal tangent: the point is **neither** a maximum nor a minimum. Reading the coefficients off the expansion gives $f''(1) = 0$ and $f'''(1) = -16\pi^2$, which is the value the official solution reports.

Remark 11.d.49: The contrast between the two points is the whole content of part **(d)**. Both are critical, and at both the second derivative vanishes, so the Hessian test says nothing at either. What decides the question is the **order** of the first non-vanishing derivative: odd order at $x = 1$ (no extremum), even order at $x = 0$ (extremum, with the sign of the coefficient choosing which).

Remark 11.d.50 (*A shortcut for the odd coefficients*): The vanishing of a_1 and a_3 needs no computation at all. Substituting $s = -t$ shows $f(-x) = f(x)$, so f is even, and the Taylor coefficients of an even function at 0 vanish in every odd degree. This is the route the official solution takes, and it is worth preferring: it settles two of the four coefficients before a single derivative is computed.

AI-Note 11.6: Cross-checked against the official solution, which agrees on every value: $a_1 = a_2 = a_3 = 0$, $a_4 = \pi^2$, $f^{(4)}(0) = 24\pi^2$, the two families $\pm\sqrt{n}$ and $\frac{1}{2}\mathbb{Z}$ of critical points, and $f'''(1) = -16\pi^2$. One divergence is presentational, and is flagged here so that the two can be read side by side: the official solution displays the product formula as $2\sin(\pi(x^2 + 1))\sin(2\pi x)$, which is $-f'(x)$ rather than $f'(x)$, since $\sin(\pi(x^2 + 1)) = -\sin(\pi x^2)$. It comes from applying $2\sin a \sin b = \cos(a - b) - \cos(a + b)$ to $\cos(\pi(x - 1)^2) - \cos(\pi(x + 1)^2)$, which is f' with its two terms interchanged. The sign never surfaces in that solution's conclusions, because it uses the formula only to locate zeros and then classifies $x = 0$ and $x = 1$ from $f^{(4)}(0)$ and $f'''(1)$ instead. It would matter to anyone running a sign argument off the displayed identity, as the classification of $x = 1$ above does.

Q.E.D.

Solution to Exercise 11.a.12:

No, f is not necessarily bounded on U.

The Mean Value Inequality (Theorem 11.a.3) asserts that if the straight line segment $[x, y]$ lies in U, then $|f(x) - f(y)| \leq M\|x - y\|$. On a convex or star-shaped domain, this bounds $|f(x) - f(y)|$ by $M \operatorname{diam}(U)$. But on a general non-convex domain, points x, y that are close in $\mathbb{R}^2$ may only be connectable by paths inside U of immense length, or not connectable at all.

Counterexample: the spiral corridor. In polar coordinates (r, θ) on $\mathbb{R}^2 \setminus \{0\}$, define the open set

$$U := \left\{ (r\cos\theta, r\sin\theta) \in \mathbb{R}^2 \;\middle|\; \theta > 2,\ 1 - \frac{1}{\theta} < r < 1 - \frac{1}{\theta + \pi} \right\}.$$

(a) U is bounded and open: For all points in U, $\frac{1}{2} < 1 - \frac{1}{2} < r < 1$, so $U \subset B_1(0)$ is bounded in $\mathbb{R}^2$. It is open because it is the pre-image of the open set $\{(\theta, r) : \theta > 2, 1 - 1/\theta < r < 1 - 1/(\theta + \pi)\}$ under the smooth local diffeomorphism $(\theta, r) \mapsto (r\cos\theta, r\sin\theta)$.

(b) No self-intersection: At angles differing by $2\pi k$ ($k \in \mathbb{N}_{\geq 1}$), the radial interval for $\theta + 2\pi k$ is strictly higher than for θ:

$$1 - \frac{1}{\theta + \pi} < 1 - \frac{1}{\theta + 2\pi} < 1 - \frac{1}{\theta + 2\pi k}.$$

Thus successive coils of the spiral never overlap, and each point in U corresponds to a unique angle θ.

(c) Unbounded function with bounded gradient: Define $f : U \to \mathbb{R}$ by

$$f(r\cos\theta, r\sin\theta) := \sqrt{\theta}.$$

Since f depends only on the angular coordinate θ, its gradient in orthogonal polar coordinates is

$$\nabla f = \frac{\partial f}{\partial r} e_r + \frac{1}{r}\frac{\partial f}{\partial \theta} e_\theta = 0 \cdot e_r + \frac{1}{r}\frac{1}{2\sqrt{\theta}} e_\theta.$$

Since $r > \frac{1}{2}$ and $\theta > 2$ on U, the norm of the gradient satisfies

$$\|\nabla f(x)\| = \frac{1}{2r\sqrt{\theta}} \leq \frac{1}{2 \cdot \frac{1}{2} \cdot \sqrt{2}} = \frac{1}{\sqrt{2}} < 1 \qquad \text{for all } x \in U.$$

Thus $\|\nabla f\|$ is uniformly bounded on U by $M = 1$. However, as $\theta \to \infty$ within U, $f(x) = \sqrt{\theta} \to +\infty$. Therefore f is unbounded on the bounded open set U.

Remark 11.d.51 (*Intrinsic metric versus Euclidean metric*): The failure occurs because the **intrinsic path distance** $d_U(x, y)$ inside U (the infimum of lengths of paths joining x and y within U) is not comparable to the Euclidean distance $\|x - y\|$. Integrating $\|\nabla f\| \leq M$ along a curve γ inside U yields $|f(x) - f(y)| \leq M \operatorname{Length}(\gamma)$, which controls variation only in terms of intrinsic path length. Because the spiral wraps infinitely many times inside $B_1(0)$, its intrinsic diameter is infinite while its Euclidean diameter is at most 2.

Q.E.D.

PART IV

Optimization and Convexity

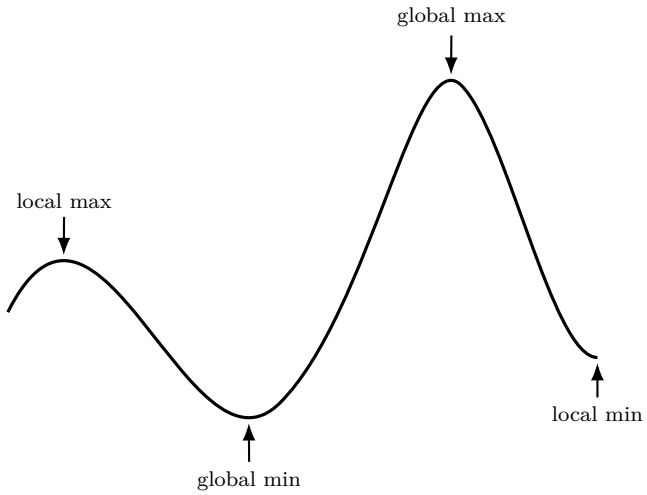
Differentiability turns the search for extrema into algebra. An interior extremum forces the gradient to vanish, so the candidates can be found by solving equations, and the sign behaviour of the Hessian then decides which kind of point each candidate is. Two complications occupy the rest of the part. When the variables are constrained to a level set, the gradient need not vanish at an extremum at all, and the method of Lagrange multipliers supplies the condition that replaces it, together with a hypothesis on the constraint gradients that is easy to drop and expensive to lose. When the Hessian test comes out inconclusive, or when the question asked is global rather than local, convexity is the hypothesis that settles it, because for a convex function a critical point is not a candidate but an answer. It is worth watching what optimization gets used **for** along the way. Two of the theorems proved in this part are not optimization problems at all: the fundamental theorem of algebra, which falls out of minimising $|p|$, and the spectral theorem, which falls out of maximising a quadratic form on the unit sphere.

Extrema and the Hessian Test (Chapter 12)

The Taylor expansion of the previous chapter stops being a formal exercise the moment one asks what it is **for**. Its first-order term locates the candidates for a maximum or a minimum, namely the points where the gradient vanishes. Its second-order term decides which of them are which. Carrying out the second half is the business of the Hessian test, and most of the work turns out to be reading the sign of a quadratic form off a matrix. A closing section asks what is left when both orders come out inconclusive, and answers it with the fundamental theorem of algebra.

Optimization (Section 12.a)

Four kinds of point on one graph, and telling them apart is the whole vocabulary of this section. Two of them are extreme only in comparison with their immediate neighbours, and two are extreme against the entire domain.



Definition 12.a.1 (*Local extrema and critical points*):

- "*Local minimum*" (German: „*lokales Minimum*") of $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$: f has a local minimum at $x_0 \in U$ if
$$\exists\, \varepsilon > 0 \text{ such that } \forall x \in B_\varepsilon(x_0) \cap U : f(x) \geq f(x_0).$$
- **Local maxima** are defined analogously with $f(x) \leq f(x_0)$.
- The extremum is **strict** if the inequality is strict for $x \neq x_0$: $f(x) > f(x_0)$ for a strict minimum, $f(x) < f(x_0)$ for a strict maximum.
- A "*critical point*" (German: „*kritischer Punkt*") of $f \in C^1(U, \mathbb{R})$, $U \subseteq \mathbb{R}^n$ open, is a point $x_0 \in U$ where $\nabla f(x_0) = 0$.

Proposition 12.a.2 (*Local extremum $\implies$ critical point*): If $f \in C^1(U, \mathbb{R})$ and $x_0 \in \operatorname{int}(U)$ is a local extremum, then $\nabla f(x_0) = 0$.

Proof:

Fix a direction $v \in \mathbb{R}^n$ and consider the one-variable function $\varphi(t) := f(x_0 + tv)$, defined for $|t|$ small enough that $x_0 + tv$ stays in U. This is where $x_0 \in \operatorname{int}(U)$ is used, and it is the only place. If f has a local extremum at x_0, then φ has one at the **interior** point $t = 0$, so by the one-variable result from Analysis I, $\varphi'(0) = 0$. By the chain rule along a path (Section 10.d),
$$0 = \varphi'(0) = \langle \nabla f(x_0), v \rangle.$$

Since v was arbitrary, $\nabla f(x_0)$ is orthogonal to every vector of $\mathbb{R}^n$, including itself: $\langle \nabla f(x_0), \nabla f(x_0) \rangle = |\nabla f(x_0)|^2 = 0$, hence $\nabla f(x_0) = 0$. Q.E.D.

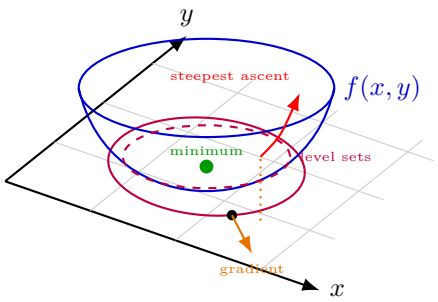
Remark 12.a.3 (*Intuition, and two standing warnings*): The proof says: at an interior extremum every direction is a *"flat"* direction, and the only vector orthogonal to everything is the zero vector,

$$\langle \nabla f(x_0), v \rangle = 0 \text{ for all } v \in \mathbb{R}^n \iff \nabla f(x_0) = 0.$$

Two things this proposition does **not** say, both of which the rest of the chapter is built around.

- **The converse is false.** A critical point need not be an extremum: $f(x, y) := x^2 - y^2$ has $\nabla f(0,0) = 0$ but a saddle there, and even in one variable $f(x) := x^3$ has $f'(0) = 0$ with no extremum. Critical points are **candidates**; separating them is exactly what Theorem 12.b.11 is for.
- **Interiority is essential.** On $U := [0,1]$ the function $f(x) := x$ attains both its minimum and its maximum, at $x = 0$ and $x = 1$, and $f' \equiv 1$ never vanishes. The hypothesis $x_0 \in \operatorname{int}(U)$ is what fails. This is why every optimization over a closed region in this chapter is done in two steps. Interior critical points come first, and the boundary is then treated separately, by the method of Lagrange multipliers.

Geometrically, the proposition says that at an interior extremum the direction of steepest ascent ceases to exist. Anywhere else the gradient is a genuine non-zero vector, perpendicular to the level set of f through the point and pointing the way f grows fastest. That perpendicularity is worth remembering past this chapter, since it is the whole geometric content of the Lagrange condition in Chapter 13.



So, the gradient produces a list of candidates and then falls silent. Which of them are minima, which are maxima, and which are neither? The first derivative cannot say, since it vanishes at all of them alike, so the question has to be put to the second.

The Hessian test (Section 12.b)

The **Hessian** (Remark 11.b.30) of $f \in C^2(U, \mathbb{R})$, $U \subseteq \mathbb{R}^n$, is given by

$$\operatorname{Hess} f(x) = (\partial_i \partial_j f)_{i,j=1,\dots,n}$$

with respect to the standard basis of $\mathbb{R}^n$. By Schwarz's theorem (Theorem 11.b.13), $\operatorname{Hess} f(x)$ is symmetric.

Remark 12.b.4 (*Why symmetry is proved here*): Everything in this section rests on symmetry of the Hessian: the spectral theorem in Remark 12.b.8, the eigenvalue criteria, the Hurwitz minors. Yet Theorem 11.b.13 was quoted without proof where it first appeared, so the proof is supplied now. It comes with the attempt that fails, which is the more informative half.

Proof of Theorem 11.b.13:

It suffices to treat $m = 1$ (apply the result to each component), $i < j$ (there is nothing to prove when $i = j$), and $n = 2$ with $i = 1$, $j = 2$ (freeze all the other variables). So, let $f \in C^2$ near (x, y) and write ∂_1, ∂_2 for the two partial derivatives.

The attempt that fails. Unwinding both mixed derivatives as iterated limits gives

$$\partial_2\partial_1 f(x,y) = \lim_{h_2\to 0}\lim_{h_1\to 0} \frac{f(x+h_1,y+h_2) - f(x+h_1,y) - f(x,y+h_2) + f(x,y)}{h_1 h_2},$$

and one would like to apply the mean value theorem in each variable and be done. The obstruction is that the two intermediate points the mean value theorem returns depend on h_1 and h_2 **separately** and need not agree, so forcing them together leaves a stray factor h_2/h_1, which blows up when $h_1 \to 0$ at fixed h_2. The symmetric-looking expression above is not enough on its own; one has to keep the two increments coupled.

The proof. Couple them by fixing a single $h \neq 0$ and setting

$$F(h) := f(x+h,y+h) - f(x+h,y) - \big(f(x,y+h) - f(x,y)\big),$$

which is the second difference of f over a square of side h, and $\varphi(t) := f(x+th, y+h) - f(x+th, y)$, so that $F(h) = \varphi(1) - \varphi(0)$. The mean value theorem gives $\xi_1 \in (0,1)$ with

$$F(h) = \varphi'(\xi_1) = h\,\partial_1 f(x+\xi_1 h,\ y+h) - h\,\partial_1 f(x+\xi_1 h,\ y).$$

Now apply the mean value theorem a second time, to $\psi(t) := \partial_1 f(x+\xi_1 h,\ y+th)$. This is legitimate precisely because $f \in C^2$, so $\partial_1 f$ is itself continuously differentiable in the second variable. Since $F(h) = h\big(\psi(1) - \psi(0)\big)$, there is $\xi_2 \in (0,1)$ with

$$F(h) = h\,\psi'(\xi_2) = h^2\,\partial_2\partial_1 f(x+\xi_1 h,\ y+\xi_2 h).$$

Grouping the four terms of $F(h)$ the other way (first and third together, second and fourth together) runs the identical argument with the roles of the variables exchanged and produces $\tilde\xi_1, \tilde\xi_2 \in (0,1)$ with $F(h) = h^2\,\partial_1\partial_2 f(x+\tilde\xi_1 h,\ y+\tilde\xi_2 h)$. Dividing both expressions by $h^2 \neq 0$,

$$\partial_2\partial_1 f(x+\xi_1 h,\ y+\xi_2 h) \;=\; \partial_1\partial_2 f(x+\tilde\xi_1 h,\ y+\tilde\xi_2 h).$$

All four numbers $\xi_1, \xi_2, \tilde\xi_1, \tilde\xi_2$ lie in $(0,1)$. They depend on h, but they stay bounded, so both arguments converge to (x,y) as $h \to 0$. Continuity of the mixed second partials (again $f \in C^2$) lets us pass to the limit on both sides, giving $\partial_2\partial_1 f(x,y) = \partial_1\partial_2 f(x,y)$. Q.E.D.

Remark 12.b.5 (*What the failed attempt shows*): The two attempts differ in exactly one respect: the failed one lets h_1 and h_2 go to zero **independently**, the successful one ties them together as a single h and only takes a limit at the very end. That is why the second difference $F(h)$ over a **square** is the right object. It is symmetric in the two variables by construction, so the two groupings of its four terms compute the two mixed partials, and the symmetry of the answer is inherited from the symmetry of the square rather than proved by force.

It also locates precisely where continuity is used: twice, once to apply the mean value theorem to $\partial_1 f$ and once to pass to the limit at the end. Drop it and the conclusion genuinely fails. Remark 11.b.14 gives the standard counterexample.

Definition 12.b.6 (*Sign of a symmetric matrix*): A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is:

- "*positive definite*" (German: „*positiv definit*") if $v^\mathsf{T} A v > 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$;
- "*negative definite*" (German: „*negativ definit*") if $v^\mathsf{T} A v < 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$;
- **indefinite** if there exist $v, u \in \mathbb{R}^n \setminus \{0\}$ such that $v^\mathsf{T} A v > 0$ and $u^\mathsf{T} A u < 0$;
- **degenerate** (German: „*entartet*") if there exists $v \in \mathbb{R}^n \setminus \{0\}$ such that $Av = 0$; equivalently, if $\det A = 0$, or if 0 is an eigenvalue of A.

Remark 12.b.7 (*Degenerate means a nontrivial kernel*): It is tempting to call A degenerate when $v^\mathsf{T} A v = 0$ for some $v \neq 0$. That condition is much weaker than the definition above: if A is

indefinite, with $v^\mathsf{T}Av > 0$ and $u^\mathsf{T}Au < 0$, then $t \mapsto (tv + (1-t)u)^\mathsf{T}A(tv + (1-t)u)$ is continuous and changes sign, so it vanishes at some $w = tv + (1-t)u$, and $w \neq 0$ because u is not a negative multiple of v. Every indefinite matrix would then count as degenerate, and the distinction drawn in Theorem 12.b.11 would collapse. The right notion is a nontrivial kernel ($\det A = 0$), which is also what Exercise 12.b.22 uses.

Remark 12.b.8 (*Linear algebra interlude: diagonalizing the Hessian*): Since the Hessian $\operatorname{Hess} f(x_0)$ is a symmetric matrix, it is diagonalizable over $\mathbb{R}$ by the spectral theorem. There exist n real eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ and a corresponding orthonormal basis of eigenvectors $v_1, \dots, v_n \in \mathbb{R}^n$ such that $\operatorname{Hess} f(x_0)v_i = \lambda_i v_i$ for all i.

Every vector $x \in \mathbb{R}^n$ can be written in this basis as $x = \sum_{i=1}^n a_i v_i$ with $a_i \in \mathbb{R}$. Applying the Hessian yields:

$$\operatorname{Hess} f(x_0)x = \operatorname{Hess} f(x_0) \sum_{i=1}^n a_i v_i = \sum_{i=1}^n a_i \operatorname{Hess} f(x_0)v_i = \sum_{i=1}^n \lambda_i a_i v_i.$$

In this eigenbasis, the Hessian matrix becomes diagonal. Consequently, the quadratic form evaluates to $x^\mathsf{T} \operatorname{Hess} f(x_0)x = \sum_{i=1}^n \lambda_i a_i^2$, which immediately links the definiteness of the Hessian to the signs of its eigenvalues.

Exercise 12.b.9 (*Polarisation formula*): Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth. Prove that the map

$$\mathbb{R}^n \ni e \longmapsto \left.\frac{d^2}{dt^2}\right|_{t=0} f(te) = e^\mathsf{T} \operatorname{Hess} f(0)\, e$$

determines all the second partial derivatives $\partial_i \partial_j f(0)$.

Proof of Exercise 12.b.9:

By the chain rule, $\frac{d}{dt} f(te) = \sum_i \partial_i f(te) e_i$, and differentiating again,

$$\left.\frac{d^2}{dt^2}\right|_{t=0} f(te) = \sum_{i,j} \partial_i \partial_j f(0)\, e_i e_j = e^\mathsf{T} \operatorname{Hess} f(0)\, e =: Q(e),$$

which is the displayed identity, with Q a quadratic form in e.

Taking $e := e_i$, the i-th standard basis vector, gives $Q(e_i) = \partial_i^2 f(0)$: the diagonal entries directly. For the off-diagonal entries, expand

$$Q(e_i + e_j) = (e_i + e_j)^\mathsf{T} \operatorname{Hess} f(0)(e_i + e_j) = Q(e_i) + 2\,\partial_i \partial_j f(0) + Q(e_j),$$

using symmetry of $\operatorname{Hess} f(0)$ (Theorem 11.b.13) to combine the two mixed terms into one. Solving,

$$\partial_i \partial_j f(0) = \tfrac{1}{2}\big(Q(e_i + e_j) - Q(e_i) - Q(e_j)\big),$$

so every entry of $\operatorname{Hess} f(0)$ is recovered from Q alone. Q.E.D.

Remark 12.b.10: This is the same phenomenon as recovering an inner product from its induced norm via the parallelogram-style polarisation identity: a symmetric bilinear form is determined by its values on the diagonal. Here it says that a single-variable computation, differentiating $t \mapsto f(te)$ twice at $t = 0$ for every direction e, already contains everything the full Hessian matrix does.

Theorem 12.b.11 (*The Hessian test*): If $f \in C^2(U, \mathbb{R})$ has a critical point $x_0 \in \operatorname{int}(U)$, then:

- $\operatorname{Hess} f(x_0)$ **positive definite** $\Rightarrow$ local **min** at x_0;
- $\operatorname{Hess} f(x_0)$ **negative definite** $\Rightarrow$ local **max** at x_0;
- $\operatorname{Hess} f(x_0)$ **indefinite** $\Rightarrow$ "*saddle point*" (German: „*Sattelpunkt*“) at x_0.

If $\operatorname{Hess} f(x_0)$ is degenerate but not indefinite (i.e. positive or negative semidefinite with 0 an eigenvalue), the test gives **no information**, and all three behaviours are possible.

Remark 12.b.12 (*Indefiniteness needs no further hypothesis*): Indefiniteness alone forces a saddle. If $v^{\mathsf{T}} \operatorname{Hess} f(x_0) v > 0$ and $u^{\mathsf{T}} \operatorname{Hess} f(x_0) u < 0$, the second-order Taylor expansion makes $t \mapsto f(x_0 + tv)$ increase and $t \mapsto f(x_0 + tu)$ decrease away from x_0, so x_0 is neither a local minimum nor a local maximum. The lecture script states this case as *"indefinite and non-degenerate"* (Prop. 11.15(3)), one clause more than it needs. The genuinely inconclusive case is the other one, semidefinite **and** degenerate, and it is the one students actually run into: Exercise 12.b.24 is exactly this trap.

Remark 12.b.13 (*Why positive definite means minimum*): The statement is easy to memorise and easy to find unmotivated: it is not obvious what a condition on $v^{\mathsf{T}} \operatorname{Hess} f(x_0) v$ has to do with f going up. The following two-step reading, which recovers the one-dimensional test as the case $n = 1$, makes the connection direct.

Step one: the Hessian moves the gradient. The entries of $\operatorname{Hess} f$ are derivatives of the entries of ∇f, so the Hessian is to ∇f what ∇f is to f. Applying the linear approximation to ∇f at the critical point, where $\nabla f(x_0) = 0$,

$$\nabla f(x_0 + v) \approx \underbrace{\nabla f(x_0)}_{=0} + \operatorname{Hess} f(x_0)\, v = \operatorname{Hess} f(x_0)\, v.$$

So, once we step off x_0 in the direction v, the gradient we find there is approximately $\operatorname{Hess} f(x_0) v$.

Step two: the gradient moves f. The gradient measures the rate of change of f, so travelling the displacement v against a gradient of $\operatorname{Hess} f(x_0) v$ changes f by roughly the inner product of the two,

$$f(x_0 + v) - f(x_0) \;\approx\; \tfrac{1}{2}\, v^{\mathsf{T}} \operatorname{Hess} f(x_0)\, v,$$

the factor $\frac{1}{2}$ appearing because the gradient grows linearly from 0 along the way, so its average over the step is half its final value. This is of course the second-order Taylor expansion of Remark 11.b.30 with the first-order term deleted by criticality; the point of the derivation is that it explains **why** that expansion has the form it does.

Now the theorem reads itself. If $\operatorname{Hess} f(x_0)$ is positive definite, then $v^{\mathsf{T}} \operatorname{Hess} f(x_0) v > 0$ for every $v \neq 0$, so f increases in **every** direction away from x_0, and that is a strict local minimum. Negative definite reverses every inequality. Indefinite means some directions give a positive value and others a negative one, so f rises along some and falls along others: a saddle. The degenerate-but-not-indefinite case is now inconclusive for a visible reason. Along an eigenvector with eigenvalue 0 the quadratic term vanishes entirely, so the approximation says nothing and the behaviour is decided by terms this expansion has discarded.

Proof of Theorem 12.b.11:

Write $A := \operatorname{Hess} f(x_0)$ and choose $r > 0$ with $B_r(x_0) \subseteq U$. For $|h| < r$, Taylor's theorem with Lagrange remainder (Theorem 11.b.19 with $k = 1$) and $\nabla f(x_0) = 0$ give a $\theta \in (0, 1)$ with

$$f(x_0 + h) - f(x_0) = \tfrac{1}{2}\, h^{\mathsf{T}} \operatorname{Hess} f(x_0 + \theta h)\, h. \tag{12.1}$$

- **Positive definite.** The function $v \mapsto v^{\mathsf{T}} A v$ is continuous on the unit sphere S^{n-1}, which is compact (Theorem 7.b.28), so it attains a minimum λ there, and $\lambda > 0$ because A is positive definite. By homogeneity, $h^{\mathsf{T}} A h \geq \lambda |h|^2$ for all h. The second partial derivatives are continuous at x_0, so there is $\delta \in (0, r)$ with $\|\operatorname{Hess} f(y) - A\|_2 \leq \lambda/2$ whenever $|y - x_0| < \delta$, where $\|\cdot\|_2$ is the Hilbert–Schmidt norm. For $0 < |h| < \delta$ and $B := \operatorname{Hess} f(x_0 + \theta h)$, Lemma 2.a.4 gives $|h^{\mathsf{T}}(B - A)h| \leq |h|\,|(B - A)h| \leq \frac{\lambda}{2}|h|^2$, hence
$$h^{\mathsf{T}} B h \;\geq\; h^{\mathsf{T}} A h - \tfrac{\lambda}{2}|h|^2 \;\geq\; \tfrac{\lambda}{2}|h|^2 \;>\; 0.$$
By (12.1), $f(x_0 + h) > f(x_0)$: a strict local minimum.
- **Negative definite.** Apply the previous case to $-f$, whose Hessian is $-A$.
- **Indefinite.** Pick v, u with $v^{\mathsf{T}} A v > 0 > u^{\mathsf{T}} A u$. The function $g(t) := f(x_0 + tv)$ is C^2 near 0 with $g'(0) = \langle \nabla f(x_0), v \rangle = 0$ and $g''(0) = v^{\mathsf{T}} A v > 0$ (chain rule twice), so by the one-variable test g has a strict local minimum at 0: $f(x_0 + tv) > f(x_0)$ for all small $t \neq 0$. The same

argument along u gives $f(x_0+tu) < f(x_0)$ for all small $t \neq 0$. Every ball around x_0 therefore contains points with larger and with smaller values, so x_0 is neither a local minimum nor a local maximum, which is what a saddle point is.

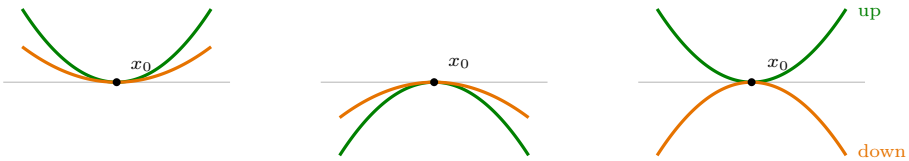
For the degenerate case, the three functions x^4+y^2, $-x^4-y^2$ and x^3+y^2 all have the semidefinite Hessian $\operatorname{diag}(0,2)$ or $\operatorname{diag}(0,-2)$ at the origin and exhibit a minimum, a maximum and a saddle respectively. Q.E.D.

AI-Example 12.b.14 (*Classifying saddle points via the Hessian*): Consider $f(x,y) := x^2 - y^2$ on $\mathbb{R}^2$. The gradient $\nabla f(x,y) = (2x, -2y)^\mathsf{T}$ vanishes at $(0,0)$, and the Hessian is constant:

$$\operatorname{Hess} f(0,0) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Its eigenvalues are $\lambda_1 = 2 > 0$ and $\lambda_2 = -2 < 0$, so $\operatorname{Hess} f(0,0)$ is indefinite and $(0,0)$ is a saddle point by Theorem 12.b.11. This is the model case the word *"saddle"* is named after, since f rises along the x-axis and falls along the y-axis.

Remark 12.b.15 (*Mnemonic*): positive $=\smile \ldots$ min; negative $=\frown \ldots$ max; indefinite = saddle.



positive definite
$\lambda_1, \lambda_2 > 0$
local **min**

negative definite
$\lambda_1, \lambda_2 < 0$
local **max**

indefinite
opposite signs
saddle

The picture is the whole theorem. By Remark 12.b.8 the second-order Taylor term is $\frac{1}{2}\sum_i \lambda_i a_i^2$ in the eigenbasis, so along the i-th eigendirection f behaves like the parabola $\frac{1}{2}\lambda_i t^2$: it curves **up** when $\lambda_i > 0$ and **down** when $\lambda_i < 0$. All up gives a minimum, all down a maximum. One of each gives a saddle, since there is then a direction of ascent and a direction of descent through the same point. And when some $\lambda_i = 0$ that direction is **flat** to second order, the quadratic term says nothing about it, and the test must fall silent. That is exactly the degenerate case, illustrated by $x^4 + y^4$ against $x^4 - y^4$ below.

AI-Note 12.1: The lecture script's Prop. 11.15 states Theorem 12.b.11 for $f \in C^3$. The test needs only C^2, which is what the Hessian's definition assumes, and the script's own proof uses nothing beyond the second-order Taylor expansion with an $o(|h|^2)$ remainder. The statement above is both the weaker hypothesis and the standard form of the theorem.

AI-Example 12.b.16 (*Inconclusive Hessian Test at a Degenerate Critical Point*): Consider $f(x,y) := x^4 + y^4$ and $g(x,y) := x^4 - y^4$ at the critical point $(0,0)$.

- For both functions, all second-order partial derivatives at $(0,0)$ vanish, so $\operatorname{Hess} f(0,0) = \operatorname{Hess} g(0,0) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.
- The Hessian test is **inconclusive** since $\det \operatorname{Hess} = 0$.
- However, direct inspection reveals that $f(x,y) \geq 0 = f(0,0)$ for all (x,y), so $(0,0)$ is a strict local minimum for f, whereas $g(x,0) > 0$ and $g(0,y) < 0$ for non-zero x, y, so $(0,0)$ is a saddle point for g.

Remark 12.b.17: Script Exercise 11.16 makes the same point with a two-parameter family: $f_{a,b}(x,y) := ax^4 + by^4$ has $\operatorname{Hess} f_{a,b}(0,0)$ equal to the zero matrix **regardless of** a, b, yet the behaviour at 0 ranges over every possible case as a, b vary. Taking $a, b > 0$ recovers the strict

minimum of f above ($a = b = 1$), $a, b < 0$ gives a strict maximum, and opposite signs recover the saddle of g above ($a = 1$, $b = -1$). It is the sharpest available statement of *"degenerate ⇒ no information"*: the identical (zero) Hessian is compatible with every possible outcome, depending only on quartic terms the second-order test cannot see.

How do I determine positive/negative definiteness? (Subsection 12.b.1)

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$. Note: $\det(A) = \lambda_1 \cdots \lambda_n$ and $\operatorname{Tr}(A) = \lambda_1 + \cdots + \lambda_n$.

For $n = 2$:

- $\det(A) = \lambda_1\lambda_2 < 0 \implies$ **indefinite**
- $\det(A) > 0,\ \operatorname{Tr}(A) > 0 \implies$ **positive definite**
- $\det(A) > 0,\ \operatorname{Tr}(A) < 0 \implies$ **negative definite**

For $n \geq 3$:

- Solve the characteristic equation $\det(A - \lambda\,\mathrm{id}) = 0$ for λ and check all eigenvalues.
- *"Hurwitz criterion"* (German: *„Hurwitz-Kriterium"*): for

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \\ a_{n1} & \cdots & & a_{nn} \end{pmatrix}$$

define the leading principal minors

$$A_1 = \det(a_{11}), \quad A_2 = \det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \dots, \quad A_n = \det(A).$$

Then:

- $A_1 > 0,\ A_2 > 0,\ A_3 > 0, \dots$ if and only if the matrix is **positive definite**
- $A_1 < 0,\ A_2 > 0,\ A_3 < 0, \dots$ if and only if the matrix is **negative definite**
- if $\det A \neq 0$ and neither sign pattern holds, A is **indefinite**.

AI-Note 12.2: The leading minors decide definiteness, but they do **not** decide semidefiniteness, so the third bullet needs $\det A \neq 0$. Without it the test misfires: $A = \operatorname{diag}(1, 0, -1)$ has $A_1 = 1$, $A_2 = A_3 = 0$, which fits the pattern $A_1 \geq 0, A_2 \geq 0, A_3 \geq 0$, and yet $e_1^{\mathsf{T}} A e_1 = 1 > 0 > -1 = e_3^{\mathsf{T}} A e_3$, so A is indefinite. For a degenerate matrix one has to look at the eigenvalues, or at **all** principal minors, not only the leading ones.

Example 12.b.18 (*Definiteness calculations*):

Matrix	Computation	Conclusion
$\left(\begin{smallmatrix} 2 & 2 \\ 2 & 5 \end{smallmatrix}\right)$	$\det = 6 > 0$, $\operatorname{Tr} = 7 > 0$	positive definite
$\left(\begin{smallmatrix} -4 & 2 \\ 2 & -4 \end{smallmatrix}\right)$	$\det = 12 > 0$, $\operatorname{Tr} = -8 < 0$	negative definite
$\left(\begin{smallmatrix} 4 & 2 \\ 2 & -4 \end{smallmatrix}\right)$	$\det = -20 < 0$	indefinite
$\left(\begin{smallmatrix} -5 & 0 & 0 \\ 0 & -4 & -2 \\ 0 & -2 & -4 \end{smallmatrix}\right)$	$A_1 = -5 < 0,\ A_2 = 20 > 0,\ A_3 = -60 < 0$	negative definite
$\left(\begin{smallmatrix} -1 & 0 & -4 \\ 0 & 2 & -2 \\ -4 & -2 & 22 \end{smallmatrix}\right)$	$A_1 = -1 < 0,\ A_2 = -2 < 0,\ A_3 = -72 < 0$	indefinite

Example 12.b.19 (*A two-parameter family running through all four Hessian cases*): Let $a, b \in \mathbb{R}$ and $f(x, y) := x\sin(y) + ax^2 + by^2$. Then $Df(0,0) = (0,0)$, and

$$\operatorname{Hess} f(0,0) = \begin{pmatrix} 2a & 1 \\ 1 & 2b \end{pmatrix}, \qquad \det \operatorname{Hess} f(0,0) = 4ab - 1.$$

Using the determinant/trace rule above:

- If $a > 0$ and $4ab - 1 > 0$: positive definite, so $(0,0)$ is a local minimum.
- If $a < 0$ and $4ab - 1 > 0$: negative definite, so $(0,0)$ is a local maximum.
- If $4ab - 1 < 0$: indefinite, so $(0,0)$ is a saddle point.
- If $4ab - 1 = 0$: degenerate, and the test says nothing.

Unlike the fixed examples above, a single formula with two free parameters realises every outcome of Theorem 12.b.11 at once, simply by moving (a,b) across the curve $4ab = 1$ in parameter space.

Exercise 12.b.20 (*Critical values of $x^3 - y^3 + 3\alpha xy$*): Find the critical values of

$$f_\alpha : \mathbb{R}^2 \to \mathbb{R}, \qquad (x,y) \mapsto x^3 - y^3 + 3\alpha xy$$

for $\alpha \in \mathbb{R} \setminus \{0\}$ and determine min/max/saddle points.

Solution to Exercise 12.b.20:

$$\frac{\partial f_\alpha}{\partial x} = 3x^2 + 3\alpha y \stackrel{!}{=} 0 \qquad \Longrightarrow \; y = -\frac{x^2}{\alpha}$$
$$\frac{\partial f_\alpha}{\partial y} = -3y^2 + 3\alpha x \stackrel{!}{=} 0 \qquad \Longrightarrow \; -\frac{x^4}{\alpha^2} + \alpha x = 0$$

(a) $x = y = 0$

(b) $x \neq 0 \implies x^3 = \alpha^3 \implies x = \alpha,\ y = -\alpha$

So, we have critical points $(0,0)$ and $(\alpha, -\alpha)$.

Compute the Hessian:

$$\begin{pmatrix} \partial_x^2 f & \partial_x \partial_y f \\ \partial_x \partial_y f & \partial_y^2 f \end{pmatrix} = \begin{pmatrix} 6x & 3\alpha \\ 3\alpha & -6y \end{pmatrix} = \operatorname{Hess} f(x,y)$$

$$\operatorname{Hess} f(0,0) = \begin{pmatrix} 0 & 3\alpha \\ 3\alpha & 0 \end{pmatrix}, \qquad \det = -9\alpha^2 < 0 \implies \text{indefinite}$$

$$\operatorname{Hess} f(\alpha, -\alpha) = \begin{pmatrix} 6\alpha & 3\alpha \\ 3\alpha & 6\alpha \end{pmatrix} = 3\alpha \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$
$$\det = 27\alpha^2 > 0, \qquad \operatorname{Tr} = 12\alpha.$$

For $\alpha > 0$ this is positive definite; for $\alpha < 0$ it is negative definite. So:

- $(0,0) \longrightarrow$ **saddle point**
- $(\alpha, -\alpha) \longrightarrow$ **local minimum** for $\alpha > 0$; **local maximum** for $\alpha < 0$

Q.E.D.

AI-Exercise 12.b.21 (*Global Extrema on a Compact Triangle*): Find the global maximum and minimum values of $f(x,y) := xy$ on the compact triangular region $T := \{(x,y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, x + y \leq 2\}$.

Solution: *See solution on Page 184*

Exercise 12.b.22 (*The signature of a 2×2 matrix* (important)): Despite the definition, it is not necessary to compute the eigenvalues of a matrix to find its signature. Prove that for a symmetric 2×2 matrix M we have the following simple rule to determine the signature in terms of $\det M$ and $\operatorname{Tr} M$:

- If $\det M > 0$, $\operatorname{Tr} M > 0$ then M is positive definite,
- If $\det M > 0$, $\operatorname{Tr} M < 0$ then M is negative definite,
- If $\det M < 0$ then M is indefinite,

- If $\det M = 0$, then M is degenerate.

Info: *This exercise is Problem 6.2 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 184*

Exercise 12.b.23 (*Classifying critical points with the Hessian*): Find and classify the critical points of $f(x,y) := xy - x^2 - y^2 - x$ on $\mathbb{R}^2$. (cf. Michaels Bsp 7.1.3)

Solution to Exercise 12.b.23:

First, we compute the gradient to find the critical points:

$$\nabla f(x,y) = \begin{pmatrix} y - 2x - 1 \\ x - 2y \end{pmatrix}.$$

Setting $\nabla f(x,y) = (0,0)$ gives the system:

$$\begin{aligned} -2x + y &= 1 \\ x - 2y = 0 &\implies x = 2y. \end{aligned}$$

Substituting into the first equation: $-2(2y) + y = 1 \implies -3y = 1 \implies y = -1/3$. Then $x = -2/3$. So, the only critical point is $(-2/3, -1/3)$.

To classify it, we compute the Hessian matrix:

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The Hessian is constant. We apply the second derivative test by examining its eigenvalues (or determinant and trace). The determinant is $\det \operatorname{Hess} f = (-2)(-2) - 1(1) = 3 > 0$. The trace is $\operatorname{Tr} \operatorname{Hess} f = -2 + (-2) = -4 < 0$. Since $\det > 0$ and $\operatorname{Tr} < 0$, the Hessian is negative definite at the critical point. Therefore, $(-2/3, -1/3)$ is a strict local maximum. Q.E.D.

Exercise 12.b.24 (*Multiple choice (what a positive semidefinite Hessian does not give you)* (important)): The Hessian matrix of $f \in C^2(\mathbb{R}^n)$ is positive semidefinite at a critical point x_0 of f, i.e.,

$$\langle v, \operatorname{Hess} f(x_0) v \rangle \geq 0 \quad \text{for all } v \in \mathbb{R}^n.$$

Which of the following statements necessarily hold? (There may be more than one).

(a) x_0 is a strict local minimum of f.
(b) x_0 is a local minimum of f.
(c) x_0 is not a local maximum of f.
(d) None of the above statements.

Info: *This exercise is Problem 6.8 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 185*

Exercise 12.b.25 (*Minimization* (important)): The function $f : \mathbb{R}^2 \to \mathbb{R}$ is given by $f(x,y) = 2x^2 + y^2 - x$. Determine the extrema of f on...

(a) ...the unit circle $S^1 = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$;
(b) ...the closed unit disk $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$.

Info: *This exercise is Problem 6.9 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution to (a): *See solution on Page 185*

Solution to (b): *See solution on Page 185*

Exercise 12.b.26 (*Critical points* (important)): Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function $f(x,y) = (ax^2 + by^2)e^{-x^2-y^2}$ with real parameters $a, b \in \mathbb{R}$. Find all critical points and determine their nature with the Hessian test, depending on a, b.

Info: *This exercise is Problem 6.12 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 185*

Exercise 12.b.27 (*Critical points and Hessian test for* $f_\alpha(x,y) = e^x(x^2 + \alpha y^2)$): For a parameter $\alpha \in \mathbb{R}$, consider the function $f_\alpha : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f_\alpha(x,y) := e^x(x^2 + \alpha y^2).$$

(a) Determine all critical points of f_α, for every $\alpha \in \mathbb{R}$.

(b) For every $\alpha \neq 0$, run the Hessian test at all critical points found in **(a)**.

(c) Determine the range $f_\alpha(\mathbb{R}^2) \subseteq \mathbb{R}$, for every $\alpha \in \mathbb{R}$, and prove your claim.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), Short Problem 1.*

Solution: *See solution on Page 187*

The fundamental theorem of algebra, by minimization (Section 12.c)

The optimality conditions of this chapter came in two orders. A critical point is a first-order condition (Proposition 12.a.2); the Hessian test is a second-order one (Theorem 12.b.11), and it is silent when the Hessian is degenerate. What happens when both are silent?

The answer is that one keeps expanding. The following proof is the standard illustration, and it is worth the page for two reasons: the order at which the expansion finally decides the question is not known in advance, and the theorem it proves is not an analysis theorem at all.

Theorem 12.c.28 (*Fundamental theorem of algebra*): Every non-constant polynomial $f \in \mathbb{C}[z]$ has a root in $\mathbb{C}$. Consequently, by the division algorithm, f has $n := \deg f$ complex roots counted with multiplicity.

Proof:

Write $f(z) = \sum_{k=0}^{n} a_k z^k$ with $n > 0$ and $a_n \neq 0$. Dividing by a_n changes no roots, so assume $a_n = 1$. Consider

$$\mu := \inf\{|f(z)| : z \in \mathbb{C}\} \geq 0.$$

The proof has two halves: first that the infimum is attained, and then that its value is 0.

The infimum is attained. Put $C := \sum_{k=0}^{n} |a_k|$. For $|z| \geq 1$ every $|z|^k$ with $k \leq n-1$ is at most $|z|^{n-1}$, so the triangle inequality gives

$$|f(z)| \geq |z|^n - \sum_{k=0}^{n-1} |a_k|\,|z|^k \geq |z|^{n-1}\big(|z| - C\big).$$

Fix any $R \geq C + 1 + 10\mu$. For $|z| \geq R$ both factors are controlled, since $|z|^{n-1} \geq 1$ and $|z| - C \geq 1 + 10\mu$, so

$$|f(z)| \geq 1 + 10\mu \qquad \text{whenever } |z| \geq R.$$

Consequently nothing outside the closed ball $K := \{z : |z| \leq R\}$ can compete for the infimum: off K the values of $|f|$ exceed μ by at least 1. Now K is compact and $|f|$ is continuous, so by the extreme value theorem (Item **(b)**) the minimum over K is attained, say at z_0, and

$$|f(z_0)| = \min_K |f| = \mu.$$

Since $|f| \geq 1 + 10\mu > \mu$ on ∂K, the point z_0 lies in the interior of K, and is therefore a genuine local minimum of $|f|$ on an open set.

The minimum value is 0. Recentre at z_0 by setting

$$g(z) := f(z_0 + z) = \sum_{k=0}^{n} b_k z^k,$$

which is again a polynomial of degree n, and note $b_0 = g(0) = f(z_0)$. Suppose, for contradiction, that $b_0 \neq 0$. Let $\ell \geq 1$ be the smallest index with $b_\ell \neq 0$, which exists because g is non-constant. Writing $z = re^{i\varphi}$ and collecting the higher-order terms,

$$g\big(re^{i\varphi}\big) = b_0 + b_\ell r^\ell e^{i\ell\varphi} + O\big(r^{\ell+1}\big) = b_0\left(1 + \frac{b_\ell}{b_0} r^\ell e^{i\ell\varphi}\right) + O\big(r^{\ell+1}\big) \qquad \text{as } r \downarrow 0.$$

The direction φ is still free, and this is where it is spent. Write $b_\ell/b_0 = se^{i\psi}$ with $s > 0$, and choose

$$\varphi := \frac{\pi - \psi}{\ell}, \qquad \text{so that} \qquad e^{i(\ell\varphi + \psi)} = e^{i\pi} = -1.$$

Along that ray the bracket becomes $1 - sr^\ell$, which is positive for small r, so for all sufficiently small $r > 0$ and some constant $M > 0$,

$$|b_0| = |g(0)| \leq \big|g\big(re^{i\varphi}\big)\big| = \Big|b_0\big(1 - sr^\ell\big) + O\big(r^{\ell+1}\big)\Big| \leq |b_0|\big(1 - sr^\ell\big) + Mr^{\ell+1},$$

the first inequality because z_0 minimises $|f|$, hence 0 minimises $|g|$. Cancelling $|b_0|$ and dividing by r^ℓ,

$$0 < s\,|b_0| \leq Mr,$$

which fails as soon as $r < s|b_0|/M$. This contradiction forces $b_0 = 0$, that is $f(z_0) = 0$. **Q.E.D.**

Remark 12.c.29 (*Why this proof sits in this chapter*): The second half is an optimality argument, and it is the reason the section is here. At the minimiser one expands and asks which term first detects a direction of descent. The first-order term is b_1, the second-order term is b_2, and **either may vanish**: the deciding index is ℓ, the first non-zero coefficient after b_0, and nothing in the hypotheses bounds it. Expanding $f(z) = z^7 + 1$ about the origin, for instance, gives $b_0 = 1 \neq 0$ and $\ell = 7$: six whole orders pass before any term detects a direction of descent. (That the origin is not in fact a minimiser of $|f|$ is precisely what the descent construction goes on to show: along $\varphi = \pi/7$ the modulus drops to $1 - r^7$.)

So the pattern of Theorem 12.b.11 does not merely fail to apply here, it could not: a test that stops at second order cannot see an obstruction of order seven. What replaces it is the observation that once a lowest non-vanishing term exists, one can **aim** at it, choosing φ so that the term points directly against b_0 and shortens it.

Two further features are worth noticing. The argument never differentiates anything: $|f|$ is not holomorphic and is not even differentiable at its zeros, and no derivative is taken anywhere above. And the choice of φ has no real-variable counterpart. On $\mathbb{R}$ there are two directions to try, and $x \mapsto x^2 + 1$ has a strict positive minimum precisely because neither works; on $\mathbb{C}$ there is a whole circle of directions and ℓ of them do the job. That is the sense in which this is a theorem about $\mathbb{C}$ that happens to be proved by optimization.

AI-Note 12.3: The name is a historical accident worth flagging, since it invites a wrong expectation of the proof above. The statement is not a theorem of algebra and has no purely algebraic proof: any argument must use some completeness or connectedness property of $\mathbb{R}$, because the theorem is false over fields that look algebraically similar but are not complete. Over the rationals, for example, $z^2 - 2$ has no root. In the proof above, the analytic input enters at exactly one place, the extreme value theorem, and that is where completeness is hiding.

AI-Exercise 12.c.30 (*Where each hypothesis is used*): The proof of Theorem 12.c.28 has more moving parts than it appears.

(a) The growth estimate concludes $|f(z)| \geq |z|^{n-1}(|z| - C)$ only for $|z| \geq 1$. Where is that restriction used, and what goes wrong without it?

(b) Show that the theorem fails for $f(z) := \overline{z}\, z + 1 = |z|^2 + 1$, viewed as a function $\mathbb{C} \to \mathbb{C}$, and identify the single step of the proof that breaks. (This function is continuous and its modulus does attain a minimum.)

(c) The choice $\varphi := (\pi - \psi)/\ell$ is one of ℓ possible choices. Write down the others, and explain why the argument does not care which is taken.

Solution: *See solution on Page 186*

Solutions (Section 12.d)

Not every solution in this chapter is collected here. Exercise 12.b.9 (Page 176), Exercise 12.b.20 (Page 180) and Exercise 12.b.23 (Page 181) are solved where they appear.

Proof of AI-Exercise 12.b.21:

Since T is compact and f is continuous, global extrema exist by the extreme value theorem (Item **(b)**).

- **Interior of T:** Critical points satisfy $\nabla f(x,y) = (y,x)^{\mathsf{T}} = (0,0)^{\mathsf{T}}$, and therefore $(x,y) = (0,0)$, which lies on the boundary. Thus, there are no critical points in $\operatorname{int}(T)$.
- **Boundary of T:**
  - Along the leg $x = 0$ $(0 \leq y \leq 2)$, $f(0,y) = 0$.
  - Along the leg $y = 0$ $(0 \leq x \leq 2)$, $f(x,0) = 0$.
  - Along the hypotenuse $x + y = 2$ $(0 \leq x \leq 2)$, substitute $y = 2 - x$:
    $$h(x) := x(2-x) = 2x - x^2, \qquad h'(x) = 2 - 2x,$$
    which vanishes at $x = 1$. This gives the candidate point $(1,1)$ with $f(1,1) = 1$. The endpoints $(2,0)$ and $(0,2)$ yield $f = 0$.

Comparing values, the **global maximum** is 1 at $(1,1)$, and the **global minimum** is 0, attained everywhere on the coordinate axes legs of T. Q.E.D.

Proof of Exercise 12.b.22:

By the spectral theorem, since M is symmetric there exists an orthogonal matrix O (so $OO^{\mathsf{T}} = \mathrm{id}$, hence $\det O = \pm 1$) and $\lambda_1, \lambda_2 \in \mathbb{R}$ such that

$$OMO^{\mathsf{T}} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

Using $\det(AB) = \det(A)\det(B)$ and $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$,

$$\det M = \det(OMO^{\mathsf{T}}) = \lambda_1\lambda_2, \qquad \operatorname{Tr} M = \operatorname{Tr}(MO^{\mathsf{T}}O) = \operatorname{Tr}(OMO^{\mathsf{T}}) = \lambda_1 + \lambda_2.$$

So, $\det M$ and $\operatorname{Tr} M$ are exactly the product and sum of the eigenvalues, and the four cases follow immediately:

- $\det M > 0$: λ_1, λ_2 have the same sign. If additionally $\operatorname{Tr} M > 0$, both are positive, so M is **positive definite**; if $\operatorname{Tr} M < 0$, both are negative, so M is **negative definite**.
- $\det M < 0$: λ_1, λ_2 have opposite signs, so M is **indefinite**.
- $\det M = 0$: one of λ_1, λ_2 is zero, so M is **degenerate**.

Q.E.D.

Solution to Exercise 12.b.24:

None of **(a)**–**(c)** hold in general:

- **(a) False:** $f \equiv 0$ has $\operatorname{Hess} f(x_0) = 0$ positive semidefinite at every x_0, but x_0 is not a **strict** local minimum (every nearby value equals $f(x_0)$).
- **(b) False:** $f(x) = x_1^3$ at $x_0 = 0$ has $\operatorname{Hess} f(0) = 0$ (positive semidefinite), but f takes negative values arbitrarily close to 0, so x_0 is not a local minimum at all.
- **(c) False:** $f(x) = -|x|^4$ at $x_0 = 0$ has $\operatorname{Hess} f(0) = 0$ (positive semidefinite, vacuously, since it is the zero matrix), yet x_0 **is** a strict local maximum of f.

Since **(a)**–**(c)** all fail, **(d) is the correct answer**: none of the above statements necessarily hold.

Q.E.D.

Solution to Exercise 12.b.25:

(a) With $g(x,y) := x^2 + y^2 - 1$, the Lagrangian is $L(x,y,\lambda) := f(x,y) - \lambda g(x,y)$, giving the system

$$0 = \partial_x L = 2x(2-\lambda) - 1, \qquad 0 = \partial_y L = 2y(1-\lambda), \qquad 0 = \partial_\lambda L = -(x^2+y^2-1).$$

From the middle equation, either $y = 0$ or $\lambda = 1$.

If $y = 0$: then $x^2 = 1$, so $x = \pm 1$, giving the points $(1,0)$ and $(-1,0)$ with $f(1,0) = 1$ and $f(-1,0) = 3$.

If $\lambda = 1$: the first equation gives $2x(1) = 1$, i.e. $x = \frac{1}{2}$, and the constraint gives $y = \pm\frac{\sqrt{3}}{2}$, with $f\left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right) = \frac{1}{2} + \frac{3}{4} - \frac{1}{2} = \frac{3}{4}$.

Comparing all four values, f attains a **global maximum** at $(-1,0)$ (value 3) and **global minima** at $\left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right)$ (value $\frac{3}{4}$); $(1,0)$ is a local (non-global) extremum with value 1.

Q.E.D.

AI-Note 12.4: The official solution lists $(1,0)$ alongside $(-1,0)$ as *"local extrema"* from the Lagrange system, then only mentions it again implicitly by omission from the final global-extrema summary. It is worth flagging explicitly that **Lagrange's condition finds all critical points of $f|_{S^1}$, not automatically the global extrema**. Here $(1,0)$ is a genuine local maximum along the circle (value 1) that is simply not the largest one; only comparing all candidate values at the end distinguishes it from the global maximum at $(-1,0)$ (value 3).

Solution to Exercise 12.b.25, continued:

(b) Having handled the boundary $S^1 = \partial D$ in **(a)**, we look for interior critical points: $Df(x,y) = (4x-1, 2y) = 0$ gives $(x,y) = \left(\frac{1}{4}, 0\right) \in \operatorname{int}(D)$. The Hessian $\operatorname{Hess} f = \left(\begin{smallmatrix} 4 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ is positive definite, so this is a local minimum, with value $f\left(\frac{1}{4}, 0\right) = \frac{1}{8} - \frac{1}{4} = -\frac{1}{8}$. Since $-\frac{1}{8}$ is smaller than every boundary value found in **(a)**, f on D attains a **global minimum** at $\left(\frac{1}{4}, 0\right)$ and a **global maximum** at $(-1,0)$.

Q.E.D.

Solution to Exercise 12.b.26:

Writing $r^2 := x^2 + y^2$, the gradient is

$$\nabla f(x,y) = \left(2x(a - ax^2 - by^2),\ 2y(b - ax^2 - by^2)\right)e^{-r^2}.$$

Case $a = b = 0$: $f \equiv 0$, so every point of $\mathbb{R}^2$ is critical, and (being the constant function) each is simultaneously a global maximum and minimum.

Case $a = 0, b \neq 0$ (and symmetrically $a \neq 0, b = 0$): the gradient equations reduce to $-2xby^2 = 0$ and $2by(1 - y^2) = 0$, whose common solutions are the entire line $\{(x, 0) : x \in \mathbb{R}\}$ together with the two points $(0, \pm 1)$. On the critical line the Hessian is singular (degenerate, as expected since a whole line is critical); at $(0, \pm 1)$, $\operatorname{Hess} f(0, \pm 1) = \operatorname{diag}(-2b, -4b)\, e^{-1}$, giving **minima** if $b < 0$ and **maxima** if $b > 0$. The case $a \neq 0, b = 0$ is symmetric, with critical line $\{(0, y)\}$ and extra points $(\pm 1, 0)$: minima if $a < 0$, maxima if $a > 0$.

Case $a, b \neq 0$, $a \neq b$: setting $x = 0$ or $y = 0$ in the gradient equations gives the five critical points $(0, 0)$, $(0, \pm 1)$, $(\pm 1, 0)$ (checking $x, y \neq 0$ simultaneously forces $a = b$, excluded here). The Hessians are

$$\operatorname{Hess} f(0, 0) = \begin{pmatrix} 2a & 0 \\ 0 & 2b \end{pmatrix},$$

$$\operatorname{Hess} f(0, \pm 1) = e^{-1} \begin{pmatrix} 2(a - b) & 0 \\ 0 & -4b \end{pmatrix},$$

$$\operatorname{Hess} f(\pm 1, 0) = e^{-1} \begin{pmatrix} -4a & 0 \\ 0 & 2(b - a) \end{pmatrix}.$$

Reading off signs for each ordering of a, b against 0:

- $a > b > 0$: min at $(0, 0)$, saddle at $(0, \pm 1)$, max at $(\pm 1, 0)$.
- $b > a > 0$: min at $(0, 0)$, max at $(0, \pm 1)$, saddle at $(\pm 1, 0)$.
- $a > 0 > b$: saddle at $(0, 0)$, min at $(0, \pm 1)$, max at $(\pm 1, 0)$.
- $b > 0 > a$: saddle at $(0, 0)$, max at $(0, \pm 1)$, min at $(\pm 1, 0)$.
- $0 > a > b$: max at $(0, 0)$, min at $(0, \pm 1)$, saddle at $(\pm 1, 0)$.
- $0 > b > a$: max at $(0, 0)$, saddle at $(0, \pm 1)$, min at $(\pm 1, 0)$.

Case $a = b \neq 0$: beyond $(0, 0)$, $(0, \pm 1)$, $(\pm 1, 0)$, setting $x, y \neq 0$ now gives $a - ax^2 - ay^2 = 0$, that is, $x^2 + y^2 = 1$, so the entire unit circle is critical. On that circle,

$$\operatorname{Hess} f(x, y) = -4a\, e^{-1} \begin{pmatrix} x^2 & xy \\ xy & y^2 \end{pmatrix},$$

whose determinant is $16a^2 e^{-2}(x^2 y^2 - x^2 y^2) = 0$. The Hessian is therefore degenerate (Definition 12.b.6) at every such point, and the test is inconclusive, as it must be, since a whole curve of critical points cannot consist of isolated extrema. **Q.E.D.**

Solution to AI-Exercise 12.c.30:

(a) The restriction is what makes $|z|^k \leq |z|^{n-1}$ for every $k \leq n - 1$, which is the step that collapses the whole sum into a single power. For $|z| < 1$ the inequality reverses, the low powers dominate, and the estimate is simply false: at $z = 0$ it would read $|a_0| \geq 0 \cdot (0 - C)$ only by accident of the right-hand side being negative, and it gives nothing usable.

Nothing is lost, because the estimate is only ever applied for $|z| \geq R$, and $R \geq 1$ automatically: $C = \sum_{k=0}^{n} |a_k|$ includes the term $|a_n| = 1$, so $C \geq 1$ and hence $R \geq C + 1 \geq 2$. The hypothesis is free.

(b) Here $f(z) = |z|^2 + 1 \geq 1$, so f has no root, and indeed $|f|$ attains its minimum 1 at $z_0 = 0$. Both halves of the first part of the proof therefore go through unchanged: $|f|$ grows without bound, the infimum is attained, and the minimiser is interior.

The step that breaks is the expansion. The proof writes $g(z) := f(z_0 + z) = \sum_k b_k z^k$, and f here is not a polynomial in z: it is $z\overline{z} + 1$, and $\overline{z}$ is not expressible in powers of z. Concretely,

at $z_0 = 0$ and $z = re^{i\varphi}$,

$$g(re^{i\varphi}) = 1 + r^2,$$

in which the perturbation carries **no** factor $e^{i\ell\varphi}$. The direction φ has vanished from the expression, so there is no direction to aim, and the term r^2 increases $|g|$ whichever way one leaves the origin.

That is the sharpest way to say what the proof actually uses: not that f is smooth, nor that it grows, but that the lowest-order correction to b_0 has a **phase** that the free parameter φ can rotate onto $-b_0$. The function $z\overline{z}$ is the standard example of a perturbation with no phase to rotate.

(c) The requirement is $e^{i(\ell\varphi+\psi)} = -1$, i.e. $\ell\varphi + \psi \equiv \pi \pmod{2\pi}$, so

$$\varphi_j := \frac{\pi - \psi + 2\pi j}{\ell}, \qquad j = 0, 1, \dots, \ell - 1,$$

which are ℓ distinct directions in $[0, 2\pi)$, equally spaced by $2\pi/\ell$. The argument does not care which is used, because it needs only **one** ray along which $|g|$ falls below $|b_0|$; the contradiction is reached from a single such ray. That ℓ of them exist rather than one is a bonus of working over $\mathbb{C}$, and it is exactly the freedom that **(b)** shows to be indispensable.

Q.E.D.

Proof of Exercise 12.b.27:

(a) The gradient of f_α is

$$\nabla f_\alpha(x, y) = \begin{pmatrix} e^x(x^2 + \alpha y^2 + 2x) \\ 2\alpha y e^x \end{pmatrix},$$

and we look for the points where it vanishes. Since $e^x > 0$ everywhere, the second component vanishes when $\alpha y = 0$, which splits the problem in two.

- **If $\alpha = 0$:** the second component vanishes identically, and the first reduces to $x^2 + 2x = 0$, so $x = 0$ or $x = -2$. The critical set is the pair of vertical lines $\{(0, y) \mid y \in \mathbb{R}\}$ and $\{(-2, y) \mid y \in \mathbb{R}\}$.
- **If $\alpha \neq 0$:** the second component forces $y = 0$, and the first then reads $x^2 + 2x = 0$ again. The critical points are $(0, 0)$ and $(-2, 0)$.

(b) For $\alpha \neq 0$, differentiating once more gives

$$\operatorname{Hess} f_\alpha(x, y) = e^x \begin{pmatrix} x^2 + 4x + 2 + \alpha y^2 & 2\alpha y \\ 2\alpha y & 2\alpha \end{pmatrix}.$$

- **At $(0, 0)$:** $\operatorname{Hess} f_\alpha(0, 0) = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2\alpha \end{smallmatrix}\right)$, with determinant 4α. For $\alpha > 0$ the determinant is positive and $\operatorname{Tr} = 2 + 2\alpha > 0$, so the matrix is positive definite and $(0, 0)$ is a strict local minimum. For $\alpha < 0$ the determinant is negative, so $(0, 0)$ is a saddle point.
- **At $(-2, 0)$:** $\operatorname{Hess} f_\alpha(-2, 0) = e^{-2}\left(\begin{smallmatrix} -2 & 0 \\ 0 & 2\alpha \end{smallmatrix}\right)$, with determinant $-4\alpha e^{-4}$. For $\alpha > 0$ this is negative, so $(-2, 0)$ is a saddle point. For $\alpha < 0$ it is positive, and the entry $\partial_1^2 f_\alpha(-2, 0) = -2e^{-2} < 0$ makes the matrix negative definite, so $(-2, 0)$ is a strict local maximum.

Notice that the two critical points swap roles as α changes sign, and that neither test is inconclusive: $\det \operatorname{Hess}$ vanishes only at $\alpha = 0$, the case excluded here.

(c) Each range is determined by exhibiting the extreme values and then filling in between by the intermediate value theorem, which applies because $\mathbb{R}^2$ is connected and f_α is continuous.

- **If $\alpha > 0$:** both x^2 and αy^2 are non-negative, so $f_\alpha \geq 0$, with $f_\alpha(0, 0) = 0$ attained. Along the x-axis, $f_\alpha(x, 0) = x^2 e^x \to \infty$ as $x \to \infty$, so values grow without bound. Hence $f_\alpha(\mathbb{R}^2) = [0, \infty)$.

- **If $\alpha < 0$:** along the x-axis $f_\alpha(x, 0) = x^2 e^x$ sweeps out $[0, \infty)$ as above, while along the y-axis $f_\alpha(0, y) = \alpha y^2$ sweeps out $(-\infty, 0]$. Together these give $f_\alpha(\mathbb{R}^2) = \mathbb{R}$.
- **If $\alpha = 0$:** $f_0(x, y) = x^2 e^x$ does not depend on y and is non-negative, with value 0 at $x = 0$ and unbounded above. Hence $f_0(\mathbb{R}^2) = [0, \infty)$.

AI-Note 12.5: It is worth pausing on the $\alpha > 0$ case: $(0, 0)$ is a strict local minimum **and** the global minimum, but $(-2, 0)$ is only a saddle, and f_α has no global maximum at all. Part **(c)** is not a corollary of part **(b)**: the Hessian test is purely local, and reading a range off it is a standard way to go wrong.

Q.E.D.

Lagrange Multipliers (Chapter 13)

The previous chapter looked for extrema of f on an open set, where the condition $\nabla f(x_0) = 0$ is available. Restrict f to a lower-dimensional constraint set and that condition is generally too strong to hold anywhere. What replaces it is the requirement that ∇f be a linear combination of the gradients of the constraints, which says geometrically that at a constrained extremum no direction **along** the constraint set is left in which f still increases. The method carries one hypothesis on those gradients, and dropping it does not merely weaken the conclusion. It lets the method miss an extremum outright.

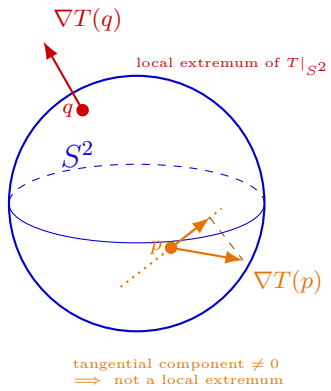
Two further sections follow the method itself. The first supplies the constrained counterpart of the Hessian test, which is the Hessian of the Lagrangian read only along the constraint set. The second turns the method on linear algebra: maximising a quadratic form on the unit sphere proves the spectral theorem, which the Hessian test of the previous chapter had been borrowing.

Lagrange multipliers (Section 13.a)

Suppose we want to find the hottest place on the surface of the earth. In theory, temperature is defined everywhere, giving some function

$$T \in C^\infty(\mathbb{R}^3, (0, \infty)) \qquad \text{(absolute zero is not attainable)}$$

If we model the earth as $S^2 := \{x \in \mathbb{R}^3 : |x| = 1\}$, then we want to find the local maxima of $T|_{S^2}$, which **are not necessarily critical points of** T!



Read the picture as follows. At p the gradient $\nabla T(p)$ has a non-zero component **along** the surface, so walking on S^2 in that direction increases T: the point p cannot be a local extremum of $T|_{S^2}$. So, if $T|_{S^2}$ **does** have a local extremum at a point, then ∇T there must have no tangential component left. It must be **perpendicular to the surface**, as at q.

Note the direction of that statement carefully: perpendicularity is **necessary, not sufficient**. Plenty of points have a perpendicular gradient without being extrema at all. On the earth, the coldest place has a perpendicular gradient just as the hottest does, and so does many a saddle. What the condition delivers is a shortlist of **candidates**. Deciding which of them are maxima, which minima, and which neither is a separate step, and in practice it is done by evaluating T at each and comparing.

Note also that S^2 is the "*level set*" (German: „*Niveaumenge*“) $g(x) = 0$ for $g(x) := |x|^2 - 1$ (i.e., $S^2 = g^{-1}\{0\}$).

Remark 13.a.1 (*Gradient perpendicular to level sets*): For $g(x) := |x|^2 - 1$ we have $\nabla g(x) = 2x$, which at a point $x \in S^2$ is the outward radial direction, and therefore perpendicular to S^2. In general, the gradient is perpendicular to the level sets.

This intuition is made rigorous in the following.

Proposition 13.a.2 (*Constrained optimization problems*): Suppose $f : U \to \mathbb{R}$, $g : U \to \mathbb{R}^k$ are C^1 and $U \subseteq \mathbb{R}^n$ is open, and write $g := (g_1, \dots, g_k)$. If $f|_{g^{-1}\{0\}}$ has a local extremum at $x_0 \in g^{-1}\{0\}$, then

$$\{\nabla f(x_0), \nabla g_1(x_0), \dots, \nabla g_k(x_0)\} \text{ are linearly dependent.}$$

Proof:

If $\nabla g_1(x_0), \dots, \nabla g_k(x_0)$ are already linearly dependent, there is nothing to prove. So assume they are independent, which says that Dg_{x_0} has rank k. Some $k \times k$ minor of $\mathrm{J}\, g(x_0)$ is then non-zero, and by continuity it stays non-zero on a ball $B \subseteq U$ around x_0. By the regular value theorem (Theorem 17.a.3, applied to $g|_B$), $M := g^{-1}\{0\} \cap B$ is a C^1 submanifold of dimension $n - k$, so its tangent space $T_{x_0}M$ has dimension $n - k$.

The tangent space is $\ker Dg_{x_0}$. If γ is a C^1 curve in M with $\gamma(0) = x_0$, then $g(\gamma(t)) = 0$ for all t, and differentiating at $t = 0$ gives $Dg_{x_0}\gamma'(0) = 0$. By Proposition 17.c.33 every tangent vector is such a $\gamma'(0)$, so $T_{x_0}M \subseteq \ker Dg_{x_0}$. The kernel also has dimension $n - k$, because Dg_{x_0} has rank k, so the two spaces are equal.

Now take any $v \in T_{x_0}M$. By Proposition 17.c.33 there is a C^1 curve $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = x_0$ and $\gamma'(0) = v$. Since $f|_M$ has a local extremum at x_0, the one-variable function $t \mapsto f(\gamma(t))$ has a local extremum at $t = 0$, so its derivative vanishes there. By the chain rule,

$$0 = \frac{d}{dt}\Big|_{t=0} f(\gamma(t)) = \langle \nabla f(x_0), v \rangle.$$

Hence $\nabla f(x_0)$ is orthogonal to $\ker Dg_{x_0}$. But $\ker Dg_{x_0}$ is exactly the orthogonal complement of $W := \operatorname{span}\{\nabla g_1(x_0), \dots, \nabla g_k(x_0)\}$, because $Dg_{x_0}v = \big(\langle \nabla g_j(x_0), v\rangle\big)_j$. So $\nabla f(x_0) \in (W^\perp)^\perp = W$, which is the asserted linear dependence. In fact we have shown more: $\nabla f(x_0) = \sum_j \lambda_j \nabla g_j(x_0)$ for some $\lambda \in \mathbb{R}^k$. Q.E.D.

Remark 13.a.3: Once $(\nabla g_1, \dots, \nabla g_k)$ are linearly independent everywhere on $g^{-1}\{0\}$ (the standing assumption below), the constraint set $g^{-1}\{0\}$ is in fact a submanifold of $\mathbb{R}^n$. This is made precise by Theorem 17.a.3, which also explains Proposition 13.a.2 geometrically: $\nabla f(x_0)$ must be normal to the tangent space of the constraint set at a constrained extremum.

For $k = 1$, this corresponds to our intuition above of $\nabla f(x_0)$ **parallel to** $\nabla g(x_0)$.

In practice, we assume $(\nabla g_1, \dots, \nabla g_k)$ nowhere linearly dependent, since otherwise the dependence relation may give $\nabla f(x_0)$ the coefficient $\lambda_0 = 0$, which tells us nothing about f (see the remark on λ_0 below).

AI-Example 13.a.4 (*What goes wrong when ∇g vanishes*): The standing assumption is not bookkeeping. Drop it and the method can miss an extremum outright. Let

$$f(x, y) := x, \qquad M := \{(x, y) \in \mathbb{R}^2 : y^2 = x^3\},$$

the "*cuspidal cubic*" (German: „*Neilsche Parabel*"). Since $y^2 \geq 0$ forces $x^3 \geq 0$, every point of M has $x \geq 0$, and $(0, 0) \in M$. So, f attains a clear **global minimum on M at the origin**.

Now run the method with $g(x, y) := y^2 - x^3$. The Lagrange condition $\nabla f = \lambda \nabla g$ reads

$$(1, 0) = \lambda\,(-3x^2,\ 2y),$$

whose first component demands $1 = -3\lambda x^2$. At the origin this says $1 = 0$: no multiplier λ exists. The method returns **no candidates at all**, while the minimum sits there in plain sight.

The culprit is visible in $\nabla g(0,0) = (0,0)$: the standing assumption fails at exactly the point that matters. Geometrically, M has a **cusp** at the origin. It is not a smooth curve there, so it has no tangent line, and "∇f *perpendicular to the constraint*" has nothing to be perpendicular to.

The practical moral: before trusting a Lagrange computation, check that $\nabla g \neq 0$ (or, for several constraints, that the ∇g_j are independent) at every point of the constraint set. Where it fails, add those points to your candidate list by hand.

Remark 13.a.5 (*Where the coefficient λ_0 went*): Proposition 13.a.2 asserts linear **dependence** of $\{\nabla f(x_0), \nabla g_1(x_0), \dots, \nabla g_k(x_0)\}$, i.e. scalars $\lambda_0, \lambda_1, \dots, \lambda_k$, not all zero, with $\lambda_0 \nabla f(x_0) + \lambda_1 \nabla g_1(x_0) + \cdots + \lambda_k \nabla g_k(x_0) = 0$. The standing assumption above, that $(\nabla g_1, \dots, \nabla g_k)$ are linearly independent, is exactly what rules out $\lambda_0 = 0$ (else the ∇g_j's alone would already be dependent), so $\lambda_0 \neq 0$ and we may divide by it, silently renaming λ_j/λ_0 as λ_j. This is why the Lagrangian below carries no separate λ_0. Adrien Martelli's notes keep λ_0 explicit instead of dividing it away, pairing the dependence relation with a normalization constraint $\lambda_0^2 + \cdots + \lambda_k^2 = 1$ (any nonzero vector of multipliers can be rescaled to satisfy this). The two formulations are equivalent whenever $\lambda_0 \neq 0$; keeping λ_0 explicit only matters in degenerate settings where the independence assumption on $(\nabla g_1, \dots, \nabla g_k)$ fails and $\lambda_0 = 0$ is genuinely forced, e.g. if x_0 is itself a critical point of the constraint map g.

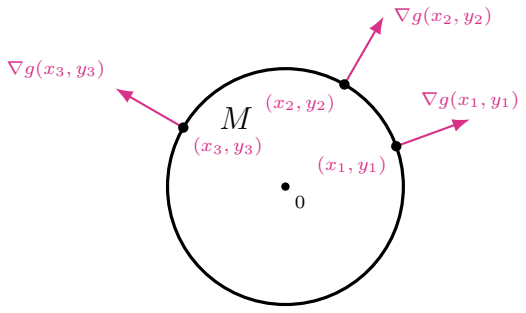
Remark 13.a.6 (*What the multiplier itself measures*): So far λ has been bookkeeping, a number produced by the method and then discarded once the candidate points are found. It carries information of its own. Consider the constrained problem with the constraint level left free,

$$m(c) := \min\{f(x) : g(x) = c\},$$

and suppose the minimiser $x^*(c)$ depends differentiably on c. Differentiating $g(x^*(c)) = c$ gives $\langle \nabla g(x^*(c)), (x^*)'(c) \rangle = 1$, while the Lagrange condition $\nabla f = \lambda \nabla g$ at $x^*(c)$ turns the chain rule for m into

$$m'(c) = \big\langle \nabla f(x^*(c)),\ (x^*)'(c) \big\rangle = \lambda \big\langle \nabla g(x^*(c)),\ (x^*)'(c) \big\rangle = \lambda.$$

So, λ is the rate at which the optimal value responds to relaxing the constraint by one unit. In economics this reading is standard enough to have its own name, the "*shadow price*" (German: „*Schattenpreis*") of the constraint. A large $|\lambda|$ says the constraint is expensive, and $\lambda = 0$ says it is not binding at all.



As the figure suggests, $\nabla g(x)$ is perpendicular to the constraint set $M := g^{-1}\{0\}$. To find extrema along M, we basically only care about the tangential part of ∇f. We can then find a multiplier λ to "*erase*" the perpendicular component of $\nabla f(x_0)$. We do this by forming the "*Lagrangian*" (German: „*Lagrange-Funktion*"):

$$\begin{aligned} L : U \times \mathbb{R}^k &\to \mathbb{R} \\ (x, \lambda) &\mapsto f(x) - \lambda \cdot g(x) = f(x) - \lambda_1 g_1(x) - \cdots - \lambda_k g_k(x) \end{aligned}$$

We then find local extrema by forcing $\nabla L(x,\lambda)=0$, i.e.

$$\underbrace{\frac{\partial L}{\partial x_i}=0 \ \ \forall i=1,\dots,n}_{\nabla f(x)-\lambda\cdot\nabla g(x)=0} \qquad \text{and} \qquad \underbrace{\frac{\partial L}{\partial \lambda_j}=0 \ \ \forall j=1,\dots,k}_{g_j(x)=0}$$

Remark 13.a.7 (*Why the Lagrangian is built this way*): The construction can look like a trick pulled out of nowhere. It is not: L is assembled precisely so that setting **one** gradient to zero encodes **both** halves of the problem at once.

Read the two underbraces. Differentiating in the x_i reproduces the geometric condition, namely ∇f parallel to the ∇g_j, with no tangential component left. Differentiating in λ_j returns $g_j(x)=0$, the constraint itself, because λ_j appears in L only in the term $-\lambda_j g_j(x)$, linearly. So, the multipliers are not auxiliary clutter: they are exactly the bookkeeping that lets a **constrained** problem in n variables be written as an **unconstrained** critical-point problem in $n+k$ variables, which is the one kind of problem we already know how to solve (Proposition 12.a.2).

That also explains the minus sign, which is arbitrary: writing $f+\lambda\cdot g$ merely flips the sign of every λ_j and changes nothing. And it explains why λ is discarded at the end. It is scaffolding, not part of the answer. That does not make it meaningless, and Remark 13.a.6 above says what it measures.

Three worked examples follow, each needing something the multiplier rule alone does not supply. The first has a constraint set with an interior, so it needs the unconstrained theory as well; the second is the smallest non-trivial case and gets reopened later with the second-order test; the third has to be turned into a constrained problem before any of this applies.

Example 13.a.8 (*Extrema of xyz on the unit ball*): Find the local extrema of

$$f:K\to\mathbb{R}, \quad (x,y,z)\mapsto xyz, \qquad K=\{(x,y,z)\in\mathbb{R}^3 : x^2+y^2+z^2\le 1\}.$$

We first find the **interior critical points**:

$$\nabla f(x,y,z)=(yz,\ xz,\ xy)\overset{!}{=}0 \implies x=y=0 \ \text{ or } \ x=z=0 \ \text{ or } \ y=z=0.$$

Then we look for local extrema **on the boundary**. We need to find the local extrema of $f|_{g^{-1}\{0\}}$, where $g:\mathbb{R}^3\to\mathbb{R}$ is given by $g(x)=|x|^2-1$.

The Lagrangian is

$$L(x,y,z,\lambda):=f(x,y,z)-\lambda g(x,y,z)=xyz-\lambda(x^2+y^2+z^2-1),$$

giving the system of equations

$$\begin{aligned}\frac{\partial L}{\partial x}&=yz-2\lambda x=0, \quad \frac{\partial L}{\partial \lambda}=1-(x^2+y^2+z^2)=0.\\ \frac{\partial L}{\partial y}&=xz-2\lambda y=0,\\ \frac{\partial L}{\partial z}&=xy-2\lambda z=0,\end{aligned}$$

If x, y, z are all non-zero, we then have

$$2\lambda=\frac{yz}{x}=\frac{xz}{y}=\frac{xy}{z} \qquad (*)$$

and multiplying these in pairs gives

$$\underbrace{y^2z^2=x^2z^2}_{y^2=x^2}=\underbrace{x^2y^2}_{y^2=z^2},$$

and therefore $x^2=y^2=z^2$. Then $x^2+y^2+z^2=1 \implies 3x^2=1 \implies x=\pm\frac{1}{\sqrt{3}}$, giving **8 solutions**:

$$(x,y,z)\in\left\{\tfrac{1}{\sqrt{3}}(\alpha,\beta,\gamma):\alpha,\beta,\gamma\in\{\pm1\}\right\}$$

plus the solutions $\{(x_1,x_2,x_3)\in\partial K : x_i = x_j = 0,\ i\neq j\}$, that is, the six points $\pm e_1, \pm e_2, \pm e_3$ (with $\lambda = 0$).

To justify division by z, x or y in $(*)$: if $z = 0$, then $-\lambda x = -\lambda y = xy = 0$, giving $x = 0$ or $y = 0$, and $\lambda = 0$, since $x = y = z = 0$ does not satisfy $x^2+y^2+z^2 = 1$. Then, since $\lambda = 0$, $\nabla f(x,y,z) = 0$ and we have already found that point in the first step. Similarly for $y = 0$ or $x = 0$.

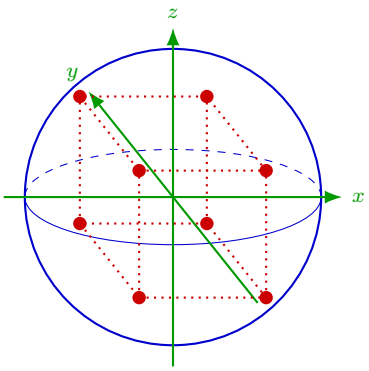
Classifying the candidates. The work so far produced candidates, not answers (Proposition 12.a.2 and the warning after it). Since K is compact and f is continuous, f does attain a global maximum and minimum on K, so it is enough to evaluate.

At a boundary candidate $\frac{1}{\sqrt{3}}(\alpha,\beta,\gamma)$ with $\alpha,\beta,\gamma\in\{\pm1\}$,

$$f\Big(\tfrac{1}{\sqrt{3}}(\alpha,\beta,\gamma)\Big) = \frac{\alpha\beta\gamma}{3\sqrt{3}},$$

so the value depends only on the sign $\alpha\beta\gamma$. Four of the eight points have $\alpha\beta\gamma = +1$ and give the maximal value $\frac{1}{3\sqrt{3}}$; the other four have $\alpha\beta\gamma = -1$ and give $-\frac{1}{3\sqrt{3}}$. On the coordinate axes found in the first step, $f\equiv 0$.

Hence f attains its **global maximum** $\frac{1}{3\sqrt{3}}$ at the four points with an even number of minus signs, and its **global minimum** $-\frac{1}{3\sqrt{3}}$ at the four with an odd number. The interior candidates are **not extrema at all**: at any point of a coordinate axis $f = 0$, yet every neighbourhood contains points where $xyz > 0$ and points where $xyz < 0$, since flipping the sign of one coordinate flips the sign of the product. They are the three-variable analogue of a saddle.



8 critical points $\frac{1}{\sqrt{3}}(\pm1,\pm1,\pm1)$ on S^2

Example 13.a.9 (*Extrema on a circle*): Find the maximum and minimum of $f(x,y) = xy$ under the condition that (x,y) lies on the circle with radius $\sqrt{2}$. (cf. Michaels Bsp 7.2.3)

We want to optimize $f(x,y) = xy$ subject to the constraint $g(x,y) = x^2+y^2-2 = 0$. The Lagrangian is

$$L(x,y,\lambda) := f(x,y) - \lambda g(x,y) = xy - \lambda(x^2+y^2-2).$$

Setting the gradient of L to zero yields the system:

$$\begin{aligned}\partial_x L &= y - 2\lambda x = 0\\ \partial_y L &= x - 2\lambda y = 0\\ \partial_\lambda L &= 2-(x^2+y^2) = 0.\end{aligned}$$

From the first two equations, $y = 2\lambda x$ and $x = 2\lambda y$. Substituting the second into the first gives $y = 4\lambda^2 y$, or $y(1-4\lambda^2) = 0$. If $y = 0$, then $x = 0$, which contradicts $x^2+y^2 = 2$. Hence $y\neq 0$ and $4\lambda^2 = 1 \implies \lambda = \pm 1/2$. If $\lambda = 1/2$, then $y = x$. The constraint gives $2x^2 = 2 \implies x = \pm1$, giving points $(1,1)$ and $(-1,-1)$. At these points, $f(\pm1,\pm1) = 1$. If $\lambda = -1/2$, then $y = -x$.

The constraint gives $2x^2 = 2 \implies x = \pm 1$, giving points $(1,-1)$ and $(-1,1)$. At these points, $f(\pm 1, \mp 1) = -1$. Since the circle is compact, these must be the global extrema. The maximum is 1 and the minimum is -1.

Example 13.a.10 (*Largest rectangle in an ellipse*): Given an ellipse $E = \{(x,y) \in \mathbb{R}^2 \mid x^2/A^2 + y^2/B^2 = 1\}$, find the rectangle (with sides parallel to the axes) whose vertices lie on E and has the largest area. (cf. Michaels Bsp 7.2.7)

Nothing here is constrained until we make it so. The first step is to name the unknowns and read the constraint off the geometry. Let the vertices of the rectangle in the four quadrants be $(\pm x, \pm y)$ with $x, y \geq 0$. The area of the rectangle is $f(x,y) = 4xy$. We maximize $f(x,y)$ subject to the constraint $g(x,y) = \frac{x^2}{A^2} + \frac{y^2}{B^2} - 1 = 0$ in the region $x > 0, y > 0$. The Lagrangian is

$$L(x,y,\lambda) := 4xy - \lambda\left(\frac{x^2}{A^2} + \frac{y^2}{B^2} - 1\right).$$

The critical point equations are:

$$\partial_x L = 4y - \frac{2\lambda x}{A^2} = 0 \implies \lambda = \frac{2yA^2}{x}$$
$$\partial_y L = 4x - \frac{2\lambda y}{B^2} = 0 \implies \lambda = \frac{2xB^2}{y}$$
$$\partial_\lambda L = 1 - \frac{x^2}{A^2} - \frac{y^2}{B^2} = 0.$$

Equating the expressions for λ yields $\frac{2yA^2}{x} = \frac{2xB^2}{y} \implies y^2A^2 = x^2B^2$, or $\frac{x^2}{A^2} = \frac{y^2}{B^2}$. Substituting this into the constraint gives $2\frac{x^2}{A^2} = 1 \implies x = \frac{A}{\sqrt{2}}$. Similarly, $y = \frac{B}{\sqrt{2}}$. The maximal area is $f\left(\frac{A}{\sqrt{2}}, \frac{B}{\sqrt{2}}\right) = 4 \cdot \frac{A}{\sqrt{2}} \cdot \frac{B}{\sqrt{2}} = 2AB$. Note that the answer does not depend on the shape of the ellipse in the way one might expect: the optimal rectangle occupies the fraction $2AB/(\pi AB) = 2/\pi$ of the ellipse's area whatever A and B are, because the substitution $(x,y) = (Au, Bv)$ turns the whole problem into the same question about the unit circle.

Exercise 13.a.11 (*Antipodal points maximise distance on the sphere*): Let $n \geq 2$. Prove that two points $x, y \in S^{n-1}$ have maximal distance $|x-y|$ if and only if $x = -y$. Consider the function $(x,y) \mapsto |x-y|^2$ on $S^{n-1} \times S^{n-1} \subset \mathbb{R}^{2n}$.

Hint: *The constraint set is a product of two spheres, so there are two constraints, and Proposition 13.a.2 asks for their gradients to be linearly independent before it will produce any multipliers at all.*

Solution: *See solution on Page 199*

AI-Exercise 13.a.12 (*Extrema on a Circle via Lagrange Multipliers*): Find the maximum and minimum values of $f(x,y) := x + 2y$ subject to the constraint $g(x,y) := x^2 + y^2 - 5 = 0$.

Solution: *See solution on Page 200*

Exercise 13.a.13 (*Extrema on a closed ball, inside and on the boundary* (important)): Consider the ball $U := \{(x,y,z) : x^2 + y^2 + z^2 < 1\}$ and the function

$$f(x,y,z) := 1 - x^2 - y^2 + 4x.$$

(a) Motivate rigorously why f attains its absolute maximum and minimum in $\overline{U}$.

(b) Find all the critical points of f which lie in the interior of U. Say whether $\sup_U f$ (or $\inf_U f$) is attained at some point in U.

(c) Say whether $\max_{\partial U} f$ and $\min_{\partial U} f$ exist.

(d) With the method of Lagrange multipliers, find the possible critical points for the constrained problem $\max\{f(x,y,z) : (x,y,z) \in \partial U\}$, and same for min.

(e) Among all the points you have found, clarify at which points the maximum/minimum of f is attained.

Info: *This exercise is Problem 5.2 of Problem Sheet 5, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 199*

Exercise 13.a.14 (*Extrema on a sphere cut by a plane* (important)): Consider the function $f(x,y,z) = 3x - y + 2z$ and the set

$$M = \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1,\ x + y = 0\}.$$

Determine the extrema of f on M and their nature.

Info: *This exercise is Problem 6.10 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 200*

Exercise 13.a.15 (*Closest point on a hyperbolic paraboloid* (important)): Find the points on the submanifold

$$M = \{(x,y,z) \in \mathbb{R}^3 : z = x^2 - y^2\}$$

that are closest to the point $(0,0,1)$.

Info: *This exercise is Problem 6.11 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 200*

Exercise 13.a.16 (*Extrema of a linear form on an ellipsoid*): Let $a_1, a_2, a_3 > 0$ and let $b = (b_1, b_2, b_3)^\mathsf{T} \in \mathbb{R}^3 \setminus \{0\}$. Find the maximum and minimum values of the linear function $f(x) := b_1x_1 + b_2x_2 + b_3x_3$ on the ellipsoid

$$E := \left\{ x \in \mathbb{R}^3 \;\middle|\; \frac{x_1^2}{a_1^2} + \frac{x_2^2}{a_2^2} + \frac{x_3^2}{a_3^2} = 1 \right\},$$

and determine the points of E at which these extreme values are attained.

Info: *This exercise is taken from the Analysis I & II exam of 9 August 2021 (Prof. Giovanni Felder), Part B, Problem 4.*

Solution: *See solution on Page 201*

Exercise 13.a.17 (*Minimising entropy on the simplex*): Let $n \geq 1$ and let $A := \{x \in (0,1]^n \mid x_1 + \cdots + x_n = 1\}$. Consider

$$f : A \to \mathbb{R}, \qquad f(x) := \sum_{i=1}^{n} x_i \log(x_i).$$

(a) Show that f has exactly one critical point p in the interior of A, and determine $f(p)$.

(b) Prove that f attains its minimum on A at p.

Info: *This exercise is taken from the Analysis I & II exam of 24 January 2022 (Prof. Giovanni Felder), Part A, Problem 8.*

Solution: *See solution on Page 201*

Exercise 13.a.18 (*Single choice: a constrained problem with no critical point*): Consider a function f of three variables, of which all we know is that

$$\nabla f(x,y,z) = \begin{pmatrix} e^x \\ 2y \\ 2z \end{pmatrix},$$

together with the constraint $g(x,y,z) := ay^2 + bz^2 - \pi = 0$ for parameters $a, b \in \mathbb{R} \setminus \{0\}$. Exactly one of the following is correct. Which?

(a) Independently of the values of a and b, f has no extremal point under the given constraint.
(b) The constraint defines a compact set in space independently of the values of a and b, so both the constrained minimum and the constrained maximum have to be attained.
(c) In the case $a = b$, the constrained minimum is attained.
(d) Without further knowledge of f, no assertion about the constrained optimisation problem is possible.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 4.*

Solution: *See solution on Page 202*

Exercise 13.a.19 (*Two constraints that leave nothing for Lagrange to do*): Determine the maximal value of

$$f : \mathbb{R}^3 \to \mathbb{R}, \qquad f(x, y, z) := x^2 + y + z^2$$

subject to the two constraints $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 = 1/2$.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Problem 13, a box problem where only the final answer was graded.*

Solution: *See solution on Page 203*

Second-order conditions on the constraint set (Section 13.b)

Every example so far has closed the same gap in the same way. The Lagrange condition returns a list of candidates, and compactness plus a comparison of values decides among them. That works only when the constraint set is compact, and it decides only the global extrema. Is there a second-order test, of the kind Theorem 12.b.11 provides in the unconstrained case? There is, and it is that same test, applied to the Lagrangian rather than to f, and read only along directions that stay on the constraint set.

Theorem 13.b.20 (*Second-order sufficient conditions for constrained extrema*): Let $U \subseteq \mathbb{R}^n$ be open, $f \in C^2(U, \mathbb{R})$, $g = (g_1, \dots, g_k) \in C^2(U, \mathbb{R}^k)$, and suppose $x_0 \in g^{-1}\{0\}$ is a **regular** point, meaning that $\nabla g_1(x_0), \dots, \nabla g_k(x_0)$ are linearly independent, and a **Lagrange point**, meaning that $\nabla f(x_0) = \sum_{j=1}^{k} \lambda_j \nabla g_j(x_0)$ for some $\lambda \in \mathbb{R}^k$ (as furnished by Proposition 13.a.2). Write $L(x, \lambda) := f(x) - \sum_{j=1}^{k} \lambda_j g_j(x)$ for the Lagrangian, and let

$$T_{x_0} := \{v \in \mathbb{R}^n : Dg(x_0)v = 0\}$$

be the tangent space of the constraint set $g^{-1}\{0\}$ at x_0 (an $(n-k)$-dimensional subspace, by regularity). Then:

(a) If $v^{\mathsf{T}} \operatorname{Hess}_x L(x_0, \lambda) v > 0$ for all $v \in T_{x_0} \setminus \{0\}$, then x_0 is a **strict local minimum** of $f|_{g^{-1}\{0\}}$.
(b) If $v^{\mathsf{T}} \operatorname{Hess}_x L(x_0, \lambda) v < 0$ for all $v \in T_{x_0} \setminus \{0\}$, then x_0 is a **strict local maximum** of $f|_{g^{-1}\{0\}}$.
(c) If $v^{\mathsf{T}} \operatorname{Hess}_x L(x_0, \lambda) v$ takes both signs on $T_{x_0} \setminus \{0\}$, then x_0 is **not** a local extremum of $f|_{g^{-1}\{0\}}$.

Here $\operatorname{Hess}_x L(x_0, \lambda)$ denotes the Hessian of $L(\cdot, \lambda)$ in the x-variable alone, evaluated at x_0.

Proof:

By regularity and the implicit function theorem (Theorem 16.a.1), after relabelling coordinates so that the $k \times k$ block $\partial g / \partial(x_{n-k+1}, \dots, x_n)$ is invertible at x_0, there is an open $V \subseteq \mathbb{R}^{n-k}$ and a C^2 map $\varphi : V \to g^{-1}\{0\}$ with $\varphi(0) = x_0$, solving the constraint locally, such that $D\varphi(0) : \mathbb{R}^{n-k} \to T_{x_0}$ is an isomorphism (this is the standard local parametrization of the submanifold $g^{-1}\{0\}$; see also Theorem 17.a.3 for why it is one).

Since $g(\varphi(t)) \equiv 0$ on V, the terms $-\lambda_j g_j(\varphi(t))$ vanish identically, so

$$h(t) := f(\varphi(t)) = L(\varphi(t), \lambda) \qquad \text{for all } t \in V.$$

By the chain rule, $Dh(0) = D_x L(x_0, \lambda)\, D\varphi(0) = 0$, since $D_x L(x_0, \lambda) = \nabla f(x_0)^{\mathsf{T}} - \sum_j \lambda_j \nabla g_j(x_0)^{\mathsf{T}} = 0$ because x_0 is a Lagrange point. So h has a critical point at 0, which is what being a Lagrange point means in the coordinates φ.

Differentiating a second time, for $w \in \mathbb{R}^{n-k}$,

$$\operatorname{Hess} h(0)[w, w] = \big(D\varphi(0)w\big)^{\mathsf{T}} \operatorname{Hess}_x L(x_0, \lambda)\big(D\varphi(0)w\big) \;+\; D_x L(x_0, \lambda) \cdot D^2\varphi(0)[w, w],$$

and the second term vanishes because $D_x L(x_0, \lambda) = 0$, regardless of what $D^2\varphi(0)$ is. This is the same cancellation that already made $Dh(0) = 0$. So, writing $v := D\varphi(0)w$, which ranges over all of T_{x_0} as w ranges over $\mathbb{R}^{n-k}$ (since $D\varphi(0)$ is an isomorphism onto T_{x_0}),

$$\operatorname{Hess} h(0)[w, w] = v^{\mathsf{T}} \operatorname{Hess}_x L(x_0, \lambda) v.$$

The three cases now follow by reading the unconstrained Hessian test (Theorem 12.b.11) off h on the open set $V \subseteq \mathbb{R}^{n-k}$, and transporting the conclusion back through φ, which is a local homeomorphism from V onto a neighbourhood of x_0 in $g^{-1}\{0\}$.

- **(a)** $\operatorname{Hess}_x L(x_0, \lambda)$ positive definite on T_{x_0} is exactly $\operatorname{Hess} h(0)$ positive definite, so the Hessian test gives a strict local minimum of h at 0, which transports to a strict local minimum of $f|_{g^{-1}\{0\}}$ at x_0.
- **(b)** Identical, with every inequality reversed.
- **(c)** If the quadratic form takes both signs on T_{x_0}, then $\operatorname{Hess} h(0)$ is indefinite, so h increases along one direction through $0 \in V$ and decreases along another (Theorem 12.b.11 again), and transporting those two directions through φ shows $f|_{g^{-1}\{0\}}$ does the same near x_0.

Q.E.D.

Remark 13.b.21: This is exactly Theorem 12.b.11 run on the Lagrangian instead of on f itself, with the Hessian test's ordinary $\mathbb{R}^n$ replaced by the tangent space T_{x_0}. That is also why L, not f, is the right function to differentiate: only L has a genuine critical point at x_0 (in the full, unrestricted sense), since f itself typically does not. In practice **(a)**–**(c)** are checked by picking any basis $v_1, \dots, v_{n-k}$ of T_{x_0} (e.g. a basis of $\ker Dg(x_0)$) and testing the definiteness of the $(n-k) \times (n-k)$ matrix $\big(v_i^{\mathsf{T}} \operatorname{Hess}_x L(x_0, \lambda) v_j\big)_{i,j}$ by the eigenvalue, determinant/trace or Hurwitz recipes at the end of Section 12.b. There is no need to construct φ explicitly, which was only a proof device.

Example 13.b.22 (*The circle example, revisited with the second-order test*)**:** Apply Theorem 13.b.20 to Example 13.a.9 ($f(x, y) = xy$ on $g(x, y) := x^2 + y^2 - 2 = 0$) to see **why** $(1, 1)$ is a maximum and $(1, -1)$ a minimum, without appealing to compactness. Here

$$\operatorname{Hess}_x L(x, y, \lambda) = \operatorname{Hess} f - \lambda \operatorname{Hess} g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} -2\lambda & 1 \\ 1 & -2\lambda \end{pmatrix},$$

with eigenvalues $-2\lambda \pm 1$. At the two values $\lambda = \pm\frac{1}{2}$ that actually occur here, one eigenvalue is 0, so on all of $\mathbb{R}^2$ the matrix is only semidefinite and the Hessian test (Theorem 12.b.11) would say nothing. Restricted to the 1-dimensional tangent space it is a different story.

At $(1, 1)$, $\lambda = \frac{1}{2}$: $\operatorname{Hess}_x L = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and $T_{(1,1)} = \ker \begin{pmatrix} 2 & 2 \end{pmatrix} = \operatorname{span}\{(1, -1)\}$. For $v = (1, -1)$,

$$v^{\mathsf{T}} \operatorname{Hess}_x L\, v = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 2 \end{pmatrix} = -4 < 0,$$

negative definite on $T_{(1,1)}$: a strict local maximum, matching $f(1, 1) = 1$.

At $(1,-1)$, $\lambda = -\frac{1}{2}$: $\operatorname{Hess}_x L = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and $T_{(1,-1)} = \ker \begin{pmatrix} 2 & -2 \end{pmatrix} = \operatorname{span}\{(1,1)\}$. For $v = (1,1)$,

$$v^{\mathsf{T}} \operatorname{Hess}_x L\, v = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 4 > 0,$$

positive definite on $T_{(1,-1)}$: a strict local minimum, matching $f(1,-1) = -1$. The same computation at $(-1,-1)$ and $(-1,1)$ (by symmetry $f(-x,-y) = f(x,y)$) reproduces the same two conclusions. This is the sharpest illustration available of the theorem's point: the **unrestricted** Hessian of L is degenerate at both critical points, yet restricting to the 1-dimensional tangent space resolves it completely in both cases.

An application: the spectral theorem (Section 13.c)

The spectral theorem has been used freely since Remark 12.b.8, on the strength of linear algebra alone. It can also be **proved** here, and the proof is a fair test of what Proposition 13.a.2 is worth. Maximise the quadratic form $x \mapsto \langle x, Ax\rangle$ over the unit sphere, and the multiplier that falls out is an eigenvalue. Nothing is circular in this, since Proposition 13.a.2 nowhere uses diagonalisation.

Theorem 13.c.23 (*Spectral theorem, via constrained optimization*): Every symmetric $A \in \mathbb{R}^{n\times n}$ admits an orthonormal basis $\{v_1, \dots, v_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of A, with real eigenvalues.

Proof:

Let $f(x) := \langle x, Ax\rangle$, a polynomial and hence continuous, and let $g(x) := |x|^2 - 1$, so that the unit sphere is $S^{n-1} = g^{-1}\{0\}$. Since A is symmetric, $\nabla f(x) = 2Ax$; also $\nabla g(x) = 2x$.

We construct the v_j one at a time. Suppose $v_1, \dots, v_k$ are already orthonormal eigenvectors of A, with eigenvalues $\lambda_1, \dots, \lambda_k$ (for $k = 0$ there is nothing to suppose). Put

$$K_k := S^{n-1} \cap \{x : \langle x, v_j\rangle = 0 \text{ for } j = 1, \dots, k\},$$

the unit sphere of the orthogonal complement of $\operatorname{span}\{v_1, \dots, v_k\}$. This is closed and bounded, hence compact by Theorem 7.b.28, and non-empty as long as $k < n$, so f attains a maximum on it at some $v_{k+1} \in K_k$.

The constraints active at v_{k+1} are g together with the k linear functions $h_j(x) := \langle x, v_j\rangle$, whose gradients are $\nabla g(v_{k+1}) = 2v_{k+1}$ and $\nabla h_j(v_{k+1}) = v_j$. These $k+1$ vectors are orthogonal and non-zero, hence linearly independent, so Proposition 13.a.2 applies and yields multipliers with

$$2Av_{k+1} = 2\lambda_{k+1} v_{k+1} + \sum_{j=1}^{k} \mu_j v_j.$$

Pair both sides with v_i for a fixed $i \le k$. On the right, $\langle v_{k+1}, v_i\rangle = 0$ and $\langle v_j, v_i\rangle = \delta_{ji}$, leaving μ_i. On the left, symmetry of A moves it across the inner product,

$$\langle Av_{k+1}, v_i\rangle = \langle v_{k+1}, Av_i\rangle = \lambda_i \langle v_{k+1}, v_i\rangle = 0,$$

using that v_i is already an eigenvector. Hence $\mu_i = 0$ for every $i \le k$, and the displayed identity collapses to $Av_{k+1} = \lambda_{k+1} v_{k+1}$. So, v_{k+1} is a unit eigenvector orthogonal to all its predecessors, and after n steps the v_j form the required basis. Each eigenvalue is real because $\lambda_j = \langle v_j, Av_j\rangle$ is a real number by construction. Q.E.D.

Corollary 13.c.24 (*Extreme eigenvalues as Rayleigh quotients*): For symmetric $A \in \mathbb{R}^{n\times n}$,

$$\lambda_{\min}(A) = \min_{x \in S^{n-1}} \langle x, Ax\rangle, \qquad \lambda_{\max}(A) = \max_{x \in S^{n-1}} \langle x, Ax\rangle.$$

Proof:

Write $x \in S^{n-1}$ in the orthonormal eigenbasis of Theorem 13.c.23 as $x = \sum_j c_j v_j$ with $\sum_j c_j^2 = 1$. Then $\langle x, Ax\rangle = \sum_j \lambda_j c_j^2$ is a convex combination of the eigenvalues, so it lies between $\lambda_{\min}$ and $\lambda_{\max}$; both bounds are attained by taking x to be the corresponding eigenvector. **Q.E.D.**

Solutions (Section 13.d)

Every exercise in this chapter is solved here. The chapter's three demonstrations of the multiplier rule (Example 13.a.8 on Page 192, Example 13.a.9 on Page 193 and Example 13.a.10 on Page 194) are worked examples, and are carried out where they stand.

Proof of Exercise 13.a.11:

Since $|x| = |y| = 1$ on S^{n-1},

$$f(x,y) := |x-y|^2 = |x|^2 - 2\langle x,y\rangle + |y|^2 = 2 - 2\langle x,y\rangle,$$

so maximising $|x-y|$ is the same as maximising f, i.e. minimising $\langle x,y\rangle$. The domain $S^{n-1}\times S^{n-1}$ is compact (a product of compact sets), so f attains a maximum somewhere by the extreme value theorem; we locate it with Lagrange multipliers under the two constraints $g_1(x,y) := |x|^2 - 1 = 0$ and $g_2(x,y) := |y|^2 - 1 = 0$.

It is $h(x,y) := -2\langle x,y\rangle$ that we optimise, not f itself. The two are interchangeable here for a reason worth stating: $f = g_1 + g_2 + 2 + h$ identically on $\mathbb{R}^{2n}$, so f and h agree on M and their gradients differ by $\nabla g_1 + \nabla g_2$, a combination the multipliers absorb, leaving the same critical points. Computing gradients in $\mathbb{R}^{2n} = \mathbb{R}^n_x \times \mathbb{R}^n_y$:

$$\nabla h = (-2y, -2x), \qquad \nabla g_1 = (2x, 0), \qquad \nabla g_2 = (0, 2y).$$

At a critical point, $\nabla g_1, \nabla g_2$ are linearly independent (they are non-zero and supported on disjoint blocks of coordinates), so Proposition 13.a.2 gives λ_1, λ_2 with $\nabla h = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$, i.e.

$$-2y = 2\lambda_1 x, \qquad -2x = 2\lambda_2 y \qquad \Longrightarrow \qquad y = -\lambda_1 x, \quad x = -\lambda_2 y.$$

So, x and y are parallel. Combined with $|x| = |y| = 1$, this forces $y = x$ or $y = -x$. These are the only two candidates, and $f(x,x) = 0$ while $f(x,-x) = |2x|^2 = 4$. The maximum of f over the compact domain is therefore 4, attained exactly when $y = -x$ (i.e. exactly at antipodal pairs, where $|x-y| = 2$). **Q.E.D.**

Remark 13.d.25 (*Squaring first is not a cosmetic step*): The problem is posed for $|x-y|$ and solved for $|x-y|^2$, and the swap is what makes the multiplier rule applicable at all. The Euclidean norm fails to be differentiable exactly where its argument vanishes, which here is the diagonal $x = y$, and that is one of the two candidate configurations the argument has to compare. Squaring removes the singularity without moving the maximiser, since $t \mapsto t^2$ is strictly increasing on $[0,\infty)$. The same manoeuvre is worth reaching for whenever a distance is being optimised.

Solution to Exercise 13.a.13:

(a) $\overline{U}$ is closed and bounded in $\mathbb{R}^3$, hence compact by Theorem 7.b.28, and f is a polynomial, hence continuous. By the extreme value theorem (Item **(b)**), f attains its absolute maximum and minimum on $\overline{U}$.

(b) $\nabla f(x,y,z) = (-2x+4,\ -2y,\ 0)$ vanishes only at $x = 2$, $y = 0$ (any z), and no such point lies in U (since $x = 2$ already violates $x^2 + y^2 + z^2 < 1$). So, f has **no critical point in the interior of** U; consequently, by Proposition 12.a.2, $\sup_U f$ and $\inf_U f$ are **not attained** at any interior point.

(c) ∂U is compact (a sphere) and f is continuous, so $\max_{\partial U} f$ and $\min_{\partial U} f$ **both exist**, again by the extreme value theorem (Item **(b)**).

(d) With $g(x,y,z) := x^2+y^2+z^2-1$, the Lagrange condition $\nabla f = \lambda \nabla g$ reads $(-2x+4,-2y,0) = \lambda(2x,2y,2z)$. The last coordinate gives $\lambda z = 0$. **If $z \neq 0$:** then $\lambda = 0$, so $-2x+4=0 \Rightarrow x=2$, impossible on ∂U. So, $z=0$. Then $x^2+y^2=1$, and the remaining equations are $4-2x = 2\lambda x$ and $-2y = 2\lambda y$, i.e. $-2y(1+\lambda)=0$. **If $\lambda = -1$:** the first equation becomes $4-2x=-2x$, i.e. $4=0$, impossible. So $y=0$, and $x^2=1$ gives $x=\pm 1$. The Lagrange candidates are $(1,0,0)$ (with $\lambda=1$) and $(-1,0,0)$ (with $\lambda=-3$).

(e) Evaluating, $f(1,0,0) = 1-1+4 = 4$ and $f(-1,0,0) = 1-1-4 = -4$. Since there are no interior critical points, these are the global extrema on $\overline{U}$: the **maximum** 4 is attained at $(1,0,0)$, and the **minimum** -4 at $(-1,0,0)$.

Q.E.D.

Proof of AI-Exercise 13.a.12:

The constraint set $S := \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2=5\}$ is compact and $\nabla g(x,y) = (2x,2y)^\mathsf{T} \neq (0,0)^\mathsf{T}$ on S. By Lagrange Multipliers, candidates satisfy $\nabla f = \lambda \nabla g$, i.e.:

$$1 = 2\lambda x \quad \text{and} \quad 2 = 2\lambda y \implies y = 2x.$$

Substituting $y=2x$ into $x^2+y^2=5$ gives $5x^2=5 \implies x=\pm 1$. The candidate points are $(1,2)$ and $(-1,-2)$. Evaluating f:

$$f(1,2) = 1+4 = 5 \quad \text{(maximum)}, \qquad f(-1,-2) = -1-4 = -5 \quad \text{(minimum)}.$$

Q.E.D.

Solution to Exercise 13.a.14:

With $g_1(x,y,z) := x^2+y^2+z^2-1$ and $g_2(x,y,z) := x+y$, the Lagrangian $L := f - \lambda_1 g_1 - \lambda_2 g_2$ gives the system

$$3-2\lambda_1 x - \lambda_2 = 0, \qquad -1-2\lambda_1 y - \lambda_2 = 0, \qquad 2-2\lambda_1 z = 0, \qquad x+y=0, \qquad x^2+y^2+z^2=1.$$

From $x+y=0$, $y=-x$. Subtracting the second equation from the first eliminates λ_2:

$$4-2\lambda_1 x + 2\lambda_1 y = 4-2\lambda_1 x - 2\lambda_1 x = 4-4\lambda_1 x = 0, \qquad \text{so} \qquad \lambda_1 x = 1.$$

From $2-2\lambda_1 z = 0$, $\lambda_1 = 1/z$ (so $z \neq 0$), hence $x=z$ (from $\lambda_1 x = 1 = \lambda_1 z$). With $y = -x = -z$, the constraint becomes $3x^2=1$, i.e. $x = \pm\frac{1}{\sqrt{3}}$, giving the two points $\frac{1}{\sqrt{3}}(1,-1,1)$ and $-\frac{1}{\sqrt{3}}(1,-1,1)$. Evaluating,

$$f\Big(\tfrac{1}{\sqrt{3}}(1,-1,1)\Big) = \frac{3+1+2}{\sqrt{3}} = 2\sqrt{3}, \qquad f\Big(-\tfrac{1}{\sqrt{3}}(1,-1,1)\Big) = -2\sqrt{3}.$$

The method is legitimate here: $\nabla g_1 = 2(x,y,z)$ and $\nabla g_2 = (1,1,0)$ are dependent only where $z=0$ and $x=y$, which together with $x+y=0$ gives the origin, not a point of M. Since M is compact (closed and bounded in $\mathbb{R}^3$) and f is continuous, f attains a maximum and a minimum on M, and both must be among the two candidates. So f attains its **maximum** $2\sqrt{3}$ at $\frac{1}{\sqrt{3}}(1,-1,1)$ and its **minimum** $-2\sqrt{3}$ at $-\frac{1}{\sqrt{3}}(1,-1,1)$.

Q.E.D.

Solution to Exercise 13.a.15:

Minimizing the distance to $(0,0,1)$ is the same as minimizing $f(x,y,z) := x^2+y^2+(z-1)^2$ subject to $g(x,y,z) := z-x^2+y^2 = 0$. Since f is a distance squared to a fixed point, its minimum over the (closed) constraint set is attained. Since $\nabla g = (-2x,2y,1)$ never vanishes, Lagrange's condition $\nabla f = \lambda \nabla g$ holds at the minimiser, and it reads

$$2x = -2\lambda x, \qquad 2y = 2\lambda y, \qquad 2(z-1) = \lambda,$$

i.e. $x(1+\lambda)=0$ and $y(1-\lambda)=0$.

Case $x=y=0$: the constraint gives $z=0$, so $(0,0,0)$ with $d^2=1$.

Case $x\neq 0$ (so $\lambda=-1$): then $2(z-1)=-1 \implies z=\frac{1}{2}$, and $y(1-\lambda)=2y=0 \implies y=0$. The constraint $z=x^2-y^2$ gives $x^2=\frac{1}{2}$, i.e. $x=\pm\frac{1}{\sqrt{2}}$, giving $\left(\pm\frac{1}{\sqrt{2}},0,\frac{1}{2}\right)$ with $d^2=\frac{1}{2}+0+\frac{1}{4}=\frac{3}{4}$.

Case $y\neq 0$ (so $\lambda=1$): then $2(z-1)=1 \implies z=\frac{3}{2}$, and $x(1+\lambda)=2x=0 \implies x=0$; the constraint becomes $-y^2=\frac{3}{2}$, which has no real solution.

Comparing $d^2=1$ against $d^2=\frac{3}{4}$, the **closest points** are $\left(\pm\frac{1}{\sqrt{2}},0,\frac{1}{2}\right)$, at minimal distance $\frac{\sqrt{3}}{2}$.

Q.E.D.

Proof of Exercise 13.a.16:

Write $S:=a_1^2b_1^2+a_2^2b_2^2+a_3^2b_3^2$, and define the constraint function

$$g(x_1,x_2,x_3):=\frac{x_1^2}{a_1^2}+\frac{x_2^2}{a_2^2}+\frac{x_3^2}{a_3^2}-1,$$

so that $E=g^{-1}(0)$. This set is closed and bounded, hence compact, and f is continuous, so f does attain a maximum and a minimum on E; it remains to find them.

The gradients are $\nabla f(x)=(b_1,b_2,b_3)^{\mathsf{T}}$ and $\nabla g(x)=\left(\frac{2x_1}{a_1^2},\frac{2x_2}{a_2^2},\frac{2x_3}{a_3^2}\right)^{\mathsf{T}}$. Since $0\notin E$, we have $\nabla g\neq 0$ on E, so every extremum satisfies Lagrange's condition $\nabla f=\lambda\nabla g$ for some λ. Componentwise this reads $b_i=2\lambda x_i/a_i^2$, and therefore

$$x_i=\frac{a_i^2b_i}{2\lambda},\qquad i=1,2,3.$$

Here $\lambda\neq 0$, since $\lambda=0$ would force $b=0$. Substituting into the constraint,

$$\sum_{i=1}^{3}\frac{1}{a_i^2}\left(\frac{a_i^2b_i}{2\lambda}\right)^2=\frac{S}{4\lambda^2}=1,$$

which gives $2\lambda=\pm\sqrt{S}$. The two candidate points are therefore

$$x^{\pm}=\pm\frac{\left(a_1^2b_1,\ a_2^2b_2,\ a_3^2b_3\right)^{\mathsf{T}}}{\sqrt{S}},$$

and evaluating f at them,

$$f(x^{\pm})=\pm\frac{1}{\sqrt{S}}\sum_{i=1}^{3}a_i^2b_i^2=\pm\frac{S}{\sqrt{S}}=\pm\sqrt{S}.$$

Since the maximum and minimum both exist and must occur among these two points, the maximum value is $\sqrt{S}$, attained at x^+, and the minimum value is $-\sqrt{S}$, attained at x^-.

Remark 13.d.26: Substituting $u_i:=x_i/a_i$ turns f into $\langle u,(a_ib_i)_i\rangle$ on the unit sphere, where the answer is Cauchy–Schwarz: the extreme values of a linear form on the unit sphere are $\pm$ the norm of its coefficient vector, attained at $\pm$ the unit vector in that direction. The Lagrange computation above is that fact in disguise, with $\sqrt{S}$ the norm of $(a_ib_i)_i$.

Q.E.D.

Proof of Exercise 13.a.17:

Throughout, write $g(x):=x_1+\cdots+x_n-1$ for the constraint function, so that $A=g^{-1}(0)\cap(0,1]^n$, and note $\nabla g\equiv(1,\dots,1)^{\mathsf{T}}\neq 0$, so the Lagrange condition is available at every point.

(a) On the interior, where every $x_i > 0$, the function $t \mapsto t \log t$ is differentiable and $\nabla f(x) = \big(\log(x_i) + 1\big)_{i=1}^{n}$. The Lagrange condition $\nabla f(x) = \lambda \nabla g(x)$ reads

$$\log(x_i) + 1 = \lambda \qquad \text{for every } i,$$

so all the x_i share the common value $e^{\lambda - 1}$; call it c. The constraint $\sum_i x_i = 1$ becomes $nc = 1$, forcing $c = 1/n$. Hence there is exactly one critical point,

$$p = \left(\tfrac{1}{n}, \dots, \tfrac{1}{n}\right), \qquad f(p) = \sum_{i=1}^{n} \tfrac{1}{n} \log \tfrac{1}{n} = \log \tfrac{1}{n} = -\log n.$$

(b) Existence needs a compact set, and A is not closed: points with some $x_i = 0$ are missing. So pass to the closed standard simplex

$$\Delta := \{x \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } i,\ \textstyle\sum_i x_i = 1\},$$

which is closed and bounded, hence compact, and extend f to it by setting $0 \log 0 := 0$. This extension is continuous, because $t \log t \to 0$ as $t \to 0^+$. Therefore f attains a minimum somewhere on Δ.

We argue by induction on n; the case $n = 1$ is immediate, since $A = \{1\}$ and $f(1) = 0 = -\log 1$. Suppose the minimum were attained at a point $x \in \Delta \setminus A$, so that exactly k of the coordinates are positive, with $1 \leq k \leq n - 1$. On that face, f is the same expression in k variables summing to 1, so the induction hypothesis applied with k in place of n gives that its smallest value there is $-\log k$. Since $k < n$ gives $-\log k > -\log n = f(p)$, and $p \in \Delta$, such a point cannot be a minimiser. The minimum is therefore attained in the interior, where by **(a)** the only candidate is p. Hence $\min_A f = f(p) = -\log n$.

Remark 13.d.27: The result is the discrete Gibbs inequality: among all probability vectors on n outcomes, the uniform one has the least $\sum x_i \log x_i$, that is, the **greatest** entropy $-\sum x_i \log x_i$, namely $\log n$. The mechanism is convexity: $\operatorname{Hess} f(x) = \operatorname{diag}(1/x_1, \dots, 1/x_n)$ is positive definite on the interior, so f is strictly convex there, and a strictly convex function on a convex set has at most one critical point. That is why part **(a)** finds exactly one and part **(b)** can then simply rule out the boundary.

Q.E.D.

Solution of Exercise 13.a.18:

The correct answer is **(a)**, and one line of the Lagrange system settles it.

We have $\nabla g(x, y, z) = (0,\, 2ay,\, 2bz)^{\mathsf{T}}$, which vanishes only where $y = z = 0$; that point is not on the constraint set, since $g(x, 0, 0) = -\pi \neq 0$. So the constraint qualification holds everywhere on $g^{-1}\{0\}$ and Proposition 13.a.2 applies without reservation: at any constrained extremum there is a λ with $\nabla f = \lambda \nabla g$, that is

$$e^x = \lambda \cdot 0, \qquad 2y = 2\lambda a y, \qquad 2z = 2\lambda b z.$$

The first equation reads $e^x = 0$, which has no real solution. The system is therefore inconsistent, there is no critical point of the constrained problem for any a and b, and by the theorem there is no constrained extremum either.

Remark 13.d.28 (*Why the other three are wrong, and what they are testing*):

- **(b)** fails on the geometry, and the phrase to distrust is **independently of the values of a and b**. Whenever the set $\{ay^2 + bz^2 = \pi\}$ in $\mathbb{R}^3$ is non-empty it is unbounded, hence not compact: the variable x is entirely unconstrained, so the set is a cylinder over a conic in the (y, z)-plane and is unbounded along the x-axis. When $a, b > 0$ that conic is an ellipse and the set is an elliptic cylinder; when a and b have opposite signs it is a hyperbola and the set is unbounded in the (y, z)-directions as well. The remaining pattern $a, b < 0$ makes the

left-hand side non-positive, so the constraint set is **empty**, which is compact but attains no extremum either. Under every sign pattern, then, **(b)** fails.

- **(c)** fails for the reason **(a)** succeeds. Setting $a = b$ makes the cross section a circle and does nothing to the first Lagrange equation, which is where the obstruction lives.
- **(d)** is the answer of a reader who notices that f itself is unknown. But Proposition 13.a.2 is a statement about ∇f, and ∇f **is** given. Knowing f only up to an additive constant is exactly enough.

The transferable point is that the constrained problem can be settled without ever writing down f, and that an inconsistent Lagrange system is a legitimate outcome, here a decisive one.

Q.E.D.

Solution of Exercise 13.a.19:

The maximum is $\boxed{5/4}$.

Subtract the second constraint from the first: $z^2 = 1 - 1/2 = 1/2$ on the whole constraint set. So z is not a free variable at all, and the objective collapses to

$$f = x^2 + y + z^2 = x^2 + y + \tfrac{1}{2}.$$

The second constraint gives $x^2 = 1/2 - y^2$, leaving a function of the single variable y:

$$\varphi(y) := \left(\tfrac{1}{2} - y^2\right) + y + \tfrac{1}{2} = 1 + y - y^2, \qquad y \in \left[-\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right],$$

the interval being forced by $x^2 = 1/2 - y^2 \geq 0$. Now $\varphi'(y) = 1 - 2y$ vanishes at $y = 1/2$, which lies in the interval since $1/2 < 1/\sqrt{2}$, and $\varphi'' = -2 < 0$, so this is the maximum. Its value is $\varphi(1/2) = 1 + 1/2 - 1/4 = 5/4$, attained where $y = 1/2$, $x^2 = 1/2 - 1/4 = 1/4$ and $z^2 = 1/2$, that is at the four points $\left(\pm\tfrac{1}{2},\ \tfrac{1}{2},\ \pm\tfrac{1}{\sqrt{2}}\right)$.

The constraint set is compact (it is the intersection of a sphere with a cylinder, two circles of radius $1/\sqrt{2}$ at heights $z = \pm 1/\sqrt{2}$), so a maximum exists and the computation above finds it.

Remark 13.d.29 (*Reach for Lagrange second*): Two constraints in three variables leave a one-dimensional set, and Proposition 13.a.2 with two multipliers will find the answer. It is more work than the problem needs. The constraints here are algebraically compatible in a way that eliminates two of the three variables outright, and noticing that turns a system of five equations in five unknowns $(x, y, z, \lambda_1, \lambda_2)$ into a one-variable parabola.

The habit worth forming: before setting up multipliers, ask whether the constraints can be **solved**. Multipliers exist for the case where they cannot, where the constraint set has no convenient parametrisation, and applying them to a constraint set you could have parametrised is a common way to lose marks to arithmetic.

Q.E.D.

Convexity (Chapter 14)

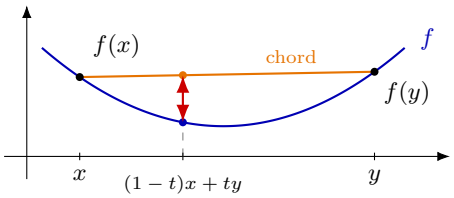
Convexity is the hypothesis under which the local information of the previous two chapters becomes global. For a convex function a critical point is not merely a local minimum but the minimum, and the Hessian test stops being a test at all, since positive semidefiniteness at every point is one of the ways of saying that a function is convex. What has to be earned is that the three natural ways of saying it (by chords, by tangent planes, by the Hessian) really are the same condition.

Convexity (Section 14.a)

Definitions and characterizations (Subsection 14.a.1)

Definition 14.a.1 (*Convex set and convex function*): A subset $A \subseteq \mathbb{R}^n$ is "*convex*" (German: „*konvex*") if for all $x, y \in A$ and every $t \in [0,1]$, the point $p := (1-t)x + ty$ is an element of A.
A function $f : A \to \mathbb{R}$ is convex if A is convex and, for all $x, y \in A$ and $t \in [0,1]$,

$$f\big((1-t)x + ty\big) \leq (1-t)f(x) + tf(y).$$



Definition 14.a.2 (*Semidefinite matrices*): Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then A is

- "*positive semidefinite*" (German: „*positiv semidefinit*"), also called **nonnegative definite**, if $v^\mathsf{T} A v \geq 0$ for all $v \in \mathbb{R}^n$;
- "*negative semidefinite*" (German: „*negativ semidefinit*"), or **nonpositive definite**, if $v^\mathsf{T} A v \leq 0$ for all $v \in \mathbb{R}^n$.

These are the boundary cases of Definition 12.b.6: positive definite implies positive semidefinite, and the two differ exactly on the degenerate matrices, where $v^\mathsf{T} A v = 0$ is allowed for some $v \neq 0$.

Proposition 14.a.3 (*Characterizations of convexity*): Let $U \subseteq \mathbb{R}^n$ be open and convex, and let $f \in C^2(U)$. Then the following are equivalent:

(a) f is convex;
(b) $\operatorname{Hess} f(x)$ is positive semidefinite for all $x \in U$;
(c) $f(y) - f(x) \geq Df_x(y-x)$ for all $x, y \in U$.

Example 14.a.4 (*The parabola*): The parabola $p : \mathbb{R}^n \to \mathbb{R}$, $x \mapsto |x|^2 = \sum_{i=1}^n x_i^2$, is convex. Indeed,

$$\partial_i \partial_j p(x) = 2\delta_{ij}, \qquad \text{and therefore} \qquad \operatorname{Hess} p(x) = 2\,\mathrm{id}_n,$$

and every eigenvalue is $2 > 0$, so $\operatorname{Hess} p(x)$ is positive definite (hence positive semidefinite) for every x, and p is convex by Proposition 14.a.3.

Exercise 14.a.5 (*Convex or not?*): Decide which of the following are convex, and prove it. Attempt each one **without** plotting it, since that is the situation in an exam. To show a function **is** convex, computing the Hessian and checking Proposition 14.a.3**(b)** is usually quickest; to show one is **not**, it is usually fastest to exhibit a single triple x, y, t violating the defining inequality.

(a) $f : [0,\infty)^2 \to \mathbb{R},\ f(x,y) := xy$.
(b) $f : \mathbb{R}^2 \to \mathbb{R},\ f(x,y) := x^2 + xy + y^2$.
(c) $f : [0,\pi]^2 \to \mathbb{R},\ f(x,y) := \cos(x)\cos(y)$.
(d) If $g, h : \mathbb{R} \to \mathbb{R}$ are convex, is $(x,y) \mapsto g(x)h(y)$ necessarily convex?

Solution: *See solution on Page 209*

Proof of the characterizations (Subsection 14.a.2)

Exercise 14.a.6 (*Equivalent characterizations of convexity*): Let $U \subseteq \mathbb{R}^n$ be open and convex.

(a) Show that $f : U \to \mathbb{R}$ is convex if and only if the functions

$$f_{x,y} : [0,1] \to \mathbb{R}, \qquad s \mapsto f(x + s(y-x))$$

are convex for all $x, y \in U$.

(b) Show that

$$f \text{ convex} \iff f(y) - f(x) \geq Df_x(y-x) \quad \text{for all } x, y \in U.$$

This also holds for $f \in C^1(U)$, $U \subseteq \mathbb{R}^n$ open and convex.

(c) Conclude that if $f \in C^1(U)$ is convex and has a critical point at $x_0 \in U$, then f has a minimum at x_0.

Solution to Exercise 14.a.6:

(a) By definition, $f_{x,y}$ is convex if and only if

$$f_{x,y}\big((1-t)s_0 + ts_1\big) \leq (1-t)f_{x,y}(s_0) + tf_{x,y}(s_1) \quad \text{for all } s_0, s_1, t \in [0,1],$$

which, unwinding the definition of $f_{x,y}$, reads

$$f\big(x + [(1-t)s_0 + ts_1](y-x)\big) \leq (1-t)f(x + s_0(y-x)) + tf(x + s_1(y-x)),$$

that is,

$$f\big((1-t)(x + s_0(y-x)) + t(x + s_1(y-x))\big) \leq (1-t)f(x + s_0(y-x)) + tf(x + s_1(y-x)).$$

Setting $s_0 = 0$, $s_1 = 1$ recovers exactly the convexity inequality for f at the points x and y, so if this holds for all $x, y \in U$, then f is convex. Conversely, if f is convex, then convexity of $f_{x,y}$ for each fixed $x, y \in U$ follows by applying convexity of f to the two points $x + s_0(y-x)$ and $x + s_1(y-x)$, which both lie in U since U is convex.

(b) Recall that $g \in C^1(I)$, for $I \subseteq \mathbb{R}$ an interval, is convex if and only if g' is increasing. Combined with **(a)**, this gives

$$f \text{ convex} \iff f'_{x,y} \text{ is increasing, for all } x, y \in U.$$

By the chain rule, for $t \in [0,1]$,

$$f'_{x,y}(t) = Df_{x+t(y-x)}(y-x),$$

and by the mean value theorem there exists $s \in (0,1)$ such that

$$f(y) - f(x) = f_{x,y}(1) - f_{x,y}(0) = f'_{x,y}(s) = Df_{x+s(y-x)}(y-x).$$

(⇒) If f is convex, then $f_{x,y}$ is convex for all $x, y \in U$ by **(a)**, so $f'_{x,y}$ is increasing, and in particular $f'_{x,y}(s) \geq f'_{x,y}(0)$. Combined with the mean value theorem identity above, this yields

$$f(y) - f(x) \geq Df_x(y-x).$$

(⇐) Suppose $f(y) - f(x) \geq Df_x(y-x)$ for all $x, y \in U$. We must show that $0 \leq s < t \leq 1 \implies f'_{x,y}(s) \leq f'_{x,y}(t)$. Set $u := x + s(y-x)$ and $v := x + t(y-x)$, so that $u - v = (s-t)(y-x)$. Applying the hypothesis to the pairs (u,v) and (v,u) gives

$$Df_u(v-u) \leq f(v) - f(u), \qquad Df_v(u-v) \leq f(u) - f(v).$$

Therefore,

$$(t-s)f'_{x,y}(s) = Df_u(v-u) \le f(v)-f(u) \le -Df_v(u-v) = Df_v(v-u) = (t-s)f'_{x,y}(t),$$

and since $t-s>0$, this gives $f'_{x,y}(s) \le f'_{x,y}(t)$, i.e., $f'_{x,y}$ is increasing, hence $f_{x,y}$ is convex for all $x, y \in U$, and so f is convex by **(a)**.

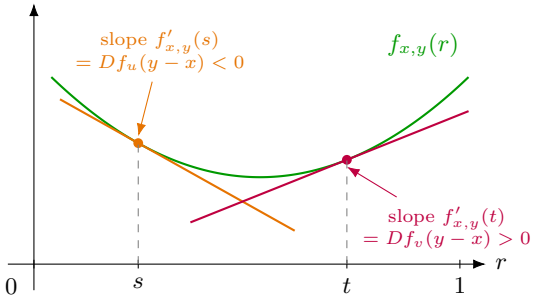
Remark 14.a.7: The argument above only ever used differentiability of f, never twice-differentiability, so the equivalence in **(b)** already holds for $f \in C^1(U)$ with $U \subseteq \mathbb{R}^n$ open and convex, one order of regularity weaker than Proposition 14.a.3, which is stated for $f \in C^2(U)$. This is exactly the fact used in Exercise 14.a.11**(b)**.

(c) Since f is convex and $x_0 \in U$ is a critical point, $Df_{x_0} = 0$, so **(b)** gives

$$0 = Df_{x_0}(y-x_0) \le f(y)-f(x_0) \quad \text{for all } y \in U,$$

i.e., $f(y) \ge f(x_0)$ for all $y \in U$. Hence f has a minimum at x_0, and in fact a global minimum on U. In the convex case this sharpens Proposition 12.a.2, which on its own only guarantees that critical points are the sole **candidates** for extrema.

Q.E.D.



The picture records both halves of Exercise 14.a.6**(b)** at once. Each tangent line lies weakly **below** the graph, which is the inequality $f(y)-f(x) \ge Df_x(y-x)$ read along the segment, and the slope grows from s to t, which is $f'_{x,y}$ increasing.

Proof of Proposition 14.a.3:

(a) $\iff$ **(c)** is Exercise 14.a.6**(b)**. For **(a)** $\iff$ **(b)**, recall from Analysis I that a C^2 function on an interval is convex if and only if its second derivative is non-negative. By Exercise 14.a.6**(a)**, f is convex if and only if every $f_{x,y}(s) = f(x+s(y-x))$ is, and differentiating twice with the chain rule,

$$f''_{x,y}(s) = (y-x)^{\mathsf{T}} \operatorname{Hess} f\big(x+s(y-x)\big)\,(y-x). \tag{14.1}$$

- **(b)⇒(a).** If every Hessian is positive semidefinite, the right-hand side of (14.1) is ≥ 0, so every $f_{x,y}$ is convex, and hence so is f.
- **(a)⇒(b).** Let $z \in U$ and $v \in \mathbb{R}^n$. Since U is open, $y := z+\varepsilon v \in U$ for some $\varepsilon > 0$. The function $f_{z,y}$ is convex, so $f''_{z,y}(0) \ge 0$, and by (14.1) with $s=0$ this reads $\varepsilon^2\, v^{\mathsf{T}} \operatorname{Hess} f(z)\, v \ge 0$. As v was arbitrary, $\operatorname{Hess} f(z)$ is positive semidefinite.

The second bullet is where openness of U is used: it lets us probe every direction v from every point, which a closed domain would not allow at its boundary. Q.E.D.

Proposition 14.a.8 (*Jensen's inequality*): Let $U \subseteq \mathbb{R}^n$ be open and convex and $f : U \to \mathbb{R}$ convex. Then for every finite collection of points $x_1, \dots, x_N \in U$ and weights $w_1, \dots, w_N > 0$ with

$\sum_{i=1}^{N} w_i = 1$,

$$f\Big(\sum_{i=1}^{N} w_i x_i\Big) \;\le\; \sum_{i=1}^{N} w_i f(x_i).$$

Proof:

Suppose first that $f \in C^1(U)$, which is the case met in practice. Let $z := \sum_i w_i x_i \in U$ (U is convex, so this weighted average lies in U). By Exercise 14.a.6**(b)** (valid already for $f \in C^1$), applied to the pair (z, x_i) for each i,

$$f(x_i) - f(z) \ge Df_z(x_i - z), \qquad i = 1, \dots, N.$$

Multiply the i-th inequality by $w_i > 0$ and sum over i:

$$\begin{aligned}\sum_i w_i f(x_i) - f(z) &= \sum_i w_i\big(f(x_i) - f(z)\big) \\ &\ge Df_z\Big(\sum_i w_i(x_i - z)\Big) = Df_z\Big(\sum_i w_i x_i - z\Big) = Df_z(0) = 0,\end{aligned}$$

using $\sum_i w_i = 1$ to pull $f(z)$ out on the left and $\sum_i w_i x_i = z$ in the last step. Rearranging gives $f(z) \le \sum_i w_i f(x_i)$, which is the claim.

If f is merely convex and not C^1, induct on N instead. For $N = 1$ there is nothing to show, and $N = 2$ is the definition. For $N \ge 3$ put $W := w_1 + \cdots + w_{N-1} = 1 - w_N$ and $z' := \sum_{i<N} (w_i/W)\, x_i$, a weighted average with weights summing to 1, so $z' \in U$. Then $z = Wz' + w_N x_N$, and convexity followed by the induction hypothesis gives

$$f(z) \le W f(z') + w_N f(x_N) \le W \sum_{i<N} \frac{w_i}{W} f(x_i) + w_N f(x_N) = \sum_{i=1}^{N} w_i f(x_i). \qquad \text{Q.E.D.}$$

Remark 14.a.9: The case $N = 2$ is exactly Definition 14.a.1 itself: Jensen's inequality is the statement that convexity, defined only for a two-point average, automatically extends to any finite weighted average. It is used silently in the solution to Exercise 2.c.43 (Chapter 2), applied to $\varphi(s) := s \log s$, and is stated explicitly here for the first time.

Example 14.a.10 (*Convex, bounded below, and yet no minimum*): Exercise 14.a.6**(c)** guarantees that **if** a convex function has a critical point, that point is a global minimum. It does not guarantee that one exists. Take

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x_1, x_2) := e^{x_1 + x_2}.$$

Its Hessian is

$$\operatorname{Hess} f(x_1, x_2) = e^{x_1 + x_2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

so for every $v \in \mathbb{R}^2$ we get $v^{\mathsf{T}} \operatorname{Hess} f(x)\, v = e^{x_1+x_2}(v_1 + v_2)^2 \ge 0$. The Hessian is positive semidefinite everywhere, so f is convex by Proposition 14.a.3. But $f > 0$ everywhere, while along the ray $(-t, -t)$ we have $f(-t, -t) = e^{-2t} \to 0$ as $t \to \infty$. Hence $\inf_{\mathbb{R}^2} f = 0$ and the infimum is attained nowhere.

What fails is not convexity but compactness: the sublevel sets $\{f \le c\}$ are unbounded, so the extreme value theorem (Lemma 7.a.7) has nothing to act on. Convexity organises the minima a function has. It does not manufacture one.

Convexity is worth the trouble because of what it removes. Under it there is no saddle to rule out, no degenerate case in which the second-order test falls silent, and no gap left between a local and a global minimum. What it does not remove is the need for a minimum to exist at all, which is the one thing Example 14.a.10 was there to show.

Exercise 14.a.11 (*Multiple choice (properties of convex functions)* (important)**):** Among the following statements about convex functions mark those (and only those) which are always true.

(a) If $f \in C^1(U)$ is convex in some open convex set $U \subset \mathbb{R}^n$ and f has a local maximum at $z \in U$, then $\nabla f \equiv 0$ in U.

(b) If $f \in C^1(U)$ is convex in some open convex set $U \subset \mathbb{R}^n$ and f has a global maximum at $z \in U$, then $\nabla f \equiv 0$ in U.

(c) Assume $f_n \in C^2(\mathbb{R})$ is a sequence of convex functions that converge pointwise to some $f : \mathbb{R} \to \mathbb{R}$. Is f necessarily convex?

(d) There exists a convex function $f \in C^\infty(\mathbb{R}^2)$ such that

$$f(x) = 1 - 2x_1 + x_2^3 + O(|x|^4) \quad \text{as } |x| \to 0.$$

(e) There exists a convex function $f \in C^\infty(\mathbb{R}^2)$ such that

$$f(x) = 1 - 2x_1 + x_2^4 + O(|x|^4) \quad \text{as } |x| \to 0.$$

(f) A convex set is not necessarily connected.

Info: *This exercise is Problem 6.7 of Problem Sheet 6, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 208*

Solutions (Section 14.b)

Not every solution in this chapter is collected here. Exercise 14.a.6 is solved where it appears, on Page 205.

Solution to Exercise 14.a.11:

(a) **False.** Let $f(x) := \max\{x_1 - \frac{1}{2}, 0\}^4$ on $U := B_1 \subset \mathbb{R}^n$, which is open and convex. The inner function $x \mapsto \max\{x_1 - \frac{1}{2}, 0\}$ is convex and takes values in $[0, \infty)$, and $t \mapsto t^4$ is convex and increasing there. A convex increasing function composed with a convex function is convex, so f is convex. It is also C^3: writing $\phi(t) := \max(t, 0)^4$, the derivatives $\phi' = 4t_+^3$, $\phi'' = 12t_+^2$ and $\phi''' = 24t_+$ all match the zero function at $t = 0$, and only the fourth derivative jumps. For all $x \in U$ with $x_1 \le \frac{1}{2}$, $f(x) = 0$ identically, so f has a (non-strict) local maximum at $x = 0$, yet ∇f does not vanish once $x_1 > \frac{1}{2}$.

Remark 14.b.12: Worth noting: $x = 0$ here is simultaneously a (non-strict) local **minimum** too, since f is locally constant on the whole region $\{x_1 \le \frac{1}{2}\} \cap U$. This is exactly why the counterexample doesn't contradict **(b)**: $x = 0$ is not a **global** maximum of f on U (any point with $x_1 > \frac{1}{2}$ has f strictly larger), so **(b)**'s stronger hypothesis simply isn't triggered.

(b) **True.** Since z is a local maximum, $\nabla f(z) = 0$ (Proposition 12.a.2). Convexity gives $f(x) \ge f(z) + \nabla f(z) \cdot (x - z) = f(z)$ for all $x \in U$; but z being a global maximum also gives $f(x) \le f(z)$ for all $x \in U$. Hence $f \equiv f(z)$ on U, so $\nabla f \equiv 0$ in U.

(c) **True.** Fix $x, y \in \mathbb{R}$ and $t \in [0, 1]$. Convexity of each f_n gives $f_n(tx + (1-t)y) \le t f_n(x) + (1-t) f_n(y)$ for every n; letting $n \to \infty$ (with x, y, t fixed) and using pointwise convergence yields the same inequality for f. Since x, y, t were arbitrary, f is convex.

(d) **False.** A convex function lies above its tangent plane, so from the Taylor data at 0, any such f would satisfy $f(x) \ge 1 - 2x_1$ for all $x \in \mathbb{R}^2$. Combined with $f(x) = 1 - 2x_1 + x_2^3 + O(|x|^4)$, this forces $x_2^3 \ge -M|x|^4$ near 0 for some $M > 0$, which fails along $x = (0, -r)$ as $r \downarrow 0$ (left side $\sim -r^3$, right side $\sim -Mr^4$, and $-r^3 < -Mr^4$ for small r). Contradiction, so no such convex f exists.

(e) **True.** Simply take $f(x) := 1 - 2x_1 + x_2^4$ itself (with the $O(|x|^4)$ remainder identically zero): it is convex, being a sum of the affine (hence convex) term $1 - 2x_1$ and the convex term x_2^4.

(f) False. A convex set is always path-connected, since for any two points the straight segment joining them lies inside the set by definition of convexity; path-connected sets are connected.

Q.E.D.

Solution to Exercise 14.a.5:

(a) Not convex. The Hessian is constant,

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with eigenvalues ± 1, hence indefinite, so Proposition 14.a.3**(b)** fails. Explicitly, take $u := (1,0)$, $v := (0,1)$ and $t := \frac{1}{2}$: the midpoint is $(\frac{1}{2},\frac{1}{2})$ with $f = \frac{1}{4}$, while $\frac{1}{2}f(u) + \frac{1}{2}f(v) = 0$. Since $\frac{1}{4} > 0$, the defining inequality is violated.

(b) Convex. Again the Hessian is constant,

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

with eigenvalues 2 ± 1, i.e. 3 and 1, both strictly positive. So, the Hessian is positive definite everywhere and f is convex, in fact strictly so.

Without eigenvalues: $x^2 + xy + y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2$ is a sum of squares of affine functions, and such a sum is always convex.

(c) Not convex. Differentiating,

$$\operatorname{Hess} f(x,y) = \begin{pmatrix} -\cos x \cos y & \sin x \sin y \\ \sin x \sin y & -\cos x \cos y \end{pmatrix}, \qquad \text{so} \qquad \operatorname{Hess} f\left(\tfrac{\pi}{2},\tfrac{\pi}{2}\right) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which is indefinite. For an explicit violation, run along the anti-diagonal: with $u := (0,\pi)$, $v := (\pi,0)$ and $t := \frac{1}{2}$ the midpoint is $(\frac{\pi}{2},\frac{\pi}{2})$, where $f = 0$, whereas $\frac{1}{2}f(u) + \frac{1}{2}f(v) = \frac{1}{2}(-1) + \frac{1}{2}(-1) = -1$. Since $0 > -1$, convexity fails.

Note that the **main** diagonal would not have exposed this: from $(0,0)$ to (π,π) the midpoint value 0 sits below the average 1, consistent with convexity. Choosing the direction well is most of the work in these problems, and the indefinite Hessian tells you which direction to choose, namely an eigenvector for the negative eigenvalue, here $(1,-1)$.

(d) No. Part **(a)** is already a counterexample: $g(x) := x$ and $h(y) := y$ are affine, hence convex, yet $g(x)h(y) = xy$ is not. Convexity is preserved by sums and by non-negative scalar multiples, but not by products, not even of non-negative increasing factors (part **(a)** lives on $[0,\infty)^2$, where x and y are both). The safe habit is to compute the Hessian of the product.

Q.E.D.

PART V

Inverse and Implicit Functions; Submanifolds

Two questions run through this part. When is a map invertible near a point, and when can an equation $f(x, y) = 0$ be solved for y as a function of x? Neither has a workable global answer, but both have clean local ones, and both rest on the same linear-algebra hypothesis, that a Jacobian, or one square block of it, is invertible. The inverse function theorem is proved here from the contraction principle of Part II, which is the first place that fixed-point machinery repays what it cost to set up, and the implicit function theorem then follows from it in a few lines. The deeper use of both theorems is definitional. A level set on which the differential has full rank is locally a graph, being locally a graph is precisely what it means to be a submanifold, and a submanifold is a subset of $\mathbb{R}^n$ that looks like $\mathbb{R}^m$ from close up, which is the object the last two parts integrate over. Once such a set carries a tangent space at each point one can ask what a straight line becomes on it, and the part closes with that question, which takes the first genuine calculus-of-variations argument in the course.

The Inverse Function Theorem (Chapter 15)

So far the differential has been used to **describe** a map near a point. The inverse function theorem turns that around. If the differential at a point is invertible, then so is the map itself near that point, and the inverse is exactly as differentiable as the map was. Everything in the rest of this part rests on that one statement.

The inverse function theorem (Section 15.a)

Theorem 15.a.1 (*Inverse function theorem*): Let $U \subseteq \mathbb{R}^n$ be open, let $f \in C^k(U, \mathbb{R}^n)$, and let $x_0 \in U$ be such that Df_{x_0} is invertible. Then there exists an open neighbourhood $U_0 \subseteq U$ of x_0 such that

$$f|_{U_0} : U_0 \to f(U_0)$$

is a C^k-"*diffeomorphism*" (German: „*Diffeomorphismus*") and

$$D\big(f|_{U_0}^{-1}\big)_{f(x)} = D\big(f|_{U_0}\big)_x^{-1} \quad \text{for all } x \in U_0.$$

Remark 15.a.2 (*Why invertibility of Df_{x_0} is the right hypothesis*): In one variable, consider $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$. The inverse $f^{-1}(y) = y^{1/3}$ exists everywhere, yet it fails to be differentiable at $y = 0$. Geometrically, the graph of f^{-1} is obtained by reflecting the graph of f across the line $y = x$. The horizontal tangent line of f at $x = 0$ (where $f'(0) = 0$) flips into a vertical tangent line for f^{-1} at $y = 0$, causing the derivative of the inverse to blow up.

In multivariable calculus, for a map $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$, the tangent space to the graph of F at $(x_0, F(x_0))$ is the affine subspace

$$\big\{(x_0 + v,\ F(x_0) + DF_{x_0}(v)) \mid v \in \mathbb{R}^n\big\} \subseteq \mathbb{R}^n \times \mathbb{R}^n.$$

A *"horizontal"* direction (a non-zero input displacement producing zero output change) corresponds to a non-zero vector $v \in \mathbb{R}^n$ such that $DF_{x_0}(v) = 0$. To eliminate vertical tangents after inversion, we must require DF_{x_0} to be injective. Since F maps between spaces of the same dimension n, injectivity of DF_{x_0} is equivalent to its invertibility.

Lemma 15.a.3 (*Lipschitz perturbations of the identity*): Let $U \subseteq \mathbb{R}^n$ be open, and let $F : U \to \mathbb{R}^n$ be a map of the form

$$F(x) := x + \phi(x),$$

where $\phi : U \to \mathbb{R}^n$ is a λ-Lipschitz function with constant $\lambda \in (0, 1)$. Then:

(a) For every open ball $B_r(x) \subseteq U$, the image $F(B_r(x))$ satisfies the inclusions

$$B_{(1-\lambda)r}(F(x)) \ \subseteq\ F(B_r(x)) \ \subseteq\ B_{(1+\lambda)r}(F(x)).$$

In particular, $F(U) \subseteq \mathbb{R}^n$ is an open set.

(b) $F : U \to F(U)$ is injective, and its inverse $F^{-1} : F(U) \to U$ is $\frac{1}{1-\lambda}$-Lipschitz continuous.

Proof:

(a) Fix $y_0 \in B_{(1-\lambda)r}(F(x))$. We seek a point $x_0 \in B_r(x)$ satisfying $F(x_0) = y_0$, which is equivalent to the fixed-point equation $x_0 = y_0 - \phi(x_0)$. Since $|y_0 - F(x)| < (1-\lambda)r$, we can choose $r' < r$ with $|y_0 - F(x)| < (1-\lambda)r'$. Then $\overline{B_{r'}(x)} \subseteq B_r(x) \subseteq U$, which matters because ϕ is defined only on U. Define $g : \overline{B_{r'}(x)} \to \mathbb{R}^n$ by $g(z) := y_0 - \phi(z)$. For any $z_1, z_2 \in \overline{B_{r'}(x)}$,

$$|g(z_1) - g(z_2)| = |\phi(z_1) - \phi(z_2)| \le \lambda |z_1 - z_2|,$$

so g is a λ-contraction. Next, we verify that g maps $\overline{B_{r'}(x)}$ into itself:

$$|g(z) - x| \le |g(z) - g(x)| + |g(x) - x| \le \lambda|z - x| + |y_0 - F(x)| < \lambda r' + (1-\lambda)r' = r'.$$

Thus $g(\overline{B_{r'}(x)}) \subseteq B_{r'}(x) \subseteq \overline{B_{r'}(x)}$. Since $\overline{B_{r'}(x)}$ is complete, the Banach fixed-point theorem (Theorem 6.c.19) yields a unique point $x_0 \in \overline{B_{r'}(x)}$ with $g(x_0) = x_0$, i.e. $F(x_0) = y_0$, and $x_0 = g(x_0) \in B_{r'}(x) \subseteq B_r(x)$. This proves $B_{(1-\lambda)r}(F(x)) \subseteq F(B_r(x))$. The outer inclusion follows directly from $|F(y) - F(x)| \leq |y - x| + |\phi(y) - \phi(x)| < (1 + \lambda)r$.

(b) For any $x_1, x_2 \in U$, applying the reverse triangle inequality gives

$$|F(x_1) - F(x_2)| = |(x_1 - x_2) + (\phi(x_1) - \phi(x_2))| \geq |x_1 - x_2| - |\phi(x_1) - \phi(x_2)| \geq (1 - \lambda)|x_1 - x_2|.$$

Hence $F(x_1) = F(x_2) \implies x_1 = x_2$, so F is injective. Setting $y_1 := F(x_1)$ and $y_2 := F(x_2)$, this bound rewrites as $|F^{-1}(y_1) - F^{-1}(y_2)| \leq \frac{1}{1-\lambda}|y_1 - y_2|$, establishing that F^{-1} is $\frac{1}{1-\lambda}$-Lipschitz continuous.

Q.E.D.

The lemma just proved is the engine of the proof of Theorem 15.a.1: near a point where the differential is invertible, a map is, after a linear change of coordinates, a small Lipschitz perturbation of the identity.

Proof of Theorem 15.a.1:

Normalisation. Put $A := Df_{x_0}$ and $\tilde{f}(x) := A^{-1}\big(f(x_0 + x) - f(x_0)\big)$ on $U - x_0$. Then $\tilde{f}(0) = 0$ and $D\tilde{f}_0 = \mathrm{id}$, and f is $\tilde{f}$ composed on both sides with affine bijections, so it suffices to prove the theorem for $\tilde{f}$ at 0. We drop the tilde.

The local inverse exists and is Lipschitz. Let $\phi(x) := f(x) - x$, so $D\phi_0 = 0$. The entries of $\mathrm{J}\,\phi$ are continuous and vanish at 0, so by Lemma 2.a.4 there is $r > 0$ with $B_r(0) \subseteq U$ and $\|D\phi_x\|_{\mathrm{op}} \leq \|\mathrm{J}\,\phi(x)\|_2 \leq \frac{1}{2}$ for all $x \in B_r(0)$. On the convex set $B_r(0)$ the mean value inequality (Remark 11.a.4) makes ϕ $\frac{1}{2}$-Lipschitz. By Lemma 15.a.3, with $U_0 := B_r(0)$, the image $V_0 := f(U_0)$ is open, $f|_{U_0} : U_0 \to V_0$ is a bijection, and its inverse $g : V_0 \to U_0$ is 2-Lipschitz. Moreover $Df_x = \mathrm{id} + D\phi_x$ is invertible for every $x \in U_0$, since $|Df_x v| \geq |v| - \frac{1}{2}|v| > 0$ for $v \neq 0$.

The inverse is differentiable. Fix $y \in V_0$, let $x := g(y)$ and $L := Df_x$. For small $k \neq 0$ put $h := g(y + k) - x$, which is non-zero because g is injective, and satisfies $|h| \leq 2|k|$. Differentiability of f at x gives $k = f(x + h) - f(x) = Lh + \rho(h)$ with $\rho(h) = o(|h|)$, hence

$$g(y + k) - g(y) = h = L^{-1}k - L^{-1}\rho(h), \qquad |L^{-1}\rho(h)| \leq \|L^{-1}\|_{\mathrm{op}} \frac{|\rho(h)|}{|h|}\, 2|k|.$$

As $k \to 0$ also $h \to 0$, so the quotient $|\rho(h)|/|h|$ tends to 0 and the error is $o(|k|)$. Thus g is differentiable at y with $Dg_y = (Df_{g(y)})^{-1}$, which is the asserted formula.

The inverse is C^k. The formula reads $Dg = \mathrm{inv} \circ Df \circ g$, where inv sends an invertible matrix to its inverse. By Cramer's rule each entry of M^{-1} is a polynomial in the entries of M divided by $\det M \neq 0$, so inv is C^∞. Now bootstrap: g is continuous, so Dg is a composition of continuous maps and g is C^1. If g is C^j for some $j < k$, then $Df \in C^{k-1}$ and $g \in C^j$ make Dg a composition of C^j maps, so g is C^{j+1}. After $k - 1$ steps g is C^k. Q.E.D.

The proof shows where each hypothesis enters. Continuity of Df is what keeps $D\phi$ small on a whole ball rather than only at the point; mere differentiability would not give a Lipschitz perturbation. Invertibility of Df_{x_0} is spent in the normalisation, and completeness of $\mathbb{R}^n$ is hidden inside the lemma, which runs on the Banach fixed-point theorem.

Remark 15.a.4 (*Checking for a global diffeomorphism*): To verify that a map $f \in C^k(U, V)$ between open sets is a global C^k-diffeomorphism, it is best to check the following two conditions:

(a) Global bijectivity: Check that f is bijective. This assures that a global inverse $f^{-1} : V \to U$ exists, and any local inverse will coincide with it.

(b) Local diffeomorphism everywhere: Check that the differential Df_x (or $\mathrm{J}\,f(x)$) is invertible at every point $x \in U$. By the inverse function theorem, this ensures f is a local C^k-diffeomorphism everywhere.

If both conditions hold, the global inverse f^{-1} inherits the C^k regularity of the local inverses, making f a global C^k-diffeomorphism.

Example 15.a.5 (*Invertible everywhere, injective nowhere near globally*): The second condition of Remark 15.a.4 does not imply the first. Let

$$f : \mathbb{R}^2 \to \mathbb{R}^2, \qquad f(x,y) := \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix},$$

which is the complex exponential written in real coordinates. Its Jacobian is

$$\mathrm{J}\, f(x,y) = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}, \qquad \det \mathrm{J}\, f(x,y) = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x} > 0,$$

so $\mathrm{J}\, f$ is invertible at **every** point and f is a local C^∞-diffeomorphism everywhere by Theorem 15.a.1. Yet $f(x, y+2\pi) = f(x,y)$ for all (x,y), so f is not injective. Indeed, every value in its image is attained infinitely often, and f is a global diffeomorphism on no horizontal strip of height greater than 2π.

Restricting the codomain does not help either: f misses the origin and nothing else, so $f : \mathbb{R}^2 \to \mathbb{R}^2 \setminus \{0\}$ is surjective but still not injective. What **does** work is restricting the domain, for instance to $\mathbb{R} \times (-\pi, \pi)$, on which f is the polar-coordinate diffeomorphism of Example 15.a.6 in disguise.

Example 15.a.6 (*Polar and spherical coordinates*): The most important examples of diffeomorphisms are transformations between coordinate systems. For instance, the map assigning polar coordinates to Cartesian coordinates on $\mathbb{R}^2 \setminus ((-\infty, 0] \times \{0\})$, and similarly for spherical coordinates on $\mathbb{R}^3$, are global C^∞-diffeomorphisms onto their images.

Question 15.a.7:

Q1. If $f : \mathbb{R}^n \to \mathbb{R}^n$ is C^k and a "*homeomorphism*" (German: „*Homöomorphismus*"), that is, bijective with continuous inverse, is it necessarily a C^k-diffeomorphism?

Q2. If $f \in C^\infty(U, \mathbb{R}^n)$ has an everywhere invertible differential, i.e., Df_x is invertible for all $x \in U$ with $U \subseteq \mathbb{R}^n$ open, then f is a C^∞-diffeomorphism onto its image. True or false?

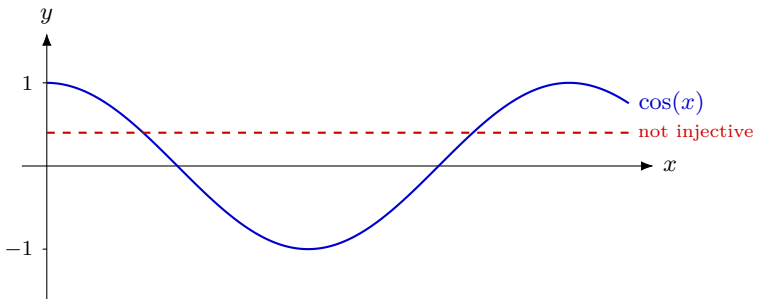
Answer 15.a.8:

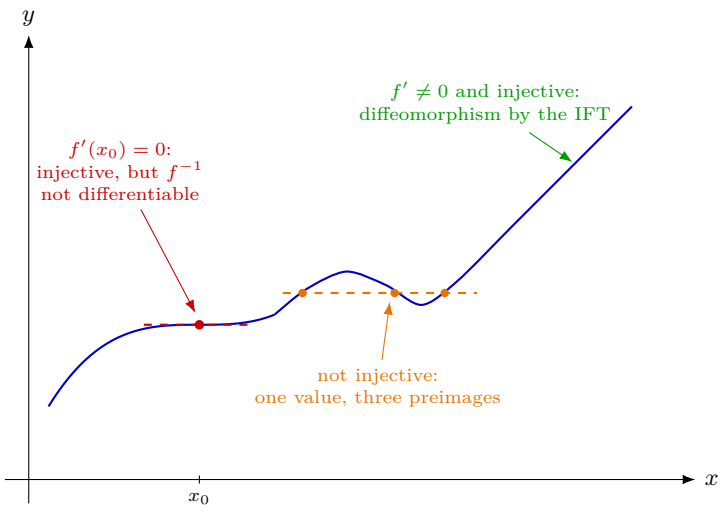
A1. False. Consider $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$. Then $f^{-1} : \mathbb{R} \to \mathbb{R}$, $x \mapsto \sqrt[3]{x}$, is not continuously differentiable at $0 \in \mathbb{R}$.

A2. False. Consider

$$\cos|_{\mathbb{R}\setminus\pi\mathbb{Z}} : \mathbb{R} \setminus \pi\mathbb{Z} \to (-1, 1), \qquad x \mapsto \cos(x).$$

For any $x \in \mathbb{R} \setminus \pi\mathbb{Z}$, $\cos'(x) = -\sin(x) \neq 0$, so the differential is everywhere invertible, but the cosine is not injective on this domain, so it fails to be a diffeomorphism onto its image.





Reading the three regions left to right makes the two answers above concrete.

- At x_0 the curve is still **strictly increasing**, so f is injective there and f^{-1} exists. But the tangent is horizontal, so the tangent to f^{-1} at $f(x_0)$ is **vertical**, and f^{-1} fails to be differentiable. Invertible, yet not a diffeomorphism: this is exactly $x \mapsto x^3$ from **A1**, and it is why Theorem 15.a.1 assumes Df_{x_0} invertible rather than merely assuming f bijective.
- On the hump, $f' \neq 0$ at every individual point, so f **is** a local diffeomorphism near each of them. Yet one value has three preimages, so f is not globally injective. This is **A2**: an everywhere-invertible differential buys locality and nothing more.
- On the final stretch both hypotheses hold, and the inverse function theorem delivers a genuine C^∞-diffeomorphism onto the image.

Exercise 15.a.9 (*Open mapping*): Let $f : U \to \mathbb{R}$ be a continuously differentiable function, where $U \subseteq \mathbb{R}^n$ is open and connected. Assume that $\nabla f(x) \neq 0$ for all $x \in U$. Prove that $f(U)$ is open in $\mathbb{R}$.

Hint: *Define an auxiliary function $F : U \to \mathbb{R}^n$ and use the Inverse Function Theorem.*

Solution to Exercise 15.a.9:

Let $x_0 \in U$. Since $\nabla f(x_0) \neq 0$, there exists an index $i \in \{1, \dots, n\}$ such that $\partial_{x_i} f(x_0) \neq 0$. Without loss of generality, assume $i = 1$. Define the auxiliary function $F : U \to \mathbb{R}^n$ by

$$F(x_1, x_2, \dots, x_n) = (f(x_1, \dots, x_n), x_2, \dots, x_n).$$

The Jacobian matrix of F at x is

$$DF(x) = \begin{pmatrix} \partial_{x_1} f(x) & \partial_{x_2} f(x) & \dots & \partial_{x_n} f(x) \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Its determinant is $\det DF(x) = \partial_{x_1} f(x)$. At $x = x_0$, we have $\det DF(x_0) = \partial_{x_1} f(x_0) \neq 0$. Thus, DF_{x_0} is invertible. By the Inverse Function Theorem (Theorem 15.a.1), there exist open neighbourhoods $V_0 \subseteq U$ of x_0 and $W_0 \subseteq \mathbb{R}^n$ of $F(x_0)$ such that $F|_{V_0} : V_0 \to W_0$ is a diffeomorphism, and in particular, $F(V_0) = W_0$ is open in $\mathbb{R}^n$.

Let $\pi_1 : \mathbb{R}^n \to \mathbb{R}$, $(y_1, \dots, y_n) \mapsto y_1$ be the projection onto the first coordinate. Projections are open mappings. Notice that $f = \pi_1 \circ F$. Therefore, $f(V_0) = \pi_1(F(V_0)) = \pi_1(W_0)$ is open

in $\mathbb{R}$. Thus, for every $x_0 \in U$, there is a neighbourhood $V_0 \subseteq U$ such that $f(V_0)$ is an open neighbourhood of $f(x_0)$ contained in $f(U)$. This proves that $f(U)$ is open in $\mathbb{R}$. Q.E.D.

Exercise 15.a.10 (*Diffeomorphism* (important)):

(a) Give a diffeomorphism between $\mathbb{R}^2$ and $(0,1)\times(0,1)$.

(b) Is $f(x) = x^5$, $x \in \mathbb{R}$ a diffeomorphism of $\mathbb{R}$ to itself? Motivate rigorously your answer.

Info: *This exercise is Problem 7.1 of Problem Sheet 7, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 216*

Exercise 15.a.11 (*True or False* (a critical point and its Hessian)): Let $f \in C^\infty(\mathbb{R}^3)$ satisfy

$$f(0) = 0, \qquad \nabla f(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \qquad \operatorname{Hess} f(0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Decide whether each of the following is true or false, with justification.

(a) $(0,0,0)$ is a local maximum point for f.

(b) $(0,0,0)$ is a saddle point for f.

(c) $f(x,y,z) = x^2 - y^2 - z^2 + o(x^2+y^2+z^2)$ as $(x,y,z) \to (0,0,0)$.

(d) $f(x,y,z) = O(x^2+y^2+z^2)$ as $(x,y,z) \to (0,0,0)$.

(e) The map $\nabla f : \mathbb{R}^3 \to \mathbb{R}^3$, $x \mapsto \nabla f(x)$, restricted to a sufficiently small ball centred at $(0,0,0)$, is a diffeomorphism onto its image.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the fifth block of true/false questions.*

Solution: *See solution on Page 217*

Exercise 15.a.12 (*True or False* (where the inverse function theorem applies)): Let $\alpha \in \mathbb{R}$ and let $\Phi \in C^1(\mathbb{R}^2, \mathbb{R}^2)$ satisfy

$$\Phi(0,0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \text{and} \qquad \mathrm{J}\,\Phi(0,0) = \begin{pmatrix} \alpha+5 & -3 \\ 3 & \alpha-5 \end{pmatrix}.$$

Decide whether each of the following is true or false, with justification.

(a) For every α with $\alpha^2 \neq 25$, the restriction of Φ to a sufficiently small ball centred at $(0,0)$ is a diffeomorphism onto its image.

(b) For every α with $\alpha^2 \neq 16$, the restriction of Φ to a sufficiently small ball centred at $(0,0)$ is a diffeomorphism onto its image.

(c) For $\alpha = 0$ and $r > 0$ small enough, the set $\Phi\big(B_r(0,0)\big)$ is open.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the sixth block of true/false questions. Its last two parts are Exercise 17.b.27.*

Solution: *See solution on Page 217*

Exercise 15.a.13 (*Single choice: which hypothesis gives local invertibility?*): Let f be twice continuously differentiable on all of $\mathbb{R}^2$. Exactly one of the following is correct. Which?

(a) If f takes values in $\mathbb{R}^n$ with $n > 2$, then f is always locally invertible around the origin.

(b) If f takes values in $\mathbb{R}^2$ and the differential Df has maximal rank at the origin, then f is locally invertible around the origin.

(c) If $\frac{\partial f}{\partial x}(0,0) \neq 0$, then f can be locally inverted around the origin.

(d) If $\frac{\partial f}{\partial y}(0,0) \neq 0$, then f can be locally inverted around the origin.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 5.*

Solution: *See solution on Page 218*

Exercise 15.a.14 (*Box question: a smooth bijection whose inverse is not C^1*): Give an explicit example of a C^∞ bijective function $g : \mathbb{R} \to \mathbb{R}$ whose inverse g^{-1} is **not** of class C^1.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), Box Problem 17, where only the final answer was graded.*

Solution: *See solution on Page 219*

Exercise 15.a.15 (*Box question: preimages of compacta are compact, and still no diffeomorphism*): Give an explicit example of a function $f \in C^2(\mathbb{R}^2, \mathbb{R}^2)$ with both of the following properties: $f^{-1}(K)$ is compact whenever $K \subseteq \mathbb{R}^2$ is compact, and f is **not** a diffeomorphism of $\mathbb{R}^2$.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), Box Problem 18, where only the final answer was graded.*

Solution: *See solution on Page 219*

Exercise 15.a.16 (*True or False* (an everywhere-invertible differential)): Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}^n$ be continuously differentiable with Df_x invertible at every $x \in U$. Decide whether each of the following is true or false, with justification.

(a) For every $y \in \mathbb{R}^n$ the preimage $f^{-1}\{y\}$ is finite or countable.

(b) f maps null sets to null sets.

(c) If f is injective, then $f : U \to f(U)$ is a diffeomorphism.

(d) If f is surjective, then $f : U \to \mathbb{R}^n$ is a diffeomorphism.

Info: *This exercise is taken from the mock examination **Probeprüfung Analysis II** of 17 August 2022 (revised), Problem 6. That paper names no lecturer.*

Solution: *See solution on Page 219*

Exercise 15.a.17 (*Expansive local diffeomorphisms and global invertibility*): Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f \in C^1(U, \mathbb{R}^n)$ be expansive: there exists a constant $\alpha > 0$ such that

$$\|f(x) - f(y)\| \geq \alpha \|x - y\| \qquad \text{for all } x, y \in U.$$

(a) Prove that f is injective, that the differential Df_x is invertible at every point $x \in U$, and that $f : U \to f(U)$ is a C^1-diffeomorphism onto an open set $f(U) \subseteq \mathbb{R}^n$.

(b) Now suppose $U = \mathbb{R}^n$. Prove that $f(\mathbb{R}^n)$ is closed in $\mathbb{R}^n$, and deduce that $f : \mathbb{R}^n \to \mathbb{R}^n$ is a surjective global C^1-diffeomorphism.

Info: *This exercise is taken from the examination of 18 August 2023 (Prof. Urs Lang), Problem 3.A3.*

Solution: *See solution on Page 220*

Solutions (Section 15.b)

Not every solution in this chapter is collected here. Exercise 15.a.9 is solved where it appears, on Page 214.

Solution to Exercise 15.a.10:

(a) First build a diffeomorphism $\mathbb{R} \to (0, 1)$: the function

$$\psi : \mathbb{R} \to (0, 1), \qquad \psi(t) := \frac{1}{2} + \frac{1}{\pi} \arctan(t)$$

is C^∞, strictly increasing (since $\arctan' > 0$), and bijective onto $(0,1)$ (as arctan is bijective onto $(-\frac{\pi}{2}, \frac{\pi}{2})$), with C^∞ inverse $\psi^{-1}(s) = \tan\big(\pi(s - \frac{1}{2})\big)$. Applying ψ in each coordinate gives the diffeomorphism

$$\phi : \mathbb{R}^2 \to (0,1)\times(0,1), \qquad \phi(x,y) := \Big(\frac{1}{2} + \frac{1}{\pi}\arctan(x),\ \frac{1}{2} + \frac{1}{\pi}\arctan(y)\Big).$$

(b) **No.** f is a C^∞ bijection of $\mathbb{R}$ with $f^{-1}(x) = \sqrt[5]{x}$, but $f'(0) = 5\cdot 0^4 = 0$. If f were a diffeomorphism, then by the chain rule applied to $f^{-1}\circ f = \mathrm{id}$,

$$1 = (f^{-1})'(f(x))\cdot f'(x) \quad \text{for all } x,$$

which is impossible at $x = 0$ since $f'(0) = 0$. Equivalently, $(f^{-1})'(x) = \frac{1}{5}x^{-4/5} \to \infty$ as $x \to 0$, so f^{-1} is not even differentiable at 0, let alone continuously so.

Q.E.D.

Solution of Exercise 15.a.11:

Write $H := \mathrm{Hess}\, f(0)$. Everything follows from its spectrum, so compute that first. The matrix is block diagonal, with the 1×1 block (1) and the 2×2 block $\left(\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right)$, whose eigenvalues are ± 1. Hence H has eigenvalues $1, 1, -1$, and in particular $\det H = -1 \neq 0$: the critical point is non-degenerate, and H is indefinite.

Since f is smooth with $f(0) = 0$ and $\nabla f(0) = 0$, Taylor's theorem gives

$$f(x) = \tfrac{1}{2}\langle Hx, x\rangle + o(|x|^2) = \tfrac{1}{2}x_1^2 + x_2x_3 + o(|x|^2) \qquad \text{as } x \to 0. \tag{15.1}$$

(a) **False.** Along the x_1-axis, (15.1) gives $f(t,0,0) = \frac{1}{2}t^2 + o(t^2) > 0 = f(0)$ for small $t \neq 0$, so f takes values larger than $f(0)$ arbitrarily close to the origin.

(b) **True.** We have just seen a direction along which f increases. In the plane $x_1 = 0$ the quadratic part is x_2x_3, so along $x_2 = -x_3 = t$ it reads $-t^2$, and $f(0,t,-t) = -t^2 + o(t^2) < 0 = f(0)$. Taking both values above and below $f(0)$ in every neighbourhood of the origin is exactly what a saddle point does.

(c) **False.** The quadratic form is correct only up to a change of coordinates, and the statement is made in the original ones. By (15.1) the second-order part of f is $\frac{1}{2}x_1^2 + x_2x_3$, which is not $x_1^2 - x_2^2 - x_3^2$: evaluating both at $(0,1,1)$ gives 1 and -2. What **is** true is that in suitable coordinates the form becomes $\frac{1}{2}(u_1^2 + u_2^2 - u_3^2)$, matching the eigenvalues $1, 1, -1$. Note in passing that this has signature $(2,1)$, not the $(1,2)$ the stated expression suggests.

(d) **True.** This is the same Taylor expansion read with a weaker conclusion: the right-hand side of (15.1) is bounded by a constant times $|x|^2$ near 0. A value and a gradient that both vanish at the point are precisely what $O(|x|^2)$ records.

(e) **True**, and this is where the chapter's theorem enters. The map ∇f is smooth, being the gradient of a smooth function, and its Jacobian at the origin is $\mathrm{J}(\nabla f)(0) = \mathrm{Hess}\, f(0) = H$, which we computed to be invertible. By the inverse function theorem (Theorem 15.a.1), ∇f is therefore a diffeomorphism from some ball $B_\varepsilon(0)$ onto an open neighbourhood of $\nabla f(0) = 0$.

Remark 15.b.18: Item **(e)** is the useful one to remember: *"non-degenerate critical point"* means exactly *"∇f is a local diffeomorphism there"*, so such points are isolated. Item **(c)** is the standard trap: diagonalising the Hessian is a statement about **some** coordinates, and carrying its conclusion back to the original ones without the change of variables is where the argument breaks.

Q.E.D.

Solution of Exercise 15.a.12:

Everything turns on one number:

$$\det \mathrm{J}\,\Phi(0,0) = (\alpha+5)(\alpha-5) - (-3)\cdot 3 = \alpha^2 - 25 + 9 = \alpha^2 - 16.$$

(a) **False.** The determinant vanishes exactly when $\alpha^2 = 16$, so $\alpha^2 \neq 25$ is the wrong condition. Take $\alpha = 4$, which satisfies $\alpha^2 = 16 \neq 25$ and so is covered by the claim. There $\det \mathrm{J}\,\Phi(0,0) = 0$, and no restriction of Φ to a neighbourhood of the origin can be a diffeomorphism: if $\Phi|_V$ had a C^1 inverse Ψ, the chain rule applied to $\Psi \circ \Phi = \mathrm{id}$ would give $\mathrm{J}\,\Psi(\Phi(0,0))\ \mathrm{J}\,\Phi(0,0) = I_2$, forcing $\mathrm{J}\,\Phi(0,0)$ to be invertible.

(b) **True.** If $\alpha^2 \neq 16$, then $\det \mathrm{J}\,\Phi(0,0) \neq 0$, so $D\Phi_{(0,0)}$ is invertible, and $\Phi \in C^1$. The inverse function theorem (Theorem 15.a.1) supplies open sets $V \ni (0,0)$ and $W \ni \Phi(0,0)$ with $\Phi|_V : V \to W$ a C^1-diffeomorphism. Any ball $B_r(0,0) \subseteq V$ then works, since a restriction of a diffeomorphism to an open subset is again one onto its image.

(c) **True.** For $\alpha = 0$ the determinant is $-16 \neq 0$, so **(b)** applies. Fix V, W as there and $r > 0$ with $B_r(0,0) \subseteq V$. A diffeomorphism is in particular a homeomorphism onto the open set W, so it maps open subsets of V to open subsets of W, and open subsets of an open set are open in $\mathbb{R}^2$. Hence $\Phi(B_r(0,0))$ is open.

Remark 15.b.19 (*Where 25 comes from, and why it is wrong*): The two thresholds are not a random pair. A reader who computes the determinant as a product of the diagonal entries and forgets the off-diagonal contribution gets $(\alpha + 5)(\alpha - 5)$ and lands on $\alpha^2 = 25$, which is exactly the false condition of **(a)**. The off-diagonal pair $-3, 3$ contributes $+9$ and moves the singular set to $\alpha^2 = 16$, that is $\alpha = \pm 4$. Item **(a)** and item **(b)** are therefore the same sentence with one arithmetic slip between them, and reading them as a pair is faster than checking either alone.

Q.E.D.

Solution of Exercise 15.a.13:

The correct answer is **(b)**.

(a) **False**, and it fails before any hypothesis on the derivative is examined. Theorem 15.a.1 needs domain and codomain of the **same** dimension, because a local inverse has to be defined on a neighbourhood of the image point. For $n > 2$ the image of f has empty interior in $\mathbb{R}^n$ (it is at best a 2-dimensional surface), so no neighbourhood of $f(0,0)$ in $\mathbb{R}^n$ lies in the image, and there is nothing for an inverse to be defined on. Being locally **injective**, which is what an immersion gives, is a strictly weaker conclusion.

(b) **True.** For a map $\mathbb{R}^2 \to \mathbb{R}^2$, maximal rank means rank 2, which for a square matrix is invertibility, that is $\det \mathrm{J}\,f(0,0) \neq 0$. That is exactly the hypothesis of Theorem 15.a.1, and $f \in C^2 \subseteq C^1$ supplies the regularity.

(c) **False.** One non-vanishing partial derivative is one column of the Jacobian, and a matrix with one non-zero column can still be singular. Take $f(x,y) := (x,0)^{\mathsf{T}}$: then $\partial f/\partial x = (1,0)^{\mathsf{T}} \neq 0$ everywhere, while $\mathrm{J}\,f = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ has determinant 0, and f is not injective on any neighbourhood of the origin, since it collapses every vertical segment to a point.

(d) **False**, by the same example with the roles of the variables exchanged: for $f(x,y) := (y,0)^{\mathsf{T}}$ the partial $\partial f/\partial y$ is non-zero and the Jacobian is again singular.

Remark 15.b.20 (*The same answer if the partials are read component-wise*): Items **(c)** and **(d)** are printed with f itself, so $\partial f/\partial x(0,0) \neq 0$ says a whole column of the Jacobian is non-zero. A reader who takes $f = (f_1, f_2)^{\mathsf{T}}$ and reads them as $\partial f_1/\partial x(0,0) \neq 0$ gets a weaker hypothesis (a single **entry** is non-zero), and the same counterexample refutes it. The question is robust to the ambiguity, which is presumably why it went unnoticed.

What both readings test is the difference between *"some derivative does not vanish"* and *"the derivative is invertible"*. Only the second is a statement about the linear map $Df_{(0,0)}$ as a whole, and only the second is what Remark 15.a.2 argues is the right hypothesis.

Q.E.D.

Solution to Exercise 15.a.14:

Take $g(x) := x^3$. It is a polynomial, hence C^∞. It is strictly increasing, because $g(b) - g(a) = (b-a)(a^2+ab+b^2)$ and the second factor is positive whenever a and b are not both zero, so g is injective; and it is surjective, since $g(x) \to \pm\infty$ as $x \to \pm\infty$ and g is continuous. Its inverse is $g^{-1}(y) = y^{1/3}$, which is not even differentiable at 0:

$$\frac{g^{-1}(h) - g^{-1}(0)}{h} = \frac{h^{1/3}}{h} = h^{-2/3} \xrightarrow[h\to 0]{} +\infty.$$

Remark 15.b.21 (*This is not a counterexample to the inverse function theorem*): It is the theorem's hypothesis failing at exactly the point where its conclusion fails. Theorem 15.a.1 asks for $g'(x_0) \neq 0$, and $g'(x) = 3x^2$ vanishes at $x = 0$ and nowhere else. Correspondingly g^{-1} is C^∞ on $\mathbb{R} \setminus \{0\}$ and fails to be differentiable only at 0. Note also what the example rules out: no amount of **global** good behaviour, bijectivity included, substitutes for the derivative condition, and this is the one-dimensional shadow of the point Example 15.a.5 makes in the other direction.

Q.E.D.

Solution to Exercise 15.a.15:

Take $f(x,y) := (x^3,\ y)$, which is C^∞ and so certainly C^2.

Preimages of compacta are compact. By the previous solution $x \mapsto x^3$ is a bijection of $\mathbb{R}$ with continuous inverse, so f is a homeomorphism of $\mathbb{R}^2$ with $f^{-1}(u,v) = (u^{1/3}, v)$. For compact K, the set $f^{-1}(K)$ is therefore the **image** of K under the continuous map f^{-1}, hence compact.

But f is not a diffeomorphism. A diffeomorphism has a differentiable inverse, and f^{-1} is not differentiable at any point of the line $\{u = 0\}$, by the previous solution again. Equivalently, and without mentioning the inverse at all, $\det \mathrm{J}\, f(x,y) = 3x^2$ vanishes along the y-axis, and the Jacobian determinant of a diffeomorphism is nowhere zero, since $\mathrm{J}(f^{-1})(f(p))\ \mathrm{J}\, f(p) = \mathrm{id}$.

Remark 15.b.22 (*Two independent ways to fail, and this example uses neither of the obvious ones*): The official key answers $f(x,y) := (x + x^2,\ y)$, which is also correct and fails for entirely different reasons: that map is neither injective (it folds the plane along $x = -\frac{1}{2}$) nor surjective (its image is $[-\frac{1}{4}, \infty) \times \mathbb{R}$), while remaining proper because $x + x^2$ is bounded only on bounded sets of x. The example above is instead a bijection, and even a homeomorphism, and still not a diffeomorphism. Both answers make the same point: the compactness condition constrains f only at infinity and says nothing whatever about the differential.

The condition itself has a name. A continuous map whose preimages of compact sets are compact is called "*proper*", and Exercise 7.c.47 shows that a continuous map need not have the property at all: unlike compactness of **images**, it is not automatic.

Q.E.D.

Solution to Exercise 15.a.16:

The hypothesis is exactly that of Theorem 15.a.1 at every point at once, so f is a **local** diffeomorphism: each $x \in U$ has an open neighbourhood that f maps diffeomorphically onto an open set. Everything below is a consequence of that one sentence.

(a) True. Local injectivity makes the fibre $f^{-1}\{y\}$ **discrete**: if $x \in f^{-1}\{y\}$, the neighbourhood on which f is injective contains no second point of the fibre. A discrete subset of $\mathbb{R}^n$ is

countable, and the reason is that $\mathbb{R}^n$ has a countable base. Concretely, around each x in the fibre choose a ball $B(a_x, \rho_x)$ with $a_x \in \mathbb{Q}^n$ and $\rho_x \in \mathbb{Q}_{>0}$ containing x and no other fibre point; distinct fibre points receive distinct pairs (a_x, ρ_x), and there are only countably many such pairs.

(b) **True**, and note that this part does not use the invertibility at all, only that f is C^1. A C^1 map is locally Lipschitz, so cover U by countably many balls B_i with $\overline{B_i} \subseteq U$ on which f is Lipschitz with some constant L_i. A set of diameter d has image of diameter at most $L_i d$, so a cube of side s lands inside a set of diameter $L_i s\sqrt{n}$ and hence inside a cube of side $L_i s\sqrt{n}$, whose volume is $(L_i\sqrt{n})^n$ times the original (apply this to the cube's intersection with B_i, whose diameter is no larger). Covering a null set $Z \cap B_i$ by cubes of total volume $< \varepsilon$ therefore covers $f(Z \cap B_i)$ by cubes of total volume $< (L_i\sqrt{n})^n \varepsilon$, so $f(Z \cap B_i)$ is null. A countable union of null sets is null, so $f(Z)$ is null.

(c) **True.** A local diffeomorphism is an open map, since it maps each of the neighbourhoods above onto an open set and every open subset of U is a union of such pieces; in particular $f(U)$ is open. Adding injectivity makes $f : U \to f(U)$ a continuous open bijection, hence a homeomorphism. Differentiability of the inverse is then local and is what Theorem 15.a.1 already supplies at each point: near any $y = f(x)$ the global inverse agrees with the local one, which is C^1.

(d) **False.** Surjectivity constrains where f goes, never how many times it goes there, and nothing in the hypotheses forbids several sheets landing on the same target. Let $\varphi : B_1(0) \to \mathbb{R}^n$, $\varphi(x) := x/(1-|x|^2)$, which is a diffeomorphism of the open unit ball onto $\mathbb{R}^n$: it is radial, and $r \mapsto r/(1-r^2)$ maps $[0,1)$ onto $[0,\infty)$ with derivative $(1+r^2)/(1-r^2)^2 > 0$. Now take the **disconnected** domain

$$U := B_1(0) \;\cup\; B_1(3e_1), \qquad f(x) := \begin{cases} \varphi(x) & x \in B_1(0), \\ \varphi(x - 3e_1) & x \in B_1(3e_1). \end{cases}$$

Then f is C^∞ with invertible differential everywhere and is surjective, but it is two-to-one, so it is not injective and cannot be a diffeomorphism.

Remark 15.b.23 (*What separates (c) from (d)*): Read as a pair, the last two parts say that injectivity is not a convenience in **(c)** but the entire remaining content. The inverse function theorem is local by nature and gives everything local for free: openness, smoothness of the inverse where it exists, the countable fibres of **(a)**. The one thing it never gives is that the local inverses fit together into a single function, and that is precisely what injectivity asserts.

The counterexample in **(d)** takes the cheapest possible route to failure, handing two disjoint pieces of U the same target, and it is worth noticing that it is disconnected for that reason rather than by accident. Example 15.a.5 makes the same point on a connected domain, where the folding is harder to see and no less real.

Q.E.D.

Solution to Exercise 15.a.17:

(a) **Injectivity:** Suppose $f(x) = f(y)$ for some $x, y \in U$. Then

$$0 = \|f(x) - f(y)\| \geq \alpha\|x - y\|,$$

which forces $\|x - y\| = 0$ since $\alpha > 0$. Hence $x = y$, proving injectivity.

Invertibility of the differential: Fix $x \in U$ and let $v \in \mathbb{R}^n \setminus \{0\}$. Since U is open, $x + tv \in U$ for all $t > 0$ sufficiently small. Applying the expansiveness inequality with $y = x + tv$ gives

$$\frac{\|f(x+tv) - f(x)\|}{t} \geq \frac{\alpha\|tv\|}{t} = \alpha\|v\|.$$

As $t \downarrow 0$, the left-hand side converges by differentiability of f to the directional derivative $\|Df_x(v)\|$. Thus

$$\|Df_x(v)\| \geq \alpha\|v\| > 0 \qquad \text{for all } v \neq 0.$$

In particular, $\ker(Df_x) = \{0\}$, so the linear map $Df_x : \mathbb{R}^n \to \mathbb{R}^n$ is injective. Since domain and codomain have the same finite dimension n, Df_x is a linear isomorphism (equivalently, $\det \mathrm{J}\, f(x) \neq 0$).

C^1-diffeomorphism onto $f(U)$: By the inverse function theorem (Theorem 15.a.1), f is a local C^1-diffeomorphism around every $x \in U$. In particular, f is an open map, so the image $f(U) \subseteq \mathbb{R}^n$ is open. Moreover, since f is globally injective on U, the inverse map $f^{-1} : f(U) \to U$ is well-defined. Around any $y_0 = f(x_0) \in f(U)$, the global inverse agrees with the local C^1-inverse provided by Theorem 15.a.1, so f^{-1} is of class C^1. Thus $f : U \to f(U)$ is a C^1-diffeomorphism.

(b) Closedness of the image: Suppose $U = \mathbb{R}^n$. Let $(y_k)_{k\in\mathbb{N}}$ be a sequence in $f(\mathbb{R}^n)$ converging to some $y \in \mathbb{R}^n$. For each k, choose $x_k \in \mathbb{R}^n$ with $f(x_k) = y_k$. Since (y_k) converges in $\mathbb{R}^n$, it is a Cauchy sequence. By expansiveness,

$$\|x_j - x_k\| \leq \frac{1}{\alpha}\|f(x_j) - f(x_k)\| = \frac{1}{\alpha}\|y_j - y_k\|.$$

Hence $(x_k)_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}^n$. Because $\mathbb{R}^n$ is complete (Proposition 6.a.5), there exists $x \in \mathbb{R}^n$ such that $x_k \to x$. By continuity of f,

$$y = \lim_{k\to\infty} y_k = \lim_{k\to\infty} f(x_k) = f(x) \in f(\mathbb{R}^n).$$

Therefore $f(\mathbb{R}^n)$ contains all its limit points, so $f(\mathbb{R}^n)$ is closed in $\mathbb{R}^n$.

Surjectivity and global diffeomorphism: The image $V := f(\mathbb{R}^n)$ is:

- non-empty (since $\mathbb{R}^n \neq \emptyset$),
- open in $\mathbb{R}^n$ (by part **(a)** with $U = \mathbb{R}^n$),
- closed in $\mathbb{R}^n$ (as just proved).

Since $\mathbb{R}^n$ is connected (being path-connected, Lemma 8.b.22), its only open and closed subsets are $\emptyset$ and $\mathbb{R}^n$ itself. Since $V \neq \emptyset$, we conclude $V = \mathbb{R}^n$. Thus f is surjective. Combined with the injectivity and local C^1-diffeomorphism established in part **(a)**, $f : \mathbb{R}^n \to \mathbb{R}^n$ is a global C^1-diffeomorphism.

Q.E.D.

The Implicit Function Theorem (Chapter 16)

The inverse function theorem asks when a map can be undone. The implicit function theorem generalizes the question: when can an equation $f(x,y)=0$ be solved for **some** of the variables in terms of the others, without f itself being invertible? The answer is the same condition applied to a submatrix of the Jacobian, and it is what turns level sets into graphs. The resemblance is not a coincidence: the proof applies the inverse function theorem to f padded out by the identity on the free variables, so the new theorem is the old one in disguise.

The implicit function theorem (Section 16.a)

Theorem 16.a.1 (*Implicit function theorem*): Let $1 \le n-d < n$, let $f \in C^k(\mathbb{R}^d \times \mathbb{R}^{n-d}, \mathbb{R}^{n-d})$ (or an open subset of $\mathbb{R}^d \times \mathbb{R}^{n-d}$), and suppose $f(x_0,y_0)=0$ for some $(x_0,y_0) \in \mathbb{R}^d \times \mathbb{R}^{n-d}$ such that the differential $D_y f(x_0,y_0)$ of $y \mapsto f(x_0,y)$ at y_0 is bijective (so $\mathrm{J}_y f(x_0,y_0)$ is invertible). Then there exist balls $B_r(x_0) \subseteq \mathbb{R}^d$ and $B_s(y_0) \subseteq \mathbb{R}^{n-d}$ and $g \in C^k(B_r(x_0), B_s(y_0))$ such that, for all $(x,y) \in B_r(x_0) \times B_s(y_0)$,

$$f(x,y)=0 \iff y=g(x).$$

Furthermore,

$$\mathrm{J}\, g(x) = -\big(\mathrm{J}_y f(x,g(x))\big)^{-1} \mathrm{J}_x f(x,g(x)).$$

In words: the equation $f(x,y)=0$ ties x and y together without telling us what y is. The theorem says that near a point where the tie is **non-degenerate in** y, the equation can be untangled. There is then an honest function g producing the right y for each nearby x, it is as smooth as f was, and its derivative is computable without ever writing g down.

Three things are worth reading off the statement before the pictures.

- **Which variables get solved for is a choice.** Splitting the n coordinates into d *"free"* ones and $n-d$ *"solved"* ones is ours to make; the theorem only demands that the block $\mathrm{J}_y f$ belonging to the solved variables be invertible. Choose a different split and a different block has to be invertible.
- **The conclusion is an equivalence, not just an inclusion.** Inside the box, $f(x,y)=0 \iff y=g(x)$. So g does not merely produce **some** solutions: within the box there are no others.
- **Nothing is claimed globally.** The level set $\{f(x,y)=0\}$ is a graph over x only near the chosen point. Section 17.a will call such a set a submanifold, and this local-graph property is exactly what that definition abstracts.

Example 16.a.2 (*A concrete example in* $\mathbb{R}^2$): Consider $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) := x^2+y^2-1$. Here $n=2$ and $d=1$, so $x \in \mathbb{R}$ and $y \in \mathbb{R}$. We want to see whether we can write y locally as a function of x, i.e., find $g(x)$ such that $f(x,g(x))=0$.

First, let us find a point (x_0,y_0) where $f(x_0,y_0)=0$. We guess $f(0,1)=0^2+1^2-1=0$, so $(0,1)$ works. Next, we check whether the y-derivative $\mathrm{J}_y f(0,1)$ is invertible. Since y is 1-dimensional, this is just the partial derivative $\partial_y f(0,1)$:

$$\partial_y f(0,1) = \partial_y (x^2+y^2-1)\big|_{(0,1)} = 2y\big|_{y=1} = 2 \neq 0.$$

Since it is non-zero, it is invertible as a 1×1 matrix. The theorem then guarantees that for small $r,s>0$, there exists $g : B_r(0) \to B_s(1)$ such that for all $(x,y) \in B_r(0) \times B_s(1)$,

$$f(x,y)=0 \iff y=g(x).$$

Since $f \in C^\infty(\mathbb{R}^2)$, we also have $g \in C^\infty(B_r(0))$. Moreover, we can compute $g'(x) = \mathrm{J}\, g(x)$ using the formula:

$$g'(x) = -\big(\partial_y f(x, g(x))\big)^{-1} \partial_x f(x, g(x)) = -(2g(x))^{-1}(2x) = -\frac{x}{g(x)}.$$

(Indeed, since $y = g(x) = \sqrt{1 - x^2}$ near $(0, 1)$, its derivative is exactly $-x/\sqrt{1 - x^2}$.)

Remark 16.a.3 (*Why* $\mathrm{J}\, g$ *has this form*): The formula is just implicit differentiation, done in blocks. By construction g satisfies $f(x, g(x)) \equiv 0$ for every $x \in B_r(x_0)$: the left-hand side is identically zero as a function of x, so its derivative is too. Differentiating with the chain rule (Theorem 10.a.1), and remembering that x enters both directly and through g,

$$0 = \underbrace{D_x f(x, g(x))}_{\text{direct}} + \underbrace{D_y f(x, g(x)) \cdot Dg(x)}_{\text{through } g},$$

and since $D_y f(x, g(x))$ is invertible near (x_0, y_0) we may move it to the other side:

$$Dg(x) = -\big(D_y f(x, g(x))\big)^{-1} D_x f(x, g(x)),$$

which is the formula in Theorem 16.a.1.

Read it as an exchange rate. Nudging x changes f by $D_x f$; to keep the equation satisfied, y must move by whatever compensates, and $\big(D_y f\big)^{-1}$ converts *"how much f moved"* back into *"how far y must travel"*. The minus sign is the compensation. This is also why invertibility of $D_y f$ is the hypothesis: if $D_y f$ were singular, some change in f could not be absorbed by moving y at all, and there would be nothing for g to do.

Proof of Theorem 16.a.1:

The trick is to turn the equation into a map between spaces of the **same** dimension, so that the inverse function theorem applies. Let U be the open domain of f and define

$$F : U \to \mathbb{R}^d \times \mathbb{R}^{n-d}, \qquad F(x, y) := \big(x,\ f(x, y)\big).$$

It keeps x and replaces y by the value of the equation. F is C^k, and in block form

$$\mathrm{J}\, F(x_0, y_0) = \begin{pmatrix} I_d & 0 \\ \mathrm{J}_x\, f(x_0, y_0) & \mathrm{J}_y\, f(x_0, y_0) \end{pmatrix}, \qquad \det \mathrm{J}\, F(x_0, y_0) = \det \mathrm{J}_y\, f(x_0, y_0) \neq 0.$$

By the inverse function theorem (Theorem 15.a.1) there is an open $W \ni (x_0, y_0)$ on which F is a C^k-diffeomorphism onto the open set $F(W)$. Shrink W to a box $B_{r'}(x_0) \times B_s(y_0) \subseteq W$; its image $V := F\big(B_{r'}(x_0) \times B_s(y_0)\big)$ is open and contains $F(x_0, y_0) = (x_0, 0)$, so there is $r \in (0, r']$ with $B_r(x_0) \times \{0\} \subseteq V$.

Let $x \in B_r(x_0)$. Since $(x, 0) \in V$, there is (x', y) in the box with $F(x', y) = (x, 0)$; comparing first components, $x' = x$, so $f(x, y) = 0$ for some $y \in B_s(y_0)$. This y is unique in $B_s(y_0)$, because F is injective on the box. Call it $g(x)$. Then for $(x, y) \in B_r(x_0) \times B_s(y_0)$,

$$f(x, y) = 0 \iff F(x, y) = (x, 0) \iff y = g(x),$$

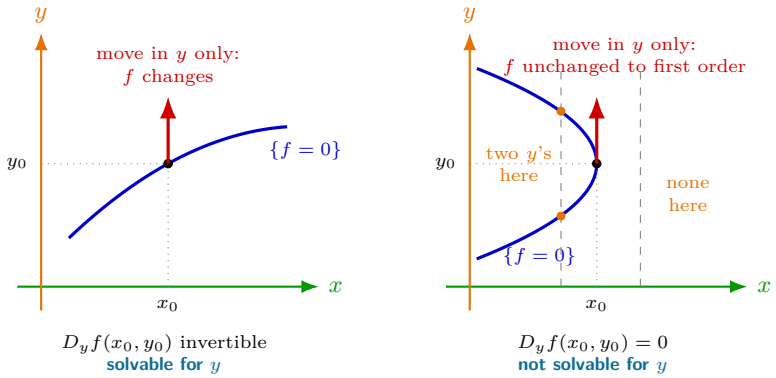
the second equivalence again by injectivity. Finally, $(x, g(x)) = (F|_W)^{-1}(x, 0)$, so g is the second component of the composition of $x \mapsto (x, 0)$ with the C^k map $(F|_W)^{-1}$, and hence $g \in C^k$. The formula for $\mathrm{J}\, g$ was derived in the remark above from $f(x, g(x)) \equiv 0$ and the chain rule. **Q.E.D.**

The proof shows what the theorem really is: the implicit function theorem is the inverse function theorem applied to $(x, y) \mapsto (x, f(x, y))$, a map that straightens the level set $\{f = 0\}$ into the flat slice $\mathbb{R}^d \times \{0\}$.

What the hypothesis means

The condition on $D_y f(x_0, y_0)$ is a statement about the **direction** in which the level set $\{f = 0\}$ runs at the point. Moving off (x_0, y_0) straight up, in the y-direction alone, changes f to first

order by $D_y f(x_0, y_0)$. If that map is invertible, no vertical motion leaves f unchanged, so the level set cannot run vertically at (x_0, y_0), and a curve that is not vertical is locally a graph over x. If instead $D_y f(x_0, y_0) = 0$, the level set is free to turn vertical, and above a single x there may sit two values of y, or none.

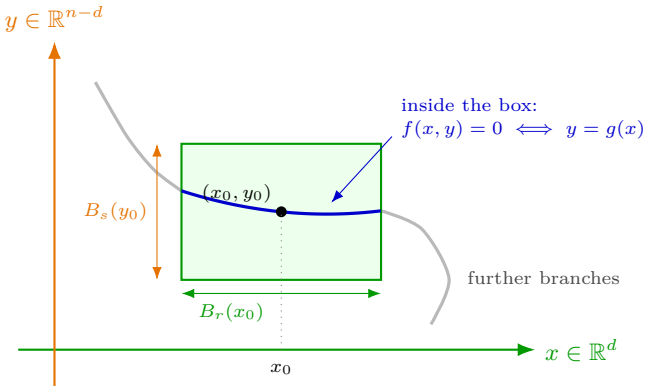


$D_y f(x_0, y_0)$ invertible
solvable for y

$D_y f(x_0, y_0) = 0$
not solvable for y

What the conclusion gives

The theorem does not say the whole level set is a graph, only that it is one **inside a small enough box**. On $B_r(x_0) \times B_s(y_0)$ the two descriptions coincide exactly: the zero set of f and the graph of g are the same set. Outside the box the level set may do anything at all, including doubling back or carrying further branches, and that is precisely why the statement is local.



Example 16.a.4 (*The circle S^1*): The circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 - 1 = 0\}$ is the level set at 0 of $F(x, y) := x^2 + y^2 - 1$, i.e., $S^1 = F^{-1}\{0\}$. Locally, there are explicit functions

$$x \mapsto \pm\sqrt{1 - x^2} = y(x),$$

defined on open intervals around x_0 whenever $y(x_0) \neq 0$. And indeed,

$$D_y F(x, y) = \frac{\partial F}{\partial y} = 2y$$

is bijective whenever $y \neq 0$, so the implicit function theorem applies there.

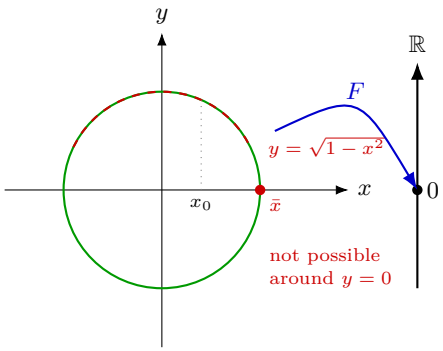
Remark 16.a.5 (*The circle realizes both panels*): The circle exhibits both cases of the figure above, and it is worth checking them against each other.

At a point with $y_0 \neq 0$, say $(0, 1)$, we have $D_y F = 2y_0 \neq 0$, the circle meets that point at a non-vertical angle, and near it the circle really is the graph $y = \sqrt{1 - x^2}$. That is the left panel.

At $\bar{x} = 1$, where $y = 0$, we get $D_y F(1, 0) = 0$. The circle is **vertical** there, and no graph $y(x)$ can exist around it: for x slightly less than 1 there are two points of the circle above x, namely

$\pm\sqrt{1-x^2}$, and for x slightly greater than 1 there are none. That is the right panel exactly, with two y's on one side and none on the other.

Note what has **not** gone wrong at $\bar{x} = 1$: the circle is a perfectly smooth curve there, with no corner and no break. What fails is only the attempt to describe it as a graph **over** x. Swap the roles of the two variables and everything works again, since $D_xF(1,0) = 2 \neq 0$: near $(1,0)$ the circle is the graph $x = \sqrt{1-y^2}$ of a function of y. This is the *"which variables get solved for is a choice"* point above, in its smallest possible instance.



Example 16.a.6 (*An implicit quintic*): Let $G(x,y) := y^5+y^3+y+x$, and consider its level set at 0, $G^{-1}\{0\} = \{(x,y) \in \mathbb{R}^2 : G(x,y) = 0\}$. Clearly $(0,0) \in G^{-1}\{0\}$, since $G(0,0) = 0^5+0^3+0+0 = 0$. Now both

$$\partial_x G(0,0) = 1 \neq 0, \qquad \partial_y G(0,0) = \left[5y^4+3y^2+1\right]_{y=0} = 1 \neq 0,$$

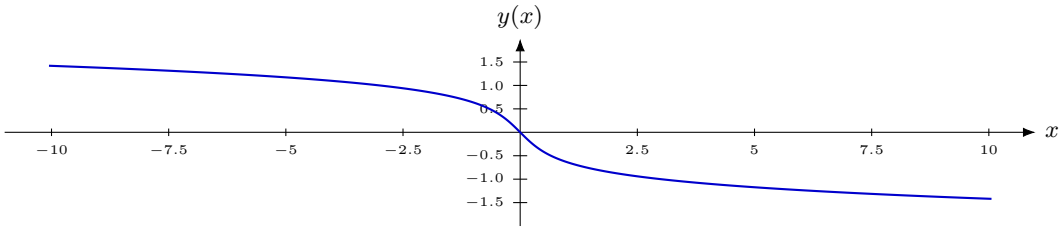
so by the implicit function theorem there are open intervals (balls) $(-r,r)$, $(-s,s)$ and functions

$$(-r,r) \to (-s,s), \quad y \mapsto x(y) = -y^5 - y^3 - y,$$
$$(-s,s) \to (-r,r), \quad x \mapsto y(x),$$

satisfying

$$y^5 + y^3 + y + x(y) = 0, \qquad y(x)^5 + y(x)^3 + y(x) + x = 0.$$

Remark 16.a.7: Do not look for a formula for $y(x)$. By the work of Abel and Galois, quintic equations have in general no solution in radicals, and the theorem does not need one: it guarantees that $y(x)$ exists (and is C^∞, since G is a polynomial), and it can be computed and plotted numerically.



Exercise 16.a.8 (*Implicitly defined functions from a system*): Show that one can solve the system of equations

$$\begin{cases} -2x^2 + y^2 + z^2 = 0 \\ x^2 + e^{y-1} - 2y = 0 \end{cases}$$

for y, z as functions of x (i.e. $y(x), z(x)$) close enough to the point $(1,1,1)$. Calculate $\begin{pmatrix} y'(1) \\ z'(1) \end{pmatrix}$.

(cf. Michaels Bsp 8.3.5)

Solution to Exercise 16.a.8:

Let $F(x,y,z) = \begin{pmatrix} -2x^2+y^2+z^2 \\ x^2+e^{y-1}-2y \end{pmatrix}$. We have $F(1,1,1) = \begin{pmatrix} -2+1+1 \\ 1+1-2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The Jacobian of F with respect to (y,z) is

$$D_{y,z}F(x,y,z) = \begin{pmatrix} \partial_y F_1 & \partial_z F_1 \\ \partial_y F_2 & \partial_z F_2 \end{pmatrix} = \begin{pmatrix} 2y & 2z \\ e^{y-1}-2 & 0 \end{pmatrix}.$$

At $(x,y,z) = (1,1,1)$, we have

$$D_{y,z}F(1,1,1) = \begin{pmatrix} 2 & 2 \\ e^0-2 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ -1 & 0 \end{pmatrix}.$$

The determinant is $\det(D_{y,z}F(1,1,1)) = 2(0) - 2(-1) = 2 \neq 0$. Thus the matrix is invertible, and by the Implicit Function Theorem, there exist functions $y(x), z(x)$ solving the system in a neighbourhood of $x=1$ such that $y(1)=1$ and $z(1)=1$.

To find the derivatives at $x=1$, we use the formula

$$\begin{pmatrix} y'(1) \\ z'(1) \end{pmatrix} = -(D_{y,z}F(1,1,1))^{-1} D_x F(1,1,1).$$

We calculate $D_x F(x,y,z) = \begin{pmatrix} -4x \\ 2x \end{pmatrix}$, so $D_x F(1,1,1) = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$. The inverse of $D_{y,z}F(1,1,1)$ is

$$\begin{pmatrix} 2 & 2 \\ -1 & 0 \end{pmatrix}^{-1} = \frac{1}{2}\begin{pmatrix} 0 & -2 \\ 1 & 2 \end{pmatrix}.$$

Then

$$\begin{pmatrix} y'(1) \\ z'(1) \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} 0 & -2 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} -4 \\ 2 \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

Therefore, $y'(1) = 2$ and $z'(1) = 0$. Q.E.D.

Exercise 16.a.9 (*Implicit solving in different variables*): Consider the equation $xyz - yz^4 + x^2y = 2$ near the point $(0,-2,1)$. Can one solve this equation for:

(a) x?
(b) y?
(c) z?

Calculate the derivative of the solution at the given point, if possible. (cf. Michaels Bsp 8.3.3)

Solution to Exercise 16.a.9:

Let $F(x,y,z) = xyz - yz^4 + x^2y - 2$. At the point $P = (0,-2,1)$, we have $F(0,-2,1) = 0 - (-2)(1) + 0 - 2 = 0$. We compute the partial derivatives of F:

$$\partial_x F(x,y,z) = yz + 2xy$$
$$\partial_y F(x,y,z) = xz - z^4 + x^2$$
$$\partial_z F(x,y,z) = xy - 4yz^3$$

Evaluating at $P = (0,-2,1)$:

$$\partial_x F(0,-2,1) = (-2)(1) + 0 = -2 \neq 0$$
$$\partial_y F(0,-2,1) = 0 - 1 + 0 = -1 \neq 0$$
$$\partial_z F(0,-2,1) = 0 - 4(-2)(1) = 8 \neq 0$$

(a) Since $\partial_x F(P) = -2 \neq 0$, the Implicit Function Theorem guarantees we can solve for $x = x(y,z)$ locally. The derivative (gradient) is

$$\nabla x(y,z)\Big|_{(-2,1)} = \begin{pmatrix} \partial_y x \\ \partial_z x \end{pmatrix} = -\frac{1}{\partial_x F(P)} \begin{pmatrix} \partial_y F(P) \\ \partial_z F(P) \end{pmatrix} = -\frac{1}{-2} \begin{pmatrix} -1 \\ 8 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 4 \end{pmatrix}.$$

(b) Since $\partial_y F(P) = -1 \neq 0$, we can solve for $y = y(x,z)$ locally. The derivative is

$$\nabla y(x,z)\Big|_{(0,1)} = \begin{pmatrix} \partial_x y \\ \partial_z y \end{pmatrix} = -\frac{1}{\partial_y F(P)} \begin{pmatrix} \partial_x F(P) \\ \partial_z F(P) \end{pmatrix} = -\frac{1}{-1} \begin{pmatrix} -2 \\ 8 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \end{pmatrix}.$$

(c) Since $\partial_z F(P) = 8 \neq 0$, we can solve for $z = z(x,y)$ locally. The derivative is

$$\nabla z(x,y)\Big|_{(0,-2)} = \begin{pmatrix} \partial_x z \\ \partial_y z \end{pmatrix} = -\frac{1}{\partial_z F(P)} \begin{pmatrix} \partial_x F(P) \\ \partial_y F(P) \end{pmatrix} = -\frac{1}{8} \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 1/8 \end{pmatrix}.$$

Q.E.D.

Exercise 16.a.10 (*Implicit function II* (important)): Show that the system of equations

$$\begin{cases} xy^5 + yu^5 + zv^5 = 1, \\ x^5 y + y^5 u + z^5 v = 1, \end{cases}$$

is solvable for the variables u and v in a neighborhood of the point $(x_0, y_0, z_0, u_0, v_0) = (0,1,1,1,0)$ and determine the derivatives $D_{(0,1,1)}u$ and $D_{(0,1,1)}v$ of the implicitly defined functions $u = u(x,y,z)$ and $v = v(x,y,z)$.

Info: *This exercise is Problem 7.5 of Problem Sheet 7, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 229*

Exercise 16.a.11 (*Implicit function theorem with arctan*): Consider the equation

$$y = x^2 \arctan(f(x,y)) + f(x,y).$$

(a) Show that there exists a unique continuously differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$ satisfying this equation for all $(x,y) \in \mathbb{R}^2$.

(b) Show that $f(x,0) = 0$ for all $x \in \mathbb{R}$ and compute the gradient $\nabla f(x,0)$ for all $x \in \mathbb{R}$.

Info: *This exercise is taken from the Analysis I & II exam of 24 January 2022 (Prof. Giovanni Felder), Part A, Problem 10.*

Solution: *See solution on Page 229*

Example 16.a.12 (*Which derivative has to be invertible*): Consider the following assertion about the roots of a polynomial whose coefficients move with a parameter t.

Claim (false). Let $n \geq 1$ and let $a_0, \ldots, a_{n-1} : (-1,1) \to \mathbb{R}$ be continuously differentiable with $a_0(0) = 0$ and $a_0'(0) \neq 0$. For $t \in (-1,1)$ put

$$p_t(x) := x^n + a_{n-1}(t)x^{n-1} + \cdots + a_1(t)x + a_0(t).$$

Then there are $\varepsilon > 0$ and a continuous $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{R}$ with $\gamma(0) = 0$ and $p_t(\gamma(t)) = 0$ for all $t \in (-\varepsilon, \varepsilon)$.

Informally: the root sitting at $x = 0$ when $t = 0$ can be followed continuously as t moves. The following argument for it is wrong.

> *For $(x,t) \in \mathbb{R} \times (-1,1)$ set $F(x,t) := p_t(x)$. Then F is a C^1 function on $\mathbb{R} \times (-1,1)$. We have $F(0,0) = 0$ and $\frac{\partial F}{\partial t}(0,0) \neq 0$. By the implicit function theorem there are $\varepsilon > 0$ and a differentiable, hence continuous, function $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{R}$ with $\gamma(0) = 0$ and $F(\gamma(t), t) = 0$ for all $t \in (-\varepsilon, \varepsilon)$. This γ is as required.*

A counterexample. Take $n = 2$ with $a_1 \equiv 0$ and $a_0(t) := t$, so that $p_t(x) = x^2 + t$. The hypotheses hold: $a_0(0) = 0$ and $a_0'(0) = 1 \neq 0$. But $p_t(x) = 0$ forces $x^2 = -t$, which has no real solution when $t > 0$. So for every $\varepsilon > 0$ the required γ fails to exist already at $t = \varepsilon/2$, and no continuity or differentiability is even in question.

Where the argument breaks. The implicit function theorem (Theorem 16.a.1) solves $F = 0$ for the variable whose partial derivative is invertible. To produce x as a function of t, what must be non-zero is

$$\frac{\partial F}{\partial x}(0,0) = a_1(0),$$

and the hypotheses say nothing whatever about $a_1(0)$. The argument instead verifies $\frac{\partial F}{\partial t}(0,0) = a_0'(0) \neq 0$, which is the derivative in the **parameter**. In the counterexample $a_1 \equiv 0$, so the condition that is actually needed fails, exactly as it must.

Remark 16.a.13 (*The hypothesis is not useless, it is just about the other variable*)**:** Reading $\partial F/\partial t(0,0) \neq 0$ as a genuine hypothesis rather than as a slip is what makes the example memorable. It is the hypothesis of the implicit function theorem with the roles of the two variables exchanged, so it does yield something: near the origin the zero set of F is the graph of a C^1 function $t = \tau(x)$, not of a function $x = \gamma(t)$. For $F(x,t) = x^2 + t$ this is the parabola $t = -x^2$, and the picture says everything. The zero set is a perfectly nice curve through the origin; it simply doubles back, so it is a graph over the x-axis and not over the t-axis. Which of the two variables can be solved for is precisely what the invertible partial derivative decides, and the answer need not be the same for both.

Info: *This example is taken from the Analysis I & II examination of summer 2016 (Prof. H. Knörrer), Problem 12, which asks for a counterexample and for the flaw in the argument. The reading of the false hypothesis in the closing remark is added here.*

Exercise 16.a.14 (*A conic solved for y, and how far the solution reaches*)**:** Consider the solution set of the equation

$$2x^2 - 4xy + y^2 - 3x + 4y = 0.$$

(a) Show that this set defines a function $y = g(x)$ with $g(1) = 1$.

(b) Compute the slope of the tangent to the solution set at $x = 1$, that is, determine $g'(1)$.

(c) Determine the maximal interval on which g can be defined.

Official hint: *For part **(c)** you do not need any particular theorem from the lecture.*

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Problem 17.*

Solution: *See solution on Page 230*

Exercise 16.a.15 (*True or False* (a graph over which variable?)**):** Following the examination this problem is taken from, call a subset of $\mathbb{R}^2$ a "*graph with respect to the x-variable*" if it has the form $\{(x, \varphi(x)) \mid x \in I\}$ for some interval I and some function $\varphi : I \to \mathbb{R}$, and likewise with the roles of the two variables exchanged. Consider

$$V := \{(x,y) \in \mathbb{R}^2 \mid y^4 + y^2 = x^3 + x\}.$$

Decide whether each of the following is true or false, with justification. "*Around p*" means: after intersecting V with a sufficiently small ball centred at p.

(a) V is a graph with respect to the x-variable around $(1,1)$.

(b) V is a graph with respect to the y-variable around $(0,0)$.

(c) V is a graph with respect to the y-variable around $(1,1)$.

(d) V is a graph with respect to the x-variable around $(0,0)$.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), where it appears as the eighth block of multiple-choice questions.*

Solution: *See solution on Page 231*

Solutions (Section 16.b)

Not every solution in this chapter is collected here. Exercise 16.a.8 (Page 225) and Exercise 16.a.9 (Page 226) are solved where they appear.

Solution to Exercise 16.a.10:

Write the system as $F(x,y,z,u,v)=0$ for $F:\mathbb{R}^5\to\mathbb{R}^2$,

$$F(x,y,z,u,v):=\begin{pmatrix} xy^5+yu^5+zv^5-1 \\ x^5y+y^5u+z^5v-1\end{pmatrix}.$$

The base point does lie on the level set, which the theorem requires before anything else:

$$F(0,1,1,1,0)=\begin{pmatrix}0+1+0-1\\0+1+0-1\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}.$$

The differential is

$$\mathrm{J}\,F(x,y,z,u,v)=\begin{pmatrix} y^5 & 5xy^4+u^5 & v^5 & 5yu^4 & 5zv^4\\ 5x^4y & x^5+5y^4u & 5z^4v & y^5 & z^5\end{pmatrix},$$

so at the base point,

$$\mathrm{J}\,F(0,1,1,1,0)=\begin{pmatrix}1&1&0&5&0\\0&5&0&1&1\end{pmatrix}.$$

By the implicit function theorem applied with (x,y,z) as the free variables and (u,v) as the variables to solve for, we need the submatrix of partial derivatives with respect to u,v to be invertible; this is

$$D_{(u,v)}F(0,1,1,1,0)=\begin{pmatrix}5&0\\1&1\end{pmatrix},\qquad \det=5\neq 0,$$

so the implicit function theorem applies, and the system is indeed solvable for u,v as C^∞ functions of (x,y,z) near $(0,1,1)$. Writing $G(x,y,z):=(u(x,y,z),v(x,y,z))$, the theorem gives

$$\begin{aligned}\mathrm{J}\,G(0,1,1)&=-\begin{pmatrix}5&0\\1&1\end{pmatrix}^{-1}\begin{pmatrix}1&1&0\\0&5&0\end{pmatrix}\\&=-\frac{1}{5}\begin{pmatrix}1&0\\-1&5\end{pmatrix}\begin{pmatrix}1&1&0\\0&5&0\end{pmatrix}\\&=-\frac{1}{5}\begin{pmatrix}1&1&0\\-1&24&0\end{pmatrix}.\end{aligned}$$

So, $D_{(0,1,1)}u=-\frac{1}{5}\,(\,1\;1\;0\,)$ and $D_{(0,1,1)}v=-\frac{1}{5}\,(\,-1\;24\;0\,)$. Q.E.D.

Proof of Exercise 16.a.11:

(a) Define $F(x,y,z):=x^2\arctan(z)+z-y$ for $(x,y,z)\in\mathbb{R}^3$. F is smooth on $\mathbb{R}^3$. For any fixed $(x,y)\in\mathbb{R}^2$, consider $h(z):=F(x,y,z)=x^2\arctan(z)+z-y$. Its derivative with respect to z is:

$$\frac{\partial F}{\partial z}(x,y,z)=\frac{x^2}{1+z^2}+1\geq 1>0\quad\text{for all } z\in\mathbb{R}.$$

Since $\partial_z F > 0$, $h(z)$ is strictly increasing on $\mathbb{R}$. As $z \to \pm\infty$, $h(z) \to \pm\infty$. By the Intermediate Value Theorem, there exists a unique $z \in \mathbb{R}$ such that $F(x,y,z) = 0$. Thus, for every (x,y), there is a unique value $f(x,y) := z$. By the Implicit Function Theorem (Theorem 16.a.1), since $\partial_z F \neq 0$ everywhere, f is continuously differentiable on $\mathbb{R}^2$.

(b) Setting $y = 0$ in the defining equation:

$$0 = x^2 \arctan(f(x,0)) + f(x,0).$$

Since $z = 0$ is a solution to $x^2 \arctan(z) + z = 0$ and the solution z is unique, we conclude $f(x,0) = 0$ for all $x \in \mathbb{R}$.

To find $\nabla f(x,0)$, differentiate the identity $x^2 \arctan(f(x,y)) + f(x,y) - y = 0$ implicitly and then evaluate at $(x,0)$, where $f = 0$ and hence $\arctan(f) = 0$ and $1/(1+f^2) = 1$:

- **With respect to x:** the identity gives $2x \arctan(f) + x^2 \frac{\partial_1 f}{1+f^2} + \partial_1 f = 0$. At $(x,0)$ the first term drops out, leaving $(x^2+1)\,\partial_1 f(x,0) = 0$, and since $x^2 + 1 > 0$ we conclude $\partial_1 f(x,0) = 0$.
- **With respect to y:** the identity gives $x^2 \frac{\partial_2 f}{1+f^2} + \partial_2 f - 1 = 0$. At $(x,0)$ this becomes $(x^2+1)\,\partial_2 f(x,0) = 1$, so $\partial_2 f(x,0) = \frac{1}{x^2+1}$.

Therefore $\nabla f(x,0) = \left(0, \frac{1}{x^2+1}\right)^{\mathsf{T}}$.

Remark 16.b.16: Part **(a)** is stronger than what the implicit function theorem alone delivers. The theorem is local: it produces an f near a given solution point. What upgrades that to a single f defined on all of $\mathbb{R}^2$ is the strict monotonicity of $z \mapsto F(x,y,z)$ together with its unbounded range, which pins down exactly one root for every (x,y). The theorem then supplies the regularity.

Q.E.D.

Solution of Exercise 16.a.14:

Write $F(x,y) := 2x^2 - 4xy + y^2 - 3x + 4y$, a polynomial and hence C^∞ on $\mathbb{R}^2$.

(a) First, $(1,1)$ lies on the set: $F(1,1) = 2 - 4 + 1 - 3 + 4 = 0$. Next,

$$\partial_y F(x,y) = -4x + 2y + 4, \qquad \partial_y F(1,1) = -4 + 2 + 4 = 2 \neq 0.$$

By Theorem 16.a.1 there are open intervals $I \ni 1$ and $J \ni 1$ and a C^∞ function $g : I \to J$ with $g(1) = 1$ whose graph is exactly $\{F = 0\} \cap (I \times J)$.

(b) Differentiating $F(x, g(x)) = 0$ and solving, or quoting the formula in Theorem 16.a.1 directly,

$$g'(x) = -\frac{\partial_x F(x, g(x))}{\partial_y F(x, g(x))}, \qquad \partial_x F(x,y) = 4x - 4y - 3.$$

At the point in question $\partial_x F(1,1) = 4 - 4 - 3 = -3$ and $\partial_y F(1,1) = 2$, so

$$g'(1) = -\frac{-3}{2} = \boxed{\tfrac{3}{2}}.$$

(c) The hint is the point of the part: F is quadratic in y, so the equation can simply be solved. Reading $F = 0$ as $y^2 + (4 - 4x)y + (2x^2 - 3x) = 0$ and applying the quadratic formula,

$$y = (2x-2) \pm \sqrt{(2x-2)^2 - (2x^2 - 3x)} = (2x-2) \pm \sqrt{2x^2 - 5x + 4}.$$

The radicand $2x^2 - 5x + 4$ has discriminant $25 - 32 = -7 < 0$ and positive leading coefficient, so it is **strictly positive for every real x**. Both branches are therefore defined on all of $\mathbb{R}$, and they never meet. At $x = 1$ they take the values 0 ± 1, so the branch through $(1,1)$ is the one with the plus sign:

$$g(x) = 2x - 2 + \sqrt{2x^2 - 5x + 4}, \qquad \text{maximal domain } \boxed{\mathbb{R}}.$$

As a check on **(b)**, $g'(x) = 2 + \dfrac{4x - 5}{2\sqrt{2x^2 - 5x + 4}}$ gives $g'(1) = 2 - \frac{1}{2} = \frac{3}{2}$, as found above.

Remark 16.b.17 (*Local theorem, global answer*): The two halves of this exercise are worth holding side by side, because they answer different questions and it is easy to think the first has answered the second.

Theorem 16.a.1 is **local**, and deliberately so: it produces some interval around $x = 1$ and says nothing whatever about how far that interval extends. Part **(c)** shows the honest answer here is *"all of* $\mathbb{R}$*"*, and no amount of squeezing the theorem would have produced it. What produced it was solving the equation.

The reverse is also worth noting. The condition $\partial_y F \neq 0$ fails along the line $y = 2x - 2$, which is precisely where the two branches would meet. They never do, because the radicand never vanishes. So the hypothesis of the theorem happens to hold along the whole branch, which is why the branch stays a graph forever. When a conic **does** have a vertical tangent, that is exactly where $\partial_y F$ vanishes and the local statement is all there is.

Q.E.D.

Solution to Exercise 16.a.15:

Set $G(x, y) := y^4 + y^2 - x^3 - x$, a polynomial and hence smooth, so that $V = G^{-1}(0)$. Its two partial derivatives are

$$\partial_x G(x, y) = -(3x^2 + 1), \qquad \partial_y G(x, y) = 4y^3 + 2y = 2y\,(2y^2 + 1).$$

Two observations settle almost everything. First, $\partial_x G < 0$ at **every** point of $\mathbb{R}^2$, since $3x^2 + 1 \geq 1$. Second, $2y^2 + 1$ is likewise never zero, so $\partial_y G$ vanishes exactly on the line $y = 0$.

(a) **True.** The point lies on V, because $1^4 + 1^2 = 2 = 1^3 + 1$, and $\partial_y G(1, 1) = 6 \neq 0$. Solving for y is what a non-vanishing $\partial_y G$ buys, so Theorem 16.a.1 gives an interval $I \ni 1$ and a C^1 function $\varphi : I \to \mathbb{R}$ with $\varphi(1) = 1$ whose graph is V near $(1, 1)$.

(b) **True.** The origin lies on V, and $\partial_x G(0, 0) = -1 \neq 0$, so the theorem applies with the roles of the variables exchanged and produces $x = \psi(y)$.

(c) **True**, for the same reason: $\partial_x G(1, 1) = -4 \neq 0$.

(d) **False.** Here $\partial_y G(0, 0) = 0$, so Theorem 16.a.1 says nothing; and unlike the usual situation, the answer really is no. Factor the defining equation as

$$y^2(y^2 + 1) = x(x^2 + 1).$$

The left-hand side is non-negative, so every point of V has $x \geq 0$: already there is no interval I around 0 over which V could be a graph. Worse, for each small $x > 0$ there are **two** points. Indeed $t \mapsto t(t + 1)$ is a strictly increasing bijection of $[0, \infty)$ onto itself, so the equation determines $t := y^2$ uniquely from x, and $t > 0$, leaving the two points $y = \pm\sqrt{t}$.

Remark 16.b.18 (*Only one of the two directions was ever in doubt*): Because $\partial_x G$ vanishes nowhere, V is a graph with respect to the y-variable around **every** one of its points, which disposes of **(b)** and **(c)** without looking at where those points are. The whole content of the problem sits in the other direction, and the contrast between **(a)** and **(d)** is the point: the same set, the same theorem, and the answer decided entirely by whether $\partial_y G$ happens to vanish at the point chosen.

This is the positive counterpart of Example 16.a.12, where the error was to check the wrong partial derivative. Here both are checked, and each of the two answers comes out somewhere.

Q.E.D.

Submanifolds, Tangent and Normal Spaces (Chapter 17)

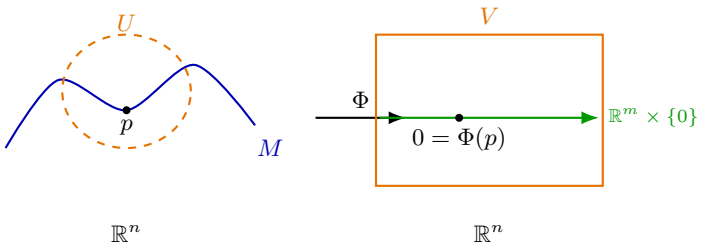
The implicit function theorem said that a level set is locally a graph. This chapter takes that as a definition. A submanifold of $\mathbb{R}^n$ is a set that looks, near each of its points, like a piece of $\mathbb{R}^m$. Three descriptions of such a set turn out to be equivalent (a level set of a submersion, a local graph, a local parametrization), and picking the convenient one is most of the work in practice. Each point then carries its own linear algebra: the tangent space of directions along the submanifold, and the normal space orthogonal to it.

Submanifolds (Section 17.a)

Recall that a "*C^k-diffeomorphism*" (Theorem 15.a.1) is a C^k bijective function whose inverse is also C^k.

Definition 17.a.1 (*Submanifold*): $M^m \subseteq \mathbb{R}^n$ is a C^k, m-dimensional "*submanifold*" (German: „*Untermannigfaltigkeit*") of $\mathbb{R}^n$ if for any point $p \in M$ there are an open neighbourhood $U \subseteq \mathbb{R}^n$ of p, an open neighbourhood $V \subseteq \mathbb{R}^n$ of 0, and a C^k-diffeomorphism $\Phi : U \to V$ with $\Phi(p) = 0$ (a "*submanifold chart*") such that

$$\Phi(U \cap M) = (\mathbb{R}^m \times \{0\}) \cap V = \{x \in V : x_{m+1} = \cdots = x_n = 0\}.$$



Remark 17.a.2 (*Why the definition is almost never used directly*): Producing such a Φ by hand is difficult even for simple M, so the definition is rarely what one works with. Two equivalent descriptions do the job instead, the **implicit** one (the regular value theorem below) and the **explicit or graphical** one (the local parametrization theorem below). Remark 17.a.10 compares all three once both are available.

The regular value theorem (Subsection 17.a.1)

Theorem 17.a.3 (*Regular value theorem*): Suppose $U \subseteq \mathbb{R}^n$ is open, $F \in C^k(U, \mathbb{R}^\ell)$, and $y \in F(U)$. If

$$DF_x : \mathbb{R}^n \to \mathbb{R}^\ell$$

is surjective (i.e., $\mathrm{J}\,F(x)$ has full rank) for all $x \in F^{-1}\{y\} = \{\bar{x} \in U : F(\bar{x}) = y\}$, then $F^{-1}\{y\}$ is a C^k submanifold of dimension $n - \ell$.

Proof:

Write $M := F^{-1}\{y\}$, $m := n - \ell$, and fix $p \in M$ (we assume $k \geq 1$). Since $\mathrm{J}\,F(p)$ has rank ℓ, some ℓ of its columns are linearly independent. After renumbering the coordinates of $\mathbb{R}^n$, which is a linear isomorphism and does not affect any of the statements, these are the last ℓ columns. Write $x = (u, w)$ with $u \in \mathbb{R}^m$, $w \in \mathbb{R}^\ell$, so that the block $\mathrm{J}_w F(p)$ is invertible, and define

$$\Phi(x) := \big(u - u_p,\ F(x) - y\big) \in \mathbb{R}^m \times \mathbb{R}^\ell, \qquad \text{where } p = (u_p, w_p).$$

Then $\Phi \in C^k$, $\Phi(p) = 0$, and $\mathrm{J}\,\Phi(p) = \begin{pmatrix} I_m & 0 \\ \mathrm{J}_u F(p) & \mathrm{J}_w F(p) \end{pmatrix}$ has determinant $\det \mathrm{J}_w F(p) \neq 0$. By the inverse function theorem there is an open $U' \ni p$ such that $\Phi : U' \to V' := \Phi(U')$ is a C^k-diffeomorphism onto an open neighbourhood of 0. Finally, for $x \in U'$ the last ℓ coordinates of $\Phi(x)$ are $F(x) - y$, so $x \in M$ if and only if $\Phi(x) \in \mathbb{R}^m \times \{0\}$. That is,

$$\Phi(U' \cap M) = (\mathbb{R}^m \times \{0\}) \cap V',$$

which is exactly a submanifold chart around p in the sense of Definition 17.a.1. Q.E.D.

The chart in the proof is worth remembering: its last ℓ coordinates are **the equations themselves**, and the first m are whichever coordinates the equations leave free. Straightening a level set is nothing more than declaring the defining equations to be new coordinates.

Important remark 17.a.4 (*Full rank means two opposite things*): *"Full rank"* is decided by the **shape** of the matrix, and the two representations of a submanifold sit on opposite sides of it:

- An **implicit** description $F : \mathbb{R}^n \to \mathbb{R}^\ell$ cuts M out by ℓ equations, so $\mathrm{J}\,F$ is **wide** ($\ell \times n$ with $\ell \leq n$). Full rank $= \ell$ means DF_x is **surjective**, so the ℓ constraints are independent, each one genuinely cutting a dimension. Here $\dim M = n - \ell$.
- A **parametrization** $f : \mathbb{R}^m \to \mathbb{R}^n$ sweeps M out by m parameters, so $\mathrm{J}\,f$ is **tall** ($n \times m$ with $m \leq n$). Full rank $= m$ means Df_x is **injective**, so the m parameters move in independent directions, none of them wasted. Here $\dim M = m$.

Same phrase, opposite condition. A wide matrix is never injective and a tall one is never surjective, so there is no ambiguity in practice. But it is worth checking which side you are on before writing down what *"full rank"* is supposed to give you.

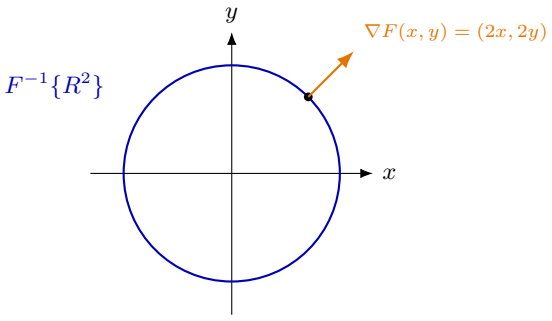
Example 17.a.5 (*Circle of radius R*): Let $F : \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Then

$$\mathrm{J}\,F(x, y) = (2x,\ 2y)$$

has full rank at every $(x, y) \in \mathbb{R}^2 \setminus \{0\}$. Here $\ell = 1$, so by Important remark 17.a.4 full rank means that $DF_{(x,y)} : \mathbb{R}^2 \to \mathbb{R}$ is **surjective**, which for a single row is simply the requirement that $(2x, 2y)$ be nonzero. For $R > 0$, $(0, 0) \notin F^{-1}\{R^2\}$, so

$$F^{-1}\{R^2\} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}$$

is a submanifold of dimension $2 - 1 = 1$ by Theorem 17.a.3. Similarly, $\{x \in \mathbb{R}^n : |x|^2 = R^2\}$ is a submanifold of dimension $n - 1$ for $n \geq 1$ (the "$(n-1)$*-sphere*" of radius R, named after its dimension, which is why the unit sphere of $\mathbb{R}^n$ is written S^{n-1}).



AI-Example 17.a.6 (*A set that is **not a submanifold, and how to prove it*):** The regular value theorem is a one-way street: if DF_x is surjective everywhere on the level set, we get a submanifold. When it is not, we get **no information**. The level set may still be a submanifold, or it may not. It is worth seeing a case where it is not.

Let $F(x,y) := xy$ and consider $C := F^{-1}\{0\}$, the union of the two coordinate axes. Here $\mathrm{J}\,F(x,y) = (y,\ x)$, which is nonzero, and hence surjective, at every point of C **except** the origin. So, Theorem 17.a.3 does apply on $C \setminus \{(0,0)\}$, telling us that the four open half-axes are 1-dimensional submanifolds, and says nothing at the origin.

And at the origin C genuinely fails to be a submanifold. The argument is a connectedness trick worth remembering, since it is the standard way to prove a negative here.

Suppose C were a 1-dimensional submanifold. By Definition 17.a.1 there would be a neighbourhood U of the origin and a diffeomorphism $\Phi : U \to V$ carrying $C \cap U$ onto a piece of a **line**, $(\mathbb{R}\times\{0\})\cap V$. Shrinking, we may take V to be a ball around 0 (replace U by the preimage of such a ball), so that this piece is an open interval. Now delete the centre point from each side and count connected components (Definition 8.a.1):

- Removing a point from an open interval leaves **two** pieces.
- Removing the origin from $C \cap U$ leaves **at least four** pieces, since U meets all four open half-axes and no connected subset of $C \setminus \{(0,0)\}$ meets two of them.

But Φ is a homeomorphism, and homeomorphisms preserve the number of connected components (Theorem 8.a.7, applied to Φ and to Φ^{-1}). Four or more cannot equal two, so no such Φ exists and C is not a submanifold.

Remark 17.a.7: The moral is that submanifolds are forbidden from having **crossings, corners or cusps**: near each of its points a submanifold must look like a flat piece of $\mathbb{R}^m$, and a crossing does not. Compare AI-Example 13.a.4, where the constraint set $\{y^2 = x^3\}$ had a cusp at exactly the point where ∇g vanished. That is the same phenomenon, seen through the Lagrange-multiplier lens back in Chapter 13.

Connection to Lagrange multipliers (Subsubsection 17.a.1.1)

Remark 17.a.8 (*Constraint sets are submanifolds*): Recall the setting of Proposition 13.a.2: we optimize $f|_M$ for $f \in C^1(U,\mathbb{R})$, where $g_1,\dots,g_k \in C^1(U,\mathbb{R})$,

$$M := \{x \in U : g_1(x) = \cdots = g_k(x) = 0\}, \qquad (\nabla g_1,\dots,\nabla g_k) \text{ linearly independent for all } x \in M$$

(hence $k \le n$). This means precisely that, for $g : U \to \mathbb{R}^k$, $x \mapsto (g_1(x),\dots,g_k(x))$, the Jacobian

$$\mathrm{J}\,g(x) = \begin{pmatrix} \nabla g_1(x) \longrightarrow \\ \vdots \\ \nabla g_k(x) \longrightarrow \end{pmatrix}$$

has full rank. Therefore, $M = g^{-1}\{0\}$ is an $(n-k)$-dimensional submanifold by the regular value theorem: the constraint sets of the Lagrange-multiplier problems in Chapter 13 were submanifolds all along.

The local parametrization theorem (Subsection 17.a.2)

Theorem 17.a.9 (*Local parametrization*): Let $M \subseteq \mathbb{R}^n$. Then M is a C^k submanifold with $\dim M = m$ if and only if for every $p \in M$ there exist an open neighbourhood V of p, an open set $U \subseteq \mathbb{R}^m$, and a C^k map $f : U \to V \cap M$ such that

- $Df_x : \mathbb{R}^m \to \mathbb{R}^n$ is injective for all $x \in U$;
- f is a homeomorphism (bijective, continuous, with continuous inverse f^{-1}).

Proof:

($\Rightarrow$) Let $\Phi : U \to V$ be a submanifold chart around p, named as in Definition 17.a.1 (so here the neighbourhood of p is U, and the parameter domain will be U_0). The set $U_0 := \{x \in \mathbb{R}^m : (x,0) \in$

$V\}$ is open, and $f(x) := \Phi^{-1}(x,0)$ is a C^k bijection of U_0 onto $U \cap M$. Its inverse is the first m coordinates of Φ, hence continuous, and Df_x is the invertible $D(\Phi^{-1})_{(x,0)}$ restricted to $\mathbb{R}^m \times \{0\}$, hence injective.

($\Leftarrow$) Fix $p = f(x_0)$. As $\mathrm{J}\, f(x_0)$ has rank m, after renumbering the coordinates of $\mathbb{R}^n$ its first m rows form an invertible block. Thicken f to a map between spaces of equal dimension,

$$G : U \times \mathbb{R}^{n-m} \to \mathbb{R}^n, \qquad G(x,z) := f(x) + (0,z),$$

whose Jacobian at $(x_0,0)$ is block lower triangular with that invertible block and I_{n-m} on the diagonal. By the inverse function theorem G restricts to a C^k-diffeomorphism from a box $W := U_1 \times B_\varepsilon(0)$ around $(x_0,0)$ onto an open set $G(W) \ni p$, and on the slice $z = 0$ it is f: $G(x,0) = f(x)$.

Here the homeomorphism clause enters. It says $f(U_1)$ is open **in** $M \cap V$, so $f(U_1) = O \cap M$ for some open $O \subseteq V$. Put $U' := G(W) \cap O$ and $\Phi := (G|_W)^{-1}$ on U', shifted by $-(x_0,0)$ so that $p \mapsto 0$. A point of U' lies in M exactly when it lies in $f(U_1) = G(U_1 \times \{0\})$, that is, exactly when its image under $(G|_W)^{-1}$ has $z = 0$. So $\Phi(U' \cap M) = (\mathbb{R}^m \times \{0\}) \cap \Phi(U')$, which is a submanifold chart around p. **Q.E.D.**

Without the homeomorphism clause the last step fails, and the failure is visible. Other parts of M may come back arbitrarily close to p without belonging to the sheet $f(U_1)$, as in a figure-eight traversed injectively: near the crossing point the image of a small parameter interval is only one branch, while M has two.

Remark 17.a.10 (*Three descriptions, and which one to reach for*): There are now three ways to certify that M is an m-dimensional submanifold, and they are equivalent. They are not equally convenient, though, and choosing badly is what makes these exercises feel hard.

- **Chart** (Definition 17.a.1) is the definition itself. Straighten M out with a diffeomorphism, so that the first m coordinates of the chart carry the submanifold and the rest vanish. It is almost never used directly on a concrete M, since producing Φ by hand is painful. Its role is theoretical, and it is what the other two are proved from.
- **Implicit, via the regular value theorem** (Theorem 17.a.3) is the one to reach for when M arrives as **equations**, such as $\{x^2 + y^2 = R^2\}$ or $\{g_1 = \cdots = g_k = 0\}$. It rests on the same idea as the chart definition, with the diffeomorphism produced implicitly rather than by hand. Check a rank condition on $\mathrm{J}\, F$ and you are done, with no parametrization ever written. This is normally the cheapest route.
- **Parametrization** (Theorem 17.a.9) is the one for an M that arrives as a **picture you can sweep out**, such as a graph, a surface of revolution, or a curve given by a formula in a parameter. It is often the most intuitive, since it builds the submanifold out of parameters explicitly. It is also the one to use when the tangent space is wanted next, because $\partial_1 f, \ldots, \partial_m f$ hand you a basis directly (Definition 17.c.30).

The rough rule: **equations → regular value theorem; a formula with parameters → local parametrization.** A graph $\{(x, f(x))\}$ satisfies both, which is why graphs are the easiest case of all.

Two cautions carried over from Important remark 17.a.4. *"Full rank"* means **surjective** in the first case and **injective** in the second, and the shape of the matrix decides which. And neither theorem yields a negative. If the rank condition fails you have learned nothing, and must argue separately (see AI-Example 17.a.6).

Example 17.a.11 (*The n-parabola*): Let $P := \{(x,t) \in \mathbb{R}^n \times \mathbb{R} : t = |x|^2\}$. One could use the regular value theorem, but instead take

$$f : \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}, \qquad x \mapsto (x, |x|^2).$$

(a) The Jacobian matrix has full rank n at every x:

$$\mathrm{J}\, f(x) = \begin{pmatrix} \mathrm{id}_{n\times n} \\ 2x^{\mathsf{T}} \end{pmatrix}.$$

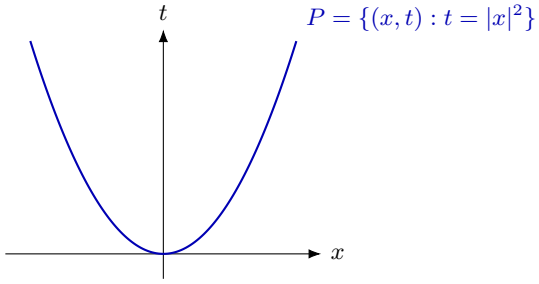
This one is tall, so by Important remark 17.a.4 full rank means that Df_x is **injective**, which is clear from the $\mathrm{id}_{n\times n}$ block alone.

(b) f is bijective onto P and continuous, with inverse $\pi : P \to \mathbb{R}^n$, $(x,t) \mapsto x$ (the canonical projection), which is also continuous.

(c) $f(\mathbb{R}^n) = P$.

So, P is an n-dimensional submanifold by Theorem 17.a.9.

Remark 17.a.12: The graphical representation of a submanifold (as the graph of a function, here $x \mapsto |x|^2$) is therefore a common, and equivalent, special case of a local parametrization: it is always injective on its domain by construction, and its inverse is simply the projection forgetting the last coordinate.



An alternative optimization technique (Subsubsection 17.a.2.1)

Remark 17.a.13 (*Parametrized optimization, revisiting Exercise 12.b.25*): Local parametrizations give a second route to constrained-extremum problems, alongside Lagrange multipliers: parametrize the constraint set and optimize the resulting unconstrained function of fewer variables. To see it at work, revisit $f(x,y) := 2x^2 + y^2 - x$ on $S^1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, which is exactly Exercise 12.b.25**(a)**, solved there via Lagrange multipliers.

Parametrize S^1 by $(x(\theta), y(\theta)) := (\cos\theta, \sin\theta)$. Then

$$f(\theta) := f(x(\theta), y(\theta)) = 2\cos^2\theta + \sin^2\theta - \cos\theta = \cos^2\theta - \cos\theta + 1,$$

using $\sin^2\theta = 1 - \cos^2\theta$. Differentiating,

$$f'(\theta) = -2\cos\theta\sin\theta + \sin\theta \overset{!}{=} 0.$$

(a) $\sin\theta = 0 \implies \theta_1 \in \{0, \pi\}$.

(b) $\sin\theta \neq 0 \implies -2\cos\theta + 1 = 0 \implies \cos\theta = \frac{1}{2} \implies \theta_2 \in \{\frac{\pi}{3}, \frac{5\pi}{3}\}$.

So, the extrema are at $(\cos\theta_1, \sin\theta_1) = (\pm 1, 0)$ and $(\cos\theta_2, \sin\theta_2) = \left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right)$. These are exactly the four points found via Lagrange multipliers in the solution to Exercise 12.b.25**(a)**, so the two methods agree.

Exercise 17.a.14 (*Dimension of a submanifold*): Show that $M := \{(x,y,z) \in \mathbb{R}^3 \mid 2x + z = 1, x^2 + y^2 - z^2 = 1\}$ is a submanifold of $\mathbb{R}^3$. What is its dimension? (cf. Michaels Bsp 9.1.2 d)

Solution to Exercise 17.a.14:

Let $F(x,y,z) = \begin{pmatrix} 2x+z-1 \\ x^2+y^2-z^2-1 \end{pmatrix}$. Then $M = F^{-1}\{0\}$. We compute the Jacobian matrix of F:

$$\mathrm{J}\,F(x,y,z) = \begin{pmatrix} 2 & 0 & 1 \\ 2x & 2y & -2z \end{pmatrix}.$$

We need to check that $\operatorname{rank}(\mathrm{J}\,F(x,y,z)) = 2$ for all $(x,y,z) \in M$. The rank is less than 2 if and only if all 2×2 subdeterminants are zero:

$$\det\begin{pmatrix} 2 & 0 \\ 2x & 2y \end{pmatrix} = 4y = 0 \implies y = 0$$

$$\det\begin{pmatrix} 2 & 1 \\ 2x & -2z \end{pmatrix} = -4z - 2x = 0 \implies x = -2z$$

$$\det\begin{pmatrix} 0 & 1 \\ 2y & -2z \end{pmatrix} = -2y = 0 \implies y = 0.$$

So, the rank is not maximal only when $y = 0$ and $x = -2z$. Let's check if such a point can belong to M. Substituting $x = -2z$ into the first equation of M gives $2(-2z) + z = -3z = 1 \implies z = -1/3$. Then $x = 2/3$. Thus the point is $(2/3, 0, -1/3)$. Now check the second equation of M: $(2/3)^2 + 0^2 - (-1/3)^2 = 4/9 - 1/9 = 3/9 = 1/3 \neq 1$. Therefore, this point is not in M. Since $\mathrm{J}\,F(x,y,z)$ has maximal rank 2 for all $(x,y,z) \in M$, M is a submanifold of $\mathbb{R}^3$. Its dimension is $3-2=1$: three coordinates minus two independent equations. **Q.E.D.**

Exercise 17.a.15 (*Submanifold of matrices*): Show that $M := \{A \in \mathrm{Mat}_{2\times 2}(\mathbb{R}) \mid \operatorname{rank}(A) = 1\}$ is a submanifold of $\mathrm{Mat}_{2\times 2}(\mathbb{R})$. What is the dimension of M?

Solution to Exercise 17.a.15:

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The condition $\operatorname{rank}(A) \leq 1$ is equivalent to $\det(A) = ad - bc = 0$. The condition $\operatorname{rank}(A) = 1$ means $\det(A) = 0$ but $A \neq 0$. So, M is the zero set of $F : \mathbb{R}^4 \setminus \{0\} \to \mathbb{R}$, $F(a,b,c,d) = ad - bc$. The space $\mathrm{Mat}_{2\times 2}(\mathbb{R})$ is isomorphic to $\mathbb{R}^4$. The Jacobian of F is

$$\mathrm{J}\,F(a,b,c,d) = \begin{pmatrix} \partial_a F & \partial_b F & \partial_c F & \partial_d F \end{pmatrix} = \begin{pmatrix} d & -c & -b & a \end{pmatrix}.$$

For $\mathrm{J}\,F$ to not have maximal rank (which is 1), we must have $d = 0$, $c = 0$, $b = 0$, and $a = 0$, which means $A = 0$. But the zero matrix is excluded from M because its rank is 0, not 1. Thus $\mathrm{J}\,F(A) \neq 0$ for all $A \in M$. By the regular value theorem, M is a submanifold. The ambient dimension is 4 and the number of independent constraints is 1. So, the dimension of M is $4 - 1 = 3$. **Q.E.D.**

Example 17.a.16 (*An immersion is locally a parametrization*): Theorem 17.a.9 asks a candidate parametrization f for two things: that Df_x be injective everywhere, and that f be a homeomorphism onto its image. The first is a condition on a matrix and is easy to check; the second is a global condition and is the one that fails in practice, as the figure-eight and the lollipop show. The following says that, locally, the second condition comes for free.

Claim. Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}^{n+1}$ be smooth with $D_{x_0} f$ injective for every $x_0 \in U$ (such an f is called an "*immersion*" (German: „*Immersion*")). Then every $x_0 \in U$ has an open neighbourhood $V \subseteq U$ such that $f(V)$ is an n-dimensional submanifold of $\mathbb{R}^{n+1}$.

Fix $x_0 \in U$. The matrix $\mathrm{J}\,f(x_0)$ has $n+1$ rows and n columns and is injective, so it has rank n: some n of its rows are linearly independent. Permuting the coordinates of $\mathbb{R}^{n+1}$ is a linear

diffeomorphism of the ambient space, and it carries submanifolds to submanifolds, so we may assume those are the first n rows. Write accordingly

$$f = \begin{pmatrix} g \\ h \end{pmatrix}, \qquad g : U \to \mathbb{R}^n, \quad h : U \to \mathbb{R},$$

so that $\mathrm{J}\, g(x_0) \in \mathbb{R}^{n\times n}$ is invertible.

By the inverse function theorem (Theorem 15.a.1) there is an open $V \ni x_0$, $V \subseteq U$, such that $g|_V : V \to W := g(V)$ is a diffeomorphism onto an open set $W \subseteq \mathbb{R}^n$. Let $\varphi := (g|_V)^{-1} : W \to V$ and set

$$\psi := h \circ \varphi : W \to \mathbb{R},$$

which is smooth as a composition of smooth maps. For $x \in V$ we have $g(x) \in W$ and

$$f(x) = \big(g(x),\, h(x)\big) = \big(g(x),\, \psi(g(x))\big),$$

using $\varphi(g(x)) = x$. As x runs over V, the point $g(x)$ runs over all of W. Therefore

$$f(V) = \{(w, \psi(w)) \mid w \in W\}$$

is exactly the graph of the smooth function ψ over the open set W, and a graph is an n-dimensional submanifold of $\mathbb{R}^{n+1}$. To see that last point, apply the regular value theorem to $(w, s) \mapsto s - \psi(w)$, whose Jacobian $(-\nabla\psi(w)^{\mathsf{T}},\ 1)$ never vanishes.

Remark 17.a.17 (*What is local here, and what is not*): The injectivity of $D_{x_0} f$ is a pointwise, linear-algebraic condition, and yet it forces f itself to be injective near x_0: on the shrunken V the map f is injective because its first block g already is. That is the whole trick, and it is why an immersion cannot fail to be a parametrization for a local reason.

What it cannot repair is a global collision. The figure-eight curve is an immersion of an open interval whose image meets itself at the crossing point, and no shrinking of the domain changes that the **image** is not a submanifold there. The claim promises a neighbourhood V in the **domain** and says nothing about a neighbourhood of $f(x_0)$ in $\mathbb{R}^{n+1}$; the two are different requests, and only the first is granted. Compare AI-Example 17.b.22, where the failure is of exactly this global kind.

> **Info:** *This example is taken from the Analysis I & II examination of winter 2018/2019 (Prof. Manfred Einsiedler), Problem 7, where it is set as a proof exercise. The framing against Theorem 17.a.9 and the closing remark are added here.*

Exercise 17.a.18 (*The orthogonal group as a submanifold*): Let $\mathrm{Mat}_2(\mathbb{R})$ denote the 2×2 real matrices, identified with $\mathbb{R}^4$ by listing the entries a, b, c, d of $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ as a column. Define

$$F : \mathrm{Mat}_2(\mathbb{R}) \to \mathrm{Mat}_2(\mathbb{R}), \qquad F(A) := AA^{\mathsf{T}},$$

and let $\mathrm{O}_2(\mathbb{R}) := \{A \in \mathrm{Mat}_2(\mathbb{R}) \mid F(A) = \mathrm{id}\}$.

- **(a)** Write F out in the coordinates a, b, c, d and explain why it is smooth.
- **(b)** Show that $F(A)$ is symmetric for every A, so that F takes values in the three-dimensional subspace $\mathrm{Sym}_2(\mathbb{R}) \subseteq \mathrm{Mat}_2(\mathbb{R})$ of symmetric matrices.
- **(c)** Compute the Jacobian matrix of F at $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. It is a 4×4 matrix.
- **(d)** Show that this Jacobian has rank 3 at every $A \in \mathrm{O}_2(\mathbb{R})$. Why does this already rule out applying the regular value theorem to F as it stands?
- **(e)** Regard F instead as a map $\widetilde{F} : \mathbb{R}^4 \to \mathbb{R}^3$ into $\mathrm{Sym}_2(\mathbb{R}) \cong \mathbb{R}^3$ and deduce from Theorem 17.a.3 that $\mathrm{O}_2(\mathbb{R})$ is a submanifold of $\mathrm{Mat}_2(\mathbb{R})$. What is its dimension?
- **(f)** Determine the tangent space $T_{\mathrm{id}}\, \mathrm{O}_2(\mathbb{R})$ from the Jacobian computed in **(c)**, and give a basis for it.

> **Info:** *This exercise is taken from the Analysis I & II examination of 22 January 2020 (Prof. Peter S. Jossen), Part B, Problem 4. The original reaches the submanifold property*

*through the constant rank theorem, which it also asks the candidate to state; parts **(d)** and **(e)** are rephrased here to use the regular value theorem instead.*

Solution: *See solution on Page 250*

AI-Note 17.1: The re-routing in **(e)** is not a simplification of Jossen's problem, it is the reason the problem is interesting. The constant rank theorem states that a C^k map whose differential has the **same** rank r throughout a neighbourhood looks, in suitable charts, like the linear projection $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_r, 0, \dots, 0)$; its level sets are then submanifolds of dimension $n - r$. This document does not state that theorem, and it does not need to here: restricting the codomain of F to the subspace $\mathrm{Sym}_2(\mathbb{R})$ that its image already lies in turns the constant rank 3 into a **full** rank 3, and the regular value theorem takes over. That manoeuvre, shrinking the target until the map becomes submersive, is the standard way to handle matrix groups without the heavier theorem, and it is worth knowing because $\mathrm{O}_n(\mathbb{R})$, $\mathrm{SL}_n(\mathbb{R})$ and their relatives all fall to it.

Three equivalent descriptions are on the table, each characterising the same class of sets. What they do not yet come with is practice, both in choosing which description to write down for a surface handed to you and, harder, in arguing that a set is **not** a submanifold at all. None of the three theorems helps with the second, since each of them only ever yields a positive.

Recognising a submanifold in practice (Section 17.b)

One surface, three descriptions (Subsection 17.b.1)

The three characterisations of a submanifold are easiest to keep apart when they are applied to the same object at the same point. Take the open upper half-sphere

$$\mathbb{S}^2_+ := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1,\ z > 0\},$$

a 2-dimensional submanifold of $\mathbb{R}^3$, and throughout fix the point

$$p := \left(0,\ \tfrac{1}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}}\right) \in \mathbb{S}^2_+.$$

Example 17.b.19 (*The upper half-sphere, three ways*):

(a) Implicit. Put $F(x, y, z) := x^2 + y^2 + z^2 - 1$ on the open set $U := \{z > 0\}$. Then $\mathbb{S}^2_+ = F^{-1}\{0\}$ and

$$\nabla F(x, y, z) = (2x,\ 2y,\ 2z), \qquad \nabla F(p) = (0,\ \sqrt{2},\ \sqrt{2}) \neq 0,$$

so $\mathrm{J}\,F$ has full rank $\ell = 1$ everywhere on U (indeed $2z > 0$ already forces the row to be nonzero). Theorem 17.a.3 gives a submanifold of dimension $3 - 1 = 2$.

(b) Graph. Over the open unit disc $V := \{(x, y) : x^2 + y^2 < 1\}$ put $f(x, y) := \sqrt{1 - x^2 - y^2}$. Then $\mathbb{S}^2_+ = \{(x, y, f(x, y)) : (x, y) \in V\}$ is the graph of f, and $f(0, 1/\sqrt{2}) = \sqrt{1 - 1/2} = 1/\sqrt{2}$ recovers the third coordinate of p. Here the half-sphere is globally a graph, which is exactly why the **upper half** is a friendlier example than the whole sphere.

(c) Parametrization. With $W := (0, \pi/2) \times (0, 2\pi)$ and

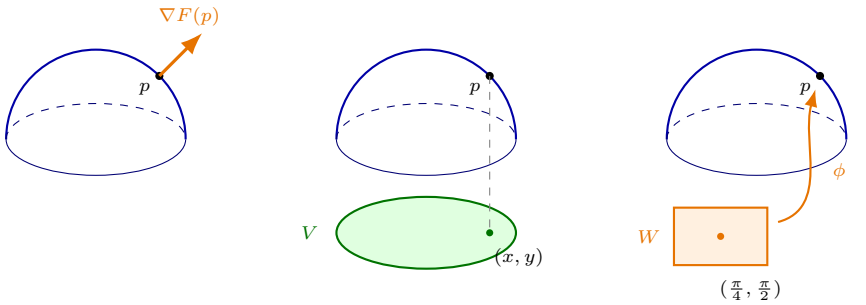
$$\phi(\theta, \psi) := (\sin\theta\cos\psi,\ \sin\theta\sin\psi,\ \cos\theta),$$

we get $p = \phi(\pi/4,\ \pi/2)$. The Jacobian

$$\mathrm{J}\,\phi(\theta, \psi) = \begin{pmatrix} \cos\theta\cos\psi & -\sin\theta\sin\psi \\ \cos\theta\sin\psi & \sin\theta\cos\psi \\ -\sin\theta & 0 \end{pmatrix}$$

is tall (3×2), and full rank $= 2$ means $D\phi$ is **injective** (Important remark 17.a.4): the bottom row forces the first column to be nonzero whenever $\sin\theta \neq 0$, and the second column

is then independent of it because $\sin\theta \neq 0$ makes $(-\sin\theta\sin\psi,\ \sin\theta\cos\psi)$ nonzero. On W we have $\theta \in (0, \pi/2)$, so $\sin\theta > 0$ throughout.



(a) implicit $\mathbb{S}^2_+ = F^{-1}\{0\}$ **(b)** graph $z = \sqrt{1 - x^2 - y^2}$ **(c)** parametrization $p = \phi(\frac{\pi}{4}, \frac{\pi}{2})$

Remark 17.b.20 (*Why all three theorems are stated **locally*):** The spherical coordinates of part **(c)** miss the north pole and the meridian arc where $\psi = 0$, but otherwise the half-sphere is unusually obliging: a single equation and a single graph each describe all of it, and the graph doubles as a global parametrization, $(x, y) \mapsto (x, y, f(x, y))$ on V. That is not the general situation, and each description fails to globalise for its own reason.

- **A parametrization need not be global.** The circle $\mathbb{S}^1$ is a 1-dimensional submanifold, yet no single parametrization covers it. Suppose $\gamma : I \to \mathbb{S}^1$ were a homeomorphism from an open interval, as Theorem 17.a.9 demands. Delete one point from each side: $I \setminus \{t\}$ falls into **two** pieces, while $\mathbb{S}^1 \setminus \{\gamma(t)\}$ is still an arc, so **one** piece. A homeomorphism preserves the number of connected components (Theorem 8.a.7), so no such γ exists. That is the counting argument of AI-Example 17.a.6, run in the opposite direction. Two overlapping arcs do cover the circle, and two is the minimum.
- **An implicit description need not be global either.** The Möbius strip is a 2-dimensional submanifold of $\mathbb{R}^3$, but it is not $g^{-1}\{0\}$ for any single g with $\nabla g \neq 0$: such a g would supply a globally consistent normal direction $\nabla g/|\nabla g|$, and the Möbius strip has none (Section 25.c).

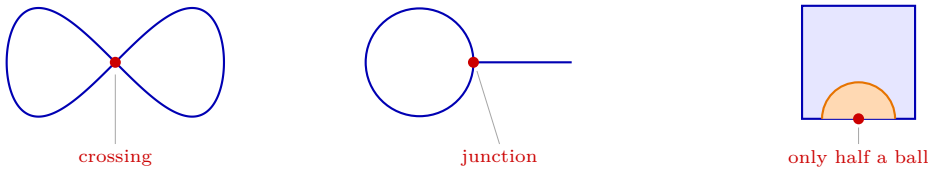
Requiring the descriptions only near each point costs nothing, since being a submanifold is itself a local condition, and it buys all the examples that matter.

A gallery of non-examples (Subsection 17.b.2)

Here is a list of sets that are **not** submanifolds, each failing for a different reason. Being a d-dimensional submanifold means looking, near every one of its points, like a flat piece of $\mathbb{R}^d$, and there are several distinct ways to violate that.

- **The figure eight** $\{(x, y) : y^2 = x^2 - x^4\}$ has a **crossing**. Away from the origin it is a perfectly good curve. At the origin four branches meet. This is the same failure as AI-Example 17.a.6, and the same connectedness count settles it.
- **The lollipop** is a circle with a segment attached. Here the trouble is not a crossing but a **branch point**: three arcs meet at the junction, where a curve has room for only two. Its three-dimensional cousin, a line sticking out of a sphere, fails for yet another reason, a **change of dimension**: near the attachment point that set is neither 1- nor 2-dimensional.
- **The closed square** $[0, 1]^2 \subseteq \mathbb{R}^2$ has a **boundary**. Its interior is a fine 2-dimensional submanifold, but no neighbourhood of a corner or edge point looks like an open piece of $\mathbb{R}^2$, because points of $[0, 1]^2$ near an edge fill only a half-plane. (Sets like this are the subject of Section 25.d; a submanifold **with boundary** is a genuinely weaker notion, not a submanifold that happens to have edges.)

- $\mathbb{Q}^n \subseteq \mathbb{R}^n$ has **no dimension at all**. It is not 0-dimensional, since a 0-dimensional submanifold is a set of isolated points and no rational point is isolated. And it is not n-dimensional, since it contains no open ball.



(a) figure eight **(b)** lollipop **(c)** closed square

Proving a negative: the tangent field test (Subsection 17.b.3)

AI-Example 17.a.6 disproved a submanifold claim by counting connected components after deleting a point. That argument is sharp but narrow: it needs the bad point to disconnect the set. A second, complementary criterion catches cases the counting argument misses.

Proposition 17.b.21 (*Continuous tangent field along a curve*): Let $M \subseteq \mathbb{R}^n$ be a 1-dimensional C^1 submanifold. Then every point of M has a neighbourhood on which there is a continuous map T assigning to each q a nonzero vector $T(q) \in T_qM$.

Proof:

Fix $p \in M$. By Theorem 17.a.9 there are an open interval $I \subseteq \mathbb{R}$, a neighbourhood V of p in $\mathbb{R}^n$, and a C^1 homeomorphism $\gamma : I \to V \cap M$ with $D\gamma_t$ injective for every t. Injectivity of a linear map $\mathbb{R} \to \mathbb{R}^n$ says exactly that $\gamma'(t) \neq 0$. Set

$$T(q) := \gamma'\big(\gamma^{-1}(q)\big), \qquad q \in V \cap M.$$

This is nonzero by the above, lies in T_qM by Proposition 17.c.33, and is continuous as a composition of the continuous maps γ^{-1} and γ'. Q.E.D.

The test is used in contrapositive form: **if no continuous nonvanishing tangent field exists near a point, the set is not a 1-dimensional submanifold there.**

AI-Example 17.b.22 (*The lollipop, disproved by tangent fields*): Let L be the lollipop of the gallery above and let q be its junction point. Deleting q leaves **three** pieces where a curve would leave two, so the counting argument of AI-Example 17.a.6 already applies. But it is instructive to see the tangent-field test reach the same verdict, since it localises the contradiction differently.

Suppose L were a 1-dimensional submanifold near q, and let T be the tangent field supplied by Proposition 17.b.21. Approach q along the stick: the tangent direction there is constant, so T tends to a vector along the stick. Approach q along the circle instead: the circle meets the stick at a right angle at q, so T tends to a vector perpendicular to the stick. A continuous map cannot have two different limits at the same point, so no such T exists.

Remark 17.b.23 (*When to reach for which*): Both tests disprove the same statement, and neither subsumes the other. Counting components is the tool when the offending point **separates** the set, and it needs no differentiability. The tangent-field test is the tool when the set stays connected but the direction of travel jumps. The corner $\{y = |x|\}$ at the origin is the standard case: removing the origin leaves the expected two pieces, so counting gives nothing away, whereas a tangent field would have to point along $(1, 1)$ as the origin is approached from the right and along $(1, -1)$ from the left, and no single nonzero vector is both.

Transporting a submanifold by a diffeomorphism (Subsection 17.b.4)

Each of the three descriptions above answers the question *"is this particular set a submanifold?"*. A different question arises as soon as a map is in play: given that M is a submanifold and Φ a diffeomorphism, what is $\Phi(M)$? Here the chart definition is the easy one to use, and it is about the only place where that is so.

Proposition 17.b.24 (*Diffeomorphisms carry submanifolds to submanifolds*): Let $U, W \subseteq \mathbb{R}^n$ be open, let $\Phi : U \to W$ be a C^k-diffeomorphism, and let $M \subseteq U$ be an m-dimensional C^k submanifold of $\mathbb{R}^n$. Then $\Phi(M)$ is an m-dimensional C^k submanifold of $\mathbb{R}^n$. In particular the dimension is unchanged.

Proof:

Fix $q \in \Phi(M)$ and write $q = \Phi(p)$ with $p \in M$. By Definition 17.a.1 there are an open neighbourhood $U_p \subseteq \mathbb{R}^n$ of p, an open $V \ni 0$, and a C^k-diffeomorphism $\Psi : U_p \to V$ with $\Psi(p) = 0$ and $\Psi(U_p \cap M) = (\mathbb{R}^m \times \{0\}) \cap V$. Replacing U_p by $U_p \cap U$, we may assume $U_p \subseteq U$.

Then $\Phi(U_p)$ is an open neighbourhood of q, and

$$\Psi \circ \Phi^{-1} : \Phi(U_p) \longrightarrow V$$

is a composition of C^k-diffeomorphisms, hence itself one, and it sends q to $\Psi(p) = 0$. Since Φ is injective, $\Phi(U_p) \cap \Phi(M) = \Phi(U_p \cap M)$, and therefore

$$(\Psi \circ \Phi^{-1})\big(\Phi(U_p) \cap \Phi(M)\big) = \Psi(U_p \cap M) = (\mathbb{R}^m \times \{0\}) \cap V.$$

That is a submanifold chart for $\Phi(M)$ at q, of the same dimension m. As q was arbitrary, $\Phi(M)$ is an m-dimensional C^k submanifold. **Q.E.D.**

Exercise 17.b.25 (*True or False* (which operations preserve submanifolds?)): Decide whether each of the following is true or false, with justification.

- **(a)** If M is a k-dimensional submanifold of $\mathbb{R}^n$ and $N \subseteq M$ is open in M, then N is itself a k-dimensional submanifold of $\mathbb{R}^n$.
- **(b)** The union of two disjoint k-dimensional submanifolds of $\mathbb{R}^n$ is a k-dimensional submanifold of $\mathbb{R}^n$.
- **(c)** The Cartesian product of a k-dimensional submanifold of $\mathbb{R}^m$ and an l-dimensional submanifold of $\mathbb{R}^n$ is a $(k+l)$-dimensional submanifold of $\mathbb{R}^{m+n}$.
- **(d)** If M and N are $(n-1)$-dimensional submanifolds of $\mathbb{R}^n$ for $n \geq 2$, and $T_pM \neq T_pN$ for every $p \in M \cap N$, then $M \cap N$ is an $(n-2)$-dimensional submanifold of $\mathbb{R}^n$.

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *(undated) in* `old_exams/fs2023/`*, Problem 7. That paper names no lecturer.*

Solution: *See solution on Page 252*

Remark 17.b.26 (*Why the dimension cannot move*): The proof transports one chart into another and never touches m, which is the whole content: a diffeomorphism can bend a submanifold arbitrarily, but it cannot change how many independent directions there are along it. This is worth stating because the tempting error runs the other way, imagining that a map into $\mathbb{R}^2$ whose image *"spreads out"* raises the dimension. It cannot. A curve stays a curve.

Exercise 17.b.27 (*True or False* (the image of a segment)): Continuing Exercise 15.a.12: let $\Phi \in C^1(\mathbb{R}^2, \mathbb{R}^2)$ satisfy $\Phi(0,0) = (0,0)^{\mathsf{T}}$ and $\mathrm{J}\,\Phi(0,0) = \left(\begin{smallmatrix} \alpha+5 & -3 \\ 3 & \alpha-5 \end{smallmatrix}\right)$, and take $\alpha = 0$. Decide whether each of the following is true or false, with justification.

- **(d)** For $r > 0$ small enough, the image of $\{(x,y) \in B_r(0,0) \mid y = 0\}$ under Φ is a C^1 submanifold of dimension 1.

(e) For $r > 0$ small enough, the image of $\{(x, y) \in B_r(0,0) \mid y = 0\}$ under Φ is a C^1 submanifold of dimension 2.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the fourth and fifth parts of the sixth block of true/false questions. Its first three parts are Exercise 15.a.12.*

Solution: *See solution on Page 250*

Exercise 17.b.28 (*True or False* (a level set in $\mathbb{R}^3$, and a curve through it)): Let $F \in C^1(\mathbb{R}^3, \mathbb{R}^2)$ satisfy

$$F(0,0,1) = \begin{pmatrix} 2 \\ -2 \end{pmatrix} \qquad \text{and} \qquad \mathrm{J}\,F(0,0,1) = \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \end{pmatrix},$$

and set $M := F^{-1}\big(\{(2,-2)^{\mathsf{T}}\}\big)$. For $t < 1$ let

$$\gamma(t) := \Big(\sin(2t),\ t + \tfrac{1}{2}t^2,\ \tfrac{\cos t}{1-t}\Big).$$

Decide whether each of the following is true or false, with justification.

(a) $(F \circ \gamma)'(0) = (0,0)^{\mathsf{T}}$.

(b) In a small neighbourhood of $(0,0,1)$, the set M is a smooth submanifold of dimension 2.

(c) In a small neighbourhood of $(0,0,1)$, the set M is a smooth submanifold of dimension 1.

(d) The vector $\gamma'(0)$ is normal to M at the point $(0,0,1)$.

(e) There exist $\delta > 0$ and $\varphi, \psi \in C^1((-\delta,\delta), \mathbb{R})$ such that $F(\varphi(y),\, y,\, \psi(y)) = (2,-2)^{\mathsf{T}}$ for all $y \in (-\delta, \delta)$.

(f) For all sufficiently small $\varepsilon > 0$, the set $F\big(B_\varepsilon((0,0,1))\big)$ is open in $\mathbb{R}^2$.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), where it appears as the ninth block of multiple-choice questions.*

Solution: *See solution on Page 253*

Exercise 17.b.29 (*Identifying 1-dimensional submanifolds in the plane*): Which of the following subsets of $\mathbb{R}^2$ are 1-dimensional C^1-submanifolds (without boundary) of $\mathbb{R}^2$? Decide each case with proof or counterexample.

(a) $M_1 := \big\{(x,y) \in \mathbb{R}^2 \;\big|\; x^4 + y^4 - 2(x^2 - y^2) = 0\big\} \setminus \{(0,0)\}$ (the lemniscate punctured at its self-intersection node).

(b) $M_2 := \big\{(\sin t,\ \sin 2t) \in \mathbb{R}^2 \;\big|\; t \in \mathbb{R}\big\}$ (the figure-eight curve).

(c) $M_3 := \big\{(e^{-t}\cos t,\ e^{-t}\sin t) \in \mathbb{R}^2 \;\big|\; t \in \mathbb{R}\big\}$ (the open logarithmic spiral).

(d) $M_4 := \big\{(x, \sin(1/x)) \in \mathbb{R}^2 \;\big|\; x \in (0,1)\big\} \cup (\{0\} \times [-1,1])$ (the topologist's sine curve with its limiting vertical segment).

Info: *This exercise is taken from the examination of 18 August 2023 (Prof. Urs Lang), Problem 2.MC3.*

Solution: *See solution on Page 255*

The tangent-field test leaned on knowing which vectors point **along** M at a given point, and so do the exercises above on transversal intersections and on normal vectors. The next section makes that space of directions an object in its own right.

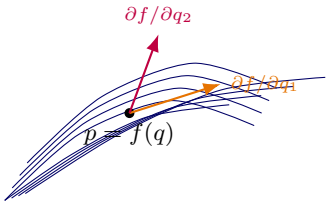
Tangent and normal spaces (Section 17.c)

Tangent space (Subsection 17.c.1)

Suppose $M^m \subseteq \mathbb{R}^n$ is a submanifold and $f : U \to f(U) \subseteq M$ is a local parametrization around some point $p \in f(U)$ with $f(q) = p$. The $n \times m$ matrix

$$\mathrm{J}\, f(q) = \begin{pmatrix} \dfrac{\partial f}{\partial q_1} & \cdots & \dfrac{\partial f}{\partial q_m} \end{pmatrix}$$

has full rank, that is, the column vectors $\partial_i f(q)$ are linearly independent. They form a basis of T_pM:



Definition 17.c.30 (*Tangent space*): For M, f as above, the "*tangent space*" (German: „*Tangentialraum*") of M at $p \in M$ is

$$T_pM := \operatorname{span}\{\partial_1 f(q), \ldots, \partial_m f(q)\} = Df_q(\mathbb{R}^m);$$

it is independent of the choice of f.

Example 17.c.31 (*Tangent space to the upper hemisphere*): Parametrize the upper hemisphere $S^2_+ := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1,\ z > 0\}$ graphically, i.e.,

$$f(x, y) = \begin{pmatrix} x \\ y \\ \sqrt{1 - x^2 - y^2} \end{pmatrix}.$$

We determine $T_{e_3} S^2_+$, where $e_3 = (0, 0, 1)$:

$$\left.\frac{\partial f}{\partial x}\right|_{(x,y)=(0,0)} = \left.\begin{pmatrix} 1 \\ 0 \\ \dfrac{-x}{\sqrt{1 - x^2 - y^2}} \end{pmatrix}\right|_{(x,y)=(0,0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = e_1,$$

$$\left.\frac{\partial f}{\partial y}\right|_{(x,y)=(0,0)} = \left.\begin{pmatrix} 0 \\ 1 \\ \dfrac{-y}{\sqrt{1 - x^2 - y^2}} \end{pmatrix}\right|_{(x,y)=(0,0)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = e_2.$$

So, $T_{e_3} S^2_+ = T_{e_3} S^2 = \operatorname{span}\{e_1, e_2\} = \mathbb{R}^2 \times \{0\}$.

Example 17.c.32 (*Tangent space to the 2-Torus*): Consider the 2-torus $\mathbb{T}^2 \subset \mathbb{R}^3$ parametrized by $\phi(t, \theta) : \mathbb{R}^2 \to \mathbb{R}^3$,

$$\phi(t, \theta) := \begin{pmatrix} (R_1 + R_2 \cos\theta) \cos t \\ (R_1 + R_2 \cos\theta) \sin t \\ R_2 \sin\theta \end{pmatrix},$$

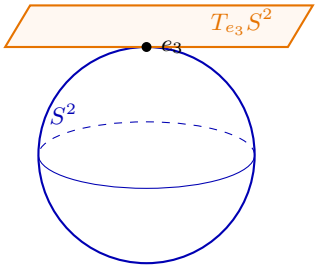
where $R_1 > R_2 > 0$ are the major and minor radii. The partial derivatives of ϕ with respect to the angles t and θ yield two linearly independent column vectors:

$$\frac{\partial \phi}{\partial t} = \begin{pmatrix} -(R_1 + R_2 \cos\theta) \sin t \\ (R_1 + R_2 \cos\theta) \cos t \\ 0 \end{pmatrix}, \qquad \frac{\partial \phi}{\partial \theta} = \begin{pmatrix} -R_2 \sin\theta \cos t \\ -R_2 \sin\theta \sin t \\ R_2 \cos\theta \end{pmatrix}.$$

By Definition 17.c.30, at any point $p = \phi(t,\theta) \in \mathbb{T}^2$, the tangent space is the two-dimensional vector subspace

$$T_p\mathbb{T}^2 = \operatorname{span}\left\{ \frac{\partial\phi}{\partial t}(t,\theta),\ \frac{\partial\phi}{\partial\theta}(t,\theta) \right\}.$$

In particular, at the outer point $p_0 := (R_1 + R_2, 0, 0)^\mathsf{T}$ (where $t = 0, \theta = 0$), we obtain $\frac{\partial\phi}{\partial t}(0,0) = (0, R_1 + R_2, 0)^\mathsf{T}$ and $\frac{\partial\phi}{\partial\theta}(0,0) = (0,0,R_2)^\mathsf{T}$, so $T_{p_0}\mathbb{T}^2 = \{0\} \times \mathbb{R}^2$.



Definition 17.c.30 builds T_pM out of a parametrization, so the space is pinned down by a formula rather than by what its elements are. There is an equivalent description that says what they are. A vector is tangent to M at p exactly when some curve running inside M passes through p with that velocity, which is both how tangent vectors are usually pictured and the quicker test in practice, since no parametrization need be written down.

Proposition 17.c.33 (*Tangent space via curves*): Let $M^m \subseteq \mathbb{R}^n$ be a C^k submanifold ($k \geq 1$) and $p \in M$. Then

$$T_pM = \big\{\gamma'(0) : \gamma \in C^1\big((-\varepsilon,\varepsilon), \mathbb{R}^n\big),\ \varepsilon > 0,\ \gamma(t) \in M \text{ for all } t,\ \gamma(0) = p\big\}.$$

Proof:

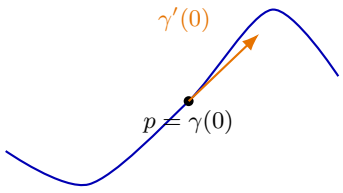
Let $\Phi : U \to V$ be a submanifold chart at p (Definition 17.a.1), so $\Phi(U \cap M) = (\mathbb{R}^m \times \{0\}) \cap V$ and $\Phi(p) = 0$; equivalently, let $f := \Phi^{-1}|_{(\mathbb{R}^m \times \{0\}) \cap V}$ be the corresponding local parametrization (Theorem 17.a.9), $f(q) = p$.

($\subseteq$) Let $v \in T_pM = Df_q(\mathbb{R}^m)$, say $v = Df_q(w)$. The curve $\gamma(t) := f(q + tw)$, defined for t small enough that $q + tw$ stays in f's domain, satisfies $\gamma(t) \in M$ for all such t, $\gamma(0) = p$, and by the chain rule $\gamma'(0) = Df_q(w) = v$.

($\supseteq$) Let γ be a curve as in the statement. Since Φ is a diffeomorphism with $\Phi(\gamma(t)) \in \mathbb{R}^m \times \{0\}$ for all t near 0, write $\Phi \circ \gamma =: (\eta, 0)$ with $\eta \in C^1\big((-\varepsilon,\varepsilon), \mathbb{R}^m\big)$. Then $\gamma = \Phi^{-1}(\eta, 0) = f(\eta(\cdot))$ near 0 (shrinking U if necessary so that f is defined on all of η's image), so by the chain rule,

$$\gamma'(0) = Df_{\eta(0)}\big(\eta'(0)\big) \in Df_q(\mathbb{R}^m) = T_pM$$

(using $\eta(0) = q$, since $\Phi(p) = 0 = \Phi(f(q))$ and Φ is injective). **Q.E.D.**



Remark 17.c.34 (*Four representations of the tangent space*): Depending on how an m-dimensional submanifold $M \subseteq \mathbb{R}^n$ is locally described near p, its tangent space T_pM can be computed algebraically in four equivalent ways:

- **Diffeomorphism** (from the definition, $\Psi(p) = 0$): $T_pM = (D\Psi_p)^{-1}(\mathbb{R}^m \times \{0\})$.
- **Implicit representation** ($M = F^{-1}\{0\}$ with DF_p surjective): $T_pM = \ker(DF_p)$.
- **Parametric representation** (local parametrization g with $g(q) = p$): $T_pM = Dg_q(\mathbb{R}^m)$.
- **Graph representation** (graph of h, $p = (z, h(z))$): $T_pM = \{(w, Dh_z(w)) \mid w \in \mathbb{R}^m\}$.

These algebraic descriptions are all equivalent to the geometric description via velocities of curves (Proposition 17.c.33).

Normal space (Subsection 17.c.2)

Suppose $F : U \to \mathbb{R}^{n-m}$ is an implicit representation of $M^m = F^{-1}\{0\}$ (via the regular value theorem) and $f : V^m \to M$ is a local parametrization, M a submanifold. Then, by definition,

(a) The Jacobian matrix:

$$\mathrm{J}\,F(x) = \begin{pmatrix} \nabla F_1(x) \longrightarrow \\ \vdots \\ \nabla F_{n-m}(x) \longrightarrow \end{pmatrix}$$

has full rank, i.e., $\{\nabla F_i(x)\}_{i=1,\dots,n-m}$ are linearly independent;

(b) $T_{f(q)}M = \operatorname{span}\{\partial_i f(q)\}$, as above;

(c) $F(f(x)) = 0$ for all $x \in V$, since $f(x) \in M = F^{-1}\{0\}$.

From **(c)**, it follows that

$$\begin{aligned} \mathrm{J}(F \circ f)(q) &= \mathrm{J}\,F(f(q)) \cdot \mathrm{J}\,f(q) \\ &= \begin{pmatrix} \nabla F_1(f(q)) \longrightarrow \\ \vdots \\ \nabla F_{n-m}(f(q)) \longrightarrow \end{pmatrix} \begin{pmatrix} \dfrac{\partial f}{\partial q_1} & \cdots & \dfrac{\partial f}{\partial q_m} \end{pmatrix} = 0. \end{aligned}$$

Now, since $\big(\mathrm{J}\,F(f(q)) \cdot \mathrm{J}\,f(q)\big)_{ij} = \langle \nabla F_i(f(q)), \partial_j f(q)\rangle = 0$ for all $i = 1, \dots, n-m$, $j = 1, \dots, m$, the vectors $\nabla F_i(f(q))$ are perpendicular, that is, "*normal*", to $T_{f(q)}M$. Since by **(a)** we have $n-m$ linearly independent vectors normal to the m-dimensional plane $T_{f(q)}M$, we can define:

Definition 17.c.35 (*Normal space*): Let $M^m \subseteq \mathbb{R}^n$ be a submanifold and $F : U^n \to \mathbb{R}^{n-m}$ an implicit representation, $M = F^{-1}\{0\}$. Then the "*normal space*" (German: „*Normalraum*") of M at $p \in M$ is

$$N_pM := \operatorname{span}\{\nabla F_1(p), \dots, \nabla F_{n-m}(p)\}.$$

By the computation above, these $n-m$ independent vectors are orthogonal to the m-dimensional T_pM, so by counting dimensions $N_pM = (T_pM)^{\perp}$. In particular N_pM does not depend on the choice of F.

Example 17.c.36 (*Normal space to the sphere*): Represent S^2 implicitly: $S^2 = F^{-1}\{1\} = \{(x, y, z) \in \mathbb{R}^3 : F(x, y, z) = 1\}$ where $F(x, y, z) = x^2 + y^2 + z^2$. We determine $N_{e_3}S^2$:

$$\nabla F(0, 0, 1) = (2x, 2y, 2z)\big|_{(x,y,z)=(0,0,1)} = 2e_3.$$

So, $N_{e_3}S^2 = \operatorname{span}\{e_3\} = \{(0, 0)\} \times \mathbb{R}$.

Proposition 17.c.37 (*The gradient is normal to the level set*): Let $U \subseteq \mathbb{R}^n$ be open, $g \in C^1(U, \mathbb{R})$, and let $S := \{x \in U : g(x) = c\}$ for some $c \in \mathbb{R}$. If $\nabla g(x_0) \neq 0$ at a point $x_0 \in S$, then near x_0 the set S is an $(n-1)$-dimensional submanifold and

$$N_{x_0}S = \operatorname{span}\{\nabla g(x_0)\}.$$

In particular the normal space is a **line**, and the affine **tangent** hyperplane to S at x_0 is $\{x \in \mathbb{R}^n : \langle x - x_0,\ \nabla g(x_0)\rangle = 0\}$.

Proof:

That S is a submanifold of dimension $n-1$ near x_0 is Theorem 17.a.3, since $\nabla g(x_0) \neq 0$ says exactly that $Dg_{x_0} : \mathbb{R}^n \to \mathbb{R}$ is surjective. For the normal space, let $v \in T_{x_0}S$ and pick a C^1 curve $\gamma : (-\varepsilon, \varepsilon) \to S$ with $\gamma(0) = x_0$ and $\gamma'(0) = v$. Then $g \circ \gamma \equiv c$, so differentiating at $t = 0$ with the chain rule along a path (Section 10.d) gives

$$0 = \frac{d}{dt}\Big|_{t=0} g(\gamma(t)) = \langle \nabla g(x_0),\ v \rangle.$$

Hence $\nabla g(x_0) \perp T_{x_0}S$, that is, $\operatorname{span}\{\nabla g(x_0)\} \subseteq N_{x_0}S$. Both are subspaces of dimension $n - \dim T_{x_0}S = n - (n-1) = 1$, so they are equal. Q.E.D.

Example 17.c.38 (*Tangent and normal vectors via parametrization*): For a parametrized surface $f : V \to \mathbb{R}^3$, the tangent space at $p = f(x)$ is $T_pM = \operatorname{span}\{\partial_1 f(x), \partial_2 f(x)\}$. A normal vector can be found via the cross product: $\vec{n} = \partial_1 f(x) \times \partial_2 f(x)$.

Let us apply this to the sphere S^2 using spherical coordinates:

$$\Phi(\theta, \varphi) = \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}, \qquad (\theta, \varphi) \in (0, \pi) \times (-\pi, \pi).$$

(Note that the image of Φ is the unit sphere minus a *"thin slice"*, namely the meridian where $\varphi = \pm\pi$ together with the two poles where $\theta \in \{0, \pi\}$.) The tangent space $T_{\Phi(\theta,\varphi)}S^2$ is spanned by the partial derivatives:

$$\partial_\theta \Phi(\theta, \varphi) = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix}, \qquad \partial_\varphi \Phi(\theta, \varphi) = \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix}.$$

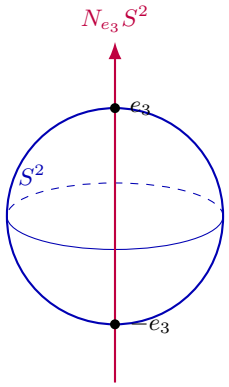
A normal vector is given by their cross product:

$$\begin{aligned} \vec{n} = \partial_\theta \Phi \times \partial_\varphi \Phi &= \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix} \times \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \sin^2\theta\cos\varphi \\ \sin^2\theta\sin\varphi \\ \sin\theta\cos\theta\cos^2\varphi + \sin\theta\cos\theta\sin^2\varphi \end{pmatrix} \\ &= \sin\theta \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}. \end{aligned}$$

Its length is $|\vec{n}| = \sin\theta$ (since $\sin\theta > 0$ for $\theta \in (0, \pi)$). The unit normal vector is therefore

$$\nu(\theta, \varphi) := \frac{\vec{n}}{|\vec{n}|} = \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix} = \Phi(\theta, \varphi).$$

This confirms that the outward normal vector at a point on the unit sphere is precisely the position vector of that point.



Exercise 17.c.39 (*Spherical coordinates* (important)): The mapping $\Phi : (0,\infty) \times (0,\pi) \times (-\pi,\pi) \to \mathbb{R}^3$ defined by

$$\Phi(r,\theta,\varphi) = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}$$

is called *spherical coordinates*.

(a) Sketch the images of $r \mapsto \Phi(r,\theta_0,\varphi_0)$, $\theta \mapsto \Phi(r_0,\theta,\varphi_0)$ and $\varphi \mapsto \Phi(r_0,\theta_0,\varphi)$ for some fixed $r_0 \in (0,\infty)$, $\theta_0 \in (0,\pi)$, $\varphi_0 \in (-\pi,\pi)$.

(b) What is the image of Φ?

(c) Show that $\det(D_{(r,\theta,\varphi)}\Phi) = r^2\sin\theta$ holds.

(d) Conclude that the mapping Φ is a diffeomorphism onto its image.

Info: *This exercise is Problem 7.6 of Problem Sheet 7, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 249*

Exercise 17.c.40 (*Quadric submanifold and tangent space basis*): Consider the set $M := \{(x_1,x_2,x_3) \in \mathbb{R}^3 \mid x_1x_2 + x_2x_3 + x_1x_3 = 1\}$.

(a) Show that M is a two-dimensional smooth submanifold of $\mathbb{R}^3$.

(b) Find an explicit basis of the tangent space T_pM at the point $p = (-1,2,3) \in M$.

Info: *This exercise is taken from the Analysis I & II exam of 9 August 2021 (Prof. Giovanni Felder), Part A, Problem 7.*

Solution: *See solution on Page 249*

Exercise 17.c.41 (*Box question: the normal to a graph in* $\mathbb{R}^4$): Let $\phi : \mathbb{R}^3 \to \mathbb{R}$ be differentiable. Give the formula for the unit normal vector $N \in \mathbb{R}^4$ to the graph of ϕ at a point $X = (x_1,x_2,x_3,\phi(x)) \in \mathbb{R}^4$.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), Box Problem 21, where only the final answer was graded.*

Solution: *See solution on Page 254*

Solutions (Section 17.d)

Not every solution in this chapter is collected here. Exercise 17.a.14 (Page 236) and Exercise 17.a.15 (Page 237) are solved where they appear.

Solution to Exercise 17.c.39:

(a) Fixing θ_0, φ_0, the image of $r \mapsto \Phi(r, \theta_0, \varphi_0)$ is the radial half-line through the point $p_0 := (\sin\theta_0 \cos\varphi_0, \sin\theta_0 \sin\varphi_0, \cos\theta_0)$ on the unit sphere, i.e., $\{rp_0 : r \in (0,\infty)\}$. Fixing r_0, φ_0, the image of $\theta \mapsto \Phi(r_0, \theta, \varphi_0)$ is a semicircle of radius r_0 from the north pole to the south pole (excluding the poles), at longitude φ_0. Fixing r_0, θ_0, the image of $\varphi \mapsto \Phi(r_0, \theta_0, \varphi)$ is a circle of latitude of radius $r_0 \sin\theta_0$ at height $r_0 \cos\theta_0$ (excluding the point at $\varphi = \pm\pi$).

(b) For each fixed r, the image sweeps out the sphere of radius r minus the half-plane at $\varphi = \pm\pi$ (the **anti**-meridian, since the prime meridian is $\varphi = 0$). Overall,

$$\operatorname{im}(\Phi) = \mathbb{R}^3 \setminus \big((-\infty, 0] \times \{0\} \times \mathbb{R}\big).$$

(c) Computing,

$$D_{(r,\theta,\varphi)}\Phi = \begin{pmatrix} \sin\theta\cos\varphi & r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ \sin\theta\sin\varphi & r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix},$$

hence, expanding along the last row,

$$\begin{aligned}\det(D_{(r,\theta,\varphi)}\Phi) &= r^2\sin\theta\cos^2\theta\cos^2\varphi + r^2\sin^3\theta\sin^2\varphi + r^2\sin\theta\cos^2\theta\sin^2\varphi + r^2\sin^3\theta\cos^2\varphi \\ &= r^2\big(\sin\theta\cos^2\theta + \sin^3\theta\big) = r^2\sin\theta.\end{aligned}$$

(d) Since $r \in (0,\infty)$ and $\theta \in (0,\pi)$, $r^2\sin\theta$ is never 0, so $\mathrm{J}\,\Phi(r,\theta,\varphi)$ is everywhere invertible and Φ is a local diffeomorphism by Theorem 15.a.1. To see it is a **global** diffeomorphism onto its image, it remains to check Φ is injective: if $\Phi(r_1,\theta_1,\varphi_1) = \Phi(r_2,\theta_2,\varphi_2)$, then $r_1 = r_2$ follows by taking the norm of both sides; since $\cos : (0,\pi) \to (-1,1)$ is injective, the third coordinate gives $\theta_1 = \theta_2$; finally $(\cos\varphi, \sin\varphi)$ with $\varphi \in (-\pi,\pi)$ determines a unique point on the unit circle, so $\varphi_1 = \varphi_2$. Hence Φ is injective, and (being a local diffeomorphism everywhere on its domain) a diffeomorphism onto its image.

Q.E.D.

Proof of Exercise 17.c.40:

(a) Define the smooth function $F : \mathbb{R}^3 \to \mathbb{R}$, $F(x_1,x_2,x_3) := x_1x_2 + x_2x_3 + x_1x_3 - 1$, so that $M = F^{-1}(0)$. Its gradient is

$$\nabla F(x) = (x_2 + x_3,\ x_1 + x_3,\ x_1 + x_2)^{\mathsf{T}}.$$

Suppose $\nabla F(x) = 0$. Adding the three equations $x_2 + x_3 = x_1 + x_3 = x_1 + x_2 = 0$ gives $2(x_1 + x_2 + x_3) = 0$, and subtracting each equation in turn then forces $x_1 = x_2 = x_3 = 0$. But $F(0,0,0) = -1 \neq 0$, so the origin is not in M. Hence ∇F is non-zero at every point of M, which is to say 0 is a regular value of F. By the regular value theorem (Theorem 17.a.3), M is a smooth submanifold of $\mathbb{R}^3$ of dimension $3 - 1 = 2$.

(b) First, $p = (-1,2,3)$ does lie on M: indeed $(-1)(2) + (2)(3) + (-1)(3) = -2 + 6 - 3 = 1$. At this point,

$$\nabla F(-1,2,3) = (2+3,\ -1+3,\ -1+2)^{\mathsf{T}} = (5,2,1)^{\mathsf{T}}.$$

The tangent space is the kernel of $DF(p)$, equivalently the orthogonal complement of $\nabla F(p)$:

$$T_pM = \big\{(v_1,v_2,v_3)^{\mathsf{T}} \in \mathbb{R}^3 \;\big|\; 5v_1 + 2v_2 + v_3 = 0\big\}.$$

Reading v_3 off from v_1 and v_2 gives two independent solutions: taking $(v_1,v_2) = (1,0)$ yields $v_3 = -5$, and taking $(v_1,v_2) = (0,1)$ yields $v_3 = -2$. Hence

$$\left\{\begin{pmatrix}1\\0\\-5\end{pmatrix}, \begin{pmatrix}0\\1\\-2\end{pmatrix}\right\}$$

is a basis of T_pM; the two vectors are independent because their first two components already are, and there are two of them, matching $\dim T_pM = 2$.

AI-Note 17.2: The official solution gives the basis $\{(2,-5,0)^\mathsf{T},\ (1,0,-5)^\mathsf{T}\}$, which looks different from the one above but is not a disagreement: both are bases of the same plane $\ker(5,2,1)$, and each vector of either list satisfies $5v_1+2v_2+v_3=0$. *"Give a basis"* has infinitely many correct answers, and no canonical one. So when checking work against a master solution here, verify that each vector is annihilated by $\nabla F(p)$ and that the two are independent, rather than comparing entries. The two lists even share the vector $(1,0,-5)^\mathsf{T}$. The official solution also invokes the constant rank theorem via surjectivity of D_xF where we cite the regular value theorem (Theorem 17.a.3); for a real-valued F these say the same thing, since a linear map $\mathbb{R}^3\to\mathbb{R}$ is surjective precisely when its gradient is non-zero.

Q.E.D.

Solution of Exercise 17.b.27:

For $\alpha=0$ the determinant computed in Exercise 15.a.12 is $-16\neq 0$, so fix $r>0$ small enough that Φ restricted to $B_r(0,0)$ is a C^1-diffeomorphism onto the open set $\Phi(B_r(0,0))$. Write

$$S:=\{(x,y)\in B_r(0,0)\mid y=0\}=(-r,r)\times\{0\}$$

for the segment in question. It is itself a 1-dimensional C^1 submanifold of $\mathbb{R}^2$: given $p=(a,0)\in S$, the translation $\Psi(x,y):=(x-a,\,y)$ is a diffeomorphism from $U_p:=B_r(0,0)$ onto the open set $V:=\Psi(U_p)$, it sends p to 0, and $\Psi(U_p\cap S)=(\mathbb{R}\times\{0\})\cap V$.

(d) True. Apply Proposition 17.b.24 to the C^1-diffeomorphism $\Phi|_{B_r(0,0)}:B_r(0,0)\to\Phi(B_r(0,0))$ and to S. The image $\Phi(S)$ is a C^1 submanifold of $\mathbb{R}^2$ of dimension 1.

(e) False. By **(d)** the set is 1-dimensional, and no non-empty set is a submanifold of two different dimensions at once. Independently of that: a 2-dimensional submanifold of $\mathbb{R}^2$ is an **open** subset of $\mathbb{R}^2$, because a chart with $m=n=2$ reads $\Psi(U\cap M)=V=\Psi(U)$, so $U\cap M=U$ and M contains the open neighbourhood U of each of its points. But $\Phi(S)$ is not open. Its complement in $\Phi(B_r(0,0))$ is the image of $B_r(0,0)\setminus S$, which is dense in $B_r(0,0)$, and a homeomorphism carries dense sets to dense sets, so every point of $\Phi(S)$ is a limit of points outside it.

Remark 17.d.42 (*The pair is a dimension check, not a submanifold check*): The two items ask about the same set, so at most one can be true, and the real question is which number a diffeomorphism is allowed to change. It changes none: Remark 17.b.26 is the whole content of the block. The picture that makes **(e)** tempting is that Φ maps a two-dimensional ball onto a two-dimensional region, so its image *"fills out"* the plane. It does, but the segment does not, and it is the segment being transported.

Q.E.D.

Solution to Exercise 17.a.18:

(a) With $A=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$,

$$AA^\mathsf{T}=\begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}a&c\\b&d\end{pmatrix}=\begin{pmatrix}a^2+b^2&ac+bd\\ac+bd&c^2+d^2\end{pmatrix},$$

so in the coordinates of the statement

$$F(a,b,c,d)=\big(a^2+b^2,\ ac+bd,\ ac+bd,\ c^2+d^2\big).$$

Every component is a polynomial, hence F is smooth.

(b) $(AA^\mathsf{T})^\mathsf{T}=(A^\mathsf{T})^\mathsf{T}A^\mathsf{T}=AA^\mathsf{T}$, so $F(A)$ is symmetric for every A. This is already visible in the coordinate formula: the second and third components of F are the same function.

(c) Differentiating the four components with respect to a, b, c, d in turn,

$$\mathrm{J}\,F(a,b,c,d) = \begin{pmatrix} 2a & 2b & 0 & 0 \\ c & d & a & b \\ c & d & a & b \\ 0 & 0 & 2c & 2d \end{pmatrix}.$$

(d) The second and third rows coincide for every A, so $\operatorname{rank}\mathrm{J}\,F(A) \le 3$ always. To see that equality holds on $\mathrm{O}_2(\mathbb{R})$, note that $AA^{\mathsf{T}} = \mathrm{id}$ says exactly that the two rows $u := (a,b)$ and $v := (c,d)$ of A are orthonormal. Suppose

$$\alpha(2a, 2b, 0, 0) + \beta(c, d, a, b) + \gamma(0, 0, 2c, 2d) = 0.$$

Its first two coordinates read $2\alpha u + \beta v = 0$, and u, v are linearly independent, so $\alpha = \beta = 0$. The last two then read $2\gamma v = 0$, so $\gamma = 0$ as well. The three distinct rows are therefore independent and the rank is exactly 3.

This is why Theorem 17.a.3 cannot be applied to F as a map $\mathbb{R}^4 \to \mathbb{R}^4$: the theorem needs $D_A F$ to be **surjective**, that is of rank 4, and the rank is 3 at every point of $\mathrm{O}_2(\mathbb{R})$. Nor is this a defect of the particular level: by **(b)** the image of F lies in a three-dimensional subspace, so $D_A F$ is nowhere surjective, for any A whatsoever.

(e) Identify $\mathrm{Sym}_2(\mathbb{R}) \cong \mathbb{R}^3$ by $\left(\begin{smallmatrix} p & q \\ q & s \end{smallmatrix}\right) \mapsto (p,q,s)$ and let

$$\widetilde{F} : \mathbb{R}^4 \to \mathbb{R}^3, \qquad \widetilde{F}(a,b,c,d) := \left(a^2 + b^2,\ ac + bd,\ c^2 + d^2\right),$$

which is F with the duplicated component dropped. Then $\mathrm{O}_2(\mathbb{R}) = \widetilde{F}^{-1}\{(1,0,1)\}$, and

$$\mathrm{J}\,\widetilde{F}(a,b,c,d) = \begin{pmatrix} 2a & 2b & 0 & 0 \\ c & d & a & b \\ 0 & 0 & 2c & 2d \end{pmatrix}$$

has rank 3 at every point of $\mathrm{O}_2(\mathbb{R})$, by the computation in **(d)**. So $(1,0,1)$ is a regular value of $\widetilde{F}$, and the regular value theorem makes $\mathrm{O}_2(\mathbb{R})$ a smooth submanifold of $\mathbb{R}^4 = \mathrm{Mat}_2(\mathbb{R})$ of dimension $4 - 3 = 1$.

(f) By Proposition 17.c.37 applied to each of the three defining equations, or directly from the description of the tangent space to a regular level set, $T_{\mathrm{id}}\,\mathrm{O}_2(\mathbb{R}) = \ker \mathrm{J}\,\widetilde{F}(\mathrm{id})$. At $(a,b,c,d) = (1,0,0,1)$,

$$\mathrm{J}\,\widetilde{F}(1,0,0,1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix},$$

so a vector (v_1, v_2, v_3, v_4) lies in the kernel precisely when $v_1 = 0$, $v_4 = 0$ and $v_2 + v_3 = 0$. Hence

$$T_{\mathrm{id}}\,\mathrm{O}_2(\mathbb{R}) = \operatorname{span}\left\{\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}\right\} = \operatorname{span}\left\{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right\},$$

a line, in agreement with $\dim \mathrm{O}_2(\mathbb{R}) = 1$ from **(e)**. Using the full 4×4 Jacobian of **(c)** gives the same kernel, since the repeated row imposes no new condition.

Remark 17.d.43 (*The tangent space names the answer*)**:** The line found in **(f)** consists of the **antisymmetric** 2×2 matrices, and that is not a coincidence: differentiating $A(t)A(t)^{\mathsf{T}} = \mathrm{id}$ along any curve in $\mathrm{O}_2(\mathbb{R})$ through id gives $\dot{A}(0) + \dot{A}(0)^{\mathsf{T}} = 0$. The same computation in n dimensions gives $\dim \mathrm{O}_n(\mathbb{R}) = \binom{n}{2}$, the dimension of the antisymmetric matrices, which for $n = 2$ is 1.

That $\mathrm{O}_2(\mathbb{R})$ is one-dimensional is worth pausing on, because it is easy to expect more. Writing $u = (\cos\theta, \sin\theta)$ for the first row, orthonormality leaves exactly two choices of second row, so

$\mathrm{O}_2(\mathbb{R})$ is a disjoint union of two circles: the rotations $\det A = 1$ and the reflections $\det A = -1$. A submanifold need not be connected, and the regular value theorem never claims it is.

Set against Exercise 17.a.15, which lives in the same $\mathbb{R}^4$, the contrast is instructive. There a single scalar equation cut out a three-dimensional submanifold and the regular value theorem applied as it stands. Here the naive count would suggest $4 - 4 = 0$ dimensions, and it is wrong precisely because the four equations $AA^{\mathsf{T}} = \mathrm{id}$ are not independent: one of them is a duplicate. Counting equations is only a shortcut for computing a rank, and when the shortcut and the rank disagree it is the rank that is right.

Q.E.D.

Solution to Exercise 17.b.25:

Throughout, *"submanifold"* is used in the sense of Definition 17.a.1: locally straightenable by a diffeomorphism of the ambient space.

(a) **True**, and there is nothing to prove beyond intersecting with one more open set. Being open in M means $N = M \cap W$ for some open $W \subseteq \mathbb{R}^n$. Fix $p \in N$ and take a submanifold chart $\Psi : U_p \to V$ for M at p. Replacing U_p by $U_p \cap W$, which is still an open neighbourhood of p, turns it into a submanifold chart for N at p of the same dimension. The definition is local, so shrinking the neighbourhood costs nothing.

(b) **False**, and this is the one that has to be met rather than reasoned about. In $\mathbb{R}^2$ take

$$M := \{(x, 0) \mid x > 0\}, \qquad N := \{(0, y) \mid y \in \mathbb{R}\}.$$

They are disjoint, since every point of M has $x > 0$ and every point of N has $x = 0$, and each is a 1-dimensional submanifold: M is an open subset of the x-axis, so part **(a)** applies, and N is a line.

But $M \cup N$ is not a submanifold at the origin. Every ball $B_\varepsilon(0,0)$ meets it in a segment of the y-axis together with a half-open segment running right, and deleting the origin from that set leaves **three** connected pieces, whereas deleting a point from a straightened $(\mathbb{R} \times \{0\}) \cap V$ leaves two. No diffeomorphism can change the number of pieces, so no submanifold chart exists at $(0,0)$.

What goes wrong is that disjointness is not enough: the two pieces are disjoint but one of them accumulates on the other. Requiring instead that M and N have disjoint **closures** makes the statement true, and then each point of the union has a neighbourhood meeting only one of the two.

(c) **True**, and it is the chart definition again. Fix $(p, q) \in M \times N$ and take submanifold charts $\Psi : U_p \to V$ for M and $\Xi : U_q \to W$ for N, with $\Psi(U_p \cap M) = (\mathbb{R}^k \times \{0\}) \cap V$ and $\Xi(U_q \cap N) = (\mathbb{R}^l \times \{0\}) \cap W$. Then $\Psi \times \Xi$ is a diffeomorphism of the open set $U_p \times U_q \subseteq \mathbb{R}^{m+n}$ onto $V \times W$, and it carries $(U_p \times U_q) \cap (M \times N)$ onto

$$\big((\mathbb{R}^k \times \{0\}) \cap V\big) \times \big((\mathbb{R}^l \times \{0\}) \cap W\big).$$

Composing with the permutation of coordinates that gathers the k and the l live directions together, which is linear and invertible and hence a diffeomorphism, turns this into $(\mathbb{R}^{k+l} \times \{0\}) \cap (V \times W)$. That is a submanifold chart of dimension $k + l$.

(d) **True**, and the hypothesis on the tangent spaces is exactly what is needed. Fix $p \in M \cap N$. Each of M and N is locally a level set, and Definition 17.a.1 supplies this directly: a submanifold chart $\Psi : U \to V$ for M at p satisfies $\Psi(U \cap M) = (\mathbb{R}^{n-1} \times \{0\}) \cap V$, so its last component $F := \Psi_n$ is C^1 with $M \cap U = F^{-1}\{0\}$, and $\nabla F(p) \neq 0$ because it is a row of the invertible matrix $\mathrm{J}\,\Psi(p)$. Do the same for N, shrink to a common neighbourhood, and write G for its function. So

$$M \cap U = F^{-1}\{0\}, \quad \nabla F(p) \neq 0, \qquad N \cap U = G^{-1}\{0\}, \quad \nabla G(p) \neq 0.$$

By Proposition 17.c.37, T_pM has $\nabla F(p)$ as its normal line, and likewise for N. Two hyperplanes in $\mathbb{R}^n$ coincide precisely when their normal lines do, so $T_pM \neq T_pN$ says exactly that $\nabla F(p)$ and $\nabla G(p)$ are linearly independent.

That makes $H := (F, G) : U \to \mathbb{R}^2$ a map whose Jacobian at p has the two independent rows $\nabla F(p)^\mathsf{T}$ and $\nabla G(p)^\mathsf{T}$, hence rank 2, so DH_p is surjective. Full rank is an open condition, so after shrinking U it holds throughout, and Theorem 17.a.3 makes

$$H^{-1}\{0\} = (M \cap U) \cap (N \cap U) = (M \cap N) \cap U$$

a submanifold of dimension $n-2$. As p was arbitrary, $M \cap N$ is one.

Remark 17.d.44 (*Transversality is the name of the hypothesis in (d)*): Two submanifolds are said to meet "*transversally*" at p if their tangent spaces together span the ambient space, $T_pM + T_pN = \mathbb{R}^n$. For two hyperplanes that is the same as being distinct, which is why part **(d)** can state the condition the way it does, and the general theorem says that a transversal intersection is a submanifold of dimension $\dim M + \dim N - n$. Here $(n-1) + (n-1) - n = n-2$, as found.

The hypothesis is doing real work, and dropping it breaks the conclusion immediately: two spheres in $\mathbb{R}^3$ meeting at a single point of tangency have the same tangent plane there, and their intersection is a point, of dimension 0 rather than 1. Two coincident planes are worse still, intersecting in a plane. Transversality is what rules both out, and the reason it is the right condition rather than a convenient one is visible in the proof: it is precisely the statement that stacking the two constraints gives a map of full rank.

Q.E.D.

Solution to Exercise 17.b.28:

Everything rests on two computations. First, $\gamma(0) = (0, 0, 1)$, and differentiating componentwise,

$$\gamma'(t) = \left(2\cos(2t),\ 1 + t,\ \frac{-\sin t\,(1-t) + \cos t}{(1-t)^2}\right), \qquad \text{so} \qquad \gamma'(0) = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

Second, $\mathrm{J}\,F(0,0,1)$ has rank 2: its leftmost 2×2 minor is $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, of determinant 1. Full rank is an open condition, since that minor is a continuous function of the point, so $\mathrm{J}\,F$ has rank 2 on a whole neighbourhood U of $(0,0,1)$, and Theorem 17.a.3 applied to $F|_U$ makes $M \cap U$ a submanifold of dimension $3 - 2 = 1$. Two independent equations cut two dimensions out of three and leave a curve; parts **(b)** and **(c)** follow at once.

(a) **False.** By the chain rule,

$$(F \circ \gamma)'(0) = \mathrm{J}\,F(\gamma(0))\,\gamma'(0) = \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 + 1 - 2 \\ 1 - 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This is worth reading as information rather than as a failed computation: had γ run inside M, then $F \circ \gamma$ would be constant and the derivative would vanish. So γ meets M at $t = 0$ and immediately leaves it.

(b) **False:** the dimension is 1, not 2.

(c) **True.**

(d) **False.** By Definition 17.c.35 the normal space is $N_{(0,0,1)}M = \mathrm{span}\{(1,1,-2)^\mathsf{T},\ (0,1,-1)^\mathsf{T}\}$, the span of the rows of $\mathrm{J}\,F(0,0,1)$. If $\gamma'(0) = a(1,1,-2)^\mathsf{T} + b(0,1,-1)^\mathsf{T}$, the first coordinate forces $a = 2$ and the second then forces $b = -1$; but the third coordinate of that combination is $-2 \cdot 2 - (-1) = -3$, not 1. Note that $\gamma'(0)$ is not tangent either: part **(a)** says exactly that it is not in $\ker \mathrm{J}\,F(0,0,1) = T_{(0,0,1)}M$. It is simply a vector in neither subspace, which is what a curve leaving M produces.

(e) True. To solve for x and z in terms of y, the block of $\mathrm{J}\,F(0,0,1)$ in the x- and z-columns must be invertible, and

$$\det\begin{pmatrix}1 & -2\\ 0 & -1\end{pmatrix} = -1 \neq 0.$$

Theorem 16.a.1 therefore supplies $\delta > 0$ and C^1 functions φ, ψ on $(-\delta, \delta)$ with $\varphi(0) = 0$, $\psi(0) = 1$ and $F(\varphi(y), y, \psi(y)) = (2, -2)^{\mathsf{T}}$ throughout. This exhibits the curve of part **(c)** concretely, as a graph over the y-axis.

(f) True. Complete F to a map of $\mathbb{R}^3$ to itself by appending the coordinate that part **(e)** solved **for**, namely y:

$$\Phi(x,y,z) := \big(F_1(x,y,z),\ F_2(x,y,z),\ y\big), \qquad \mathrm{J}\,\Phi(0,0,1) = \begin{pmatrix}1 & 1 & -2\\ 0 & 1 & -1\\ 0 & 1 & 0\end{pmatrix},$$

whose determinant is $1\cdot(1\cdot 0 - (-1)\cdot 1) = 1 \neq 0$. By Theorem 15.a.1, Φ is a C^1 diffeomorphism of some $B_\varepsilon((0,0,1))$ onto an open subset of $\mathbb{R}^3$. Writing $\pi(u,v,w) := (u,v)$ for the projection, we have $F = \pi \circ \Phi$, and π maps open sets to open sets, since $\pi(B_r(p)) = B_r(\pi(p))$. Hence $F(B_\varepsilon) = \pi(\Phi(B_\varepsilon))$ is open.

AI-Note 17.3: The examination calls M a *"smooth manifold"* in parts **(b)** and **(c)**, but it assumes only $F \in C^1$, so what the regular value theorem delivers is a C^1 submanifold. The distinction changes neither verdict, since both parts are about the dimension.

Q.E.D.

Solution to Exercise 17.c.41:

Write points of $\mathbb{R}^4$ as (x, x_4) with $x \in \mathbb{R}^3$, and present the graph as a level set: it is $G^{-1}(\{0\})$ for

$$G(x, x_4) := x_4 - \phi(x), \qquad \nabla G(x, x_4) = \begin{pmatrix}-\nabla\phi(x)\\ 1\end{pmatrix} \in \mathbb{R}^4.$$

This gradient never vanishes, its last entry being 1, so Proposition 17.c.37 applies: the graph is a 3-dimensional submanifold of $\mathbb{R}^4$ and its normal space at X is the line spanned by $\nabla G(X)$. Normalising,

$$N(X) = \frac{1}{\sqrt{1 + |\nabla\phi(x)|^2}}\begin{pmatrix}-\partial_1\phi(x)\\ -\partial_2\phi(x)\\ -\partial_3\phi(x)\\ 1\end{pmatrix},$$

determined up to sign; this is the choice pointing in the direction of increasing x_4.

The formula is worth recognising rather than deriving. For $\phi : \mathbb{R}^2 \to \mathbb{R}$ it is the familiar unit normal $(-\partial_1\phi, -\partial_2\phi, 1)/\sqrt{1 + |\nabla\phi|^2}$ of a surface in $\mathbb{R}^3$, and for $\phi : \mathbb{R} \to \mathbb{R}$ it is $(-\phi', 1)/\sqrt{1 + \phi'^2}$, perpendicular to the tangent $(1, \phi')$ as it should be. Nothing about the derivation used $n = 3$; presenting a graph as a level set of $x_{n+1} - \phi(x)$ works in every dimension, which is the point of setting the question one dimension past the picture.

AI-Note 17.4 (*A slip in the official key*): The key gives two displays for this answer. The second, $(-\nabla\phi(x), 1)^{\mathsf{T}}$ divided by $\sqrt{1 + |\nabla\phi(x)|^2}$, is the formula above. The first is not: it lists a fourth component $-\partial_4\phi(x)$ and normalises by $\sqrt{1 + \partial_1\phi(x)^2 + \partial_2\phi(x)^2 + \partial_3\phi(x)^2 + \partial_4\phi(x)^2}$. Since ϕ is a function of three variables there is no $\partial_4\phi$, and a vector with those four components plus the entry 1 would have five. Only the second display is correct.

Q.E.D.

Solution to Exercise 17.b.29:

(a) M_1 **is a 1-dimensional submanifold.** Consider the smooth function $F : \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}$ given by $F(x,y) := x^4+y^4-2x^2+2y^2$. The set M_1 is the zero level set $F^{-1}\{0\}$. Its gradient is

$$\nabla F(x,y) = \big(4x^3 - 4x,\ 4y^3 + 4y\big) = \big(4x(x^2-1),\ 4y(y^2+1)\big).$$

Critical points satisfy $\nabla F(x,y) = (0,0)$. The second component vanishes if and only if $y = 0$. Setting $y = 0$, the first component vanishes if $x \in \{0, 1, -1\}$.

- $(0,0)$ is excluded from the domain $\mathbb{R}^2 \setminus \{(0,0)\}$.
- At $(\pm 1, 0)$, we have $F(\pm 1, 0) = 1 - 2 = -1 \neq 0$, so these points do not lie in M_1.

Thus $\nabla F(x,y) \neq (0,0)$ everywhere on M_1. By the regular value theorem (Theorem 17.a.3), 0 is a regular value of F, so M_1 is an embedded 1-dimensional C^1 (indeed smooth) submanifold of $\mathbb{R}^2$.

(b) M_2 **is NOT a submanifold.** The curve passes through the origin $(0,0)$ at $t = 0$ (with velocity vector $(1,2)^\mathsf{T}$) and at $t = \pi$ (with velocity vector $(-1,2)^\mathsf{T}$). These two velocity vectors are linearly independent, so the two branches cross transversally at $(0,0)$. For any sufficiently small open ball $B_\varepsilon(0,0)$, the punctured intersection $(M_2 \cap B_\varepsilon(0,0)) \setminus \{(0,0)\}$ has four distinct connected components (the four incoming and outgoing branches; compare AI-Example 17.a.6 and Section 17.b). However, for any 1-dimensional manifold, every point has a connected neighbourhood homeomorphic to an open interval, from which deleting the point leaves at most two connected components. Thus M_2 cannot be a 1-submanifold in any neighbourhood of the origin.

(c) M_3 **IS a 1-dimensional submanifold.** Parametrize the curve by $\gamma(t) = (e^{-t}\cos t,\ e^{-t}\sin t)$ for $t \in \mathbb{R}$. The Euclidean distance to the origin is $r(t) = \|\gamma(t)\| = e^{-t}$, which is strictly monotonically decreasing on $\mathbb{R}$. In particular, $\gamma : \mathbb{R} \to M_3$ is a bijection, and each circle $\partial B_r(0)$ intersects M_3 in at most one point. Furthermore,

$$\gamma'(t) = e^{-t}(-\cos t - \sin t,\ -\sin t + \cos t)^\mathsf{T},$$

so $\|\gamma'(t)\|^2 = 2e^{-2t} > 0$ for all $t \in \mathbb{R}$, which means γ is an immersion. To show M_3 is an embedded submanifold of $\mathbb{R}^2$, fix any $p = \gamma(t_0) \in M_3$. Let $r_0 = e^{-t_0} > 0$, and consider the open annulus $A := \{(x,y) \in \mathbb{R}^2 : e^{-t_0-1} < \|(x,y)\| < e^{-t_0+1}\}$. By strict monotonicity of $r(t)$, the intersection $M_3 \cap A$ is precisely the image $\gamma((t_0-1, t_0+1))$, which is an open, non-self-intersecting arc of finite length. Since γ' is nowhere zero, and γ restricted to (t_0-1, t_0+1) has the continuous inverse $(x,y) \mapsto -\log\|(x,y)\|$, this arc is the image of a local parametrization in the sense of Theorem 17.a.9. (Note that while the origin $(0,0)$ lies in the closure $\overline{M_3}$, $(0,0) \notin M_3$, and every point of M_3 is strictly away from 0.) Hence M_3 is a 1-submanifold of $\mathbb{R}^2$.

(d) M_4 **is NOT a submanifold.** Consider any point $p = (0, y_0)$ on the limiting segment $\{0\} \times [-1,1]$. Any open ball $B_\varepsilon(p)$ in $\mathbb{R}^2$ contains the vertical segment $\{0\} \times (y_0-\varepsilon, y_0+\varepsilon)$ and also infinitely many disjoint oscillatory wave segments of the graph $y = \sin(1/x)$ for $x \in (0, \varepsilon)$. Because no path within $M_4 \cap B_\varepsilon(p)$ can connect the vertical segment to the oscillating waves without leaving a sufficiently small ball, the intersection $M_4 \cap B_\varepsilon(p)$ is not locally connected (cf. Example 8.b.26). In particular, no neighbourhood of p in M_4 is homeomorphic to an open interval. Thus M_4 is not a 1-dimensional submanifold.

Q.E.D.

Geodesics (Chapter 18)

With submanifolds and their tangent spaces in place, a natural question is what a straight line becomes once space is curved. This short chapter answers it. A geodesic is a curve of locally minimal length inside a submanifold, and characterising one takes the first genuine calculus-of-variations argument in the course. Differentiate the length functional along a perturbation, and require the derivative to vanish for every perturbation.

A short introduction to geodesics (Section 18.a)

AI-Note 18.1 (*This chapter goes beyond the course*): A warning before starting. The official lecture notes have no counterpart to this chapter at all: the word "*geodesic*" appears in them exactly once, in Example 9.14 on printed p. 5, where the great-circle distance $d_1(x,y) = R\arccos\left(x\cdot y/R^2\right)$ is offered as a more natural metric on the sphere than the Euclidean one, with no indication of why that arc is the shortest. Nothing in the course computes a geodesic, and the calculus of variations does not appear in it.

So this chapter is not lecture material and not exam material, and nothing in it can be assumed known. It is here because the calculation is short and because the idea behind it, treating length as a functional and applying optimization to it, is worth meeting once; and it is here in particular because it answers the question Example 9.14 raises and then drops.

We will try to find the shortest path (i.e., a geodesic) from $0 \in \mathbb{R}^n$ to $x \in \mathbb{R}^n$. This is, of course, the straight line, but the essentially same calculation can be used to find the shortest path between points on a sphere or any other submanifold.

Idea. Apply "*optimization*" to the map

$$L : \mathcal{A} \to [0,\infty), \qquad c \mapsto L(c) = \int_0^1 |c'(t)|\,dt, \qquad \mathcal{A} := \{\gamma \in C^1([0,1],\mathbb{R}^n) : \gamma(0) = 0,\ \gamma(1) = x\}.$$

Lemma 18.a.1 (*A few differentiation identities*): Let $\alpha, \beta : (0,1) \to \mathbb{R}^n$ be C^1. Then

$$\frac{d}{dt}\langle \alpha(t), \beta(t)\rangle = \langle \alpha'(t), \beta(t)\rangle + \langle \alpha(t), \beta'(t)\rangle,$$

and at every t with $\alpha(t) \neq 0$,

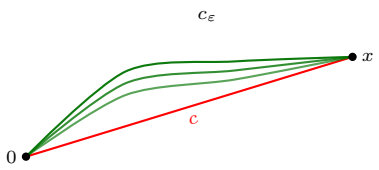
$$\frac{d}{dt}|\alpha(t)| = \frac{1}{|\alpha(t)|}\langle \alpha(t), \alpha'(t)\rangle.$$

The restriction is necessary: for $\alpha(t) = \left(t - \frac{1}{2}, 0, \ldots, 0\right)$ the function $|\alpha(t)| = |t - \frac{1}{2}|$ is not differentiable at $t = \frac{1}{2}$.

Proposition 18.a.2 (*The shortest path is a straight line*): Let $x \neq 0$, and assume that $c \in C^\infty([0,1],\mathbb{R}^n)$ with $c(0) = 0$, $c(1) = x$ and $|c'(t)| \equiv \lambda$ is the optimal path we are looking for. Then $c(t) = xt$ and $\lambda = |x|$.

Proof:

Let $g \in C^\infty([0,1],\mathbb{R}^n)$ with $g(0) = g(1) = 0$, and define, for $\varepsilon \in \mathbb{R}$, the family of paths $c_\varepsilon(t) := c(t) + \varepsilon g(t)$, so that $c_0 = c$. The boundary conditions on g are what keep the whole family admissible: $c_\varepsilon(0) = c(0) = 0$ and $c_\varepsilon(1) = c(1) = x$ for every ε, so each c_ε is a competitor in $\mathcal{A}$ and not merely a nearby curve.



By assumption, c is a minimizer of L, so it is a critical point of the family, i.e.

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(c_\varepsilon) = 0.$$

(This equality itself requires a proof, but the idea is exactly the same as that of optimization in $\mathbb{R}^n$.) First,

$$\left.\frac{\partial}{\partial\varepsilon}\right|_{\varepsilon=0} c_\varepsilon(t) = \left.\frac{\partial}{\partial\varepsilon}\right|_{\varepsilon=0} \big(c(t) + \varepsilon g(t)\big) = g(t).$$

Recalling $|c'| \equiv \lambda$ and using Lemma 18.a.1,

$$\begin{aligned}
\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(c_\varepsilon) &= \left.\frac{d}{d\varepsilon}\int_0^1 |c'_\varepsilon(t)|\,dt\right|_{\varepsilon=0} = \int_0^1 \left.\frac{\partial}{\partial\varepsilon}|c'_\varepsilon(t)|\,dt\right|_{\varepsilon=0} \\
&= \int_0^1 \left.\frac{\langle c'_\varepsilon(t),\ \partial_\varepsilon c'_\varepsilon(t)\rangle}{|c'_\varepsilon(t)|}\,dt\right|_{\varepsilon=0} = \frac{1}{\lambda}\int_0^1 \Big\langle c'(t),\ \partial_\varepsilon\big|_{\varepsilon=0} c'_\varepsilon(t)\Big\rangle\,dt \\
&= \frac{1}{\lambda}\int_0^1 \frac{d}{dt}\Big\langle c'(t), g(t)\Big\rangle - \langle c''(t), g(t)\rangle\,dt = \frac{1}{\lambda}\left[\langle c'(t), g(t)\rangle\Big|_0^1 - \int_0^1 \langle c''(t), g(t)\rangle\,dt\right] \\
&= -\frac{1}{\lambda}\int_0^1 \langle c''(t), g(t)\rangle\,dt \overset{!}{=} 0,
\end{aligned}$$

where the boundary term vanished since $g(0) = g(1) = 0$. Since g is arbitrary, it follows that $c''(t) \equiv 0$. This last step is the "*fundamental lemma of the calculus of variations*": a continuous function integrating to zero against **every** smooth g vanishing at the endpoints must itself vanish, since otherwise one could choose g to be a bump concentrated where c'' has a fixed sign and make the integral non-zero. Together with $c(0) = 0$ and $c(1) = x$ this gives $c(t) = xt$ and $\lambda = |x|$. **Q.E.D.**

Remark 18.a.3 (*The two steps that hide real work*): The proof moves quickly over $\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(c_\varepsilon) = 0$ and over the final $\overset{!}{=} 0$, and those are precisely the two places where something is being assumed. The first is the claim that a minimizer of L over the infinite-dimensional set $\mathcal{A}$ is a critical point of every one-parameter family through it. For a fixed g, the function $\varphi(\varepsilon) := L(c_\varepsilon)$ is an ordinary function of one real variable, and it has a minimum at $\varepsilon = 0$ because every c_ε lies in $\mathcal{A}$ (the differences $c_\varepsilon - c = \varepsilon g$ vanish at both endpoints). So $\varphi'(0) = 0$ by the one-variable case of Proposition 12.a.2, **provided** φ is differentiable. That proviso, together with the exchange of $d/d\varepsilon$ and $\int_0^1$ in the computation, is the real work. It holds because the integrand $|c'(t) + \varepsilon g'(t)|$ is C^1 in (t, ε) near $\varepsilon = 0$, the constant speed $\lambda > 0$ keeping it away from the kink of the norm at 0, and a C^1 integrand on a compact interval may be differentiated under the integral sign (Theorem 20.c.40).

The second is the fundamental lemma named in the proof, which upgrades "*the integral vanishes for every g*" to "*c'' vanishes at every t*", rather than merely on average.

Remark 18.a.4 (*Why constant speed is assumed, and why it costs nothing*): Proposition 18.a.2 assumes $|c'(t)| \equiv \lambda$, which looks like a restriction on the competitor curves and is not one. Length is invariant under reparametrization, since substituting $t = \varphi(s)$ for an increasing C^1 bijection φ leaves $\int |c'|\,dt$ unchanged. Every curve with nowhere-vanishing speed therefore has a constant-speed representative tracing the same image, obtained by running along it proportionally to arc length. The hypothesis chooses that representative.

It is worth choosing, because the functional L cannot see the difference and the equation can. Without it the computation still goes through, but $1/|c'(t)|$ stays inside the integral and the conclusion is that $\big(c'/|c'|\big)' \equiv 0$, which says the **direction** of travel is constant while the speed is free. Constant speed converts that into the cleaner $c'' \equiv 0$. AI-Exercise 18.a.6 carries this out.

AI-Exercise 18.a.5 (*The fundamental lemma of the calculus of variations*): The proof of Proposition 18.a.2 passes from *"the integral vanishes for every g"* to *"c'' vanishes at every t"* in one step. Prove that step. Precisely: let $h \in C^0\big([0,1],\mathbb{R}^n\big)$ satisfy

$$\int_0^1 \big\langle h(t),\, g(t)\big\rangle\, dt = 0 \qquad \text{for every } g \in C^\infty\big([0,1],\mathbb{R}^n\big) \text{ with } g(0)=g(1)=0.$$

Show that $h \equiv 0$.

Hint: *Argue by contradiction, and use a bump function of the kind constructed in Definition 25.b.8 to concentrate g where h is known not to vanish.*

Solution: *See solution on Page 260*

AI-Exercise 18.a.6 (*The variation without the constant-speed hypothesis*): Repeat the computation in the proof of Proposition 18.a.2 for a minimiser $c \in C^2\big([0,1],\mathbb{R}^n\big)$ that is **not** assumed to have constant speed, only $|c'(t)| > 0$ for every t.

- **(a)** Show that $\frac{d}{d\varepsilon}\big|_{\varepsilon=0} L(c_\varepsilon) = \int_0^1 \big\langle c'(t)/|c'(t)|,\ g'(t)\big\rangle\, dt$, and integrate by parts to bring it into the form $-\int_0^1 \langle \cdot,\, g(t)\rangle\, dt$.
- **(b)** Conclude, using AI-Exercise 18.a.5, that $\big(c'/|c'|\big)' \equiv 0$.
- **(c)** Deduce that the image of c is still the straight segment from 0 to x, and that $L(c) = |x|$. Which feature of c does the argument now fail to pin down, and why is that unavoidable?

Solution: *See solution on Page 261*

Exercise 18.a.7 (*Geodesics of the Poincaré half-plane*): Let $U \subseteq \mathbb{R}^2$ be open and convex and let $g : U \to (0,\infty)$ be smooth. For a smooth curve $\gamma : [a,b] \to U$, define its weighted length by

$$L_g(\gamma) := \int_a^b g(\gamma(t))\, |\dot\gamma(t)|\, dt.$$

We say γ is "*of minimal length*" if $L_g(\gamma) \le L_g(\sigma)$ for every smooth $\sigma : [c,d] \to U$ with $\sigma(c) = \gamma(a)$ and $\sigma(d) = \gamma(b)$.

- **(a)** Show that $L_g(\gamma \circ \psi) = L_g(\gamma)$ for every smooth diffeomorphism $\psi : [c,d] \to [a,b]$.
- **(b)** Let F be a smooth vector field on U with $|F| \equiv 1$ and $\operatorname{curl}(gF) = 0$. Show that if $\gamma : [a,b] \to U$ satisfies $\dot\gamma(t) = F(\gamma(t))$ for all t, then γ is of minimal length.

For the remaining parts, take $U := \{(x,y) \in \mathbb{R}^2 \mid y > 0\}$ and $g(x,y) := 1/y$.

- **(c)** For $F(x,y) := \frac{1}{x^2+y^2}\begin{pmatrix} 2xy \\ y^2-x^2 \end{pmatrix}$, verify that $|F| \equiv 1$ and $\operatorname{curl}(gF) = 0$.
- **(d)** Fix $a > 0$ and let C be the circle of radius a centred at $(a,0)$, so that $(0,0) \in C$. Let γ parametrise a piece of the upper half $C \cap U$, clockwise and by arclength. Deduce that γ is of minimal length.

Official hint: *For **(b)**, you may assume that the two curves meet only at their endpoints, that together they bound a domain, and that Stokes' theorem is valid on that domain.*

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Problem 3. The original closes with a bonus part: any two points of U lie on a unique circle centred on $\{y = 0\}$, and the arc joining them is the unique shortest curve between them.*

Solution: *See solution on Page 262*

Geodesics on a submanifold (Section 18.b)

What survives of *"shortest means straight"* when the curve is not free to leave a surface? On $\mathbb{R}^n$ the answer was $c'' \equiv 0$, and on a sphere that is hopeless: no curve with zero acceleration stays on a sphere at all. Something weaker has to replace it, and the computation itself says what.

The only place the ambient space entered the proof of Proposition 18.a.2 was in the choice of perturbations g. There, g ranged over **all** smooth fields vanishing at the endpoints, and that is exactly what forced $c'' \equiv 0$. On a submanifold the competitors must stay on M, so g may only push along M, and the conclusion weakens in step: c'' has to be orthogonal to the directions g is allowed to take, and to nothing else.

Proposition 18.b.8 (*The geodesic equation*): Let $M \subseteq \mathbb{R}^n$ be a C^2 submanifold and let $c \in C^2([0,1], \mathbb{R}^n)$ with $c(t) \in M$ for all t and constant speed $|c'(t)| \equiv \lambda > 0$. If c minimises length among C^1 curves in M with the same endpoints, then

$$c''(t) \perp T_{c(t)}M \qquad \text{for every } t \in [0,1],$$

that is, $c''(t) \in N_{c(t)}M$ in the sense of Definition 17.c.35.

Proof sketch:

Let $g \in C^1$ vanish at the endpoints and be "*tangential*" along c, meaning $g(t) \in T_{c(t)}M$ for every t. Take any family c_ε of C^1 curves **in** M with $c_0 = c$, with the endpoints held fixed, and with $\partial_\varepsilon\big|_{\varepsilon=0} c_\varepsilon = g$. Such a family exists: read M near $c(t)$ through a local parametrization (Theorem 17.a.9), push g back to the parameter domain, move there, and push forward again. This is the one step taken on trust here, and it is the only place the submanifold structure is used.

From there the computation of Proposition 18.a.2 runs verbatim, since it never asked where g came from (the one new ingredient is $\partial_\varepsilon c_\varepsilon' = (\partial_\varepsilon c_\varepsilon)'$ at $\varepsilon = 0$, which is Theorem 11.b.13 for a family that is C^2 in (t, ε)), and yields

$$0 = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} L(c_\varepsilon) = -\frac{1}{\lambda}\int_0^1 \big\langle c''(t),\, g(t)\big\rangle\, dt.$$

The fundamental lemma of the calculus of variations now applies to tangential g only, so it gives that the tangential **part** of $c''(t)$ vanishes for every t, rather than c'' itself. That is precisely $c''(t) \perp T_{c(t)}M$. Q.E.D.

Remark 18.b.9 (*Reading the equation*): The statement has a mechanical reading worth carrying: a geodesic is a curve that accelerates **only as much as the surface forces it to**. Any acceleration along M would be wasted motion that a shorter competitor avoids, so all that is left points out of M, and the surface supplies exactly that much to keep the curve on itself. Setting $M = \mathbb{R}^n$ recovers Proposition 18.a.2, since then $T_{c(t)}M = \mathbb{R}^n$ and orthogonality to everything means $c'' \equiv 0$.

The converse fails, and for the same reason it failed in Proposition 12.a.2: the equation is a first-order criticality condition, so it detects candidates. On the sphere the long way round a great circle satisfies it and is not shortest.

AI-Example 18.b.10 (*Great circles, and why a circle of latitude is not a geodesic*): Take $M := S^2 = \{x \in \mathbb{R}^3 : |x|^2 = 1\}$. By Proposition 17.c.37 applied to $g(x) := |x|^2$, the normal space at $p \in S^2$ is $N_pS^2 = \operatorname{span}\{p\}$, so Proposition 18.b.8 reads: c **is a geodesic candidate exactly when** $c''(t)$ **is a multiple of** $c(t)$**.**

A great circle satisfies it. Fix orthonormal $p, v \in \mathbb{R}^3$ and put

$$c(t) := \cos(\lambda t)\, p + \sin(\lambda t)\, v.$$

Then $|c(t)| = 1$, so c runs in S^2, and $|c'(t)| \equiv \lambda$, so the constant-speed hypothesis holds. Differentiating twice,

$$c''(t) = -\lambda^2\big(\cos(\lambda t)\,p + \sin(\lambda t)\,v\big) = -\lambda^2\,c(t),$$

which is a multiple of $c(t)$. The condition holds at every t.

A circle of latitude does not. Fix a height h with $0 < |h| < 1$, put $r := \sqrt{1-h^2}$, and run around the circle at that height at constant speed:

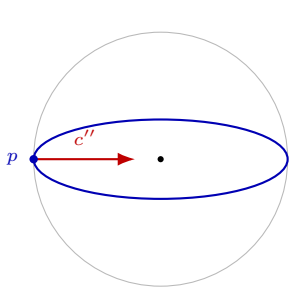
$$c(t) := \big(r\cos(\omega t),\ r\sin(\omega t),\ h\big), \qquad \omega \neq 0.$$

Again $|c(t)|^2 = r^2 + h^2 = 1$ and $|c'(t)| \equiv r|\omega|$ is constant. But

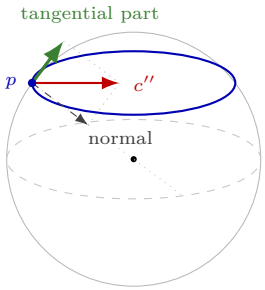
$$c''(t) = -\omega^2\big(r\cos(\omega t),\ r\sin(\omega t),\ 0\big),$$

whose third coordinate is 0 while the third coordinate of $c(t)$ is $h \neq 0$, and whose first two coordinates are not both zero. So, $c''(t)$ is not a multiple of $c(t)$, and no circle of latitude other than the equator ($h = 0$, where the two agree and the curve is a great circle) is a geodesic.

This is the flight-path fact in the form the equation gives it: flying due west along a parallel is not the shortest route between two points on it, which is why long-haul routes bend towards the pole.



great circle: c'' points at the centre, so it is purely normal



circle of latitude: a tangential part survives, so it is no geodesic

AI-Exercise 18.b.11 (*Geodesics on the cylinder*): Let $Z := \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$ be the unit cylinder, and for $a, b \in \mathbb{R}$ not both zero consider the helix

$$c(t) := \big(\cos(at),\ \sin(at),\ bt\big).$$

(a) Show that c runs in Z at constant speed, and compute that speed.

(b) Determine $N_{c(t)}Z$, and verify that c satisfies the geodesic equation of Proposition 18.b.8 for every choice of a and b.

(c) Which curves do the cases $a = 0$ and $b = 0$ describe?

Solution: *See solution on Page 262*

Solutions (Section 18.c)

Solution to AI-Exercise 18.a.5:

Suppose $h \not\equiv 0$, so that $h(t_0) \neq 0$ for some t_0. Since h is continuous, the set where $h \neq 0$ is open in $[0,1]$, so we may take $t_0 \in (0,1)$: if h were non-zero only at an endpoint it would still be non-zero just inside.

Choose a component i with $h_i(t_0) \neq 0$. Replacing h by $-h$ if necessary, which changes neither the hypothesis nor the conclusion, assume $h_i(t_0) =: 2m > 0$. By continuity there is a $\delta > 0$ with $(t_0 - \delta,\ t_0 + \delta) \subseteq (0,1)$ and

$$h_i(t) > m \qquad \text{for all } t \in (t_0 - \delta,\ t_0 + \delta).$$

Now take the bump function of Definition 25.b.8 on that interval,

$$\varphi(t) := \begin{cases} \exp\!\left(\dfrac{-1}{\delta^2 - (t-t_0)^2}\right), & |t-t_0| < \delta, \\ 0, & \text{otherwise}, \end{cases}$$

which is smooth on $\mathbb{R}$, non-negative, strictly positive on $(t_0-\delta, t_0+\delta)$ and zero outside it, and set $g := \varphi\, e_i$. Then g is smooth with $g(0) = g(1) = 0$, so it is an admissible test function, and

$$\int_0^1 \big\langle h(t),\, g(t)\big\rangle\, dt = \int_{t_0-\delta}^{t_0+\delta} h_i(t)\,\varphi(t)\, dt \;\geq\; m \int_{t_0-\delta}^{t_0+\delta} \varphi(t)\, dt \;>\; 0,$$

the last inequality because φ is continuous, non-negative and not identically zero on the interval. This contradicts the hypothesis. Consequently $h \equiv 0$.

Remark 18.c.12 (*Where each hypothesis is spent*): Two ingredients carry the proof, and it is worth seeing which does what. **Continuity** of h turns a single non-zero value into a whole interval of non-zero values of one sign, and without it the statement is false: a function equal to 1 at one point and 0 elsewhere integrates to zero against everything. **Smoothness** of the bump is what keeps g admissible, and it is the only reason the test class can be shrunk from all continuous g to all smooth g without weakening the conclusion. The conclusion itself is pointwise, which is the whole content: the hypothesis says h vanishes on average against every test function, and the lemma upgrades that to vanishing everywhere.

Q.E.D.

Solution to AI-Exercise 18.a.6:

(a) By the second identity of Lemma 18.a.1 applied to $\alpha := c'_\varepsilon$, and differentiating under the integral sign,

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} L(c_\varepsilon) = \int_0^1 \frac{\big\langle c'(t),\; \partial_\varepsilon\big|_{\varepsilon=0} c'_\varepsilon(t)\big\rangle}{|c'(t)|}\, dt = \int_0^1 \left\langle \frac{c'(t)}{|c'(t)|},\; g'(t)\right\rangle dt.$$

Nothing was pulled out of the integral, which is the one difference from the proof of Proposition 18.a.2. Since $c \in C^2$ and $|c'| > 0$, the unit tangent $c'/|c'|$ is C^1, so integration by parts is available:

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} L(c_\varepsilon) = \left[\left\langle \frac{c'}{|c'|},\, g\right\rangle\right]_0^1 - \int_0^1 \left\langle \left(\frac{c'}{|c'|}\right)'(t),\; g(t)\right\rangle dt = -\int_0^1 \left\langle \left(\frac{c'}{|c'|}\right)'(t),\; g(t)\right\rangle dt,$$

the boundary term vanishing because $g(0) = g(1) = 0$.

(b) The left-hand side is zero for every admissible g, and $\big(c'/|c'|\big)'$ is continuous, so AI-Exercise 18.a.5 applies with $h := \big(c'/|c'|\big)'$ and gives $\big(c'/|c'|\big)' \equiv 0$.

(c) Therefore $c'/|c'| \equiv u$ for a constant unit vector u, so $c'(t) = s(t)\,u$ with $s := |c'| > 0$. Integrating from 0 and using $c(0) = 0$,

$$c(t) = \sigma(t)\,u, \qquad \sigma(t) := \int_0^t s(\tau)\, d\tau,$$

with σ strictly increasing and $\sigma(0) = 0$. The endpoint condition $c(1) = x$ reads $\sigma(1)\,u = x$, and $\sigma(1) = L(c)$ by definition of the length, so $L(c) = |x|$ and $u = x/|x|$ (note $x \neq 0$, since $\sigma(1) > 0$). The image is $\{\sigma u : 0 \leq \sigma \leq |x|\}$, the straight segment from 0 to x.

What the argument no longer pins down is the **speed function** s, and with it the parametrization: any positive continuous s with $\int_0^1 s = |x|$ produces a minimiser. That is unavoidable rather than a defect of the proof, because L is invariant under reparametrization (Remark 18.a.4), so it cannot distinguish two curves with the same image, and a criterion derived from L alone cannot either. Assuming constant speed picks one representative out of that family, and $c'' \equiv 0$ is the statement for that representative.

Q.E.D.

Solution to AI-Exercise 18.b.11:

(a) The first two coordinates satisfy $\cos^2(at) + \sin^2(at) = 1$ for every t, so $c(t) \in Z$. Differentiating,

$$c'(t) = \big(-a\sin(at),\ a\cos(at),\ b\big), \qquad |c'(t)| = \sqrt{a^2\sin^2(at) + a^2\cos^2(at) + b^2} = \sqrt{a^2+b^2},$$

which is constant and non-zero, since a and b are not both zero.

(b) Write $Z = F^{-1}\{0\}$ for $F(x,y,z) := x^2 + y^2 - 1$. Then $\nabla F(x,y,z) = (2x,\ 2y,\ 0)$, which is non-zero on Z, so Theorem 17.a.3 applies and Definition 17.c.35 gives

$$N_pZ = \operatorname{span}\{(p_1,\ p_2,\ 0)\}, \qquad p \in Z.$$

The cylinder's normal at a point is horizontal and radial: it forgets the height entirely. Differentiating c a second time,

$$c''(t) = \big(-a^2\cos(at),\ -a^2\sin(at),\ 0\big) = -a^2\big(c_1(t),\ c_2(t),\ 0\big),$$

which is exactly a multiple of the spanning vector of $N_{c(t)}Z$. So $c''(t) \in N_{c(t)}Z$, i.e. $c''(t) \perp T_{c(t)}Z$, and the geodesic equation of Proposition 18.b.8 holds for every a and b.

Note that the z-component of c'' vanishes identically, which is what makes this work. The vertical direction is tangential to Z everywhere, so any vertical acceleration would have a tangential part and would violate the equation.

(c) With $a = 0$ the curve is $t \mapsto (1, 0, bt)$, a vertical straight line ruling the cylinder. With $b = 0$ it is $t \mapsto (\cos(at), \sin(at), 0)$, the horizontal unit circle. Both are limiting cases of the helix, and both are geodesics by **(b)**.

Remark 18.c.13 (*Why the cylinder has **only these geodesics*):** The three families found here are in fact all of them, and the quickest reason is that the cylinder is "*locally isometric*" to the plane: rolling a sheet of paper into a cylinder bends it without stretching, so it preserves the length of every curve drawn on it. Lengths are all the geodesic equation ever sees, so geodesics must correspond to geodesics. Unrolling sends helices to straight lines, vertical rulings to vertical lines, and horizontal circles to horizontal lines, which is every straight line in the plane and nothing else.

The sphere admits no such trick, which is the same obstruction that makes flat world maps distort areas or angles, and it is why AI-Example 18.b.10 had to be checked by computation instead.

Q.E.D.

Proof of Exercise 18.a.7:

(a) Let $\psi : [c,d] \to [a,b]$ be a diffeomorphism; assume first that it is orientation-preserving, so $\psi' > 0$. Then $(\gamma\circ\psi)'(s) = \dot\gamma(\psi(s))\psi'(s)$, and the substitution $t = \psi(s)$ gives

$$L_g(\gamma\circ\psi) = \int_c^d g(\gamma(\psi(s)))\,|\dot\gamma(\psi(s))|\,\psi'(s)\,ds = \int_a^b g(\gamma(t))\,|\dot\gamma(t)|\,dt = L_g(\gamma).$$

If instead $\psi' < 0$, then $|\psi'| = -\psi'$ and the substitution also swaps the limits of integration, so the two sign changes cancel and the value is unchanged.

(b) Let $\sigma : [c,d] \to U$ be any competitor with $\sigma(c) = \gamma(a)$ and $\sigma(d) = \gamma(b)$. By the hint we may assume γ and σ meet only at their endpoints and together bound a domain $D \subseteq U$, on which Stokes' theorem applies. Since $\operatorname{curl}(gF) = 0$ on U,

$$\int_{\partial D} gF\cdot\tau\,dL = \int_D \operatorname{curl}(gF)\,dx = 0,$$

and ∂D consists of γ and σ traversed in opposite senses. Therefore the two line integrals agree:

$$\int_{\gamma} gF \cdot \tau \, dL = \int_{\sigma} gF \cdot \tau \, dL.$$

Now compare each side with the corresponding length. Along γ we have $\dot\gamma = F(\gamma)$, which has $|\dot\gamma| = 1$, so $gF \cdot \dot\gamma = g|F|^2 = g = g|\dot\gamma|$, and the left-hand side is exactly $L_g(\gamma)$. Along σ, the Cauchy–Schwarz inequality and $|F| = 1$ give the one-sided bound $gF \cdot \dot\sigma \le g|F|\,|\dot\sigma| = g|\dot\sigma|$, so the right-hand side is at most $L_g(\sigma)$. Combining, $L_g(\gamma) \le L_g(\sigma)$.

(c) First the norm:

$$|F(x,y)|^2 = \frac{4x^2y^2 + (y^2-x^2)^2}{(x^2+y^2)^2} = \frac{(x^2+y^2)^2}{(x^2+y^2)^2} = 1.$$

Next, with $g = 1/y$,

$$gF = \frac{1}{y(x^2+y^2)} \begin{pmatrix} 2xy \\ y^2 - x^2 \end{pmatrix} = \begin{pmatrix} \dfrac{2x}{x^2+y^2} \\[2mm] \dfrac{y^2-x^2}{y(x^2+y^2)} \end{pmatrix},$$

so that $\operatorname{curl}(gF) = \partial_x (gF)_2 - \partial_y (gF)_1$ is computed from the two derivatives

- $\partial_x \left(\dfrac{y^2-x^2}{y(x^2+y^2)} \right) = \dfrac{-2xy(x^2+y^2) - (y^2-x^2)\,2xy}{y^2(x^2+y^2)^2} = \dfrac{-2xy\big[(x^2+y^2) + (y^2-x^2)\big]}{y^2(x^2+y^2)^2} = \dfrac{-4xy^3}{y^2(x^2+y^2)^2} = \dfrac{-4xy}{(x^2+y^2)^2}$,
- $\partial_y \left(\dfrac{2x}{x^2+y^2} \right) = \dfrac{-4xy}{(x^2+y^2)^2}$.

The two agree, and therefore $\operatorname{curl}(gF) = 0$ on U.

(d) Parametrise C by $x = a + a\cos\theta$, $y = a\sin\theta$, so that $\theta \in (0,\pi)$ traces out $C \cap U$. Then

$$x^2 + y^2 = 2a^2(1+\cos\theta), \qquad \begin{pmatrix} 2xy \\ y^2 - x^2 \end{pmatrix} = 2a^2(1+\cos\theta) \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix},$$

using $\sin^2\theta - (1+\cos\theta)^2 = -2\cos\theta\,(1+\cos\theta)$ for the second component. The common factor cancels, leaving

$$F(x,y) = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}.$$

Differentiating the parametrisation gives $\gamma'(\theta) = a(-\sin\theta, \cos\theta)^{\mathsf T} = -a\,F$, so F points along C in the direction of **decreasing** θ: from $(0,0)$ over the top of the circle towards $(2a,0)$, which is the clockwise sense. Hence a clockwise arclength parametrisation γ of a piece of $C \cap U$ satisfies $\dot\gamma = F(\gamma)$ exactly, and part **(b)** applies: γ is of minimal length.

Note that the radius is **forced**. Solving $\dot\gamma \parallel F(\gamma)$ for the circle through a given $(x,y) \in U$ centred at $(c,0)$ gives $c = (x^2+y^2)/(2x)$, and then $(x-c)^2 + y^2 = c^2$: the circle passes through the origin. So it is not true that every circle centred on $\{y=0\}$ is an integral curve of this particular F; only the ones through the origin are. Every such circle **is** a geodesic of the half-plane, but seeing that requires a different F for each one, which is what the bonus part of the original problem arranges.

Q.E.D.

PART VI

Integration

Integration in several variables raises a question that one variable never had to face. Before asking what $\int_A f$ is, one has to say which sets A possess a volume at all, and Jordan measure is the answer this course uses. Most of the work in the first chapter goes into the sets it fails on, and a short digression sketches the Lebesgue theory that repairs the failure. With the integral defined, the problem becomes computing one, and three tools do nearly all of it: Fubini reduces a multiple integral to iterated one-dimensional ones, the change of variables formula trades an awkward domain for a convenient one and pays $|\det \mathrm{J}\,\phi|$ for the trade, and differentiation under the integral sign reaches integrals possessing no elementary antiderivative at all. The final chapter leaves flat regions behind altogether. The Gram determinant measures how much a parametrization stretches length, area or volume, and with it one can integrate over a curve or a surface sitting inside $\mathbb{R}^n$ rather than over a piece of $\mathbb{R}^n$ itself.

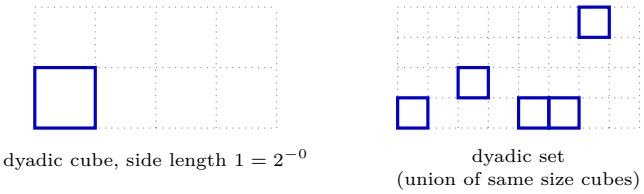
Jordan Measure and the Riemann Integral (Chapter 19)

Integration in several variables needs a notion of the volume of a domain before it can define an integral over one. This chapter sets up Jordan measure and the multidimensional Riemann integral built on it, and then spends most of its effort on the sets that are **not** Jordan measurable. That is where every counterexample on the problem sheet lives, and the reason they exist is a single one: a Jordan cover may use only finitely many cubes, all of one size. A short section extends the integral to unbounded domains, which every later chapter uses and none of them defines. The chapter closes with a digression on the Lebesgue theory, which repairs that single defect properly; it is stated without proofs and used nowhere afterwards.

Multivariate integration (Section 19.a)

Jordan measure and the Riemann integral (Subsection 19.a.1)

The rigorous definition of the multidimensional integral relies on the "*Jordan measure*" (German: „*Jordan-Mass*“). The intuition involves *"dyadic cubes"* (pixels) that approximate a region from the inside and outside.



dyadic cube, side length $1 = 2^{-0}$

dyadic set
(union of same size cubes)

Definition 19.a.1 (*Dyadic cube and dyadic set*): For $p \in \mathbb{N}$ and $a \in \mathbb{Z}^n$, the "*dyadic cube*" (German: „*dyadischer Würfel*“) of generation p at a is

$$Q_p(a) := 2^{-p}\big(a + [0,1)^n\big),$$

a half-open cube of side length 2^{-p}. A "*dyadic set*" (German: „*dyadische Menge*“) is a finite union $E = \bigcup_{\ell=1}^{N} Q_p(a(\ell))$ of distinct dyadic cubes **of one common generation** p, and its volume is

$$\mu(E) := 2^{-pn} N.$$

Exercise 19.a.2 (*Inclusion–exclusion for dyadic sets*): Show that, for two dyadic sets E_1, E_2,

$$\mu(E_1 \cup E_2) = \mu(E_1) + \mu(E_2) - \mu(E_1 \cap E_2).$$

Show more generally that, for dyadic sets $E_1, \dots, E_N$, the measure of their union is given by the principle of inclusion–exclusion,

$$\mu\Big(\bigcup_{i=1}^{N} E_i\Big) = \sum_{k=1}^{N} (-1)^{k+1} \Bigg(\sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} \mu(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_k}) \Bigg).$$

Proof of Exercise 19.a.2:

Any finite collection of dyadic sets can be refined to a common generation p without changing any of their measures (splitting a cube of generation p into 2^n cubes of generation $p+1$ replaces one term of $\mu(E) := 2^{-pn}N$ by 2^n terms of size $2^{-(p+1)n}$, which sum to the same value). So, we may assume $E_1, \dots, E_N$ all consist of cubes of one common generation p. Each such cube Q then either lies in none of the E_i, or in exactly the E_i indexed by some non-empty $J \subseteq \{1, \dots, N\}$, and Q contributes 2^{-pn} to $\mu(E_{i_1} \cap \cdots \cap E_{i_k})$ exactly when $\{i_1, \dots, i_k\} \subseteq J$.

Two sets. If Q lies in neither E_1 nor E_2, it contributes 0 to both sides. If it lies in exactly one, it contributes 2^{-pn} to $\mu(E_1)+\mu(E_2)$ and to $\mu(E_1\cup E_2)$, and 0 to $\mu(E_1\cap E_2)$: both sides agree. If it lies in both, it contributes 2^{-pn} to $\mu(E_1\cup E_2)$, and on the right $2\cdot 2^{-pn}-2^{-pn}=2^{-pn}$: both sides agree again.

N sets. It suffices to check the identity cube by cube. A cube Q lying in exactly the sets indexed by $J\neq\emptyset$, $|J|=:m\geq 1$, contributes 2^{-pn} to the left side (it lies in the union). On the right, for each k it contributes 2^{-pn} once for every k-element subset of J, of which there are $\binom{m}{k}$, so the total contribution is

$$2^{-pn}\sum_{k=1}^{m}(-1)^{k+1}\binom{m}{k}=2^{-pn}\Big(1-\sum_{k=0}^{m}(-1)^k\binom{m}{k}\Big)=2^{-pn}\big(1-(1-1)^m\big)=2^{-pn},$$

using the binomial theorem and $m\geq 1$. Both sides agree cube by cube, hence in total. Q.E.D.

The requirement that all cubes share a generation is not cosmetic: it is what separates the Jordan theory from the Lebesgue theory, as Remark 19.a.12 makes precise.

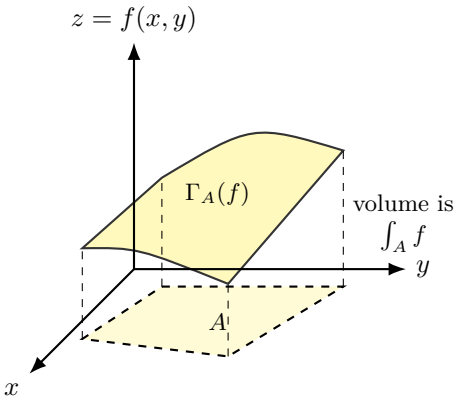
Similar to the 1D Riemann integral, we approximate the volume of a set $E\subseteq\mathbb{R}^n$ using an outer sequence of dyadic sets $G_i\supseteq E$ and an inner sequence $F_i\subseteq E$. If the limits of their volumes coincide as the pixel size $2^{-p}\to 0$, E is "*Jordan measurable*".

Definition 19.a.3 (*Inner and outer Jordan measure*): Let $E\subseteq\mathbb{R}^n$ be bounded. Its "*outer*" and "*inner Jordan measure*" are

$$\mu_{\text{out}}(E):=\inf\{\mu(G):G\text{ dyadic set},\ E\subseteq G\},\qquad \mu_{\text{in}}(E):=\sup\{\mu(F):F\text{ dyadic set},\ F\subseteq E\},$$

where the empty union counts as a dyadic set of volume 0. Always $\mu_{\text{in}}(E)\leq\mu_{\text{out}}(E)$. The set E is "*Jordan measurable*" (German: „*Jordan-messbar*“) if the two agree, and their common value $\mu(E)$ is its "*Jordan measure*". It is a "*Jordan null set*" if $\mu_{\text{out}}(E)=0$.

Furthermore, the multidimensional integral of a non-negative function $f:A\to\mathbb{R}$ is exactly the $(n+1)$-dimensional Jordan measure (volume) of its hypograph $\Gamma_A(f)$:

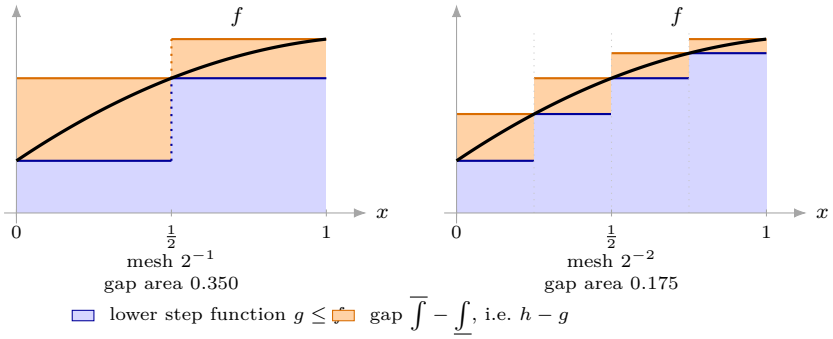


Definition 19.a.4 (*Riemann integral in $\mathbb{R}^n$*): Let $A\subseteq\mathbb{R}^n$ and $f:A\to\mathbb{R}$. Analogous to Analysis I, the integral is defined via dyadic step functions (German: „*dyadische Treppenfunktionen*“). The lower and upper integrals are defined as

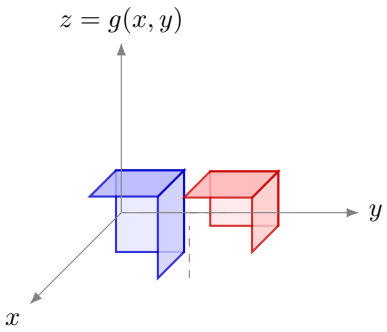
$$\underline{\int_A} f:=\sup\left\{\int_A g\;\middle|\; g:\mathbb{R}^n\to\mathbb{R}\text{ dyadic step function with }g\leq f\right\},$$
$$\overline{\int_A} f:=\inf\left\{\int_A h\;\middle|\; h:\mathbb{R}^n\to\mathbb{R}\text{ dyadic step function with }h\geq f\right\}.$$

The function f is "*Riemann integrable*" if $\underline{\int_A}f=\overline{\int_A}f$. In this case, their common value is denoted by $\int_A f$.

In one variable the construction can simply be drawn, and it is worth seeing once. Fix a dyadic mesh. On each dyadic interval let g be the largest constant still lying below f, and h the smallest constant still lying above it. Refining the mesh can only raise g and lower h, so the orange gap between them shrinks, and f is integrable exactly when that gap can be driven to 0.



Halving the mesh here halves the gap, from 0.350 to 0.175. That is not a coincidence of this example: for a monotone f on an interval the upper and lower steps differ, on each piece, by exactly the rise of f across it, and those rises telescope. In $\mathbb{R}^n$ the same construction stacks columns over dyadic cubes instead of intervals:



Integral of a
dyadic step function:
sum of column volumes

Remark 19.a.5 (*What the construction inherits from one variable*): This integral satisfies all the *"good"* properties from the one-dimensional case, such as linearity. Moreover, if $E \subseteq \mathbb{R}^n$ is Jordan measurable and $f : \overline{E} \to [0, \infty)$ is continuous, then f is Riemann integrable on E. The course states this without proving it, and the proof is the same one as in **Analysis I** with cubes in place of intervals: continuity on the compact set $\overline{E}$ gives uniform continuity (Proposition 7.a.17), which bounds the oscillation of f on every cube of a fine enough mesh simultaneously. The upper and lower sums then differ by at most that bound times the volume of the cubes meeting E, plus $\sup f$ times the volume of the cubes meeting ∂E, and the second term is small because ∂E is a Jordan null set (Theorem 19.a.6).

Theorem 19.a.6 (*Jordan measurability criterion via boundary*): A bounded subset $E \subseteq \mathbb{R}^n$ is Jordan measurable if and only if its boundary ∂E is a Jordan null set, i.e. $\mu_{\text{out}}(\partial E) = 0$.

Proof:

For a bounded S and a generation p, let $N_p(S)$ count the generation-p cubes meeting S and $K_p(S)$ those contained in S. The smallest generation-p dyadic set containing S is the union of the $N_p(S)$ cubes, and the largest one inside S is the union of the $K_p(S)$ cubes. Splitting each cube into its 2^n children leaves volumes unchanged, so any dyadic set can be moved to a finer generation, and therefore

$$\mu_{\text{out}}(S) = \lim_{p \to \infty} 2^{-pn} N_p(S), \qquad \mu_{\text{in}}(S) = \lim_{p \to \infty} 2^{-pn} K_p(S),$$

the first sequence decreasing and the second increasing.

(⇐) A cube that meets E without lying in E contains a point of E and a point of its complement. Being convex, hence connected, it must then meet ∂E: otherwise it would split into its parts in $\operatorname{int} E$ and in the exterior of E, two disjoint non-empty relatively open pieces. So $N_p(E) - K_p(E) \le N_p(\partial E)$, and letting $p \to \infty$ gives $\mu_{\text{out}}(E) - \mu_{\text{in}}(E) \le \mu_{\text{out}}(\partial E) = 0$.

(⇒) Given $\varepsilon > 0$, pick dyadic sets $F \subseteq E \subseteq G$ of one common generation with $\mu(G) - \mu(F) < \varepsilon$. A point $x \in \partial E$ lies in $\overline{E} \subseteq \overline{G}$ but not in $\operatorname{int} E \supseteq \operatorname{int} F$, so $x \in (G \setminus F) \cup \partial G \cup \partial F$. The last two sets lie in the union Z of the boundaries of the finitely many cubes of G, and Z is Jordan null: a face of a cube of side s meets at most $(2^q s + 2)^{n-1}$ cubes of generation q, of total volume at most $2^{-q}(s + 2^{1-q})^{n-1} \to 0$. Hence $\mu_{\text{out}}(\partial E) \le \mu(G \setminus F) + \mu_{\text{out}}(Z) < \varepsilon$. **Q.E.D.**

Lemma 19.a.7 (*Compact sets: Jordan null versus Lebesgue null*): Let $K \subseteq \mathbb{R}^n$ be compact. Then K is a Jordan null set if and only if K is a Lebesgue null set (in the sense of Remark 19.a.11 below: coverable by countably many cubes of arbitrarily small total volume).

Proof:

If K is Jordan null, then for every $\varepsilon > 0$, K is covered by finitely many dyadic cubes $Q_1, \dots, Q_m$ with total volume $\sum_{i=1}^{m} \mu(Q_i) < \varepsilon$, so K is a Lebesgue null set. Conversely, if K is Lebesgue null, there is a countable cover by dyadic cubes $\{Q_j\}_{j \ge 1}$ with $\sum_{j=1}^{\infty} \mu(Q_j) < \varepsilon$. Taking the interiors of slightly enlarged cubes yields an open cover of the compact set K, which admits a finite subcover of total volume $< 2\varepsilon$. These finitely many boxes are not yet a dyadic set, since their side lengths differ, but covering each of them by the dyadic cubes of one sufficiently fine common generation that meet it adds as little volume as we like. So K is Jordan null. **Q.E.D.**

Example 19.a.8 (*The Cantor set is a Jordan null set*): Consider the middle-thirds "*Cantor set*" $C \subseteq [0,1]$, defined inductively by $C_0 := [0,1]$ and

$$C_{n+1} := \tfrac{1}{3} C_n \cup \left(\tfrac{2}{3} + \tfrac{1}{3} C_n\right), \qquad C := \bigcap_{n=0}^{\infty} C_n.$$

Each C_n consists of 2^n closed intervals of length 3^{-n}, so $\mu(C_n) = (2/3)^n$. Since $C \subseteq C_n$ for all n, we have

$$\mu_{\text{out}}(C) \le \mu(C_n) = \left(\tfrac{2}{3}\right)^n \xrightarrow{n \to \infty} 0,$$

proving that the Cantor set C is a Jordan null set.

Moreover, C consists precisely of those points $x \in [0,1]$ possessing a ternary (base-3) expansion $x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$ composed entirely of digits $a_n \in \{0, 2\}$ without the digit 1.

Example 19.a.9 (*An open set that is not Jordan measurable*): Jordan measurability can fail even for open sets. Enumerate the rational numbers in $[0,1]$ as $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \dots\}$, and define the open set

$$U := \bigcup_{k=1}^{\infty} B_{2^{-(k+2)}}(q_k) \;\cap\; (0,1).$$

Since $\mathbb{Q} \cap [0,1]$ is dense in $[0,1]$, so is U. On the outside, a dyadic set $G \supseteq U$ leaves out of $[0,1]$ only finitely many half-open intervals, none of positive length since each would miss the dense set U; so $\mu(G) \ge 1$ and $\mu_{\text{out}}(U) = 1$. On the inside, let $F \subseteq U$ be a dyadic set. Shrinking each of its finitely many intervals $[a,b)$ to $[a, b-\delta]$ gives a compact subset of U, which is covered by finitely many of the open intervals $B_{2^{-(k+2)}}(q_k)$, of total length at most $\sum_{k=1}^{\infty} 2 \cdot 2^{-(k+2)} = \frac{1}{2}$; letting $\delta \to 0$ gives $\mu(F) \le \frac{1}{2}$. Thus $\mu_{\text{in}}(U) \le \frac{1}{2} < 1 = \mu_{\text{out}}(U)$, showing that U is **not** Jordan measurable. Its boundary $\partial U = [0,1] \setminus U$ is a *"fat"* boundary of positive outer measure.

Proposition 19.a.10 (*Algebra of Jordan measurable sets*): If $E, F \subset \mathbb{R}^n$ are Jordan measurable, then their union $E \cup F$, intersection $E \cap F$, and differences $E \setminus F$, $F \setminus E$ are also Jordan measurable. Furthermore, if $E \cap F = \emptyset$ (or if they only intersect in a Jordan null set), then

$$\mu(E \cup F) = \mu(E) + \mu(F).$$

Proof:

All four sets are bounded. By Theorem 19.a.6 it suffices to show that their boundaries are Jordan null, and this rests on a single observation:

$$\partial(E \cup F),\ \partial(E \cap F),\ \partial(E \setminus F) \ \subseteq\ \partial E \cup \partial F.$$

Indeed, if $x \notin \partial E$, some ball around x lies entirely in E or entirely in its complement; shrinking it, the same holds for F. On such a ball each of $E \cup F$, $E \cap F$, $E \setminus F$ is either everything or nothing, so x is not a boundary point of any of them.

A union of two Jordan null sets is Jordan null: cover them by dyadic sets G, H with $\mu(G), \mu(H) < \varepsilon/2$, refine both to a common generation (which changes no volume, see the solution of Exercise 19.a.2), and note $\mu(G \cup H) \leq \mu(G) + \mu(H) < \varepsilon$ by inclusion–exclusion. A subset of a Jordan null set is trivially Jordan null. So all three boundaries are null, which proves measurability.

For additivity, let $\mu_{\text{out}}(E \cap F) = 0$ (this includes $E \cap F = \emptyset$).

- $\geq$: take dyadic $A \subseteq E$, $B \subseteq F$ of a common generation. Then $A \cap B \subseteq E \cap F$ is a dyadic set inside a null set, so $\mu(A \cap B) = 0$, and $\mu(A) + \mu(B) = \mu(A \cup B) \leq \mu_{\text{in}}(E \cup F)$. Taking suprema, $\mu(E) + \mu(F) \leq \mu(E \cup F)$.
- $\leq$: take dyadic $G \supseteq E$, $H \supseteq F$. Then $G \cup H \supseteq E \cup F$ and $\mu(G \cup H) \leq \mu(G) + \mu(H)$. Taking infima, $\mu(E \cup F) \leq \mu(E) + \mu(F)$.

Only the first bullet used the hypothesis on $E \cap F$, as it must: the second is the subadditivity that holds for every pair of sets. **Q.E.D.**

Remark 19.a.11 (*Lebesgue null vs. Jordan null*): A set $E \subset \mathbb{R}^n$ is "*Lebesgue null*" if for every $\varepsilon > 0$, there exist **countably** many dyadic cubes Q_i covering E with $\sum \mu(Q_i) < \varepsilon$. For a "*Jordan null*" set, the cover must be formed by **finitely** many cubes (which implicitly requires E to be bounded). Because a countable cover is allowed, the set $\mathbb{Q} \cap [0,1]$ is Lebesgue null. However, it is not Jordan null (in fact, it is not even Jordan measurable, since its boundary is all of $[0,1]$; compare Theorem 19.a.6), since any finite union of intervals covering it must also cover its closure $[0,1]$, thus having measure at least 1.

Remark 19.a.12 (*The relaxation is really about **sizes, not just counts*):** Stating the difference as *"finitely many"* against *"countably many"* is correct but hides where the extra power actually comes from. A Jordan cover is a **dyadic set**, and every cube in a dyadic set of generation p has the same side length 2^{-p} (Definition 19.a.1). So, the two conditions really read:

- **Jordan null:** for each $\varepsilon > 0$, cover E by cubes that all have **one common side length**, of total volume $< \varepsilon$. Finiteness is then automatic: only finitely many cubes of a fixed size fit into a bounded region.
- **Lebesgue null:** for each $\varepsilon > 0$, cover E by cubes of **whatever side lengths you like**, of total volume $< \varepsilon$. Allowing the sizes to vary is exactly what lets the cover be infinite while the total volume stays finite.

The two relaxations are therefore one relaxation. It is permission to shrink the cubes as you go that buys the countable cover, and it is the countable cover that makes the notions differ.

$\mathbb{N} \subseteq \mathbb{R}$ shows this in its purest form. Cover the integer k by an interval of length $\varepsilon\, 2^{-k-1}$; the total length is ε, so $\mathbb{N}$ is Lebesgue null. No Jordan cover can do this: cubes of one fixed length 2^{-p} need at least one cube per integer, infinitely many of them, of infinite total length. (Indeed

$\mathbb{N}$ is unbounded, so it is not Jordan measurable at all.) The point is that the geometric series is available only when the sizes may vary.

Remark 19.a.13 (*The null sets are not the whole story*): Everything said so far compares the **null** sets of the two theories, which is as far as this chapter needs to go. It leaves the obvious question unanswered: is Jordan measure the same thing as Lebesgue measure, and if not, what is the difference? Section 19.c answers that in a few pages, without proofs and without being used anywhere later.

Theorem 19.a.14 (*Linear transformations of Jordan measurable sets*): Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear, invertible map. If $E \subset \mathbb{R}^n$ is Jordan measurable, then the image $L(E)$ is also Jordan measurable, and its measure scales by the absolute value of the determinant of L:

$$\mu(L(E)) = |\det L|\, \mu(E).$$

Exercise 19.a.15 (*True or False (null sets and outer measure)* (important)):

(a) A bounded countable set is always Jordan-null.

(b) A countable set is always Lebesgue-null.

(c) Let $D \subset [0,1]$ be a dense set (i.e. $\overline{D} = [0,1]$). Then $\mu_{out}(D) = 1$. (μ_{out} was defined in Definition 13.10 of the lecture notes; here it is Definition 19.a.3.)

(d) Let $X, Y \subset [0,1]$ Jordan measurable sets such that $\mu(X) > 1/2$ and $\mu(Y) > 1/2$. Then $X \cap Y \neq \emptyset$.

(e) Let $X, Y \subset [0,1]$ such that $\mu_{out}(X) > 1/2$ and $\mu_{out}(Y) > 1/2$. Then $X \cap Y \neq \emptyset$.

Official hint: ***8.2.5:*** *μ_{out} can be positive and large on very sparse sets...*

Sascha Brack's hints: *A toolkit of sets worth testing every part against, namely $\mathbb{Q}$, the irrationals $\mathbb{R} \setminus \mathbb{Q}$, $\mathbb{Q} \cap [0,1]$, and $(\mathbb{R} \setminus \mathbb{Q}) \cap [0,1]$. For **8.2.3** specifically: monotonicity already gives $\mu_{out}(D) \leq \mu_{out}([0,1]) = 1$, so the work is showing $\mu_{out}(D) \geq 1$, essentially that the "smallest" dyadic set containing D also contains $[0,1]$.*

Info: *This exercise is Problem 8.2 of Problem Sheet 8, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 278*

Exercise 19.a.16 (*Multiple choice (which maps preserve null sets)* (important)): Let $U \subset \mathbb{R}^n$ be a nonempty, open subset, $f : U \to \mathbb{R}^m$ a function, and $N \subset U$ a Jordan null set. In which of the following cases is the image $f(N) \subset \mathbb{R}^m$ necessarily a null set? *Attention: Only one answer is correct!*

(a) If f is uniformly continuous.

(b) If f is uniformly continuous and $m \geq n$.

(c) If f is locally Lipschitz continuous.

(d) If f is locally Lipschitz continuous and $m \geq n$.

Info: *This exercise is Problem 8.4 of Problem Sheet 8, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 279*

Exercise 19.a.17 (*Null sets: what they cannot be, and how they combine*): Throughout, "*Lebesgue null*" is as in Remark 19.a.11: for every $\varepsilon > 0$ the set admits a countable cover by cubes of total volume less than ε.

(a) Can a non-empty open subset $U \subseteq \mathbb{R}^n$ be Lebesgue null? Justify your answer.

(b) Is a bounded Lebesgue null set necessarily closed? Justify your answer.

(c) Show that a countable union of Lebesgue null sets is Lebesgue null.

(d) Let $f : [0,1] \to \mathbb{R}$ be Riemann integrable. Show that its graph, viewed as the subset $\Gamma_f := \{(x, f(x)) \mid x \in [0,1]\}$ of $\mathbb{R}^2$, is a Lebesgue null set.

Info: *This exercise is taken from the Analysis I & II examination of 27 January 2021 (Prof. Peter S. Jossen), Part B, Problem 2, whose first part asks for the definition of a Lebesgue null set.*

Solution: *See solution on Page 282*

AI-Note 19.1: Nothing in this block uses Lebesgue **measure** or the Lebesgue integral, and this course develops neither. *"Lebesgue null"* is used here only in the elementary sense of Remark 19.a.11 (coverable by countably many cubes of arbitrarily small total volume), which is the sense already in play in Exercise 19.a.15**(b)** on the official problem sheet. Part **(c)** is the one property of that notion this document repeatedly leans on without ever proving, which is why the block earns its place.

Exercise 19.a.18 (*A non-negative function with vanishing integral*): Say that a property holds "*almost everywhere*" on a set if the subset where it fails is Lebesgue null (Remark 19.a.11).

Let $Q \subseteq \mathbb{R}^n$ be a closed box and let $f : Q \to \mathbb{R}$ be Riemann integrable with $f \geq 0$ and $\int_Q f \, d\mathrm{vol}_n = 0$. Show that $f = 0$ almost everywhere on Q.

Info: *This exercise is taken from the mock examination **Probeprüfung Analysis II** of 31 January 2023, Problem 10**(c)**. That paper names no lecturer. Its first two parts ask for the definition of a null set and for closure under countable unions, which are Remark 19.a.11 and Exercise 19.a.17**(c)**.*

Solution: *See solution on Page 283*

Exercise 19.a.19 (*A null set times a null set, and a null set times an interval*): Let $X \subset \mathbb{R}$ be a Jordan null set (Definition 19.a.3).

(a) Show rigorously that $X \times X \subset \mathbb{R}^2$ is also Jordan null.

(b) Show rigorously that $X \times [0,1] \subset \mathbb{R}^2$ is also Jordan null.

Info: *This exercise is Problem 8.1 of Problem Sheet 8, Analysis II, Spring Semester 2026. The FS 2024 edition of the course sets it as the bonus problem of its own Sheet 8.*

Solution: *See solution on Page 285*

Exercise 19.a.20 (*Integrating over a ball with the lattice points removed*): Let $\overline{B}_2(0) := \{x \in \mathbb{R}^4 \mid |x| \leq 2\}$ be the closed ball of radius 2 in $\mathbb{R}^4$, and set

$$M := \overline{B}_2(0) \setminus \{x \in \mathbb{R}^4 \mid x_i \in \mathbb{Z} \text{ for } 1 \leq i \leq 4\}.$$

Show that

$$f(x) := \cos\Big(\frac{\pi}{2}\left(x_1^2 + x_2^2 + x_3^2 + x_4^2\right)^{3/2}\Big)$$

is Riemann integrable over M.

Official hint: *You may of course use any result from the lecture without proving it again.*

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Problem 16.*

Solution: *See solution on Page 286*

Exercise 19.a.21 (*Single choice: what a diffeomorphism does to measure*): Let U and V be diffeomorphic Jordan measurable sets, and suppose U is closed and bounded with $\mu(U) = 1$. Exactly one of the following necessarily holds. Which?

(a) $\mu(V) < \infty$.

(b) $\mu(V) = \mu(U)$.
(c) $\mu(V) > \mu(U)$.
(d) $\mu(V) < \mu(U)$.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 6.*

Solution: *See solution on Page 287*

Exercise 19.a.22 (*Single choice: the integral computed by slicing the range*): The Riemann integral was introduced above via step functions. Let f be a bounded, positive step function on a Jordan measurable set Ω. Exactly one of the following expressions has the same value as $\int_\Omega f$. Which?

(a) $\displaystyle\int_0^{\max f} \mu\big(\{x \in \Omega \mid f(x) > t\}\big)\, dt$.
(b) $\displaystyle\int_0^{\max f} \mu\big(\{x \in \Omega \mid f(x) < t\}\big)\, dt$.
(c) $\displaystyle\int_0^{\max f} \mu\big(\{x \in \Omega \mid f(x) = t\}\big)\, dt$.
(d) $\displaystyle\int_0^{\max f} \mu\big(\{x \in \Omega \mid f(x) \leq t\}\big)\, dt$.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 7.*

Solution: *See solution on Page 287*

Improper integrals (Section 19.b)

Everything so far has integrated a continuous function over a **bounded** Jordan measurable set. Two situations fall outside that and both occur constantly in practice: the domain may be unbounded, and the integrand may be unbounded near the edge of its domain. What should $\int_{\mathbb{R}^2} f$ mean?

The answer avoids limits entirely, at the cost of one hypothesis.

Definition 19.b.23 (*Improper integral*): Let $U \subseteq \mathbb{R}^n$ be open and let $f \in C^0(U)$ be **non-negative**. The "*improper integral*" (German: „*uneigentliches Integral*“) of f over U, still written $\int_U f$, is

$$\int_U f := \sup\left\{ \int_K f \;\middle|\; K \subseteq U \text{ compact and Jordan measurable} \right\} \in [0, +\infty].$$

Each $\int_K f$ on the right is an ordinary integral: K is compact and Jordan measurable and f is continuous on it, so Definition 19.a.4 applies. The supremum of a set of real numbers bounded below by 0 always exists in $[0, +\infty]$, so $\int_U f$ is always defined. It may be $+\infty$, and in that case one says the improper integral "*diverges*".

Remark 19.b.24 (*Why non-negativity, and what it buys*): The hypothesis $f \geq 0$ is not a technicality that a more careful definition would remove. It is what makes the supremum monotone, so that enlarging K can only increase $\int_K f$, and that monotonicity is the whole of Lemma 19.b.25 below.

Drop it and the value stops being well defined. For a sign-changing f one can arrange exhaustions that pick up the positive part faster or slower than the negative part and obtain different answers, exactly as a conditionally convergent series can be rearranged to any sum. Example 20.c.42 is the case in point: $\int_0^\infty (\sin x)/x\, dx = \pi/2$ is **not** an improper integral in the sense above, since $(\sin x)/x$

changes sign infinitely often and $\int_0^\infty |\sin x|/x\,dx = +\infty$. Its value depends on integrating in the order $\int_0^R$ and then letting $R \to \infty$, which is why that example needs a justification the others do not.

The definition is clean but useless for computing, since one cannot take a supremum over all compact subsets by hand. The next result says that any single sensible exhaustion suffices, and that it does not matter which one is chosen.

Lemma 19.b.25 (*Improper integrals as limits*): Let $U \subseteq \mathbb{R}^n$ be open and $f \in C^0(U)$ non-negative. Let $K_1 \subseteq K_2 \subseteq K_3 \subseteq \cdots$ be compact Jordan measurable sets with

$$\bigcup_{\ell=1}^{\infty} \operatorname{int}(K_\ell) = U.$$

Then

$$\int_U f = \lim_{\ell \to \infty} \int_{K_\ell} f,$$

and in particular the limit exists in $[0, +\infty]$.

Proof:

Since the K_ℓ are nested and $f \geq 0$, the sequence $\int_{K_\ell} f$ is non-decreasing, so it converges in $[0, +\infty]$; call the limit Λ.

$\int_U f \leq \Lambda$. Let $K \subseteq U$ be compact and Jordan measurable. The sets $\operatorname{int}(K_\ell)$ are open and cover U, hence cover K, so by compactness finitely many of them suffice; since they are nested, a single one does, say $K \subseteq \operatorname{int}(K_{\ell_0}) \subseteq K_{\ell_0}$. Non-negativity then gives $\int_K f \leq \int_{K_{\ell_0}} f \leq \Lambda$. Taking the supremum over all such K yields $\int_U f \leq \Lambda$.

$\Lambda \leq \int_U f$. Each K_ℓ is itself compact and Jordan measurable and contained in U, so it competes in the supremum defining $\int_U f$. Consequently $\int_{K_\ell} f \leq \int_U f$ for every ℓ, and letting $\ell \to \infty$ gives $\Lambda \leq \int_U f$. Q.E.D.

Remark 19.b.26 (*What the lemma licenses*): Read the two halves together: the value does not depend on the exhaustion, so one may choose whichever exhaustion makes the computation easy. That is the licence Example 20.a.5 and Example 20.a.3 were already using when they integrated over $\mathbb{R}^2$ in polar coordinates, and it is what AI-Exercise 19.b.27 makes explicit by running two different exhaustions against each other and comparing the answers.

AI-Exercise 19.b.27 (*The Gaussian integral, two exhaustions at once*): Let $f(x, y) := e^{-(x^2+y^2)}$ on $U := \mathbb{R}^2$, which is continuous and positive, and put $I := \int_{-\infty}^{\infty} e^{-t^2}\,dt$.

- **(a)** Check that the closed disks $D_\ell := \overline{B_\ell(0)}$ and the closed squares $Q_\ell := [-\ell, \ell]^2$ each form an exhaustion of $\mathbb{R}^2$ in the sense of Lemma 19.b.25.
- **(b)** Compute $\lim_{\ell \to \infty} \int_{D_\ell} f$ in polar coordinates.
- **(c)** Compute $\lim_{\ell \to \infty} \int_{Q_\ell} f$ using Theorem 20.b.29, leaving the answer in terms of I.
- **(d)** Conclude the value of I, and say which step would collapse if f were allowed to change sign.

Solution: *See solution on Page 280*

Exercise 19.b.28 (*True or False* (improper integrals of an inverse power)): Consider the function $f(x, y) = \frac{1}{(x^2+y^2)^\alpha}$ for $\alpha > 0$, as well as the planar regions

$$A := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \geq 1\}, \qquad B := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

Decide whether each of the following statements is true or false.

- **(a)** $\int_A f < \infty$ if and only if $\alpha \leq 1$.

(b) $\int_B f < \infty$ if $\alpha = \frac{10}{9}$.
(c) $\lim_{\alpha \to 0^+} \int_B f = \frac{\pi^2}{6} < \infty$.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the tenth block of true/false questions.*

Solution: *See solution on Page 281*

Exercise 19.b.29 (*True or False* (an oscillating numerator, decided by comparison)): Consider the function

$$f(x,y) := \frac{\cos(xy)}{(x^2+y^2)^2}$$

and the planar regions

$$A := \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \geq 1\}, \qquad B := \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\}.$$

Decide whether each of the following is true or false, with justification.

(a) $\int_A |f| < \infty$.
(b) $\int_B |f| < \infty$.
(c) $\int_{\mathbb{R}^2} |f| < \infty$.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the tenth block of true/false questions.*

Solution: *See solution on Page 281*

A digression: a glimpse of the Lebesgue integral (Section 19.c)

This course builds one theory of integration: the Riemann integral over Jordan measurable sets. It is enough for everything in the chapters that follow, and no result later in this document depends on a single line of this section. Read it as an answer to a question that Section 19.a provokes and cannot settle.

The question is what *"Lebesgue-null"* means, given that this course never defines a Lebesgue integral, and yet the term appears on the problem sheets (Exercise 19.a.15**(b)**) and in the comparison remarks above. It comes from a second theory of integration, and the sanest way to place it is to see what that theory fixes. **Nothing below is proved.** The statements are given so that the vocabulary has somewhere to attach, and so that a reader meeting measure theory later recognises what it was built for.

Two things the Riemann theory cannot do (Subsection 19.c.1)

Example 19.c.30 (*The measurable sets are too few*): Proposition 19.a.10 says the Jordan measurable sets are closed under finite unions, intersections and differences. They are **not** closed under countable unions. Each singleton $\{q\} \subseteq \mathbb{R}$ is Jordan measurable with measure 0, and yet

$$\mathbb{Q} \cap [0,1] = \bigcup_{k \geq 1} \{q_k\}$$

is not Jordan measurable at all: any finite cover of it by cubes also covers its closure $[0,1]$, so $\mu_{\text{out}} = 1$ while $\mu_{\text{in}} = 0$. Example 19.a.9 is worse still, since the set there is **open** and the volume of an open set ought not to be a question a theory declines to answer.

Example 19.c.31 (*Limits escape the class of integrable functions*): Enumerate $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \dots\}$ and let f_k be the indicator function of $\{q_1, \dots, q_k\}$. Each f_k is Riemann integrable,

having only finitely many discontinuities, and $\int_0^1 f_k = 0$. The sequence increases pointwise to the Dirichlet function

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases}$$

which is not Riemann integrable, its upper and lower sums being 1 and 0 for every partition.

A bounded increasing sequence of integrable functions therefore need not have an integrable limit. The only exchange theorem available in this course needs **uniform** convergence, and uniform convergence is a strong hypothesis that the sequences arising in practice routinely fail.

Both defects have the same shape. The Riemann–Jordan theory is built on finite covers by cubes of one size, and it is stable under finite operations and unstable under countable ones. Lebesgue's construction changes exactly that.

Lebesgue measure (Subsection 19.c.2)

Definition 19.c.32 (*Lebesgue outer measure*): For $E \subseteq \mathbb{R}^n$ the "*Lebesgue outer measure*" (German: „*äusseres Lebesgue-Mass*“) is

$$\lambda^*(E) := \inf \left\{ \sum_{i=1}^{\infty} \operatorname{vol}(Q_i) \;\middle|\; E \subseteq \bigcup_{i=1}^{\infty} Q_i,\ Q_i \text{ cubes} \right\},$$

the infimum running over all **countable** covers of E by cubes of arbitrary sizes.

This is the notion already in use above: E is Lebesgue null exactly when $\lambda^*(E) = 0$, which is Remark 19.a.11 restated. The defect of λ^* is that it is only subadditive, not additive, so one restricts it to a good class of sets.

Definition 19.c.33 (*Lebesgue measurable set*): A set $E \subseteq \mathbb{R}^n$ is "*Lebesgue measurable*" (German: „*Lebesgue-messbar*“) if it splits every test set additively:

$$\lambda^*(A) = \lambda^*(A \cap E) + \lambda^*(A \setminus E) \qquad \text{for every } A \subseteq \mathbb{R}^n.$$

On this class one writes $\lambda := \lambda^*$ and calls it "*Lebesgue measure*" (German: „*Lebesgue-Mass*“).

Theorem 19.c.34 (*What the construction delivers, without proof*): The Lebesgue measurable subsets of $\mathbb{R}^n$ form a "*σ-algebra*" (German: „*σ-Algebra*“): a family containing $\emptyset$ and closed under complements and under **countable** unions. It contains every open set, every closed set and every Lebesgue null set. On it, λ is countably additive: for pairwise disjoint measurable $E_1, E_2, \ldots$,

$$\lambda\left(\bigcup_{k \geq 1} E_k \right) = \sum_{k \geq 1} \lambda(E_k).$$

Moreover every Jordan measurable set E is Lebesgue measurable with $\lambda(E) = \mu(E)$, and the inclusion is strict.

Remark 19.c.35 (*The answer to the question, in one line*): Jordan measure and Lebesgue measure are **not** the same, and the difference is not a matter of convention. Lebesgue measure is a strict extension: it agrees with Jordan measure wherever the latter is defined, so no volume computed anywhere in this document is put at risk, and it is defined on many more sets. The single structural upgrade is from an algebra to a σ-algebra, that is, from finite to countable stability, and both defects of Examples 19.c.30 and 19.c.31 dissolve in it.

The integral: partition the range, not the domain (Subsection 19.c.3)

The Riemann integral chops the **domain** into small pieces and hopes the function is nearly constant on each. Lebesgue chops the **range** instead, and asks how large the set is on which the function

takes each value. The standard image: to count a heap of coins, a Riemann integrator picks them up in the order they lie and adds as they come, while a Lebesgue integrator first sorts them into denominations and multiplies. The second survives a much messier heap.

Concretely, a "*simple function*" is a finite combination $s = \sum_{i=1}^{N} c_i \chi_{E_i}$ with $c_i \geq 0$ and E_i measurable, where χ_E is the indicator function of E; its integral is declared to be $\int s\,d\lambda := \sum_i c_i \lambda(E_i)$. For measurable $f \geq 0$ one then sets

$$\int f\,d\lambda := \sup\left\{ \int s\,d\lambda \;\middle|\; s \text{ simple},\ 0 \leq s \leq f \right\},$$

and for general f one splits $f = f^+ - f^-$ into positive and negative parts, calling f "*Lebesgue integrable*" if $\int |f|\,d\lambda < \infty$.

The Dirichlet function of Example 19.c.31 is now trivial: it is the indicator function of $\mathbb{Q} \cap [0,1]$, a null set, so its integral is 0. Its Riemann sums never settle because they are forced to look at intervals; its Lebesgue integral is immediate because the set carrying the value 1 is negligible.

What the extra work buys (Subsection 19.c.4)

Theorem 19.c.36 (*The convergence theorems, without proof*): Let f_k be measurable functions on a measurable $E \subseteq \mathbb{R}^n$.

- **(a) Monotone convergence.** If $0 \leq f_1 \leq f_2 \leq \dots$ pointwise and $f_k \to f$ pointwise, then $\int_E f_k\,d\lambda \to \int_E f\,d\lambda$.
- **(b) Fatou's lemma.** If $f_k \geq 0$, then $\int_E \liminf_k f_k\,d\lambda \leq \liminf_k \int_E f_k\,d\lambda$.
- **(c) Dominated convergence.** If $f_k \to f$ pointwise and there is a single integrable g with $|f_k| \leq g$ for all k, then f is integrable and $\int_E f_k\,d\lambda \to \int_E f\,d\lambda$.

Remark 19.c.37 (*Why these are the point of the whole construction*): Compare the hypotheses. Ours is uniform convergence, a statement about $\sup_x |f_k(x) - f(x)|$, which fails as soon as the functions develop a moving spike. Dominated convergence asks only for pointwise convergence plus one integrable envelope, and the envelope is usually the easy part. Monotone convergence asks for nothing but monotonicity.

This is where the cost of staying inside the Riemann theory is actually paid in this document. Differentiating under the integral sign (Section 20.c) has to manufacture uniform convergence of the difference quotients out of compactness, which is why it asks for a continuous partial derivative on a compact set where a domination hypothesis is what the argument morally wants. And the improper integral (Section 19.b) is restricted to non-negative integrands, so that a monotone supremum can stand in for the monotone convergence theorem. Both become shorter and more general in the Lebesgue setting.

How the two integrals compare (Subsection 19.c.5)

Theorem 19.c.38 (*Lebesgue's criterion, and the comparison, without proof*): Let $Q \subseteq \mathbb{R}^n$ be a closed box and $f : Q \to \mathbb{R}$ bounded.

- **(a)** f is Riemann integrable if and only if its set of discontinuities is Lebesgue null.
- **(b)** If f is Riemann integrable, then it is Lebesgue integrable and the two integrals agree.

Part **(a)** is the reason a course with no Lebesgue integral still has to say *"Lebesgue null"*. It is the sharp characterisation of Riemann integrability, and no notion built from finite covers replaces it: the Dirichlet function is discontinuous everywhere, a set of measure 1, and it fails; a function with countably many jumps is discontinuous on a null set, and it succeeds.

AI-Note 19.2: It is tempting to conclude that the Lebesgue integral simply subsumes the Riemann integral. For **proper** integrals on a box, Theorem 19.c.38**(b)** says exactly that. For **improper** ones it is false in both directions, and the standard witness is

$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2},$$

which exists as the limit $\lim_{R\to\infty}\int_0^R$ of proper Riemann integrals (**not** as an improper integral in the sense of Definition 19.b.23, whose non-negativity hypothesis this integrand fails, as Remark 19.b.24 spells out), but not as a Lebesgue integral, because $\int_0^\infty |\sin x|/x\,dx = \infty$ and the Lebesgue theory integrates only absolutely. The improper Riemann integral is a limit of integrals and can exploit cancellation between the humps; the Lebesgue integral is defined through $|f|$ and cannot. So neither theory contains the other outright, and *"Lebesgue is strictly stronger"* is a half-truth worth not repeating.

Remark 19.c.39 (*What to take away*): Three sentences are enough to carry. Jordan measure is Lebesgue measure restricted to a smaller, finitely-stable class of sets, and where both are defined they agree. The Lebesgue theory exists because that class is too small to be closed under limits, and its payoff is the convergence theorems rather than any new volume. And the one place its vocabulary genuinely reaches into this course is Lebesgue's criterion, which is why *"Lebesgue null"* had to be defined here at all.

Aside 19.c.40 (*Some sets have no volume at all, and a ball can be doubled*): Theorem 19.c.34 carefully says which sets λ is defined on, and never claims that is all of them. It cannot: there exist subsets of $\mathbb{R}$ that are not Lebesgue measurable. Vitali's construction takes the quotient $\mathbb{R}/\mathbb{Q}$ and picks one representative from each class inside $[0,1]$. The resulting set V has countably many disjoint rational translates that together fill up an interval, so countable additivity would force $\lambda(V)$ to be simultaneously 0 and positive. No value works, and V is unmeasurable.

The construction needs the axiom of choice, and this is not a technicality. It is consistent with the other axioms of set theory (granted a large-cardinal assumption, by a theorem of Solovay) that **every** subset of $\mathbb{R}$ is Lebesgue measurable, so the pathology is imported by the choice principle rather than discovered in the reals.

Pushed into $\mathbb{R}^3$, the same idea gives the Banach–Tarski paradox: a solid ball can be cut into five pieces which, moved only by rotations and translations, reassemble into two solid balls of the original size. Nothing is stretched, and no volume is created, because the five pieces are unmeasurable and simply do not have volumes to add up. The paradox is a statement about the axiom of choice, not about geometry.

Exercises (Subsection 19.c.6)

AI-Exercise 19.c.41 (*Null sets from countability*): Work directly from Definition 19.c.32.

- **(a)** Show that a single point of $\mathbb{R}^n$ is Lebesgue null.
- **(b)** Show that every countable subset of $\mathbb{R}^n$ is Lebesgue null, and deduce that $\mathbb{Q}^n$ is.
- **(c)** Show that a Lebesgue null set has empty interior. (Compare Exercise 19.a.17**(a)**, which is the same statement contraposed.)

Solution: *See solution on Page 283*

AI-Exercise 19.c.42 (*An integral the Riemann theory cannot see*): Define $f:[0,1]\to\mathbb{R}$ by

$$f(x) := \begin{cases} 1, & x \in \mathbb{Q}, \\ 2, & x \notin \mathbb{Q} \text{ and } x < \tfrac{1}{2}, \\ 3, & x \notin \mathbb{Q} \text{ and } x \geq \tfrac{1}{2}. \end{cases}$$

(a) Compute $\int_{[0,1]} f\,d\lambda$.

(b) Determine the set of points at which f is continuous, and conclude from Theorem 19.c.38 that f is not Riemann integrable.

Solution: *See solution on Page 284*

AI-Exercise 19.c.43 (*Dominated convergence where uniform convergence fails*)**:** For $k \geq 1$ let $f_k : [0,1] \to \mathbb{R}$, $f_k(x) := x^k$.

(a) Determine the pointwise limit f of (f_k) and show that the convergence is not uniform.

(b) Exhibit an integrable dominating function and apply Theorem 19.c.36**(c)** to compute $\lim_{k\to\infty} \int_{[0,1]} f_k\,d\lambda$.

(c) Confirm the answer by evaluating $\int_0^1 x^k\,dx$ directly.

Solution: *See solution on Page 284*

AI-Exercise 19.c.44 (*Improper Riemann but not Lebesgue*)**:** Define $f : [1,\infty) \to \mathbb{R}$ by $f(x) := (-1)^{k+1}/k$ for $x \in [k, k+1)$, $k \in \mathbb{N}_{\geq 1}$.

(a) Show that the improper integral $\int_1^\infty f(x)\,dx$ converges, and identify its value.

(b) Show that $\int_1^\infty |f(x)|\,dx = \infty$, so that f is not Lebesgue integrable.

(c) Why does this not contradict Theorem 19.c.38**(b)**?

Solution: *See solution on Page 285*

Solutions (Section 19.d)

Not every solution in this chapter is collected here. Exercise 19.a.2 is solved where it appears, on Page 265.

Solution to Exercise 19.a.15:

The whole problem turns on one distinction: **Jordan** outer measure is an infimum over **finite** covers, **Lebesgue** outer measure over **countable** ones. Parts **(a)**–**(c)** are that distinction three times; parts **(d)** and **(e)** are the difference between a measure and a merely sub-additive outer measure.

(a) **False.** Take $X := \mathbb{Q} \cap [0,1]$: bounded and countable, yet $\mu_{out}(X) = 1$ by part **(c)**, since X is dense in $[0,1]$. Countability buys nothing here, since only finitely many intervals are allowed at a time, and finitely many intervals covering a dense set must cover the whole of $[0,1]$.

(b) **True**, and this is exactly where countable covers do buy something. Enumerate $X = \{x_1, x_2, \dots\}$ and, given $\varepsilon > 0$, cover x_k by an interval of length $\varepsilon 2^{-k}$. The total length is $\sum_{k\geq 1} \varepsilon 2^{-k} = \varepsilon$, and ε was arbitrary, so the Lebesgue outer measure is 0.

(c) **True.** Monotonicity gives $\mu_{out}(D) \leq \mu_{out}([0,1]) = 1$, so only the lower bound needs an argument. Let $I_1, \dots, I_N$ be finitely many intervals covering D. Enlarging each to its closure changes no length, so we may assume they are closed; then $\bigcup_{k=1}^N I_k$ is **closed**, which is where finiteness enters, and contains D, hence contains $\overline{D} = [0,1]$. Therefore $\sum_{k=1}^N |I_k| \geq \mu([0,1]) = 1$. Taking the infimum over such covers, $\mu_{out}(D) \geq 1$.

(d) **True.** Suppose $X \cap Y = \emptyset$. Jordan measure is additive on disjoint Jordan measurable sets, so

$$\mu(X \cup Y) = \mu(X) + \mu(Y) > \tfrac{1}{2} + \tfrac{1}{2} = 1.$$

But $X \cup Y \subseteq [0,1]$, so monotonicity forces $\mu(X \cup Y) \leq 1$, a contradiction. Hence $X \cap Y \neq \emptyset$.

(e) **False**, and this is the official hint's point. Take

$$X := \mathbb{Q} \cap [0,1], \qquad Y := (\mathbb{R} \setminus \mathbb{Q}) \cap [0,1].$$

Both are dense in $[0,1]$, so part **(c)** gives $\mu_{out}(X) = \mu_{out}(Y) = 1 > \frac{1}{2}$; yet $X \cap Y = \emptyset$ by construction. The argument of part **(d)** breaks at its first step: μ_{out} is only **sub**additive, so $\mu_{out}(X \cup Y) \le \mu_{out}(X) + \mu_{out}(Y)$, an inequality pointing the wrong way to produce a contradiction. Neither set is Jordan measurable, which is precisely why part **(d)** does not apply.

AI-Note 19.3: Parts **(a)** and **(e)** are the same counterexample used twice, and part **(c)** is the lemma both rest on. It is worth extracting what makes it work: a **finite** union of closed sets is closed, so a finite cover of D automatically covers $\overline{D}$. Every difference between Jordan and Lebesgue theory visible on this sheet traces back to that one sentence.

Q.E.D.

Solution to Exercise 19.a.16:

The answer is (d): f locally Lipschitz continuous and $m \ge n$. Both hypotheses are needed, and dropping either one admits a counterexample. One caveat about the word *"null"*: read as **Jordan** null, **(d)** also needs $\overline{N} \subset U$, which is automatic when $U = \mathbb{R}^n$, while read as **Lebesgue** null it holds exactly as stated. Both versions are proved below.

Why (a) and (b) fail: uniform continuity is not enough. Let $c : \mathbb{R} \to [0,1]$ be the Cantor function on $[0,1]$, extended by 0 to the left of 0 and by 1 to the right of 1. It is continuous, uniformly so on the compact interval $[0,1]$ and constant outside it, so it is uniformly continuous on $U := \mathbb{R}$. Let $N \subset [0,1]$ be the Cantor set. Then N is Jordan-null (it is covered by 2^k intervals of total length $(2/3)^k \to 0$), while $c(N) = [0,1]$ is not null in either sense. Here $n = m = 1$, so this single example kills **(a) and (b)**. The extra hypothesis $m \ge n$ does nothing to rescue mere continuity. Intuitively, a uniformly continuous map has no bound on how much it may stretch, only on how abruptly.

Why (c) fails: $m \ge n$ is not decoration. Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) := y$, which is Lipschitz with constant 1, and let $N := \{0\} \times [0,1]$, a segment and so Jordan-null in $\mathbb{R}^2$. Then $f(N) = [0,1]$, which is not null in $\mathbb{R}^1$. Here $m = 1 < 2 = n$: a projection loses a dimension, and what was negligible in the source stops being negligible in the smaller target.

Why (d) holds when $\overline{N} \subset U$. Let N be Jordan-null with $\overline{N} \subset U$. Being Jordan-null, N is bounded, so $\overline{N}$ is compact and has positive distance from the closed set $\mathbb{R}^n \setminus U$. Choose $\delta > 0$ below that distance (any $\delta > 0$ if $U = \mathbb{R}^n$) and put $K := \{x \in \mathbb{R}^n : \operatorname{dist}(x, \overline{N}) \le \delta\}$, a compact subset of U. On K the map f is Lipschitz with a single constant L. Otherwise there are $x_k, y_k \in K$ with $|f(x_k) - f(y_k)| > k|x_k - y_k|$; as f is bounded on K, this forces $|x_k - y_k| \to 0$, so a subsequence of both converges to one point of K, and the local Lipschitz bound near that point is violated once k exceeds its constant.

Fix $\varepsilon > 0$ and cover N by finitely many cubes $Q_1, \dots, Q_M$, each meeting N, of sides $s_i \le \min\{1, \delta/\sqrt{n}\}$ with $\sum_i s_i^n < \varepsilon$ (subdividing the cubes of a Jordan cover keeps its total volume). Each Q_i has diameter $s_i\sqrt{n} \le \delta$ and meets N, so $Q_i \subset K$. Hence $f(Q_i \cap N)$ has diameter at most $L s_i \sqrt{n}$ and therefore fits inside a cube of side $L s_i \sqrt{n}$ in $\mathbb{R}^m$, of volume $(L\sqrt{n})^m s_i^m$. Since $s_i \le 1$ and $m \ge n$ we have $s_i^m \le s_i^n$, so the images are covered with total volume at most

$$(L\sqrt{n})^m \sum_{i=1}^{M} s_i^n \;<\; (L\sqrt{n})^m \, \varepsilon.$$

As ε was arbitrary, $f(N)$ is Jordan null.

Why (d) holds in general, for Lebesgue null images. Drop the assumption $\overline{N} \subset U$ and exhaust U by the compact sets

$$K_j := \big\{x \in \mathbb{R}^n : |x| \le j,\ \operatorname{dist}(x, \mathbb{R}^n \setminus U) \ge \tfrac{1}{j}\big\}, \qquad j \ge 1,$$

reading the distance as $+\infty$ when $U = \mathbb{R}^n$. Each K_j lies in U, and every point of the open set U lies in some K_j. The set $N_j := N \cap K_j$ is Jordan null, being a subset of N, and $\overline{N_j} \subseteq K_j \subset U$. The case just proved makes every $f(N_j)$ Jordan null, hence Lebesgue null, and

$$f(N) = \bigcup_{j \geq 1} f(N_j)$$

is Lebesgue null as a countable union of Lebesgue null sets (Exercise 19.a.17**(c)**).

AI-Note 19.4: The proof shows exactly where each hypothesis is spent. **Lipschitz** is what converts a bound on the side of a cube into a bound on the diameter of its image. Continuity alone gives no such conversion. $m \geq n$ is what makes $s_i^m \leq s_i^n$, i.e. what stops the volume from being measured with too few powers of s_i. The two counterexamples attack precisely these two steps.

AI-Note 19.5 (*Jordan null or Lebesgue null?*): As printed, the problem does not say which null sets it means, and for Jordan null images **(d)** is false without $\overline{N} \subset U$. Take $n = m = 1$, $U := (0,1)$ and $N := \{1/k : k \geq 2\}$, which is Jordan null: one interval $[0, \varepsilon/2]$ covers all but finitely many of its points. Enumerate $\mathbb{Q} \cap [0,1] = \{q_2, q_3, \dots\}$ and let f be the piecewise linear function with $f(1/k) := q_k$, linear on each $[1/(k+1), 1/k]$ and equal to q_2 on $[1/2, 1)$. Every compact subset of $(0,1)$ meets only finitely many of the linear pieces, so f is locally Lipschitz, and $m \geq n$. Yet $f(N) = \mathbb{Q} \cap [0,1]$ is not Jordan null (Remark 19.a.11), although it is Lebesgue null, as the general case above says it must be. What breaks is the single Lipschitz constant. The values $f(1/k) = q_k$ do not converge, so f is Lipschitz on no interval $(0, \eta)$, and the slopes of the pieces are unbounded near $0 \in \overline{N} \setminus U$.

Q.E.D.

Solution to AI-Exercise 19.b.27:

(a) Both families are nested and consist of compact sets whose boundaries are a circle and the four sides of a square respectively, each a Jordan null set, so both are Jordan measurable by Theorem 19.a.6. For the covering condition, $\operatorname{int}(D_\ell) = B_\ell(0)$ and $\operatorname{int}(Q_\ell) = (-\ell, \ell)^2$, and every point of $\mathbb{R}^2$ lies in both once ℓ exceeds its distance from the origin. Hence $\bigcup_\ell \operatorname{int}(D_\ell) = \bigcup_\ell \operatorname{int}(Q_\ell) = \mathbb{R}^2$, and Lemma 19.b.25 applies to each.

(b) In polar coordinates, using Theorem 20.a.1 with Jacobian r,

$$\int_{D_\ell} e^{-(x^2+y^2)}\, dx\, dy = \int_0^{2\pi}\!\!\int_0^{\ell} e^{-r^2}\, r\, dr\, d\varphi = 2\pi \left[-\frac{e^{-r^2}}{2} \right]_0^{\ell} = \pi\big(1 - e^{-\ell^2}\big) \xrightarrow[\ell \to \infty]{} \pi.$$

(c) On a square the integrand factorises, so Theorem 20.b.29 splits it:

$$\int_{Q_\ell} e^{-(x^2+y^2)}\, dx\, dy = \left(\int_{-\ell}^{\ell} e^{-x^2} dx \right) \left(\int_{-\ell}^{\ell} e^{-y^2} dy \right) = \left(\int_{-\ell}^{\ell} e^{-x^2} dx \right)^2 \xrightarrow[\ell \to \infty]{} I^2.$$

(d) Both limits compute the same number $\int_{\mathbb{R}^2} f$, by Lemma 19.b.25, so $I^2 = \pi$ and $I = \sqrt{\pi}$, the positive root since the integrand is positive.

The step that collapses without $f \geq 0$ is the identification in **(d)**. Non-negativity is what makes $\ell \mapsto \int_{K_\ell} f$ monotone, and monotonicity is what forces every exhaustion to the same limit. For a sign-changing integrand the disks and the squares are free to disagree, and Remark 19.b.24 gives the standard example of an integral whose value genuinely depends on how the domain is exhausted.

Remark 19.d.45 (*This is the argument Example 20.a.3 was already making*): The classical derivation of $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$ squares the integral, reads the product as an integral over $\mathbb{R}^2$, and switches to polar coordinates. Written out, that is precisely the comparison above: the squaring step is the square exhaustion and the polar step is the disk exhaustion, and the derivation works because the

two agree. What Lemma 19.b.25 adds is the permission to move between them, which the classical presentation uses without mentioning.

Q.E.D.

Proof of Exercise 19.b.28:

Both improper integrals can be evaluated by switching to polar coordinates, where $f(r\cos\theta, r\sin\theta) = r^{-2\alpha}$ and the area element is $r\,dr\,d\theta$.

(a) **False:** Over the region $A = \{r \geq 1\}$, the integral becomes $2\pi\int_1^\infty r^{1-2\alpha}\,dr$. The integral of r^p on $[1,\infty)$ converges if and only if $p < -1$. Thus, we need $1-2\alpha < -1$, which simplifies to $2\alpha > 2$, or $\alpha > 1$. The statement claims it converges if and only if $\alpha \leq 1$, which is the exact opposite.

(b) **False:** Over the region $B = \{r \leq 1\}$, the integral becomes $2\pi\int_0^1 r^{1-2\alpha}\,dr$. The integral of r^p on $(0,1]$ converges if and only if $p > -1$. This requires $1-2\alpha > -1$, meaning $\alpha < 1$. For $\alpha = 10/9 > 1$, the integral diverges.

(c) **False:** As established above, for $\alpha < 1$, the integral evaluates to

$$\int_B f = 2\pi\left[\frac{r^{2-2\alpha}}{2-2\alpha}\right]_0^1 = \frac{2\pi}{2-2\alpha} = \frac{\pi}{1-\alpha}.$$

Taking the limit as $\alpha \to 0^+$, we get $\lim_{\alpha\to 0^+}\frac{\pi}{1-\alpha} = \pi$. The statement incorrectly claims the limit is $\frac{\pi^2}{6}$.

Q.E.D.

Solution of Exercise 19.b.29:

The integrand is undefined at the origin, so $|f|$ is a continuous non-negative function on the open set $\mathbb{R}^2 \setminus \{(0,0)\}$ and Definition 19.b.23 applies there. The two boundary circles are Jordan null sets, so it makes no difference whether A and B are read as closed or open. Throughout, write $r := \sqrt{x^2+y^2}$, so that $|f| = |\cos(xy)|/r^4$, and recall from Remark 20.a.7 that a radial bound integrates as $2\pi\int g(r)\,r\,dr$.

Nothing here can be computed exactly, because $\cos(xy)$ has no antiderivative in polar coordinates. Both verdicts come from bounding that numerator, once from above and once from below.

(a) **True.** On all of $\mathbb{R}^2$, $|\cos(xy)| \leq 1$, so on A

$$\int_A |f| \;\leq\; \int_A \frac{1}{r^4} \;=\; 2\pi\int_1^\infty \frac{1}{r^4}\,r\,dr \;=\; 2\pi\int_1^\infty r^{-3}\,dr \;=\; 2\pi\left[\frac{-1}{2r^2}\right]_1^\infty = \pi \;<\; \infty.$$

(b) **False.** Here the numerator must be bounded **below**, and on B it is bounded below by a positive constant. Indeed $|xy| \leq \frac{1}{2}(x^2+y^2) \leq \frac{1}{2}$ for $(x,y) \in B$ by the inequality of arithmetic and geometric means, and cos is positive and decreasing on $[0,\frac{1}{2}] \subseteq [0,\pi/2)$, so $\cos(xy) \geq \cos\frac{1}{2} > 0$ throughout B. Consequently

$$\int_B |f| \;\geq\; \cos\tfrac{1}{2}\int_B \frac{1}{r^4} \;=\; 2\pi\cos\tfrac{1}{2}\int_0^1 r^{-3}\,dr \;=\; +\infty,$$

since $\int_0^1 r^p\,dr$ diverges for $p \leq -1$ and here $p = -3$.

(c) **False.** $B\setminus\{(0,0)\}$ is contained in $\mathbb{R}^2\setminus\{(0,0)\}$, so monotonicity of the improper integral in the domain gives $\int_{\mathbb{R}^2}|f| \geq \int_B|f| = +\infty$ by **(b)**. One divergent piece is enough; the convergence found in **(a)** cannot repair it.

Remark 19.d.46 (*The singularity is at the origin, not at infinity*): Set against Exercise 19.b.28, which asks the same two questions for $r^{-2\alpha}$, this block is the case $\alpha = 2$: the radial integrand is

$r^{-4} \cdot r = r^{-3}$, which converges at infinity and diverges at 0, and no exponent does both. That is the general picture in the plane. On A convergence needs $r^{-2\alpha+1}$ integrable at infinity, that is $\alpha > 1$; on B it needs the same expression integrable at 0, that is $\alpha < 1$. The two conditions are complementary, so $\int_{\mathbb{R}^2} r^{-2\alpha}$ diverges for every α, and the bounded oscillating factor $\cos(xy)$ changes none of it.

Q.E.D.

Solution to Exercise 19.a.17:

(a) **No.** Pick $p \in U$ and a closed cube $K \subseteq U$ centred at p, of side length $\ell > 0$; this exists because U is open. Suppose U were Lebesgue null and cover it by countably many cubes $Q_1, Q_2, \dots$ with $\sum_i \mu(Q_i) < \varepsilon$. Enlarge each Q_i to an **open** cube $Q_i' \supseteq Q_i$ concentric with it and with $\mu(Q_i') \le 2\mu(Q_i)$. The Q_i' form an open cover of the compact set K, so finitely many of them, say $Q_{i_1}', \dots, Q_{i_r}'$, already cover K. Subadditivity of the volume on finite unions then gives

$$\ell^n = \mu(K) \le \sum_{j=1}^{r} \mu(Q_{i_j}') \le 2 \sum_i \mu(Q_i) < 2\varepsilon.$$

Choosing $\varepsilon := \ell^n/2$ makes this read $\ell^n < \ell^n$, a contradiction. So U is not Lebesgue null.

(b) **No.** The set $E := \{1/k \mid k \in \mathbb{N}_{\ge 1}\} \subseteq \mathbb{R}$ is bounded and countable, hence Lebesgue null by **(c)** below applied to its one-point subsets. It is not closed: $1/k \to 0$ and $0 \notin E$.

(c) Let $E = \bigcup_{k \ge 1} E_k$ with each E_k Lebesgue null, and let $\varepsilon > 0$. Applying the definition to E_k with the tolerance $\varepsilon\, 2^{-k}$ produces countably many cubes $\{Q_{k,j}\}_{j \ge 1}$ covering E_k with $\sum_j \mu(Q_{k,j}) < \varepsilon\, 2^{-k}$. The doubly indexed family $\{Q_{k,j}\}_{k,j \ge 1}$ is a countable union of countable families, hence countable; it covers E; and its total volume is

$$\sum_{k \ge 1} \sum_{j \ge 1} \mu(Q_{k,j}) < \sum_{k \ge 1} \varepsilon\, 2^{-k} = \varepsilon,$$

all terms being non-negative, so the order of summation is immaterial. Therefore E is Lebesgue null.

(d) Let $\varepsilon > 0$. Since f is Riemann integrable it is bounded, and there is a partition $0 = t_0 < t_1 < \cdots < t_N = 1$ whose upper and lower sums differ by less than $\varepsilon/2$:

$$\sum_{i=1}^{N} (M_i - m_i)\, \Delta t_i < \frac{\varepsilon}{2}, \qquad M_i := \sup_{[t_{i-1}, t_i]} f, \quad m_i := \inf_{[t_{i-1}, t_i]} f, \quad \Delta t_i := t_i - t_{i-1}.$$

For $x \in [t_{i-1}, t_i]$ the point $(x, f(x))$ lies in the rectangle $R_i := [t_{i-1}, t_i] \times [m_i - \delta,\, M_i + \delta]$, where $\delta > 0$ is chosen below. Hence $\Gamma_f \subseteq \bigcup_{i=1}^{N} R_i$ and

$$\sum_{i=1}^{N} \mu(R_i) = \sum_{i=1}^{N} (M_i - m_i + 2\delta)\, \Delta t_i < \frac{\varepsilon}{2} + 2\delta,$$

using $\sum_i \Delta t_i = 1$. Taking $\delta := \varepsilon/8$ makes the total less than ε. Since ε was arbitrary and the cover is finite, Γ_f is in fact a **Jordan** null set, and a Jordan null set is Lebesgue null.

Remark 19.d.47 (*Each part is one hypothesis doing one job*)**:** Part **(a)** is where the enlargement trick belongs. A countable cover cannot be summed against a volume directly, because volume is only finitely additive here; compactness converts the countable cover into a finite one, and the factor 2 is the price of making the cubes open first. This is the same manoeuvre as in the proof of Lemma 19.a.7, and it is the standard way to get from Lebesgue-style covers back to Jordan-style ones.

Part **(d)** answers more than it was asked: the graph is Jordan null, not merely Lebesgue null, because the cover produced is finite. The integrability of f is exactly what supplies that. Boundedness

alone would not do, since there are bounded $f : [0,1] \to [0,1]$ whose graph is dense in the unit square, and a dense subset of the square has outer measure 1. Recall why: a finite cover by closed rectangles also covers the closure.

Read the four parts together and they say that null sets are large enough to be closed under countable unions, and small enough never to contain a ball. That pair is what makes *"negligible"* a usable notion, and **(c)** is the half this document uses most often without comment.

Q.E.D.

Proof of Exercise 19.a.18:

The set where the conclusion fails is $\{f > 0\}$, and the whole proof is the observation that it is a countable union of sets on which f is bounded below. For $t > 0$ put

$$A_t := \{x \in Q \mid f(x) \geq t\}, \qquad \text{so that} \qquad \{f > 0\} = \bigcup_{k=1}^{\infty} A_{1/k}.$$

By Exercise 19.a.17**(c)** it therefore suffices to show that each A_t is null.

Fix $t > 0$ and $\varepsilon > 0$. Since f is Riemann integrable with integral 0, there is a partition P of Q into boxes whose upper sum satisfies $U(f,P) < \varepsilon t$. Let $\mathcal{R}$ be the collection of boxes of P that meet A_t. On any $R \in \mathcal{R}$ there is a point where $f \geq t$, so $\sup_R f \geq t$; and $f \geq 0$ makes every term of the upper sum non-negative, so discarding the boxes outside $\mathcal{R}$ only decreases it. Hence

$$t \sum_{R \in \mathcal{R}} \operatorname{vol}_n(R) \;\leq\; \sum_{R \in \mathcal{R}} \Big(\sup_R f \Big) \operatorname{vol}_n(R) \;\leq\; U(f,P) \;<\; \varepsilon t,$$

and dividing by $t > 0$ gives $\sum_{R \in \mathcal{R}} \operatorname{vol}_n(R) < \varepsilon$. The boxes of $\mathcal{R}$ cover A_t by construction, so A_t admits a **finite** cover by boxes of total volume less than ε. As ε was arbitrary, A_t is null.

Remark 19.d.48 (*Each level set is Jordan null; only the union needs Lebesgue*): The covers produced above are finite, so the argument proves more than was asked: every A_t with $t > 0$ is **Jordan** null, not merely Lebesgue null (Remark 19.a.11). The final set $\{f > 0\}$ need not be, because Jordan content is not countably additive and a countable union of Jordan null sets can fail to be Jordan null; $\mathbb{Q} \cap [0,1]$ is the standing example, a countable union of points whose Jordan outer content is 1. So the countable union in the first display is exactly the step that forces the weaker notion, and it is the reason the conclusion is stated *"almost everywhere"* rather than on the complement of a Jordan null set.

Note also where the hypotheses are used, since two of them are doing quite different jobs. Non-negativity is what lets the upper sum be truncated to $\mathcal{R}$; without it the discarded boxes could contribute an arbitrarily negative amount and the estimate collapses. Riemann integrability is what makes an upper sum small at all. And the conclusion really cannot be improved to $f \equiv 0$: the function on $[0,1]$ that is 1 at each of finitely many points and 0 elsewhere is Riemann integrable with integral 0, and is not identically zero. For **continuous** f the stronger conclusion does hold, and needs no null sets: if $f(p) > 0$, continuity gives a whole box on which $f > f(p)/2$, forcing a positive integral.

Q.E.D.

Solution to AI-Exercise 19.c.41:

(a) Let $p \in \mathbb{R}^n$ and $\varepsilon > 0$. The single cube of side length $\delta := (\varepsilon/2)^{1/n}$ centred at p covers $\{p\}$ and has volume $\delta^n = \varepsilon/2 < \varepsilon$. A finite cover is in particular a countable one, so $\lambda^*(\{p\}) = 0$.

(b) Let $E = \{p_1, p_2, \dots\}$ be countable and let $\varepsilon > 0$. Cover p_k by a cube Q_k of volume $\varepsilon\, 2^{-k-1}$, as in **(a)**. Then $E \subseteq \bigcup_k Q_k$ and

$$\sum_{k\geq 1} \mathrm{vol}(Q_k) = \varepsilon \sum_{k\geq 1} 2^{-k-1} = \frac{\varepsilon}{2} < \varepsilon.$$

Hence $\lambda^*(E) = 0$. Since $\mathbb{Q}^n$ is a finite product of countable sets it is countable, so it is Lebesgue null.

(c) Suppose E is Lebesgue null and had an interior point p. Then E contains a cube K of some side length $\ell > 0$, and λ^* is monotone, so $\lambda^*(E) \geq \lambda^*(K) = \ell^n > 0$, contradicting $\lambda^*(E) = 0$. Hence $\mathrm{int}(E) = \emptyset$.

Remark 19.d.49: Part **(b)** is the whole difference between the two theories in three lines. The geometric series is what lets infinitely many cubes have finite total volume, and it is available only because the cubes are allowed different sizes, which is exactly the point of Remark 19.a.12. Part **(c)** then says a null set is small in the topological sense too, though the converse fails badly: there are nowhere dense sets of positive measure (*"fat"* Cantor sets). Nor is *"null"* the same as *"countable"*: the Cantor set of Example 19.a.8 is null and uncountable.

Q.E.D.

Solution to AI-Exercise 19.c.42:

(a) Split $[0,1]$ into the three sets on which f is constant:

$$E_1 := \mathbb{Q} \cap [0,1], \qquad E_2 := [0, \tfrac{1}{2}) \setminus \mathbb{Q}, \qquad E_3 := [\tfrac{1}{2}, 1] \setminus \mathbb{Q}.$$

These are disjoint, measurable and cover $[0,1]$, and $f = 1 \cdot \chi_{E_1} + 2 \cdot \chi_{E_2} + 3 \cdot \chi_{E_3}$ is a simple function. By AI-Exercise 19.c.41**(b)**, $\lambda(E_1) = 0$, so removing it from $[0, \frac{1}{2})$ and $[\frac{1}{2}, 1]$ changes nothing: $\lambda(E_2) = \lambda(E_3) = \frac{1}{2}$. Therefore

$$\int_{[0,1]} f \, d\lambda = 1 \cdot 0 + 2 \cdot \tfrac{1}{2} + 3 \cdot \tfrac{1}{2} = \frac{5}{2}.$$

(b) f is continuous at **no** point of $[0,1]$. Fix x and any $\delta > 0$. The interval $(x-\delta, x+\delta) \cap [0,1]$ contains both rational and irrational points, since each set is dense, so f takes the value 1 there and also a value in $\{2,3\}$. The oscillation of f on every neighbourhood of x is therefore at least 1, and f is discontinuous at x.

The discontinuity set is thus all of $[0,1]$, which has measure 1 and is not Lebesgue null. By Theorem 19.c.38**(a)**, f is not Riemann integrable.

Remark 19.d.50: The number $5/2$ is what one gets by ignoring the rationals entirely, and that is the honest summary of what the Lebesgue integral does here: it reads f as the step function $2 \cdot \chi_{[0,\frac{1}{2})} + 3 \cdot \chi_{[\frac{1}{2},1]}$, which it is, away from a null set. The Riemann theory cannot perform that identification, because its upper sums must take the supremum over whole intervals and every interval meets $\mathbb{Q}$.

Q.E.D.

Solution to AI-Exercise 19.c.43:

(a) For $0 \leq x < 1$ we have $x^k \to 0$, and $1^k = 1$ for every k. So the pointwise limit is $f = \chi_{\{1\}}$. The convergence is not uniform: taking $x_k := 2^{-1/k} \in (0,1)$ gives $f_k(x_k) = \frac{1}{2}$, so $\sup_{x\in[0,1]} |f_k(x) - f(x)| \geq \frac{1}{2}$ for every k.

(b) Every f_k satisfies $0 \le f_k \le 1$ on $[0,1]$, and the constant $g \equiv 1$ is integrable there, with $\int_{[0,1]} g\,d\lambda = 1 < \infty$. Dominated convergence applies and gives

$$\lim_{k\to\infty} \int_{[0,1]} f_k\,d\lambda = \int_{[0,1]} \chi_{\{1\}}\,d\lambda = \lambda(\{1\}) = 0.$$

(c) Directly, $\int_0^1 x^k\,dx = \frac{1}{k+1} \to 0$, in agreement.

Remark 19.d.51: The point is the contrast in hypotheses, not the value. The limit here is a function this course would rather not integrate at all, and the uniform-convergence theorem does not apply, yet a single constant bound settles the exchange. The dominating function is almost always this easy to find, which is why dominated convergence, and not any new set function, is what practitioners actually use the Lebesgue theory for.

Q.E.D.

Solution to AI-Exercise 19.c.44:

(a) f is constant on each $[k, k+1)$, an interval of length 1, so for integer $N \ge 1$

$$\int_1^N f(x)\,dx = \sum_{k=1}^{N-1} \frac{(-1)^{k+1}}{k}.$$

These are the partial sums of the alternating harmonic series, which converges to $\log 2$. For non-integer upper limits the integral differs from the nearest such partial sum by at most $1/\lfloor R \rfloor \to 0$, so the improper integral converges with $\int_1^\infty f(x)\,dx = \log 2$.

(b) $|f| = 1/k$ on $[k, k+1)$, hence

$$\int_1^N |f(x)|\,dx = \sum_{k=1}^{N-1} \frac{1}{k} \xrightarrow{N\to\infty} \infty,$$

the harmonic series diverging. Since Lebesgue integrability of f means $\int |f|\,d\lambda < \infty$, the function f is not Lebesgue integrable.

(c) That theorem compares the two integrals for a **bounded** function on a **box**, where the Riemann integral is proper. Here the domain $[1,\infty)$ is unbounded and the integral exists only as a limit $\lim_{R\to\infty} \int_1^R$, which is a different construction: it is allowed to exploit cancellation between the positive and negative humps, and it is that cancellation the absolute-value requirement forbids.

Remark 19.d.52 (*Neither theory contains the other*): This is the standard corrective to *"Lebesgue generalises Riemann"*. On a box the Lebesgue integral really is the stronger notion. Once improper integrals enter, the two constructions answer different questions, and $\int_0^\infty \sin(x)/x\,dx = \pi/2$ is the famous instance of the same phenomenon. In practice one keeps both: the Lebesgue integral for limits and for several-variable arguments, the improper Riemann integral for conditionally convergent ones.

Q.E.D.

Solution of Exercise 19.a.19:

Throughout, unwind the definition in Remark 19.a.11: X is Jordan null means that for every $\varepsilon > 0$ there are **finitely many** intervals $I_1, \dots, I_N \subset \mathbb{R}$ with

$$X \subseteq \bigcup_{k=1}^{N} I_k \qquad \text{and} \qquad \sum_{k=1}^{N} |I_k| < \varepsilon.$$

Both parts are proved by covering the product with the products of such a cover; what differs is only how the total area is estimated.

(a) Let $\varepsilon > 0$ and put $\delta := \sqrt{\varepsilon}$. Choose intervals $I_1, \dots, I_N$ covering X with $\sum_k |I_k| < \delta$. Then

$$X \times X \subseteq \bigcup_{j=1}^{N} \bigcup_{k=1}^{N} I_j \times I_k,$$

a cover by N^2 rectangles, still finitely many. Their total area factorises:

$$\sum_{j=1}^{N} \sum_{k=1}^{N} |I_j|\,|I_k| = \left(\sum_{j=1}^{N} |I_j| \right) \left(\sum_{k=1}^{N} |I_k| \right) < \delta^2 = \varepsilon.$$

Since ε was arbitrary, $X \times X$ is Jordan null.

(b) Let $\varepsilon > 0$ and choose $I_1, \dots, I_N$ covering X with $\sum_k |I_k| < \varepsilon$, with no square root this time. Then

$$X \times [0,1] \subseteq \bigcup_{k=1}^{N} I_k \times [0,1],$$

again a finite cover by rectangles, of total area

$$\sum_{k=1}^{N} |I_k| \cdot 1 = \sum_{k=1}^{N} |I_k| < \varepsilon.$$

Remark 19.d.53 (*Where the two parts differ, and what the difference is telling you*): The two estimates are δ^2 and $\delta \cdot 1$, and the exponent is the whole story: the first factor's total length enters **linearly** in both, while the second factor contributes its own total size as a multiplier.

That is why **(b)** works with a factor that is emphatically **not** null. Being null in $\mathbb{R}$ is not required of $[0,1]$; being **bounded** is, and that is all. The same proof gives $X \times B$ Jordan null for any bounded $B \subseteq \mathbb{R}$, with the total area bounded by $\varepsilon \cdot \operatorname{diam}(B)$.

Drop boundedness and the statement fails: $X \times \mathbb{R}$ is not Jordan null for any non-empty X, and not for the interesting reason. Jordan null sets are bounded by definition (a finite cover by intervals is bounded), so $X \times \mathbb{R}$ is disqualified before any area is computed. Compare Remark 19.a.11: the Lebesgue notion, allowing countable covers, does call $\{0\} \times \mathbb{R}$ null, and this is one of the cleanest places to see the two notions come apart.

Q.E.D.

Solution of Exercise 19.a.20:

Two facts do all the work: M is Jordan measurable, and f is continuous and bounded. The criterion of Remark 19.a.5 then applies.

M is bounded, being a subset of $\overline{B}_2(0)$.

The removed set is finite. Write $L := \{x \in \mathbb{R}^4 \mid x_i \in \mathbb{Z}\} = \mathbb{Z}^4$ and $S := L \cap \overline{B}_2(0)$, so that $M = \overline{B}_2(0) \setminus S$. A lattice point $x \in S$ has $\sum_i x_i^2 \leq 4$ with each x_i an integer, so each $|x_i| \leq 2$ and $S \subseteq \{-2,-1,0,1,2\}^4$. In particular S is finite.

M is Jordan measurable. A single point is a Jordan null set (cover it by one cube of side ε), hence Jordan measurable, and a finite union of Jordan measurable sets is Jordan measurable, so S is. The closed ball is Jordan measurable: its boundary is the sphere $\partial B_2(0)$, a Jordan null set, so Theorem 19.a.6 applies. Then $M = \overline{B}_2(0) \setminus S$ is Jordan measurable by Proposition 19.a.10, and since $\mu(S) = 0$ we even get $\mu(M) = \mu(\overline{B}_2(0))$.

f is continuous and bounded on $\overline{M}$. It is a composition of polynomials, the square root and cos: the map $x \mapsto |x|^2$ is a polynomial, $t \mapsto t^{3/2}$ is continuous on $[0,\infty)$, and cos is continuous on $\mathbb{R}$.

So f is continuous on all of $\mathbb{R}^4$, in particular on $\overline{M} = \overline{B}_2(0)$, and $|f| \le 1$ everywhere because it is a cosine.

Conclusion. M is Jordan measurable and f is continuous on $\overline{M}$. The criterion is stated for non-negative functions, so apply it to the continuous non-negative parts $f^{\pm} := \max\{\pm f, 0\}$ and use linearity: $f = f^+ - f^-$ is Riemann integrable over M.

Remark 19.d.54 (*What the exercise is actually testing*): Neither the dimension 4 nor the elaborate formula for f matters; both are there to discourage computation. The formula is chosen so that f is manifestly continuous but has no closed-form integral, and the removed set is chosen to look infinite ($\mathbb{Z}^4$ is) while being finite **after intersection with the ball**. Establishing that is the one step of the problem that is not bookkeeping.

Note also what the removal does **not** affect. Deleting a Jordan null set changes neither Jordan measurability nor the value of the integral, so $\int_M f = \int_{\overline{B}_2(0)} f$. It is precisely this insensitivity that fails for countable removals: $[0,1] \setminus \mathbb{Q}$ is not Jordan measurable at all (Remark 19.a.11).

Q.E.D.

Solution of Exercise 19.a.21:

The answer is **(a)**, $\mu(V) < \infty$.

A diffeomorphism Φ carrying U onto V is in particular continuous, and U is compact (closed and bounded in $\mathbb{R}^n$, Theorem 7.b.28). A continuous image of a compact set is compact, so V is compact, hence bounded, and a bounded Jordan measurable set has finite measure.

Options **(b)**, **(c)** and **(d)** all assert something about the **size** of $\mu(V)$, and a diffeomorphism controls nothing of the kind. The change-of-variables formula (Theorem 20.a.1) gives $\mu(V) = \int_U |\det \mathrm{J}\,\Phi|$, and the Jacobian determinant is at the diffeomorphism's disposal: the dilation $\Phi(x) := \lambda x$ turns $\mu(U) = 1$ into $\mu(V) = \lambda^n$, which can be made equal to, greater than, or less than 1 at will. So all three fail, and they fail for the same reason. Q.E.D.

Solution of Exercise 19.a.22:

The answer is **(a)**, the "*layer-cake formula*"

$$\int_\Omega f = \int_0^{\max f} \mu\big(\{x \in \Omega \mid f(x) > t\}\big)\, dt.$$

For a positive step function the identity is a rearrangement of a finite sum. Let f take the distinct positive values $c_1 < \cdots < c_N$ on Jordan measurable pieces $A_1, \dots, A_N$ partitioning Ω, so that $\int_\Omega f = \sum_k c_k\, \mu(A_k)$. For $t \in [c_{k-1}, c_k)$ the superlevel set $\{f > t\}$ is exactly $A_k \cup \cdots \cup A_N$, whose measure is $\sum_{j \ge k} \mu(A_j)$. Hence

$$\int_0^{\max f} \mu(\{f > t\})\, dt = \sum_{k=1}^{N} (c_k - c_{k-1}) \sum_{j \ge k} \mu(A_j) = \sum_{j=1}^{N} \mu(A_j) \sum_{k \le j} (c_k - c_{k-1}) = \sum_{j=1}^{N} c_j\, \mu(A_j),$$

with $c_0 := 0$, the middle step being an exchange of the order of summation and the last a telescoping sum. The two sides agree.

Of the alternatives, **(c)** is zero for all but finitely many t and so integrates to 0; **(b)** and **(d)** use **sub**level sets, which run the wrong way: their integral is $\mu(\Omega) \max f - \int_\Omega f$, not $\int_\Omega f$.

Remark 19.d.55 (*Why the layer-cake formula is worth remembering*): It computes an integral by slicing the **range** rather than the domain, which is exactly the move that separates the Lebesgue integral from the Riemann integral (Section 19.c). Here it is proved for step functions only, but that is the general case in disguise: the Lebesgue integral of a non-negative measurable function satisfies

the same formula, with μ the Lebesgue measure and the right-hand side an improper Riemann integral of a monotone function of t, and some treatments take it as the definition.

Q.E.D.

Change of Variables, Fubini and Feynman's Trick (Chapter 20)

The previous chapter was about which integrals exist. This one is about computing them, and three tools do nearly all of that work. Change of variables trades an awkward domain for a convenient one and pays a Jacobian factor for the trade. Fubini reduces an n-dimensional integral to n one-dimensional ones. Differentiating under the integral sign reaches integrals with no elementary antiderivative at all, by turning the integral into an ODE in a parameter it never had. A short closing section puts all three to work at once on the centroid of a region, which is the first quantity in the course that is defined by an integral rather than computed by one.

Change of variables comes first, and its examples already evaluate the transformed integral one variable at a time; what licenses that is Fubini's theorem (Theorem 20.b.29) in the section after. Nothing circular is going on, and a reader who prefers the logical order can read Section 20.b first.

Change of variables (Section 20.a)

Theorem 20.a.1 (*Change of variables*): Suppose $U, V \subseteq \mathbb{R}^n$ are open sets and $\Phi : U \to V$ is a C^1-diffeomorphism. If $A \subseteq U$ is Jordan measurable with $\overline{A} \subseteq U$, and $f : A \to \mathbb{R}$ is continuous, then

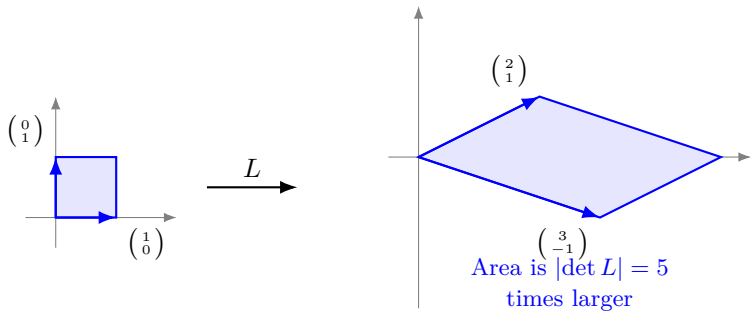
$$\int_A f(x)\,dx = \int_{\Phi(A)} \frac{f \circ \Phi^{-1}(y)}{\left|\det \mathrm{J}\,\Phi\big(\Phi^{-1}(y)\big)\right|}\,dy.$$

AI-Note 20.1: Jordan measurability is defined in Definition 19.a.3, in the previous chapter. For this chapter it is enough to know that boxes, and the bounded regions cut out by finitely many smooth curves that appear in the examples, are all Jordan measurable, because their boundaries are Jordan null (Theorem 19.a.6).

On the shape of the formula: it is often quoted the other way round, as $\int_{\Phi(A)} f = \int_A (f\circ\Phi)\,|\det \mathrm{J}\,\Phi|$. The version above is the same statement with Φ pushed to the other side, which is why the determinant appears **divided** rather than multiplied. Which form is convenient depends on whether it is Φ or Φ^{-1} that you can write down. In Example 20.b.37 below it is Φ, so the dividing form is the one that saves work.

The theorem is not proved here, but the two steps that carry its proof are, and both sit at the end of this section. Exercise 20.a.26 settles the linear case, where the determinant enters and everything else is bookkeeping. Example 20.a.27 settles the geometric step that a Lipschitz estimate cannot reach, namely that a diffeomorphism close to the identity cannot let a cube collapse.

Remark 20.a.2 (*Intuition for the Jacobian determinant*): From linear algebra, we know that for a linear map given by a matrix L, the absolute value of its determinant $|\det L|$ describes how much the map scales volumes. For example, the map $L = \left(\begin{smallmatrix} 2 & 3 \\ 1 & -1 \end{smallmatrix}\right)$ expands areas by a factor of $|\det L| = |-2-3| = 5$.



The change of variables formula generalizes this to non-linear maps: the Jacobian determinant acts as the local volume scaling factor.

Example 20.a.3 (*Gaussian integral via polar coordinates*): A classic application of the change of variables formula is computing the Gaussian integral $I := \int_{-\infty}^{\infty} e^{-x^2}\,dx$. We consider its square:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,dy\right) = \int_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{dvol}(x,y).$$

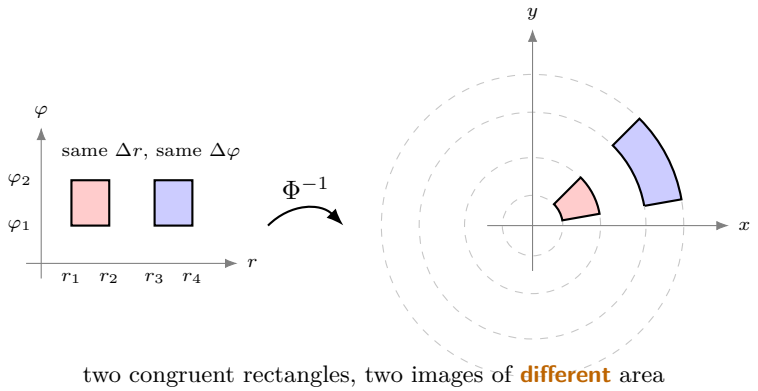
We use polar coordinates, but strictly following Theorem 20.a.1, we work on the slit plane

$$\mathbb{R}^2_* := \mathbb{R}^2 \setminus \{(x,0) : x \le 0\}$$

and define the forward map $\Phi : \mathbb{R}^2_* \to (0,\infty)\times(-\pi,\pi)$ by

$$\Phi(x,y) := \left(\sqrt{x^2+y^2}, \operatorname{atan2}(y,x)\right) =: (r,\varphi).$$

Both sets are open and Φ is a C^∞-diffeomorphism between them, with the slit removed precisely because the angle cannot be chosen continuously across it.



two congruent rectangles, two images of **different** area
⇒ we must compensate with the (Jacobian) determinant

The Jacobian matrix is

$$\mathrm{J}\,\Phi(x,y) = \begin{pmatrix} \frac{x}{\sqrt{x^2+y^2}} & \frac{y}{\sqrt{x^2+y^2}} \\ \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{pmatrix}.$$

Its determinant is

$$\det \mathrm{J}\,\Phi(x,y) = \frac{x^2}{(x^2+y^2)^{3/2}} + \frac{y^2}{(x^2+y^2)^{3/2}} = \frac{1}{\sqrt{x^2+y^2}} = \frac{1}{r}.$$

Applying the theorem (the discarded slit is vol_2-null, so removing it changes neither side), we get

$$I^2 = \int_{\Phi(\mathbb{R}^2_*)} \frac{e^{-r^2}}{|1/r|}\,dr\,d\varphi = \int_{-\pi}^{\pi} d\varphi \int_0^{\infty} r e^{-r^2}\,dr = 2\pi\cdot\left[-\frac{1}{2}e^{-r^2}\right]_0^{\infty} = 2\pi\cdot\frac{1}{2} = \pi.$$

Since $I > 0$, we conclude that $I = \sqrt{\pi}$.

AI-Note 20.2: One hypothesis is still being stretched, and it is worth naming rather than hiding. Theorem 20.a.1 asks for a Jordan measurable A with $\overline{A}$ inside the domain, and $\mathbb{R}^2_*$ is unbounded, hence not Jordan measurable at all: every integral in this computation is improper. What licenses it is Section 19.b: the integrand is non-negative, so Lemma 19.b.25 says any exhaustion by compact Jordan measurable sets gives the same value, and one may run the change of variables on each piece and pass to the limit. AI-Exercise 19.b.27 does exactly this for this integral, with two different exhaustions. The slit is genuinely necessary; the unboundedness is genuinely harmless, but only because $e^{-(x^2+y^2)} \ge 0$.

Example 20.a.4 (*Change of variables on a diamond*): Compute $I := \int_E e^{x+y}(x-y)\,d\mathrm{vol}(x,y)$, where E is the region bounded by the lines $y=x$, $y=-x+2$, $y=x-2$, and $y=-x$.

Writing out the inequalities, $E = \{y \le x\} \cap \{y \le -x+2\} \cap \{y \ge -x\} \cap \{y \ge x-2\}$. Rearranging these gives $0 \le x-y \le 2$ and $0 \le x+y \le 2$. This suggests the substitution $u := x+y$ and $v := x-y$. The new domain is the square $V = [0,2] \times [0,2]$. The inverse map is $\Phi^{-1}(u,v) = (x,y) = \left(\frac{u+v}{2}, \frac{u-v}{2}\right)$. Its Jacobian matrix is $\mathrm{J}\,\Phi^{-1}(u,v) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$, so $\left|\det \mathrm{J}\,\Phi^{-1}\right| = \left|-\frac{1}{4}-\frac{1}{4}\right| = \frac{1}{2}$. By the change of variables formula:

$$I = \int_0^2 \int_0^2 e^u v \left|\det \mathrm{J}\,\Phi^{-1}\right| du\,dv = \frac{1}{2}\int_0^2 \int_0^2 e^u v\,du\,dv = \frac{1}{2}\left[e^u\right]_0^2 \left[\frac{v^2}{2}\right]_0^2 = \frac{1}{2}(e^2-1)\cdot 2 = e^2-1.$$

Example 20.a.5 (*Improper integral via polar coordinates*): Compute $\int_{\mathbb{R}^2} \frac{1}{(1+x^2+y^2)^2}\,d\mathrm{vol}(x,y)$. This integral is improper because the domain is unbounded. Since the integrand is non-negative and continuous, its Riemann improper integral is well-defined, and we can compute it using polar coordinates $x = r\cos\theta, y = r\sin\theta$. The Jacobian determinant is r. The domain $\mathbb{R}^2$ corresponds to $r \in [0,\infty)$ and $\theta \in [0,2\pi)$.

$$\int_{\mathbb{R}^2} \frac{1}{(1+x^2+y^2)^2}\,dx\,dy = \int_0^{2\pi}\int_0^{\infty} \frac{r}{(1+r^2)^2}\,dr\,d\theta.$$

Substituting $u = r^2$, $du = 2r\,dr$, the inner integral is $\frac{1}{2}\int_0^\infty \frac{1}{(1+u)^2}\,du = \frac{1}{2}\left[\frac{-1}{1+u}\right]_0^\infty = \frac{1}{2}$. The outer integral then gives $\frac{1}{2}\int_0^{2\pi} d\theta = \pi$.

Example 20.a.6 (*Volume of a sphere*): Let $A := \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 \le 1\}$ be the unit ball. Given that A is Jordan measurable, its volume is $\mu(A) = \int_A 1\,d\mathrm{vol}(x,y,z)$. We can compute this using spherical coordinates:

$$\Phi(r,\theta,\varphi) = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}.$$

Up to a Jordan null set (which does not change the volume), the domain A corresponds to the box $V = (0,1)\times(0,\pi)\times(-\pi,\pi)$. The Jacobian determinant is $\det \mathrm{J}\,\Phi(r,\theta,\varphi) = r^2\sin\theta$, which is positive on this domain. Applying the change of variables formula and Fubini's theorem (which allows slicing up the integral):

$$\begin{aligned}
\mu(A) &= \int_{\Phi^{-1}(A)} \left|\det \mathrm{J}\,\Phi(r,\theta,\varphi)\right| dr\,d\theta\,d\varphi \\
&= \int_{-\pi}^{\pi}\int_0^{\pi}\int_0^1 r^2\sin\theta\,dr\,d\theta\,d\varphi \\
&= \left(\int_{-\pi}^{\pi} d\varphi\right)\left(\int_0^{\pi}\sin\theta\,d\theta\right)\left(\int_0^1 r^2\,dr\right) \\
&= (2\pi)\cdot(2)\cdot\left(\frac{1}{3}\right) = \frac{4\pi}{3}.
\end{aligned}$$

Remark 20.a.7 (*Polar coordinates as slicing into shells*): The factor r^2 in the computation above, and the r in the two-dimensional case, are usually produced by grinding out a Jacobian determinant. There is a reading that gets them without any computation, and that generalises to $\mathbb{R}^n$ at no extra cost.

Slice $\mathbb{R}^n$ into spheres centred at the origin. The sphere of radius r is an $(n-1)$-dimensional submanifold, and it is the unit sphere scaled by r; since scaling by r multiplies $(n-1)$-dimensional volume by r^{n-1}, its volume is

$$\mathrm{vol}_{n-1}\left(r\,\mathbb{S}^{n-1}\right) = C_n\,r^{n-1}, \qquad C_n := \mathrm{vol}_{n-1}\left(\mathbb{S}^{n-1}\right).$$

A shell between radii r and $r+dr$ is that sphere thickened by dr, so it contributes $C_n r^{n-1}\,dr$ of n-dimensional volume. Adding the shells,

$$\int_{B_R(0)} f(|x|)\,dx = C_n \int_0^R f(r)\,r^{n-1}\,dr \qquad \text{for radial } f.$$

The powers now need no memorising: $n=2$ gives r^1 and $n=3$ gives r^2, matching the two Jacobians above, and $C_2=2\pi$, $C_3=4\pi$ are the circumference of the unit circle and the area of the unit sphere. Taking $f\equiv 1$, $n=3$, $R=1$ reproduces $4\pi\int_0^1 r^2\,dr = 4\pi/3$ in one line.

This is a genuine shortcut only for **radial** integrands, where the angular integral is a constant that can be pulled out; when f depends on the angles as well, the full change of variables is unavoidable. But it explains where the radial power comes from, which the determinant computation does not.

Remark 20.a.8 (*Reference: the standard coordinate Jacobians*): The polar, cylindrical and spherical substitutions recur throughout this chapter and the next several; collected here in one place, with $r\geq 0$ throughout and each parametrizing its target domain up to a Jordan null set:

Coordinates	Transformation	$\lvert\det \mathrm{J}\,\Phi\rvert$
Polar ($\mathbb{R}^2$)	$\Phi(r,\varphi)=(r\cos\varphi,\ r\sin\varphi)$	r
Cylindrical ($\mathbb{R}^3$)	$\Phi(r,\varphi,z)=(r\cos\varphi,\ r\sin\varphi,\ z)$	r
Spherical ($\mathbb{R}^3$)	$\Phi(r,\theta,\varphi)=(r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$	$r^2\sin\theta$

The angular ranges are $\varphi\in(-\pi,\pi)$ throughout, and $\theta\in(0,\pi)$ ("*colatitude*", measured from the positive z-axis) for the spherical case. Cylindrical coordinates are just polar coordinates in the first two variables, carried along unchanged in z, which is exactly why the Jacobian is the same r as in the planar case: the extra dz contributes a factor of 1. Spherical coordinates are used already in Example 20.a.6 above; cylindrical coordinates are the natural choice whenever a region has rotational symmetry about an axis without being a ball, e.g. a solid cylinder or cone.

Example 20.a.9 (*Polar coordinates, checked item by item*): Consider $\Psi:\mathbb{R}\times[0,\infty)\to\mathbb{R}^2$, $\Psi(\theta,r):=(r\cos\theta,\ r\sin\theta)$, and note the argument order: the angle first. Four statements, of which two are true.

(a) Ψ **restricted to** $(0,2\pi]\times(0,\infty)$ **is injective: true.** Suppose $\Psi(\theta_1,r_1)=\Psi(\theta_2,r_2)$. Taking Euclidean norms gives $r_1=r_2=:r>0$, and dividing by r leaves $(\cos\theta_1,\sin\theta_1)=(\cos\theta_2,\sin\theta_2)$, so $\theta_1-\theta_2\in 2\pi\mathbb{Z}$. A half-open interval of length 2π contains at most one representative of each such class, so $\theta_1=\theta_2$.

(b) Ψ **restricted to** $(0,2\pi]\times(0,\infty)$ **is surjective: false.** Its image is $\mathbb{R}^2\setminus\{(0,0)\}$: every non-zero point is reached, by **(a)** exactly once, and the origin is reached only from $r=0$, which the restriction excludes.

(c) $\det \mathrm{J}\,\Psi(\theta,r)=r^2\sin\theta\cos\theta$**: false.** The columns of the Jacobi matrix are the partial derivatives in θ and in r, in that order:

$$\mathrm{J}\,\Psi(\theta,r)=\begin{pmatrix}-r\sin\theta & \cos\theta\\ r\cos\theta & \sin\theta\end{pmatrix},\qquad \det \mathrm{J}\,\Psi(\theta,r)=-r\sin^2\theta-r\cos^2\theta=-r.$$

The magnitude $|\det \mathrm{J}\,\Psi|=r$ is the entry in Remark 20.a.8; the minus sign appears only because the exam lists the angle before the radius, which swaps two columns.

(d) $\mathrm{vol}_2(B_R)=2\pi\int_0^R r\,dr$ **for** $B_R=\{x^2+y^2<R^2\}$**: true.** By Theorem 20.a.1 with the weight $|\det \mathrm{J}\,\Psi|=r$, and since the integrand 1 does not depend on θ, the angular integral contributes a factor 2π. The right-hand side is $2\pi\cdot R^2/2=\pi R^2$, as it should be. This is also the $n=2$ case of Remark 20.a.7.

The proposed formula in **(c)** is worth a second look, because it is not a random wrong answer: $r^2 \sin\theta$ is the **spherical** Jacobian, and $\cos\theta$ is what a misremembered product rule adds to it. There is a cheap guard against importing one row of the table into another. Every system in it has the form $\Phi(r,\omega) = r\,u(\omega)$ for a map u into the unit sphere, so Φ is homogeneous of degree 1 in r. The radial column of $\mathrm{J}\,\Phi$ is then $u(\omega)$, free of r, while each of the $n-1$ angular columns is $r\,\partial u/\partial\omega_i$ and so carries exactly one factor r. Multilinearity of the determinant pulls all $n-1$ of them out, leaving $|\det \mathrm{J}\,\Phi| = r^{n-1}$ times a function of the angles alone. In $\mathbb{R}^2$ that power is r^1, and the proposal carries r^2.

Info: *This example is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the eighth block of true/false questions.*

Example 20.a.10 (*Volume of an ellipse*): Compute the area (2-dimensional volume) of the ellipse $E := \{(x,y) \in \mathbb{R}^2 \mid (x/a)^2 + (y/b)^2 < 1\}$ for $a, b > 0$. We use the transformation $\Phi(r,\varphi) = (ar\cos\varphi, br\sin\varphi)$. This maps the rectangle $V = (0,1)\times(0,2\pi)$ bijectively onto E minus a null set (the positive x-axis). The Jacobian matrix is

$$\mathrm{J}\,\Phi(r,\varphi) = \begin{pmatrix} a\cos\varphi & -ar\sin\varphi \\ b\sin\varphi & br\cos\varphi \end{pmatrix},$$

and its determinant is $\det \mathrm{J}\,\Phi(r,\varphi) = abr(\cos^2\varphi + \sin^2\varphi) = abr$. Thus, by the change of variables formula,

$$\int_E 1\,dx\,dy = \int_0^{2\pi}\int_0^1 abr\,dr\,d\varphi = \left(\int_0^{2\pi} ab\,d\varphi\right)\left(\int_0^1 r\,dr\right) = 2\pi ab\cdot\frac{1}{2} = \pi ab.$$

Example 20.a.11 (*Area of a cardioid*): Compute the area of the region Ω given in polar coordinates by $\Omega := \{(r,\varphi) \mid \varphi \in [0,2\pi),\ 0 \le r \le 1+\cos\varphi\}$. Using polar coordinates $\Phi(r,\varphi) = (r\cos\varphi, r\sin\varphi)$, the area is

$$\int_\Omega 1\,dx\,dy = \int_0^{2\pi}\int_0^{1+\cos\varphi} r\,dr\,d\varphi = \int_0^{2\pi}\left[\frac{r^2}{2}\right]_{r=0}^{r=1+\cos\varphi} d\varphi = \frac{1}{2}\int_0^{2\pi}(1+\cos\varphi)^2\,d\varphi.$$

Expanding the integrand: $(1+\cos\varphi)^2 = 1 + 2\cos\varphi + \cos^2\varphi$. We know $\int_0^{2\pi}\cos\varphi\,d\varphi = 0$ and $\int_0^{2\pi}\cos^2\varphi\,d\varphi = \pi$. Thus,

$$\int_\Omega 1\,dx\,dy = \frac{1}{2}(2\pi + 0 + \pi) = \frac{3\pi}{2}.$$

Example 20.a.12 (*Combining substitution and Fubini*): Compute $\int_\Omega (x^3y + xy^3)\,dx\,dy$ where $\Omega := \{(x,y) \in \mathbb{R}^2 \mid x,y>0,\ 1/x \le y \le 3/x,\ 1 \le x^2 - y^2 \le 4\}$. The integrand factors as $xy(x^2+y^2)$. The boundary inequalities are equivalent to $1 \le xy \le 3$ and $1 \le x^2 - y^2 \le 4$, which strongly suggests defining $u := xy$ and $v := x^2 - y^2$. The domain in the (u,v)-plane is the rectangle $R = [1,3]\times[1,4]$. Let $\Psi(x,y) = (xy, x^2-y^2) = (u,v)$. The Jacobian of this inverse transformation is

$$\mathrm{J}\,\Psi(x,y) = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix}, \qquad \det \mathrm{J}\,\Psi(x,y) = -2y^2 - 2x^2 = -2(x^2+y^2).$$

For the forward transformation $\Phi = \Psi^{-1}$, we have $\left|\det \mathrm{J}\,\Phi(u,v)\right| = \frac{1}{|\det \mathrm{J}\,\Psi(x,y)|} = \frac{1}{2(x^2+y^2)}$. Applying the change of variables formula:

$$\int_\Omega xy(x^2+y^2)\,dx\,dy = \int_R u(x^2+y^2)\frac{1}{2(x^2+y^2)}\,du\,dv = \frac{1}{2}\int_1^4\int_1^3 u\,du\,dv = \frac{1}{2}\cdot 3\cdot\left[\frac{u^2}{2}\right]_1^3 = \frac{3}{2}\cdot 4 = 6.$$

Exercise 20.a.13 (*Change of variables and Jacobians* (important)): For each of the following domains and change of variables find the jacobian and the appropriate transformed domain. There is no need to actually compute the integrals!

(a) $A := \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0,\ x_2 < x_1,\ 1 < x_1^2 + x_2^2 < 4\}$ and $x_1 = r\cos\theta,\ x_2 = r\sin\theta$. Using the change of variables formula, complete the dots in the following formula

$$\int_A x_1^2 \sin(x_2)\, dx_1 dx_2 = \int_{\dots} \dots\, dr\, d\theta$$

(b) $B := \{(x, y) \in \mathbb{R}^2 \mid 1 < xy < 2,\ x^2 < y < 2x^2\}$ and $u := xy,\ v := x^2$. Using the change of variables formula, complete the dots in the following formula

$$\int_B y^2 e^{-xy}\, dx\, dy = \int_{\dots} \dots\, du\, dv$$

(c) $C := \{(x, y, z) \in \mathbb{R}^3 \mid 1 < z - 2y < 2,\ 0 < z < 1,\ -2 < 3x + y + z < 0\}$ and $u := z$, $v := z - 2y$, $w := 3x + y + z$. Using the change of variables formula, complete the dots in the following formula

$$\int_C xyz\, dx\, dy = \int_{\dots} \dots\, du\, dv\, dw.$$

Sascha Brack's hint on 8.5.2: *First pin down the signs. The constraints force $x \neq 0$ and $y \neq 0$; then $y > 0$ forces $x > 0$, so $u > 0$ and $v > 0$ throughout B. That is what lets you divide by x and y and rearrange the inequalities without their directions flipping.*

Info: *This exercise is Problem 8.5 of Problem Sheet 8, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 307*

AI-Note 20.3: The integral in part **(c)** is written $\int_C xyz\, dx\, dy$ on the official sheet, but C is a subset of $\mathbb{R}^3$ and the right-hand side carries three differentials, so this must read $dx\, dy\, dz$. Quoted verbatim above, as the sheet is a primary source.

AI-Exercise 20.a.14 (*A substitution read off the integrand*): Let T be the closed triangle with vertices $(0,0)$, $(1,0)$ and $(0,1)$, and consider

$$I := \int_T \exp\left(\frac{y - x}{y + x}\right) dx\, dy.$$

(a) Substitute $u := y - x$ and $v := y + x$. Show that the interior of T corresponds to $\{(u, v) : 0 < v < 1,\ -v < u < v\}$, and find the factor relating $dx\, dy$ to $du\, dv$.

(b) Evaluate I. The inner integral is the one to do first, and it is elementary.

(c) The integrand is undefined at the origin, which lies in T. Why does this not affect the answer?

Solution: *See solution on Page 308*

Exercise 20.a.15 (*Volume of an ellipsoid via linear change of variables*): Let $a, b, c > 0$ and consider the solid ellipsoid

$$U := \left\{(x, y, z) \in \mathbb{R}^3 \;\middle|\; \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 \leq 1\right\}.$$

(a) Define a linear change of variables $\Phi : \mathbb{R}^3 \to \mathbb{R}^3$ mapping the closed unit ball $\overline{B_1(0)} \subseteq \mathbb{R}^3$ bijectively onto U.

(b) Compute the Jacobian determinant $\det \mathrm{J}\,\Phi(u, v, w)$ of this transformation.

(c) Compute the three-dimensional Jordan volume $\mathrm{vol}_3(U)$ using the change of variables formula and the known volume of the unit ball.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Short Problem 1, part 1, which also asks for a labelled 3D sketch of U.*

Solution: *See solution on Page 310*

AI-Note 20.4: The original exam writes the ellipsoid with an equality, $(x/a)^2+(y/b)^2+(z/c)^2 = 1$, and then asks for its volume, but that set is the ellipsoid **surface**, whose Jordan volume is 0. The intended object is the solid region it bounds, so the inequality is used here.

Exercise 20.a.16 (*A cubic shear, and the area it sweeps out*): Let $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $\varphi(x) := (x_1^3 - x_2,\ x_2 + x_1)$, where $x = (x_1, x_2)$.

(a) Compute the derivative $D_x\varphi$ for every $x \in \mathbb{R}^2$.

(b) Show that $\varphi : \mathbb{R}^2 \to \varphi(\mathbb{R}^2)$ is a diffeomorphism.

(c) What is the area of the image $\varphi(Q)$ of the square $Q := [0,1] \times [0,1]$?

Info: *This exercise is taken from the Analysis I & II exam of 9 August 2021 (Prof. Giovanni Felder), Part A, Problem 5.*

Solution: *See solution on Page 310*

Exercise 20.a.17 (*A Gaussian second moment in polar coordinates*): For $\alpha > 0$, compute the integral

$$I_\alpha := \int_{\mathbb{R}^2} x^2 e^{-\alpha(x^2+y^2)}\, dx\, dy.$$

Official hint: *Notice that, by symmetry, $I_\alpha = \int_{\mathbb{R}^2} y^2 e^{-\alpha(x^2+y^2)}\, dx\, dy$ as well. Compute $2I_\alpha$ using polar coordinates.*

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the ninth block of true/false questions, there phrased as a series of checkpoints along this computation.*

Solution: *See solution on Page 311*

Exercise 20.a.18 (*An exponential of the radius, and how it decays in α*): For $\alpha > 0$, compute

$$I_\alpha := \int_{\mathbb{R}^2} e^{-\alpha\sqrt{x^2+y^2}}\, dx\, dy,$$

and then decide whether each of the following is true or false: $I_1 = \pi$; $I_2 = \pi/2$; $\lim_{\alpha\to+\infty} \alpha I_\alpha = 0$.

Official hint: *You may use polar coordinates and integration by parts.*

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the ninth block of true/false questions. The instruction to compute I_α is the exam's own preamble to those three statements.*

Solution: *See solution on Page 314*

Exercise 20.a.19 (*True or False* (the image of a small set under a C^1 map)): Consider $\Phi \in C^1(\mathbb{R}^2, \mathbb{R}^2)$ with

$$\Phi(0,0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathrm{J}\,\Phi(0,0) = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix},$$

and let $X := \Phi^{-1}\big((0,0)^{\mathsf{T}}\big)$. Decide whether each of the following statements is true or false.

(a) For $r > 0$ small enough, the set $\Phi(B_r(0,0))$ is open.

(b) $\mathrm{vol}_2\big(\Phi(B_r(0,0))\big) = 2\pi r^2 + o(r^2)$ as $r \to 0^+$.

(c) $\mathrm{vol}_2\big(\Phi([0,r] \times [0, ar^{1/2}])\big) = 3ar^{3/2} + o(r^{3/2})$ as $r \to 0^+$, where $a > 0$ is a constant.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the sixth block of true/false questions.*

Solution: *See solution on Page 311*

Exercise 20.a.20 (*True or False* (volumes under an affine map)): Consider the map

$$\Psi : \mathbb{R}^3 \to \mathbb{R}^3, \qquad \Psi(x, y, z) := \big(-x + e^\pi,\ y + \alpha z,\ \alpha y + z \big),$$

where $\alpha \in \mathbb{R}$ is a parameter. Decide whether each of the following statements is true or false.

(a) $\mathrm{vol}_3(\Psi(E)) = 0$ for all Jordan measurable $E \subseteq \mathbb{R}^3$ if and only if $\alpha = 1$.

(b) For $\alpha = -2$, $\mathrm{vol}_3(\Psi^k(E)) = 3^k \,\mathrm{vol}_3(E)$ for all Jordan measurable $E \subseteq \mathbb{R}^3$ and all $k \geq 0$, where $\Psi^0 := \mathrm{id}$ and $\Psi^{k+1} := \Psi \circ \Psi^k$.

(c) For $\alpha = 0$, $\mathrm{vol}_3(\Psi^{2025}(E)) = \mathrm{vol}_3(E)$ for all Jordan measurable $E \subseteq \mathbb{R}^3$.

(d) For all $\alpha \in (0, 1)$, $\mathrm{vol}_3\left(\Psi^{-1}([0,1]^3)\right) = -1/(1-\alpha^2)$.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the seventh block of true/false questions.*

Solution: *See solution on Page 312*

Example 20.a.21 (*One determinant, three volume statements*): Consider $\Psi : \mathbb{R}^3 \to \mathbb{R}^3$, $\Psi(x, y, z) := (z - 1,\ x + 2y,\ x - y)$. Its linear part has matrix

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 1 & -1 & 0 \end{pmatrix}, \qquad \det A = 1 \cdot \det \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} = -3,$$

expanding along the first row. The translation by $(-1, 0, 0)$ does not affect volumes, so $\mathrm{vol}_3(\Psi(E)) = 3\,\mathrm{vol}_3(E)$ for every Jordan measurable E, by Theorem 20.a.1. Everything else follows from the factor 3.

(a) $\mathrm{vol}_3(\Psi^{-1}(E)) = \mathrm{vol}_3(E)$ **for all Jordan measurable E: false.** The inverse scales by $|\det A^{-1}| = 1/3$, so $\mathrm{vol}_3(\Psi^{-1}(E)) = \mathrm{vol}_3(E)/3$.

(b) $\mathrm{vol}_3(\Psi(E)) = 2\,\mathrm{vol}_3(E)$**: false.** The factor is 3, not 2.

(c) $\mathrm{vol}_3(\Psi(\Psi(E))) = 9\,\mathrm{vol}_3(E)$**: true.** Applying **(b)**'s correct factor twice gives $3 \cdot 3 = 9$, which is $|\det A^2| = |\det A|^2$.

Only **(b)** requires the determinant to be computed. Items **(a)** and **(c)** then follow from multiplicativity of the determinant alone, without recomputing anything, which is the point of the block.

Official hint: *To answer these three, it can be useful to compute the Jacobian determinant of Ψ first.*

Info: *This example is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the seventh block of true/false questions.*

Exercise 20.a.22 (*True or False* (spherical coordinates)): Consider the polar (spherical) coordinates $\Psi\colon (0, 2\pi) \times (0, \pi) \times (0, \infty) \to \mathbb{R}^3$ defined by

$$\Psi(\phi, \theta, r) = (r\cos\phi\sin\theta, r\sin\phi\sin\theta, r\cos\theta).$$

Decide whether each of the following statements is true or false.

(a) Ψ is injective.

(b) Ψ is surjective.

(c) The Jacobian determinant $\det \mathrm{J}\,\Psi(\theta, r)$ is given by $r^2 \sin\theta$.

(d) $\mathrm{vol}_3(B_R \setminus B_r) = \pi(R^3 - r^3)$.

(e) $\mathrm{vol}_3(B_R) = 2\pi \int_0^\pi \int_0^R r^2 \sin\theta \, dr \, d\theta$.

(f) $\lim_{R \to 1^+} \frac{\mathrm{vol}_3(B_R \setminus B_1)}{R-1} = 4\pi$.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the eighth block of true/false questions.*

Solution: *See solution on Page 313*

Exercise 20.a.23 (*Box question: Linear map scaling volume*): Give an explicit example of a linear map $\Phi\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\mathrm{vol}_2(\Phi(E)) = 2\,\mathrm{vol}_2(E) \quad \text{for all } E \subset \mathbb{R}^2 \text{ Jordan measurable.}$$

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Box Problem 2, where only the final answer was graded.*

Solution: *See solution on Page 314*

Exercise 20.a.24 (*Box question: a linear map that preserves every volume*): Give an explicit example of a linear map $\Phi : \mathbb{R}^2 \to \mathbb{R}^2$, **different from the identity**, such that

$$\mathrm{vol}_2(\Phi(E)) = \mathrm{vol}_2(E) \quad \text{for all Jordan measurable } E \subseteq \mathbb{R}^2.$$

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), Box Problem 2, where only the final answer was graded.*

Solution: *See solution on Page 315*

AI-Exercise 20.a.25 (*The volume of a spanned ellipsoid*): Let $v_1, v_2, v_3 \in \mathbb{R}^3$ be vectors with $\det(v_1, v_2, v_3) = 2$, and consider the region

$$A := \{t_1v_1 + t_2v_2 + t_3v_3 \mid (t_1, t_2, t_3) \in \mathbb{R}^3 \text{ with } t_1^2 + t_2^2 + t_3^2 \le 1\}.$$

Decide whether each of the following statements is true or false.

(a) $\mathrm{vol}_3(A) = \frac{8\pi}{3}$.

(b) A is Jordan measurable.

(c) The volume of A depends on the choice of the vectors v_1, v_2, v_3.

Solution: *See solution on Page 314*

Exercise 20.a.26 (*The substitution rule in the linear case*): Let A be an invertible $n \times n$ matrix. Theorem 19.a.14 asserts that for every Jordan measurable $B \subseteq \mathbb{R}^n$ the image $A(B)$ is again Jordan measurable, with

$$\mathrm{vol}\big(A(B)\big) = |\det A|\ \mathrm{vol}(B). \tag{20.1}$$

This exercise proves it.

(a) Prove (20.1) directly in the case that A is a diagonal matrix.

(b) Explain in a few sentences why it now suffices to prove (20.1) for permutation matrices and for shear matrices.

(c) Prove (20.1) when A is a shear matrix. You may assume for simplicity that $B = Q$ is a closed axis-parallel box.

Info: *This exercise is taken from the Analysis I & II examination of winter 2018/2019 (Prof. Manfred Einsiedler), Problem 13.*

Solution: *See solution on Page 315*

Example 20.a.27 (*The covering lemma behind the change of variables formula*): Theorem 20.a.1 is stated in this chapter without proof. The step that makes the proof hard is not the Jacobian factor but a prior, purely geometric one: a diffeomorphism that is close to the identity cannot let a cube shrink away. Made precise, that is the following.

Claim. Let $\ell > 0$, let $Q_0 := [-\ell, \ell]^n \subseteq \mathbb{R}^n$, let $X \supseteq Q_0$ be open and let $\Phi : X \to \Phi(X) \subseteq \mathbb{R}^n$ be a diffeomorphism with $\Phi(0) = 0$ and $D_0\Phi = I_n$. Suppose $\sigma > 0$ is such that

$$\big|(D_x\Phi - I_n)v\big| \le \sigma|v| \qquad \text{for all } x \in Q_0,\ v \in \mathbb{R}^n,$$

and assume $s := \sigma\sqrt{n} < 1$. Then

$$(1 - s)\, Q_0 \subseteq \Phi(Q_0).$$

Fix $y \in (1-s)Q_0$ and define $F_y : Q_0 \to \mathbb{R}^n$ by $F_y(x) := y - \big(\Phi(x) - x\big)$.

(a) Why a fixed point is what we want. By construction $F_y(x) = x$ holds if and only if $y - \Phi(x) + x = x$, that is, if and only if $\Phi(x) = y$. So a fixed point of F_y in Q_0 is exactly a preimage of y under Φ lying in Q_0, which is what $y \in \Phi(Q_0)$ means.

(b) The displacement $\Phi - \mathrm{id}$ is a contraction. Write $\Psi(x) := \Phi(x) - x$, so that $D_x\Psi = D_x\Phi - I_n$ and the hypothesis reads $|D_x\Psi(v)| \le \sigma|v|$ on Q_0, that is, $\|D_x\Psi\|_{\mathrm{op}} \le \sigma$ there. Since Q_0 is convex, the segment from x_1 to x_2 stays inside it, and the mean value inequality (Theorem 11.a.3) gives

$$\big|\Phi(x_2) - x_2 - \big(\Phi(x_1) - x_1\big)\big| = |\Psi(x_2) - \Psi(x_1)| \le \sigma|x_2 - x_1|$$

for all $x_1, x_2 \in Q_0$.

(c) F_y maps Q_0 into itself. Applying **(b)** with $x_1 := 0$, and using $\Psi(0) = \Phi(0) - 0 = 0$, we get $|\Psi(x)| \le \sigma|x|$ for $x \in Q_0$. For $x \in Q_0$ we have $\|x\|_\infty \le \ell$ and hence $|x| \le \sqrt{n}\,\ell$, so

$$\|\Psi(x)\|_\infty \le |\Psi(x)| \le \sigma\sqrt{n}\,\ell = s\ell.$$

Since $y \in (1-s)Q_0$ means $\|y\|_\infty \le (1-s)\ell$, the triangle inequality gives

$$\|F_y(x)\|_\infty \le \|y\|_\infty + \|\Psi(x)\|_\infty \le (1-s)\ell + s\ell = \ell,$$

that is, $F_y(x) \in Q_0$. So $F_y : Q_0 \to Q_0$.

(d) Conclusion. Q_0 is a closed subset of $\mathbb{R}^n$, hence complete, and by **(b)** the map F_y satisfies $|F_y(x_2) - F_y(x_1)| = |\Psi(x_2) - \Psi(x_1)| \le \sigma|x_2 - x_1|$ with $\sigma \le s < 1$. It is therefore a contraction of the complete metric space Q_0 into itself, and the Banach fixed point theorem (Theorem 6.c.19) supplies a fixed point $x \in Q_0$. By **(a)**, $\Phi(x) = y$, so $y \in \Phi(Q_0)$. As $y \in (1-s)Q_0$ was arbitrary, $(1-s)Q_0 \subseteq \Phi(Q_0)$.

Remark 20.a.28 (*What the lemma is doing in the proof of the change of variables formula*): The formula is proved by cutting the domain into tiny cubes and comparing $\mathrm{vol}(\Phi(Q))$ with $|\det D\Phi|\,\mathrm{vol}(Q)$ on each. The upper bound is the easy half: Φ is Lipschitz on a compact piece, so $\Phi(Q)$ cannot be much bigger than Q. The lower bound is where a naive argument stalls, because nothing about a Lipschitz map prevents an image from collapsing. This lemma is that lower bound. First normalise so that $D_0\Phi = I_n$, which is always possible by composing with a linear map, the case already settled in Exercise 20.a.26. The lemma then says the image of a cube still contains a cube shrunk only by the factor $1 - s$, and $s \to 0$ as the cubes shrink, because $D\Phi$ is continuous. Letting the two bounds close is the proof.

Notice which two theorems do the work: the mean value inequality on a convex set in **(b)**, and the Banach fixed point theorem in **(d)**. The same pair proves the inverse function theorem (Theorem 15.a.1), and that is not a coincidence. This claim is a quantitative version of the statement that Φ is locally surjective near a point where its differential is invertible.

Info: *This example is taken from the Analysis I & II examination of winter 2017/2018 (Prof. Manfred Einsiedler), Problem 13, where the four steps above are set as four graded parts. The framing and the closing remark are added here.*

Fubini's theorem (Section 20.b)

Theorem 20.b.29 (*Fubini*): Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$ be intervals (*"boxes"*), and let $f : X \times Y \to \mathbb{R}$, $(x, y) \mapsto f(x, y)$, be continuous. Then

$$\int_{X\times Y} f(x,y)\,dx\,dy = \int_X \left(\int_Y f(x,y)\,dy\right) dx = \int_Y \left(\int_X f(x,y)\,dx\right) dy,$$

where we look at the expression in brackets as a function, e.g. $x \mapsto \int_Y f(x,y)\,dy$.

Proof for closed boxes:

Let X and Y be closed, so that f is uniformly continuous on the compact box $X \times Y$ (Proposition 7.a.17). First, $F(x) := \int_Y f(x,y)\,dy$ is continuous, since $|F(x)-F(x')| \le \mathrm{vol}(Y)\sup_y |f(x,y)-f(x',y)|$ and the supremum tends to 0 as $x' \to x$; so $\int_X F$ exists.

Let $\varepsilon > 0$ and cut X into boxes $X_1, \dots, X_p$ and Y into boxes $Y_1, \dots, Y_q$, so finely that f varies by less than ε on each $X_i \times Y_j$. Let m_{ij} and M_{ij} be the minimum and maximum of f there. For $x \in X_i$, integrating $m_{ij} \le f(x,y) \le M_{ij}$ over each Y_j and adding gives $\sum_j m_{ij}\,\mathrm{vol}(Y_j) \le F(x) \le \sum_j M_{ij}\,\mathrm{vol}(Y_j)$, and integrating that over each X_i and adding,

$$L := \sum_{i,j} m_{ij}\,\mathrm{vol}(X_i)\,\mathrm{vol}(Y_j) \;\le\; \int_X F \;\le\; \sum_{i,j} M_{ij}\,\mathrm{vol}(X_i)\,\mathrm{vol}(Y_j) =: R.$$

But L and R are a lower and an upper sum of f for the partition of $X \times Y$ into the boxes $X_i \times Y_j$, so $\int_{X\times Y} f$ lies in $[L,R]$ as well. The interval has length $R - L \le \varepsilon\,\mathrm{vol}(X)\,\mathrm{vol}(Y)$, and ε was arbitrary, so $\int_X F = \int_{X\times Y} f$. The other order is the same argument with the roles of X and Y exchanged. Q.E.D.

The proof also shows what Fubini is: a double sum $\sum_{i,j}$ may be added row by row or column by column, and the integral inherits that freedom from its Riemann sums.

Proposition 20.b.30 (*Cavalieri's principle*): Let $A, B \subseteq \mathbb{R}^{n+1}$ be Jordan measurable sets. For each height $t \in \mathbb{R}$, denote by $A_t := \{x \in \mathbb{R}^n \mid (x,t) \in A\}$ and $B_t := \{x \in \mathbb{R}^n \mid (x,t) \in B\}$ the n-dimensional cross-sections at t, and suppose every one of these slices is itself Jordan measurable. If the cross-sectional volumes agree for all $t \in \mathbb{R}$, i.e.

$$\mu_n(A_t) = \mu_n(B_t) \quad \forall t \in \mathbb{R},$$

then the total $(n+1)$-dimensional volumes are equal: $\mu_{n+1}(A) = \mu_{n+1}(B)$.

Proof:

Both sets are bounded, so fix a box $Q \times [c,d]$ containing them. Integrating the indicator function over slices gives:

$$\mu_{n+1}(A) = \int_{\mathbb{R}^{n+1}} \chi_A(x,t)\,dx\,dt = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^n} \chi_{A_t}(x)\,dx \right) dt = \int_{\mathbb{R}} \mu_n(A_t)\,dt.$$

Replacing $\mu_n(A_t)$ with $\mu_n(B_t)$ yields $\mu_{n+1}(B)$. Q.E.D.

AI-Note 20.5: Two things in that proof are being used in slightly stronger form than Theorem 20.b.29 states them, and both are worth naming.

First, χ_A is not continuous, so the version of Fubini in play is the one for **Riemann integrable** functions on a box: χ_A is Riemann integrable there precisely because A is Jordan measurable, and the iterated identity holds provided the inner integral exists at every t, which is what the slice hypothesis buys.

Second, that hypothesis is not free. Take

$$A := \big([0,1] \times (0,1]\big) \;\cup\; \big((\mathbb{Q} \cap [0,1]) \times \{0\}\big) \;\subseteq\; \mathbb{R}^2.$$

Its interior is $(0,1)^2$ and its closure is $[0,1]^2$, so ∂A is the frame of the unit square, which is vol_2-null: A is Jordan measurable with $\mu_2(A) = 1$. But $A_0 = \mathbb{Q} \cap [0,1]$ is not Jordan measurable, and $\mu_1(A_0)$ simply does not exist. A single bad slice is enough to make the inner integral undefined, which is why the hypothesis is stated on every t rather than derived.

Theorem 20.b.31 (*Volume of a cone over a measurable base*): Let $\Omega \subseteq \mathbb{R}^n$ be a bounded, Jordan measurable domain. The "*cone over* Ω" (German: „*Kegel über* Ω") with vertex at $(0,1) \in \mathbb{R}^n \times \mathbb{R}$

and base $\Omega \times \{0\}$ is defined by

$$C\Omega := \big\{(x,t) \in \mathbb{R}^n \times [0,1] \mid x \in (1-t)\Omega\big\}.$$

Then $C\Omega$ is Jordan measurable in $\mathbb{R}^{n+1}$, and its volume is given by

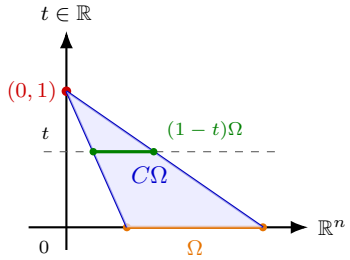
$$\mu_{n+1}(C\Omega) = \frac{\mu_n(\Omega)}{n+1}.$$

Proof:

For any $t \in [0,1]$, the cross-section of $C\Omega$ at height t is $(C\Omega)_t = (1-t)\Omega \subseteq \mathbb{R}^n$. By the scaling property of n-dimensional Jordan measure, $\mu_n\big((1-t)\Omega\big) = (1-t)^n \mu_n(\Omega)$. Applying Cavalieri's Principle (Proposition 20.b.30):

$$\mu_{n+1}(C\Omega) = \int_0^1 \mu_n\big((1-t)\Omega\big)\, dt = \mu_n(\Omega) \int_0^1 (1-t)^n\, dt = \mu_n(\Omega) \left[-\frac{(1-t)^{n+1}}{n+1}\right]_0^1 = \frac{\mu_n(\Omega)}{n+1}.$$

Q.E.D.



Example 20.b.32 (*Fubini on the unit square*):

$$\int_{[0,1]^2} x^{y+1}\, dx\, dy = \int_0^1 dy \int_0^1 dx\; x^{y+1} = \int_0^1 dy \left[\frac{x^{y+2}}{y+2}\right]_0^1$$
$$= \int_0^1 dy\, \frac{1}{y+2} = \log|y+2|\,\Big|_0^1 = \log\left(\tfrac{3}{2}\right).$$

Remark 20.b.33 (*Pick the easier order*): Integrating x^{y+1} in y first would mean integrating an exponential in the exponent, which is far worse. That is Fubini's real use: you may pick the easier order, and here only one of the two is comfortable.

Example 20.b.34 (*Changing the order of integration*): Consider the iterated integral

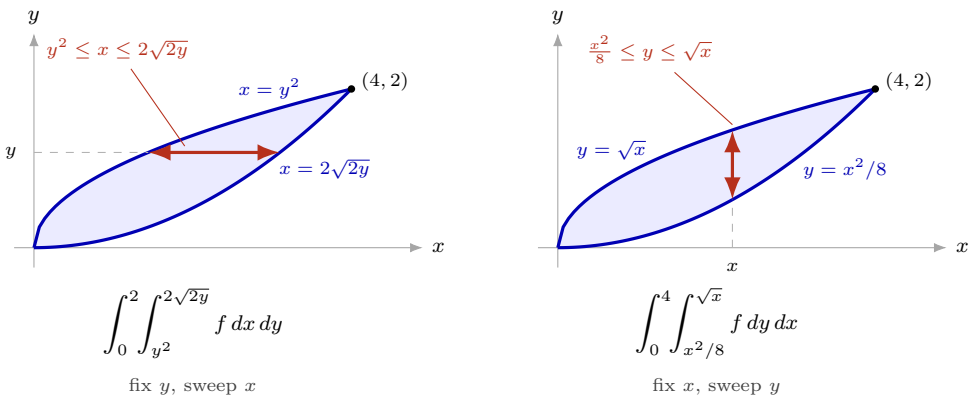
$$\int_0^2 \int_{y^2}^{2\sqrt{2y}} f(x,y)\, dx\, dy.$$

Let us rewrite it with the integration order reversed (first y, then x).

The region is defined by $0 \le y \le 2$ and $y^2 \le x \le 2\sqrt{2y}$. Notice that for $y \in [0,2]$, the lower bound is $x \ge 0$ and the upper bound is $x \le 2\sqrt{4} = 4$, so x ranges from 0 to 4. We now solve the inequalities for y: $x \le 2\sqrt{2y} \implies x^2 \le 8y \implies y \ge \frac{1}{8}x^2$. $x \ge y^2 \implies y \le \sqrt{x}$ (since $x, y \ge 0$). Thus, the new bounds are $0 \le x \le 4$ and $\frac{1}{8}x^2 \le y \le \sqrt{x}$. The reversed integral is:

$$\int_0^4 \int_{\frac{1}{8}x^2}^{\sqrt{x}} f(x,y)\, dy\, dx.$$

Info: *The same region, with the same two bounds, is Problem 14 of the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), a box problem where only the reversed integral was graded.*



Remark 20.b.35 (*Reading the swap off the picture*): The region is the same in both panels; only the direction of the slice changes, and with it which variable is the outer one. That is the whole of Fubini's theorem for a non-rectangular domain, and the two mechanical rules follow from looking:

- **The outer limits are constants, and they are the shadow of the region on the outer axis.** On the left the outer variable is y and the region's shadow on the y-axis is $[0, 2]$; on the right the outer variable is x and its shadow is $[0, 4]$. The two are different numbers, which is the most common slip.
- **The inner limits are functions of the outer variable, read off where the slice enters and leaves the region.** Entering and leaving happens on the **same two curves** in both panels, since there is only one pair of boundary curves here. What changes is which of $x = y^2$ and $y = \sqrt{x}$ is the useful way to write the first of them.

The reason the algebra in the example above is not merely a relabelling is the second bullet: each boundary has to be re-solved for the new inner variable. A region whose boundary cannot be solved for one of the variables cannot be swapped at all without first splitting it, and that is the situation Example 20.b.37 runs into.

Example 20.b.36 (*Surface parametrization and Fubini*): Let $f(u, v) := (u\cos v, u\sin v, u^2)^\mathsf{T} \in \mathbb{R}^3$ for $(u, v) \in \mathbb{R}^2$. Write the third coordinate $z(u, v)$ as a function of the first two, $\tilde{z}(x, y)$, and compute $\int_0^y \int_0^x \tilde{z}(x', y')\,dx'\,dy'$.

We have $x(u, v) = u\cos v$ and $y(u, v) = u\sin v$. Then $x^2+y^2 = u^2\cos^2 v+u^2\sin^2 v = u^2 = z(u, v)$. So, we define $\tilde{z}(x, y) := x^2 + y^2$. The integral is:

$$\begin{aligned}\int_0^y \int_0^x (x'^2 + y'^2)\,dx'\,dy' &= \int_0^y \left[\frac{x'^3}{3} + x'y'^2\right]_{x'=0}^{x'=x} dy' \\ &= \int_0^y \left(\frac{x^3}{3} + xy'^2\right) dy' \\ &= \left[\frac{x^3y'}{3} + x\frac{y'^3}{3}\right]_{y'=0}^{y'=y} \\ &= \frac{xy}{3}(x^2 + y^2). \end{aligned} \tag{20.2}$$

Example 20.b.37 (*A domain that Fubini alone cannot handle*): Consider

$$\int_A \frac{y}{\sqrt{x}}\,dy\,dx, \qquad A := \left\{(x, y) \in \mathbb{R}^2 \;:\; x, y > 0,\;\; 1 \le \frac{y}{\sqrt{x}} \le 2,\;\; 1 \le xy \le 2\right\}.$$

This is not easily integrated with Fubini alone. But we can apply a transformation of variables to simplify the integration domain A: take

$$\Phi : U \to V, \qquad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u(x, y) \\ v(x, y) \end{pmatrix} = \begin{pmatrix} \frac{y}{\sqrt{x}} \\ xy \end{pmatrix},$$

with U and V still to be determined, subject only to their being open.

Check injectivity. Suppose, for $x, y, x', y' > 0$,

$$\textbf{(1)}\quad \frac{y}{\sqrt{x}} = \frac{y'}{\sqrt{x'}}, \qquad \textbf{(2)}\quad xy = x'y'.$$

From **(2)**, $\dfrac{x}{x'} = \dfrac{y'}{y}$. Inserting into **(1)**,

$$\frac{1}{\sqrt{x}} = \frac{y'}{y}\cdot\frac{1}{\sqrt{x'}} = \frac{x}{x'}\cdot\frac{1}{\sqrt{x'}} \quad\Longrightarrow\quad x^{3/2} = (x')^{3/2} \quad\Longrightarrow\quad x = x',$$

and consequently $\dfrac{y'}{y} = \dfrac{x}{x'} = 1$. So, for $U := (0,\infty)^2$ and $V := \Phi(U)$, the map Φ is a bijection between open sets.

Functional determinant. We calculate $|\det \mathrm{J}\,\Phi|$; if it is strictly positive we may proceed, since Φ is then a local diffeomorphism as well as a bijection, hence a diffeomorphism. Now

$$\mathrm{J}\,\Phi(x,y) = \begin{pmatrix} -\dfrac{y}{2\sqrt{x}^{\,3}} & \dfrac{1}{\sqrt{x}} \\ y & x \end{pmatrix},$$

so that

$$\left|\det \mathrm{J}\,\Phi(x,y)\right| = \left|-\frac{3}{2}\frac{y}{\sqrt{x}}\right| = \frac{3}{2}\frac{y}{\sqrt{x}} = \frac{3}{2}\,u(x,y).$$

So, Φ is a diffeomorphism and $du\,dv = \frac{3}{2}u\,dx\,dy$. Furthermore, by design,

$$\Phi(A) = [1,2]^2.$$

So, with the change of variables (Theorem 20.a.1), followed by Fubini (Theorem 20.b.29),

$$\int_A \frac{y}{\sqrt{x}}\,dy\,dx = \int_{[1,2]^2} u\,\frac{1}{\frac{3}{2}u}\,du\,dv = \frac{2}{3}\int_1^2 dv\int_1^2 du = \frac{2}{3}.$$

AI-Note 20.6: The design principle is worth stating, because it is the reverse of how substitution is usually taught. Here Φ was not chosen to simplify the **integrand** but to straighten the **domain**: the two constraints defining A are literally $1 \le u \le 2$ and $1 \le v \le 2$, so Φ turns an awkward curvilinear region into a square, and only then does Fubini become available. That the integrand $y/\sqrt{x}$ is itself u, and cancels against most of the Jacobian factor, is a bonus of the same choice.

AI-Exercise 20.b.38 (*An order of integration that cannot be done*): Consider

$$I := \int_0^1\int_x^1 e^{y^2}\,dy\,dx.$$

(a) Explain what goes wrong if one attempts the inner integral as written.

(b) Describe the region of integration by inequalities, and rewrite I with the two integrations in the opposite order.

(c) Evaluate I.

Solution: *See solution on Page 309*

Exercise 20.b.39 (*True or False* (a Gaussian slab, and what happens at $\alpha = 0$)): For $\alpha \ge 0$ consider

$$I_\alpha := \int_{\mathbb{R}^3} e^{-x^2-y^2-\alpha|z|}\,dx\,dy\,dz.$$

Decide whether each of the following is true or false, with justification.

(a) $I_0 = \sqrt{\pi}$.

(b) $\alpha I_\alpha = I_1$.

(c) $\lim\limits_{\alpha\to+\infty}\left(\alpha+\sqrt{\alpha}\right)I_\alpha = 2\pi$.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), where it appears as the twelfth block of multiple-choice questions. Its paper supplies* $\int_{\mathbb{R}} e^{-t^2}\,dt = \sqrt{\pi}$ *as a formula that may be used without proof; Example 20.a.3 derives it.*

Solution: *See solution on Page 316*

We now want to look at a powerful new integration technique using ODEs.

Feynman's trick (Section 20.c)

Theorem 20.c.40 (*Differentiation under the integral sign*): Let $U \subseteq \mathbb{R}^n \times \mathbb{R}$ be open, $f \in C^1(U)$, and let $K \subseteq \mathbb{R}^n$ be compact and Jordan measurable such that $K \times [a,b] \subseteq U$. Then the function

$$(a,b) \to \mathbb{R}, \qquad y \mapsto \int_K f(x,y)\,dx$$

is continuously differentiable, and

$$\frac{d}{dy}\int_K f(x,y)\,dx = \int_K \frac{\partial}{\partial y} f(x,y)\,dx \qquad \text{for all } y \in (a,b).$$

Proof:

Write $\partial_y f$ for the partial derivative in the last variable, and $\mu(K)$ for the Jordan measure of K. The set $K \times [a,b]$ is compact, so $\partial_y f$ is uniformly continuous on it (Proposition 7.a.17): for every $\varepsilon > 0$ there is $\delta > 0$ such that $|\partial_y f(x,s) - \partial_y f(x,t)| \le \varepsilon$ whenever $x \in K$, $s,t \in [a,b]$ and $|s-t| < \delta$. All integrands below are continuous on the compact Jordan measurable set K, hence integrable.

Fix $y \in (a,b)$ and $h \neq 0$ with $|h| < \delta$ and $y + h \in [a,b]$. For each $x \in K$, the one-variable mean value theorem gives $\theta_x \in (0,1)$ with $f(x,y+h) - f(x,y) = h\,\partial_y f(x, y+\theta_x h)$, and therefore

$$\left| \frac{f(x,y+h) - f(x,y)}{h} - \partial_y f(x,y) \right| = \left| \partial_y f(x, y+\theta_x h) - \partial_y f(x,y) \right| \le \varepsilon.$$

The left-hand side is a continuous function of x (the point θ_x only served to bound it), so integrating over K,

$$\left| \frac{1}{h}\left(\int_K f(x,y+h)\,dx - \int_K f(x,y)\,dx \right) - \int_K \partial_y f(x,y)\,dx \right| \le \varepsilon\,\mu(K).$$

As ε was arbitrary, the difference quotient converges to $\int_K \partial_y f(x,y)\,dx$, which is the asserted formula. The same uniform continuity gives $\left| \int_K \partial_y f(x,s)\,dx - \int_K \partial_y f(x,t)\,dx \right| \le \varepsilon\,\mu(K)$ for $|s-t| < \delta$, so the derivative is continuous. Q.E.D.

Both hypotheses are visible in the proof. Compactness of K is what makes the convergence of the difference quotients uniform in x, and continuity of $\partial_y f$ **jointly** in (x,y), not merely in y for each fixed x, is what uniform continuity was applied to.

Example 20.c.41 ($\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$ *by Feynman's trick*): We want to compute the integral

$$I(1) = \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx,$$

which has no convenient antiderivative. By introducing a parameter $\alpha \in [0,1]$:

$$I(\alpha) := \int_0^1 \frac{\log(1+\alpha x)}{1+x^2}\,dx.$$

Note that the integrand is a C^1 function on $U := \left(-\frac{1}{1+\varepsilon}, \frac{1}{\varepsilon}\right) \times (-\varepsilon,\, 1+\varepsilon)$ in the variables (x,α), with $K := [0,1]$ and $[a,b] := [-\frac{\varepsilon}{2}, 1+\frac{\varepsilon}{2}]$ satisfying $K \times [a,b] \subseteq U$ for $0 < \varepsilon < 1$, so that I is C^1 on

an open interval containing $[0,1]$ (pay attention to the logarithm, which is what forces $1+\alpha x > 0$). So, we may differentiate under the integral!

We obtain an ordinary differential equation. By Theorem 20.c.40,

$$I'(\alpha) = \frac{d}{d\alpha}\int_0^1 \frac{\log(1+\alpha x)}{1+x^2}\,dx = \int_0^1 \frac{x}{(1+\alpha x)(1+x^2)}\,dx.$$

Partial fractions: writing

$$\frac{x}{(1+\alpha x)(1+x^2)} = \frac{A}{1+\alpha x} + \frac{Bx+C}{1+x^2}$$

and clearing denominators gives $A + Ax^2 + Bx + B\alpha x^2 + C + C\alpha x = x$, so comparing coefficients,

$$-A = C, \qquad -\tfrac{A}{\alpha} = B, \qquad B + C\alpha = 1 \implies -\tfrac{A}{\alpha} - A\alpha = 1,$$

whence

$$A = -\frac{\alpha}{1+\alpha^2}, \qquad B = \frac{1}{1+\alpha^2}, \qquad C = \frac{\alpha}{1+\alpha^2}.$$

Therefore

$$\begin{aligned} I'(\alpha) &= \left(\int_0^1 \frac{-\alpha}{1+\alpha x} + \frac{x+\alpha}{1+x^2}\,dx\right)\frac{1}{1+\alpha^2} \\ &= \frac{1}{1+\alpha^2}\Big[-\log(1+\alpha x) + \tfrac{1}{2}\log(1+x^2) + \alpha\arctan(x)\Big]_0^1 \\ &= \frac{1}{1+\alpha^2}\Big[\log\sqrt{2} + \tfrac{\alpha\pi}{4} - \log(1+\alpha)\Big]. \end{aligned}$$

With initial value $I(0) = \int_0^1 \frac{\log(1)}{1+x^2}\,dx = 0$, we obtain from the fundamental theorem of calculus

$$\begin{aligned} \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx = I(1) &= I(1) - I(0) \\ &= \int_0^1 \left(\frac{\log\sqrt{2}}{1+\alpha^2} + \frac{\pi}{4}\,\frac{\alpha}{1+\alpha^2} - \underbrace{\frac{\log(1+\alpha)}{1+\alpha^2}}_{\to\, I(1)}\right)d\alpha. \end{aligned}$$

The last term is $I(1)$ again, the integral we started from with α merely renamed, so bringing it to the left-hand side,

$$2\,I(1) = \Big[\log\sqrt{2}\,\arctan(\alpha) + \tfrac{\pi}{8}\log(1+\alpha^2)\Big]_0^1 = \frac{\pi\log\sqrt{2}}{4} + \frac{\pi\log\sqrt{2}}{4},$$

$$\implies \quad I(1) = \frac{\pi\log\sqrt{2}}{4} = \boxed{\frac{\pi\log 2}{8}}.$$

Example 20.c.42 (*Dirichlet integral via Feynman's trick*): Consider the integral $\int_0^\infty \frac{\sin x}{x}\,dx$. We can evaluate it by introducing a damping factor e^{-tx} and defining

$$I(t) := \int_0^\infty e^{-tx}\frac{\sin x}{x}\,dx \qquad \text{for } t > 0.$$

Differentiating under the integral sign with respect to t (which is justified since the integrand and its derivative decay rapidly for $t > 0$):

$$I'(t) = \int_0^\infty \frac{\partial}{\partial t}\left(e^{-tx}\frac{\sin x}{x}\right)dx = \int_0^\infty -xe^{-tx}\frac{\sin x}{x}\,dx = -\int_0^\infty e^{-tx}\sin x\,dx.$$

This standard integral can be evaluated using integration by parts twice, or by using Euler's formula $e^{ix} = \cos x + i\sin x$ and noting that $e^{-tx}\sin x = \operatorname{Im}(e^{(-t+i)x})$:

$$\int_0^\infty e^{(-t+i)x}\,dx = \left[\frac{e^{(-t+i)x}}{-t+i}\right]_0^\infty = 0 - \frac{1}{-t+i} = \frac{t+i}{t^2+1}.$$

Taking the imaginary part gives $\int_0^\infty e^{-tx}\sin x\,dx = \frac{1}{t^2+1}$. Thus,

$$I'(t) = -\frac{1}{t^2+1},$$

which integrates to $I(t) = -\arctan(t) + C$. Since $|\sin x| \le x$, we have $|I(t)| \le \int_0^\infty e^{-tx}\,dx = \frac{1}{t}$, so $\lim_{t\to\infty} I(t) = 0$ and we must have $C = \lim_{t\to\infty}\arctan(t) = \frac{\pi}{2}$. Therefore, $I(t) = \frac{\pi}{2} - \arctan(t)$. The original integral is the improper integral $I(0)$, and it remains to show that $I(t) \to I(0)$ as $t \to 0^+$. The tail can be bounded directly. For $t \ge 0$ the function $h_t(x) := e^{-tx}/x$ is positive and decreasing on $(0,\infty)$, so integrating by parts on $[R,S]$ with $0 < R < S$ gives

$$\begin{aligned}\left|\int_R^S h_t(x)\sin x\,dx\right| &= \left|\Big[-h_t(x)\cos x\Big]_R^S + \int_R^S h_t'(x)\cos x\,dx\right| \\ &\le h_t(R) + h_t(S) + \int_R^S \big(-h_t'(x)\big)\,dx = 2h_t(R) \le \frac{2}{R}.\end{aligned}$$

By the Cauchy criterion the improper integral $I(t)$ therefore exists for every $t \ge 0$, and its tail beyond R is at most $2/R$ in absolute value, uniformly in t. On $[0,R]$ we use $|e^{-tx}-1| \le tx$ and $|\sin x| \le 1$, so that

$$|I(t) - I(0)| \le \int_0^R |e^{-tx}-1|\,\frac{|\sin x|}{x}\,dx + \frac{2}{R} + \frac{2}{R} \le tR + \frac{4}{R}.$$

Choosing $R = t^{-1/2}$ gives $|I(t) - I(0)| \le 5\sqrt{t} \to 0$, and hence

$$\int_0^\infty \frac{\sin x}{x}\,dx = \lim_{t\to 0^+}\left(\frac{\pi}{2} - \arctan(t)\right) = \frac{\pi}{2}.$$

AI-Note 20.7 (*Why this is called a trick, and when to reach for it*): The move that makes this work is not the differentiation. It is noticing that the **antiderivative** of the resulting integrand is elementary while the original one is not. The pattern is worth naming:

- **(a)** Insert a parameter α so that the target is $I(\alpha_1)$ for some α_1, and so that $I(\alpha_0)$ is trivial for some other α_0. Here $I(0) = 0$, because $\log 1 = 0$.
- **(b)** Differentiate under the integral. The log disappears, leaving a rational function that partial fractions can handle.
- **(c)** Integrate I' back from α_0 to α_1.

The self-reference in step **(c)**, where the answer reappears on the right-hand side so that one solves for $I(1)$ rather than computing it, is specific to this integral and is the reason it is a classic. Do not expect it in general. Usually step **(c)** is an honest integration.

AI-Exercise 20.c.43 (*Frullani's integral*): Fix $0 < a < b$. The aim is to show that

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x}\,dx = \log\left(\frac{b}{a}\right),$$

an answer that depends on a and b only through their ratio.

- **(a)** The integrand is undefined at $x = 0$. Compute $\lim_{x\to 0^+}$ of it, so that the integrand extends continuously to $[0,\infty)$.
- **(b)** Hold a fixed and let the second exponent be the parameter: define

$$I(t) := \int_0^\infty \frac{e^{-ax} - e^{-tx}}{x}\,dx \qquad \text{for } t > 0.$$

  Differentiate under the integral sign and evaluate the resulting integral, which is elementary.
- **(c)** Identify the value of t for which $I(t)$ is trivially zero, and integrate I' from there to $t = b$.

Solution: *See solution on Page 309*

Exercise 20.c.44 (*Area of a truncated moving ellipse and parameter-dependent integrals*): Let $b > 0$ and consider the ellipse $E := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2/b^2 \le 1\}$. For $t \in [-1, 1]$, let $E_t := \{(x, y) \in E \mid x \ge t\}$ be the region cut off by the vertical line $x = t$, and let $A(t) := \mathrm{vol}_2(E_t)$ denote its area.

(a) Express $A(t)$ as a one-variable integral with respect to x.

(b) Compute $A'(t)$ for $t \in (-1, 1)$ using the fundamental theorem of calculus.

(c) Determine the one-sided derivatives $A'_+(-1)$ and $A'_-(1)$ at the endpoints. Is A of class C^1 on the closed interval $[-1, 1]$?

(d) Now let the ellipse itself deform with the parameter: define

$$\Omega_t := \left\{(x, y) \in \mathbb{R}^2 \;\middle|\; x^2 + \frac{y^2}{1 + t^2} \le 1,\ x \ge t\right\} \qquad \text{for } t \in [0, 1].$$

Express $F(t) := \mathrm{vol}_2(\Omega_t)$ as an integral, and compute $F'(t)$ for $t \in (0, 1)$ by differentiating under the integral sign with parameter-dependent limits.

Info: *This exercise is adapted from the Analysis I & II examination of 22 January 2020 (Prof. Peter S. Jossen), Part B, Problem 4.*

Solution: *See solution on Page 318*

Centroids (Section 20.d)

Every integral in this chapter so far has been computed for its own sake. The simplest thing one actually does with an integral over a region is average a coordinate over it, and the resulting point is worth a name.

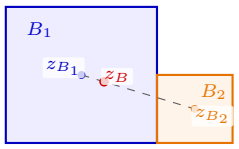
Definition 20.d.45 (*Centroid of a plane region*): Let $B \subseteq \mathbb{R}^2$ be Jordan measurable with $\mathrm{vol}_2(B) > 0$. The "*centroid*" (German: „*Flächenschwerpunkt*“) of B is the point $z_B := (x_B, y_B)$ with

$$x_B := \frac{1}{\mathrm{vol}_2(B)} \int_B x \, d\,\mathrm{vol}_2(x, y), \qquad y_B := \frac{1}{\mathrm{vol}_2(B)} \int_B y \, d\,\mathrm{vol}_2(x, y).$$

The same formula defines the centroid of a Jordan measurable set in any $\mathbb{R}^n$, one coordinate at a time; nothing below uses the restriction to the plane except the triangle.

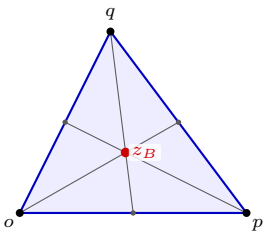
Remark 20.d.46 (*Why the normalisation is there*): Without the factor $1/\mathrm{vol}_2(B)$ the two integrals would scale with the size of B as well as with its position, and doubling a region would move its *"centre"*. Dividing by the area makes z_B an **average** of the coordinate functions rather than a total, and the immediate consequence is that z_B is where one expects it to be: for a region symmetric about a line, the centroid lies on that line, because the coordinate perpendicular to it integrates to zero.

Centroids add



$\mathrm{vol}_2(B_2)/\mathrm{vol}_2(B) = \frac{1}{5}$, and z_B sits one fifth of the way along

. . . and a triangle's



the three medians meet at $z_B = \frac{1}{3}(o + p + q)$

Exercise 20.d.47 (*Centroids add, and the centroid of a triangle*): Let $B \subseteq \mathbb{R}^2$ be Jordan measurable with $\mathrm{vol}_2(B) > 0$.

(a) Suppose $B = \bigcup_{i=1}^{n} B_i$ for pairwise disjoint Jordan measurable sets $B_1, \dots, B_n \subseteq \mathbb{R}^2$, each with $\mathrm{vol}_2(B_i) > 0$. Show that

$$z_B = \sum_{i=1}^{n} \frac{\mathrm{vol}_2(B_i)}{\mathrm{vol}_2(B)}\, z_{B_i}.$$

(b) Show that if B is a triangle with vertices $o = (0,0)$, $p = (r,0)$ and $q = (s,t)$, where $r, s, t > 0$, then $z_B = \frac{1}{3}(o + p + q)$.

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *of 31 January 2023, Problem 11. That paper names no lecturer.*

Solution: *See solution on Page 317*

Exercise 20.d.48 (*What the centroid minimises*): Let $B := \{(x,y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0,\ x^2 + y^2 \leq 1\}$ be the closed quarter disc, and for $(s,t) \in \mathbb{R}^2$ put

$$I(s,t) := \int_B (x-s)^2 + (y-t)^2 \, d\,\mathrm{vol}_2(x,y).$$

(a) Determine the pair (s,t) at which I is minimal.

(b) Show that the answer to part **(a)** is no accident: for **any** Jordan measurable $B \subseteq \mathbb{R}^2$ with $\mathrm{vol}_2(B) > 0$, the function I is minimised exactly at $(s,t) = z_B$, and at no other point.

Info: *Part* ***(a)*** *is taken from the mock examination* ***Probeprüfung Analysis II*** *of 17 August 2022 (revised), Problem 1; that paper names no lecturer. Part* ***(b)*** *is added here, since the general statement is what the computation is really about.*

Solution: *See solution on Page 318*

Solutions (Section 20.e)

Solution to Exercise 20.a.13:

In each part the work is the same two steps: find $|\det \mathrm{J}|$, then translate every constraint. As the exercise says, no integral is actually evaluated.

(a) **Polar coordinates.** Here $\Phi(r,\theta) := (r\cos\theta, r\sin\theta)$ has $\det \mathrm{J}\,\Phi = r > 0$ (Remark 20.a.8), so $dx_1 dx_2 = r\,dr\,d\theta$. The three constraints translate one at a time:

- $1 < x_1^2 + x_2^2 < 4$ becomes $1 < r < 2$;
- $x_1 > 0$ becomes $\cos\theta > 0$, i.e. $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$;
- $x_2 < x_1$ becomes $r\sin\theta < r\cos\theta$; since $r > 0$ and $\cos\theta > 0$ this is $\tan\theta < 1$, i.e. $\theta < \frac{\pi}{4}$.

So the transformed domain is the rectangle $(r,\theta) \in (1,2) \times \left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$ and

$$\int_A x_1^2 \sin(x_2)\, dx_1 dx_2 = \int_1^2 \int_{-\pi/2}^{\pi/4} r^3 \cos^2\theta\, \sin(r\sin\theta)\, d\theta\, dr,$$

the extra power of r coming from $x_1^2 = r^2\cos^2\theta$ times the Jacobian factor r.

(b) $u := xy$, $v := x^2$. Following the hint, fix the signs first. On B the constraint $x^2 < y$ forces $y > 0$, and then $xy > 1 > 0$ forces $x > 0$; so $u > 0$ and $v > 0$ throughout, and $x = \sqrt{v}$, $y = u/\sqrt{v}$.

Differentiating in the convenient direction,

$$\frac{\partial(u,v)}{\partial(x,y)} = \det \begin{pmatrix} y & x \\ 2x & 0 \end{pmatrix} = -2x^2 = -2v, \qquad \text{so} \qquad dx\,dy = \frac{du\,dv}{2v}.$$

Now the constraints. $1 < xy < 2$ becomes $1 < u < 2$ at once. The second is the interesting one: $x^2 < y < 2x^2$ reads $v < u/\sqrt{v} < 2v$, i.e.

$$v^{3/2} < u < 2\,v^{3/2}, \qquad \text{equivalently} \qquad \left(\frac{u}{2}\right)^{2/3} < v < u^{2/3}.$$

The integrand becomes $y^2e^{-xy} = (u^2/v)\,e^{-u}$, so

$$\int_B y^2e^{-xy}\,dx\,dy = \int_1^2 \int_{(u/2)^{2/3}}^{u^{2/3}} \frac{u^2e^{-u}}{2v^2}\,dv\,du.$$

Note what did **not** happen: straightening $1 < xy < 2$ into a pair of constant bounds did nothing for the other constraint, which stays curved. Straightening one constraint does not oblige the rest to cooperate.

(c) A linear change of variables. With $u := z$, $v := z - 2y$, $w := 3x + y + z$,

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \det\begin{pmatrix} 0 & 0 & 1 \\ 0 & -2 & 1 \\ 3 & 1 & 1 \end{pmatrix} = 6, \qquad \text{so} \qquad dx\,dy\,dz = \frac{du\,dv\,dw}{6},$$

a constant, since the map is linear. Each constraint already **is** one of the new coordinates, so the transformed domain is literally the box

$$(u,v,w) \in (0,1) \times (1,2) \times (-2,0).$$

Inverting the substitution gives $z = u$, $y = \frac{u-v}{2}$ and $x = \frac{1}{6}\big(2w - 3u + v\big)$, so $xyz = \frac{1}{12}\,u\,(u-v)\,(2w - 3u + v)$ and

$$\int_C xyz\,dx\,dy\,dz = \int_0^1 \int_1^2 \int_{-2}^0 \frac{u\,(u-v)\,(2w-3u+v)}{72}\,dw\,dv\,du.$$

AI-Note 20.8: The three parts are graded deliberately, and comparing them is the lesson. In part **(c)** the change of variables was **chosen** to be the constraints, so the domain is a box and the Jacobian a constant, and nothing is left to do. In part **(a)** the domain is a box in the new coordinates but the Jacobian is not constant. In part **(b)** neither holds: the Jacobian varies and one of the two constraints stays curved. Reading the substitution off the constraints, as in part **(c)**, is the move worth remembering, and it is available far more often than it looks.

Q.E.D.

Solution to AI-Exercise 20.a.14:

(a) The new domain and the Jacobian factor. The map $\Psi(x,y) := (y - x,\ y + x)$ is linear and invertible, with

$$\Psi^{-1}(u,v) = (x,y) = \left(\frac{v-u}{2},\ \frac{u+v}{2}\right), \qquad \det \mathrm{J}\,\Psi^{-1} = \det\begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} = -\frac{1}{2},$$

so $dx\,dy = \frac{1}{2}\,du\,dv$. The three sides of T translate one at a time: $x \geq 0$ reads $u \leq v$, the condition $y \geq 0$ reads $u \geq -v$, and $x + y \leq 1$ reads $v \leq 1$. Taking strict inequalities for the interior gives exactly $0 < v < 1$ and $-v < u < v$, a triangle again, but one whose horizontal slices run in the direction the integrand likes.

(b) The value. On that region the integrand is $e^{u/v}$, and v is constant along the inner integration, so the inner integral is an exponential in u:

$$I = \frac{1}{2}\int_0^1 \int_{-v}^{v} e^{u/v}\,du\,dv = \frac{1}{2}\int_0^1 \Big[v\,e^{u/v}\Big]_{u=-v}^{u=v}\,dv = \frac{e - e^{-1}}{2}\int_0^1 v\,dv = \frac{e - e^{-1}}{4}.$$

(c) The origin. A single point is a Jordan null set, so removing it changes neither the Jordan measurability of T nor the value of the integral. What does need checking is that the integral

exists at all, and it does, because the integrand is **bounded**: on T one has $|y-x| \le y+x$, so the exponent u/v lies in $[-1,1]$ and the integrand lies between e^{-1} and e. It is the boundedness, and not the smallness of the bad set on its own, that rescues the computation.

Remark 20.e.49 (*Two ways to choose a substitution*): Example 20.b.37 chose Φ to straighten the **domain**. Here the choice comes from the **integrand**, which is a function of $y-x$ and $y+x$ and of nothing else. That the domain also becomes pleasant is a bonus, and a mild one: the new region is still a triangle. What made the integral collapse is that the inner integration now runs along the direction in which the integrand is a plain exponential.

Q.E.D.

Solution to AI-Exercise 20.b.38:

(a) The inner integral asks for an antiderivative of $y \mapsto e^{y^2}$, and there is none in elementary terms. Nothing about the outer integral repairs this, so the order as written is a dead end.

(b) The region is the triangle below the diagonal, and it admits two descriptions:

$$R = \{(x,y) : 0 \le x \le 1,\ x \le y \le 1\} = \{(x,y) : 0 \le y \le 1,\ 0 \le x \le y\}.$$

The first reads it by vertical slices, the second by horizontal ones. Taking the second and applying Theorem 20.b.29,

$$I = \int_0^1 \int_0^y e^{y^2}\, dx\, dy.$$

(c) Now e^{y^2} is constant in x, so the inner integral contributes a factor of y, and that factor is exactly what the substitution $s := y^2$ needs:

$$I = \int_0^1 y\, e^{y^2}\, dy = \left[\frac{e^{y^2}}{2}\right]_0^1 = \frac{e-1}{2}.$$

Remark 20.e.50 (*What the swap actually bought*): The reversal did not find a cleverer antiderivative. It produced a factor of y that was not there before, and $y\,e^{y^2}$ integrates where e^{y^2} does not. That is the pattern worth recognising: when the length of a slice depends on the variable that is causing the trouble, swapping the order hands you that length as a factor in the integrand.

Q.E.D.

Solution to AI-Exercise 20.c.43:

(a) Both exponentials equal 1 at $x = 0$, so the quotient is a difference quotient in disguise. Expanding, $e^{-ax} - e^{-bx} = (b-a)x + O(x^2)$, and therefore

$$\lim_{x \to 0^+} \frac{e^{-ax} - e^{-bx}}{x} = b - a.$$

Assigning that value at $x = 0$ makes the integrand continuous on $[0,\infty)$, and it decays exponentially, so the improper integral converges.

(b) Only the second exponential depends on t, and the x that the chain rule produces cancels the one in the denominator:

$$I'(t) = \int_0^\infty \frac{\partial}{\partial t}\left(\frac{e^{-ax} - e^{-tx}}{x}\right) dx = \int_0^\infty \frac{x\, e^{-tx}}{x}\, dx = \int_0^\infty e^{-tx}\, dx = \frac{1}{t}.$$

That cancellation is the whole trick. Differentiating in the parameter removes the $1/x$ which made the integrand hard, and leaves an integral of one line.

(c) At $t = a$ the two exponentials coincide and the integrand vanishes identically, so $I(a) = 0$. Consequently, by the fundamental theorem of calculus,

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x}\,dx = I(b) = I(b) - I(a) = \int_a^b \frac{dt}{t} = \log\left(\frac{b}{a}\right).$$

That the answer sees only the ratio b/a is now visible in the derivation rather than a surprise, since $I'(t) = 1/t$ is scale-invariant.

Remark 20.e.51 (*The hypotheses of Theorem 20.c.40 do not literally cover this*): As stated, Theorem 20.c.40 differentiates under an integral over a **compact** set K, and here the domain is $[0,\infty)$. The step is still legitimate, by the same reasoning Example 20.c.42 appeals to. Fix $0 < t_0 < t_1$. For $t \in [t_0, t_1]$ the t-derivative of the integrand is $e^{-tx} \le e^{-t_0 x}$, a bound that is independent of t and integrable on $[0,\infty)$, so the differentiation may be carried out on $[0,R]$ and the limit $R \to \infty$ taken afterwards. The compact statement is the one the course proves; the dominated version is what the applications actually use.

Q.E.D.

Proof of Exercise 20.a.15:

(a) Define $\Phi(u,v,w) := (au, bv, cw)^{\mathsf{T}}$, and write $(x,y,z) = \Phi(u,v,w)$. Then $(x/a)^2 + (y/b)^2 + (z/c)^2 = u^2 + v^2 + w^2$, so (u,v,w) lies in the closed unit ball if and only if $(x,y,z) \in U$. Since $a,b,c > 0$, the map Φ is a linear bijection, and therefore $\Phi(\overline{B_1(0)}) = U$.

(b) The Jacobian matrix of Φ is the diagonal matrix

$$\mathrm{J}\,\Phi = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

constant in (u,v,w), with determinant $\det \mathrm{J}\,\Phi = abc > 0$.

(c) By the change of variables theorem (Theorem 20.a.1),

$$\mathrm{vol}_3(U) = \int_U 1\,d\,\mathrm{vol}_3 = \int_{\overline{B_1(0)}} |\det \mathrm{J}\,\Phi|\,d\,\mathrm{vol}_3 = abc \int_{\overline{B_1(0)}} 1\,d\,\mathrm{vol}_3\,.$$

The unit ball in $\mathbb{R}^3$ has volume $\frac{4}{3}\pi$, and therefore

$$\mathrm{vol}_3(U) = \frac{4}{3}\pi abc.$$

Q.E.D.

Proof of Exercise 20.a.16:

(a) Differentiating each component,

$$D_x\varphi = \mathrm{J}\,\varphi(x) = \begin{pmatrix} 3x_1^2 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \det \mathrm{J}\,\varphi(x) = 3x_1^2 + 1 \ge 1 > 0.$$

(b) The determinant never vanishes, so by the inverse function theorem (Theorem 15.a.1) φ is a local diffeomorphism at every point; in particular $\varphi(\mathbb{R}^2)$ is open. It remains to prove global injectivity, and then φ is a bijection onto its image which is locally a diffeomorphism, hence a diffeomorphism onto its image.

Suppose $\varphi(x) = \varphi(y)$. The second components give $x_2 + x_1 = y_2 + y_1$, that is, $x_2 - y_2 = -(x_1 - y_1)$. Substituting this into the equality of first components, $x_1^3 - x_2 = y_1^3 - y_2$, yields

$$x_1^3 - y_1^3 = x_2 - y_2 = -(x_1 - y_1), \qquad \text{that is,} \qquad (x_1 - y_1)\big(x_1^2 + x_1 y_1 + y_1^2 + 1\big) = 0.$$

The second factor is at least 1, because $x_1^2 + x_1y_1 + y_1^2 = \left(x_1 + \frac{y_1}{2}\right)^2 + \frac{3}{4}y_1^2 \geq 0$. Hence $x_1 = y_1$, and then $x_2 = y_2$.

(c) By part **(a)** the Jacobian determinant is positive, so the change of variables theorem (Theorem 20.a.1) gives

$$\mathrm{vol}_2(\varphi(Q)) = \int_Q |\det \mathrm{J}\,\varphi|\,dx = \int_0^1\int_0^1 \left(3x_1^2+1\right) dx_2\,dx_1 = \int_0^1 \left(3x_1^2+1\right) dx_1 = 1+1 = 2.$$

Remark 20.e.52: The map is not linear, so the image of the square is not a parallelogram and there is no single factor by which areas scale. The change of variables formula handles that by scaling **pointwise** and integrating the result, which is the whole reason the determinant appears inside the integral rather than outside it.

Q.E.D.

Solution of Exercise 20.a.17:

Following the hint, the substitution $(x, y) \mapsto (y, x)$ is a linear change of variables with Jacobian determinant -1, and it carries the integrand $x^2e^{-\alpha(x^2+y^2)}$ to $y^2e^{-\alpha(x^2+y^2)}$. Hence the two integrals agree, and adding them gives

$$2I_\alpha = \int_{\mathbb{R}^2} (x^2+y^2)\, e^{-\alpha(x^2+y^2)}\,dx\,dy.$$

The integrand is now radial, which is what makes polar coordinates pay. Writing $x = r\cos\theta$, $y = r\sin\theta$, with Jacobian determinant r,

$$2I_\alpha = \int_0^{2\pi}\int_0^\infty r^2e^{-\alpha r^2}\, r\,dr\,d\theta = 2\pi\int_0^\infty r^3e^{-\alpha r^2}\,dr.$$

Substituting $u := r^2$, so that $du = 2r\,dr$, turns the remaining integral into one that can be done by parts:

$$\int_0^\infty r^3e^{-\alpha r^2}\,dr = \frac{1}{2}\int_0^\infty ue^{-\alpha u}\,du = \frac{1}{2\alpha^2}.$$

Therefore $2I_\alpha = 2\pi\cdot\frac{1}{2\alpha^2} = \frac{\pi}{\alpha^2}$, and

$$I_\alpha = \frac{\pi}{2\alpha^2}.$$

Remark 20.e.53: Two sanity checks come free. Scaling $(x, y) \mapsto (x, y)/\sqrt{\alpha}$ predicts $I_\alpha = \alpha^{-2}I_1$, which the answer satisfies; and the integrand factorises as $x^2e^{-\alpha x^2}\cdot e^{-\alpha y^2}$, so I_α is the product of the two one-dimensional moments,

$$\left(\int_{\mathbb{R}} x^2e^{-\alpha x^2}dx\right)\left(\int_{\mathbb{R}} e^{-\alpha y^2}dy\right) = \frac{1}{2\alpha}\sqrt{\frac{\pi}{\alpha}}\cdot\sqrt{\frac{\pi}{\alpha}} = \frac{1}{2\alpha}\cdot\frac{\pi}{\alpha} = \frac{\pi}{2\alpha^2},$$

as found. The symmetry trick in the first line is the same one that evaluates $\int_{\mathbb{R}} e^{-x^2}dx$ by squaring it: an integral that resists in Cartesian coordinates becomes radial once it is paired with its own mirror image.

Q.E.D.

Proof of Exercise 20.a.19:

Everything turns on a single number:

$$\det \mathrm{J}\,\Phi(0,0) = 2\cdot 1 - 1\cdot(-1) = 3.$$

In particular $\mathrm{J}\,\Phi(0,0)$ is invertible, so the inverse function theorem (Theorem 15.a.1) supplies an open ball V around the origin on which Φ is a C^1-diffeomorphism onto an open set. Since Φ is

C^1, the map $x \mapsto |\det \mathrm{J}\,\Phi(x)|$ is continuous, and therefore

$$|\det \mathrm{J}\,\Phi(x)| = 3 + \varepsilon(x), \qquad \text{where } \varepsilon(x) \to 0 \text{ as } x \to 0. \tag{20.3}$$

For a Jordan measurable E with $\overline{E} \subseteq V$, the change of variables formula (Theorem 20.a.1), taken in the multiplied form $\int_{\Phi(E)} f = \int_E (f \circ \Phi)\,|\det \mathrm{J}\,\Phi|$ recorded just below it and applied to $f \equiv 1$, reads

$$\mathrm{vol}_2\big(\Phi(E)\big) = \int_E |\det \mathrm{J}\,\Phi(x)|\,dx. \tag{20.4}$$

Parts **(b)** and **(c)** are both this identity, read against (20.3).

(a) True. For r small enough we have $B_r(0,0) \subseteq V$, and a diffeomorphism is in particular an open map. Hence $\Phi(B_r(0,0))$ is open.

(b) False. Applying (20.4) to $E = B_r(0,0)$ and substituting (20.3),

$$\mathrm{vol}_2\big(\Phi(B_r(0,0))\big) = \int_{B_r(0,0)} \big(3 + \varepsilon(x)\big)\,dx = 3\pi r^2 + \int_{B_r(0,0)} \varepsilon(x)\,dx.$$

The remaining integral is bounded in absolute value by $\pi r^2 \sup_{B_r(0,0)} |\varepsilon|$, and that supremum tends to 0 with r, so the term is $o(r^2)$. The correct expansion is $3\pi r^2 + o(r^2)$. The stated coefficient 2 is the top-left entry of the Jacobian rather than its determinant.

(c) True. Write $R_r := [0, r] \times [0, ar^{1/2}]$, a rectangle of area $ar^{3/2}$. Every point of R_r has norm at most $\sqrt{r^2 + a^2 r}$, which tends to 0, so $R_r \subseteq V$ for r small enough and (20.4) applies:

$$\mathrm{vol}_2\big(\Phi(R_r)\big) = \int_{R_r} \big(3 + \varepsilon(x)\big)\,dx = 3ar^{3/2} + \int_{R_r} \varepsilon(x)\,dx,$$

and the error is bounded by $ar^{3/2} \sup_{R_r} |\varepsilon| = o(r^{3/2})$.

Remark 20.e.54 (*Why part (c) is not simply part (b) again*): Under (20.4) the two parts really are the same computation, and that is the point of the pair. The tempting alternative route is not. One is inclined to replace Φ by its linearisation, $\Phi(x) = \mathrm{J}\,\Phi(0,0)x + o(|x|)$, and read the volume off the linear map. For the ball that argument can be made to work, since the error $o(|x|) = o(r)$ is small against the radius r in every direction. For R_r it breaks: the rectangle has sides r and $ar^{1/2}$, so its diameter is of order $r^{1/2}$ while its short side is of order r, and the error term $o(r^{1/2})$ is then far **larger** than the side it would have to be weighed against. The change of variables formula never approximates Φ at all. It integrates the exact local scaling factor, and the only approximation left is to that factor's value, uniformly over a set shrinking to a point.

AI-Note 20.9: The set $X := \Phi^{-1}((0,0)^{\mathsf{T}})$ is introduced in the exam's statement and then used by none of the three items. It is reproduced in Exercise 20.a.19 for fidelity to the source, and nothing above depends on it. A little can be said about it in passing: the inverse function theorem makes Φ injective near the origin, so X meets a small ball around $(0,0)$ in that single point, while away from the origin the hypotheses constrain X not at all.

Q.E.D.

Proof of Exercise 20.a.20:

The map is affine, $\Psi(v) = Mv + b$, with

$$M = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & \alpha & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} e^{\pi} \\ 0 \\ 0 \end{pmatrix}.$$

Expanding along the first row gives $\det M = -(1 - \alpha^2) = \alpha^2 - 1$. A translation does not change volume, so for every Jordan measurable E,

$$\mathrm{vol}_3\big(\Psi(E)\big) = |\det M|\ \mathrm{vol}_3(E) = |\alpha^2 - 1|\ \mathrm{vol}_3(E). \tag{20.5}$$

Iterating Ψ iterates its linear part, so Ψ^k is affine with linear part M^k and (20.5) gives $\mathrm{vol}_3(\Psi^k(E)) = |\det M|^k \,\mathrm{vol}_3(E)$. Every item is now a matter of evaluating $|\alpha^2 - 1|$.

(a) **False.** By (20.5), $\mathrm{vol}_3(\Psi(E))$ vanishes for every E precisely when $\det M = 0$, that is when $\alpha^2 = 1$. That holds for $\alpha = 1$ **and** for $\alpha = -1$, so the *"only if"* direction fails. The correct criterion is $\alpha = \pm 1$.

(b) **True.** For $\alpha = -2$ we get $\det M = 4 - 1 = 3$, and therefore $\mathrm{vol}_3(\Psi^k(E)) = 3^k \,\mathrm{vol}_3(E)$ for every $k \geq 0$.

(c) **True.** For $\alpha = 0$ we get $\det M = -1$, so $|\det M| = 1$ and every iterate preserves volume, the 2025th included.

(d) **False.** The inverse of an affine map with invertible linear part M is affine with linear part M^{-1}, so for $\alpha \in (0,1)$, where $1 - \alpha^2 > 0$,

$$\mathrm{vol}_3\left(\Psi^{-1}([0,1]^3)\right) = \frac{\mathrm{vol}_3([0,1]^3)}{|\det M|} = \frac{1}{1-\alpha^2}.$$

The stated value is the negative of this, and no volume is negative.

Remark 20.e.55: Item **(d)** can be dismissed before any computation: the left-hand side is a volume and the right-hand side is negative throughout $(0,1)$. Item **(a)** is the one that rewards care. Writing the scaling factor as $|\alpha^2 - 1|$ puts both roots in view at once, and a criterion of the form *"if and only if $\alpha = 1$"* then fails for the cheapest of reasons: it forgets that squaring loses a sign.

Q.E.D.

Proof of Exercise 20.a.22:

(a) **True:** On the restricted domain $(0, 2\pi) \times (0, \pi) \times (0, \infty)$, the map Ψ is injective. The angles $\phi \in (0, 2\pi)$ and $\theta \in (0, \pi)$ uniquely parameterise points away from the half-plane $\{(x, 0, z) \mid x \geq 0\}$ and the z-axis.

(b) **False:** The image of Ψ omits the set where $\phi = 0$ (which corresponds to the half-plane $y = 0, x \geq 0$) and the z-axis (where $\theta = 0$ or $\theta = \pi$, or $r = 0$). Therefore, it does not cover all of $\mathbb{R}^3$.

(c) **False:** The order of the variables in the definition of Ψ is (ϕ, θ, r). Computing the Jacobian matrix for this specific ordering gives a determinant of $-r^2 \sin\theta$, not $+r^2 \sin\theta$. While the **absolute value** of the determinant (which appears in the change of variables formula) is $r^2 \sin\theta$, the determinant itself is negative because (ϕ, θ, r) is not an orientation-preserving ordering of the standard spherical coordinates.

(d) **False:** The volume of a ball of radius R in $\mathbb{R}^3$ is $\frac{4}{3}\pi R^3$. Therefore, $\mathrm{vol}_3(B_R \setminus B_r) = \frac{4}{3}\pi(R^3 - r^3)$, not $\pi(R^3 - r^3)$.

(e) **True:** By the change of variables theorem, integrating the absolute value of the Jacobian determinant over the domain of Ψ yields the volume. The integral with respect to ϕ separates and gives a factor of $\int_0^{2\pi} d\phi = 2\pi$, leaving $2\pi \int_0^{\pi} \int_0^{R} r^2 \sin\theta \, dr \, d\theta$.

(f) **True:** This limit represents the derivative of the volume $V(R) = \frac{4}{3}\pi R^3$ with respect to R, evaluated at $R = 1$. We have $\frac{d}{dR}\left(\frac{4}{3}\pi R^3\right) = 4\pi R^2$, which at $R = 1$ evaluates to 4π (this is exactly the surface area of the unit sphere).

AI-Note 20.10: Item **(c)** is answered *"false"* above on the sign, and that answer is what the statement literally says: with the variables ordered (ϕ, θ, r) as the exam itself orders them, the determinant is $-r^2 \sin\theta$. A marking scheme may nonetheless have intended *"true"*, reading the item as being about the factor $|\det \mathrm{J}\,\Psi|$ that enters the change of variables formula, which is $+r^2 \sin\theta$. No official solution to this exam was available to settle it, so the divergence is recorded

rather than resolved. Note also that the exam writes $\det \mathrm{J}\,\Psi(\theta, r)$, naming only two of the three variables; that is the source's own notation, kept in Exercise 20.a.22 as it stands.

Q.E.D.

Solution to Exercise 20.a.23:

For any linear map $\Phi\colon \mathbb{R}^2 \to \mathbb{R}^2$, the change of variables formula implies that the volume of the image of a Jordan measurable set E is given by

$$\mathrm{vol}_2(\Phi(E)) = |\det \Phi| \cdot \mathrm{vol}_2(E).$$

To satisfy the condition $\mathrm{vol}_2(\Phi(E)) = 2\,\mathrm{vol}_2(E)$ for all E, we must have $|\det \Phi| = 2$. Any matrix with determinant ± 2 will work. An explicit and simple choice is the diagonal matrix

$$\Phi(x, y) = (2x, y) \quad \text{or equivalently, the matrix } \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Other valid choices include $\Phi(x, y) = (x, 2y)$ or $\Phi(x, y) = (\sqrt{2}x, \sqrt{2}y)$. Q.E.D.

Proof of AI-Exercise 20.a.25:

Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the linear map $T(t_1, t_2, t_3) := t_1 v_1 + t_2 v_2 + t_3 v_3$. Its matrix in the standard basis has columns v_1, v_2, v_3, so $\det T = \det(v_1, v_2, v_3) = 2$, and A is precisely the image under T of the closed unit ball $B := \{t \in \mathbb{R}^3 \mid t_1^2 + t_2^2 + t_3^2 \le 1\}$.

(a) True: By the change of variables formula (Theorem 20.a.1), or equally by the geometric reading of the determinant,

$$\mathrm{vol}_3(A) = \mathrm{vol}_3(T(B)) = |\det T| \cdot \mathrm{vol}_3(B) = 2 \cdot \frac{4}{3}\pi = \frac{8\pi}{3}.$$

(b) True: The unit ball B is Jordan measurable, being bounded with boundary the unit sphere, a smooth surface and hence of volume zero. The image of a Jordan measurable set under an invertible linear map is again Jordan measurable.

(c) False: By part **(a)** the volume depends on the v_i only through $\det(v_1, v_2, v_3)$, which the hypothesis fixes at 2. Every admissible choice therefore yields the same volume $\frac{8\pi}{3}$.

Q.E.D.

Solution of Exercise 20.a.18:

The integrand is continuous and positive on $\mathbb{R}^2$, so the improper integral of Definition 19.b.23 is well defined, and by Lemma 19.b.25 it may be computed along the exhaustion by closed disks. In polar coordinates the integrand depends only on $r = \sqrt{x^2 + y^2}$, so the angular integral contributes 2π and

$$I_\alpha = 2\pi \int_0^\infty e^{-\alpha r}\, r\, dr.$$

Integrating by parts with $u := r$ and $v' := e^{-\alpha r}$,

$$\int_0^R r\, e^{-\alpha r}\, dr = \Big[-\frac{r}{\alpha} e^{-\alpha r}\Big]_0^R + \frac{1}{\alpha} \int_0^R e^{-\alpha r}\, dr = -\frac{R}{\alpha} e^{-\alpha R} + \frac{1}{\alpha^2}\big(1 - e^{-\alpha R}\big).$$

Both exponential terms vanish as $R \to \infty$ because $\alpha > 0$, so $\int_0^\infty r e^{-\alpha r}\, dr = 1/\alpha^2$ and

$$\boxed{I_\alpha = \frac{2\pi}{\alpha^2}}$$

The three statements now follow.

(a) $I_1 = \pi$**: false.** $I_1 = 2\pi$.

(b) $I_2 = \pi/2$**: true.** $I_2 = 2\pi/4 = \pi/2$.

(c) $\lim_{\alpha\to+\infty} \alpha I_\alpha = 0$**: true.** $\alpha I_\alpha = 2\pi/\alpha \to 0$.

Remark 20.e.56 (*Why the r changes the answer so much*): Without the Jacobian factor r the radial integral would be $\int_0^\infty e^{-\alpha r} dr = 1/\alpha$, and the decay in α would be one power slower. The factor r suppresses the region near the origin, where the integrand is largest, and it is exactly what turns $1/\alpha$ into $1/\alpha^2$. Item **(c)** is a check on that power and nothing else: it is αI_α rather than I_α precisely so that the answer 0 distinguishes α^{-2} from α^{-1}.

Q.E.D.

Solution of Exercise 20.a.24:

By Theorem 20.a.1, a linear Φ scales every Jordan measurable set's volume by $|\det \Phi|$, so the requirement is exactly $|\det \Phi| = 1$ together with $\Phi \neq \mathrm{id}$. The shear

$$\Phi(x, y) := (x + y,\ y), \qquad \Phi = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \det \Phi = 1,$$

does it. So do the reflection $(x, y) \mapsto (y, x)$, of determinant -1, and every rotation, of determinant $+1$. The point reflection $(x, y) \mapsto (-x, -y)$ is one of the latter: in the plane it **is** the rotation by π, with determinant $(-1)(-1) = +1$, which is worth noticing because in $\mathbb{R}^3$ the same formula gives determinant -1 instead.

Remark 20.e.57 (*What the extra clause is for*): Dropping *"different from the identity"* would make the problem vacuous, and adding it makes the answer informative: the volume-preserving linear maps of the plane form a large group, namely $\{A \in \mathrm{GL}_2(\mathbb{R}) \mid |\det A| = 1\}$, and the identity is one point of it. The shear above is the example worth remembering, because it is the one that visibly distorts a square into a parallelogram while changing nothing about its area. Length, angle and shape are all destroyed; only area survives, and area is all the change of variables formula ever sees.

Q.E.D.

Solution to Exercise 20.a.26:

(a) Let $A = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ with all $\lambda_i \neq 0$, so that $\det A = \lambda_1 \cdots \lambda_n$. For a box $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ the image $A(Q)$ is again a box, the i-th edge running between $\lambda_i a_i$ and $\lambda_i b_i$, of length $|\lambda_i|(b_i - a_i)$. Hence

$$\mathrm{vol}\big(A(Q)\big) = \prod_{i=1}^{n} |\lambda_i|(b_i - a_i) = |\det A|\ \mathrm{vol}(Q).$$

For general Jordan measurable B, approximate from inside and outside by dyadic sets $F \subseteq B \subseteq G$ with $\mathrm{vol}(G) - \mathrm{vol}(F) < \varepsilon$ (Definition 19.a.1). Since A is a bijection, $A(F) \subseteq A(B) \subseteq A(G)$, and both $A(F)$ and $A(G)$ are finite disjoint unions of boxes, to which the computation above applies cube by cube. Therefore

$$|\det A|\ \mathrm{vol}(F) \le \mu_{\mathrm{in}}\big(A(B)\big) \le \mu_{\mathrm{out}}\big(A(B)\big) \le |\det A|\ \mathrm{vol}(G),$$

and the two outer quantities differ by less than $|\det A|\,\varepsilon$. Letting $\varepsilon \to 0$ shows $A(B)$ is Jordan measurable with $\mathrm{vol}(A(B)) = |\det A|\ \mathrm{vol}(B)$.

(b) Two observations combine. First, (20.1) is multiplicative: if it holds for A_1 and for A_2, then applying it twice gives

$$\mathrm{vol}\big(A_1 A_2(B)\big) = |\det A_1|\ \mathrm{vol}\big(A_2(B)\big) = |\det A_1|\,|\det A_2|\ \mathrm{vol}(B) = |\det(A_1 A_2)|\ \mathrm{vol}(B),$$

using $\det(A_1A_2) = \det A_1 \det A_2$. So the set of matrices satisfying (20.1) is closed under products. Second, Gaussian elimination writes any invertible A as a product of elementary matrices, and each elementary matrix is a diagonal matrix, a permutation matrix or a shear. Part **(a)** handles the diagonal factors, so proving the permutation and shear cases finishes the proof for every invertible A.

(c) A permutation matrix P_π sends a box to the box with its edges relabelled, so it preserves volume, and $|\det P_\pi| = 1$: the case is immediate.

For the shear, take A to be the identity with one off-diagonal entry c, say $Ae_j = e_j$ for $j \neq 2$ and $Ae_2 = e_2 + c\,e_1$; every shear is of this form after a permutation. Then A is triangular with unit diagonal, so $\det A = 1$, and the claim is that A preserves volume. Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$. Writing points as (x_1, x_2, x') with $x' \in \mathbb{R}^{n-2}$, we have $A(x_1, x_2, x') = (x_1 + cx_2,\ x_2,\ x')$. Compute $\mathrm{vol}(A(Q))$ by Fubini (Theorem 20.b.29), integrating in x_1 first with x_2 and x' held fixed. For fixed $x_2 \in [a_2, b_2]$ and x', the slice of $A(Q)$ is the interval $[a_1 + cx_2,\ b_1 + cx_2]$, a **translate** of $[a_1, b_1]$, of the same length $b_1 - a_1$. Since every slice has the length of the corresponding slice of Q, integrating over the remaining variables gives $\mathrm{vol}(A(Q)) = \mathrm{vol}(Q) = |\det A|\ \mathrm{vol}(Q)$.

Remark 20.e.58 (*Where the determinant actually enters*): Only part **(a)** ever multiplies anything by a determinant, and there it is nothing but the product of the edge-scaling factors, the determinant of a diagonal matrix being visible to the naked eye. The shear contributes the factor 1, and the permutation contributes 1: neither case requires knowing what a determinant is. So the whole content of (20.1) is the trivial diagonal computation, plus the fact that det is multiplicative and elementary matrices generate $\mathrm{GL}_n(\mathbb{R})$. That is a good answer to the question of why $|\det A|$, of all functions of A, is the volume distortion factor: it is the only multiplicative function that scales correctly on diagonal matrices and is blind to shears.

Part **(c)** is worth reading once more for the shape of the argument. A shear preserves volume because it slides each line parallel to e_1 along itself, and a translated interval has the same length. Cavalieri's principle in one sentence, and the reason a leaning stack of coins has the volume of a straight one.

Q.E.D.

Solution to Exercise 20.b.39:

The integrand factors as $e^{-x^2} \cdot e^{-y^2} \cdot e^{-\alpha|z|}$, one factor per variable. It is continuous and non-negative, so Fubini's theorem splits the integral into a product of three one-dimensional ones, and the value in $[0, +\infty]$ is the same whichever exhaustion of $\mathbb{R}^3$ computes it. Two of the three factors are Gaussians, so

$$I_\alpha = \left(\int_{\mathbb{R}} e^{-x^2} dx\right)\left(\int_{\mathbb{R}} e^{-y^2} dy\right)\int_{\mathbb{R}} e^{-\alpha|z|} dz = \pi \int_{\mathbb{R}} e^{-\alpha|z|} dz.$$

The remaining factor is where α finally matters. For $\alpha > 0$ the integrand is even and integrable,

$$\int_{\mathbb{R}} e^{-\alpha|z|} dz = 2\int_0^\infty e^{-\alpha z} dz = \frac{2}{\alpha}, \qquad \text{so} \qquad I_\alpha = \frac{2\pi}{\alpha};$$

for $\alpha = 0$ the integrand is the constant 1 and the integral is $+\infty$.

(a) False. $I_0 = +\infty$: the z-direction is not damped at all, so the integrand is constant along every vertical line. The value $\sqrt{\pi}$ is what a **single** Gaussian factor contributes, which is the trap.

(b) True for every $\alpha > 0$, since $\alpha I_\alpha = \alpha \cdot 2\pi/\alpha = 2\pi$ and $I_1 = 2\pi$. Note that the product is independent of α, which is the content of the statement.

(c) True. Substituting the closed form,

$$(\alpha+\sqrt{\alpha})\frac{2\pi}{\alpha} = 2\pi\left(1+\frac{1}{\sqrt{\alpha}}\right) \xrightarrow[\alpha\to+\infty]{} 2\pi.$$

The $\sqrt{\alpha}$ is there only to be negligible; the statement is that I_α decays exactly like $2\pi/\alpha$ and not faster.

AI-Note 20.11: Part **(b)** is stated for the whole range $\alpha \geq 0$ the problem opens with, and at $\alpha = 0$ it reads $0 \cdot (+\infty)$, which is not a real number. The identity is to be read for $\alpha > 0$, where it holds; the official key marks it true without comment.

Q.E.D.

Proof of Exercise 20.d.47:

(a) Jordan content is additive on pairwise disjoint Jordan measurable sets, so $\mathrm{vol}_2(B) = \sum_i \mathrm{vol}_2(B_i)$, and the integral is additive over the same decomposition. Applying the latter to the coordinate function x and then reading each summand through Definition 20.d.45,

$$\int_B x\, d\,\mathrm{vol}_2 = \sum_{i=1}^{n}\int_{B_i} x\, d\,\mathrm{vol}_2 = \sum_{i=1}^{n}\mathrm{vol}_2(B_i)\, x_{B_i}.$$

Dividing by $\mathrm{vol}_2(B)$ gives the first coordinate of the claim, and y gives the second.

(b) Parametrise the triangle by the affine map that carries the standard triangle $T := \{(u,v) \mid u,v \geq 0,\ u+v \leq 1\}$ onto B, namely

$$\Phi(u,v) := u\,p + v\,q = (ur+vs,\ vt), \qquad \mathrm{J}\,\Phi = \begin{pmatrix} r & s \\ 0 & t \end{pmatrix}, \qquad \det \mathrm{J}\,\Phi = rt > 0.$$

It is injective, since $\det \mathrm{J}\,\Phi \neq 0$ and Φ is affine, so Theorem 20.a.1 applies. First the area,

$$\mathrm{vol}_2(B) = \int_T rt\, du\, dv = rt\ \mathrm{vol}_2(T) = \tfrac{rt}{2},$$

and then the two coordinate integrals at once, writing them as the single $\mathbb{R}^2$-valued integral $\int_B (x,y)\, d\,\mathrm{vol}_2$:

$$\int_B (x,y)\, d\,\mathrm{vol}_2 = \int_T \Phi(u,v)\, rt\, du\, dv = rt\left(p\int_T u\, du\, dv + q\int_T v\, du\, dv\right).$$

Both remaining integrals equal $\int_0^1 u(1-u)\, du = \tfrac{1}{2} - \tfrac{1}{3} = \tfrac{1}{6}$, the second by the symmetry of T in u and v. Hence

$$z_B = \frac{rt \cdot \frac{1}{6}(p+q)}{\frac{rt}{2}} = \tfrac{1}{3}(p+q) = \tfrac{1}{3}(o+p+q),$$

the last step because $o = 0$.

Remark 20.e.59 (*Part (a) is what makes part (b) useful*): On its own, the triangle formula is a curiosity. Together with the additivity it computes the centroid of **any** polygon, since every polygon can be cut into triangles: take the area-weighted average of the vertex averages. That is the whole algorithm, and it needs no integral once part **(b)** is known.

Note also that nothing in the computation used the particular placement of the vertices. Any triangle is the image of T under some invertible affine map, and $z_B = \tfrac{1}{3}(o+p+q)$ says that the centroid is carried along by that map: an affine map sends the centroid of a region to the centroid of the image, because it commutes with averaging. The same is not true of the circumcentre or of any other centre a triangle has.

Q.E.D.

Solution to Exercise 20.d.48:

One expansion settles both parts, so it is done first, for a general Jordan measurable B with $V := \mathrm{vol}_2(B) > 0$. Expanding the integrand and integrating term by term,

$$\begin{aligned} I(s,t) &= \int_B (x^2+y^2)\,d\,\mathrm{vol}_2 - 2s\int_B x\,d\,\mathrm{vol}_2 - 2t\int_B y\,d\,\mathrm{vol}_2 + (s^2+t^2)V \\ &= \int_B (x^2+y^2)\,d\,\mathrm{vol}_2 - 2sVx_B - 2tVy_B + (s^2+t^2)V, \end{aligned}$$

using Definition 20.d.45 in the second line. Completing the square in s and in t,

$$I(s,t) = V\Big[(s-x_B)^2 + (t-y_B)^2\Big] + \underbrace{\int_B (x^2+y^2)\,d\,\mathrm{vol}_2 - V\big(x_B^2+y_B^2\big)}_{\text{independent of } (s,t)}. \tag{20.6}$$

(a) For the quarter disc the area is $V = \pi/4$, and in polar coordinates, with θ running over $[0,\pi/2]$,

$$\int_B x\,d\,\mathrm{vol}_2 = \int_0^{\pi/2}\!\!\int_0^1 (r\cos\theta)\,r\,dr\,d\theta = \left(\int_0^1 r^2\,dr\right)\left(\int_0^{\pi/2}\cos\theta\,d\theta\right) = \frac{1}{3}.$$

Hence $x_B = (1/3)/(\pi/4) = 4/(3\pi)$, and $y_B = 4/(3\pi)$ by the symmetry of B in the diagonal. By part **(b)** the minimising pair is

$$(s,t) = z_B = \left(\frac{4}{3\pi},\ \frac{4}{3\pi}\right) \approx (0.4244,\ 0.4244).$$

(b) In (20.6) the bracket is non-negative, vanishes exactly when $(s,t) = z_B$, and is multiplied by $V > 0$; the remaining term does not depend on (s,t) at all. So I is minimised at z_B and at no other point.

Remark 20.e.60 (*The leftover constant is the parallel axis theorem*): Equation (20.6) says more than where the minimum is. Setting $z := (s,t)$ and evaluating the constant term by putting $z = z_B$, it reads

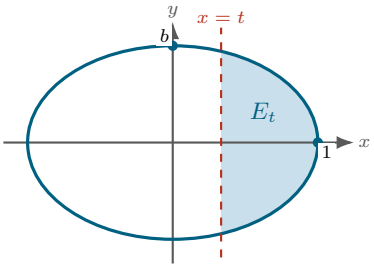
$$I(z) = \mathrm{vol}_2(B)\,|z - z_B|^2 + I(z_B),$$

so the second moment about **any** point is the second moment about the centroid plus the area times the squared distance between the two. That is Steiner's parallel axis theorem, and it is the reason the centroid is the point mechanics cares about: it is the unique point about which a body is easiest to spin, and the penalty for choosing any other grows quadratically with how far off you are.

AI-Note 20.12: The official key answers part **(a)** by differentiating under the integral sign to get $\int_B -2(x-s)\,d\,\mathrm{vol}_2 = 0$, then solving. That is correct and shorter, but it produces only a critical point, so convexity still has to be invoked to know it is a minimum, and it never records that the answer is the centroid of B. Completing the square gives the location, the uniqueness, the general statement and Steiner's theorem in one line, which is why it is used above.

Q.E.D.

Solution to Exercise 20.c.44:



(a) For each fixed $x \in [t,1]$, the vertical cross-section of the ellipse consists of $y \in [-b\sqrt{1-x^2},\, b\sqrt{1-x^2}]$. By Fubini's theorem:

$$A(t) = \int_t^1 \left(\int_{-b\sqrt{1-x^2}}^{b\sqrt{1-x^2}} dy \right) dx = 2b \int_t^1 \sqrt{1-x^2}\, dx.$$

(b) Since the integrand $g(x) := 2b\sqrt{1-x^2}$ is continuous on $[-1,1]$, the fundamental theorem of calculus gives

$$A'(t) = \frac{d}{dt}\left(2b \int_t^1 \sqrt{1-x^2}\, dx \right) = -2b\sqrt{1-t^2} \qquad \text{for all } t \in (-1,1).$$

(c) At the right endpoint $t = 1$, we have $A(1) = 0$. The left-sided difference quotient satisfies

$$\lim_{t\to 1^-} \frac{A(t) - A(1)}{t-1} = \lim_{t\to 1^-} A'(t) = \lim_{t\to 1^-} \left(-2b\sqrt{1-t^2} \right) = 0,$$

so $A'_-(1) = 0$. Similarly at the left endpoint $t = -1$, where $A(-1) = \pi b$ is the full area of the ellipse:

$$\lim_{t\to -1^+} \frac{A(t) - A(-1)}{t-(-1)} = \lim_{t\to -1^+} A'(t) = \lim_{t\to -1^+} \left(-2b\sqrt{1-t^2} \right) = 0,$$

so $A'_+(-1) = 0$. Since $A'(t) = -2b\sqrt{1-t^2}$ is continuous on the open interval $(-1,1)$ and its limits as $t \to \pm 1$ equal the one-sided derivatives 0, the derivative A' extends continuously to the closed interval $[-1,1]$. Hence $A \in C^1([-1,1])$.

(d) The domain Ω_t has horizontal width $x \in [t,1]$ and vertical bounds $y \in [-\sqrt{1+t^2}\sqrt{1-x^2},\, \sqrt{1+t^2}\sqrt{1-x^2}]$. Integrating out y:

$$F(t) = \int_t^1 2\sqrt{1+t^2}\sqrt{1-x^2}\, dx = 2\sqrt{1+t^2} \int_t^1 \sqrt{1-x^2}\, dx.$$

Here the parameter t enters both the integrand and the lower integration limit. Applying the product rule (or Leibniz's integral rule with moving limits):

$$\begin{aligned} F'(t) &= \left(\frac{d}{dt} 2\sqrt{1+t^2} \right) \int_t^1 \sqrt{1-x^2}\, dx + 2\sqrt{1+t^2}\, \frac{d}{dt}\left(\int_t^1 \sqrt{1-x^2}\, dx \right) \\ &= \frac{2t}{\sqrt{1+t^2}} \int_t^1 \sqrt{1-x^2}\, dx - 2\sqrt{1+t^2}\sqrt{1-t^2} \\ &= \frac{2t}{\sqrt{1+t^2}} \int_t^1 \sqrt{1-x^2}\, dx - 2\sqrt{1-t^4}. \end{aligned}$$

The first term accounts for the dilation of the vertical semi-axis, while the second term accounts for the loss of area as the vertical cutting line sweeps rightward at unit speed.

Q.E.D.

Gram Determinants, Area and Arc Length (Chapter 21)

The previous chapters integrated over subsets of $\mathbb{R}^n$, where the volume element is just dx. This one asks what to integrate over a d-dimensional surface sitting inside $\mathbb{R}^n$, where $d < n$ and there is no such element to start from. The answer runs from the area of a parallelogram, through the Gram determinant, which measures the volume spanned by d vectors in $\mathbb{R}^n$ when the matrix they form is not square and has no determinant of its own, to the d-volume of a parametrized submanifold, with the length of a curve falling out as the case $d = 1$. A last section integrates functions, not just the constant 1, over such a surface.

Length, area, and volume (Section 21.a)

The area of a parallelogram in $\mathbb{R}^2$ (Subsection 21.a.1)

The relationship between the determinant and volume is a key concept in this part of the lecture. We illustrate the special case of $\mathbb{R}^2$.

Proposition 21.a.1 (*Area of a parallelogram*): Let $x, y \in \mathbb{R}^2$. The area of the parallelogram $(0, x, x+y, y)$ is

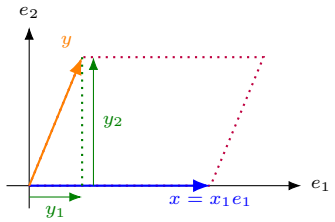
$$A = \big|\det(x \mid y)\big|.$$

Proof:

Assume w.l.o.g. that $x = x_1 e_1$, a reduction justified below. Then, writing $y = (y_1, y_2)^{\mathsf{T}}$,

$$A = |x_1 y_2| = \left|\det \begin{pmatrix} x_1 & y_1 \\ 0 & y_2 \end{pmatrix}\right|,$$

directly from the base-times-height formula for a parallelogram with one side on the e_1-axis.



For the reduction, let $R \in \mathrm{O}(2)$ be any rotation carrying x, y to a pair Rx, Ry with Rx along e_1; rotations preserve area, so

$$A = \big|\det(Rx \mid Ry)\big| = \big|\det(R)\det(x \mid y)\big| = \big|\det(x \mid y)\big|,$$

using $|\det R| = 1$ and multiplicativity of $\det$. So, the *"w.l.o.g."* above was justified: the formula $A = |\det(x \mid y)|$ holds for the rotated pair, and equals $|\det(x \mid y)|$ for the original one. **Q.E.D.**

The Gram determinant (Subsection 21.a.2)

Lemma 21.a.2 (*Gram determinant of a square matrix*): For $M \in \mathrm{Mat}_{n\times n}(\mathbb{R})$,

$$|\det(M)| = \sqrt{\det(M)\det(M^{\mathsf{T}})} = \sqrt{\det(M^{\mathsf{T}})\det(M)} = \sqrt{\det(M^{\mathsf{T}}M)}.$$

Proof:

The first equality is $|\det M| = \sqrt{(\det M)^2}$ together with $\det(M^\mathsf{T}) = \det(M)$; the last is multiplicativity of the determinant, $\det(M^\mathsf{T})\det(M) = \det(M^\mathsf{T}M)$. Q.E.D.

The quantity $\det(M^\mathsf{T}M)$ is called the "*Gram determinant*" (German: „*Gram-Determinante*“) of M. The n-parallelotope spanned by the columns of M is $M([0,1]^n)$, whose volume is $|\det M|$ by Theorem 19.a.14 (and Proposition 21.a.1 is the case $n = 2$). So that volume is the **square root** of the Gram determinant.

AI-Note 21.1: Terminology diverges from the official script here, worth flagging since it is exactly the kind of thing that costs a mark in an exam. The script's Definition 13.55 calls $\sqrt{\det(L^\mathsf{T}L)}$, the square root and so the **volume** itself, the *"Gram determinant"*. This document instead follows the more standard usage and calls the bare $\det(M^\mathsf{T}M)$ the Gram determinant, with the square root then giving the volume (Definition 21.a.3 below). Every downstream formula agrees regardless of which name is attached to which quantity, so this is terminology only, not a mathematical disagreement.

The script's own statement of Definition 13.55 is, independently, internally inconsistent: it introduces a linear map $L : \mathbb{R}^m \to \mathbb{R}^n$ with $m < n$, but then writes the formula using an undefined d (*"the $d \times d$ matrix $L^\mathsf{T}L$"*) where the surrounding text means m; and the prose describes *"the determinant of the ... matrix LL^T"* while the displayed formula correctly uses $L^\mathsf{T}L$. Only $L^\mathsf{T}L$ (size $m \times m$) is invertible for $m < n$, since LL^T is $n \times n$ and singular. The displayed formula is therefore the reliable half of the definition, and the surrounding prose is not.

AI-Note 21.2: The point of restating $|\det M|$ this way is that the middle expression, $\det(M^\mathsf{T}M)$, still makes sense once $M \in \mathrm{Mat}_{n\times m}(\mathbb{R})$ is rectangular: $M^\mathsf{T}M \in \mathrm{Mat}_{m\times m}(\mathbb{R})$ is square even when M is not, so its determinant is always defined, whereas $\det(M)$ itself is not. This is what lets the notion of *"volume spanned by vectors"* survive dropping the requirement $m = n$. A set of fewer than n vectors in $\mathbb{R}^n$ still spans a lower-dimensional parallelotope, and that parallelotope still has an m-dimensional volume, even though no square matrix and no ordinary determinant is available to compute it with.

Definition 21.a.3 (*Volume of a parallelotope via the Gram determinant*): For m vectors $v_1, \dots, v_m \in \mathbb{R}^n$, the "*area*"/"*length*"/ "*volume*" of the parallelotope spanned by them is

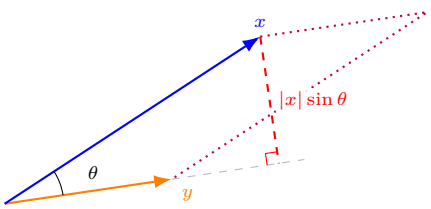
$$\sqrt{\det\left(\langle v_i, v_j\rangle\right)} = \sqrt{\det(V^\mathsf{T}V)}, \qquad V := (v_1 \mid v_2 \mid \cdots \mid v_m) \in \mathrm{Mat}_{n\times m}(\mathbb{R}).$$

Example 21.a.4 ($m = 1$: *the length of a vector*): For $m = 1$, $\sqrt{\det(v_1^\mathsf{T}v_1)} = \sqrt{\langle v_1, v_1\rangle} = |v_1|$, which is the length of v_1.

Example 21.a.5 ($m = 2$, $n = 3$: *area of a parallelogram in* $\mathbb{R}^3$): For $m = 2$, $n = 3$, we hope that the Gram determinant of

$$M = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix}$$

gives the area of the parallelogram spanned by x and y in $\mathbb{R}^3$. If θ is the angle between x and y, this should be $A = |x|\,|y|\,|\sin\theta|$.



And indeed:

$$\begin{aligned}\sqrt{\det(M^\mathsf{T}M)} &= \sqrt{\det\begin{pmatrix}\langle x,x\rangle & \langle x,y\rangle\\ \langle x,y\rangle & \langle y,y\rangle\end{pmatrix}}\\ &= \sqrt{\langle x,x\rangle\langle y,y\rangle - \langle x,y\rangle^2}\\ &= \sqrt{|x|^2|y|^2 - |x|^2|y|^2\cos^2\theta}\\ &= |x|\,|y|\,|\sin\theta| \qquad (= |x\times y|).\end{aligned}$$

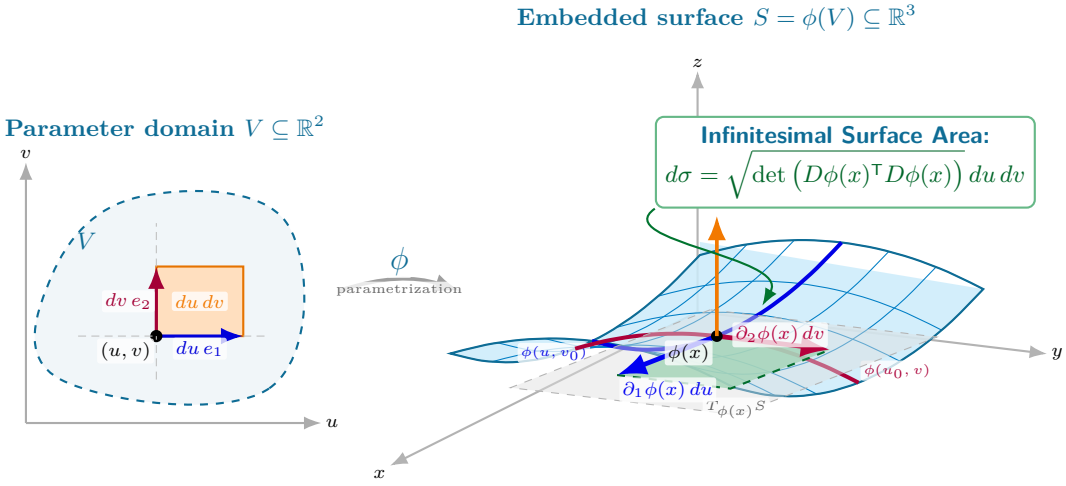
Volume of embedded surfaces (Section 21.b)

Near a point $\phi(x)$, a parametrized surface is approximately the image of a small cube under the linear map $D\phi(x)$, a parallelepiped spanned by the partial derivatives $\partial_1\phi(x),\dots,\partial_d\phi(x)$. The previous section measured exactly that, so the volume of the surface is what one gets by adding these pieces up.

Definition 21.b.6 (*d-volume on a parametrized d-submanifold*): Let $V\subseteq\mathbb{R}^d$ be open, and let $\phi: V\to\mathbb{R}^n$ be a parametrized submanifold. Given a Jordan measurable set $E\subseteq V$ with $\overline{E}\subseteq V$, we define the d-volume of $\phi(E)$ by

$$\mathrm{vol}_d(\phi(E)) := \int_E \sqrt{\det\left(D\phi(x)^\mathsf{T}D\phi(x)\right)}\,dx.$$

Here the integrand, the square root of the Gram determinant, is the volume of the parallelotope spanned by the vectors $\partial_1\phi(x),\partial_2\phi(x),\dots,\partial_d\phi(x)$, which by assumption (see *"parametrized submanifold"*) are linearly independent, i.e. $D\phi(x)$ has full rank, so the volume is strictly positive.



This factor gives a measure for the stretching of E as it is *"glued"* to the surface. If

$$D\phi(x)^\mathsf{T}D\phi(x) = I_d \quad \text{for all } x,$$

that is, if the columns $\partial_i\phi(x)$ are orthonormal, then ϕ preserves lengths of curves and angles between them and is called an "*isometry*" (German: „*Isometrie*"); the factor is then 1. The weaker condition $\sqrt{\det(D\phi^\mathsf{T}D\phi)}\equiv 1$ only says that ϕ preserves d-volume: $(x,y)\mapsto(2x,y/2,0)$ satisfies it and doubles every length along the x-axis. For $d=1$ the two conditions coincide, and an isometry is a curve parametrized by arc length, treated in Section 21.c below.

Remark 21.b.7 (*Why the transpose, and why the square root*): The shape of the formula looks arbitrary until one asks what could possibly stand in its place. The honest question is: **by what**

factor does ϕ stretch d-dimensional volume near x? Near x the map ϕ is its differential, so the question is really about the linear map $D\phi(x)$, and the answer runs in three steps.

(a) The obvious candidate does not exist. For a map $\mathbb{R}^n \to \mathbb{R}^n$ the stretching factor is $|\det|$. Here $D\phi(x)$ is $n \times d$ with $d < n$, a **rectangular** matrix, and a rectangular matrix has no determinant. So, the one-line answer is unavailable, and something has to replace it.

(b) A factor nevertheless exists. A linear map still scales all d-volumes in its domain by one and the same constant: it sends the unit cube of $\mathbb{R}^d$ to a fixed parallelotope, and every other region is built from scaled copies of that cube. Call this constant $c \geq 0$. It is what we want; we only need a formula for it.

(c) The Gram matrix supplies it. The product $D\phi(x)^{\mathsf{T}} D\phi(x)$ is $d \times d$, square at last, so it **does** have a determinant, and that determinant turns out to be exactly c^2. To see why, write $D\phi(x) = QM$, where the d columns of $Q \in \mathrm{Mat}_{n\times d}(\mathbb{R})$ are an orthonormal basis of a d-dimensional subspace containing the image and $M := Q^{\mathsf{T}} D\phi(x)$ is square. The map Q carries $\mathbb{R}^d$ onto that subspace without distorting anything, so $c = |\det M|$, while $D\phi^{\mathsf{T}} D\phi = M^{\mathsf{T}} Q^{\mathsf{T}} QM = M^{\mathsf{T}} M$ has determinant $(\det M)^2$. Hence $c = \sqrt{\det(D\phi^{\mathsf{T}} D\phi)}$, and the square root is there for no deeper reason than to undo that squaring.

The two familiar formulas are the small cases. For $d = 1$ the matrix $D\phi(t)$ is the single column $\phi'(t)$, so $D\phi^{\mathsf{T}} D\phi = |\phi'(t)|^2$ and $c = |\phi'(t)|$, which is the arc-length integrand of Section 21.c. For $d = 2$, $n = 3$ one computes $c = |\partial_1\phi \times \partial_2\phi|$, the area of the parallelogram the two tangent vectors span, which is the surface-area formula used throughout this chapter.

Remark 21.b.8 (*Two objects, three notations*): From this chapter onwards the lecture notes write $D\phi(x)$ for the Jacobian **matrix** of ϕ at x, the object earlier chapters of this document call $\mathrm{J}\,\phi(x)$. That collides with the notation $D\phi_x$ used since Chapter 9 for the **differential**, the linear map itself. The two are not the same kind of object, even though they carry the same information:

$$\underbrace{D\phi_x}_{\text{linear map}}(v) \;=\; \underbrace{\mathrm{J}\,\phi(x)}_{\text{matrix}}\, v \;=\; \underbrace{D\phi(x)}_{\text{same matrix}}\, v.$$

The subscript is the whole distinction: $D\phi_x$ has x downstairs, $D\phi(x)$ has it in brackets. Both spellings of the matrix appear in the chapters ahead; where it matters, read $D\phi(x)$ and $\mathrm{J}\,\phi(x)$ as synonyms.

Example 21.b.9 (*Integral on the graph of a function*): Given an open set $U \subseteq \mathbb{R}^2$ and a C^1 function $f : U \to \mathbb{R}$, we can define the parametrized surface $\phi : U \to \mathbb{R}^3$ by $\phi(x, y) = (x, y, f(x, y))^{\mathsf{T}}$. Assuming $M := \mathrm{Im}\,\phi$ is a C^1 submanifold, its area is given by $\int_M 1\, d\mathrm{vol}_2$.

We first find the Jacobian matrix of the parametrization:

$$\mathrm{J}\,\phi(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \partial_1 f(x, y) & \partial_2 f(x, y) \end{pmatrix}.$$

Then we compute the square root of the Gram determinant:

$$\begin{aligned}\sqrt{|\det(\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi)|} &= \sqrt{\det\begin{pmatrix} 1+(\partial_1 f)^2 & \partial_1 f\partial_2 f \\ \partial_2 f\partial_1 f & 1+(\partial_2 f)^2 \end{pmatrix}} \\ &= \sqrt{\big(1+(\partial_1 f)^2\big)\big(1+(\partial_2 f)^2\big) - (\partial_1 f)^2(\partial_2 f)^2} \\ &= \sqrt{1+(\partial_1 f)^2+(\partial_2 f)^2} = \sqrt{1+|\nabla f(x,y)|^2}.\end{aligned}$$

By definition, we get the surface area formula for a graph:

$$\int_M 1\, d\mathrm{vol}_2 = \int_U \sqrt{1+|\nabla f(x,y)|^2}\, dx\, dy.$$

Example: area of a paraboloid patch (Subsection 21.b.1)

Example 21.b.10 (*Area of a paraboloid patch*): Compute the area of the surface

$$F = \left\{(x,y,z) \in \mathbb{R}^3 \ : \ x^2 + 4y^2 < 8, \quad z = \tfrac{1}{2}(x^2+2y^2)\right\}.$$

Since $z = \frac{1}{2}(x^2+2y^2)$ means F is a graph, we can conveniently write F as the image of the parametrization

$$\phi : A \to F, \qquad (x,y) \mapsto \left(x,\ y,\ \tfrac{1}{2}(x^2+2y^2)\right), \qquad A := \{(x,y) \in \mathbb{R}^2 : x^2+4y^2 < 8\}.$$

Gram determinant. We compute

$$D\phi(x,y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ x & 2y \end{pmatrix} \quad \Longrightarrow \quad D\phi(x,y)^{\mathsf{T}} D\phi(x,y) = \begin{pmatrix} 1+x^2 & 2xy \\ 2xy & 1+4y^2 \end{pmatrix},$$

so that

$$\sqrt{\det D\phi(x,y)^{\mathsf{T}} D\phi(x,y)} = \sqrt{(1+x^2)(1+4y^2) - 4x^2y^2} = \sqrt{1+x^2+4y^2}.$$

The integral. By Definition 21.b.6,

$$\mathrm{vol}_2(F) = \int_A \sqrt{1+x^2+4y^2}\, dx\, dy, \qquad A = \{(x,y) \in \mathbb{R}^2 : x^2+4y^2 < 8\}.$$

We should choose more convenient coordinates. Polar coordinates come to mind, but then we still have problems, since A is not radially symmetric. We want to write $A = \Psi\big((0,R)\times(0,2\pi)\big)$ (up to a set of measure zero). Note that for $y = 0$, $|x| < \sqrt{8} = 2\sqrt{2}$, and for $x = 0$, $|y| < \sqrt{2}$. Choosing $R := \sqrt{2}$ and weighing x, y accordingly,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2r\cos\theta \\ r\sin\theta \end{pmatrix} =: \Psi(r,\theta).$$

Change of variables.

$$dx\, dy = |\det D\Psi(r,\theta)|\, dr\, d\theta = \left|\det \begin{pmatrix} 2\cos\theta & -2r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}\right| dr\, d\theta = 2r\, dr\, d\theta,$$

giving

$$\begin{aligned} \mathrm{vol}_2(F) &= \int_A \sqrt{1+x^2+4y^2}\, dx\, dy = \int_0^{\sqrt{2}} dr \int_0^{2\pi} d\theta\ 2r\sqrt{1+4r^2} \\ &= 4\pi \int_0^{\sqrt{2}} dr\ r(1+4r^2)^{1/2} = \frac{\pi}{2}\int_0^{\sqrt{2}} dr\ 8r(1+4r^2)^{1/2} \\ &= \frac{\pi}{2}\left[(1+4r^2)^{3/2}\cdot\frac{2}{3}\right]_0^{\sqrt{2}} = \frac{\pi}{3}(27-1) = \frac{26\pi}{3}. \end{aligned}$$

AI-Note 21.3: Worth flagging what makes A awkward: it is an ellipse, not a disc, so naive polar coordinates $(x,y) = (\rho\cos\theta, \rho\sin\theta)$ would force ρ to depend on θ along the boundary. The fix is to build the eccentricity into the change of variables itself. Scaling x by a factor of 2 relative to y turns the ellipse into the disc $r < \sqrt{2}$ before polar coordinates are even applied. This is the same idea as Example 20.b.37: choose Ψ to straighten the domain, not to simplify the integrand.

Example 21.b.11 (*Surface area of a sphere*): Given the sphere $M := \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = r^2\}$, we can use the parametrization $\phi : (0,\pi)\times(-\pi,\pi) \to \mathbb{R}^3$,

$$\phi(\theta,\varphi) = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}.$$

The Jacobian matrix is

$$\mathrm{J}\,\phi(\theta,\varphi) = \begin{pmatrix} r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ -r\sin\theta & 0 \end{pmatrix},$$

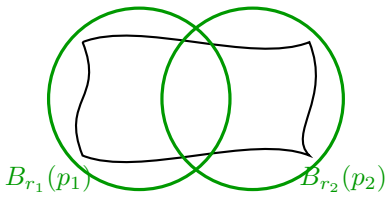
which has rank 2 for all (θ,φ) in the domain. Then we compute the Gram determinant:

$$\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi = \begin{pmatrix} r^2 & 0 \\ 0 & r^2\sin^2\theta \end{pmatrix} \quad\Longrightarrow\quad \sqrt{\det(\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi)} = r^2\sin\theta.$$

The surface area is given by the integral:

$$\begin{aligned} \int_M 1\, d\mathrm{vol}_2 &= \int_{-\pi}^{\pi}\int_0^{\pi} \sqrt{\det(\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi)}\, d\theta\, d\varphi \\ &= \int_{-\pi}^{\pi}\int_0^{\pi} r^2\sin\theta\, d\theta\, d\varphi \\ &= r^2\left(\int_{-\pi}^{\pi} d\varphi\right)\left(\int_0^{\pi}\sin\theta\, d\theta\right) \\ &= r^2\cdot 2\pi\cdot 2 = 4\pi r^2. \end{aligned}$$

Note that no partition of unity was needed here, since the parametrization covers the sphere up to a null set. Partitions of unity, which glue several charts together, are the subject of Section 21.d; the sketch below shows the idea.



the weights η_ℓ are chosen so that integrating in each ball separately and adding the integrals gives exactly the integral over the whole submanifold.

Exercise 21.b.12 (*True or False* (two maps and the area formula)): The examination this problem is taken from supplies, as a formula to be used without proof, the statement that for $\Phi \in C^1(\mathbb{R}^n,\mathbb{R}^m)$ with $m \geq n$ and $K \subseteq \mathbb{R}^n$ compact,

$$\mathrm{vol}_n(\Phi(K)) = \int_K \sqrt{\det\left(\mathrm{J}\,\Phi(x)^{\mathsf{T}}\,\mathrm{J}\,\Phi(x)\right)}\, dx. \tag{21.1}$$

For $\alpha \in \mathbb{R}$ consider the maps

$$\Phi : (x,y) \mapsto (xy,\ x^2-y^2), \qquad \Psi : (x,y) \mapsto (x+y,\ \alpha x - y,\ y+1).$$

Decide whether each of the following is true or false, for every compact $E \subseteq \mathbb{R}^2$.

(a) $\mathrm{vol}_2(\Phi(E)) = 2\int_E (x^2+y^2)\, dx\, dy$.
(b) $\mathrm{vol}_2(\Phi(E)) = 2\int_E (x^2+2y^2)\, dx\, dy$.
(c) $\mathrm{vol}_2(\Psi(E)) = 2\alpha\ \mathrm{vol}_2(E)$.
(d) $\mathrm{vol}_2(\Psi(E)) = \sqrt{2}\,|\alpha+1|\ \mathrm{vol}_2(E)$.
(e) $\mathrm{vol}_2(\Psi(E)) = \sqrt{2(\alpha^2+\alpha+1)}\ \mathrm{vol}_2(E)$.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), where it appears as the eleventh block of multiple-choice questions. Formula (21.1) is quoted from that paper's own list of results that candidates may use without proof.*

Solution: *See solution on Page 331*

Exercise 21.b.13 (*Volume* (important)): Calculate the volume of the region $B \subset \mathbb{R}^3$ enclosed by the surfaces $x^2+y^2+z^2=8$ and $2z=x^2+y^2$.

Official hint: *Use cylindrical coordinates.*

Corsin Nick's hint:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix} \qquad \text{for } (r,\theta,\varphi) \in (0,\infty)\times(0,\pi)\times(0,2\pi).$$

Info: *This exercise is Problem 9.1 of Problem Sheet 9, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 332*

AI-Note 21.4 (*Two hints, two coordinate systems*): The second hint above is the **spherical** parametrization, while the official hint recommends **cylindrical** coordinates (r,θ,z). The cylindrical ones are the more natural choice for this problem, and the solution on Page 332 uses them: both surfaces are surfaces of revolution about the z-axis, which cylindrical coordinates respect directly, whereas spherical coordinates mix r and θ into both defining equations.

Exercise 21.b.14 (*Spherical quadrilateral* (important)): Calculate the area of the spherical quadrilateral $S \subset \mathbb{S}^2$ bounded by the meridians $-\pi < \varphi_1 < \varphi_2 < \pi$ and parallels $-\pi/2 < \theta_1 < \theta_2 < \pi/2$.

Info: *This exercise is Problem 10.1 of Problem Sheet 10, Analysis II, Spring Semester 2026.*

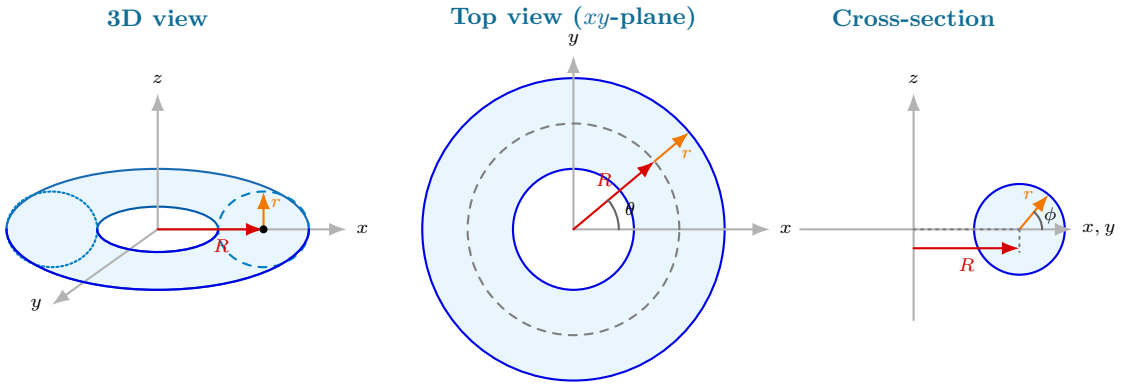
Solution: *See solution on Page 334*

Exercise 21.b.15 (*The surface area of a torus* (important)): Let $S := \{(x,y,0) \in \mathbb{R}^3 \mid x^2+y^2=4\}$ and $T := \{x \in \mathbb{R}^3 \mid d_S(x) \le 1\}$, where $d_S(x) = \inf\{|x-y| \mid y \in S\}$ denotes the minimal distance from S to $x \in \mathbb{R}^3$. Parametrize ∂T and then calculate the area.

Corsin Nick's hint, general parametrization of a torus surface:

$$x = (R + r\cos\phi)\cos\theta, \qquad y = (R + r\cos\phi)\sin\theta, \qquad z = r\sin\phi,$$

for the torus swept by a tube of radius r at distance R from the axis.



Info: *This exercise is Problem 10.2 of Problem Sheet 10, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 334*

Length of a curve (Section 21.c)

The case $d = 1$ deserves its own section, because there the Gram determinant collapses. For a single vector $v = \gamma'(t)$ it is $\det(v^{\mathsf{T}} v) = |v|^2$, so the volume factor of Section 21.b is just the speed $|\gamma'(t)|$.

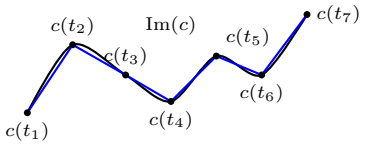
Definition 21.c.16 (*Length of a C^1-curve*): The "*length*" (German: „*Länge*") of a curve or path $\gamma \in C^1([a,b], \mathbb{R}^n)$ is given by

$$L(\gamma) := \int_a^b |\gamma'(t)|\, dt.$$

Definition 21.c.17 (*Length of a C^0-curve*): For a C^0-curve $c \in C^0([a,b], \mathbb{R}^n)$, the length is defined as

$$L(c) := \sup \left\{ \sum_{i=1}^{N-1} |c(t_{i+1}) - c(t_i)| \ : \ N \geq 2,\ a = t_1 < \cdots < t_N = b \right\},$$

the supremum over all lengths of inscribed polygons.



Proposition 21.c.18 (*The two lengths agree*): For $\gamma \in C^1([a,b], \mathbb{R}^n)$, the lengths of Definitions 21.c.16 and 21.c.17 coincide.

Proof:

For a vector-valued integral, $\left|\int v\right| \leq \int |v|$: pair $\int v$ with the unit vector u in its direction and use $\langle u, v(s)\rangle \leq |v(s)|$. Hence each chord satisfies $|\gamma(t_{i+1}) - \gamma(t_i)| = \left|\int_{t_i}^{t_{i+1}} \gamma'\right| \leq \int_{t_i}^{t_{i+1}} |\gamma'|$, and summing shows that every inscribed polygon is at most $\int_a^b |\gamma'|$.

Conversely, let $\varepsilon > 0$. Since γ' is uniformly continuous on $[a,b]$, there is $\delta > 0$ with $|\gamma'(s) - \gamma'(t)| < \varepsilon$ whenever $|s - t| < \delta$. Take a partition with all $\Delta t_i := t_{i+1} - t_i < \delta$. Then

$$|\gamma(t_{i+1}) - \gamma(t_i)| \geq |\gamma'(t_i)|\,\Delta t_i - \int_{t_i}^{t_{i+1}} |\gamma'(s) - \gamma'(t_i)|\, ds \geq \big(|\gamma'(t_i)| - \varepsilon\big)\Delta t_i \geq \int_{t_i}^{t_{i+1}} |\gamma'| - 2\varepsilon\,\Delta t_i,$$

and summing, the polygon is at least $\int_a^b |\gamma'| - 2\varepsilon(b-a)$. As ε was arbitrary, the supremum equals $\int_a^b |\gamma'|$. Q.E.D.

So the integral is not a new definition of length but a formula for the old one: on a short enough interval a C^1 curve is nearly straight, and the speed times the time is the chord.

Remark 21.c.19: Notice that

$$|\gamma'(t)| = \sqrt{D\gamma(t)^{\mathsf{T}} \cdot D\gamma(t)}$$

is exactly the square root of the Gram determinant of $D\gamma(t)$, that is, the 1-dimensional volume of Definition 21.a.3, the $m = 1$ case of Example 21.a.4, now applied pointwise along the curve. This is why the length of a curve is the special case $d = 1$ of Definition 21.b.6: a curve is a 1-dimensional parametrized submanifold, and $L(\gamma) = \mathrm{vol}_1(\gamma([a,b]))$.

Example 21.c.20 (*Line integral*): Let $\phi : [-\pi/2, \pi/2] \to \mathbb{R}^2$ be given by $\phi(\theta) = \begin{pmatrix} 3\sin\theta \\ 4\sin\theta \end{pmatrix}$. Compute the length of the curve $M := \mathrm{Im}\,\phi$, i.e. $\int_M 1\, d\mathrm{vol}_1$.

First, we compute the Jacobian (which here is just the derivative vector):

$$\mathrm{J}\,\phi(\theta) = \begin{pmatrix} 3\cos\theta \\ 4\cos\theta \end{pmatrix}.$$

Then we compute the square root of the Gram determinant (which is the length of the tangent vector):

$$\sqrt{\det(\mathrm{J}\,\phi(\theta)^{\mathsf{T}}\,\mathrm{J}\,\phi(\theta))} = \sqrt{9\cos^2\theta + 16\cos^2\theta} = \sqrt{25\cos^2\theta} = 5|\cos\theta|.$$

Since $\cos\theta \geq 0$ for $\theta \in [-\pi/2, \pi/2]$, this simplifies to $5\cos\theta$. The length is:

$$\int_{-\pi/2}^{\pi/2} 5\cos\theta\, d\theta = 5\big[\sin\theta\big]_{-\pi/2}^{\pi/2} = 5(1-(-1)) = 10.$$

Proposition 21.c.21 (*Length of a graph*): Let $f \in C^1([a,b],\mathbb{R})$. The graph $\Gamma_f := \{(x, f(x)) : x \in [a,b]\} \subseteq \mathbb{R}^2$ has length

$$\mathrm{vol}_1(\Gamma_f) = \int_a^b \sqrt{1 + f'(x)^2}\, dx.$$

Proof:

Parametrize by $\phi(t) := (t, f(t))^{\mathsf{T}}$ on $[a,b]$, which is injective and C^1 with $\phi'(t) = (1, f'(t))^{\mathsf{T}} \neq 0$. The volume factor of the single tangent vector is

$$\sqrt{\det\big(\mathrm{J}\,\phi(t)^{\mathsf{T}}\,\mathrm{J}\,\phi(t)\big)} = |\phi'(t)| = \sqrt{1+f'(t)^2},$$

and integrating over $[a,b]$ is the definition of vol_1. Q.E.D.

Example 21.c.22 (*A quarter of an astroid*): Let $\gamma(t) := (\cos^3 t,\ \sin^3 t)^{\mathsf{T}}$ for $t \in [0, \frac{\pi}{2}]$. Then

$$\gamma'(t) = \begin{pmatrix} -3\cos^2 t\,\sin t \\ 3\sin^2 t\,\cos t \end{pmatrix},$$

so, using $\cos t \geq 0$ and $\sin t \geq 0$ on this interval,

$$|\gamma'(t)| = 3\cos t\,\sin t\sqrt{\cos^2 t + \sin^2 t} = 3\cos t\,\sin t = \tfrac{3}{2}\sin(2t).$$

The length is therefore

$$\int_0^{\pi/2} \tfrac{3}{2}\sin(2t)\, dt = \Big[-\tfrac{3}{4}\cos(2t)\Big]_0^{\pi/2} = -\tfrac{3}{4}\big((-1)-1\big) = \tfrac{3}{2}.$$

Note that $\gamma'(0) = \gamma'(\frac{\pi}{2}) = 0$: the astroid has cusps at its endpoints, so this is a curve whose parametrization fails the immersion condition exactly at the two boundary points. The length integral is unaffected, since two points contribute nothing to it.

Example 21.c.23 (*Spiral length*): Consider the spiral parametrized by $\phi : (0, 2\pi) \to \mathbb{R}^3$,

$$\phi(t) = \begin{pmatrix} \cos(\omega t) \\ \sin(\omega t) \\ vt \end{pmatrix},$$

where $v \in \mathbb{R}\setminus\{0\}$ and $\omega \in (0,\infty)$. Compute the length of the spiral $\mathrm{Im}\,\phi$.

We compute the Jacobian:

$$\mathrm{J}\,\phi(t) = \begin{pmatrix} -\omega\sin(\omega t) \\ \omega\cos(\omega t) \\ v \end{pmatrix}.$$

The volume factor is:

$$\sqrt{\mathrm{J}\,\phi(t)^{\mathsf{T}}\,\mathrm{J}\,\phi(t)} = \sqrt{\omega^2\sin^2(\omega t) + \omega^2\cos^2(\omega t) + v^2} = \sqrt{\omega^2 + v^2}.$$

The length of the spiral is the integral over the parameter domain:

$$\int_0^{2\pi} \sqrt{\omega^2 + v^2}\, dt = 2\pi\sqrt{\omega^2 + v^2}.$$

Example 21.c.24 (*Line integral of a function on a circle*)**:** Given a function $f : \mathbb{R}^3 \to \mathbb{R}$, $f(x,y,z) = x^2 + z^2$. We want to integrate this function on the circle $M := \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 = 1,\ z = 2\}$. We parametrize the circle by $\phi : (0, 2\pi) \to \mathbb{R}^3$,

$$\phi(\varphi) = \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 2 \end{pmatrix}.$$

The Jacobian is $\mathrm{J}\,\phi(\varphi) = \phi'(\varphi) = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix}$. The volume factor is $\sqrt{\mathrm{J}\,\phi^{\mathsf{T}}\,\mathrm{J}\,\phi} = |\phi'(\varphi)| = \sqrt{\sin^2\varphi + \cos^2\varphi} = 1$. The function evaluated on the curve is $f(\phi(\varphi)) = \cos^2\varphi + 4$. The line integral is thus:

$$\begin{aligned} \int_M f \, d\underset{1}{\mathrm{vol}} &= \int_0^{2\pi} f(\phi(\varphi))|\phi'(\varphi)|\, d\varphi \\ &= \int_0^{2\pi} (\cos^2\varphi + 4)\cdot 1\, d\varphi \\ &= \int_0^{2\pi} \cos^2\varphi\, d\varphi + \int_0^{2\pi} 4\, d\varphi \\ &= \pi + 8\pi = 9\pi. \end{aligned}$$

Exercise 21.c.25 (*Multiple integrals*)**:**

(a) Let $D := [0,2] \times [0,1]$. Calculate $\iint_D (x^3 + 3x^2y + y^3)\, dx\, dy$.

(b) Let $D \subset \mathbb{R}^2$ be the interior of the triangle with vertices $(0,0)$, $(0,\pi)$, and (π,π). Calculate $\iint_D x\cos(x+y)\, dx\, dy$.

(c) Let $D := \{(x,y) \in \mathbb{R}^2 \mid x > 1,\ y > 1,\ x + y < 3\}$. Calculate $\iint_D \frac{1}{(x+y)^3}\, dx\, dy$.

Corsin Nick's hint on part (b): *You can parametrize the triangle as*

$$D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le y \le \pi\}.$$

Info: *This exercise is Problem 9.2 of Problem Sheet 9, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 333*

Exercise 21.c.26 (*Solids of revolution* (important)**):** Given $f \in C^1((a,b)) \cap C([a,b])$ such that $f \ge 0$, define the rotational body around the x-axis as

$$\Omega := \left\{(x,y,z) \in \mathbb{R}^3 \mid a < x < b,\ \sqrt{y^2+z^2} < f(x)\right\},$$
$$\partial_{\text{side}}\Omega := \left\{(x,y,z) \in \mathbb{R}^3 \mid a < x < b,\ \sqrt{y^2+z^2} = f(x)\right\}.$$

(a) Say whether Ω, $\partial_{\text{side}}\Omega$ are open/closed/connected subsets of $\mathbb{R}^3$.

(b) Show that $\mathrm{vol}_3(\Omega) = \pi \int_a^b f(x)^2\, dx$.

(c) Show that $\mathrm{vol}_2(\partial_{\text{side}}\Omega) = 2\pi \int_a^b \sqrt{1 + f'(x)^2}\, f(x)\, dx$. *Hint:* parametrize $(x,\theta) \mapsto (x, f(x)\cos\theta, f(x)\sin\theta)$.

(d) Calculate the volume and surface area of the 'improper rotational body' that arises when $f(x) = 1/x$, $a = 1$, $b = \infty$.

(e) (*) Show that Ω is a bounded C^1 domain (as in Definition 22.a.1) if and only if

$$f(x) > 0 \text{ in } (a,b), \qquad f'(a^+) = +\infty, \qquad f'(b^-) = -\infty. \tag{$*$}$$

Info: *This exercise is Problem 10.3 of Problem Sheet 10, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 334*

Integration over submanifolds (Section 21.d)

While Section 21.b explains how to compute the d-volume of a parametrized submanifold, it requires the entire relevant portion of the surface to be covered by a single parametrization ϕ. For more complicated submanifolds $M \subseteq \mathbb{R}^n$, we need to integrate scalar functions $f : M \to \mathbb{R}$ globally.

To achieve this, we cover the m-dimensional submanifold M with the images of local parametrizations $\Phi_l : V_l \to \mathbb{R}^n$, called an "*atlas*". Each image is relatively open in M, that is, $\mathrm{Im}(\Phi_l) = M \cap U_l$ for an open set $U_l \subseteq \mathbb{R}^n$. By employing a smooth "*partition of unity*" $\eta_l \in C_c^\infty(\mathbb{R}^n, [0,1])$ (see AI-Note 25.3 and Proposition 25.b.9 for a rigorous construction) subordinate to this cover, we ensure that:

(a) $\sum_{l=1}^N \eta_l \equiv 1$ on M,

(b) $\mathrm{supp}(\eta_l) \subset U_l$, so that on M the function η_l vanishes outside $\mathrm{Im}(\Phi_l)$.

Definition 21.d.27 (*Integral of a scalar function over a submanifold*): Given an m-dimensional submanifold $M \subseteq \mathbb{R}^n$, an atlas of local parametrizations $\Phi_l : V_l \to \mathbb{R}^n$, and a subordinate partition of unity $\{\eta_l\}$, the integral of a function $f : M \to \mathbb{R}$ over M is defined as:

$$\int_M f \, \underset{m}{\mathrm{dvol}} := \sum_{l=1}^N \int_{V_l} f(\Phi_l(x)) \eta_l(\Phi_l(x)) \sqrt{\det(\mathrm{J}\,\Phi_l^{\mathsf{T}}(x)\,\mathrm{J}\,\Phi_l(x))}\, dx.$$

This definition pieces together the local integrals (weighted by η_l) to compute the global integral over the entire submanifold, generalizing the volume integral from Definition 21.b.6.

The definition names two things that are not canonical (an atlas, and a partition of unity subordinate to it), so it defines nothing until those choices are shown not to matter.

Proposition 21.d.28 (*The integral depends on neither the atlas nor the partition*): The right-hand side of Definition 21.d.27 is unchanged when the atlas $\{\Phi_l\}_{l=1}^N$ is replaced by any other finite atlas of M and $\{\eta_l\}$ by any partition of unity subordinate to it.

Proof:

Two steps, and the first carries the whole argument.

One patch, two parametrizations. Let $\Phi : V \to \mathbb{R}^n$ and $\Psi : W \to \mathbb{R}^n$ be local parametrizations with the same image, and let g be a continuous function on M that vanishes outside that image. The transition map $\tau := \Psi^{-1} \circ \Phi : V \to W$ is a C^1-diffeomorphism and $\Phi = \Psi \circ \tau$, so the chain rule gives $\mathrm{J}\,\Phi(x) = \mathrm{J}\,\Psi(\tau(x))\,\mathrm{J}\,\tau(x)$ and therefore

$$\mathrm{J}\,\Phi^{\mathsf{T}}\,\mathrm{J}\,\Phi = \mathrm{J}\,\tau^{\mathsf{T}}\left(\mathrm{J}\,\Psi^{\mathsf{T}}\,\mathrm{J}\,\Psi\right)\mathrm{J}\,\tau.$$

Both outer factors are $m \times m$, so taking determinants and using multiplicativity twice,

$$\sqrt{\det\left(\mathrm{J}\,\Phi^{\mathsf{T}}\,\mathrm{J}\,\Phi\right)(x)} = \left|\det \mathrm{J}\,\tau(x)\right| \sqrt{\det\left(\mathrm{J}\,\Psi^{\mathsf{T}}\,\mathrm{J}\,\Psi\right)(\tau(x))}.$$

The factor $|\det \mathrm{J}\,\tau|$ is precisely what Theorem 20.a.1 consumes, so substituting $y = \tau(x)$ gives

$$\int_V g(\Phi(x)) \sqrt{\det\left(\mathrm{J}\,\Phi^{\mathsf{T}}\,\mathrm{J}\,\Phi\right)}\, dx = \int_W g(\Psi(y)) \sqrt{\det\left(\mathrm{J}\,\Psi^{\mathsf{T}}\,\mathrm{J}\,\Psi\right)}\, dy.$$

The Gram determinant was built to make exactly this cancellation happen, and this is where that pays.

Two atlases. Let $\{(\Phi_l, \eta_l)\}_{l=1}^{N}$ and $\{(\Psi_k, \theta_k)\}_{k=1}^{K}$ be two systems as in Definition 21.d.27, producing values I and J. Since $\sum_k \theta_k \equiv 1$ on M, inserting that sum into I and exchanging the two finite sums,

$$I = \sum_{l=1}^{N}\sum_{k=1}^{K} \int_{V_l} (\eta_l\, \theta_k\, f)(\Phi_l(x)) \sqrt{\det\left(\mathrm{J}\,\Phi_l^{\mathsf{T}}\,\mathrm{J}\,\Phi_l\right)}\, dx.$$

On M the function $\eta_l \theta_k f$ vanishes outside $\mathrm{Im}(\Phi_l) \cap \mathrm{Im}(\Psi_k)$; restricting Φ_l and Ψ_k to the preimages of that intersection makes them two parametrizations of one patch, so the first step applies and replaces Φ_l by Ψ_k in that term. Doing so for every (l,k) and then summing over l first, using $\sum_l \eta_l \equiv 1$, collapses the double sum to J. Hence $I = J$. **Q.E.D.**

Remark 21.d.29 (*When a single chart already does, this costs nothing*): If M is covered by a single parametrization Φ_1, a partition subordinate to that atlas has $\eta_1 \equiv 1$ on M, and Definition 21.d.27 collapses to

$$\int_M f \, d\mathrm{vol}_m = \int_{V_1} f(\Phi_1(x)) \sqrt{\det\left(\mathrm{J}\,\Phi_1^{\mathsf{T}}\,\mathrm{J}\,\Phi_1\right)}\, dx,$$

which is Definition 21.b.6 with f carried along; at $f \equiv 1$ it returns $\mathrm{vol}_m(M)$ exactly. So the general definition extends the earlier one rather than competing with it.

In practice one almost never builds a partition of unity, because a single chart usually covers all of M but a null set (spherical coordinates miss only a half-meridian of S^2, which carries no vol_2), and a null set contributes nothing to the integral. That is why every surface integral in Section 21.b could be evaluated from one parametrization without comment. The partition of unity is what makes the definition **exist** for an arbitrary M; it is not a device for computing with.

AI-Exercise 21.d.30 (*Two atlases on the circle, and one that is not an atlas*): Let $M := S^1 \subseteq \mathbb{R}^2$ and take $f \equiv 1$, so that the integral is the length.

(a) Evaluate $\int_{S^1} 1 \, d\mathrm{vol}_1$ from the single parametrization $\Phi(t) := (\cos t, \sin t)$ on $(0, 2\pi)$, whose image omits one point of S^1.

(b) Now take the two-arc atlas $\Phi_1 := \Phi$ on $(0, 2\pi)$ and $\Phi_2(t) := (\cos t, \sin t)$ on $(-\pi, \pi)$. Show that **every** partition of unity subordinate to it returns the same value, without choosing one.

(c) The two graph parametrizations $\Psi_\pm(u) := \left(u, \pm\sqrt{1-u^2}\right)$ on $(-1,1)$ satisfy $\sqrt{\det(\mathrm{J}\,\Psi_\pm^{\mathsf{T}}\,\mathrm{J}\,\Psi_\pm)} = (1-u^2)^{-1/2}$, and each semicircle has length π. Why does $\{\Psi_+, \Psi_-\}$ nonetheless fail to be an atlas of S^1, and what goes wrong if one uses it in Definition 21.d.27 anyway?

Solution: *See solution on Page 337*

Solutions (Section 21.e)

Solution to Exercise 21.b.12:

Take the second map first, because there (21.1) applies without any difficulty. Write $\Psi = L + c$ with

$$L := \begin{pmatrix} 1 & 1 \\ \alpha & -1 \\ 0 & 1 \end{pmatrix}, \qquad c := \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

so $\mathrm{J}\,\Psi(x,y) = L$ at every point. The Gram matrix of its two columns $c_1 = (1,\alpha,0)^{\mathsf{T}}$ and $c_2 = (1,-1,1)^{\mathsf{T}}$ is

$$L^{\mathsf{T}}L = \begin{pmatrix} \langle c_1,c_1\rangle & \langle c_1,c_2\rangle \\ \langle c_2,c_1\rangle & \langle c_2,c_2\rangle \end{pmatrix} = \begin{pmatrix} 1+\alpha^2 & 1-\alpha \\ 1-\alpha & 3 \end{pmatrix},$$

whose determinant is

$$3(1+\alpha^2) - (1-\alpha)^2 = 3 + 3\alpha^2 - 1 + 2\alpha - \alpha^2 = 2(\alpha^2+\alpha+1).$$

The integrand of (21.1) is therefore the constant $\sqrt{2(\alpha^2+\alpha+1)}$, and integrating a constant over E gives

$$\mathrm{vol}_2(\Psi(E)) = \sqrt{2(\alpha^2+\alpha+1)}\ \mathrm{vol}_2(E).$$

So **(e)** is true and **(c)** and **(d)** are false. Note that $\alpha^2+\alpha+1$ has discriminant -3 and so is strictly positive for every real α: the two columns are independent whatever α is, Ψ is an injective affine map, and the formula is exact. Statement **(c)** is what an α-linear guess produces, and **(d)** is what one gets by taking a square root before, rather than after, forming the Gram determinant.

For Φ the Jacobian is square, and there $\sqrt{\det(\mathrm{J}\,\Phi^{\mathsf{T}}\,\mathrm{J}\,\Phi)} = |\det \mathrm{J}\,\Phi|$. Since

$$\mathrm{J}\,\Phi(x,y) = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix}, \qquad \det \mathrm{J}\,\Phi(x,y) = -2y^2 - 2x^2,$$

the integrand is $2(x^2+y^2)$. Applied literally, (21.1) therefore returns $2\int_E (x^2+y^2)$, which is statement **(a)**, and never $2\int_E(x^2+2y^2)$, so **(b)** is false. Q.E.D.

AI-Note 21.5 (*The quoted area formula needs injectivity, and Φ does not have it*): Exercise 21.b.12**(a)** is marked **true** in the official key, and it is what (21.1) gives. But (21.1) is false as printed, because it carries no injectivity hypothesis, and Φ is exactly the map that exposes this: $\Phi(-p) = \Phi(p)$ for every p.

Concretely, take $E := \overline{B_1(0)}$ and put $u := xy$, $v := x^2 - y^2$. Then

$$4u^2 + v^2 = 4x^2y^2 + (x^2-y^2)^2 = (x^2+y^2)^2,$$

so Φ maps E into the ellipse $\{4u^2+v^2 \le 1\}$, and onto it: given such a (u,v), set $s := \sqrt{4u^2+v^2} \le 1$ and solve $x^2 = (s+v)/2$, $y^2 = (s-v)/2$, both non-negative because $v^2 \le s^2$. These satisfy $x^2+y^2 = s \le 1$ and $(xy)^2 = (s^2-v^2)/4 = u^2$, so a choice of signs gives $xy = u$. Hence

$$\Phi(E) = \{(u,v) \mid 4u^2 + v^2 \le 1\},$$

an ellipse with semi-axes $\frac{1}{2}$ and 1, of area $\pi/2$. The right-hand side of (21.1), meanwhile, is

$$\int_E 2(x^2+y^2)\,dx\,dy = 2\int_0^{2\pi}\int_0^1 r^2 \cdot r\,dr\,d\theta = \pi.$$

The two differ by a factor of 2, which is precisely the number of preimages. What is true in general is that the integral counts multiplicity,

$$\int_E |\det \mathrm{J}\,\Phi|\,dx = \int_{\mathbb{R}^n} \#\big(\Phi^{-1}(w) \cap E\big)\,dw \ \ge\ \mathrm{vol}_n(\Phi(E)),$$

with equality exactly when Φ is injective off a null set. Theorem 20.a.1 carries the injectivity hypothesis for this reason, and so does Definition 21.b.6, which is stated for a **parametrization** rather than for an arbitrary C^1 map.

The verdict on part **(a)** is therefore: true as the examination intends it, false as mathematics. It is kept here because the gap is more instructive than the computation.

Solution to Exercise 21.b.13:

Following the official hint rather than the spherical one (AI-Note 21.4), we use cylindrical coordinates $(x,y,z) = (r\cos\theta, r\sin\theta, z)$, under which $x^2+y^2 = r^2$. The two bounding surfaces become

$r^2+z^2=8$ and $z=r^2/2$; intersecting them,

$$z^2+2z-8=0 \quad\Longrightarrow\quad z=2 \text{ or } z=-4,$$

and since $z=r^2/2\ge 0$ on the paraboloid, the surfaces meet at $z=2$, $r=2$. For $r\in(0,2)$ the paraboloid $z=r^2/2$ lies below the upper half of the sphere $z=\sqrt{8-r^2}$, so

$$B=\left\{(r,\theta,z): 0<r<2,\ 0<\theta<2\pi,\ \tfrac12 r^2<z<\sqrt{8-r^2}\right\}.$$

Hence, with $dx\,dy\,dz=r\,dr\,d\theta\,dz$,

$$\operatorname{vol}(B)=\int_0^{2\pi}\!\!\int_0^2 r\left(\sqrt{8-r^2}-\frac{r^2}{2}\right)dr\,d\theta=2\pi\int_0^2\left(r\sqrt{8-r^2}-\frac{r^3}{2}\right)dr.$$

For the first term, substitute $u=8-r^2$, $du=-2r\,dr$:

$$\int_0^2 r\sqrt{8-r^2}\,dr=\left[-\frac{u^{3/2}}{3}\right]_{u=8}^{u=4}=\frac{8^{3/2}-4^{3/2}}{3}=\frac{16\sqrt2-8}{3}.$$

For the second, $\int_0^2 \frac12 r^3\,dr=\frac12\cdot\frac{16}{4}=2$. So

$$\operatorname{vol}(B)=2\pi\left(\frac{16\sqrt2-8}{3}-2\right)=2\pi\cdot\frac{16\sqrt2-14}{3}=\boxed{\frac{4\pi}{3}\left(8\sqrt2-7\right)}.$$

Q.E.D.

Solution to Exercise 21.c.25:

(a) By Fubini (Theorem 20.b.29),

$$\iint_D (x^3+3x^2y+y^3)\,dx\,dy=\int_0^2\int_0^1 (x^3+3x^2y+y^3)\,dy\,dx=\int_0^2\left(x^3+\frac32x^2+\frac14\right)dx.$$

The last integral is $\left[\frac{x^4}{4}+\frac{x^3}{2}+\frac{x}{4}\right]_0^2=4+4+\frac12=\boxed{\frac{17}{2}}$.

(b) Using the parametrization from the hint, $D=\{0\le x\le y\le\pi\}$,

$$\iint_D x\cos(x+y)\,dx\,dy=\int_0^\pi\int_x^\pi x\cos(x+y)\,dy\,dx=\int_0^\pi x\big[\sin(x+y)\big]_{y=x}^{y=\pi}dx.$$

Since $\sin(x+\pi)=-\sin x$, the bracket equals $-\sin x-\sin(2x)$, so

$$\iint_D x\cos(x+y)\,dx\,dy=-\int_0^\pi x\sin x\,dx-\int_0^\pi x\sin(2x)\,dx.$$

Integrating by parts, $\int_0^\pi x\sin x\,dx=[-x\cos x+\sin x]_0^\pi=\pi$, and $\int_0^\pi x\sin(2x)\,dx=\left[-\frac{x}{2}\cos(2x)+\frac14\sin(2x)\right]_0^\pi=-\frac{\pi}{2}$. Hence

$$\iint_D x\cos(x+y)\,dx\,dy=-\pi-\left(-\frac{\pi}{2}\right)=\boxed{-\frac{\pi}{2}}.$$

(c) Here $D=\{1<x<2,\ 1<y<3-x\}$, so

$$\iint_D\frac{1}{(x+y)^3}\,dx\,dy=\int_1^2\int_1^{3-x}(x+y)^{-3}\,dy\,dx=\int_1^2\left[-\frac12(x+y)^{-2}\right]_{y=1}^{y=3-x}dx.$$

Since $x+(3-x)=3$, the bracket is $-\frac12\big(3^{-2}-(x+1)^{-2}\big)=\frac12(x+1)^{-2}-\frac{1}{18}$, and

$$\int_1^2\left(\frac{1}{2(x+1)^2}-\frac{1}{18}\right)dx=\left[-\frac{1}{2(x+1)}\right]_1^2-\frac{1}{18}=\left(-\frac16+\frac14\right)-\frac{1}{18}=\frac{1}{12}-\frac{1}{18}=\boxed{\frac{1}{36}}.$$

Q.E.D.

Solution to Exercise 21.b.14:

We parametrize $\mathbb{S}^2$ (up to a null set) using latitude/longitude spherical coordinates

$$\Phi : (-\pi, \pi) \times \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right) \to \mathbb{S}^2, \qquad (\varphi, \theta) \mapsto \begin{pmatrix} \cos\varphi\cos\theta \\ \sin\varphi\cos\theta \\ \sin\theta \end{pmatrix}.$$

Note that this is **not** the $(r \equiv 1)$ parametrization used elsewhere in this chapter, whose polar angle runs over $(0, \pi)$. Here θ is measured from the equator, matching the exercise's range $\theta \in (-\pi/2, \pi/2)$. Computing

$$\partial_\varphi\Phi \times \partial_\theta\Phi = \begin{pmatrix} -\sin\varphi\cos\theta \\ \cos\varphi\cos\theta \\ 0 \end{pmatrix} \times \begin{pmatrix} -\cos\varphi\sin\theta \\ -\sin\varphi\sin\theta \\ \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\varphi\cos^2\theta \\ \sin\varphi\cos^2\theta \\ \sin\theta\cos\theta \end{pmatrix},$$

its length is $|\partial_\varphi\Phi \times \partial_\theta\Phi| = \cos\theta$ (for $\theta \in (-\pi/2, \pi/2)$, so $\cos\theta > 0$). By Proposition 23.b.3**(b)**, this is the square root of the Gram determinant, so $dA = \cos\theta\, d\theta\, d\varphi$, and

$$\int_S dA = \int_{\varphi_1}^{\varphi_2}\int_{\theta_1}^{\theta_2} \cos\theta\, d\theta\, d\varphi = \boxed{(\varphi_2 - \varphi_1)\big(\sin\theta_2 - \sin\theta_1\big)}.$$

Q.E.D.

Solution to Exercise 21.b.15:

The circle S has radius 2, and T is the solid torus swept by a tube of radius 1 around it, so ∂T is the torus surface of the hint with $R = 2$, $r = 1$. Using that parametrization,

$$\phi(\theta, \varphi) = \begin{pmatrix} (2 + \cos\varphi)\cos\theta \\ (2 + \cos\varphi)\sin\theta \\ \sin\varphi \end{pmatrix}, \qquad \theta, \varphi \in (0, 2\pi).$$

Then $\partial_\theta\phi = \big(-(2 + \cos\varphi)\sin\theta,\ (2 + \cos\varphi)\cos\theta,\ 0\big)$ has length $2 + \cos\varphi$, and $\partial_\varphi\phi = (-\sin\varphi\cos\theta,\ -\sin\varphi\sin\theta,\ \cos\varphi)$ has length 1; the two are orthogonal (their dot product is $\sin\varphi\cos\varphi\sin\theta\cos\theta - \sin\varphi\cos\varphi\sin\theta\cos\theta = 0$), so

$$\sqrt{\det D\phi^{\mathsf{T}} D\phi} = |\partial_\theta\phi|\,|\partial_\varphi\phi| = 2 + \cos\varphi.$$

Hence

$$\mathrm{vol}_2(\partial T) = \int_0^{2\pi}\int_0^{2\pi} (2 + \cos\varphi)\, d\theta\, d\varphi = 2\pi\left(2\cdot 2\pi + \underbrace{\int_0^{2\pi}\cos\varphi\, d\varphi}_{=0}\right) = \boxed{8\pi^2}.$$

Q.E.D.

Solution to Exercise 21.c.26:

(a) $\partial_{\text{side}}\Omega$ is not closed (it omits, e.g., the limit point $(a, f(a), 0)$ as $x \to a^+$) and not open (a 2-dimensional surface has empty interior in $\mathbb{R}^3$), but it **is** path-connected: from any point, first move along the graph of f in the x-direction, then move around the circle in the (y, z)-variables. Ω is open (as the strict sublevel set of a continuous function), not closed, and its connectedness depends on f: if f vanishes somewhere in (a, b), the cross-section pinches to a point there and Ω is disconnected into the pieces on either side; if $f > 0$ throughout (a, b), Ω is connected.

(b) In cylindrical coordinates (x,ρ,θ) with $y=\rho\cos\theta$, $z=\rho\sin\theta$, $dx\,dy\,dz=\rho\,d\rho\,d\theta\,dx$, the domain becomes the box-like region $\{a<x<b,\ 0<\rho<f(x),\ \theta\in[0,2\pi)\}$, so

$$\mathrm{vol}_3(\Omega)=\int_a^b\int_0^{2\pi}\int_0^{f(x)}\rho\,d\rho\,d\theta\,dx=\int_a^b\int_0^{2\pi}\frac12 f(x)^2\,d\theta\,dx=\boxed{\pi\int_a^b f(x)^2\,dx}.$$

(c) Using the suggested parametrization $\Phi(x,\theta)=(x,f(x)\cos\theta,f(x)\sin\theta)$, write the orthonormal frame $\xi_1:=(1,0,0)$, $\xi_2:=(0,\cos\theta,\sin\theta)$, $\xi_3:=(0,-\sin\theta,\cos\theta)$, so that

$$\partial_x\Phi=\xi_1+f'(x)\xi_2,\qquad \partial_\theta\Phi=f(x)\xi_3.$$

Since ξ_1,ξ_2,ξ_3 are orthonormal and right-handed, $\xi_1\times\xi_3=-\xi_2$ and $\xi_2\times\xi_3=\xi_1$, so

$$\partial_x\Phi\times\partial_\theta\Phi=f(x)\big(\xi_1\times\xi_3+f'(x)\,\xi_2\times\xi_3\big)=f(x)\big(-\xi_2+f'(x)\xi_1\big),$$
$$|\partial_x\Phi\times\partial_\theta\Phi|=f(x)\sqrt{1+f'(x)^2}.$$

By Proposition 23.b.3**(b)**, this is the square root of the Gram determinant, so

$$\mathrm{vol}_2(\partial_{\text{side}}\Omega)=\int_a^b\int_0^{2\pi}f(x)\sqrt{1+f'(x)^2}\,d\theta\,dx=\boxed{2\pi\int_a^b f(x)\sqrt{1+f'(x)^2}\,dx}.$$

(d) With $f(x)=1/x$ on $[1,\infty)$, both integrands are non-negative, so the improper integrals are unambiguous (possibly $+\infty$). The volume is finite:

$$\pi\int_1^\infty\frac{1}{x^2}\,dx=\pi\Big[-\frac1x\Big]_1^\infty=\boxed{\pi}.$$

For the surface area, $f'(x)=-1/x^2$, so the integrand is $\sqrt{1+x^{-4}}/x\geq 1/x$ pointwise, and since $\int_1^\infty\frac1x\,dx$ diverges, so does

$$2\pi\int_1^\infty\frac{\sqrt{1+x^{-4}}}{x}\,dx=\boxed{+\infty}.$$

This is the classical "*Gabriel's horn*" (German: „*Torricellis Trompete*"): finite volume, infinite surface area.

(e) ($\Leftarrow$) Suppose $(*)$ holds, and write $\rho:=\sqrt{y^2+z^2}$. Away from the two ends, $\partial\Omega$ is the side surface $\{\rho=f(x)\}$, regularly parametrized as in part **(c)** because $f>0$, and $\Omega=\{\rho<f(x)\}$ lies on one side of it: $G(x,y,z):=\rho-f(x)$ is C^1 off the axis with $\nabla G=\big(-f'(x),\ y/\rho,\ z/\rho\big)\neq 0$, as condition **(b)** of Definition 22.a.1 asks.

At the end $x=a$, suppose first that $f(a)>0$. Since $f'(a^+)=+\infty$, the inverse function theorem applies to $f|_{(a,a+\varepsilon)}$ for ε small, giving a C^1, increasing inverse there; extend it by

$$g(y):=\begin{cases}f^{-1}(y) & f(a)<y<f(a+\varepsilon),\\ a & y\leq f(a).\end{cases}$$

Since $f'(a^+)=+\infty$, $\lim_{y\to f(a)^+}g'(y)=\lim_{x\to a^+}1/f'(x)=0=\lim_{y\to f(a)^-}g'(y)$, so $g\in C^1$ across $y=f(a)$. Around a boundary point $P:=(a,f(a)\cos\phi,f(a)\sin\phi)$, the map

$$(t,\theta)\mapsto\big(g(t),\ t\cos\theta,\ t\sin\theta\big)$$

is then a regular C^1 parametrization of $\partial\Omega$. For $t\leq f(a)$ it traces the end disc $\{x=a\}$, and for $t>f(a)$ the side surface, where $t=f(g(t))$. Its partial derivatives $\big(g'(t),\cos\theta,\sin\theta\big)$ and $\big(0,-t\sin\theta,t\cos\theta\big)$ are linearly independent because t is close to $f(a)>0$. Near P, moreover, $\Omega=\{x>g(\rho)\}$, so $G:=g(\rho)-x$ has $\partial_xG=-1$ and exhibits Ω on one side of its boundary.

If instead $f(a)=0$, that parametrization degenerates and the boundary point to examine is $P:=(a,0,0)$, on the axis. Since $f'(a^+)=+\infty$, f is increasing near a, so its inverse $h:=f^{-1}$ is defined on $(0,\delta)$ with $h(0^+)=a$ and $h'(y)=1/f'(h(y))\to 0$ as $y\to 0^+$. Setting $h(0):=a$, the map $(y,z)\mapsto h\big(\sqrt{y^2+z^2}\big)$ has partial derivatives $h'(r)\,y/r$ and $h'(r)\,z/r$ with $r:=\sqrt{y^2+z^2}$, which are continuous and tend to 0 as $r\to 0$. So it is C^1 near

the origin and exhibits $\partial\Omega$ near P as a C^1 graph over the (y,z)-plane, with $\Omega = \{x > h(\rho)\}$ on one side of it. The symmetric argument at b, using $f'(b^-) = -\infty$, handles the other end. So $\partial\Omega$ is a C^1 submanifold with Ω locally on one side of it, and Ω is a bounded C^1 domain.

($\Rightarrow$) Suppose Ω is a bounded C^1 domain, and $\Omega \neq \emptyset$, that is, $f \not\equiv 0$. Only condition **(a)** of Definition 22.a.1 is needed, through two standard facts about a C^1 surface: near each of its points it is the graph of a C^1 function over its tangent plane there, and its tangent planes depend continuously on the point. One more observation is used twice. Rotations about the x-axis map Ω, and hence $\partial\Omega$, to itself. At a point $P \in \partial\Omega$ on the axis they therefore map $T_P(\partial\Omega)$ to itself, and the only 2-plane in $\mathbb{R}^3$ invariant under all of them is $\operatorname{span}\{e_2, e_3\}$. Near such a P, then, $\partial\Omega$ is a graph $x = h(y,z)$ with h of class C^1 and $\nabla h = 0$ at the centre.

First, $f > 0$ on (a,b). Suppose not. The zero set of f in (a,b) is then closed in (a,b), non-empty, and not all of (a,b), so, as (a,b) is connected, it is not open. Hence there are a zero $x_0 \in (a,b)$ of f and points $x_k \to x_0$ with $f(x_k) > 0$, and $f'(x_0) = 0$ because x_0 is a minimum of $f \geq 0$. The points $P := (x_0, 0, 0)$ and $Q_k := (x_k, f(x_k), 0)$ lie in $\partial\Omega$: each is a limit of points of Ω, namely of $(x_k, 0, 0)$ and of $\big(x_k, (1 - \frac{1}{j})f(x_k), 0\big)$ as $j \to \infty$, and none lies in Ω. As $Q_k \to P$, for large k the point Q_k lies on the graph above, so $x_k = h(f(x_k), 0)$, while $x_0 = h(0,0)$. The mean value theorem, applied to h and to f, gives

$$|x_k - x_0| \leq \eta_k\, f(x_k) \qquad \text{and} \qquad f(x_k) = |f(x_k) - f(x_0)| \leq \eta'_k\, |x_k - x_0|,$$

where η_k is the largest value of $|\partial_y h|$ on the segment from $(0,0)$ to $(f(x_k), 0)$, and η'_k the largest value of $|f'|$ between x_0 and x_k. Both tend to 0, because $\partial_y h(0,0) = 0 = f'(x_0)$. Hence $|x_k - x_0| \leq \eta_k \eta'_k\, |x_k - x_0|$, which is impossible once $\eta_k \eta'_k < 1$, since $x_k \neq x_0$. Geometrically, where f vanishes the body is pinched, and its boundary comes to a cusp tangent to the axis, which no C^1 surface does.

Next, the condition at a. Let $P := (a, f(a), 0)$, which lies in $\partial\Omega$, and first show $T_P(\partial\Omega) = \operatorname{span}\{e_2, e_3\}$. If $f(a) = 0$, then P lies on the axis and the rotation observation gives it. If $f(a) > 0$, then $T_P(\partial\Omega)$ is 2-dimensional and contains

$$\frac{d}{dt}\Big|_{t=0^+} \big(a, f(a) - t, 0\big) = -e_2, \qquad \frac{d}{dt}\Big|_{t=0^-} \big(a, f(a)\cos t, f(a)\sin t\big) = f(a)e_3,$$

the first curve running along the end disc and the second around its rim, so again $T_P(\partial\Omega) = \operatorname{span}\{e_2, e_3\}$.

The side points $P_x := (x, f(x), 0)$, $a < x < b$, lie in $\partial\Omega$ and converge to P as $x \to a^+$, and the curve $s \mapsto (s, f(s), 0)$ shows that $(1, f'(x), 0)$ is tangent to $\partial\Omega$ at P_x. As tangent planes depend continuously on the point, the unit vectors $(1, f'(x), 0)/\sqrt{1 + f'(x)^2}$ must approach $\operatorname{span}\{e_2, e_3\}$, so their e_1-component $1/\sqrt{1 + f'(x)^2}$ tends to 0: $|f'(x)| \to \infty$ as $x \to a^+$. Being continuous and eventually non-zero, f' then has a single sign on some interval $(a, a + \delta)$, by the intermediate value theorem, and that sign cannot be negative. If $f(a) = 0$, a decreasing f would turn negative. If $f(a) > 0$, a decreasing f would put, for every ρ slightly below $f(a)$, the two points $(a, \rho, 0)$ of the end disc and $\big(f^{-1}(\rho), \rho, 0\big)$ of the side into $\partial\Omega$ near P. They are different points with the same (y,z)-coordinates, which contradicts $\partial\Omega$ being a graph over $\operatorname{span}\{e_2, e_3\}$ near P. Hence $f'(a^+) = +\infty$. Geometrically, a finite slope leaves an edge along the rim of the end disc, or a conical vertex or cusp when $f(a) = 0$. The symmetric argument at b gives $f'(b^-) = -\infty$.

Q.E.D.

AI-Note 21.6: All parts of Exercises 21.b.14, 21.b.15 and 21.c.26 were worked out independently before consulting `Sol10_Analysis2_eng.pdf`. The three main answers ($(\varphi_2 - \varphi_1)(\sin\theta_2 - \sin\theta_1)$, $8\pi^2$, $\pi \int_a^b f(x)^2 dx$ / $2\pi \int_a^b f(x)\sqrt{1 + f'(x)^2}dx$ / volume π, area $+\infty$) match exactly. Part **(e)** was reconstructed independently along the same two-directional strategy (build an explicit C^1 parametrization near each endpoint using the inverse function theorem; for the converse, show that the tangent plane at the rim is $\operatorname{span}\{e_2, e_3\}$ and that the tangent directions $(1, f'(x), 0)$ of the side

must approach it) and agrees with the official solution in substance, though the write-up above is condensed relative to the official one.

Solution to AI-Exercise 21.d.30:

Throughout, $\mathrm{J}\,\Phi$ is the 2×1 matrix Φ', so $\mathrm{J}\,\Phi^{\mathsf{T}}\,\mathrm{J}\,\Phi = |\Phi'|^2$ is a 1×1 Gram determinant and its square root is the familiar speed $|\Phi'|$ of Section 21.c.

(a) Here $\Phi'(t) = (-\sin t, \cos t)$ has $|\Phi'| \equiv 1$, so

$$\int_{S^1} 1 \, d\underset{1}{\mathrm{vol}} = \int_0^{2\pi} 1 \, dt = 2\pi.$$

The omitted point $(1,0)$ is a single point, hence vol_1-null, and contributes nothing, which is the shortcut Remark 21.d.29 describes.

(b) Let η_1, η_2 be **any** partition of unity subordinate to this atlas, so that $\eta_1 + \eta_2 \equiv 1$ on S^1. Since $|\Phi_i'| \equiv 1$ for both,

$$\int_{S^1} 1 \, d\underset{1}{\mathrm{vol}} = \int_0^{2\pi} \eta_1(\cos t, \sin t)\, dt + \int_{-\pi}^{\pi} \eta_2(\cos t, \sin t)\, dt.$$

The second integrand is a 2π-periodic function of t, and the integral of such a function over any interval of length 2π is the same, so the second integral may be taken over $(0, 2\pi)$ as well. Adding the two under one integral sign,

$$\int_{S^1} 1 \, d\underset{1}{\mathrm{vol}} = \int_0^{2\pi} \big(\eta_1 + \eta_2\big)(\cos t, \sin t)\, dt = \int_0^{2\pi} 1 \, dt = 2\pi,$$

and no particular partition was ever chosen. That is Proposition 21.d.28 in miniature: the η_l disappear because they sum to 1, and everything else was already parametrization-independent.

(c) The two images miss $(\pm 1, 0)$. $\mathrm{Im}(\Psi_+)$ is the open upper semicircle and $\mathrm{Im}(\Psi_-)$ the open lower one, so their union is $S^1 \setminus \{(1,0), (-1,0)\}$ and the family covers no neighbourhood of either point.

What fails is then not the arithmetic but the definition. Subordinacy asks for open sets $U_\pm \subseteq \mathbb{R}^2$ with $U_\pm \cap S^1 = \mathrm{Im}(\Psi_\pm)$ and $\mathrm{supp}(\eta_\pm) \subseteq U_\pm$. Neither U_+ nor U_- contains $(1,0)$ or $(-1,0)$, so both η_+ and η_- vanish at those two points; their sum vanishes there too and cannot be the constant 1 on S^1. No partition of unity subordinate to this family exists, and Definition 21.d.27 never gets started.

It is worth noticing that the recipe nevertheless returns the right number. Taking $\eta_\pm$ summing to 1 off those two points and substituting $u = \cos t$, which turns $(1-u^2)^{-1/2}\,|du|$ into dt, gives $\pi + \pi = 2\pi$ again. The reason is the one from part **(a)**: the uncovered set is vol_1-null. So this atlas is illegitimate and harmless at once, and only the first of those is a statement about S^1; the second is an accident of the set that was missed.

Q.E.D.

PART VII

Vector Calculus and Differential Forms

The classical integral theorems of vector calculus look at first like three unrelated results. Green's theorem converts a line integral into a double integral, the divergence theorem converts a flux through a closed surface into a triple integral, and the classical theorem of Stokes converts a circulation into a flux. This part states all three, checks the divergence theorem by hand on a square, and then explains why they are one theorem. The apparatus that achieves this consists of differential forms and the exterior derivative d, a single operator absorbing grad, curl and div at once, which satisfies the one identity $d \circ d = 0$ in place of the two unrelated-looking ones $\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$. In that language each of the statements above reads $\int_M d\omega = \int_{\partial M} \omega$, and the classical versions are what it becomes when M is a plane region, a solid, or a surface. Everything built earlier is spent here. Submanifolds supply what one integrates over, orientation fixes the sign, compactly supported partitions of unity patch local computations into global ones, and simple connectedness decides when a field whose curl vanishes is a gradient after all.

Vector Calculus Operators (Chapter 22)

The integral theorems of this part relate what happens inside a domain to what happens on its boundary, so two things have to come first. One is a class of domains whose boundary is well behaved enough to integrate over. The other is the differential operators (gradient, divergence, curl) whose interplay the theorems express. This chapter supplies both, and shows why a smooth boundary alone does not yet make a domain one can integrate over.

Bounded C^1 domains and why the definition in the script is so terrible (Section 22.a)

Definition 22.a.1 (*Bounded C^k domain*): Given $k \geq 1$, a bounded open set $\Omega \subset \mathbb{R}^n$ is called a "*bounded C^k domain*" if

- **(a)** $M := \partial\Omega$ is an $(n-1)$-dimensional submanifold of class C^k, and
- **(b)** Ω lies locally on one side of M: every $p \in M$ has an open neighbourhood V with a function $g \in C^k(V)$ such that $\nabla g \neq 0$ on V and $\Omega \cap V = \{x \in V : g(x) < 0\}$.

Some machinery is needed to actually **use** this definition. To define "*flux*" we need the concept of interior and exterior normal vectors. For Ω as above, a normal vector $v \in N_p(\partial\Omega)$ is "*interior*" if

$$p + hv \in \Omega \quad \text{for all } h > 0 \text{ small,}$$

and "*exterior*" if

$$p + hv \in \mathbb{R}^n \setminus \Omega \quad \text{for all } h > 0 \text{ small.}$$

A "*Gauss map*" is a continuous map of unit normal vectors on $\partial\Omega$. There exists a unique interior and exterior Gauss map; write

$$\nu \in C^0(\partial\Omega, \mathbb{S}^{n-1}) \text{ such that } \nu(p) \text{ is exterior normal}$$

for the exterior unit normal map.

Condition **(b)** is what makes this true. With V and g as there, $\partial\Omega \cap V = \{x \in V : g(x) = 0\}$, so $\nabla g(p)$ spans the normal line $N_p(\partial\Omega)$, and the unit normal $\nabla g(p)/|\nabla g(p)|$ is exterior: g increases in that direction, so $g(p+hv) > 0$ and $p + hv \notin \Omega$ for small $h > 0$. Its negative is interior for the same reason. A normal line holds only these two unit vectors, so $\nu = \nabla g/|\nabla g|$ near p, whichever g was chosen, and this is continuous.

Remark 22.a.2 (*Why a smooth boundary is not enough*): Condition **(a)** alone admits sets on which none of the above makes sense. Take

$$\Omega := B_1(0) \setminus \partial B_{1/2}(0) \subseteq \mathbb{R}^3,$$

the open unit ball with the sphere of radius $\frac{1}{2}$ removed. It is bounded and open, and $\partial\Omega = \partial B_1(0) \cup \partial B_{1/2}(0)$ is a smooth surface, so **(a)** holds. But Ω lies on **both** sides of the inner sphere: at a point p of it, both unit normals $\pm 2p$ are interior. There is no exterior normal at p, hence no exterior Gauss map, and the flux of Definition 23.b.2 is undefined.

No choice of normal on the inner sphere repairs this. For $F(x) := x$, with $\operatorname{div} F = 3$, the volume side of the divergence theorem (Theorem 23.b.5) is $\int_\Omega 3 \, d\,\mathrm{vol}_3 = 4\pi$, the removed sphere being a null set. The outer sphere alone already contributes $\int_{\partial B_1(0)} \langle x, x \rangle \, d\,\mathrm{vol}_2 = 4\pi$ to the boundary side, and the inner sphere adds $\pm\frac{1}{2}\,\mathrm{vol}_2\left(\partial B_{1/2}(0)\right) = \pm\frac{\pi}{2} \neq 0$, whichever unit normal it is given.

Condition **(b)** excludes the example. Since $\nabla g(p) \neq 0$, it puts points with $g > 0$, which lie outside $\overline{\Omega}$, arbitrarily close to every boundary point, and near the inner sphere there are none.

AI-Note 22.1: This definition is sometimes stated with condition **(a)** alone, together with the assertion that a unique interior and exterior Gauss map exists, which Remark 22.a.2 refutes. Condition **(b)** has been added here so that the assertion, and with it the flux and the divergence theorem built on it, holds. The domains to which this document applies those results all satisfy it.

Theorem 22.a.3 (*Jordan–Brouwer separation theorem*): If $M \subseteq \mathbb{R}^{n+1}$ is an n-dimensional compact and connected submanifold, then $\mathbb{R}^{n+1} \setminus M$ has exactly two connected components, an "*inside*" and an "*outside*".

Vector calculus operators (Section 22.b)

Definition 22.b.4 (*Vector field*): Let $U \subseteq \mathbb{R}^n$ be open. A "C^k *vector field*" (German: „*Vektorfeld*“) is a function $F : U \to \mathbb{R}^n$ of class $C^k(U, \mathbb{R}^n)$. The scalar functions $F_1, \dots, F_n$ in $F(x) = (F_1(x), \dots, F_n(x))^\mathsf{T}$ are called the "*components*".

Definition 22.b.5 (*Gradient*): Let $f : U \to \mathbb{R}$ be a function of class C^k on an open set $U \subseteq \mathbb{R}^n$. The "*gradient*" of f is the C^{k-1} vector field

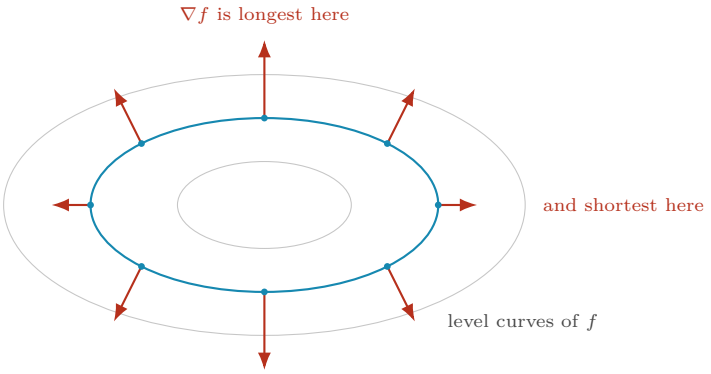
$$\operatorname{grad} f(x) = \nabla f(x) := \begin{pmatrix} \partial_1 f(x) \\ \vdots \\ \partial_n f(x) \end{pmatrix}.$$

Definition 22.b.6 (*Divergence*): For a C^1 vector field $F : U \to \mathbb{R}^n$, the "*divergence*" is the scalar function

$$\operatorname{div} F(x) := \sum_{i=1}^{n} \partial_i F_i(x).$$

In 3D, physicists write this as $\vec{\nabla} \cdot \vec{F}(x) = \partial_1 F_1(x) + \partial_2 F_2(x) + \partial_3 F_3(x)$.

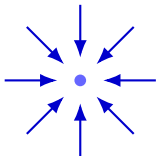
Remark 22.b.7 (*The gradient, read off the level curves*): Two properties of ∇f are easier to see than to prove, and a contour picture shows both at once. The gradient at a point is **perpendicular to the level curve** through that point, and its **length measures how tightly the level curves are packed** there.



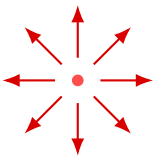
Both readings are the same fact. Moving a distance ε in the direction ∇f changes f by about $\varepsilon|\nabla f|$, so where $|\nabla f|$ is large one crosses from one level curve to the next after a short step, and where it is small one has further to walk. Moving **along** a level curve does not change f at all, so the directional derivative vanishes there, which says exactly that ∇f is orthogonal to the curve.

Remark 22.b.8 (*Sources and sinks*): Intuitively, $\operatorname{div} F(x)$ tells us how much flow of F is generated at x. If $\operatorname{div} F(x) = 0$, then the total generated flow is zero. If $\operatorname{div} F(x) > 0$, then x is a

"*source*" (German: „*Quelle*"), meaning more flow is generated than removed. If $\operatorname{div} F(x) < 0$, then x is a "*sink*" (German: „*Senke*"), meaning more flow is removed than generated.



sink: $\operatorname{div} F < 0$



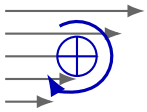
source: $\operatorname{div} F > 0$

Definition 22.b.9 (*Rotation/Curl*): Let F be a C^k vector field on an open set $U \subseteq \mathbb{R}^3$. The "*rotation*" or "*curl*" of F is the C^{k-1} vector field

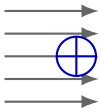
$$\operatorname{curl} F(x) := \begin{pmatrix} \partial_2 F_3(x) - \partial_3 F_2(x) \\ \partial_3 F_1(x) - \partial_1 F_3(x) \\ \partial_1 F_2(x) - \partial_2 F_1(x) \end{pmatrix}.$$

Physicists write $\operatorname{curl} F(x) = \vec{\nabla} \times \vec{F}(x)$. In $\mathbb{R}^2$, the rotation of a vector field $F = (F_1, F_2)^\mathsf{T}$ is the scalar function $\operatorname{curl} F(x) := \partial_1 F_2(x) - \partial_2 F_1(x)$.

Remark 22.b.10 (*The paddle-wheel reading of the curl*): Intuitively, $\operatorname{curl} F(x)$ tells us how much the field tends to rotate around x. If $\operatorname{curl} F(x) \neq 0$, a small wheel centered at x would rotate.



rotates



does not rotate

Example 22.b.11 (*Computing vector calculus operators*): Given $\phi : \mathbb{R}^3 \to \mathbb{R}$, $\phi(x,y,z) := x^2 + y^2 + z^2 e^{-xy}$. Let us compute the gradient, the divergence of the gradient, and the rotation of the gradient.

- **Gradient:**

$$\nabla\phi(x,y,z) = \begin{pmatrix} 2x - yz^2 e^{-xy} \\ 2y - xz^2 e^{-xy} \\ 2ze^{-xy} \end{pmatrix}.$$

- **Divergence of the gradient** (denoted $\operatorname{div}(\nabla\phi)$ or $\Delta\phi$):

$$\begin{aligned} \operatorname{div}(\nabla\phi) &= \partial_x(2x - yz^2e^{-xy}) + \partial_y(2y - xz^2e^{-xy}) + \partial_z(2ze^{-xy}) \\ &= (2 + y^2z^2e^{-xy}) + (2 + x^2z^2e^{-xy}) + (2e^{-xy}) \\ &= 4 + (x^2z^2 + y^2z^2 + 2)e^{-xy}. \end{aligned}$$

- **Rotation of the gradient** ($\operatorname{curl}(\nabla\phi)$):

$$\begin{aligned} \operatorname{curl}(\nabla\phi) &= \begin{pmatrix} \partial_y(2ze^{-xy}) - \partial_z(2y - xz^2e^{-xy}) \\ \partial_z(2x - yz^2e^{-xy}) - \partial_x(2ze^{-xy}) \\ \partial_x(2y - xz^2e^{-xy}) - \partial_y(2x - yz^2e^{-xy}) \end{pmatrix} \\ &= \begin{pmatrix} -2xze^{-xy} - (-2xze^{-xy}) \\ -2yze^{-xy} - (-2yze^{-xy}) \\ (xyz^2 - z^2)e^{-xy} - (xyz^2 - z^2)e^{-xy} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

This is not a coincidence: for any C^2 function f, $\operatorname{curl}(\nabla f) = 0$ by Schwarz's theorem.

Proposition 22.b.12 (curl ∘ grad = 0 *and* div ∘ curl = 0): Let f be a twice continuously differentiable function and F a twice continuously differentiable vector field on an open set $U \subseteq \mathbb{R}^3$. Then

$$\operatorname{curl}(\operatorname{grad} f) = 0 \qquad \text{and} \qquad \operatorname{div}(\operatorname{curl} F) = 0.$$

Proof:

Both identities follow from Schwarz's theorem (Theorem 11.b.13), which applies since $f, F \in C^2$.

$\operatorname{curl}(\operatorname{grad} f) = 0$**.** The first component of $\operatorname{curl}(\operatorname{grad} f)$ is $\partial_2(\partial_3 f) - \partial_3(\partial_2 f) = 0$ by Schwarz, and the other two components vanish the same way, cyclically permuting the indices. This is exactly Example 22.b.11's observation above, now stated and proved in general.

$\operatorname{div}(\operatorname{curl} F) = 0$**.**

$$\operatorname{div}(\operatorname{curl} F) = \partial_1(\partial_2 F_3 - \partial_3 F_2) + \partial_2(\partial_3 F_1 - \partial_1 F_3) + \partial_3(\partial_1 F_2 - \partial_2 F_1),$$

and every one of the six terms cancels against another by Schwarz: $\partial_1\partial_2 F_3$ cancels $-\partial_2\partial_1 F_3$, $-\partial_1\partial_3 F_2$ cancels $\partial_3\partial_1 F_2$, and $\partial_2\partial_3 F_1$ cancels $-\partial_3\partial_2 F_1$. Q.E.D.

Remark 22.b.13: This is the classical-vector-calculus shadow of "$d^2 = 0$" (Theorem 24.b.15**(b)**, Chapter 24): grad, curl and div are exactly the 0-, 1- and 2-form incarnations of the exterior derivative in $\mathbb{R}^3$ (Example 24.b.16), and both identities above are the single fact $d \circ d = 0$ read off once in each degree.

Proposition 22.b.14 (*Product rule for the curl*): Let $U \subseteq \mathbb{R}^3$ be open, $F : U \to \mathbb{R}^3$ a C^1 vector field, and $\varphi : U \to \mathbb{R}$ a C^1 function. Then

$$\operatorname{curl}(\varphi F) = \varphi \operatorname{curl}(F) + (\operatorname{grad}\varphi) \times F.$$

Proof:

By the product rule applied componentwise, e.g. for the first component,

$$\partial_2(\varphi F_3) - \partial_3(\varphi F_2) = \varphi(\partial_2 F_3 - \partial_3 F_2) + (\partial_2\varphi)F_3 - (\partial_3\varphi)F_2,$$

and the first term is $\varphi \operatorname{curl}(F)$'s first component, the last two are $((\operatorname{grad}\varphi) \times F)$'s first component. The other two components follow identically, cyclically permuting the indices. Q.E.D.

AI-Note 22.2: The official script sets this identity as an exercise (Ex. 14.61) but states it without the first term, as $\operatorname{curl}(\varphi F) = (\operatorname{grad}\varphi) \times F$, which is false in general. Taking $\varphi \equiv 1$ would force $\operatorname{curl} F \equiv 0$ for every F. The missing term is exactly $\varphi \operatorname{curl}(F)$, restored above. (The script's exercise also names the vector field f in its statement and F in its conclusion, as though the two were interchangeable.)

AI-Exercise 22.b.15 (*Testing a product rule against a special case*): The identity the official script sets as an exercise, without the first term of Proposition 22.b.14, is

$$\operatorname{curl}(\varphi F) \overset{?}{=} (\operatorname{grad}\varphi) \times F.$$

(a) Find a φ and an F for which this fails, and say what both sides evaluate to. The cheapest choice needs no differentiation of φ at all.

(b) Verify Proposition 22.b.14 directly for $\varphi(x, y, z) := z$ and $F(x, y, z) := (-y,\ x,\ 0)$, by computing $\operatorname{curl}(\varphi F)$, $\varphi \operatorname{curl} F$ and $(\operatorname{grad}\varphi) \times F$ separately and checking that the last two add up to the first.

Solution: *See solution on Page 344*

AI-Exercise 22.b.16 (*Which fields are gradients?*): On $\mathbb{R}^3$, consider

$$F(x, y, z) := (yz,\ xz,\ xy), \qquad G(x, y, z) := (y,\ -x,\ 0).$$

(a) Use Proposition 22.b.12 to rule one of the two out as a gradient field.
(b) For the other, find a function f with $\nabla f = F$.
(c) Confirm the verdict of part **(a)** by a second, independent route: compute the line integral of the ruled-out field along the unit circle $\gamma(t) := (\cos t,\ \sin t,\ 0)$, $t \in [0, 2\pi]$, and compare with Proposition 26.b.14.

Solution: *See solution on Page 344*

Exercise 22.b.17 (*True or False* (four candidate identities)): Which of the following formulas are correct for **all** smooth functions $u, v : \mathbb{R}^n \to (0, \infty)$? Decide each, with justification.

(a) $\partial_i\big(e^{u^2}\big) = e^{u^2}\, 2u\, \partial_i u$.
(b) $\partial_i \nabla u = \nabla \partial_i u$.
(c) $\partial_i(u/v) = (\partial_i u/v) + (u/\partial_i v)$.
(d) $\operatorname{div}(u\nabla u) = |\nabla u|^2 + u \operatorname{Hess} u$, where $\operatorname{Hess} u$ is the Hessian matrix of u.

Info: *This exercise is taken from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), where it appears as the fourth block of multiple-choice questions.*

Solution: *See solution on Page 345*

Exercise 22.b.18 (*True or False* (what a gradient field cannot do)): Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth, $n \geq 2$, and consider its gradient field ∇f. Decide whether each of the following is true or false, with justification.

(a) If $\nabla f(x_0) \neq 0$ at some $x_0 \in \mathbb{R}^n$, then there is an open neighbourhood U of x_0 such that $\{x \in U \mid f(x) = f(x_0)\}$ is an $(n-1)$-dimensional submanifold of $\mathbb{R}^n$.
(b) There is no regular smooth curve $\gamma : [0,1] \to \mathbb{R}^n$ with $\gamma(0) = \gamma(1)$ satisfying $\gamma'(t) = \nabla f(\gamma(t))$ for all $t \in [0,1]$.
(c) If $\nabla f(x) \neq 0$ for every $x \in \mathbb{R}^n$, then f is unbounded.
(d) If $\emptyset \neq B \subset \mathbb{R}^n$ is a compact, smoothly bounded region with outward unit normal field ν, and $\langle \nabla f(x), \nu(x) \rangle > 0$ for all $x \in \partial B$, then f has a local minimum in the interior of B.

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *of 17 August 2022 (revised), Problem 5. That paper names no lecturer.*

Solution: *See solution on Page 345*

Exercise 22.b.19 (*The divergence is unchanged by a linear change of coordinates*): Let $A \in \mathrm{GL}_n(\mathbb{R})$ and let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a differentiable vector field. Prove that for every $x \in \mathbb{R}^n$,

$$\operatorname{div}\big(A^{-1} \circ f \circ A\big)(x) = (\operatorname{div} f)(Ax).$$

Official hint: *Trace.*

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *of 31 January 2023, Problem 12. That paper names no lecturer.*

Solution: *See solution on Page 346*

Exercise 22.b.20 (*Flow of vector field* (important)): Compute the flow of the vector field

$$V : (x, y, z) \mapsto (x, y, z - x^2 - y^2)$$

through the upper hemisphere $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1,\ z \geq 0\}$ from *"inside"* to *"outside"*. That is the number

$$\int_S V(x) \cdot \nu(x)\, d\,\mathrm{vol}_2\,.$$

Info: *This exercise is Problem 11.1 of Problem Sheet 11, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 346*

Solutions (Section 22.c)

Every solution in this chapter is collected here.

Solution to AI-Exercise 22.b.15:

(a) Take $\varphi \equiv 1$ and $F(x,y,z) := (-y,\, x,\, 0)$. The right-hand side is $(\operatorname{grad}\varphi) \times F = 0 \times F = 0$, since a constant function has vanishing gradient. The left-hand side is

$$\operatorname{curl}(\varphi F) = \operatorname{curl} F = \big(\partial_y 0 - \partial_z x,\ \partial_z(-y) - \partial_x 0,\ \partial_x x - \partial_y(-y)\big) = (0,\, 0,\, 2) \neq 0.$$

So the two sides differ. The choice needs no work because $\varphi \equiv 1$ kills the right-hand side outright while leaving the left-hand side equal to $\operatorname{curl} F$, and any field with non-zero curl then finishes the job.

(b) With $\varphi(x,y,z) = z$ and the same F, we have $\varphi F = (-yz,\, xz,\, 0)$, so

$$\operatorname{curl}(\varphi F) = \big(\partial_y 0 - \partial_z(xz),\ \partial_z(-yz) - \partial_x 0,\ \partial_x(xz) - \partial_y(-yz)\big) = (-x,\, -y,\, 2z).$$

The two terms on the right of Proposition 22.b.14 are

$$\varphi \operatorname{curl} F = z\,(0,0,2) = (0,\, 0,\, 2z), \qquad (\operatorname{grad}\varphi) \times F = (0,0,1) \times (-y,x,0) = (-x,\, -y,\, 0),$$

and they add to $(-x,\, -y,\, 2z)$, which is $\operatorname{curl}(\varphi F)$.

Q.E.D.

Solution to AI-Exercise 22.b.16:

(a) By Proposition 22.b.12, a gradient field has vanishing curl, so a non-zero curl rules a field out. Computing both,

$$\operatorname{curl} F = \big(\partial_y(xy) - \partial_z(xz),\ \partial_z(yz) - \partial_x(xy),\ \partial_x(xz) - \partial_y(yz)\big) = (x-x,\ y-y,\ z-z) = 0,$$

$$\operatorname{curl} G = \big(\partial_y 0 - \partial_z(-x),\ \partial_z y - \partial_x 0,\ \partial_x(-x) - \partial_y y\big) = (0,\ 0,\ -2) \neq 0.$$

So G is not a gradient field. Note that this settles G and does **not** yet settle F: the test is necessary, not by itself sufficient.

(b) Integrating the first component in x suggests $f(x,y,z) := xyz$, and checking all three at once,

$$\nabla f = (yz,\ xz,\ xy) = F.$$

(c) Along $\gamma(t) = (\cos t, \sin t, 0)$ we have $\gamma'(t) = (-\sin t, \cos t, 0)$ and $G(\gamma(t)) = (\sin t,\, -\cos t,\, 0)$, so the integrand is $\langle G(\gamma), \gamma' \rangle = -\sin^2 t - \cos^2 t = -1$ and

$$\int_\gamma G \cdot d\vec{s} = \int_0^{2\pi} (-1)\, dt = -2\pi.$$

The curve is closed, $\gamma(2\pi) = \gamma(0)$, so a potential would force the integral to be $f(\gamma(2\pi)) - f(\gamma(0)) = 0$ by Proposition 26.b.14. Since $-2\pi \neq 0$, no potential exists, confirming part **(a)** without differentiating anything.

Remark 22.c.21 (*On $\mathbb{R}^3$ the curl test is not merely necessary*): Part **(a)** was careful to say that $\operatorname{curl} F = 0$ does not by itself produce a potential, and in general it does not. On the domain here it does: $\mathbb{R}^3$ is convex, hence simply connected, so Corollary 26.c.25 applies and every curl-free field on it is a gradient (Remark 26.c.27). The guess in part **(b)** was therefore guaranteed to succeed before it was made.

What the caution in **(a)** protects against is a curl-free field on a smaller domain. Delete the z-axis from $\mathbb{R}^3$ and there are curl-free fields with no potential, such as the vortex field $(-y,x,0)/(x^2+y^2)$, whose circulation around the axis is 2π; this is the point of Question C.b.21. The hypothesis that matters is a property of the domain, not of the curl.

Q.E.D.

Solution to Exercise 22.b.17:

(a) **True.** The chain rule applied twice: $\partial_i(e^w) = e^w \partial_i w$ with $w := u^2$, and $\partial_i(u^2) = 2u\,\partial_i u$.

(b) **True.** Both sides are vectors in $\mathbb{R}^n$, and the j-th component of the left-hand side is $\partial_i \partial_j u$ while that of the right-hand side is $\partial_j \partial_i u$. These agree by Theorem 11.b.13, which applies because u is smooth.

(c) **False.** The quotient rule reads $\partial_i(u/v) = (v\,\partial_i u - u\,\partial_i v)/v^2$, and the printed formula is not that. It is worth noting that the second summand is not even well defined at a point where $\partial_i v$ vanishes, which already settles the matter. For a counterexample where both sides do exist, take $n = 1$, $u \equiv 1$ and $v(x) := e^x$. Then $u/v = e^{-x}$, so the left-hand side is $-e^{-x}$, whereas the right-hand side is $0 + 1/e^x = e^{-x}$. The two differ in sign.

(d) **False**, and it fails before any computation: $\operatorname{Hess} u$ is an $n \times n$ matrix, so for $n \geq 2$ the right-hand side is not a real number, while the left-hand side is. What the product rule for the divergence actually gives is

$$\operatorname{div}(u\nabla u) = \nabla u \cdot \nabla u + u \operatorname{div}(\nabla u) = |\nabla u|^2 + u\,\Delta u,$$

and $\Delta u = \operatorname{Tr}(\operatorname{Hess} u)$ is the **trace** of the Hessian. Replacing $\operatorname{Hess} u$ by $\operatorname{Tr}(\operatorname{Hess} u)$ in the printed statement makes it true.

Remark 22.c.22 (*The identity in part (d) is worth keeping*): Read from right to left, $|\nabla u|^2 = \operatorname{div}(u\nabla u) - u\Delta u$ is the differential half of integration by parts: integrating it over a bounded C^1 domain Ω and applying Theorem 23.b.5 to the first term turns the left-hand side into a boundary integral minus $\int_\Omega u\Delta u$. That is Green's first identity, and it is how $|\nabla u|^2$ is traded for $u\Delta u$ throughout the theory of the Laplacian.

Q.E.D.

Solution to Exercise 22.b.18:

(a) **True.** The condition $\nabla f(x_0) \neq 0$ says exactly that $Df_{x_0} : \mathbb{R}^n \to \mathbb{R}$ is surjective, and it persists on a neighbourhood U of x_0 because ∇f is continuous. Theorem 17.a.3, applied to $f|_U$ and the value $f(x_0)$, then makes $\{x \in U \mid f(x) = f(x_0)\}$ a submanifold of dimension $n-1$. The restriction to a neighbourhood is not a technicality: the full level set $\{f = f(x_0)\}$ may contain critical points far from x_0 and be no manifold at all.

(b) **True**, and this is the statement worth remembering. Suppose such a γ existed. Integrating f along it and using the chain rule along a path (Section 10.d) together with the hypothesis $\gamma' = \nabla f \circ \gamma$,

$$f(\gamma(1)) - f(\gamma(0)) = \int_0^1 \big\langle \nabla f(\gamma(t)),\ \gamma'(t) \big\rangle\, dt = \int_0^1 |\gamma'(t)|^2\, dt.$$

The curve is regular, so γ' never vanishes and the integrand is continuous and strictly positive; the right-hand side is therefore strictly positive. But $\gamma(0) = \gamma(1)$ makes the left-hand side zero. No such curve exists.

(c) **False.** Take $f(x) := \arctan(x_1)$, so that $\nabla f(x) = \big(1/(1+x_1^2), 0, \dots, 0\big)$, which is nowhere zero, while $|f| < \pi/2$ everywhere. A non-vanishing gradient forces f to keep increasing in some direction, but says nothing about how fast, and a bounded function can increase forever.

(d) **True.** Since B is compact and f continuous, $f|_B$ attains its minimum at some $p \in B$. Suppose $p \in \partial B$. Moving inward from p means moving in the direction $-\nu(p)$, and the directional derivative there is

$$D_{-\nu(p)} f(p) = \big\langle \nabla f(p),\ -\nu(p) \big\rangle = -\big\langle \nabla f(p),\ \nu(p) \big\rangle < 0$$

by hypothesis. So $f(p - t\nu(p)) < f(p)$ for all small $t > 0$, and those points lie in B because ∂B is smooth. That contradicts minimality, so p lies in the interior, and a minimum over B attained at an interior point is in particular a local minimum.

Remark 22.c.23 (*Two of these are restrictions that only gradients obey*): Parts **(b)** and **(d)** both say a gradient field cannot do something, and both are proved the same way: integrate f rather than reason about ∇f directly. That is the whole advantage of being a gradient, and the contrast is sharp. A general vector field has closed orbits routinely ($F(x, y) := (-y, x)$ is the rotation field of Example A.b.4, whose orbits are circles), and part **(b)** says no gradient field ever does. In physical terms a conservative field cannot drive a cycle, because going round would return the potential to its starting value while the field did strictly positive work.

Q.E.D.

Proof of Exercise 22.b.19:

Write $g := A^{-1} \circ f \circ A$, so that $g(x) = A^{-1} f(Ax)$. A linear map is its own differential, so the chain rule gives

$$\mathrm{J}\, g(x) = A^{-1}\ \mathrm{J}\, f(Ax)\, A.$$

The divergence of a vector field is the trace of its Jacobian, $\operatorname{div} g = \sum_i \partial_i g_i = \operatorname{Tr}(\mathrm{J}\, g)$. Since $\operatorname{Tr}(MN) = \operatorname{Tr}(NM)$, taking $M := A^{-1}\, \mathrm{J}\, f(Ax)$ and $N := A$,

$$\operatorname{div} g(x) = \operatorname{Tr}\left(A^{-1}\ \mathrm{J}\, f(Ax)\, A\right) = \operatorname{Tr}\left(\mathrm{J}\, f(Ax)\right) = (\operatorname{div} f)(Ax).$$

Remark 22.c.24 (*What is really invariant, and how much of it*): The proof used nothing about $\mathrm{J}\, f$ beyond the fact that conjugation does not move the trace, and the same is true of every coefficient of the characteristic polynomial: $A^{-1}MA$ and M have the same one. So the argument proves more than it was asked to. Not only the divergence but the whole characteristic polynomial of the Jacobian is unchanged by a linear change of coordinates, and in particular so is $\det \mathrm{J}\, f$, which is the factor Theorem 20.a.1 integrates against.

That is worth knowing because the individual summands are **not** invariant. There is no sense in which $\partial_1 f_1$ survives a change of coordinates; it is only the sum that does, and the divergence is therefore a property of the vector field rather than of the axes chosen to write it in. A quantity defined by a formula in coordinates has to earn that status, and this exercise is the proof that the divergence has.

Q.E.D.

Solution of Exercise 22.b.20:

The vector field $V(x, y, z) = (x, y, z - x^2 - y^2)$ has divergence

$$\operatorname{div} V = \partial_x(x) + \partial_y(y) + \partial_z(z - x^2 - y^2) = 1 + 1 + 1 = 3.$$

S is not itself the boundary of a bounded C^1 domain, so close it up with the equatorial disk $D := \{(x, y, 0) : x^2 + y^2 \le 1\}$: then $\Omega := \{(x, y, z) \in B_1(0) : z \ge 0\}$, the upper half ball, is a bounded **piecewise** C^1 domain with $\partial\Omega = S \cup D$ (the two faces meet along the equator, a circle, which carries no vol_2-measure), and S's *"inside to outside"* normal agrees with the outward normal of Ω on S. By Theorem 23.b.7,

$$\int_S V \cdot \nu \, d\,\mathrm{vol}_2 + \int_D V \cdot \nu_D \, d\,\mathrm{vol}_2 = \int_\Omega \operatorname{div} V \, dx = 3\,\mathrm{vol}_3(\Omega) = 3 \cdot \frac{2\pi}{3} = 2\pi,$$

using $\mathrm{vol}_3(\Omega) = \frac{1}{2}\cdot\frac{4}{3}\pi = \frac{2\pi}{3}$. On D the outward normal is $\nu_D = -e_3$, and $V(x,y,0) = (x, y, -x^2 - y^2)$, so $V \cdot \nu_D = x^2 + y^2$; in polar coordinates,

$$\int_D V \cdot \nu_D \, d\,\mathrm{vol}_2 = \int_0^{2\pi}\int_0^1 r^2 \cdot r \, dr \, d\varphi = 2\pi \left[\frac{r^4}{4}\right]_0^1 = \frac{\pi}{2}.$$

Hence

$$\int_S V \cdot \nu \, d\,\mathrm{vol}_2 = 2\pi - \frac{\pi}{2} = \boxed{\frac{3\pi}{2}}.$$

Q.E.D.

Flux and the Divergence Theorem (Chapter 23)

The divergence theorem is the first of the integral theorems: the flux of a vector field out through the boundary of a domain equals the integral of its divergence inside. The chapter opens with the case $n = 1$, where the statement degenerates into the fundamental theorem of calculus and the mechanism is visible without any geometry, and then works up to the general theorem and its physical reading as a conservation law.

Gauss' theorem in low dimensions: the case $n = 1$ and the unit square (Section 23.a)

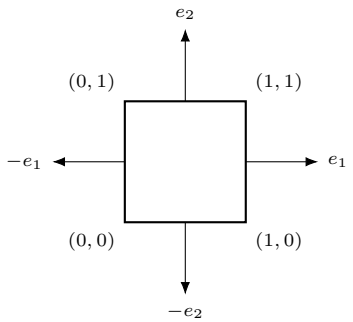
Gauss' theorem (Theorem 23.b.5), stated in Section 23.b below, is, in a precise sense, a higher-dimensional fundamental theorem of calculus: it trades a derivative integrated over a region for the original quantity evaluated on the region's boundary. Setting $n = 1$ makes the analogy literal. Let $f \in C^1([a,b])$, viewed as a (one-dimensional) vector field. Since $\operatorname{div} f = f'$ in dimension 1, Gauss' formula becomes

$$\int_a^b \frac{\partial f}{\partial x}\, dx = \int_{[a,b]} \operatorname{div} f \, d\operatorname{vol}_1 = \int_{\partial[a,b]} \langle f, \nu \rangle \, d\operatorname{vol}_0 = f(b) - f(a) \qquad \text{(FTC, almost)}.$$

The last step identifies $\int_{\partial[a,b]} \langle f, \nu \rangle \, d\operatorname{vol}_0$ with $f(b) - f(a)$ by treating $\partial[a,b] = \{a, b\}$ as a 0-dimensional submanifold with outward unit normals $\nu(a) = -1$, $\nu(b) = +1$, and vol_0 as counting measure.

AI-Note 23.1: The calculation has gaps: it needs a 0-dimensional Jordan measure and a 0-dimensional submanifold, which the definitions of this course do not provide, although they can be extended to include this case. Hence the *"almost"*. The point of the calculation is the analogy with the FTC, not a rigorous $n = 1$ instance of Theorem 23.b.5.

Example 23.a.1 (*The divergence theorem on the unit square, by hand*): Conversely, the fundamental theorem of calculus can be used to verify Gauss' theorem directly in the simplest two-dimensional case, the unit square. This is a preview of Theorem 23.b.5, which is stated in the next section; for $B = (0,1)^2$ the computation below proves it. Let $F : [0,1]^2 \to \mathbb{R}^2$ be a C^1 vector field and $B := (0,1)^2$. The boundary ∂B splits into four edges, with outward unit normals $-e_1$, e_1, $-e_2$, e_2 on the left, right, bottom, and top edges respectively.



Parametrizing each edge by $t \mapsto (0,t)$, $(1,t)$, $(t,0)$, $(t,1)$ for $t \in [0,1]$,

$$\int_{\partial B} \langle F, \nu \rangle \, d\operatorname{vol}_1 = \int_0^1 \langle F(0,t), -e_1 \rangle \, dt + \int_0^1 \langle F(1,t), e_1 \rangle \, dt + \int_0^1 \langle F(t,0), -e_2 \rangle \, dt + \int_0^1 \langle F(t,1), e_2 \rangle \, dt.$$

Grouping the first pair and the second pair,

$$\int_{\partial B} \langle F, \nu \rangle \, d\,\mathrm{vol}_1 = \int_0^1 \big[F_1(1,t) - F_1(0,t)\big]\, dt + \int_0^1 \big[F_2(t,1) - F_2(t,0)\big]\, dt.$$

By the fundamental theorem of calculus applied to $s \mapsto F_1(s,t)$ and to $s \mapsto F_2(t,s)$, each on $[0,1]$,

$$\int_{\partial B} \langle F, \nu \rangle \, d\,\mathrm{vol}_1 = \int_0^1 \left(\int_0^1 \partial_1 F_1(s,t)\, ds \right) dt + \int_0^1 \left(\int_0^1 \partial_2 F_2(t,s)\, ds \right) dt,$$

and by Fubini,

$$\int_{\partial B} \langle F, \nu \rangle \, d\,\mathrm{vol}_1 = \int_B \partial_1 F_1 \, d\,\mathrm{vol}_2 + \int_B \partial_2 F_2 \, d\,\mathrm{vol}_2 = \int_B \operatorname{div} F \, d\,\mathrm{vol}_2 \,.$$

Flux and the divergence theorem (Section 23.b)

Definition 23.b.2 (*Flux*): Suppose $F : U \to \mathbb{R}^n$ is a C^1 vector field, $U \subseteq \mathbb{R}^n$ open, and $\Omega \subseteq U$ is a C^1 bounded domain. The "*flux*" (German: „*Fluss*") of F through $\partial\Omega$ is defined as

$$\int_{\partial\Omega} \langle F, \nu \rangle \, d\,\mathrm{vol}_{n-1} \qquad \text{(analysis notation)} \quad = \quad \int_{\partial\Omega} \vec{F}(x) \cdot d\vec{A} \qquad \text{(physics notation)}$$

$$= \int_B \big\langle F \circ \phi(x),\ \nu \circ \phi(x) \big\rangle \sqrt{\det D\phi(x)^{\mathsf{T}} D\phi(x)}\, dx \qquad \text{(explicit)},$$

where $\nu : \partial\Omega \to \mathbb{S}^{n-1}$ is the exterior unit normal, i.e. $\nu(y)$ is the outward-facing unit normal of $\partial\Omega$ at $y \in \partial\Omega$, and $\phi : B \to \partial\Omega$ is a (local) parametrization of $\partial\Omega$. In general, multiple integrals in the explicit formulation may be necessary.

Proposition 23.b.3 (*Flux in $\mathbb{R}^3$*): Let $\Omega \subseteq \mathbb{R}^3$ be a bounded C^1 domain and $F \in C^1(\overline{\Omega}, \mathbb{R}^3)$ a vector field, and let $\phi : B \to \partial\Omega \setminus N$ be a parametrization of $\partial\Omega \setminus N$, where $N \subseteq \partial\Omega$ is a null set, $B \subseteq \mathbb{R}^2$. Then:

(a) $\nu(x) := \dfrac{\partial_1\phi(x) \times \partial_2\phi(x)}{|\partial_1\phi(x) \times \partial_2\phi(x)|}$ is an (interior or exterior) Gauss map (more precisely, $\nu \circ \phi^{-1}$ is a Gauss map, i.e. $\nu(x) \in N_{\phi(x)}(\partial\Omega \setminus N)$).

(b) The square root of the Gram determinant is given by $\sqrt{\det D\phi(x)^{\mathsf{T}} D\phi(x)} = |\partial_1\phi(x) \times \partial_2\phi(x)|$ (Example 21.a.5).

(c) Therefore,

$$\int_{\partial\Omega} \vec{F} \cdot d\vec{A} = \pm \int_B \big\langle F \circ \phi(x),\ \partial_1\phi(x) \times \partial_2\phi(x) \big\rangle \, dx,$$

where the sign depends on whether $\nu(x)$ is interior $(-)$ or exterior $(+)$.

Example 23.b.4 (*Flux through the unit sphere*): Compute the flux of $F(x,y,z) := (xz,\, z,\, y)$ through the surface of $\mathbb{S}^2$ with interior $B_1(0)$.

We use spherical coordinates with $r \equiv 1$:

$$\phi : (0, 2\pi) \times (0, \pi) \to \mathbb{R}^3, \qquad (\varphi, \theta) \mapsto \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}.$$

Normal vector.

$$\partial_\varphi\phi = \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix}, \qquad \partial_\theta\phi = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix} \quad \Longrightarrow \quad \partial_\varphi\phi \times \partial_\theta\phi = \begin{pmatrix} -\sin^2\theta\cos\varphi \\ -\sin^2\theta\sin\varphi \\ -\sin\theta\cos\theta \end{pmatrix}.$$

Sanity check: does this make sense, and is it exterior or interior? We have

$$\partial_\varphi\phi \times \partial_\theta\phi = -\sin\theta\ \phi(\varphi,\theta) = -\sin\theta\,(x,y,z),$$

and the exterior normal of $\mathbb{S}^2$ is $+(x,y,z)$, so $\partial_\varphi\phi \times \partial_\theta\phi$ is an **interior** normal (for $\theta \in (0,\pi)$, $\sin\theta > 0$). So we multiply by -1 to switch the order in the cross product, matching the exterior sign in Proposition 23.b.3.

The integral.

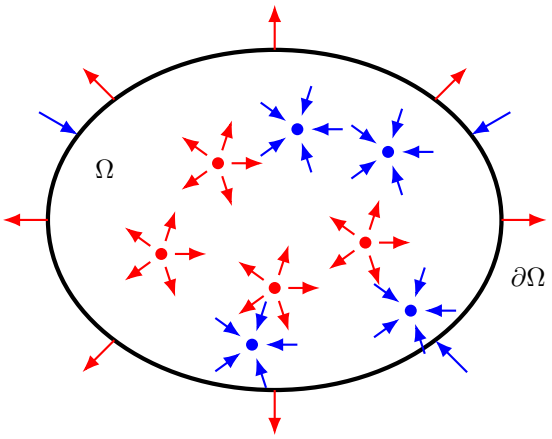
$$\begin{aligned}\int_{\mathbb{S}^2} \vec{F}\cdot d\vec{A} &= \int_0^{2\pi}\int_0^{\pi} \begin{pmatrix} xz \\ z \\ y \end{pmatrix} \cdot \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix} \sin\theta\, d\theta\, d\varphi \\ &= \int_0^{2\pi}\int_0^{\pi} \Big[\sin^3\theta\cos\theta\cos^2\varphi + \sin^2\theta\cos\theta\sin\varphi + \sin^2\theta\cos\theta\sin\varphi\Big]\, d\theta\, d\varphi \\ &= \pi\int_0^{\pi} \sin^3\theta\cos\theta\, d\theta = \pi\left[\frac{\sin^4\theta}{4}\right]_0^{\pi} = 0,\end{aligned}$$

using $\int_0^{2\pi}\cos^2\varphi\, d\varphi = \pi$ and $\int_0^{2\pi}\sin\varphi\, d\varphi = 0$ for the middle two terms.

Theorem 23.b.5 (*Gauss*): Let $\Omega \subseteq \mathbb{R}^n$ be a bounded C^1 domain and $F : \overline{\Omega} \to \mathbb{R}^n$ a C^1 vector field. Then

$$\int_\Omega \operatorname{div} F\, d\operatorname{vol}_n = \int_{\partial\Omega} \langle F,\nu\rangle\, d\operatorname{vol}_{n-1}, \qquad \text{or, in physics notation,} \qquad \int_\Omega \vec{\nabla}\cdot\vec{F}\, dx = \int_{\partial\Omega}\vec{F}\cdot d\vec{A}.$$

Remark 23.b.6 (*Where the theorem comes from*): The theorem goes back to Lagrange (discovered 1762) and Ostrogradsky (proven 1828). It reduces a volume integral of a derivative to a boundary integral of the original field, which is the shape of the fundamental theorem of calculus. In the picture, what the sources inside Ω emit and the sinks absorb is exactly what crosses $\partial\Omega$.



Theorem 23.b.7 (*Divergence theorem on a piecewise C^1 domain*): Let $\Omega \subseteq \mathbb{R}^n$ be a bounded open set whose boundary can be written as a finite union

$$\partial\Omega = S_1 \cup \cdots \cup S_N$$

of C^1 hypersurfaces meeting only along a set of vol_{n-1}-measure zero, and let $F : \overline{\Omega} \to \mathbb{R}^n$ be a C^1 vector field. Then the conclusion of Theorem 23.b.5 still holds,

$$\int_\Omega \operatorname{div} F\, d\operatorname{vol}_n = \int_{\partial\Omega}\langle F,\nu\rangle\, d\operatorname{vol}_{n-1} = \sum_{i=1}^{N}\int_{S_i}\langle F,\nu\rangle\, d\operatorname{vol}_{n-1},$$

where the exterior unit normal ν is defined at every boundary point off the exceptional set.

AI-Note 23.2: Theorem 23.b.7 is stated here without proof, as the course states it: the August 2024 exam grants it explicitly, telling candidates they may use the divergence theorem in a piecewise smooth domain without proof. It is worth having written down, because this document was already

using it in five places. A cube has edges and corners and is not a C^1 domain, so Example 23.b.10 below needs this version and not Theorem 23.b.5; so does Exercise 23.b.19, whose region has an edge where the paraboloid meets its lid. The same goes for Example 23.b.11, a solid cylinder with two edge circles; for Section 22.c, an upper half ball whose hemisphere meets its equatorial disc at a right angle; and for Section 23.c, capped on the rim $x^2 + y^2 = 1$.

The reason it is true is worth a sentence even without a proof. Cut Ω along finitely many surfaces into pieces that **are** C^1 domains and apply Theorem 23.b.5 to each. Every interior cut is traversed by exactly two pieces, whose exterior normals there point in opposite directions, so those contributions cancel in pairs and only $\partial\Omega$ survives. The edges themselves contribute nothing because they carry no $(n-1)$-dimensional volume, which is the same reason the normal being undefined along them does no harm.

Example 23.b.8 (*Recomputing Example 23.b.4 via the divergence theorem*): With $\Omega = B_1(0)$ and $F(x,y,z) = (xz, z, y)$ as before, $\operatorname{div} F = \partial_x(xz) + \partial_y(z) + \partial_z(y) = z + 0 + 0 = z$. So, in spherical coordinates ($z = r\cos\theta$),

$$\int_{\partial\Omega} \vec{F}\cdot d\vec{A} = \int_\Omega \operatorname{div} F\, dx = \int_\Omega z\, dx = \int_0^1\int_0^{2\pi}\int_0^{\pi} \underbrace{\cos\theta\sin\theta}_{=\frac{1}{2}\sin(2\theta),\ \pi\text{-periodic}} r^3\, d\theta\, d\varphi\, dr = 0,$$

since $\int_0^\pi \sin(2\theta)\, d\theta = 0$. This confirms Example 23.b.4 with none of its cross-product bookkeeping.

Remark 23.b.9 (*Why the divergence theorem saves work*): The two computations of the same flux differ in character: the direct definition needs an explicit Gauss map, an orientation sanity check, and three cross terms in the integrand, while the divergence theorem needs only $\operatorname{div} F$ and the spherical coordinates already used in Example 20.a.6. This is the theorem's main practical use, and it is not conceptual elegance but the plain fact that $\operatorname{div} F$ is often far simpler than F restricted to a curved boundary.

Example 23.b.10 (*Computing flux using the divergence theorem*): Let $\Omega := [-1,1]^3 \subset \mathbb{R}^3$ and $F(x,y,z) := (x^2, y^2, z^2)^\mathsf{T}$. We compute the flux of F out of Ω using the divergence theorem. First, we find the divergence of F:

$$\operatorname{div} F(x,y,z) = \partial_x(x^2) + \partial_y(y^2) + \partial_z(z^2) = 2x + 2y + 2z.$$

Then, integrating over Ω:

$$\begin{aligned}\int_{\partial\Omega} \langle F(p),\ \nu(p)\rangle \operatorname*{dvol}_2(p) &= \int_\Omega \operatorname{div} F(x,y,z)\operatorname*{dvol}_3(x,y,z)\\ &= \int_{-1}^1\int_{-1}^1\int_{-1}^1 (2x+2y+2z)\, dx\, dy\, dz.\end{aligned}$$

By symmetry, the integral of $2x$ over $[-1,1]$ is 0, and similarly for y and z. Therefore, the total flux is 0.

Example 23.b.11 (*Flux out of a cylinder*): Let $\Omega := \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 \le R^2,\ 0 \le z \le H\}$ be a cylinder of radius R and height H. Consider the vector field $F(x,y,z) := (x^3 + \sin(z), y^3 + xz, z^3 + xy)^\mathsf{T}$. We compute the outward flux of F through $\partial\Omega$ using the divergence theorem in the piecewise form Theorem 23.b.7, needed here because the two rim circles where the lateral surface meets the lids are edges, so Ω is not a C^1 domain. First, we compute the divergence of F:

$$\operatorname{div} F(x,y,z) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2).$$

By the divergence theorem, the flux is the integral of the divergence over Ω. We compute this using

cylindrical coordinates (r, θ, z):

$$\begin{aligned}\int_{\partial\Omega} \langle F(p),\ \nu(p)\rangle \, d\underset{2}{\mathrm{vol}}(p) &= \int_{\Omega} 3(x^2+y^2+z^2)\, d\underset{3}{\mathrm{vol}}(x,y,z) \\ &= \int_0^H \int_0^{2\pi} \int_0^R 3(r^2+z^2) r \, dr \, d\theta \, dz \\ &= 2\pi \int_0^H \left[\frac{3}{4}r^4 + \frac{3}{2}r^2 z^2\right]_0^R dz \\ &= 2\pi \int_0^H \left(\frac{3}{4}R^4 + \frac{3}{2}R^2 z^2\right) dz \\ &= 2\pi \left[\frac{3}{4}R^4 z + \frac{1}{2}R^2 z^3\right]_0^H \\ &= 2\pi \left(\frac{3}{4}R^4 H + \frac{1}{2}R^2 H^3\right) = \frac{\pi}{2} R^2 H (3R^2 + 2H^2).\end{aligned}$$

Example 23.b.12 (*Archimedes' principle*): A solid $\Omega \subset \mathbb{R}^3$ (a bounded C^1 domain) is fully submerged in water, with the coordinates chosen so that $x_3 = 0$ is the water's surface. At depth $-x_3$ the water pressure is $P(x) := -\rho g x_3$, where ρ is the density of water and g the gravitational acceleration. The force the water exerts on a surface element near $q \in \partial\Omega$ is proportional to the pressure and points inward, along $-\nu(q)$, so the total buoyant force is

$$\vec{F} := -\int_{\partial\Omega} P\, \nu \, dS.$$

Substituting the pressure and applying the divergence theorem to each component,

$$F_i = \int_{\partial\Omega} \rho g x_3 \nu_i \, dS = \rho g \int_{\partial\Omega} \langle x_3 e_i, \nu\rangle \, dS = \rho g \int_{\Omega} \mathrm{div}(x_3 e_i)\, dV.$$

Since $\mathrm{div}(x_3 e_i) = \partial_i x_3$ equals 1 if $i = 3$ and 0 otherwise, this gives $F_1 = F_2 = 0$ and

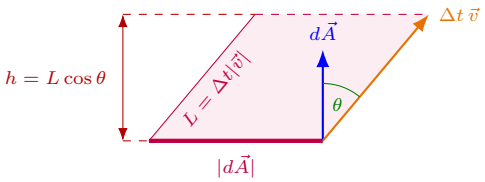
$$F_3 = \rho g \int_{\Omega} 1 \, dV = \rho g \ \mathrm{vol}_3(\Omega).$$

The buoyant force is vertical, with magnitude $\rho g \,\mathrm{vol}_3(\Omega)$, which is exactly the weight of the water displaced by Ω, confirming Archimedes' principle. The divergence theorem converts a surface integral of an unknown, position-dependent pressure into a volume integral that only sees the shape of Ω through its volume.

Gauss' theorem is also a powerful tool in physics. Consider a fluid of density ϱ moving with velocity $\vec{v}$, and a surface element $d\vec{A}$ whose normal makes an angle θ with $\vec{v}$. The fluid crossing that element in time Δt is exactly the fluid filling the column of slant length $L = \Delta t\, |\vec{v}|$ erected on it along $\vec{v}$, and that column stands at perpendicular height $L\cos\theta$. Its volume is therefore

$$V = |d\vec{A}|\, L \cos\theta = \Delta t\, \vec{v} \cdot d\vec{A},$$

and the mass of fluid carried through the element is ϱV.



Therefore, the fluid flowing out of any bounded C^1 domain Ω per unit time is

$$\int_{\partial\Omega} \varrho \vec{v} \cdot d\vec{A} = \int_{\Omega} \vec{\nabla} \cdot (\varrho \vec{v})\, dx.$$

Of course, since the total amount of fluid is conserved, this is also the change in the amount of fluid in Ω over time:

$$\int_\Omega \vec{\nabla}\cdot(\varrho\vec{v})\,dx = -\frac{d}{dt}\int_\Omega \varrho\,dx = -\int_\Omega \frac{\partial\varrho}{\partial t}\,dx.$$

Since this holds for any Ω, the "*continuity equation*" (German: „*Kontinuitätsgleichung*“) follows:

$$\vec{\nabla}\cdot(\varrho\vec{v}) + \frac{\partial\varrho}{\partial t} = 0.$$

Exercise 23.b.13 (*Radial vector fields* (important)): For $\alpha\in\mathbb{R}$ consider the vector field

$$V_\alpha(x) = |x|^\alpha x, \qquad x\in\mathbb{R}^n\setminus\{0\},$$

where $|x|$ is the usual Euclidean norm.

(a) Compute $\operatorname{div} V_\alpha(x)$ for all $x\neq 0$, and show that it vanishes identically for $\alpha=-n$.

(b) Using the Divergence theorem on V_0, show the following relation between the n-volume of a sphere and the $n-1$ volume of its boundary:

$$r\ \operatorname{vol}_{n-1}(\partial B_r) = n\ \operatorname{vol}_n(B_r), \qquad \forall r>0.$$

(c) Show that

$$\int_{\partial B_1} V_{-n}\cdot\nu\,d\operatorname{vol}_{n-1} > 0, \qquad \int_{B_1}\operatorname{div} V_{-n}(x)\,dx = 0.$$

Why doesn't this contradict the Divergence Theorem?

Info: *This exercise is Problem 11.5 of Problem Sheet 11, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 357*

AI-Exercise 23.b.14 (*Flux of a cubic field through the sphere*): Let $F(x,y,z) := (x^3,\,y^3,\,z^3)$ and let $\Omega := B_1(0)\subseteq\mathbb{R}^3$. Compute the flux of F out of $\partial\Omega$.

(a) Say in one sentence what the integrand of Definition 23.b.2 would look like here if the sphere were parametrized directly, and why that route is unattractive.

(b) Compute $\operatorname{div} F$, and observe that it depends on the point only through its distance r from the origin.

(c) Evaluate the volume integral. The shell reading of polar coordinates in Remark 20.a.7 does it without a Jacobian determinant.

Solution: *See solution on Page 358*

AI-Exercise 23.b.15 (*An open surface has to be closed first*): Let $S := \{(x,y,z)\in\mathbb{R}^3 : z = 1-x^2-y^2,\ z\geq 0\}$ be the piece of the paraboloid lying above the xy-plane, oriented by the upward-pointing normal, and let $F(x,y,z) := (x,\,y,\,z)$.

(a) Theorem 23.b.5 applies to the boundary of a domain, and S is not one. Which set D must be adjoined to S so that $S\cup D = \partial\Omega$ for a bounded domain Ω, and what is the exterior unit normal along D?

(b) Compute $\int_{\partial\Omega}\langle F,\nu\rangle\,d\operatorname{vol}_2$ using the divergence theorem. You will need $\operatorname{vol}_3(\Omega)$, which is a one-line computation in cylindrical coordinates.

(c) Compute the flux through D alone, directly from Definition 23.b.2, and deduce the flux of F through S.

Solution: *See solution on Page 358*

Exercise 23.b.16 (*Flux of a field involving a curl across an ellipsoid*): For given positive numbers a,b,c, consider the solid ellipsoid

$$U := \left\{(x,y,z)\in\mathbb{R}^3 \;\middle|\; \left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2+\left(\frac{z}{c}\right)^2\leq 1\right\},$$

and the vector fields

$$A(x,y,z) := (\cosh(x), \sinh(x), e^{-\pi(y^2+z^2)})^{\mathsf{T}},$$
$$B(x,y,z) := (x - \sinh(y), y + \cosh(x), z + e^{-\pi(x^2+y^2)})^{\mathsf{T}} + \operatorname{curl} A.$$

Compute the outer flow of B across the boundary of U (take the normal vector to ∂U to point outside of U).

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Short Problem 1, part 2.*

Solution: *See solution on Page 359*

Exercise 23.b.17 (*Gauss' law: outer flux of an inverse-square field across an ellipsoid*): Let $a, b, c > 0$ and consider the solid ellipsoid

$$U := \left\{ (x,y,z) \in \mathbb{R}^3 \;\middle|\; \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 \le 1 \right\},$$

together with the electric point-charge field $E : \mathbb{R}^3 \setminus \{0\} \to \mathbb{R}^3$,

$$E(x,y,z) := \frac{(x,y,z)^{\mathsf{T}}}{(x^2+y^2+z^2)^{3/2}}.$$

(a) Verify that $\operatorname{div} E(x,y,z) = 0$ for all $(x,y,z) \neq (0,0,0)$.

(b) Compute the outer flux of E through the boundary ∂U of the ellipsoid.

Official hint: *In order to exploit the symmetries of the field E, consider the flow of E across the boundary of the domain $U \setminus B_r(0)$, where $0 < r < \min\{a,b,c\}$.*

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Short Problem 1, part 3. Part **(a)** is added here as a step; the original asks only for the flux.*

Solution: *See solution on Page 359*

Exercise 23.b.18 (*Tuning a parameter to prescribe a flux*): Let $U \subseteq \mathbb{R}^2$ be a smooth bounded domain with $\operatorname{vol}_2(U) = 1$, and consider the vector field

$$F_\alpha(x,y) := \big(3y^2 + \alpha x,\ 2x - y\big), \qquad \alpha \in \mathbb{R}.$$

Determine the value of α for which the outer flux of F_α across ∂U equals 1.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Box Problem 3, where only the final answer was graded.*

Solution: *See solution on Page 360*

Exercise 23.b.19 (*Outer flux across a truncated paraboloid*): Consider the region, the surface and the two vector fields

$$U := \{(x,y,z) \in \mathbb{R}^3 \mid 1 + x^2 + y^2 < z,\ z < 5\},$$
$$M := \{(x,y,z) \in \mathbb{R}^3 \mid 1 + x^2 + y^2 = z,\ z < 5\},$$
$$E(x,y,z) := (xz,\ 2y - yz,\ x^2 + y^2)^{\mathsf{T}}, \qquad A(x,y,z) := (-ye^z,\ xe^z,\ e^z)^{\mathsf{T}}.$$

(a) Sketch M in 3D perspective and label the axes.

(b) Compute the outer flux of E across M, taking the normal vector to M that points out of U.

(c) Compute the outer flux of $\operatorname{curl} A$ across M, with the same choice of normal.

Official hint: *If needed, you may use the divergence theorem in a piecewise smooth domain without proof.*

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), Short Problem 2.*

Solution: *See solution on Page 360*

Exercise 23.b.20 (*A family of integrals that does not depend on its parameter*): Let $B_1(0) = \{x \in \mathbb{R}^3 \mid |x| \le 1\}$ be the closed unit ball. Show that for every continuously differentiable vector field v on $\mathbb{R}^3$, the quantity

$$I_n := \int_{B_1(0)} \operatorname{div}\left(\left(x_1^2 + x_2^2 + x_3^2\right)^n v(x)\right) d\,\mathrm{vol}_3(x)$$

does not depend on n: that is, $I_n = I_m$ for all $n, m \in \mathbb{Z}_{\ge 0}$.

Info: *This exercise is taken from the Analysis I & II exam of 9 August 2021 (Prof. Giovanni Felder), Part A, Problem 10, which opens by asking for a statement of the divergence theorem on $B_1(0)$.*

Solution: *See solution on Page 360*

Exercise 23.b.21 (*A divergence-free field that dies at the boundary*): Let $B := \{x \in \mathbb{R}^3 : |x| \le 1\}$ be the closed unit ball and let $K = (K_1, K_2, K_3)$ be a smooth vector field on $\mathbb{R}^3$ which vanishes outside B and satisfies $\operatorname{div} K = 0$ on all of $\mathbb{R}^3$.

(a) Let f be a smooth real-valued function on $\mathbb{R}^3$. Show that

$$\int_B \langle \nabla f, K\rangle \, d\,\mathrm{vol}_3 = 0.$$

(b) Deduce that $\displaystyle\int_B K_1 \, d\,\mathrm{vol}_3 = 0$.

Info: *This exercise is taken from the Analysis I & II examination of winter 2016 (Prof. H. Knörrer), Problem 3.*

Solution: *See solution on Page 362*

Example 23.b.22 (*The flux through a closed surface sees only the oscillation*): Let $U \subseteq \mathbb{R}^3$ be an open neighbourhood of the closed unit ball $\overline{B_1(0)}$ and let $f : U \to \mathbb{R}^3$ be a continuously differentiable vector field.

Claim. There is a constant $C > 0$, independent of f, with

$$\left|\int_{\partial B_1(0)} \langle f, \nu\rangle \, d\,\mathrm{vol}_2\right| \le C \sup_{x,y \in \overline{B_1(0)}} |f(x) - f(y)|.$$

The right-hand side does not change when a constant vector is added to f, so any proof must produce a left-hand side with the same property. That observation is the entire argument, because a constant field is invisible to the flux: if $c \in \mathbb{R}^3$ is constant then $\operatorname{div} c = 0$, and Theorem 23.b.5 gives

$$\int_{\partial B_1(0)} \langle c, \nu\rangle \, d\,\mathrm{vol}_2 = \int_{B_1(0)} \operatorname{div} c \, d\,\mathrm{vol}_3 = 0.$$

So fix any $y_0 \in \overline{B_1(0)}$, put $c := f(y_0)$, and subtract:

$$\int_{\partial B_1(0)} \langle f, \nu\rangle \, d\,\mathrm{vol}_2 = \int_{\partial B_1(0)} \langle f - c,\ \nu\rangle \, d\,\mathrm{vol}_2 \,.$$

Now estimate crudely. Since $|\nu| = 1$, the Cauchy–Schwarz inequality bounds the integrand by $|f(x) - f(y_0)|$, and every such value is at most the supremum in the claim. Therefore

$$\left|\int_{\partial B_1(0)} \langle f, \nu\rangle \, d\,\mathrm{vol}_2\right| \le \int_{\partial B_1(0)} |f(x) - f(y_0)| \, d\,\mathrm{vol}_2(x) \le \mathrm{vol}_2\left(\partial B_1(0)\right) \cdot \sup_{x,y \in \overline{B_1(0)}} |f(x) - f(y)|.$$

The surface area of the unit sphere is 4π, so $C := 4\pi$ works, and it depends on nothing but the domain.

Remark 23.b.23 (*Why the estimate has to look like this*): A bound by $\sup|f|$ would be easy and useless: it is false that the flux is small when f is small in a relative sense, and true but uninformative that it is bounded by $4\pi\sup|f|$. The point of the stated inequality is the invariance. Adding a constant wind to a fluid changes $\sup|f|$ arbitrarily while changing the net flux through a closed surface not at all, so the correct majorant is a quantity blind to constants: the oscillation.

Once that is seen, the proof writes itself, and this is the general shape: to prove an estimate invariant under some family of perturbations, first normalise using the invariance, then be as crude as you like. Note also how little was used. The divergence theorem was applied only to the constant field c, so the hypothesis that f is C^1 is never used; continuity, which makes the flux integral exist, is all the proof needs. No derivative of f appears in the final bound, and the constant 4π is just the area of the sphere. The same statement holds for any bounded C^1 domain, with C the area of its boundary.

> **Info:** *This example is taken from the Analysis I & II examination of summer 2018 (Prof. Manfred Einsiedler), Problem 9, where it is set as a proof exercise. The closing remark is added here.*

Exercise 23.b.24 (*Green's first identity, and what vanishing data forces*): Recall that the "*Laplacian*" (German: „*Laplace-Operator*“) of a C^2 function u is $\Delta u := \operatorname{div}(\nabla u) = \sum_i \partial_i^2 u$, and that u is called "*harmonic*" (German: „*harmonisch*“) if $\Delta u = 0$.

Let $U \subseteq \mathbb{R}^3$ be open, let $B \subset U$ be a compact, connected, smoothly bounded region with outward unit normal field $\nu : \partial B \to \mathbb{R}^3$, and let $u : U \to \mathbb{R}$ be harmonic. Write $\partial_\nu u := \langle \nabla u, \nu \rangle$ for the "*outward normal derivative*" (German: „*Aussennormalableitung*“) of u on ∂B.

(a) Show that

$$\int_B |\nabla u|^2 \, d\operatorname{vol}_3 = \int_{\partial B} u \, \partial_\nu u \, d\operatorname{vol}_2 .$$

(b) Deduce that u is constant on B if either $u|_{\partial B} = 0$ or $\partial_\nu u = 0$.

> **Info:** *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *(undated) in* `old_exams/fs2023/`*, Problem 12. That paper names no lecturer.*

Solution: *See solution on Page 363*

Exercise 23.b.25 (*The mean value property and maximum principle for harmonic functions*): Let $\Omega \subseteq \mathbb{R}^n$ $(n \geq 2)$ be an open connected set, and let $u \in C^2(\Omega)$ be harmonic, so that $\Delta u = \operatorname{div}(\nabla u) = 0$. For a point $x_0 \in \Omega$ and radius $R > 0$ such that $\overline{B_R(x_0)} \subset \Omega$, define for $r \in (0, R]$ the spherical average

$$\phi(r) := \frac{1}{\omega_{n-1} r^{n-1}} \int_{\partial B_r(x_0)} u(x) \, d\operatorname{vol}_{n-1}(x),$$

where $\omega_{n-1} r^{n-1} = \operatorname{vol}_{n-1}(\partial B_r(x_0))$ is the surface area of the $(n-1)$-sphere.

(a) By writing $\phi(r) = \frac{1}{\omega_{n-1}} \int_{\partial B_1(0)} u(x_0 + r\xi) \, d\operatorname{vol}_{n-1}(\xi)$ and differentiating with respect to r, apply Gauss's divergence theorem on $B_r(x_0)$ to show that $\phi'(r) = 0$ for all $r \in (0, R]$.

(b) Conclude that $\phi(r) = u(x_0)$ for all $r \in (0, R]$ (the spherical "*mean value property*"), and deduce the solid ball mean value property:

$$u(x_0) = \frac{1}{\alpha_n r^n} \int_{B_r(x_0)} u(x) \, d\operatorname{vol}_n(x),$$

where $\alpha_n r^n = \operatorname{vol}_n(B_r(x_0))$.

(c) Prove the "*strong maximum principle*": if u attains its minimum (or its maximum) over Ω at some point $x_0 \in \Omega$, then u is constant on Ω.

Info: *This exercise is taken from the examination of 18 August 2023 (Prof. Urs Lang), Problem 3.A1 (c).*

Solution: *See solution on Page 363*

Exercise 23.b.26 (*True or False* (fluxes through a family of spheres)): Let $U := \mathbb{R}^3 \setminus \{0\}$, let $f : U \to \mathbb{R}^3$ be a continuously differentiable vector field, and for $r > 0$ write

$$I_r := \int_{\partial B_r(0)} \langle f, \nu \rangle \, d\,\mathrm{vol}_2$$

for the flux through the sphere of radius r with respect to the outward normal. Decide whether each of the following is true or false, with justification.

(a) If f is divergence free, then $I_r = 0$ for every $r > 0$.
(b) If f is divergence free, then I_r does not depend on r.
(c) If f is curl free, then I_r does not depend on r.
(d) If $f = \operatorname{curl} g$ for some vector field g, then $I_r = 0$ for every $r > 0$.

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *of 31 January 2023, Problem 6. That paper names no lecturer.*

Solution: *See solution on Page 365*

Solutions (Section 23.c)

Solution of Exercise 23.b.13:

(a) Writing $V_\alpha^i(x) = |x|^\alpha x_i$,

$$\partial_i\big(|x|^\alpha x_i\big) = \alpha |x|^{\alpha-2} x_i^2 + |x|^\alpha \qquad \text{(no sum)},$$

so, summing over $i = 1, \dots, n$,

$$\operatorname{div} V_\alpha(x) = \sum_{i=1}^n \Big(\alpha |x|^{\alpha-2} x_i^2 + |x|^\alpha\Big) = \alpha |x|^{\alpha-2} |x|^2 + n|x|^\alpha = (\alpha + n)|x|^\alpha.$$

Since $|x|^\alpha \neq 0$ for $x \neq 0$, this vanishes identically exactly when $\alpha = -n$.

(b) For $\alpha = 0$, $V_0(x) = x$ and, by part **(a)**, $\operatorname{div} V_0 \equiv n$. Applying Theorem 23.b.5 on B_r,

$$\int_{B_r} \operatorname{div} V_0 \, dx = n \, \mathrm{vol}_n(B_r).$$

On ∂B_r the outward unit normal is $\nu(x) = x/r$, so $V_0(x) \cdot \nu(x) = x \cdot x / r = |x|^2 / r = r$ (using $|x| = r$ on ∂B_r), giving

$$\int_{\partial B_r} V_0 \cdot \nu \, d\,\mathrm{vol}_{n-1} = r \, \mathrm{vol}_{n-1}(\partial B_r).$$

Equating the two expressions of $\int_{B_r} \operatorname{div} V_0 \, dx = \int_{\partial B_r} V_0 \cdot \nu \, d\,\mathrm{vol}_{n-1}$ gives $r \, \mathrm{vol}_{n-1}(\partial B_r) = n \, \mathrm{vol}_n(B_r)$.

(c) By part **(a)**, $\operatorname{div} V_{-n}(x) = 0$ for every $x \neq 0$, so $\int_{B_1} \operatorname{div} V_{-n}(x) \, dx = 0$ trivially (the integrand vanishes on $B_1 \setminus \{0\}$, a set of full measure in B_1). On ∂B_1, $\nu(x) = x$ and $|x| = 1$, so $V_{-n}(x) = |x|^{-n} x = x$, giving

$$\int_{\partial B_1} V_{-n} \cdot \nu \, d\,\mathrm{vol}_{n-1} = \int_{\partial B_1} x \cdot x \, d\,\mathrm{vol}_{n-1} = \int_{\partial B_1} 1 \, d\,\mathrm{vol}_{n-1} = \mathrm{vol}_{n-1}(\partial B_1) > 0.$$

This does not contradict Theorem 23.b.5: the theorem requires F to be C^1 on all of $\overline{B_1}$, but $V_{-n}(x) = |x|^{-n} x$ is not even defined, let alone continuously differentiable, at $x = 0 \in \overline{B_1}$. Its hypotheses simply do not hold on this domain.

Q.E.D.

Solution to AI-Exercise 23.b.14:

(a) On the unit sphere the exterior unit normal is $\nu(x,y,z) = (x,y,z)$, so the integrand of Definition 23.b.2 is $\langle F, \nu\rangle = x^4 + y^4 + z^4$. In the spherical parametrization that becomes $\sin^4\theta\cos^4\varphi + \sin^4\theta\sin^4\varphi + \cos^4\theta$, multiplied by the surface element $\sin\theta$: a fourth-power trigonometric integral in two variables, where the divergence theorem offers a radial integral in one.

(b) Writing r for the distance to the origin, $\operatorname{div} F = 3x^2 + 3y^2 + 3z^2 = 3r^2$, which is radial.

(c) Because the integrand is radial, Remark 20.a.7 applies with $n = 3$ and $C_3 = 4\pi$:

$$\int_{\partial\Omega} \langle F, \nu\rangle \, d\operatorname{vol}_2 = \int_{B_1(0)} 3r^2 \, d\operatorname{vol}_3 = 4\pi \int_0^1 3r^2 \cdot r^2 \, dr = 12\pi \int_0^1 r^4 \, dr = \frac{12\pi}{5}.$$

Remark 23.c.27 (*The direct route, for once, is also available*): The surface integral abandoned in part **(a)** is not actually intractable, and working it out is a worthwhile check. By the symmetry of the sphere under permuting the coordinates, the three terms of $x^4 + y^4 + z^4$ contribute equally, so

$$\int_{\mathbb{S}^2} \left(x^4 + y^4 + z^4\right) d\operatorname{vol}_2 = 3\int_{\mathbb{S}^2} z^4 \, d\operatorname{vol}_2 = 3\int_0^{2\pi}\int_0^{\pi} \cos^4\theta \, \sin\theta \, d\theta \, d\varphi = 6\pi \left[-\frac{\cos^5\theta}{5}\right]_0^{\pi} = \frac{12\pi}{5},$$

in agreement. Note that what made this bearable was a symmetry argument, applied **before** any trigonometry: expanding all three fourth powers first is the version that is unpleasant.

Q.E.D.

Solution to AI-Exercise 23.b.15:

(a) Take $\Omega := \{(x,y,z) : x^2 + y^2 < 1,\ 0 < z < 1 - x^2 - y^2\}$, the solid region between the paraboloid and the plane $z = 0$. Its boundary is S together with the flat disk

$$D := \{(x,y,0) : x^2 + y^2 \le 1\},$$

whose exterior unit normal is $\nu \equiv -e_3$, since Ω lies above it. On S the exterior normal is the upward-pointing one, which is the orientation the exercise already fixes, so no sign has to be corrected later.

(b) Here $\operatorname{div} F = 1 + 1 + 1 = 3$, so the flux out of $\partial\Omega$ is three times a volume. In cylindrical coordinates,

$$\operatorname{vol}_3(\Omega) = \int_0^{2\pi}\int_0^1\int_0^{1-r^2} r \, dz \, dr \, d\varphi = 2\pi \int_0^1 \left(r - r^3\right) dr = 2\pi\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{\pi}{2},$$

and therefore

$$\int_{\partial\Omega} \langle F, \nu\rangle \, d\operatorname{vol}_2 = 3\,\operatorname{vol}_3(\Omega) = \frac{3\pi}{2}.$$

(c) Along D the third coordinate is $z = 0$, so $\langle F, \nu\rangle = \langle (x,y,0), -e_3\rangle = -z = 0$ at every point: the field is tangent to the cap, and contributes nothing. Splitting the boundary integral,

$$\int_S \langle F, \nu\rangle \, d\operatorname{vol}_2 = \int_{\partial\Omega} \langle F, \nu\rangle \, d\operatorname{vol}_2 - \int_D \langle F, \nu\rangle \, d\operatorname{vol}_2 = \frac{3\pi}{2} - 0 = \frac{3\pi}{2}.$$

Remark 23.c.28 (*The cap vanished because of this field, not because it is a cap*): It is tempting to read part **(c)** as licence to ignore the lid. Change F to $\widetilde{F}(x,y,z) := (x,\ y,\ z+1)$ and nothing about the first two parts moves, since $\operatorname{div}\widetilde{F} = 3$ as before and the closed flux is still $3\pi/2$. But now $\langle \widetilde{F}, \nu\rangle = -1$ on D, so the cap carries a flux of $-\pi$, and the flux through the paraboloid is $3\pi/2 + \pi = 5\pi/2$ instead. The subtraction in part **(c)** is doing real work; it merely had nothing to subtract.

Exercise 22.b.20 is the same technique on a hemisphere, and the two are worth reading as a pair: there the cap contributes $\pi/2$ and must be subtracted, here it contributes nothing. The method is identical and only the arithmetic differs, which is the point.

Q.E.D.

Solution to Exercise 23.b.16:

Both fields are C^1 on all of $\mathbb{R}^3$, so the divergence theorem (Theorem 23.b.5) converts the flux into a volume integral:

$$\int_{\partial U} \langle B, \nu \rangle \, d\,\mathrm{vol}_2 = \int_U \operatorname{div} B \, d\,\mathrm{vol}_3 \,.$$

The summand $\operatorname{curl} A$ contributes nothing, since the divergence of a curl vanishes identically. Only the first summand has to be differentiated:

$$\begin{aligned} \operatorname{div} B &= \partial_x\big(x - \sinh y\big) + \partial_y\big(y + \cosh x\big) + \partial_z\big(z + e^{-\pi(x^2+y^2)}\big) \\ &= 1 + 1 + 1 = 3. \end{aligned}$$

The flux is therefore $3\,\mathrm{vol}_3(U)$, and the solid ellipsoid with semi-axes a, b, c has volume $\frac{4}{3}\pi abc$. Consequently, the outer flow of B across ∂U is

$$3 \cdot \tfrac{4}{3}\pi abc = 4\pi abc.$$

Remark 23.c.29: Every intimidating term in B is decoration. Each of $\sinh y$, $\cosh x$ and $e^{-\pi(x^2+y^2)}$ depends only on the variables the matching partial derivative does not differentiate, so all three vanish, and $\operatorname{curl} A$ is invisible to the divergence whatever A is. What survives is the radial field (x, y, z), whose divergence is 3.

Q.E.D.

Proof of Exercise 23.b.17:

(a) Write $r := |x| = \sqrt{x^2 + y^2 + z^2}$, so that $E(x) = x/r^3$. Since $\partial_i r = x_i/r$, the product rule gives

$$\operatorname{div} E = \sum_{i=1}^{3} \partial_i \left(x_i r^{-3}\right) = \sum_{i=1}^{3} \left(r^{-3} - 3x_i r^{-4}\frac{x_i}{r}\right) = 3r^{-3} - 3r^{-5}|x|^2 = 0 \qquad (r > 0).$$

(b) Choose $r_0 > 0$ small enough that $\overline{B_{r_0}(0)} \subseteq U$, and set $\Omega := U \setminus \overline{B_{r_0}(0)}$, so that $\partial\Omega = \partial U \cup \partial B_{r_0}(0)$. The point 0 has been removed, so E is smooth on $\overline{\Omega}$ and part **(a)** applies throughout Ω. The divergence theorem (Theorem 23.b.5) therefore gives

$$0 = \int_\Omega \operatorname{div} E \, d\,\mathrm{vol}_3 = \int_{\partial U} \langle E, \nu \rangle \, d\,\mathrm{vol}_2 + \int_{\partial B_{r_0}(0)} \langle E, \nu_{\mathrm{in}} \rangle \, d\,\mathrm{vol}_2,$$

where on the inner sphere the outward normal **of** Ω points towards the origin, $\nu_{\mathrm{in}} = -x/r_0$. Consequently

$$\int_{\partial U} \langle E, \nu \rangle \, d\,\mathrm{vol}_2 = \int_{\partial B_{r_0}(0)} \left\langle E, \frac{x}{r_0} \right\rangle d\,\mathrm{vol}_2 \,.$$

On $\partial B_{r_0}(0)$ we have $E(x) = x/r_0^3$, so the integrand is the constant $|x|^2/r_0^4 = r_0^2/r_0^4 = r_0^{-2}$, and

$$\int_{\partial U} \langle E, \nu \rangle \, d\,\mathrm{vol}_2 = \frac{1}{r_0^2}\,\mathrm{vol}_2(\partial B_{r_0}(0)) = \frac{4\pi r_0^2}{r_0^2} = 4\pi.$$

Remark 23.c.30: The answer does not depend on a, b, c, nor for that matter on the shape of U at all, only on whether it encloses the origin. That is Gauss' law: the flux counts the charge

inside, and nothing else. The r_0 in the argument is a device that never appears in the answer, which is the usual sign that the quantity computed is a topological one.

Q.E.D.

Solution of Exercise 23.b.18:

The field is polynomial, hence C^1, and U is a smooth bounded domain, so the divergence theorem (Theorem 23.b.5) converts the flux into a volume integral. The divergence is a constant:

$$\operatorname{div} F_\alpha = \partial_x\big(3y^2 + \alpha x\big) + \partial_y\big(2x - y\big) = \alpha - 1.$$

Therefore

$$\int_{\partial U} \langle F_\alpha, \nu\rangle \, d\,\mathrm{vol}_1 = \int_U \operatorname{div} F_\alpha \, d\,\mathrm{vol}_2 = (\alpha - 1)\ \mathrm{vol}_2(U) = \alpha - 1,$$

using $\mathrm{vol}_2(U) = 1$. Setting this equal to 1 gives $\boxed{\alpha = 2}$.

Remark 23.c.31: The terms $3y^2$ and $2x$ are decoration: they contribute nothing to the divergence, so nothing to the flux, however violently they stir the field around inside U. Only the parts of F_α that create or destroy field lines (here αx and $-y$) can be seen from the boundary. Note also that the answer is independent of the shape of U and uses only its area.

Q.E.D.

Proof of Exercise 23.b.20:

Write $w_n(x) := \big(|x|^2\big)^n = (x_1^2 + x_2^2 + x_3^2)^n$. This is a polynomial, so $w_n v$ is a C^1 vector field on $\mathbb{R}^3$ and the divergence theorem (Theorem 23.b.5) applies on the unit ball:

$$I_n = \int_{B_1(0)} \operatorname{div}\big(w_n v\big)\, d\,\mathrm{vol}_3 = \int_{\partial B_1(0)} \big\langle w_n v, \nu\big\rangle \, d\,\mathrm{vol}_2\,.$$

The whole point is now visible: the integration has been moved to the unit sphere, and there $|x| = 1$, so $w_n(x) = 1^n = 1$ for **every** n. Hence

$$I_n = \int_{\partial B_1(0)} \langle v, \nu\rangle \, d\,\mathrm{vol}_2,$$

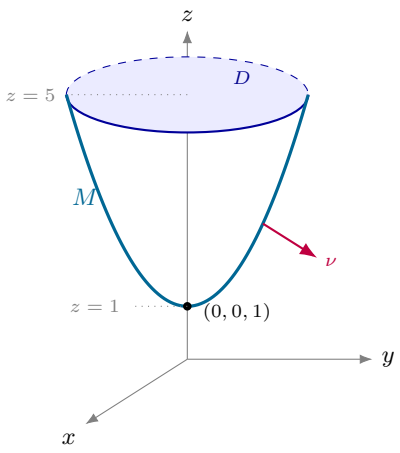
an expression in which n no longer appears. Since this holds for every $n \in \mathbb{Z}_{\geq 0}$, we get $I_n = I_m$ for all n, m, all of them equal to the flux of v itself through the unit sphere.

AI-Note 23.3: It is worth seeing how much work is avoided. Expanding $\operatorname{div}(w_n v) = w_n \operatorname{div} v + \langle \nabla w_n, v\rangle$ with $\nabla w_n = 2n|x|^{2n-2}x$ gives a genuinely messy integrand whose n-dependence is not remotely obvious, and no amount of manipulating it inside the ball will make the cancellation appear. The divergence theorem removes the difficulty by moving to a surface on which the offending factor is identically 1. The radius 1 is doing real work here: on $\partial B_R(0)$ the same argument would give $I_n = R^{2n}\int\langle v, \nu\rangle$, which does depend on n unless $R = 1$.

Q.E.D.

Solution of Exercise 23.b.19:

(a) M is the paraboloid of revolution $z = 1 + x^2 + y^2$, opening upwards, with vertex at $(0, 0, 1)$ and cut off at height $z = 5$. The cut happens where $x^2 + y^2 = 4$, so the rim is the circle of radius 2 at height 5.



(b) The field E has

$$\operatorname{div} E = \partial_x(xz) + \partial_y(2y - yz) + \partial_z(x^2 + y^2) = z + (2 - z) + 0 = 2.$$

The boundary of U consists of M together with the flat lid $D := \{(x, y, 5) \mid x^2 + y^2 < 4\}$, and the divergence theorem in its piecewise form (Theorem 23.b.7) gives

$$\int_M \langle E, \nu \rangle \, d\,\mathrm{vol}_2 + \int_D \langle E, \nu \rangle \, d\,\mathrm{vol}_2 = \int_U \operatorname{div} E \, d\,\mathrm{vol}_3 = 2\,\mathrm{vol}_3(U).$$

In cylindrical coordinates the region lies over the disk of radius 2, between the paraboloid and the plane $z = 5$, so

$$\mathrm{vol}_3(U) = \int_0^{2\pi}\int_0^2 \big(5 - (1 + r^2)\big)\, r \, dr \, d\theta = 2\pi \int_0^2 (4 - r^2)\, r \, dr = 2\pi \Big[2r^2 - \tfrac{r^4}{4}\Big]_0^2 = 2\pi(8 - 4) = 8\pi,$$

and therefore the total outer flux across ∂U is 16π.

On D the exterior normal is $\nu = e_3$, so the integrand is the third component of E, namely $x^2 + y^2$, evaluated at $z = 5$:

$$\int_D \langle E, \nu \rangle \, d\,\mathrm{vol}_2 = \int_0^{2\pi}\int_0^2 r^2 \cdot r \, dr \, d\theta = 2\pi \Big[\tfrac{r^4}{4}\Big]_0^2 = 8\pi.$$

Subtracting,

$$\boxed{\int_M \langle E, \nu \rangle \, d\,\mathrm{vol}_2 = 16\pi - 8\pi = 8\pi.}$$

(c) First compute the curl, using the formula of Definition 22.b.9:

$$\operatorname{curl} A = \big(\partial_y A_3 - \partial_z A_2,\ \partial_z A_1 - \partial_x A_3,\ \partial_x A_2 - \partial_y A_1\big)^{\mathsf{T}} = \big(-xe^z,\ -ye^z,\ 2e^z\big)^{\mathsf{T}},$$

since $\partial_y A_3 = 0$ and $\partial_z A_2 = xe^z$; $\partial_z A_1 = -ye^z$ and $\partial_x A_3 = 0$; and $\partial_x A_2 = e^z$ against $\partial_y A_1 = -e^z$.

A curl is divergence-free, and here that is visible directly: $\operatorname{div}(\operatorname{curl} A) = -e^z - e^z + 2e^z = 0$. Consequently the divergence theorem gives **zero** total flux across ∂U, so the flux through M is the negative of the flux through D. On D the normal is again e_3 and $z = 5$ throughout, so the integrand is the constant $2e^5$ and

$$\int_D \langle \operatorname{curl} A, \nu \rangle \, d\,\mathrm{vol}_2 = 2e^5 \cdot \mathrm{vol}_2(D) = 2e^5 \cdot \pi \cdot 2^2 = 8\pi e^5.$$

Therefore

$$\boxed{\int_M \langle \operatorname{curl} A, \nu \rangle \, d\,\mathrm{vol}_2 = -8\pi e^5.}$$

AI-Note 23.4: The version of the divergence theorem this needs is Theorem 23.b.7, not Theorem 23.b.5: the region U has an edge along the rim circle $\{x^2 + y^2 = 4,\ z = 5\}$, where the

paraboloid meets the lid at a nonzero angle and the exterior normal jumps, so it is not a C^1 domain. That is precisely what the exam's hint licenses, by allowing the divergence theorem on a piecewise smooth domain without proof. Nothing in the computation is affected, because the rim carries no vol_2.

Remark 23.c.32 (*Checking part (c) against Stokes*): Part **(c)** never used the shape of M, only that M and D share a boundary, which is a sign that Stokes' theorem should give the same answer directly. The rim is $c(t) := (2\cos t,\ 2\sin t,\ 5)$, and on it

$$A(c(t)) = \big(-2\sin t\, e^5,\ 2\cos t\, e^5,\ e^5\big), \qquad \dot c(t) = (-2\sin t,\ 2\cos t,\ 0),$$

so $\langle A(c), \dot c\rangle = 4e^5(\sin^2 t + \cos^2 t) = 4e^5$ and $\oint_c A\cdot dc = 8\pi e^5$. That is the flux of $\operatorname{curl} A$ through any surface spanning the rim whose orientation induces **this** traversal, which for the lid means $\nu = e_3$: it matches the $8\pi e^5$ computed above. The exterior normal on M points downwards and outwards, inducing the opposite traversal, and the sign flips to $-8\pi e^5$.

The same shortcut does not apply to part **(b)**, since E is not a curl, and one really does have to compute a volume there.

Q.E.D.

Solution to Exercise 23.b.21:

(a) The product rule for the divergence of a scalar multiple of a vector field reads

$$\operatorname{div}(fK) = \langle \nabla f, K\rangle + f \operatorname{div} K.$$

The hypothesis $\operatorname{div} K = 0$ kills the second term, so $\langle \nabla f, K\rangle = \operatorname{div}(fK)$ everywhere. The field fK is smooth, so Theorem 23.b.5 applies on the unit ball and gives

$$\int_B \langle \nabla f, K\rangle \, d\,\mathrm{vol}_3 = \int_B \operatorname{div}(fK)\, d\,\mathrm{vol}_3 = \int_{\partial B} f\,\langle K, \nu\rangle \, d\,\mathrm{vol}_2\,.$$

It remains to see that the boundary integral vanishes. K vanishes on $\mathbb{R}^3 \setminus B$, and every point of ∂B is a limit of points outside B, for instance $x = \lim_{t\downarrow 1} tx$ for $|x| = 1$. Since K is continuous, $K(x) = \lim_{t\downarrow 1} K(tx) = 0$ for every $x \in \partial B$. The integrand is therefore identically zero on ∂B, and $\int_B \langle \nabla f, K\rangle \, d\,\mathrm{vol}_3 = 0$.

(b) Apply **(a)** to the smooth function $f(x) := x_1$, whose gradient is the constant field $\nabla f = e_1$. Then $\langle \nabla f, K\rangle = \langle e_1, K\rangle = K_1$, so

$$\int_B K_1 \, d\,\mathrm{vol}_3 = \int_B \langle \nabla f, K\rangle \, d\,\mathrm{vol}_3 = 0.$$

Q.E.D.

Remark 23.c.33 (*The hypotheses are used one each, and the second part is free*): Part **(a)** spends $\operatorname{div} K = 0$ on turning the integrand into a divergence, and spends the vanishing of K outside B on killing the boundary term. Neither hypothesis does anything else, and dropping either breaks the statement: without $\operatorname{div} K = 0$ a term $\int_B f \operatorname{div} K$ survives, and without the support condition the flux of fK across the sphere need not vanish.

Part **(b)** is the reason to prove **(a)** in the generality of an arbitrary f. Taking $f(x) = x_1$ is a device: the coordinate function is the cheapest smooth function whose gradient is a fixed basis vector, so a statement about all $\langle \nabla f, K\rangle$ specialises to one about a single component of K. Running the same trick on x_2 and x_3 gives $\int_B K = 0$ as a vector identity, which is the real content: **a divergence-free field with compact support has zero mean.** That is a standard fact about incompressible flows and is worth remembering in that form.

Proof of Exercise 23.b.24:

(a) Apply Theorem 23.b.5 to the vector field $F := u\nabla u$, which is C^1 on U because u is C^2. Its divergence is computed by the product rule,

$$\operatorname{div}(u\nabla u) = \langle \nabla u, \nabla u\rangle + u\,\operatorname{div}(\nabla u) = |\nabla u|^2 + u\,\Delta u = |\nabla u|^2,$$

the last step because u is harmonic. On the boundary, $\langle F, \nu\rangle = u\,\langle \nabla u, \nu\rangle = u\,\partial_\nu u$. The divergence theorem therefore reads

$$\int_B |\nabla u|^2\, d\,\mathrm{vol}_3 = \int_B \operatorname{div} F\, d\,\mathrm{vol}_3 = \int_{\partial B} \langle F, \nu\rangle\, d\,\mathrm{vol}_2 = \int_{\partial B} u\,\partial_\nu u\, d\,\mathrm{vol}_2\,.$$

(b) Either hypothesis makes the integrand $u\,\partial_\nu u$ vanish identically on ∂B, so part **(a)** gives

$$\int_B |\nabla u|^2\, d\,\mathrm{vol}_3 = 0.$$

The integrand is continuous and non-negative, so it vanishes identically on B: if $|\nabla u(p)|^2 = c > 0$ at some p in the interior, continuity would give a ball $B_r(p) \subseteq B$ on which the integrand exceeds $c/2$, forcing the integral to be at least $\frac{c}{2}\,\mathrm{vol}_3(B_r(p)) > 0$; and the boundary contributes nothing to a volume integral. Hence $\nabla u \equiv 0$ on B.

A function with vanishing gradient on a connected set is constant there. Concretely, join two points p, q of B by a C^1 path γ inside B and differentiate along it (Section 10.d):

$$u(q) - u(p) = \int_0^1 \frac{d}{dt} u(\gamma(t))\, dt = \int_0^1 \big\langle \nabla u(\gamma(t)),\ \gamma'(t)\big\rangle\, dt = 0.$$

Remark 23.c.34 (*What this buys: uniqueness for the two boundary value problems*): Read the two hypotheses of part **(b)** as data prescribed on ∂B and the statement becomes a uniqueness theorem. Suppose u_1 and u_2 are harmonic on U and agree on ∂B. Their difference $u := u_1 - u_2$ is harmonic with $u|_{\partial B} = 0$, so **(b)** makes it constant, and the constant is 0 because it vanishes on the boundary: **a harmonic function is determined on B by its boundary values**. That is uniqueness for the Dirichlet problem. The second hypothesis gives the same for the Neumann problem, where the normal derivative is prescribed instead, except that there the conclusion is weaker by exactly one constant, and necessarily so, since adding a constant changes neither Δu nor $\partial_\nu u$.

The mechanism is worth naming. Part **(a)** is "*Green's first identity*", and it is integration by parts: the product rule for $\operatorname{div}(u\nabla u)$, which is Exercise 22.b.17**(d)**, trades a derivative from one factor to the other at the cost of a boundary term. The harmonicity is what kills the term that would otherwise survive, and the sign is what makes the argument work at all: $|\nabla u|^2$ has one sign, so a vanishing integral forces a vanishing integrand, which no indefinite quantity would allow.

Q.E.D.

Solution to Exercise 23.b.25:

(a) Change variables on the sphere via $x = x_0 + r\xi$ with $\xi \in \partial B_1(0)$. Then

$$\phi(r) = \frac{1}{\omega_{n-1}} \int_{\partial B_1(0)} u(x_0 + r\xi)\, d\,\mathrm{vol}_{n-1}(\xi).$$

Since u is C^2 and the sphere is compact, we may differentiate under the integral sign:

$$\phi'(r) = \frac{1}{\omega_{n-1}} \int_{\partial B_1(0)} \langle \nabla u(x_0 + r\xi),\, \xi\rangle\, d\,\mathrm{vol}_{n-1}(\xi).$$

Re-scaling back to $\partial B_r(x_0)$, where $\xi = \frac{x - x_0}{r} = \nu(x)$ is the outward unit normal,

$$\phi'(r) = \frac{1}{\omega_{n-1} r^{n-1}} \int_{\partial B_r(x_0)} \langle \nabla u(x),\, \nu(x)\rangle\, d\,\mathrm{vol}_{n-1}(x).$$

By Gauss's divergence theorem (Theorem 23.b.5) applied to the vector field $F = \nabla u$ on the ball $B_r(x_0)$:

$$\int_{\partial B_r(x_0)} \langle \nabla u, \nu \rangle \, d\,\mathrm{vol}_{n-1} = \int_{B_r(x_0)} \operatorname{div}(\nabla u) \, d\,\mathrm{vol}_n = \int_{B_r(x_0)} \Delta u \, d\,\mathrm{vol}_n = 0,$$

because u is harmonic ($\Delta u \equiv 0$). Therefore $\phi'(r) = 0$ for all $r \in (0, R]$.

(b) Since $\phi'(r) = 0$ on $(0, R]$, $\phi(r)$ is constant. By continuity of u at x_0,

$$\lim_{r \downarrow 0} \phi(r) = \lim_{r \downarrow 0} \frac{1}{\omega_{n-1} r^{n-1}} \int_{\partial B_r(x_0)} u(x) \, d\,\mathrm{vol}_{n-1}(x) = u(x_0).$$

Thus $\phi(r) = u(x_0)$ for every $r \in (0, R]$. This establishes the spherical mean value property.

To obtain the volume mean value property, integrate the spherical identity in polar/spherical coordinates. Spherical slicing (Fubini) writes an integral over the ball as an integral of its sphere integrals, and part **(b)** evaluates each of those: since $\phi(\rho) = u(x_0)$, the sphere $\partial B_\rho(x_0)$ contributes $\omega_{n-1}\rho^{n-1}u(x_0)$. Hence

$$\begin{aligned} \int_{B_r(x_0)} u(x) \, d\,\mathrm{vol}_n(x) &= \int_0^r \left(\int_{\partial B_\rho(x_0)} u \, d\,\mathrm{vol}_{n-1} \right) d\rho \\ &= \int_0^r \omega_{n-1}\rho^{n-1} u(x_0) \, d\rho = u(x_0) \frac{\omega_{n-1} r^n}{n}. \end{aligned}$$

Note that no extra factor of ρ^{n-1} enters: the surface measure $d\,\mathrm{vol}_{n-1}$ on $\partial B_\rho(x_0)$ already carries it, which is exactly why the sphere integral above is proportional to ρ^{n-1} rather than constant. Since $\alpha_n = \omega_{n-1}/n$ is the volume of the unit ball, this yields

$$u(x_0) = \frac{1}{\alpha_n r^n} \int_{B_r(x_0)} u(x) \, d\,\mathrm{vol}_n(x).$$

(c) Suppose u attains its minimum over Ω at x_0, and write $m := u(x_0)$, so that $u \geq m$ on all of Ω. Let

$$S := \{x \in \Omega \mid u(x) = m\}.$$

- S is non-empty since $x_0 \in S$.
- S is closed in Ω by continuity of u, since $S = u^{-1}(\{m\})$.
- S is open. Let $y \in S$ and choose $R > 0$ with $\overline{B_R(y)} \subset \Omega$. For every $0 < r < R$, the solid mean value property of part **(b)**, applied at y, gives

$$0 = \frac{1}{\alpha_n r^n} \int_{B_r(y)} \big(u(x) - m\big) \, d\,\mathrm{vol}_n(x).$$

  The integrand $u - m$ is continuous, and it is non-negative because m is the minimum of u over all of Ω. If it were strictly positive at some point $x_1 \in B_r(y)$, by continuity it would remain strictly positive on a small ball around x_1, forcing the integral to be strictly positive. Hence $u \equiv m$ on $B_r(y)$ for every $r < R$, so $B_R(y) \subseteq S$.

Since Ω is connected, the only non-empty clopen subset of Ω is Ω itself. Therefore $S = \Omega$, which proves that u is constant on Ω. The case of a maximum follows by replacing u with $-u$.

Remark 23.c.35 (*Why part (c) asks for an extremum over all of Ω*): The openness step is where this is spent: a point $y \in S$ shares the value m, and $u - m \geq 0$ near y only because m is the minimum of u over all of Ω. If x_0 is merely a local minimum, the same ball argument still shows that u is constant on a ball around x_0. At a point $y \in S$ far from x_0, however, nothing prevents u from taking values below m nearby, so the argument cannot be repeated there.

The conclusion is nevertheless true for a local extremum too. Harmonic functions are real analytic, so a harmonic function that is constant on a ball is constant on the whole connected set Ω. That fact is not developed in this course, which is why part **(c)** is stated for the extremum over Ω, the form in which the strong maximum principle is usually given.

Q.E.D.

Solution to Exercise 23.b.26:

(a) **False**, and the counterexample is the one field this situation was invented for. Take the inverse-square field

$$f(x) := \frac{x}{|x|^3}, \qquad x \in U.$$

It is divergence free on U:

$$\operatorname{div} f = \sum_i \partial_i\big(x_i|x|^{-3}\big) = \sum_i \Big(|x|^{-3} - 3x_i^2|x|^{-5}\Big) = 3|x|^{-3} - 3|x|^{-3} = 0.$$

But on $\partial B_r(0)$ the outward normal is $\nu = x/r$, so $\langle f, \nu\rangle = |x|^2/r^4 = 1/r^2$, a constant, and

$$I_r = \frac{1}{r^2}\operatorname{vol}_2\big(\partial B_r(0)\big) = \frac{4\pi r^2}{r^2} = 4\pi \neq 0.$$

(b) **True**, and note that the field above already illustrates it: $I_r = 4\pi$ for every r. In general, fix $0 < r_1 < r_2$ and integrate over the shell $A := \{x \mid r_1 \le |x| \le r_2\}$, which is a compact smoothly bounded region contained in U. This is the step the puncture at the origin does not obstruct, and it is why the hypothesis is enough. Its boundary is the two spheres, and the outward normal of A on the inner one is $-\nu$, pointing towards the origin. So Theorem 23.b.5 gives

$$0 = \int_A \operatorname{div} f \, d\operatorname{vol}_3 = I_{r_2} - I_{r_1}.$$

(c) **False.** Take $f(x) := x$, which has $\operatorname{curl} f = 0$ since it is the gradient of $|x|^2/2$. Here $\langle f, \nu\rangle = \langle x, x/r\rangle = r$ on $\partial B_r(0)$, so

$$I_r = r \cdot 4\pi r^2 = 4\pi r^3,$$

which depends on r as much as anything could. Being curl free constrains circulation, not flux, and the two conditions answer different questions.

(d) **True.** By Proposition 22.b.12 the field $\operatorname{curl} g$ is divergence free, so part **(b)** already makes I_r independent of r. That alone does not give the value 0, as part **(a)** shows, and it cannot be obtained by letting $r \to 0$, since g need not extend across the origin. What settles it is that a sphere has no boundary. Cut $\partial B_r(0)$ along its equator C into the upper and lower caps $S^{\pm}$. Applying Theorem 25.c.15 to each,

$$\int_{S^+} \langle \operatorname{curl} g, \nu\rangle \, d\operatorname{vol}_2 = \oint_C g \cdot d\gamma, \qquad \int_{S^-} \langle \operatorname{curl} g, \nu\rangle \, d\operatorname{vol}_2 = -\oint_C g \cdot d\gamma,$$

the sign differing because the outward normal orients the equator oppositely as the boundary of the two caps. Adding them gives $I_r = 0$.

Remark 23.c.36 (*Divergence free is weaker than being a curl, and this is what measures the gap*)**:** Parts **(a)**, **(b)** and **(d)** together say something sharper than any of them says alone. On $U = \mathbb{R}^3 \setminus \{0\}$ every divergence-free field has a flux independent of r, every field of the form $\operatorname{curl} g$ has that constant equal to zero, and the inverse-square field has it equal to 4π. Therefore

$$\frac{x}{|x|^3} \text{ is divergence free on } U \quad \text{but is } \textbf{not} \ \operatorname{curl} g \text{ for any } g \text{ on } U.$$

The number 4π is an obstruction, and it is detecting the hole: on a ball, or on any star-shaped domain, no such field exists and every divergence-free field is a curl. That is the degree-two case of the Poincaré lemma in its star-shaped form (Remark 26.c.27, stated there without proof), whose degree-one case Example 26.c.28 exhibits with the same domain punctured along a line instead of a point.

In physics the same computation is Gauss's law, and 4π is why the charge enclosed can be read off a flux at any radius whatever.

Q.E.D.

Alternating Forms, the Wedge Product and the Exterior Derivative

(Chapter 24)

Gradient, divergence and curl look like three unrelated operators that happen to satisfy two unrelated identities ($\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$). This chapter builds the machinery that makes them one operator satisfying one identity. Antisymmetric k-linear forms and the wedge product supply the objects; the exterior derivative d supplies the operator; and $d \circ d = 0$ supplies the identity.

Antisymmetric k-linear forms and the wedge product (Section 24.a)

AI-Note 24.1 (*Motivation*): Suppose one studies a submanifold $\mathcal{M} \subseteq \mathbb{R}^n$ via a parametrization $\phi : D \to \mathcal{M}$, and integrates some quantity over $\mathcal{M}$ (for instance the Gauss curvature, as in the Gauss–Bonnet theorem). The natural question is whether the resulting number is a property of ϕ, or a property of $\mathcal{M}$ itself: if it is invariant under reparametrization, it must be the latter. "*Differential forms*" are built to satisfy exactly this property. For a diffeomorphism $\Psi : V \to D$ with $\det D\Psi > 0$ and a form ω,

$$\begin{aligned}\int_{\mathcal{M}} \omega &= \int_D \omega_{\phi(x)}(D\phi_x)\, dx \\ &\overset{\text{change of var.}}{=} \int_V \omega_{\phi(\Psi(y))}(D\phi_{\Psi(y)})|\det D\Psi_y|\, dy \\ &= \int_V \omega_{(\phi\circ\Psi)(y)}(D(\phi\circ\Psi)_y)\, dy = \int_{\mathcal{M}} \omega,\end{aligned}$$

where the middle equality uses the chain rule $D(\phi\circ\Psi)_y = D\phi_{\Psi(y)}\, D\Psi_y$ together with multilinearity and $\det D\Psi_y = |\det D\Psi_y|$. This calculation is precisely why ω must be built from multilinear, antisymmetric maps: multilinearity lets the Jacobian $D\Psi_y$ factor out as $\det D\Psi_y$, matching the $|\det D\Psi_y|$ produced by the change-of-variables formula, exactly when $\det D\Psi_y > 0$.

Definition 24.a.1 (*Antisymmetric k-linear form*): Let V be an n-dimensional real vector space and let $L : V^k \to \mathbb{R}$. Then L is an "*antisymmetric k-linear form*" (German: „*antisymmetrische k-lineare Form*“) if

(a) (k-linear): L is linear in every entry:

$$L(v_1, \dots, \alpha v_i + \beta w_i, \dots, v_k) = \alpha L(v_1, \dots, v_i, \dots, v_k) + \beta L(v_1, \dots, w_i, \dots, v_k);$$

(b) (antisymmetric): swapping any two entries negates the value:

$$L(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = -L(v_1, \dots, v_j, \dots, v_i, \dots, v_k).$$

Remark 24.a.2: For $k = n$ and $V = \mathbb{R}^n$, the normalization $L(e_1, \dots, e_n) = 1$ pins down a **unique** such L, namely $L = \det$.

Notation 24.a.3: For $v \in V$ and a fixed basis $(e_1, \dots, e_n)$ of V, superscripts index the coordinates of a single vector, $v = \sum_{i=1}^n v^i e_i = (v^1, \dots, v^n)$, while subscripts index a list of vectors $v_1, \dots, v_k \in V$. This distinguishes "*the i-th coordinate of v*" from "*the i-th vector in the list*". From here on, $V = \mathbb{R}^n$ with the standard basis.

AI-Example 24.a.4 (*Two families of examples*):

(a) Fix $A \in \operatorname{Mat}_{k\times n}(\mathbb{R})$. For $v_1, \dots, v_k \in \mathbb{R}^n$, write $V := (v_1 \mid \cdots \mid v_k) \in \operatorname{Mat}_{n\times k}(\mathbb{R})$ for the matrix with these columns; then $AV \in \operatorname{Mat}_{k\times k}(\mathbb{R})$, and $L(v_1, \dots, v_k) := \det(AV)$ is an antisymmetric k-linear form, since $\det$ is multilinear and antisymmetric in the columns of its argument, and $v \mapsto Av$ is linear.

(b) For $k=1$, $L:\mathbb{R}^n\to\mathbb{R}$ is a "*covector*", $L\in(\mathbb{R}^n)^*$. The standard basis $(e_1^*,\dots,e_n^*)$ of $(\mathbb{R}^n)^*$, satisfying $e_i^*(e_j)=\delta_{ij}$, consists of 1-forms; in particular $e_i^*(v)=v^i$.

Remark 24.a.5 (*An equivalent characterization*): In the lecture, antisymmetry was replaced by the equivalent condition

$$L(VA)=L(V)\det(A),\qquad V:=(v_1,\dots,v_k)\in\mathrm{Mat}_{n\times k}(\mathbb{R}),\quad A\in\mathrm{Mat}_{k\times k}(\mathbb{R}),$$

writing $L(V)$ for $L(v_1,\dots,v_k)$. One direction is immediate: taking for A the matrix that swaps two columns, with $\det A=-1$, gives back antisymmetry. We verify the other in the special case $n=3$, $k=2$: let $V=(v\mid w)\in\mathrm{Mat}_{3\times2}(\mathbb{R})$ and $A=\begin{pmatrix}a&b\\c&d\end{pmatrix}$, so that $VA=(av+cw\mid bv+dw)$. Rewriting $L(VA)$ in bilinear form and using $L(v,v)=-L(v,v)=0$ (from antisymmetry) together with $L(w,v)=-L(v,w)$,

$$\begin{aligned}L(VA)&=L(av+cw,\,bv+dw)\\&=ab\,L(v,v)+ad\,L(v,w)+cb\,L(w,v)+cd\,L(w,w)\\&=(ad-bc)\,L(v,w)=L(V)\det(A).\end{aligned}$$

Definition 24.a.6 (*Wedge product*): For a k-form α and an m-form β on $\mathbb{R}^n$, the "*wedge product*" (German: „*Dachprodukt*“) $\alpha\wedge\beta$ is the $(k+m)$-form

$$(\alpha\wedge\beta)(v_1,\dots,v_k,v_{k+1},\dots,v_{k+m})=\sum_{\sigma\in S_{k,m}}\operatorname{sgn}(\sigma)\,\alpha(v_{\sigma(1)},\dots,v_{\sigma(k)})\,\beta(v_{\sigma(k+1)},\dots,v_{\sigma(k+m)}),$$

where $S_{k,m}$ is the set of permutations σ of $\{1,\dots,k+m\}$ satisfying $\sigma(1)<\dots<\sigma(k)$ and $\sigma(k+1)<\dots<\sigma(k+m)$ (the "*(k,m)-shuffles*").

The formula is rarely needed in practice. The following properties of $\wedge$ are the ones actually used:

(a) $\wedge$ is bilinear.

(b) For α a k-form and β an m-form, $\alpha\wedge\beta$ is a $(k+m)$-form.

(c) $(\alpha\wedge\beta)\wedge\gamma=\alpha\wedge(\beta\wedge\gamma)=\alpha\wedge\beta\wedge\gamma$ (associativity).

(d) For 1-forms $L_1,\dots,L_k$ and vectors $v_1,\dots,v_k\in\mathbb{R}^n$,

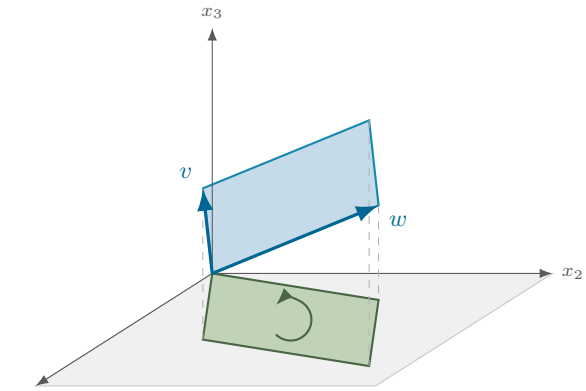
$$L_1\wedge L_2\wedge\dots\wedge L_k(v_1,\dots,v_k)=\det\big(L_i(v_j)\big)_{1\le i,j\le k}.$$

(e) For $\alpha(T):=\det(AT)$, $\beta(S):=\det(BS)$ as in example **(a)** of the two families above,

$$\alpha\wedge\beta(X)=\det\left(\begin{bmatrix}A\\B\end{bmatrix}X\right).$$

Example 24.a.7 (*A wedge of basis covectors is a minor determinant*): By property **(d)**, for $1\le i_1<\dots<i_k\le n$,

$$e_{i_1}^*\wedge\dots\wedge e_{i_k}^*(v_1,\dots,v_k)=\det\left(\begin{pmatrix}v_1^{i_1}&\cdots&v_k^{i_1}\\\vdots&\ddots&\vdots\\v_1^{i_k}&\cdots&v_k^{i_k}\end{pmatrix}\right)=\det\left(\begin{bmatrix}e_{i_1}^{\mathsf T}\\\vdots\\e_{i_k}^{\mathsf T}\end{bmatrix}\begin{bmatrix}v_1&\cdots&v_k\end{bmatrix}\right).$$

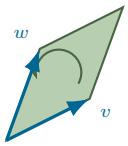


shadow on the x_1x_2-plane,
signed area $= dx^1 \wedge dx^2(v, w)$

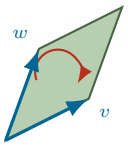
Important remark 24.a.8 (*A k-form measures shadows*): The determinant in the example above is not a bookkeeping device. It is the **signed k-dimensional volume of the shadow** that the parallelepiped spanned by $v_1, \dots, v_k$ casts on the coordinate plane $x_{i_1} \cdots x_{i_k}$, obtained by deleting every other coordinate. Everything in this chapter follows from reading it that way.

- **Why antisymmetric.** Swapping two vectors reverses the sense in which the shadow is traversed, and a signed area that is not sensitive to that is not measuring orientation at all. In particular $L(\dots, v, \dots, v, \dots) = 0$: a parallelepiped with a repeated edge is flat, and casts a shadow of zero volume.
- **Why k can exceed neither n nor the rank.** There is no k-dimensional shadow to cast if the vectors span fewer than k dimensions, which is Proposition 24.a.9's reason for discarding every repeated index.
- **Why the coefficients are indexed by increasing tuples.** One shadow per coordinate plane, and $i_1 < \cdots < i_k$ names each plane once. A k-form on $\mathbb{R}^n$ is a recipe for weighting the $\binom{n}{k}$ shadows and adding them up.

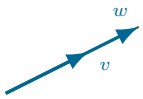
In the picture, v and w have different third coordinates and the shadow does not notice: $dx^1 \wedge dx^2$ deletes x_3 before measuring anything.



$dx^1 \wedge dx^2(v, w)$
$= +3.6$



$dx^1 \wedge dx^2(w, v)$
$= -3.6$



w parallel to v
$= 0$

Proposition 24.a.9 (*Every k-linear form is a combination of wedge products of covectors*): Every antisymmetric k-linear form L on $\mathbb{R}^n$ can be written as

$$L = \sum_{1 \le i_1 < \cdots < i_k \le n} \ell_{i_1 \dots i_k} \, e^*_{i_1} \wedge \cdots \wedge e^*_{i_k}$$

for uniquely determined scalars $\ell_{i_1 \dots i_k} \in \mathbb{R}$.

Proof:

Expand each argument $v_i = \sum_{m=1}^n v_i^m e_m$ and use k-linearity:

$$L(v_1, \dots, v_k) = L\left(\sum_{i_1=1}^n v_1^{i_1} e_{i_1}, \dots, \sum_{i_k=1}^n v_k^{i_k} e_{i_k} \right) = \sum_{1 \le i_1, \dots, i_k \le n} L(e_{i_1}, \dots, e_{i_k}) \, v_1^{i_1} \cdots v_k^{i_k}.$$

By antisymmetry, $L(e_{i_1},\dots,e_{i_k})=0$ whenever two indices coincide, so the sum may be restricted to $i_1,\dots,i_k$ pairwise distinct; regrouping each such tuple by the permutation σ that sorts it into increasing order $i_{\sigma(1)}<\dots<i_{\sigma(k)}$,

$$\begin{aligned} L(v_1,\dots,v_k) &= \sum_{1\le i_1<\dots<i_k\le n} L(e_{i_1},\dots,e_{i_k}) \sum_{\sigma\in S_k} \operatorname{sgn}(\sigma)\, v_1^{i_{\sigma(1)}}\cdots v_k^{i_{\sigma(k)}} \\ &= \sum_{1\le i_1<\dots<i_k\le n} \ell_{i_1\dots i_k}\, e_{i_1}^*\wedge\dots\wedge e_{i_k}^*(v_1,\dots,v_k), \end{aligned}$$

with $\ell_{i_1\dots i_k}:=L(e_{i_1},\dots,e_{i_k})$, using that the inner sum over σ is exactly the determinant expansion of $e_{i_1}^*\wedge\dots\wedge e_{i_k}^*(v_1,\dots,v_k)$ from property **(d)** above.

For uniqueness, evaluate both sides at $(e_{j_1},\dots,e_{j_k})$ with $j_1<\dots<j_k$. By property **(d)**, $e_{i_1}^*\wedge\dots\wedge e_{i_k}^*(e_{j_1},\dots,e_{j_k})$ is the determinant of the matrix $\big(\delta_{i_a j_b}\big)_{a,b}$, which is 1 if the two increasing tuples agree and has a zero row otherwise. So the right-hand side returns $\ell_{j_1\dots j_k}$, and the coefficients are the values of L on increasing tuples of basis vectors, whatever expansion one started from. **Q.E.D.**

AI-Exercise 24.a.10 (*Reading off all three shadows*): Let $v:=(1,2,0)$ and $w:=(0,1,3)$ in $\mathbb{R}^3$.

- **(a)** Compute $dx^1\wedge dx^2(v,w)$, $dx^1\wedge dx^3(v,w)$ and $dx^2\wedge dx^3(v,w)$ directly from property **(d)**.
- **(b)** Compute the cross product $v\times w$.
- **(c)** The three numbers in part **(a)** and the three components of $v\times w$ are the same three numbers, but not in the same order and not all with the same sign. Find the correspondence, and say which of the three shadows needs its sign flipped to make the pattern uniform.

Solution: *See solution on Page 374*

AI-Exercise 24.a.11 (*A 2-form whose square does not vanish*): On $\mathbb{R}^4$, let $\alpha:=dx^1\wedge dx^2+dx^3\wedge dx^4$.

- **(a)** Compute $\alpha\wedge\alpha$.
- **(b)** For a 1-form β one always has $\beta\wedge\beta=0$. Explain why the same argument does not apply to α, and state for which degrees k a k-form must satisfy $\alpha\wedge\alpha=0$.

Solution: *See solution on Page 375*

Differential forms and the exterior derivative (Section 24.b)

Definition 24.b.12 (*Differential form*): A "*differential p-form*" ω (German: „*Differentialform*“) is defined on a domain $D\subseteq\mathbb{R}^n$ if, for every $x\in D$, $\omega(x)=\omega_x:(\mathbb{R}^n)^p\to\mathbb{R}$ is an antisymmetric p-linear map. Since $(e_1(x),\dots,e_n(x))$ is a basis of $\mathbb{R}^n$ for every $x\in D$, Proposition 24.a.9 lets us write

$$\omega(x)=\sum_{1\le i_1<\dots<i_p\le n}\omega_{i_1\dots i_p}(x)\, e_{i_1}^*(x)\wedge\dots\wedge e_{i_p}^*(x)$$

for functions $\omega_{i_1\dots i_p}:D\to\mathbb{R}$.

Notation 24.b.13: We abbreviate $e_{i_1}^*\wedge\dots\wedge e_{i_p}^*=:e_I^*$ for $I=(i_1,\dots,i_p)$, and write $\Omega^p(U)$ for the set of p-forms on $U\subseteq\mathbb{R}^n$. By convention, $\Omega^0(U)=C^\infty(U)$.

Example 24.b.14 (*The forms attached to a function and to a vector field*): Let $U\subseteq\mathbb{R}^n$.

- **(a)** Any $f\in C^\infty(U)$ is a 0-form.
- **(b)** A vector field $F:U\to\mathbb{R}^n$ has an associated 1-form $\omega_F(x):v\mapsto\langle F(x),v\rangle$, that is,

$$\omega_F(x)(v)=\sum_{i=1}^n F^i(x)\,v^i=\sum_{i=1}^n F^i(x)\,e_i^*(x)(v)\qquad\Longrightarrow\qquad\omega_F=\sum_{i=1}^n F^i\,e_i^*.$$

(c) $F : U \to \mathbb{R}^n$ also naturally induces an $(n-1)$-form

$$\omega_F^{n-1}(x)(v_1, \dots, v_{n-1}) := \det\big(F(x) \mid v_1 \mid \cdots \mid v_{n-1}\big).$$

Developing this determinant along the first column,

$$\omega_F^{n-1}(x)(v_1, \dots, v_{n-1}) = \sum_{i=1}^{n} (-1)^{i-1} F^i(x) \det \begin{pmatrix} v_1^1 & \cdots & v_{n-1}^1 \\ \vdots & & \vdots \\ v_1^{i-1} & \cdots & v_{n-1}^{i-1} \\ v_1^{i+1} & \cdots & v_{n-1}^{i+1} \\ \vdots & & \vdots \\ v_1^n & \cdots & v_{n-1}^n \end{pmatrix}$$

$$= \sum_{i=1}^{n} (-1)^{i-1} F^i(x)\, e_1^* \wedge \cdots \wedge \widehat{e_i^*} \wedge \cdots \wedge e_n^*(v_1, \dots, v_{n-1}),$$

where $\widehat{e_i^*}$ denotes an omitted factor, so that $\omega_F^{n-1} = \sum_{i=1}^{n} (-1)^{i-1} F^i\, e_1^* \wedge \cdots \wedge \widehat{e_i^*} \wedge \cdots \wedge e_n^*$.

(d) In particular, for $F : \mathbb{R}^3 \to \mathbb{R}^3$,

$$\omega_F^2(x)(v_1, v_2) = \langle F(x), v_1 \times v_2 \rangle = \det\big(F(x) \mid v_1 \mid v_2\big).$$

AI-Note 24.2: Item **(d)** identifies ω_F^2 with the flux integrand from Section 23.b: the flux $\int_S \langle F, \nu \rangle \, d\,\mathrm{vol}_2$ over a parametrized surface is exactly the integral, in the sense of the motivating change-of-variables calculation above, of the 2-form ω_F^2.

Theorem 24.b.15 (*Exterior derivative*): There exists a unique linear operator $d : \Omega^k(U) \to \Omega^{k+1}(U)$ satisfying

(a) for $f \in C^\infty(U) = \Omega^0(U)$: $df = Df$, the ordinary differential;

(b) for $\omega \in \Omega^k(U)$, $\theta \in \Omega^s(U)$: $d(\omega \wedge \theta) = d\omega \wedge \theta + (-1)^k\, \omega \wedge d\theta$;

(c) $d \circ d\,\omega = d^2\omega = 0$.

AI-Note 24.3: The existence of d is not proved here. The proof can be found in the official script (Analysis_II_Script_v1.pdf, §14.3, pp. 136–138), where the operator is constructed explicitly in coordinates and properties **(a)**–**(c)** are verified.

Denote by dx^i the differential of $x \mapsto x^i$ in a moving frame $(e_1(x), \dots, e_n(x))$, that is, $dx^i = e_i^*(x)$. Every p-form ω on $U \subseteq \mathbb{R}^n$ then takes the shape

$$\omega = \sum_{1 \le i_1 < \cdots < i_p \le n} \omega_{i_1 \cdots i_p}\, dx^{i_1} \wedge \cdots \wedge dx^{i_p} = \sum_I \omega_I\, dx^I, \qquad \omega_I \in C^\infty(U),$$

with exterior derivative

$$d\omega = \sum_I d\omega_I \wedge dx^I.$$

This formula is not a further choice: properties **(a)**–**(c)** force it. Since $dx^i = d(x^i)$ by **(a)**, property **(c)** gives $d(dx^i) = 0$, and then **(b)**, applied repeatedly, gives $d(dx^I) = 0$ for every wedge of coordinate differentials. One more application of **(b)** yields $d(\omega_I\, dx^I) = d\omega_I \wedge dx^I + \omega_I\, d(dx^I) = d\omega_I \wedge dx^I$, and linearity does the rest. That is the uniqueness half of Theorem 24.b.15; existence amounts to checking that the formula does satisfy **(b)** and **(c)**. For a function f, the check of **(c)** is Schwarz's theorem in disguise:

$$d(df) = \sum_{i,j} \partial_j \partial_i f\, dx^j \wedge dx^i = 0,$$

because the coefficient $\partial_j \partial_i f$ is symmetric in (i, j) (Theorem 11.b.13) while $dx^j \wedge dx^i$ is antisymmetric, so the terms cancel in pairs.

Example 24.b.16 (*The exterior derivative recovers the gradient and the curl*):

(a) For $f \in C^\infty(\mathbb{R}^n) = \Omega^0(\mathbb{R}^n)$,

$$df_x(v) = \begin{pmatrix} \dfrac{\partial f}{\partial x^1} & \cdots & \dfrac{\partial f}{\partial x^n} \end{pmatrix} \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix} = \frac{\partial f}{\partial x^1}\, dx^1(v) + \cdots + \frac{\partial f}{\partial x^n}\, dx^n(v),$$

so $df = \frac{\partial f}{\partial x^1}\, dx^1 + \cdots + \frac{\partial f}{\partial x^n}\, dx^n = \langle \nabla f, \cdot \rangle$: the exterior derivative of a 0-form is exactly the 1-form associated (as in the example above) to the gradient vector field ∇f. This is the shape that makes Stokes' theorem specialize to the FTC, as in Section 23.a.

(b) Define a 1-form on $U \subseteq \mathbb{R}^3$ by $\omega(x,y,z) = \langle F(x,y,z), \cdot \rangle = F^1\, dx + F^2\, dy + F^3\, dz$ for a vector field F. By property **(b)** of Theorem 24.b.15 and $d(dx^i) = 0$ (since $dx^i = d(x^i)$ and $d^2 = 0$),

$$d\omega = dF^1 \wedge dx + dF^2 \wedge dy + dF^3 \wedge dz,$$

and expanding each $dF^i = \partial_x F^i\, dx + \partial_y F^i\, dy + \partial_z F^i\, dz$, only the cross terms survive the wedge (since $dx^i \wedge dx^i = 0$ by antisymmetry):

$$d\omega = \Big(\frac{\partial F^3}{\partial y} - \frac{\partial F^2}{\partial z}\Big) dy \wedge dz + \Big(\frac{\partial F^1}{\partial z} - \frac{\partial F^3}{\partial x}\Big) dz \wedge dx + \Big(\frac{\partial F^2}{\partial x} - \frac{\partial F^1}{\partial y}\Big) dx \wedge dy.$$

Comparing with example item **(d)** in Section 24.b, this is exactly $\omega^2_{\nabla \times F} = \langle \nabla \times F, \cdot \times \cdot \rangle = \det(\nabla \times F \mid \cdot \mid \cdot)$: the exterior derivative of the 1-form associated to F is the 2-form associated to $\operatorname{curl} F$.

(c) Recall the $(n-1)$-form $\omega^{n-1}_F = \sum_{i=1}^n (-1)^{i-1} F^i\, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$ from Section 24.b (example item **(c)**), with $\widehat{dx^i}$ denoting an omitted factor. In $d(F^i\, dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n) = dF^i \wedge dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n$, only the $\partial_i F^i\, dx^i$ summand of $dF^i = \sum_j \partial_j F^i\, dx^j$ survives, since every $j \neq i$ repeats a factor already present and so kills the wedge by antisymmetry. Therefore

$$d\omega^{n-1}_F = \sum_{i=1}^n (-1)^{i-1} \partial_i F^i\; dx^i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n.$$

Moving dx^i past the $i-1$ factors $dx^1, \ldots, dx^{i-1}$ ahead of it, into its natural position, costs another sign of $(-1)^{i-1}$, cancelling the one already there:

$$dx^i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n = (-1)^{i-1}\, dx^1 \wedge \cdots \wedge dx^n$$

$$\implies \quad d\omega^{n-1}_F = \sum_{i=1}^n \partial_i F^i\; dx^1 \wedge \cdots \wedge dx^n = (\operatorname{div} F)\, dx^1 \wedge \cdots \wedge dx^n.$$

So, the exterior derivative of ω^{n-1}_F is $\operatorname{div} F$ times the volume form. That completes the pattern of items **(a)** and **(b)** above: grad, curl and div are exactly the 0-, 1- and $(n-1)$-form faces of the single operator d.

(d) Let $\omega = x_3\, dx_3 + x_1 x_2\, dx_2$ be a 1-form on $\mathbb{R}^3$. Let us compute its exterior derivative using the rules:

$$\begin{aligned} d\omega &= d(x_3) \wedge dx_3 + d(x_1 x_2) \wedge dx_2 \\ &= dx_3 \wedge dx_3 + (x_2\, dx_1 + x_1\, dx_2) \wedge dx_2 \\ &= 0 + x_2\, dx_1 \wedge dx_2 + x_1 \underbrace{dx_2 \wedge dx_2}_{=0} \\ &= x_2\, dx_1 \wedge dx_2. \end{aligned}$$

$$\begin{array}{ccccccc} \Omega^0(U) & \xrightarrow{d} & \Omega^1(U) & \xrightarrow{d} & \Omega^2(U) & \xrightarrow{d} & \Omega^3(U) \\ {\scriptstyle =}\uparrow & & \uparrow{\scriptstyle F \mapsto \omega_F} & & \uparrow{\scriptstyle F \mapsto \omega^2_F} & & \uparrow{\scriptstyle g \mapsto g\, dx \wedge dy \wedge dz} \\ C^\infty(U) & \xrightarrow[\text{grad}]{} & C^\infty(U, \mathbb{R}^3) & \xrightarrow[\text{curl}]{} & C^\infty(U, \mathbb{R}^3) & \xrightarrow[\text{div}]{} & C^\infty(U) \end{array}$$

Important remark 24.b.17 (*One operator, one identity*): For $U \subseteq \mathbb{R}^3$ the diagram above commutes, and each square is one of the three computations just performed: the left square is item **(a)**, the middle is item **(b)**, the right is item **(c)**. Read downward it says that the classical vector operators are d in disguise, and that the disguise is nothing but the choice of which object to call a vector field.

The payoff is the bottom row's two mysteries collapsing into one line of the top row. Property **(c)** of Theorem 24.b.15 is $d \circ d = 0$, a single statement about a single operator, and reading it through the two identifications gives

$$\operatorname{curl} \operatorname{grad} = 0 \qquad \text{and} \qquad \operatorname{div} \operatorname{curl} = 0.$$

These are the same fact twice, seen from two rungs of the same ladder. That is the whole reason for the machinery: in $\mathbb{R}^3$ there happen to be two identities because there happen to be two interior rungs, and in $\mathbb{R}^n$ there are $n-1$ of them, none of which needs a separate proof.

Two cautions the picture invites. The vertical arrows are **not** canonical: $F \mapsto \omega_F$ uses the inner product and $F \mapsto \omega_F^2$ uses the determinant, so both depend on $\mathbb{R}^3$ carrying a metric and an orientation, which is exactly what a general manifold does not hand you for free. And the ladder is this short only because $n = 3$: $\Omega^4(U)$ on $\mathbb{R}^3$ is zero, since there is no fourth coordinate to build a shadow on (Important remark 24.a.8), which is why div is the last operator in the row and not because anyone chose to stop there.

AI-Exercise 24.b.18 (*The angle form is closed*): On $U := \mathbb{R}^2 \setminus \{0\}$ define the 1-form

$$\omega := \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

(a) Show that $d\omega = 0$.

(b) By part **(a)** and Theorem 24.b.15, one might hope that $\omega = df$ for some $f \in C^\infty(U)$. Compute $\int_\gamma \omega$ along the unit circle $\gamma(t) := (\cos t, \sin t)$, $t \in [0, 2\pi]$, and explain what this rules out. **Hint:** Proposition 26.b.14, read through the identification $F \leftrightarrow \omega_F$ of the diagram above.

Solution: *See solution on Page 375*

AI-Exercise 24.b.19 (*Exterior derivatives by hand*): Compute $d\omega$ for each of the following, on the largest domain where the expression makes sense.

(a) $\omega = xy\,dx + yz\,dy + zx\,dz$ on $\mathbb{R}^3$. Which vector field's curl have you just computed?

(b) $\omega = x_1 x_2 x_3\,dx_2 \wedge dx_3$ on $\mathbb{R}^3$.

(c) $\omega = f\,dg$, where $f, g \in C^\infty(\mathbb{R}^n)$. Express the answer using $\wedge$ and deduce that $d(f\,dg + g\,df) = 0$.

Solution: *See solution on Page 376*

Exercise 24.b.20 (*Six wedge products $df \wedge \lambda$*): All forms below live on $\mathbb{R}^3$, and in every case λ is a 1-form. Compute

$$\omega := df \wedge \lambda$$

in simplified form. In parts **(a)**–**(c)** the symbol f denotes a function, that is a 0-form, and the answer should be presented as

$$\omega = F\,dx \wedge dy + G\,dx \wedge dz + H\,dy \wedge dz;$$

in parts **(d)**–**(f)** the symbol f denotes a 1-form instead, and the answer should be presented as

$$\omega = F\,dx \wedge dy \wedge dz.$$

(a) $f(x,y) = x^2 y + \sin y$, $\quad \lambda = x\,dx + y\,dy$.

(b) $f(x,y,z) = xyz$, $\quad \lambda = x\,dy + y\,dz + z\,dx$.

(c) $f(x,y,z) = e^x y + z^2$, $\quad \lambda = (x+y)\,dx + z\,dy + xz\,dz$.

(d) $f = x^2\,dy + y^2\,dz\,, \qquad \lambda = x\,dx + z\,dy\,.$
(e) $f = xy\,dx + z\,dy + x^2 z\,dz\,, \qquad \lambda = y\,dx + x\,dy + z\,dz\,.$
(f) $f = e^y\,dx + xz\,dy + y^2 z\,dz\,, \qquad \lambda = dx + y\,dy + z\,dz\,.$

Info: *This exercise is the whole of* ***Some differential form computations 1****, an ungraded example sheet issued by Prof. Joaquim Serra on 13 May 2026 for Analysis II, Spring Semester 2026. The sheet prints full worked solutions, which are the basis of the solutions given here.*

Solution: *See solution on Page 376*

AI-Note 24.4: Serra uses the letter f for a 0-form in **(a)**–**(c)** and for a 1-form in **(d)**–**(f)**, and the sheet says so explicitly rather than changing the name. Keep the distinction in view while computing, because df means two different things across the two halves: the differential of a function in the first, and the exterior derivative of a 1-form in the second (Theorem 24.b.15). The degree bookkeeping is what makes the answers land in different spaces: $df \wedge \lambda$ is a 2-form in **(a)**–**(c)** and a 3-form in **(d)**–**(f)**, which is why the requested output shapes differ.

Solutions (Section 24.c)

Solution to AI-Exercise 24.a.10:

(a) The three shadows. Property **(d)** says each is the determinant of the 2×2 block of $(v \mid w)$ cut out by the two indices, so with $v = (1, 2, 0)$ and $w = (0, 1, 3)$:

$$dx^1 \wedge dx^2(v, w) = \det \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = 1,$$
$$dx^1 \wedge dx^3(v, w) = \det \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} = 3,$$
$$dx^2 \wedge dx^3(v, w) = \det \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} = 6.$$

(b) The cross product.

$$v \times w = \big(2 \cdot 3 - 0 \cdot 1,\ 0 \cdot 0 - 1 \cdot 3,\ 1 \cdot 1 - 2 \cdot 0\big) = (6,\ -3,\ 1).$$

(c) The correspondence. Component by component,

$$(v \times w)_1 = 6 = dx^2 \wedge dx^3(v, w),$$
$$(v \times w)_2 = -3 = -\,dx^1 \wedge dx^3(v, w),$$
$$(v \times w)_3 = 1 = dx^1 \wedge dx^2(v, w).$$

So the 1st and 3rd match on the nose and the 2nd needs a sign flip. The pattern becomes uniform once that middle shadow is written in the cyclic order $3, 1$ instead of the increasing order $1, 3$: since $dx^3 \wedge dx^1 = -\,dx^1 \wedge dx^3$, one has

$$v \times w = \big(dx^2 \wedge dx^3(v, w),\ dx^3 \wedge dx^1(v, w),\ dx^1 \wedge dx^2(v, w)\big).$$

Remark 24.c.21 (*The cross product is a 2-form in disguise*): This is not a coincidence about these two vectors. Expanding $\det(u \mid v \mid w)$ along its first column, as in example item **(c)** of Section 24.b, gives $\det(u \mid v \mid w) = \langle u,\ v \times w \rangle$ for every u, with exactly the three shadows above, in cyclic order, as the components of $v \times w$. So the cross product **is** the 2-form $(v, w) \mapsto \det(\,\cdot \mid v \mid w)$, read as a vector through the inner product of $\mathbb{R}^3$; example item **(d)** records the same fact.

Two things fall out. The alternating sign that everyone memorises in the cofactor expansion of $v \times w$ is the same sign as $dx^3 \wedge dx^1 = -\,dx^1 \wedge dx^3$, so it is antisymmetry rather than a mnemonic. And the cross product exists only in $\mathbb{R}^3$ because that is the only dimension where $\binom{n}{2} = n$, i.e.

where there are exactly as many 2-form shadows as coordinates to store them in. In $\mathbb{R}^4$ there are six shadows and four slots, and no cross product.

Q.E.D.

Solution to AI-Exercise 24.a.11:

(a) Expand by bilinearity, writing $dx^{ij} := dx^i \wedge dx^j$:

$$\alpha \wedge \alpha = dx^{12} \wedge dx^{12} + dx^{12} \wedge dx^{34} + dx^{34} \wedge dx^{12} + dx^{34} \wedge dx^{34}.$$

The first and last vanish, since each repeats a factor and $dx^i \wedge dx^i = 0$. For the two middle terms, moving the block dx^{34} past the block dx^{12} means moving two 1-forms past two 1-forms, which costs $(-1)^{2\cdot 2} = +1$, so $dx^{34} \wedge dx^{12} = dx^{12} \wedge dx^{34}$ and the two middle terms add rather than cancel:

$$\alpha \wedge \alpha = 2\, dx^1 \wedge dx^2 \wedge dx^3 \wedge dx^4 \neq 0.$$

(b) The argument for a 1-form is $\beta \wedge \beta = -\beta \wedge \beta$, which forces $\beta \wedge \beta = 0$. That commutation sign comes from the graded rule: for a k-form α and an m-form θ,

$$\alpha \wedge \theta = (-1)^{km}\, \theta \wedge \alpha.$$

With $\theta = \alpha$ and $m = k$ this reads $\alpha \wedge \alpha = (-1)^{k^2} \alpha \wedge \alpha$, and $(-1)^{k^2} = (-1)^k$. So the argument gives $\alpha \wedge \alpha = 0$ exactly when k is **odd**, and says nothing at all when k is even. The 2-form above is the standard witness that *"nothing at all"* is the correct reading.

Q.E.D.

Solution to AI-Exercise 24.b.18:

(a) Write $\omega = P\, dx + Q\, dy$ with $P = \dfrac{-y}{x^2+y^2}$ and $Q = \dfrac{x}{x^2+y^2}$. Then $d\omega = (\partial_x Q - \partial_y P)\, dx \wedge dy$, and by the quotient rule

$$\partial_x Q = \frac{(x^2+y^2) - x \cdot 2x}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}, \qquad \partial_y P = \frac{-(x^2+y^2) + y \cdot 2y}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}.$$

They are equal, so $d\omega = 0$ on all of U.

(b) Along $\gamma(t) = (\cos t, \sin t)$ the denominator is 1, and $-y\, dx + x\, dy$ pulls back to $(-\sin t)(-\sin t)\, dt + (\cos t)(\cos t)\, dt = dt$. Hence

$$\int_\gamma \omega = \int_0^{2\pi} dt = 2\pi \neq 0.$$

Under the identification $F \leftrightarrow \omega_F$, an exact form $\omega = df$ corresponds to a conservative field with potential f, and Proposition 26.b.14 makes its integral around any closed loop $f(\gamma(1)) - f(\gamma(0)) = 0$. Since $2\pi \neq 0$, no such f exists: ω is **closed but not exact**.

Remark 24.c.22 (*What the counterexample is really about*): The obstruction is not analytic but topological. On any simply connected open subset of U (the plane slit along a ray, say) the form **is** exact, with f the polar angle θ, which is why ω is called the "*angle form*". What fails on all of $\mathbb{R}^2 \setminus \{0\}$ is that θ cannot be defined continuously all the way round: it gains 2π per circuit, and 2π is exactly the integral computed above.

So *"closed"* and *"exact"* differ precisely by the holes in the domain. That is the content of Corollary 26.c.25, which says the two coincide for 1-forms on simply connected domains, and of the Poincaré lemma in the form most texts give it, on star-shaped domains (Remark 26.c.27). This one-line example is the reason those theorems need a hypothesis on the domain at all.

Q.E.D.

Solution to AI-Exercise 24.b.19:

(a) Using $d\omega = \sum_I d\omega_I \wedge dx^I$ and discarding every wedge with a repeated factor,

$$\begin{aligned} d\omega &= d(xy) \wedge dx + d(yz) \wedge dy + d(zx) \wedge dz \\ &= (y\,dx + x\,dy) \wedge dx + (z\,dy + y\,dz) \wedge dy + (z\,dx + x\,dz) \wedge dz \\ &= x\,dy \wedge dx + y\,dz \wedge dy + z\,dx \wedge dz \\ &= -x\,dx \wedge dy - y\,dy \wedge dz - z\,dz \wedge dx. \end{aligned}$$

By item **(b)** of Example 24.b.16 this is $\omega^2_{\operatorname{curl} F}$ for $F = (xy,\, yz,\, zx)$, so the computation is $\operatorname{curl} F = (-y,\, -z,\, -x)$. (Check directly: $\operatorname{curl} F = (\partial_y(zx) - \partial_z(yz),\, \partial_z(xy) - \partial_x(zx),\, \partial_x(yz) - \partial_y(xy)) = (-y,\, -z,\, -x)$.)

(b) Only the dx_1 part of $d(x_1x_2x_3)$ survives, the other two repeating a factor already present:

$$d\omega = d(x_1x_2x_3) \wedge dx_2 \wedge dx_3 = x_2x_3\,dx_1 \wedge dx_2 \wedge dx_3.$$

(c) By property **(b)** of Theorem 24.b.15 with $k = 0$, and then $d(dg) = 0$ by property **(c)**,

$$d(f\,dg) = df \wedge dg + f\,d(dg) = df \wedge dg.$$

Swapping the roles of f and g gives $d(g\,df) = dg \wedge df = -\,df \wedge dg$, since both are 1-forms. Adding,

$$d(f\,dg + g\,df) = df \wedge dg - df \wedge dg = 0.$$

This is the Leibniz rule in disguise: $f\,dg + g\,df = d(fg)$, and $d \circ d = 0$.

Q.E.D.

Solution to Exercise 24.b.20:

Throughout, the only two facts used are that $\wedge$ is bilinear and that it is alternating on 1-forms, so $dx \wedge dx = 0$ and $dy \wedge dx = -dx \wedge dy$.

(a) $df = 2xy\,dx + (x^2 + \cos y)\,dy$, so

$$\begin{aligned} \omega &= \big(2xy\,dx + (x^2 + \cos y)\,dy\big) \wedge \big(x\,dx + y\,dy\big) \\ &= 2xy^2\,dx \wedge dy + x(x^2 + \cos y)\,dy \wedge dx \\ &= \big(2xy^2 - x^3 - x\cos y\big)\,dx \wedge dy. \end{aligned}$$

(b) $df = yz\,dx + xz\,dy + xy\,dz$, so

$$\begin{aligned} \omega &= \big(yz\,dx + xz\,dy + xy\,dz\big) \wedge \big(x\,dy + y\,dz + z\,dx\big) \\ &= xyz\,dx \wedge dy + y^2z\,dx \wedge dz + xyz\,dy \wedge dz \\ &\quad + xz^2\,dy \wedge dx + xyz\,dz \wedge dx + x^2y\,dz \wedge dy. \end{aligned}$$

Reordering each product into the standard order gives

$$\omega = xz(y - z)\,dx \wedge dy + yz(y - x)\,dx \wedge dz + xy(z - x)\,dy \wedge dz.$$

(c) $df = e^x y\,dx + e^x\,dy + 2z\,dz$, so

$$\begin{aligned} \omega &= \big(e^x y\,dx + e^x\,dy + 2z\,dz\big) \wedge \big((x + y)\,dx + z\,dy + xz\,dz\big) \\ &= e^x yz\,dx \wedge dy + e^x xyz\,dx \wedge dz + e^x(x + y)\,dy \wedge dx \\ &\quad + e^x xz\,dy \wedge dz + 2z(x + y)\,dz \wedge dx + 2z^2\,dz \wedge dy, \end{aligned}$$

and after reordering,

$$\omega = e^x(yz - x - y)\,dx \wedge dy + z(e^x xy - 2x - 2y)\,dx \wedge dz + z(e^x x - 2z)\,dy \wedge dz.$$

(d) Here f is a 1-form, so $df = d(x^2) \wedge dy + d(y^2) \wedge dz = 2x\,dx \wedge dy + 2y\,dy \wedge dz$. Therefore

$$\omega = \big(2x\,dx \wedge dy + 2y\,dy \wedge dz\big) \wedge \big(x\,dx + z\,dy\big).$$

Every term repeats a factor and dies except $2y\,dy\wedge dz\wedge x\,dx$. Since $dy\wedge dz\wedge dx = dx\wedge dy\wedge dz$, this leaves

$$\omega = 2xy\,dx\wedge dy\wedge dz.$$

(e) Again f is a 1-form. Using $d(xy) = y\,dx + x\,dy$, $d(z) = dz$ and $d(x^2z) = 2xz\,dx + x^2\,dz$,

$$\begin{aligned} df &= (y\,dx + x\,dy)\wedge dx + dz\wedge dy + (2xz\,dx + x^2\,dz)\wedge dz \\ &= -x\,dx\wedge dy - dy\wedge dz + 2xz\,dx\wedge dz. \end{aligned}$$

Wedging with $\lambda = y\,dx + x\,dy + z\,dz$, only three products survive:

$$\omega = -xz\,dx\wedge dy\wedge dz - y\,dy\wedge dz\wedge dx + 2x^2z\,dx\wedge dz\wedge dy,$$

and reordering the last two gives

$$\omega = -\big(xz + y + 2x^2z\big)\,dx\wedge dy\wedge dz.$$

(f) With $d(e^y) = e^y\,dy$, $d(xz) = z\,dx + x\,dz$ and $d(y^2z) = 2yz\,dy + y^2\,dz$,

$$\begin{aligned} df &= e^y\,dy\wedge dx + (z\,dx + x\,dz)\wedge dy + (2yz\,dy + y^2\,dz)\wedge dz \\ &= (z - e^y)\,dx\wedge dy + (2yz - x)\,dy\wedge dz. \end{aligned}$$

Wedging with $\lambda = dx + y\,dy + z\,dz$, the surviving terms are $z(z - e^y)\,dx\wedge dy\wedge dz$ and $(2yz - x)\,dy\wedge dz\wedge dx$, so

$$\omega = \big(z^2 - ze^y + 2yz - x\big)\,dx\wedge dy\wedge dz.$$

Remark 24.c.23 (*Two habits that make these quick*): First, **expand before you reorder.** Wedging two sums of 1-forms in $\mathbb{R}^3$ produces nine terms, of which three vanish immediately; sorting each survivor into the standard order $dx\wedge dy$, $dx\wedge dz$, $dy\wedge dz$ only at the end keeps the sign bookkeeping in one place. Reordering as you go is where signs get lost.

Second, **count degrees first.** In **(d)**–**(f)** the form df is already a 2-form, so $df\wedge\lambda$ is a 3-form, and on $\mathbb{R}^3$ there is only one basis 3-form. That means every term either dies or lands on $dx\wedge dy\wedge dz$, and one can discard any product repeating a differential without writing it down at all. Parts **(d)**–**(f)** become three-line computations once that is used, whereas **(a)**–**(c)** genuinely need all three components.

Worth noticing in **(a)**: the answer has no $dx\wedge dz$ or $dy\wedge dz$ component, and it could not have. Neither f nor λ involves z at all, so the whole computation happens in the xy-plane and dz never enters.

Q.E.D.

Pullbacks, Integration of Forms and Orientability (Chapter 25)

To integrate a differential form over a submanifold one has to pull it back to a parameter domain, and then check that the answer does not depend on which parametrization was chosen. It does not, provided the change of parameters between any two of them has positive Jacobian determinant; otherwise the sign flips. Choosing parametrizations that all agree in this way is an orientation, whether one exists is orientability, and that is what this chapter is really about. The last section fixes the induced orientation on a boundary, which is the convention that makes the sign in Stokes' theorem come out right.

Pullback of forms (Section 25.a)

Definition 25.a.1 (*Pullback of a form*): Let $U \subseteq \mathbb{R}^m$, $V \subseteq \mathbb{R}^n$ be open, $F : U \to V$ a smooth map, and $\omega \in \Omega^p(V)$ a differential p-form on V. The "*pullback*" (German: „*Zurückziehung*") of ω under F is the differential p-form on U defined by

$$(F^*\omega)_x(v_1, \dots, v_p) := \omega_{F(x)}\big(DF_x(v_1), \dots, DF_x(v_p)\big).$$

The pullback satisfies two properties that make it the right tool for transporting forms along smooth maps:

(a) $F^*(\omega \wedge \theta) = F^*\omega \wedge F^*\theta$;

(b) $F^*(d\omega) = d(F^*\omega)$ (the pullback commutes with the exterior derivative).

AI-Note 25.1: This is precisely the operation used, informally, in the reparametrization-invariance motivation for Definition 24.a.1: writing $\phi^*\omega$ and $(\phi \circ \Psi)^*\omega$ for the two sides of that calculation makes it read as $\int_D \phi^*\omega = \int_V \Psi^*(\phi^*\omega) = \int_V (\phi \circ \Psi)^*\omega$. The first step is the change of variables formula for a top-degree form, $\int_D \eta = \int_V \Psi^*\eta$ when $\det D\Psi > 0$ (Exercise 25.b.10), applied to $\eta := \phi^*\omega$; the second is the fact that pullback along a composition is the composition of pullbacks, $(\phi \circ \Psi)^* = \Psi^* \circ \phi^*$, which is the chain rule.

Example 25.a.2 (*Computing a pullback*): Let $\omega = x_3\,dx_3 + x_1x_2\,dx_2$ be a 1-form on $\mathbb{R}^3$. Consider the map $F : \mathbb{R}^2 \to \mathbb{R}^3$ given by

$$F(u, v) = \begin{pmatrix} u + v \\ e^u \\ u^2 - v \end{pmatrix}.$$

We want to compute the pullback $F^*\omega$, which will be a 1-form on $\mathbb{R}^2$. Using the properties of the pullback, we have:

$$\begin{aligned} F^*\omega &= F^*(x_3\,dx_3 + x_1x_2\,dx_2) \\ &= F^*(x_3)F^*(dx_3) + F^*(x_1x_2)F^*(dx_2) \\ &= (x_3 \circ F)d(F^*(x_3)) + (x_1x_2 \circ F)d(F^*(x_2)) \\ &= (u^2 - v)d(u^2 - v) + (u + v)e^u d(e^u) \\ &= (u^2 - v)(2u\,du - dv) + (u + v)e^u(e^u\,du) \\ &= \big(2u(u^2 - v) + (u + v)e^{2u}\big)\,du - (u^2 - v)\,dv. \end{aligned}$$

AI-Exercise 25.a.3 (*The pullback of the area form is the Jacobian*): Let $\Phi(r, \theta) := (r\cos\theta,\ r\sin\theta)$ be the polar coordinate map, and write (x, y) for the coordinates on the target.

(a) Compute $\Phi^*(dx)$ and $\Phi^*(dy)$, and wedge them together to obtain $\Phi^*(dx \wedge dy)$. Compare the answer with the polar entry of Remark 20.a.8.

(b) Exercise 25.b.10 proves the general identity behind part **(a)**, namely that a pullback of a top-degree form produces the factor $\det \mathrm{J}\,\Phi$, and it assumes $\det \mathrm{J}\,\Phi > 0$ throughout. Theorem 20.a.1, by contrast, carries $|\det \mathrm{J}\,\Phi|$ and needs no such hypothesis. Explain what the two are measuring differently, and which of them notices that $\Phi(u_1, u_2) := (u_2, u_1)$ reverses orientation.

(c) Using part **(b)**, say what would go wrong in Exercise 25.b.10 if its positivity hypothesis were dropped, and why the change of variables formula is untroubled by the same map.

Solution: *See solution on Page 385*

Integration of differential forms on submanifolds (Section 25.b)

If $\mathcal{M}$ is an m-dimensional submanifold of $\mathbb{R}^m$ (i.e. simply an open set) and $\omega \in \Omega^m(\mathcal{M})$ is an m-form, then

$$\int_{\mathcal{M}} \omega := \int_{\mathcal{M}} \omega_x(e_1, \dots, e_m)\, dx;$$

in particular, for $f \in C^\infty(\mathcal{M})$,

$$\int_{\mathcal{M}} f\, dx^1 \wedge \cdots \wedge dx^m = \int_{\mathcal{M}} f(x)\, dx,$$

recovering the ordinary (Riemann) integral of f.

Definition 25.b.4 (*Integral of a form over a parametrized submanifold*)**:** If $L^k \subseteq \mathbb{R}^n$ is a k-dimensional submanifold, $n > k$, and $\phi : U \to L$ is a parametrization with $U \subseteq \mathbb{R}^k$ open, then the integral of $\omega \in \Omega^k(L)$ is defined as

$$\int_L \omega := \int_U \phi^*\omega = \int_U \omega_{\phi(x)}\left(\frac{\partial \phi}{\partial x^1}, \dots, \frac{\partial \phi}{\partial x^k}\right) dx.$$

In particular, for $f \in C^\infty(L)$ and $1 \le i_1 < \cdots < i_k \le n$,

$$\int_L f\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \int_U f(\phi(x)) \det\left(\frac{\partial \phi^{i_s}}{\partial x^t}\right)_{1 \le s,t \le k} dx,$$

because, by property **(d)** of Definition 24.a.6,

$$dx^{i_1} \wedge \cdots \wedge dx^{i_k}(\partial_1 \phi, \dots, \partial_k \phi) = \det \begin{pmatrix} dx^{i_1}(\partial_1 \phi) & \cdots & dx^{i_1}(\partial_k \phi) \\ \vdots & \ddots & \vdots \\ dx^{i_k}(\partial_1 \phi) & \cdots & dx^{i_k}(\partial_k \phi) \end{pmatrix} = \det\left(\frac{\partial \phi^{i_s}}{\partial x^t}\right)_{1 \le s,t \le k}.$$

AI-Example 25.b.5 (*Evaluating a 2-form on a concrete pair of vectors*)**:** Everything above is stated symbolically, and the mechanics are easy to lose track of. Here is the whole computation on numbers. It is worth doing once, because every integral in this chapter is an integral of exactly this evaluation.

The recipe implicit in the displays above is: **a wedge $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ evaluated on k vectors selects rows $i_1, \dots, i_k$ from the matrix whose columns are those vectors, and takes the determinant.** The coefficient function is then evaluated at the point and multiplied on.

Take $\omega := 3\cos(x^1)\, dx^1 \wedge dx^3 \in \Omega^2(\mathbb{R}^3)$, the point $p := (0, 1, 1)$, and the two vectors

$$v_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad v_2 := \begin{pmatrix} 1 \\ 1 \\ \frac{1}{2} \end{pmatrix}, \qquad \text{so} \qquad \begin{pmatrix} v_1 & v_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

The index set is $I = (1, 3)$, so we keep rows 1 and 3 and discard row 2:

$$dx^1 \wedge dx^3(v_1, v_2) = \det \begin{pmatrix} dx^1(v_1) & dx^1(v_2) \\ dx^3(v_1) & dx^3(v_2) \end{pmatrix} = \det \begin{pmatrix} 1 & 1 \\ 0 & \frac{1}{2} \end{pmatrix} = 1 \cdot \tfrac{1}{2} - 1 \cdot 0 = \tfrac{1}{2}.$$

The coefficient at p is $3\cos(0) = 3$, and therefore

$$\omega_p(v_1, v_2) = 3 \cdot \tfrac{1}{2} = \tfrac{3}{2}.$$

Two things are worth noticing. The second components of v_1 and v_2 never entered the computation, since a form built from dx^1 and dx^3 is blind to motion in the x^2 direction, which is what makes it the natural thing to integrate over a surface in the x^1x^3-plane. And the answer is **signed**: swapping v_1 and v_2 swaps the determinant's columns and returns $-\frac{3}{2}$. That sign is exactly the orientation data of Section 25.c, and it is why $\int_L \omega$ needs an oriented L to be well defined.

AI-Note 25.2: Applying Definition 25.b.4 to ω_F^{n-1} or ω_F^2 from Chapter 24 recovers exactly the flux integrals of Chapter 23. Those were, all along, integrals of forms in this sense. And Definition 25.a.1 is what makes the motivating reparametrization-invariance calculation precise: $\int_L \omega$ does not depend on the parametrization ϕ chosen, only on L and its orientation.

AI-Note 25.3 (*Integrating over non-graphical submanifolds via partitions of unity*): When a submanifold $\mathcal{M} \subseteq \mathbb{R}^n$ cannot be covered by a single local parametrization chart (for instance, a sphere S^2 or a compact manifold with boundary), one cannot directly evaluate $\int_{\mathcal{M}} \omega$ using a single coordinate domain U.

The solution is to decompose any continuous function f on $\mathcal{M}$ into a sum of functions $f = \sum_{l=1}^{N} \eta_l f$, each supported inside a single coordinate chart domain U_l. Integrating each term $\eta_l f$ separately and summing the results yields the total integral over $\mathcal{M}$. The key technical tool for this decomposition is a "*smooth partition of unity*" (German: „*glatte Zerlegung der Eins*").

Definition 25.b.6 (*Support of a function*): Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$ a continuous function. The "*support*" (German: „*Träger*") of f is the closure of the set where f is non-zero:

$$\operatorname{supp}(f) := \overline{\{x \in U \mid f(x) \neq 0\}}.$$

If $\operatorname{supp}(f)$ is compact, that is, bounded and with no limit point on the boundary of U, we say that f is "*compactly supported*" (German: „*kompakt getragen*").

Example 25.b.7 (*Supports of common functions*):

(a) For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sin(x)$, the function is non-zero on $\mathbb{R} \setminus \pi\mathbb{Z}$. The closure of this set is all of $\mathbb{R}$, so $\operatorname{supp}(f) = \mathbb{R}$. This function is not compactly supported.

(b) Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) = 0$ for $x \leq 0$ and $g(x) = e^{-1/x}$ for $x > 0$. The set where $g(x) \neq 0$ is $(0, \infty)$. Its closure is $[0, \infty)$. Thus $\operatorname{supp}(g) = [0, \infty)$. This is also not compactly supported (since it is not bounded).

Definition 25.b.8 (*Standard smooth bump function*): The canonical C^∞ "*bump function*" (German: „*Höckerfunktion*") $\psi : \mathbb{R}^n \to \mathbb{R}$ supported on the unit ball $B_1(0)$ is defined by

$$\psi(x) := \begin{cases} \exp\left(-\dfrac{1}{1 - |x|^2}\right) & \text{if } |x| < 1, \\ 0 & \text{if } |x| \geq 1. \end{cases}$$

Since all higher derivatives of $t \mapsto \exp(-1/(1-t))$ approach 0 faster than any polynomial as $t \to 1^-$, $\psi \in C^\infty(\mathbb{R}^n)$ with $\operatorname{supp}(\psi) = \overline{B_1(0)}$.

Proposition 25.b.9 (*Existence of a smooth partition of unity subordinate to a finite cover*): Let $K \subseteq \mathbb{R}^n$ be compact, and let $\{B_{r_l}(p_l)\}_{l=1}^N$ be a finite open cover of K. Then there exist an open set $U \supset K$ and $N+1$ non-negative smooth functions $\eta_1, \ldots, \eta_N, \tilde{\eta} \in C^\infty(\mathbb{R}^n, [0,1])$ such that:

(a) $\tilde{\eta}(x) + \sum_{l=1}^{N} \eta_l(x) = 1$ for all $x \in \mathbb{R}^n$;

(b) $\operatorname{supp}(\eta_l) \subseteq B_{r_l}(p_l)$ for each $l \in \{1, \ldots, N\}$;

(c) $\operatorname{supp}(\tilde{\eta}) \subseteq \mathbb{R}^n \setminus U$.

In particular, on the neighborhood U of K, $\tilde{\eta} \equiv 0$ so that $\sum_{l=1}^{N} \eta_l(x) = 1$.

Proof:

Step 1: shrink the balls. The function $x \mapsto \max_l \big(r_l - |x - p_l|\big)$ is continuous and positive on K, because every point of K lies in some $B_{r_l}(p_l)$. On the compact K it therefore has a positive minimum δ. Every $x \in K$ has some l with $|x-p_l| \le r_l - \delta$. Discard the balls with $r_l \le \delta/2$ and set $\eta_l := 0$ for them: no point of K needs them, since for those l the inequality would force $|x-p_l| < 0$. For the remaining balls put $\rho_l := r_l - \delta/2 > 0$ and $\sigma_l := r_l - \delta/4$, so that $0 < \rho_l < \sigma_l < r_l$ and still $K \subseteq U := \bigcup_l B_{\rho_l}(p_l)$, an open set. From here on, l ranges over the remaining balls only.

Step 2: one bump per ball. With ψ from Definition 25.b.8, let $\varphi_l(x) := \psi\big((x-p_l)/\sigma_l\big)$. Each φ_l is smooth, non-negative, positive exactly on $B_{\sigma_l}(p_l)$, and has support $\overline{B_{\sigma_l}(p_l)} \subseteq B_{r_l}(p_l)$. Their sum $\Phi := \sum_l \varphi_l$ is positive on the compact set $\overline{U} = \bigcup_l \overline{B_{\rho_l}(p_l)}$, so $\Phi \ge c$ there for some $c > 0$.

Step 3: divide, but only where it is safe. Choose a smooth $h : \mathbb{R} \to [0,1]$ with $h(s) = 1$ for $s \le c/2$ and $h(s) = 0$ for $s \ge c$. For instance, with g from Example 25.b.7**(b)**, $h(s) := g(c-s)/\big(g(c-s) + g(s-c/2)\big)$; the denominator never vanishes because at least one of $c - s > 0$ and $s - c/2 > 0$ holds. Now set

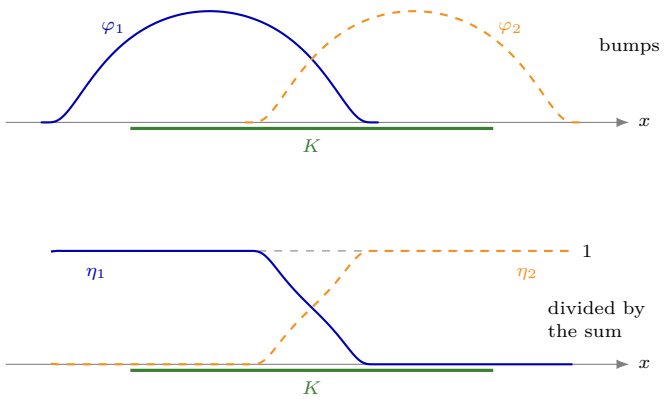
$$\tilde\eta := h \circ \Phi, \qquad \eta_l := \begin{cases} (1 - h(\Phi))\,\dfrac{\varphi_l}{\Phi} & \text{where } \Phi > 0, \\ 0 & \text{where } \Phi < c/2. \end{cases}$$

The two cases overlap only where $0 < \Phi < c/2$, and there $1 - h(\Phi) = 0$, so they agree. Each is smooth on an open set, and the two open sets cover $\mathbb{R}^n$, so η_l is smooth. All values lie in $[0,1]$.

(a) Where $\Phi > 0$: $\tilde\eta + \sum_l \eta_l = h(\Phi) + (1 - h(\Phi)) \cdot \Phi/\Phi = 1$. Where $\Phi = 0$: $\tilde\eta = h(0) = 1$ and every $\eta_l = 0$.

(b) η_l vanishes wherever φ_l does, so $\operatorname{supp}\eta_l \subseteq \operatorname{supp}\varphi_l \subseteq B_{r_l}(p_l)$.

(c) On U we have $\Phi \ge c$, hence $\tilde\eta = 0$. So $\{\tilde\eta \ne 0\}$ lies in the closed set $\mathbb{R}^n \setminus U$, and so does its closure. **Q.E.D.**



Exercise 25.b.10 (*Change of variables with differential forms* (important)): Let $U, V \subseteq \mathbb{R}^n$ be open sets, and let $\psi = (\psi^1, \dots, \psi^n) : U \to V$ be a diffeomorphism with $\det\Big(\frac{\partial \psi^j}{\partial x^i}\Big) > 0$. Define an n-differential form ω on U by $\omega = d\psi^1 \wedge d\psi^2 \wedge \cdots \wedge d\psi^n$.

(a) Show that $\omega = \det\Big(\frac{\partial \psi^j}{\partial x^i}\Big)\, dx^1 \wedge \cdots \wedge dx^n$.

(b) From the relation in (a), rewrite the change of variables formula using differential forms.

Info: *This exercise is Problem 12.9 of Problem Sheet 12, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 384*

AI-Exercise 25.b.11 (*Integrating $dx \wedge dy$ over a hemisphere*)**:** Let $D := \{(u,v) : u^2+v^2 < 1\}$ and let S^+ and S^- be the open upper and lower halves of the unit sphere in $\mathbb{R}^3$, each oriented by the **outward** normal. Consider the 2-form $dx \wedge dy$ on $\mathbb{R}^3$.

(a) Parametrize S^+ as a graph, $\phi(u,v) := \big(u,\ v,\ \sqrt{1-u^2-v^2}\big)$ over D. Compute $\phi^*(dx \wedge dy)$ and deduce $\int_{S^+} dx \wedge dy$.

(b) Do the same for S^- with $\psi(u,v) := \big(u,\ v,\ -\sqrt{1-u^2-v^2}\big)$. The pullback is the same as in **(a)**, yet the answer is not. Which hypothesis of Definition 25.b.4 accounts for the difference?

(c) Deduce $\int_{S^2} dx \wedge dy$, and confirm the value a second time by writing $dx \wedge dy = d(x\,dy)$ and applying Theorem 26.d.31.

Solution: *See solution on Page 385*

Orientable manifolds (Section 25.c)

AI-Note 25.4: The two definitions below are those of the lecture script (Definitions 14.68 and 14.65).

Definition 25.c.12 (*Support of a k-form*)**:** Let $U \subseteq \mathbb{R}^n$ be open and $\alpha \in \Omega^k(U)$. The "*support*" of α is the closed set

$$\operatorname{supp}\alpha := \overline{\{x \in U : \alpha_x \neq 0\}} \subseteq \overline{U}.$$

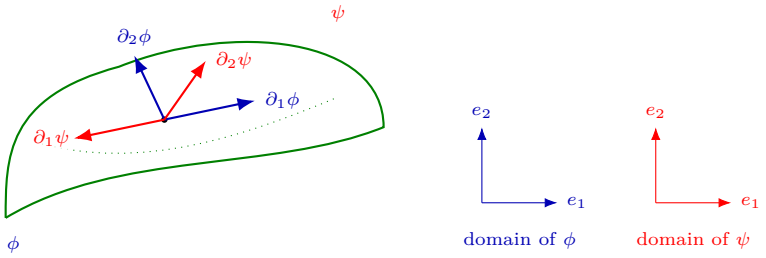
We say α has "*compact support*" in U if $\operatorname{supp}\alpha$ is bounded and contained in U.

Definition 25.c.13 (*Orientable manifold*)**:** Let $M \subseteq \mathbb{R}^n$ be a k-dimensional C^∞ submanifold. We say M is "*orientable*" (German: „*orientierbar*“) if there exists a family of local parametrizations $\phi_i : V_i \to M \cap U_i$, $i \in I$, whose images cover M, and such that for every $i,j \in I$ with $M \cap U_i \cap U_j \neq \emptyset$, the transition map $\widetilde{\phi}_i^{-1} \circ \widetilde{\phi}_j : \widetilde{V}_j \to \widetilde{V}_i$ has positive Jacobian determinant:

$$\det D(\widetilde{\phi}_i^{-1} \circ \widetilde{\phi}_j) > 0 \qquad \text{on } \widetilde{V}_j.$$

Here the tilde denotes restriction to the overlap: $\widetilde{V}_j := \phi_j^{-1}(M \cap U_i \cap U_j)$, $\widetilde{\phi}_j := \phi_j|_{\widetilde{V}_j}$, and likewise for i. Such a family $\{\phi_i\}$ is called an "*atlas*" of M; if it satisfies the determinant condition above it is called an "*orientation*" of M. Once such a family is chosen, M is said to be "*oriented*".

The figure below shows how the orientation condition fails: two overlapping local parametrizations ϕ, ψ of the same patch of a surface, whose coordinate frames $(\partial_1\phi, \partial_2\phi)$ and $(\partial_1\psi, \partial_2\psi)$ at a shared point disagree in handedness.



Remark 25.c.14 (*An atlas that is not an orientation*)**:** At the shared point, $(\partial_1\phi, \partial_2\phi)$ turns counterclockwise while $(\partial_1\psi, \partial_2\psi)$ turns clockwise, so the two frames are mirror images of each other, and $\det(D(\phi^{-1} \circ \psi)) < 0$ there. This is exactly the atlas $\{\phi, \psi\}$ failing Definition 25.c.13: it covers the patch, but its transition map has **negative** Jacobian determinant there, so $\{\phi, \psi\}$ is an atlas but not an orientation.

AI-Note 25.5: Exercise 26.e.44 and Exercise 12.6 (the latter not reproduced below, see the priority table) invoke "*Stokes' theorem*" in its classical $\mathbb{R}^3$ form, which this book derives only in Section 26.d.3, as a consequence of the general theorem. It is recorded here ahead of that, since the exercise-sheet solutions rely on it.

Theorem 25.c.15 (*Stokes' theorem, classical form in $\mathbb{R}^3$*): Let $S \subseteq \mathbb{R}^3$ be a compact, oriented C^1 surface with boundary, oriented by a continuous unit normal field ν, and let F be a C^1 vector field on a neighbourhood of S. Let ∂S carry the induced orientation, given by the unit tangent τ for which, walking along ∂S in the direction of τ with ν pointing up, S lies to the left (the *"right-hand rule"*). Then

$$\int_S \langle \operatorname{curl} F, \nu \rangle \, d\operatorname{vol}_2 = \int_{\partial S} \langle F, \tau \rangle \, d\operatorname{vol}_1, \qquad \text{in physics notation} \qquad \int_S \operatorname{curl} \vec{F} \cdot d\vec{A} = \int_{\partial S} \vec{F} \cdot d\vec{s}.$$

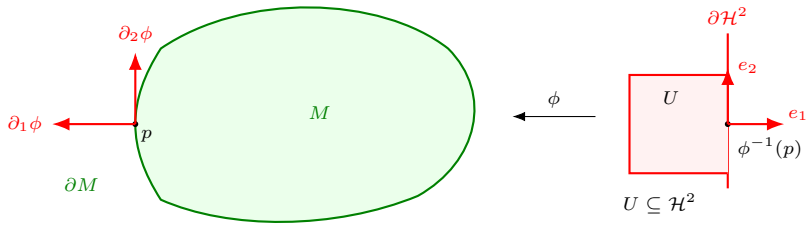
Submanifolds with boundary and induced orientation (Section 25.d)

Recall from Theorem 17.a.9 that near each point p a submanifold M is the image of a smooth map $\phi : U \to V \cap M$ from an open $U \subseteq \mathbb{R}^k$, with $D\phi_x$ injective and ϕ a homeomorphism onto $V \cap M$. The definition below extends this to submanifolds with boundary.

Definition 25.d.16 (*Submanifold with boundary*): For a k-dimensional submanifold **with boundary**, we require similarly that for a local parametrization $\phi : U \to V \cap M$ as in Theorem 17.a.9, either $U \subseteq \mathbb{R}^k$ is open, or

$$U = U' \cap \mathcal{H}^k, \qquad U' \subseteq \mathbb{R}^k \text{ open}, \qquad \mathcal{H}^k := \{(x_1, \dots, x_k) \in \mathbb{R}^k : x_1 \le 0\}$$

the (closed) "*left half-space*". If $p \in \partial M$ and $\phi : U \subseteq \mathcal{H}^k \to V \cap M$ is a local parametrization of M around p, then $\phi|_{\partial \mathcal{H}^k}$ (identifying $\partial \mathcal{H}^k = \{0\} \times \mathbb{R}^{k-1} \cong \mathbb{R}^{k-1}$) is a local parametrization of ∂M around p.



Definition 25.d.17 (*Induced orientation*): If $\{\partial_1\phi, \dots, \partial_k\phi\}$ is the oriented basis of T_pM given by a local parametrization ϕ around $p \in \partial M$, then $\{\partial_2\phi, \dots, \partial_k\phi\}$ is the "*induced*" oriented basis of $T_p(\partial M)$ around p.

Example: cylinder (Subsection 25.d.1)

Consider the cylindrical surface

$$C = \{(x, y, z) \in \mathbb{R}^3 : y^2 + z^2 = r^2, \ 0 \le x \le L\},$$

a 2-dimensional submanifold with boundary, and the two parametrizations $\phi, \psi : [-L, 0] \times [0, 2\pi] \to C$ (both domains $\subseteq \mathcal{H}^2$),

$$\phi(x, \theta) = \begin{pmatrix} -x \\ r\cos\theta \\ -r\sin\theta \end{pmatrix}, \qquad \psi(x, \theta) = \begin{pmatrix} x + L \\ r\cos\theta \\ r\sin\theta \end{pmatrix}.$$

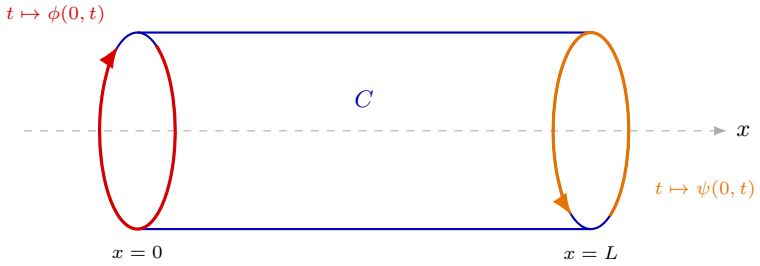
Both cover all of C: as the local x-coordinate ranges over $[-L, 0]$, $-x$ and $x + L$ both range over $[0, L]$.

The two are positively oriented with respect to each other: writing out the composition,

$$\phi^{-1} \circ \psi(x, \theta) = \phi^{-1}(x + L,\, r\cos\theta,\, r\sin\theta) = (-x - L, -\theta) = -(x + L, \theta),$$

so $D(\phi^{-1} \circ \psi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\det D(\phi^{-1} \circ \psi) = 1 > 0$: $\{\phi, \psi\}$ is a valid orientation of C by Definition 25.c.13.

But restricted to the line $\partial\mathcal{H}^2 = \{(x, \theta) : x = 0\}$, ϕ and ψ parametrize **opposite** boundary components of the cylinder. Here $\phi(0, \cdot)$ traces the circle at $x = 0$ and $\psi(0, \cdot)$ the circle at $x = L$, and Definition 25.d.17 gives them opposite rotational senses when both are drawn in the same (y, z)-view.



one cylinder, both rims seen from the same side:
the two induced senses turn opposite ways

Remark 25.d.18 (*Why the opposite senses are not a contradiction*): Nothing has gone wrong. The pair $\{\phi, \psi\}$ is a perfectly good orientation of C as a whole, since the overlap check above is exactly what Definition 25.c.13 demands and it passes. What the picture shows is that a **consistently** oriented cylinder induces boundary orientations on its two ends that turn opposite ways when both circles are viewed from the same side.

That is the familiar fact from the divergence theorem, in another costume: the outward normal at the two ends of a solid cylinder points in opposite x-directions. Induced orientation is doing its job here, not failing at it.

Solutions (Section 25.e)

Solution of Exercise 25.b.10:

(a) Writing each $d\psi^j$ in the coordinate basis, $d\psi^j = \sum_{i=1}^n \frac{\partial \psi^j}{\partial x^i}\, dx^i$, so

$$\omega = \left(\sum_{i_1=1}^n \frac{\partial \psi^1}{\partial x^{i_1}} dx^{i_1}\right) \wedge \cdots \wedge \left(\sum_{i_n=1}^n \frac{\partial \psi^n}{\partial x^{i_n}} dx^{i_n}\right) = \sum_{i_1,\ldots,i_n} \frac{\partial \psi^1}{\partial x^{i_1}} \cdots \frac{\partial \psi^n}{\partial x^{i_n}}\, dx^{i_1} \wedge \cdots \wedge dx^{i_n}$$

by bilinearity of $\wedge$ (property **(a)** of Definition 24.a.6). Any term with a repeated index vanishes, since $dx^i \wedge dx^i = 0$ by antisymmetry, so only tuples $(i_1, \ldots, i_n) = (\sigma(1), \ldots, \sigma(n))$ for a permutation $\sigma \in S_n$ contribute; using $dx^{\sigma(1)} \wedge \cdots \wedge dx^{\sigma(n)} = \operatorname{sgn}(\sigma)\, dx^1 \wedge \cdots \wedge dx^n$,

$$\omega = \left(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \frac{\partial \psi^1}{\partial x^{\sigma(1)}} \cdots \frac{\partial \psi^n}{\partial x^{\sigma(n)}}\right) dx^1 \wedge \cdots \wedge dx^n.$$

The coefficient is exactly the Leibniz expansion of $\det(D\psi)$, so

$$\boxed{d\psi^1 \wedge \cdots \wedge d\psi^n = \det(D\psi)\, dx^1 \wedge \cdots \wedge dx^n.}$$

(b) Let $\eta = f(y)\, dy^1 \wedge \cdots \wedge dy^n$ be an n-form on V. Its pullback under ψ is $\psi^*\eta = f(\psi(x))\, d\psi^1 \wedge \cdots \wedge d\psi^n$, which by part **(a)** equals $f(\psi(x)) \det(D\psi(x))\, dx^1 \wedge \cdots \wedge dx^n$. By Definition 25.b.4

(applied with U itself as the n-dimensional submanifold),

$$\int_U \psi^*\eta = \int_U f(\psi(x)) \det(D\psi(x))\, dx.$$

Since $\det(D\psi) = |\det(D\psi)| > 0$ by hypothesis, the right-hand side is exactly the ordinary change-of-variables formula, so

$$\int_U \psi^*\eta = \int_U f(\psi(x))\, |\det(D\psi(x))|\, dx = \int_V f(y)\, dy = \int_V \eta.$$

Thus $\boxed{\int_U \psi^*\eta = \int_V \eta}$: the change-of-variables formula is exactly the statement that integration of forms is invariant under pullback by an orientation-preserving diffeomorphism.

Q.E.D.

Solution to AI-Exercise 25.a.3:

(a) The pullback of a coordinate differential is the differential of the corresponding component, so from $x = r\cos\theta$ and $y = r\sin\theta$,

$$\Phi^*(dx) = \cos\theta\, dr - r\sin\theta\, d\theta, \qquad \Phi^*(dy) = \sin\theta\, dr + r\cos\theta\, d\theta.$$

Wedging, and using $dr \wedge dr = d\theta \wedge d\theta = 0$ together with $d\theta \wedge dr = -\, dr \wedge d\theta$,

$$\begin{aligned}\Phi^*(dx \wedge dy) &= \big(\cos\theta \cdot r\cos\theta\big)\, dr \wedge d\theta + \big(-\, r\sin\theta \cdot \sin\theta\big)\, d\theta \wedge dr \\ &= r\big(\cos^2\theta + \sin^2\theta\big)\, dr \wedge d\theta = r\, dr \wedge d\theta.\end{aligned}$$

The factor is exactly the r tabulated for polar coordinates in Remark 20.a.8, obtained here without computing a determinant: the antisymmetry of the wedge did the cofactor expansion.

(b) They measure different things, and the absolute value is the difference. Theorem 20.a.1 is a statement about **volume**, which is unsigned: a region and its mirror image have the same area, so only $|\det \mathrm{J}\,\Phi|$ can appear. The pullback of a top-degree form is a statement about **oriented** volume, which carries a sign, and it records $\det \mathrm{J}\,\Phi$ itself.

The swap $\Phi(u_1, u_2) = (u_2, u_1)$ separates them. Its Jacobian is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with determinant -1, so

$$\Phi^*(dx \wedge dy) = du_2 \wedge du_1 = -\, du_1 \wedge du_2,$$

while $|\det \mathrm{J}\,\Phi| = 1$ and the change of variables formula does not change at all. The form notices the reversal; the volume cannot.

(c) Exercise 25.b.10 concludes that $\int_U \psi^*\eta = \int_V \eta$, with no sign. Drop the hypothesis $\det > 0$ and take ψ to be the swap above: the pullback acquires a factor -1, so the two integrals differ by a sign and the conclusion is false as stated. The honest statement without the hypothesis carries $\operatorname{sgn}(\det \mathrm{J}\,\psi)$.

Theorem 20.a.1 is untroubled by the same map because it never claimed anything about orientation. This is the precise sense in which the positivity hypothesis in Exercise 25.b.10 is not a technical convenience: it is the price of upgrading an unsigned identity to a signed one.

Q.E.D.

Solution to AI-Exercise 25.b.11:

(a) Along ϕ the first two coordinates are the parameters themselves, $x = u$ and $y = v$, so $\phi^*(dx) = du$ and $\phi^*(dy) = dv$, giving

$$\phi^*(dx \wedge dy) = du \wedge dv.$$

The third coordinate never enters, which is the whole reason this integral is easy. It remains to check that ϕ induces the outward orientation: for a graph $z = g(u, v)$,

$$\partial_u\phi \times \partial_v\phi = (-g_u,\ -g_v,\ 1),$$

whose third component is $+1$, so the induced normal points **upward**, and on the upper hemisphere upward is outward. Hence

$$\int_{S^+} dx \wedge dy = \int_D du \wedge dv = \mathrm{vol}_2(D) = \pi.$$

(b) The pullback computation is identical, since ψ also has $x = u$ and $y = v$, so again $\psi^*(dx \wedge dy) = du \wedge dv$. But ψ is still a graph, so its induced normal still points upward, and on the **lower** hemisphere upward is **inward**. So ψ is orientation-reversing for the prescribed outward orientation, and Definition 25.b.4 requires a parametrization compatible with the chosen orientation. Correcting by the sign,

$$\int_{S^-} dx \wedge dy = -\int_D du \wedge dv = -\pi.$$

(c) The equator is a null set, so the two halves add:

$$\int_{S^2} dx \wedge dy = \pi + (-\pi) = 0.$$

For the second route, $d(x\,dy) = dx \wedge dy$, and S^2 is a compact orientable 2-manifold with $\partial S^2 = \emptyset$, so Theorem 26.d.31 gives

$$\int_{S^2} dx \wedge dy = \int_{S^2} d(x\,dy) = \int_{\partial S^2} x\,dy = 0.$$

Remark 25.e.19 (*$dx \wedge dy$ measures the shadow*): Part **(a)** says something worth keeping: integrating $dx \wedge dy$ over an oriented surface in $\mathbb{R}^3$ returns the **signed** area of its shadow on the xy-plane. The hemisphere has area 2π, but its shadow is the unit disc, and π is what the computation returns. A 2-form does not measure surface area; the square root of the Gram determinant of Chapter 21 does that, and the two answers differ by exactly the tilt of the surface.

Read that way, part **(c)** is obvious before any computation. The sphere's two halves cast the same shadow, and a closed surface presents each patch of shadow once from above and once from below, so everything cancels. That is the geometric content of $\int_M d\omega = 0$ on a closed M, and Question C.b.27 draws the other standard consequence from it.

Q.E.D.

Stokes' and Green's Theorems (Chapter 26)

Everything in this part converges here. Stokes' theorem says that integrating $d\omega$ over a submanifold is the same as integrating ω over its boundary. That one statement contains the fundamental theorem of calculus, Green's theorem, the divergence theorem and the classical Stokes theorem as special cases. The chapter takes the special cases first, where the content is visible, then the general theorem, then what it buys. Among other things it buys a formula computing an area from a boundary integral alone.

Sneak peek at Stokes' theorem (Section 26.a)

Before the general theorem of Section 26.d, a few examples build intuition for its classical form (Theorem 25.c.15): for a surface $A \subset \mathbb{R}^3$ and a vector field F,

$$\int_A (\operatorname{curl} F) \cdot \nu \, d\operatorname{vol}_2 = \int_{\partial A} F \cdot \tau \, dL,$$

where ν is the unit normal to A and τ is the unit tangent vector to the boundary curve ∂A.

Example 26.a.1 (*Stokes' theorem on a hemisphere: a quadratic field*): Let $A := \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 1,\ z \geq 0\}$ be the upper hemisphere. Given the vector field $F(x,y,z) := (z^2, x^2, y^2)^\mathsf{T}$, we can evaluate both sides of Stokes' theorem.

(a) The rotation of F: $\operatorname{curl} F = \nabla \times F = (2y, 2z, 2x)^\mathsf{T}$.

(b) The surface normal: On the unit sphere, the outward normal is simply $\nu(x,y,z) = (x,y,z)^\mathsf{T}$.

(c) The boundary ∂A: The boundary of the upper hemisphere is the unit circle in the xy-plane, $x^2+y^2 = 1$ with $z = 0$. This can be parametrized by the closed path $\gamma : [0, 2\pi] \to \mathbb{R}^3$,

$$\gamma(t) = \begin{pmatrix} \cos t \\ \sin t \\ 0 \end{pmatrix}.$$

Evaluating the path integral (the right-hand side of Stokes' theorem):

$$\int_{\partial A} F \cdot \tau \, dL = \int_0^{2\pi} F(\gamma(t)) \cdot \gamma'(t) \, dt.$$

Since $z = 0$ on γ, we have $F(\gamma(t)) = (0, \cos^2 t, \sin^2 t)^\mathsf{T}$. The derivative is $\gamma'(t) = (-\sin t, \cos t, 0)^\mathsf{T}$.

$$\int_0^{2\pi} \begin{pmatrix} 0 \\ \cos^2 t \\ \sin^2 t \end{pmatrix} \cdot \begin{pmatrix} -\sin t \\ \cos t \\ 0 \end{pmatrix} dt = \int_0^{2\pi} \cos^3 t \, dt = 0.$$

The surface side agrees without a parametrization. Its integrand is $\langle \operatorname{curl} F, \nu \rangle = 2(xy + yz + zx)$, and each of the three terms changes sign under $x \mapsto -x$ or under $y \mapsto -y$, reflections that map the hemisphere to itself, so each integrates to 0 over A.

Example 26.a.2 (*Path integral using Stokes' theorem*): Compute $\oint_\Gamma F \cdot \tau \, dL$ where $F(x,y,z) := (x,y,z)^\mathsf{T}$ and Γ is the unit circle in the xy-plane (oriented counter-clockwise).

Instead of computing the line integral directly, we can use Stokes' theorem. The curve Γ bounds the unit disk $A = \{(x,y,0) \mid x^2+y^2 \leq 1\}$. First, we compute the rotation of F:

$$\operatorname{curl} F = \nabla \times F = \begin{pmatrix} \partial_y(z) - \partial_z(y) \\ \partial_z(x) - \partial_x(z) \\ \partial_x(y) - \partial_y(x) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

By Stokes' theorem applied to the disk A (or to any other surface with boundary Γ, such as the upper hemisphere), the line integral equals a surface integral:

$$\oint_\Gamma F \cdot \tau \, dL = \int_A (\operatorname{curl} F) \cdot \nu \, d\operatorname{vol}_2 = 0.$$

Green's theorem and conservative fields (Section 26.b)

Line integrals (Wegintegrale) (Subsection 26.b.1)

Definition 26.b.3 (*Work / Line integral*): Let $U \subseteq \mathbb{R}^n$ be an open set, $F : U \to \mathbb{R}^n$ a continuous vector field, and $\gamma : [a,b] \to U$ a piecewise C^1 path. We define the "*work*" (or "*line integral*") of F along γ as

$$\int_\gamma F \cdot d\gamma := \int_a^b F(\gamma(t)) \cdot \gamma'(t) \, dt.$$

The work measures how much the vector field *"pushes"* along the path.

Exercise 26.b.4 (*Work adds when two paths are concatenated* (semi-important)): Let $U \subseteq \mathbb{R}^n$ be open and connected and let $\gamma, \sigma \in C^1([0,1], U)$ be paths with $\gamma(0) = \sigma(1)$. Define the "*concatenation of σ and γ*" (German: „*Verkettung*") as the path

$$(\gamma * \sigma)(t) := \begin{cases} \sigma(2t) & \text{for } 0 \le t \le \frac{1}{2}, \\ \gamma(2t-1) & \text{for } \frac{1}{2} \le t \le 1. \end{cases}$$

Prove rigorously that for every $F \in C^1(U, \mathbb{R}^n)$,

$$\int_{\gamma * \sigma} F \cdot d(\gamma * \sigma) = \int_\gamma F \cdot d\gamma + \int_\sigma F \cdot d\sigma.$$

Explain also what the formula says about physical work.

> **Info:** *This exercise is Problem 12.1 of Problem Sheet 12, Analysis II, Spring Semester 2026, where the closing question is marked as an extra worth no points. The FS 2024 edition of the course sets it as the bonus problem of its own Sheet 12.*

Solution: *See solution on Page 404*

AI-Note 26.1: Read the definition before trusting the name: $\gamma * \sigma$ traverses $\boldsymbol{\sigma}$ **first** and γ second, which is why the compatibility condition is $\gamma(0) = \sigma(1)$ rather than the other way round. The ordering is the one familiar from composition of maps, where $f \circ g$ applies g first, and the sheet's own phrase *"concatenation of σ and γ"* names them in the order they are traversed even though the symbol does not. Nothing below depends on the convention (the sum on the right is symmetric), but a diagram drawn with the paths the wrong way round will not close up.

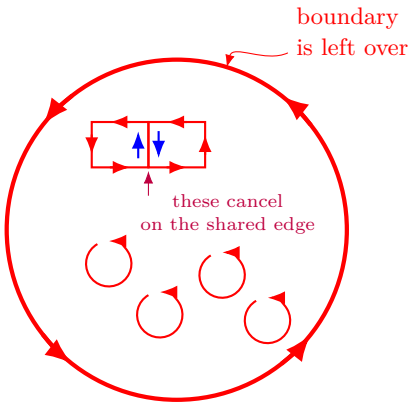
Green's theorem (Subsection 26.b.2)

Theorem 26.b.5 (*Green's Theorem*): Let $F : U \to \mathbb{R}^2$ be a continuously differentiable vector field on an open set $U \subseteq \mathbb{R}^2$. Then, for any bounded C^1 domain Ω (Definition 22.a.1) with $\overline{\Omega} \subseteq U$,

$$\int_\Omega \operatorname{curl} F \, d\operatorname{vol}_2 = \int_{\partial\Omega} F \cdot \tau \, dL,$$

where τ is the unit tangent vector to the boundary $\partial\Omega$ oriented so that Ω lies to its left, which is counter-clockwise on the outer boundary curve (i.e. $\tau = (-\nu_2, \nu_1)^\mathsf{T}$, with ν being the outward unit normal).

The theorem is used from here on and proved in Section 26.d.2, where it falls out of the general Stokes theorem in three lines. The picture below shows why it should be true.



AI-Note 26.2:

We integrate the rotation inside the domain, cell by cell. Every interior edge is traversed twice, once by each of the two cells that share it and in opposite directions, so its contributions cancel. Only the boundary is left over in the end!

Example 26.b.6 (*Area of a region via Green's theorem*): Consider a region $\Omega \subseteq \mathbb{R}^2$ defined by the region inside the curve $x^{2/3} + y^{2/3} = 1$ in the first quadrant, bounded by the axes. The curve can be parametrized as $\varphi(t) = (\cos^3 t, \sin^3 t)^\mathsf{T}$ for $t \in [0, \pi/2]$.

The area is $\mathrm{vol}_2(\Omega) = \int_\Omega 1 \, d\mathrm{vol}_2$. We look for a vector field F such that $\operatorname{curl} F = \partial_x F_2 - \partial_y F_1 \equiv 1$ on Ω. A simple choice is $F(x, y) := (0, x)^\mathsf{T}$. By Green's theorem, we have:

$$\mathrm{vol}_2(\Omega) = \int_\Omega \operatorname{curl} F \, d\mathrm{vol}_2 = \int_{\partial\Omega} F \cdot \tau \, dL = \int_0^{\pi/2} F(\varphi(t)) \cdot \varphi'(t) \, dt.$$

Since $\tau(\varphi(t)) \, dL = \varphi'(t) \, dt$ along the parametrization, we compute the dot product:

$$F(\varphi(t)) \cdot \varphi'(t) = \begin{pmatrix} 0 \\ \varphi_1(t) \end{pmatrix} \cdot \begin{pmatrix} \varphi_1'(t) \\ \varphi_2'(t) \end{pmatrix} = \varphi_1(t)\varphi_2'(t).$$

The two straight pieces of $\partial\Omega$ contribute nothing: on the y-axis $F = 0$, and on the x-axis F is vertical while the boundary runs horizontally. So the area is a single one-dimensional integral,

$$\mathrm{vol}_2(\Omega) = \int_0^{\pi/2} \cos^3 t \cdot 3 \sin^2 t \cos t \, dt = 3 \int_0^{\pi/2} \cos^4 t \, \sin^2 t \, dt = 3 \cdot \frac{\pi}{32} = \frac{3\pi}{32},$$

a quarter of the area $3\pi/8$ enclosed by the whole astroid.

Example 26.b.7 (*Line integral via Green's theorem*): We want to compute the work of $F(x, y) := (0, xy^2)^\mathsf{T}$ along the ellipse γ given by $x^2 + (y/2)^2 = 1$, oriented counter-clockwise. Let $\Omega = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2/4 \le 1\}$. By Green's theorem,

$$\int_\gamma F \cdot d\gamma = \int_{\partial\Omega} F \cdot \tau \, dL = \int_\Omega \operatorname{curl} F \, d\mathrm{vol}_2.$$

We have $\operatorname{curl} F = \partial_x(xy^2) - \partial_y(0) = y^2$. We compute the double integral using modified polar coordinates $x = r\cos\theta$, $y = 2r\sin\theta$ with Jacobian $2r$:

$$\int_\Omega y^2 \, d\mathrm{vol}_2 = \int_0^{2\pi} \int_0^1 (2r\sin\theta)^2 (2r) \, dr \, d\theta = 8 \int_0^{2\pi} \sin^2\theta \, d\theta \int_0^1 r^3 \, dr = 8(\pi)\left(\frac{1}{4}\right) = 2\pi.$$

Example 26.b.8 (*Line integral of a curl-free field*): Let Ω be the unit disk $B_1(0) \subseteq \mathbb{R}^2$ and $F(x, y) := (xy^2, x^2y)^\mathsf{T}$. We compute $\int_{\partial\Omega} F \cdot \tau \, dL$. By Green's theorem, we need to integrate the curl of F over Ω:

$$\operatorname{curl} F = \partial_x(x^2y) - \partial_y(xy^2) = 2xy - 2xy = 0.$$

Thus, the integral is simply $\int_\Omega 0 \, d\mathrm{vol}_2 = 0$.

Example 26.b.9 (*2D Divergence theorem*): Green's theorem can also be viewed as a 2D version of the divergence theorem. Consider the vector field $F(x,y) := (x-y^2, y-x^2)^\mathsf{T}$. We want to compute the outward flux of F through the curve γ given by the graph of $y = \cos x$ for $x \in [-\pi/2, \pi/2]$.

We can close the curve by adding the segment σ on the x-axis from $\pi/2$ to $-\pi/2$, forming a closed region Ω (the area under the cosine bump). By the 2D divergence theorem:

$$\int_\gamma F \cdot \nu \, dL + \int_\sigma F \cdot \nu \, dL = \int_\Omega \operatorname{div} F \, d\operatorname{vol}_2.$$

The divergence is $\operatorname{div} F = \partial_x(x-y^2) + \partial_y(y-x^2) = 1+1 = 2$. The integral of the divergence is $2\operatorname{vol}_2(\Omega) = 2\int_{-\pi/2}^{\pi/2} \cos x \, dx = 2(2) = 4$. For the segment $\sigma(x) = (x,0)$ from $\pi/2$ to $-\pi/2$, the outward normal from Ω is $\nu = (0,-1)^\mathsf{T}$. The flux through σ is:

$$\int_\sigma F \cdot \nu \, dL = \int_{-\pi/2}^{\pi/2} \begin{pmatrix} x \\ -x^2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} dx = \int_{-\pi/2}^{\pi/2} x^2 \, dx = \left[\frac{x^3}{3}\right]_{-\pi/2}^{\pi/2} = \frac{\pi^3}{12}.$$

Therefore, the flux through γ is $4 - \frac{\pi^3}{12}$.

Example 26.b.10 (*The work of a field can depend on the path*): Let $F(x,y) := (-y, x)^\mathsf{T}$, and let $\gamma_0, \gamma_1, \gamma_2 : [0,1] \to \mathbb{R}^2$ be the three paths from $(0,0)$ to $(1,1)$ given by $\gamma_0(t) := (t,t)$ (the diagonal), and

$$\gamma_1(t) := \begin{cases} (2t, 0) & t \in [0, \frac{1}{2}] \\ (1, 2t-1) & t \in [\frac{1}{2}, 1] \end{cases} \qquad \gamma_2(t) := \begin{cases} (0, 2t) & t \in [0, \frac{1}{2}] \\ (2t-1, 1) & t \in [\frac{1}{2}, 1] \end{cases}$$

(right-then-up, and up-then-right, along the sides of the unit square). Direct computation gives three different answers:

$$\int_{\gamma_0} F \cdot d\gamma_0 = \int_0^1 \langle (-t,t), (1,1) \rangle \, dt = 0, \qquad \int_{\gamma_1} F \cdot d\gamma_1 = 1, \qquad \int_{\gamma_2} F \cdot d\gamma_2 = -1.$$

Moving **perpendicular** to F costs no work, moving **with** F does positive work, and moving **against** F does negative work. Since the three paths share endpoints but disagree on the work done, F cannot be conservative (Definition 26.b.11 below): Proposition 26.b.14 would otherwise force all three integrals to agree.

Conservative vector fields (Subsection 26.b.3)

Definition 26.b.11 (*Conservative vector field*): Given a C^k vector field F on an open set $U \subseteq \mathbb{R}^n$, we call it "*conservative*" if there exists a scalar field $f : U \to \mathbb{R}$ of class C^{k+1} such that $F \equiv \nabla f$. The function f is called a "*potential*" for F.

Lemma 26.b.12 (*Integrability conditions*): Let $U \subseteq \mathbb{R}^n$ be open, and let F be a C^1 conservative vector field on U with components $F_1, \ldots, F_n$. Then we necessarily have the integrability conditions:

$$\partial_j F_k = \partial_k F_j \qquad \text{for all } j,k \in \{1,\ldots,n\}.$$

Proof:

Let f be a potential such that $F = \nabla f = (\partial_1 f, \ldots, \partial_n f)^\mathsf{T}$. Then by Schwarz's theorem (symmetry of second derivatives),

$$\partial_j F_k = \partial_j(\partial_k f) = \partial_j \partial_k f = \partial_k \partial_j f = \partial_k(\partial_j f) = \partial_k F_j. \qquad \text{Q.E.D.}$$

Example 26.b.13 (*A non-conservative field*): Not every vector field is conservative. For example, a frictional force is not conservative. Mathematically, consider $F : \mathbb{R}^3 \to \mathbb{R}^3$ given by $F(x,y,z) :=$

$(xz - y, x^2 + z^3, 3xz^3 - xy)^\mathsf{T}$. If F were conservative, it would have to satisfy the integrability conditions (Lemma 26.b.12). But

$$\partial_x F_2 = 2x \quad \text{and} \quad \partial_y F_1 = -1.$$

Since they are not equal, F cannot be conservative.

Proposition 26.b.14 (*Work along a path as difference of potential*): Let $U \subseteq \mathbb{R}^n$ be open, and let $F : U \to \mathbb{R}^n$ be a continuous conservative vector field with potential f. Then for any C^1 path $\gamma : [0,1] \to U$, the line integral of F along γ depends only on the endpoints:

$$\int_\gamma F \cdot d\gamma = f(\gamma(1)) - f(\gamma(0)).$$

Proof:

By definition of the path integral and the chain rule,

$$\begin{aligned}\int_\gamma F \cdot d\gamma &= \int_0^1 F(\gamma(t)) \cdot \gamma'(t)\, dt = \int_0^1 \nabla f(\gamma(t)) \cdot \gamma'(t)\, dt \\ &= \int_0^1 \mathrm{J}\, f(\gamma(t))\ \mathrm{J}\, \gamma(t)\, dt = \int_0^1 \mathrm{J}(f \circ \gamma)(t)\, dt = \int_0^1 (f \circ \gamma)'(t)\, dt.\end{aligned}$$

By the fundamental theorem of calculus, this equals $(f \circ \gamma)(1) - (f \circ \gamma)(0) = f(\gamma(1)) - f(\gamma(0))$. Q.E.D.

Example 26.b.15 (*Line integral of a conservative field*): Consider the vector field $F : [0,1] \times [0,2] \to \mathbb{R}^2$ given by $F(x,y) := (y^2, 2xy)^\mathsf{T}$. First, we show that F is conservative by finding a potential f such that $\nabla f = F$. We need $\partial_x f = y^2$ and $\partial_y f = 2xy$. Integrating the first equation with respect to x gives $f(x,y) = xy^2 + c(y)$. Differentiating with respect to y gives $\partial_y f = 2xy + c'(y)$, which must equal $2xy$. Thus $c'(y) = 0$, and we can choose $c(y) = 0$. The potential is $f(x,y) = xy^2$.

Now let Γ be a closed path in the rectangle, for instance its boundary, parametrized by γ with $\gamma(0) = \gamma(1) = (0,0)^\mathsf{T}$. (The formula for F makes sense on all of $\mathbb{R}^2$, which supplies the open set that Proposition 26.b.14 asks for.) By Proposition 26.b.14, the line integral around this closed loop is simply the difference in potential at the endpoints:

$$\oint_\Gamma F \cdot d\gamma = f(\gamma(1)) - f(\gamma(0)) = f(0,0) - f(0,0) = 0.$$

Example 26.b.16 (*Line integral of a conservative field via potential*): Consider the vector field $F(x,y) := (10x^4 - 2xy^3, -3x^2y^2)^\mathsf{T}$ and a path γ connecting $(0,0)$ to $(2,1)$, for instance, the curve defined by $x^4 - 6xy^3 = 4y^2$.

Since F is defined on all of $\mathbb{R}^2$, which is simply connected, and satisfies $\partial_y F_1 = -6xy^2 = \partial_x F_2$, it is conservative by Corollary 26.c.25 in the next section. Here we need not rely on that, since a potential can be found directly by integration:

$$\begin{aligned}\partial_x f &= 10x^4 - 2xy^3 \implies f(x,y) = 2x^5 - x^2y^3 + c(y), \\ \partial_y f &= -3x^2y^2 + c'(y).\end{aligned}$$

Equating this with $F_2 = -3x^2y^2$, we get $c'(y) = 0$, so we can choose $c(y) = 0$. The potential is $f(x,y) = 2x^5 - x^2y^3$. By Proposition 26.b.14, the work along γ is path-independent and given by:

$$\int_\gamma F \cdot d\gamma = f(2,1) - f(0,0) = (2(32) - 4(1)) - 0 = 60.$$

Notice that the equation of the curve $x^4 - 6xy^3 = 4y^2$ was irrelevant, as long as it connects the two endpoints!

Exercise 26.b.17 (*Choosing parameters for an irrotational field*): Determine real parameters α, β, γ such that the vector field

$$F(x,y,z) := (x + 2y + \alpha z,\ \beta x - 3y - z,\ 4x + \gamma y + 2z)$$

on $\mathbb{R}^3$ becomes irrotational, i.e. $\operatorname{curl}(F) = 0$. Determine, after choosing these parameters, a potential for F.

Proof of Exercise 26.b.17:

Setting $\operatorname{curl} F = 0$ componentwise (Definition 22.b.9):

$$\partial_y F_3 - \partial_z F_2 = \gamma - (-1) \overset{!}{=} 0, \qquad \partial_z F_1 - \partial_x F_3 = \alpha - 4 \overset{!}{=} 0, \qquad \partial_x F_2 - \partial_y F_1 = \beta - 2 \overset{!}{=} 0,$$

giving $\gamma = -1$, $\alpha = 4$, $\beta = 2$, so $F(x,y,z) = (x + 2y + 4z,\ 2x - 3y - z,\ 4x - y + 2z)$. This F is C^1 on the simply connected $\mathbb{R}^3$ and curl-free, hence conservative (the 3-dimensional case of Corollary 26.c.25). Integrating $\partial_x f = x + 2y + 4z$ gives $f = \frac{1}{2}x^2 + 2xy + 4xz + c(y,z)$; matching $\partial_y f = 2x + c_y$ against $F_2 = 2x - 3y - z$ gives $c_y = -3y - z$, so $c = -\frac{3}{2}y^2 - yz + d(z)$; matching $\partial_z f = 4x - y + d'(z)$ against $F_3 = 4x - y + 2z$ gives $d'(z) = 2z$, so $d(z) = z^2$. A potential is

$$f(x,y,z) = \tfrac{1}{2}x^2 + 2xy + 4xz - \tfrac{3}{2}y^2 - yz + z^2.$$

Q.E.D.

Exercise 26.b.18 (*True or False* (the angle field)): Consider the vector field

$$X := \frac{1}{x^2 + y^2}(-y,\ x) \qquad \text{defined on } U := \mathbb{R}^2 \setminus \{(0,0)\},$$

let $V := \{(x,y) \in U \mid x > 0\}$ be the open right half-plane, and let $\gamma : [0,1] \to U$ be a C^1 curve with $\gamma(0) = \gamma(1) = (0,1)$. Decide whether each of the following is true or false, with justification.

(a) There exists $u \in C^1(U)$ with $\nabla u = X$ on U.
(b) There exists $v \in C^1(V)$ with $\nabla v = X$ on V.
(c) $\int_\gamma X \cdot d\gamma = 2\pi$, provided γ restricted to $[0,1)$ is injective.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), where it appears as the eleventh and last block of true/false questions.*

Solution: *See solution on Page 409*

Exercise 26.b.19 (*True or False* (two closed fields against two circles)): Consider the vector fields

$$X := \sin(x_1)\, e_1 \qquad \text{and} \qquad Y := \frac{-x_2 e_1 + x_1 e_2}{x_1^2 + x_2^2},$$

both defined on $U := \mathbb{R}^2 \setminus \{(0,0)\}$, and note that $\partial_1 X_2 = \partial_2 X_1$ and $\partial_1 Y_2 = \partial_2 Y_1$ throughout U. Consider also the two circles

$$\gamma(t) := (10 + \cos t,\ \sin t) \qquad \text{and} \qquad \sigma(t) := (\cos t,\ \sin t), \qquad t \in [0, 2\pi].$$

Decide whether each of the following is true or false, with justification.

(a) There exists $u \in C^1(U)$ with $\nabla u = X$ on U.
(b) There exists $v \in C^1(U)$ with $\nabla v = Y$ on U.
(c) $\int_\gamma X \cdot d\gamma = 0$.
(d) $\int_\sigma Y \cdot d\sigma = 0$.
(e) $\int_\gamma Y \cdot d\gamma = 0$.
(f) $\int_\sigma X \cdot d\sigma = 0$.

Info: *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), where it appears as the eleventh and last block of true/false questions.*

Solution: *See solution on Page 410*

Exercise 26.b.20 (*Box question: a nonconstant conservative field in space*)**:** Give an explicit example of a nonconstant conservative vector field $V \in C^1(U, \mathbb{R}^3)$ on some open set $U \subseteq \mathbb{R}^3$.

> **Info:** *This exercise is taken from the exam of 21 August 2024 (Prof. Joaquim Serra), Box Problem 3, where only the final answer was graded. The exam's formula sheet defines "conservative" as it is defined here: V is conservative if and only if there is $u \in C^1(U)$ with $\nabla u = V$.*

Solution: *See solution on Page 411*

Exercise 26.b.21 (*Single choice: a work integral around a circle off the axis*)**:** Let

$$V(x, y, z) := \begin{pmatrix} 1 \\ x \\ -2 \end{pmatrix}.$$

The work of V along the positively oriented unit circle in the yz-plane centred at the origin, *"positively oriented"* meaning as seen from the positive x-axis, equals exactly one of the following. Which?

(a) 2π.
(b) π.
(c) 0.
(d) -2π.

> **Info:** *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 8.*

Solution: *See solution on Page 413*

Example 26.b.22 (*Differentiating an integral over a moving rectangle*)**:** Fix $a < b$ and $c < d$, and for $t \in \mathbb{R}$ let

$$Q_t := [a + t,\, b + t] \times [c + t,\, d + t]$$

be the rectangle Q_0 translated along the diagonal direction $(1, 1)$.

Claim. For every continuous $f : \mathbb{R}^2 \to \mathbb{R}$ and every $t \in \mathbb{R}$,

$$\frac{d}{dt} \int_{Q_t} f \, d\,\mathrm{vol}_2 = \int_{\partial Q_t} f \, (-dx + dy),$$

the boundary being traversed counterclockwise.

Write the left-hand side as an iterated integral and differentiate the limits. Put

$$H(t, x) := \int_{c+t}^{d+t} f(x, y)\, dy, \qquad \text{so that} \qquad G(t) := \int_{Q_t} f \, d\,\mathrm{vol}_2 = \int_{a+t}^{b+t} H(t, x)\, dx$$

by Fubini (Theorem 20.b.29). The fundamental theorem of calculus applies to H in its second argument, f being continuous, and gives $\partial_t H(t, x) = f(x, d + t) - f(x, c + t)$. Differentiating G combines the two mechanisms (moving limits and a moving integrand) and yields

$$G'(t) = \underbrace{H(t, b + t) - H(t, a + t)}_{\text{the limits move}} + \underbrace{\int_{a+t}^{b+t} \big(f(x, d + t) - f(x, c + t)\big) dx}_{\text{the integrand moves}},$$

that is,

$$G'(t) = \int_{c+t}^{d+t} f(b + t, y)\, dy \; - \int_{c+t}^{d+t} f(a + t, y)\, dy \\ + \int_{a+t}^{b+t} f(x, d + t)\, dx \; - \int_{a+t}^{b+t} f(x, c + t)\, dx.$$

Now read off the right-hand side of the claim. Traversing ∂Q_t counterclockwise, the four edges contribute:

- **Bottom** ($y = c+t$, x increasing): $dy = 0$, so the contribution is $-\int_{a+t}^{b+t} f(x, c+t)\,dx$.
- **Right** ($x = b+t$, y increasing): $dx = 0$, contributing $+\int_{c+t}^{d+t} f(b+t, y)\,dy$.
- **Top** ($y = d+t$, x decreasing): $dy = 0$ and the orientation reverses the sign twice, contributing $+\int_{a+t}^{b+t} f(x, d+t)\,dx$.
- **Left** ($x = a+t$, y decreasing): $dx = 0$, contributing $-\int_{c+t}^{d+t} f(a+t, y)\,dy$.

The four terms are exactly the four in $G'(t)$, which proves the claim.

Remark 26.b.23 (*Green's theorem is the shadow, not the substance*): The combination $-dx + dy$ is the line integral of the vector field $(-f, f)$, so for $f \in C^1$ Green's theorem (Theorem 26.b.5) rewrites the right-hand side as

$$\int_{\partial Q_t} (-f)\,dx + f\,dy = \int_{Q_t} \big(\partial_1 f + \partial_2 f\big)\,d\,\mathrm{vol}_2,$$

and the identity becomes the chain rule: substituting $x = u+t$, $y = v+t$ turns G into $\int_{Q_0} f(u+t, v+t)\,d\,\mathrm{vol}_2(u,v)$, whose t-derivative is exactly $\int_{Q_0}(\partial_1 f + \partial_2 f)(u+t, v+t)$. That is the quickest route, and it is worth knowing which direction $(1,1)$ is doing: the rectangle moves diagonally, so both partial derivatives appear, and a purely horizontal motion would leave only $\partial_1 f$.

But that route needs $f \in C^1$, and the claim assumes only continuity. The argument given above never differentiates f; it differentiates only the **limits** of integration, which the fundamental theorem of calculus permits for a merely continuous integrand. Preferring the longer proof here is not pedantry: the identity is true in a regularity range where Green's theorem has nothing to say.

The statement is a two-dimensional instance of what is called the Reynolds transport theorem, which says that the rate of change of an integral over a moving region is the flux of the integrand through its boundary. Here the region moves rigidly with velocity $(1,1)$, and $-dx + dy$ is precisely the pairing of that velocity with the outward normal.

Info: *This example is taken from the Analysis I & II examination of summer 2016 (Prof. H. Knörrer), Problem 10, where it is set as a proof exercise. The closing remark is added here.*

The Poincaré lemma and simply connected domains (Section 26.c)

Lemma 26.b.12 gives a necessary condition for a potential. When is it also sufficient? Proposition 26.b.14 points the way: a potential exists as soon as the integral of F along a path depends only on the path's endpoints, for then $f(x) := \int_{x_0}^{x} F$ is well defined. The question becomes when two paths with the same endpoints give the same integral. For a field satisfying the integrability conditions, which in the language of Chapter 24 says that the 1-form $\alpha = \sum_k F_k\,dx^k$ is closed, the answer is: whenever one path can be deformed into the other.

Theorem 26.c.24 (*Poincaré Lemma*): Let $U \subseteq \mathbb{R}^n$ be open, and let $\alpha \in \Omega^1(U)$ be a C^1 closed 1-form, that is, $d\alpha = 0$. Let $\gamma_0, \gamma_1 : [0,1] \to U$ be piecewise C^1 paths with the same initial point x_0 and the same endpoint x_1. Suppose that γ_0 and γ_1 are "*homotopic*" in U with fixed endpoints. For simplicity, assume that the homotopy $H : [0,1]^2 \to U$ is smooth. Then

$$\int_{\gamma_0} \alpha = \int_{\gamma_1} \alpha.$$

Proof:

Write $H(s,\cdot)$ for the intermediate paths, so that $H(0,\cdot)=\gamma_0$ and $H(1,\cdot)=\gamma_1$, and pull α back to the square (Definition 25.a.1):

$$H^*\alpha = P\,ds + Q\,dt, \qquad P := \alpha_H(\partial_s H), \quad Q := \alpha_H(\partial_t H).$$

The pullback commutes with d, so $(\partial_s Q - \partial_t P)\,ds\wedge dt = d(H^*\alpha) = H^*(d\alpha) = 0$, that is, $\partial_s Q = \partial_t P$. Now $I(s) := \int_0^1 Q(s,t)\,dt$ is the integral of α along the path $H(s,\cdot)$, so the claim is $I(0)=I(1)$. Differentiating under the integral sign (Theorem 20.c.40),

$$I'(s) = \int_0^1 \partial_s Q(s,t)\,dt = \int_0^1 \partial_t P(s,t)\,dt = P(s,1) - P(s,0).$$

Both terms vanish, because the endpoints are fixed: $H(s,0)=x_0$ and $H(s,1)=x_1$ for every s, so $\partial_s H = 0$ at $t=0$ and at $t=1$. Hence I is constant. Q.E.D.

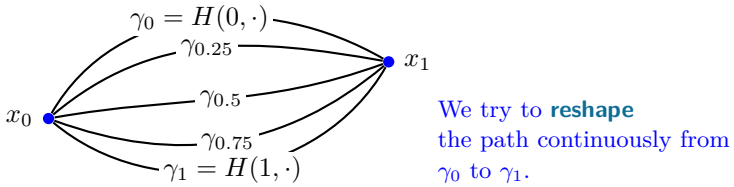
The proof shows what *"closed"* means geometrically. Deforming the path changes the integral only through what happens at its ends, and with the ends pinned there is nothing left to change.

Corollary 26.c.25 (*Existence of potentials in simply connected domains*): Let $U\subseteq\mathbb{R}^n$ be an open "*simply connected*" set. Then every closed C^1 1-form on U is exact. That is, if $\alpha\in\Omega^1(U)$ satisfies $d\alpha=0$, then there exists a function $f\in C^2(U)$ such that $\alpha = df$. Equivalently, if $F\in C^1(U,\mathbb{R}^n)$ satisfies the integrability conditions

$$\partial_j F_k = \partial_k F_j \qquad \text{for all } j,k\in\{1,\dots,n\},$$

then F is conservative: there exists a potential $f\in C^2(U)$ such that $\nabla f = F$.

AI-Note 26.3 (*Intuition on simply connected spaces*): Recall from Definition 8.b.27 that U is simply connected when any two paths in U with the same endpoints are homotopic, or equivalently (Remark 8.b.28) when any closed loop in U can be continuously contracted to a point within U.



Examples of simply connected spaces:

- $\mathbb{R}^n$
- $\mathbb{R}^3\setminus\{(0,0,0)\}$

Examples of spaces that are **not** simply connected:

- $S^1\subseteq\mathbb{R}^2$ (the unit circle)
- $\mathbb{R}^3\setminus \operatorname{span}\{(0,0,1)^\mathsf{T}\}$ (the z-axis removed)
- The torus

This topological property is what allows the integrability conditions to globally imply the existence of a potential.

Proof of Corollary 26.c.25:

Write $\alpha = \sum_i F_i\,dx_i$, so that $\int_\gamma \alpha = \int_\gamma F\cdot d\gamma$ and $d\alpha=0$ is exactly the integrability condition. Fix a base point $x_0\in U$.

Polygonal paths reach everything. Run the proof of Proposition 8.b.24 with *"polygonal path"* (finitely many straight segments, traversed one after another) in place of *"path"*. Every step survives:

the constant path is polygonal, and the only paths the argument appends are straight segments inside a ball. So in a connected open set any two points are joined by a polygonal path, which is piecewise C^1. As U is path-connected, hence connected, every $x \in U$ is reached from x_0 by such a path γ_x.

The candidate potential is well defined. Put

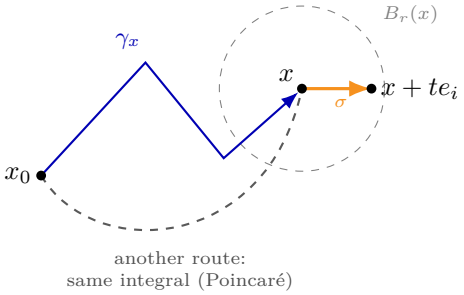
$$f(x) := \int_{\gamma_x} \alpha.$$

Any two piecewise C^1 paths from x_0 to x are homotopic with fixed endpoints, because U is simply connected, so by Theorem 26.c.24 they give the same integral, and $f(x)$ does not depend on which γ_x was chosen. (The lemma as stated assumes a smooth homotopy. That a continuous homotopy between piecewise C^1 paths can be smoothed without leaving the open set U is a standard approximation fact, which we use without proof.)

Its gradient is F. Let $x \in U$, $B_r(x) \subseteq U$, $i \in \{1, \dots, n\}$ and $0 < |t| < r$. A path from x_0 to $x + te_i$ is γ_x followed by the segment $\sigma(s) := x + s\,te_i$, $s \in [0,1]$, which stays in the ball. Since the integral over a concatenation is the sum of the integrals (Exercise 26.b.4),

$$f(x + te_i) - f(x) = \int_\sigma F \cdot d\sigma = \int_0^1 F_i(x + ste_i)\, t\, ds = \int_0^t F_i(x + ue_i)\, du.$$

Dividing by t and letting $t \to 0$, the fundamental theorem of calculus gives $\partial_i f(x) = F_i(x)$. Hence $\nabla f = F$, which is C^1, so $f \in C^2(U)$ and $df = \alpha$. Q.E.D.



Remark 26.c.26 (*Computing a potential in practice*): The proof is also a recipe. On a star-shaped U take straight segments from the centre x_0, so $f(x) = \int_0^1 F\big(x_0 + s(x - x_0)\big) \cdot (x - x_0)\, ds$; on a box, the staircase of coordinate steps used in the proof of Theorem 10.c.14 is often quicker, since each step integrates a single component F_i in a single variable.

Remark 26.c.27 (*Which statement is "the" Poincaré lemma*): In most texts the name **Poincaré lemma** refers to a different-looking statement. A set U is **star-shaped** if it contains a point x_0 such that the segment from x_0 to any point of U lies in U; on a star-shaped open $U \subseteq \mathbb{R}^n$, every closed C^1 k-form is exact, for every $k \geq 1$. How much of that this document covers:

- For $k = 1$ and U convex, it follows from what is proved here. The straight-line homotopy $H(s,t) := (1-s)\gamma_0(t) + s\gamma_1(t)$ stays in U and fixes the endpoints, so U is simply connected and Corollary 26.c.25 applies. A star-shaped set is simply connected as well, by a homotopy routed through x_0, and Remark 26.c.26 then writes the potential down.
- For $k \geq 2$ the statement is true but not proved here. For $k = 2$ on a star-shaped domain in $\mathbb{R}^3$ it says that every divergence-free field there is a curl.

Example 26.c.28 (*Integrability without a potential: why simple connectedness is needed*): Let $U := \mathbb{R}^2 \setminus \{0\}$ (which is **not** simply connected, unlike its 3-dimensional analogue $\mathbb{R}^3 \setminus \{0\}$ listed above) and

$$F(x,y) := \left(\frac{-y}{x^2 + y^2},\ \frac{x}{x^2 + y^2} \right), \qquad (x,y) \in U.$$

A direct computation gives

$$\partial_1 F_2 = \partial_x\left(\frac{x}{x^2+y^2}\right) = \frac{-x^2+y^2}{(x^2+y^2)^2} = \partial_y\left(\frac{-y}{x^2+y^2}\right) = \partial_2 F_1,$$

so F satisfies the integrability conditions (Lemma 26.b.12) **everywhere** on U. Yet F is not conservative: for the loop $\gamma(t) := (\cos(2\pi t), \sin(2\pi t))$, $t \in [0,1]$, tracing the unit circle once counterclockwise,

$$\oint_\gamma F \cdot d\gamma = 2\pi \int_0^1 \langle(-\sin(2\pi t), \cos(2\pi t)), (-\sin(2\pi t), \cos(2\pi t))\rangle \, dt = 2\pi \neq 0,$$

which by Proposition 26.b.14 is impossible for a conservative field (a closed loop would have to return $f(\gamma(1)) - f(\gamma(0)) = 0$). This is exactly the witness that Corollary 26.c.25 needs the hypothesis it has: the integrability conditions are necessary but, on a domain that is not simply connected, not sufficient.

Exercise 26.c.29 (*Integrability conditions on a domain that is not connected*): Let $g : \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable and let F be the vector field defined on $U := \{(x,y,z) \in \mathbb{R}^3 : y \neq 0\}$ by

$$F(x,y,z) := \left((1+x)\, e^{x+\frac{1}{y}}, \quad -\frac{x}{y^2}\, e^{x+\frac{1}{y}}, \quad g'(z)\right).$$

(a) Is U open? Is it connected? Is it simply connected?

(b) Does F satisfy the integrability conditions?

(c) Is F conservative? Justify your answer.

(d) Does F admit a potential? If so, give one; if not, explain why not.

(e) Compute the work integral $\int_\gamma F \cdot d\gamma$ along the path $\gamma : [0,1] \to U$, $\gamma(t) := \big(t\cos(2\pi t),\ 2 + t\sin(2\pi t),\ t^2 - t + 1\big)$. Check first that γ really does take values in U.

Info: *This exercise is taken from the Analysis I & II examination of 27 January 2021 (Prof. Peter S. Jossen), Part B, Problem 3, one part of which asks for the definition of a work integral and is omitted here.*

Solution: *See solution on Page 411*

Exercise 26.c.30 (*Single choice: what irrotational buys you, and what it does not*): Let V be an irrotational vector field on $G \subseteq \mathbb{R}^2$, that is, $\operatorname{curl}(V) = 0$ throughout G. Exactly one of the following is correct. Which?

(a) The line integral over any closed curve lying entirely in G must be zero.

(b) If in addition G is connected, then V must have a potential: there must exist $\varphi(x,y)$ with $\nabla\varphi = V$ on all of G.

(c) It must also hold that $\operatorname{div}(V) = 0$.

(d) It may be that there exists a potential $\varphi(x,y)$ on G with $\nabla\varphi = V$.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 9.*

Solution: *See solution on Page 413*

Stokes' theorem (Section 26.d)

Three integral theorems are now on the table: the divergence theorem (Theorem 23.b.5), Green's theorem (Theorem 26.b.5) and the classical Stokes theorem (Theorem 25.c.15). Each trades an integral over a region for one over its boundary, and each was stated in its own notation. The forms and orientations of the last two chapters were built so that all three can be written as one formula.

Theorem 26.d.31 (*Stokes' theorem*): Let $M \subseteq \mathbb{R}^n$ be an oriented m-manifold with (possibly empty) boundary, with ∂M carrying the induced orientation of Definition 25.d.17, and let $\omega \in \Omega^{m-1}(\mathbb{R}^n)$ be a C^1 form for which $\operatorname{supp}\omega \cap M$ is compact. Then

$$\int_{\partial M} \omega = \int_M d\omega.$$

AI-Note 26.4: Both hypotheses earn their place, and it is worth seeing what each one stops.

Compactness of $\operatorname{supp}\omega \cap M$, not of $\operatorname{supp}\omega$. Requiring only that ω be compactly supported in $\mathbb{R}^n$ is not enough, because M need not be closed in $\mathbb{R}^n$ and the support can then run off the end of M without ever leaving a compact set of $\mathbb{R}^n$. Take $n = m = 1$ and $M = (0,1)$, which has $\partial M = \emptyset$ under Definition 25.d.16, since every one of its points has an open chart. Let $\omega = f \in \Omega^0(\mathbb{R})$ be compactly supported with $f \equiv 0$ near 0 and $f \equiv 1$ near 1. Then $\int_M d\omega = \int_0^1 f' = 1$, while $\int_{\partial M} \omega$ is an integral over the empty set and vanishes. What fails is that $\operatorname{supp}\omega \cap M = (0,1)$ is not compact. When M itself is compact, which is the case in every application below, the condition is automatic.

An orientation, not merely orientability. Both integrals change sign when an orientation is reversed, so a theorem that only asserts one **exists** cannot pin down the sign of either side. The two orientations must moreover be matched, which is what Definition 25.d.17 does: it is the outward-vector-first convention, and it is the same one that produces $\tau = (-\nu_2, \nu_1)$ in Green's theorem and the right-hand rule in Remark 26.d.33.

AI-Note 26.5: Theorem 25.c.15 was stated earlier, as background needed to solve Exercise 26.e.44, in its classical $\mathbb{R}^3$ curl-theorem form. It is exactly the case $m = 2$, $n = 3$, $\omega = \omega_F$ of the general theorem above, as Section 26.d.3 below confirms.

Example: Green's theorem (Subsection 26.d.1)

Example 26.d.32 (*The exterior derivative of a 1-form on $\mathbb{R}^2$*): Let $\omega(x,y) = P(x,y)\,dx + Q(x,y)\,dy$ be a 1-form on $\mathbb{R}^2$. Then

$$d\omega = dP \wedge dx + dQ \wedge dy = \left(\frac{\partial P}{\partial x}dx + \frac{\partial P}{\partial y}dy\right) \wedge dx + \left(\frac{\partial Q}{\partial x}dx + \frac{\partial Q}{\partial y}dy\right) \wedge dy = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy,$$

using $dx \wedge dx = dy \wedge dy = 0$ and $dy \wedge dx = -dx \wedge dy$.

Example: proof of Green's theorem (Subsection 26.d.2)

Green's theorem (Theorem 26.b.5, Theorem 14.24 of the lecture script) has been used since Section 26.b. With the general theorem in hand it takes three lines.

Proof of Theorem 26.b.5:

Write $F(x,y) = (P(x,y), Q(x,y))$, so that $\operatorname{curl} F = \partial_x Q - \partial_y P$ (the scalar 2-dimensional curl), and apply Theorem 26.d.31 to $\omega = P\,dx + Q\,dy$, whose exterior derivative was computed in Example 26.d.32:

$$\int_\Omega \operatorname{curl} F\,dx\,dy = \int_\Omega \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy = \int_\Omega d\omega \overset{\text{Stokes}}{=} \int_{\partial\Omega} P(x,y)\,dx + Q(x,y)\,dy = \int_{\partial\Omega} F \cdot \tau\,dL.$$

The last step needs a word. Let $\gamma : [0, L] \to \partial\Omega$ be a C^1 curve of unit speed and positive orientation, i.e. $\gamma'(t) = \tau(\gamma(t))$. Then

$$\langle F, \tau \rangle = \langle F, \gamma'(t) \rangle = (P\,dx + Q\,dy)(\gamma'(t)),$$

since $\tau = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \nu$ is the tangent vector consistent with the induced orientation, so integrating $(P\,dx + Q\,dy)(\gamma'(t))$ over $t \in [0, L]$ is exactly $\int_{\partial\Omega} P\,dx + Q\,dy$ by Definition 25.b.4. **Q.E.D.**

Example: Stokes' theorem in $\mathbb{R}^3$ (Subsection 26.d.3)

Recall from Example 24.b.16 that for $\omega_F = \langle F, \cdot \rangle$, $d\omega_F = \langle \operatorname{curl} F, \cdot \times \cdot \rangle$. So, if $\Omega \subseteq \mathbb{R}^3$ is a 2-dimensional submanifold with boundary, parametrized by ϕ with tangent vectors $\tau_i := \partial_i \phi$ and oriented by the unit normal $\nu := \tau_1 \times \tau_2 / |\tau_1 \times \tau_2|$, then

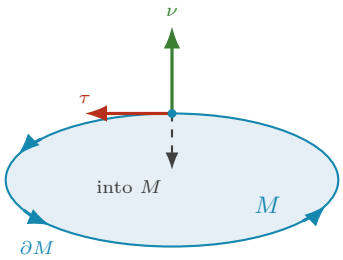
$$\int_\Omega d\omega_F = \int_\Omega \langle \operatorname{curl} F, \tau_1 \times \tau_2 \rangle \, dx = \int_\Omega \langle \operatorname{curl} F, \nu \rangle \, d\operatorname{vol}_2 = \int_\Omega \vec{\nabla} \times \vec{F} \cdot d\vec{A},$$

and

$$\int_{\partial\Omega} \omega_F = \int_{\partial\Omega} \langle F, \tau \rangle \, d\operatorname{vol}_1 = \int_{\partial\Omega} \vec{F} \cdot d\vec{s}.$$

By Theorem 26.d.31, these are equal, which recovers Theorem 25.c.15.

The last sentence of Theorem 25.c.15, the right-hand rule fixing the direction of τ from ν, is what the rest of this section computes with, and it is the one convention in this part that is easy to get backwards.



walking along τ with ν upright, M lies to the left: the leftward direction is $\nu \times \tau$

Remark 26.d.33 (*The convention as a formula*): *"To the left"* is a reliable instruction only once one knows which way is up, and ν is what supplies that. The picture above turns the sentence into an identity that can be checked instead of visualised: at a boundary point, the direction pointing from ∂M into M is

$$\text{inward} = \nu \times \tau.$$

Equivalently $\tau = \text{inward} \times \nu$, so any two of the three determine the third. On the unit disc in the xy-plane with $\nu = e_3$, at the point $(1, 0, 0)$, the positively oriented τ is $(0, 1, 0)$ and $\nu \times \tau = (-1, 0, 0)$ does point at the origin, confirming the familiar counterclockwise convention.

Reversing ν reverses τ, which is why both sides of Theorem 25.c.15 change sign together and the theorem is insensitive to the choice. What it is **not** insensitive to is choosing the two independently, and that is the error the formula above rules out.

Example 26.d.34 (*Stokes' theorem on a hemisphere: a linear field*): Let $F(x, y, z) := (z, x, y)^\mathsf{T}$. We want to compute the flux of the curl of F through the upper hemisphere $H := \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1,\ z \geq 0\}$.

First, the curl of F is:

$$\operatorname{curl}(F) = \begin{pmatrix} \partial_y F_3 - \partial_z F_2 \\ \partial_z F_1 - \partial_x F_3 \\ \partial_x F_2 - \partial_y F_1 \end{pmatrix} = \begin{pmatrix} 1 - 0 \\ 1 - 0 \\ 1 - 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

By Theorem 25.c.15, the flux of $\operatorname{curl}(F)$ through H equals the line integral of F along the boundary ∂H. The boundary is the unit circle in the xy-plane, which we parametrize by $\gamma(t) = (\cos t, \sin t, 0)^\mathsf{T}$ for $t \in [0, 2\pi]$. The tangent vector is $\gamma'(t) = (-\sin t, \cos t, 0)^\mathsf{T}$. The

line integral is:

$$\begin{aligned}\int_{\partial H} F\cdot d\gamma &= \int_0^{2\pi} F(\gamma(t))\cdot\gamma'(t)\,dt\\ &= \int_0^{2\pi}\begin{pmatrix}0\\ \cos t\\ \sin t\end{pmatrix}\cdot\begin{pmatrix}-\sin t\\ \cos t\\ 0\end{pmatrix}dt\\ &= \int_0^{2\pi}\cos^2 t\,dt = \pi.\end{aligned}$$

Thus, $\int_H \operatorname{curl}(F)\cdot\nu\,d\operatorname{vol}_2 = \pi$.

Exercise 26.d.35 (*Flux through a cylinder, direct and via Stokes*): Let $F(x,y,z) := (yz, x^2, 1)^{\mathsf{T}}$ on $\mathbb{R}^3$, and let M be the lateral surface of the cylinder $M := \{(x,y,z) : x^2+y^2=1,\ 0<z<1\}$, with the outward-pointing normal $\nu(x,y,z) = (x,y,0)$. Compute $\int_M \operatorname{curl}(F)\cdot\nu\,d\operatorname{vol}_2$ directly, and again using Stokes' theorem.

Proof of Exercise 26.d.35:

First, $\operatorname{curl}(F) = (\partial_y F_3 - \partial_z F_2,\ \partial_z F_1 - \partial_x F_3,\ \partial_x F_2 - \partial_y F_1) = (0,\ y,\ 2x-z)$.

Directly. Parametrize M by $\phi(\theta,z) := (\cos\theta, \sin\theta, z)$, $(\theta,z)\in[0,2\pi)\times(0,1)$; then $\partial_\theta\phi\times\partial_z\phi = (\cos\theta,\sin\theta,0) = \nu(\phi(\theta,z))$, already a unit vector, so $d\operatorname{vol}_2 = d\theta\,dz$. Then $\operatorname{curl}(F)(\phi(\theta,z)) = (0,\sin\theta,2\cos\theta - z)$, and

$$\int_M \operatorname{curl}(F)\cdot\nu\,d\operatorname{vol}_2 = \int_0^1\int_0^{2\pi}\big\langle(0,\sin\theta,2\cos\theta-z),(\cos\theta,\sin\theta,0)\big\rangle\,d\theta\,dz = \int_0^1\int_0^{2\pi}\sin^2\theta\,d\theta\,dz = \pi.$$

Via Stokes. The boundary ∂M consists of the two circles $z=0$ and $z=1$; the orientation induced by the outward normal ν (right-hand rule) traverses the **bottom** circle ($z=0$) with θ increasing and the **top** circle ($z=1$) with θ **decreasing**. The parameter rectangle $[0,2\pi]\times[0,1]$ has $z=0$ as part of its positively-oriented boundary traversed left to right and $z=1$ traversed right to left, and the two vertical edges $\theta = 0, 2\pi$ cancel since they are the same curve in M. This is the phenomenon drawn in Section 25.d: one consistently oriented cylinder induces senses on its two rims that turn opposite ways when both are viewed from the same side, and Remark 25.d.18 explains why that is the convention working rather than failing. With $\gamma_0(\theta) := (\cos\theta,\sin\theta,0)$ and $\gamma_1(\theta) := (\cos\theta,\sin\theta,1)$, both for $\theta\in[0,2\pi]$,

$$F(\gamma_0(\theta)) = (0,\cos^2\theta,1), \qquad F(\gamma_1(\theta)) = (\sin\theta,\cos^2\theta,1),$$

and $\gamma_0'(\theta) = \gamma_1'(\theta) = (-\sin\theta,\cos\theta,0)$, so

$$\int_0^{2\pi} F(\gamma_0(\theta))\cdot\gamma_0'(\theta)\,d\theta = \int_0^{2\pi}\cos^3\theta\,d\theta = 0,$$

$$\int_0^{2\pi} F(\gamma_1(\theta))\cdot\gamma_1'(\theta)\,d\theta = \int_0^{2\pi}\big(-\sin^2\theta+\cos^3\theta\big)\,d\theta = -\pi.$$

The bottom circle is traversed with θ increasing (contributing $+0$) and the top circle with θ **decreasing** (contributing $-(-\pi)=\pi$), so $\int_{\partial M} F\cdot\tau\,d\operatorname{vol}_1 = 0+\pi = \pi$, agreeing with the direct computation. Q.E.D.

Exercise 26.d.36 (*Flux through a spherical cap, via Stokes and via Gauss*): Let $0<h<1$ and let $S := \{(x,y,z)\in S^2 : z > -h\}$ be the part of the unit sphere above height $-h$. Consider $F(x,y,z) := (-y,x,0)^{\mathsf{T}}$. Calculate the flux $\int_S \operatorname{curl}(F)\cdot\nu\,d\operatorname{vol}_2$ (outward normal) directly, once using Stokes' theorem, and once using the divergence theorem.

Proof of Exercise 26.d.36:

Here $\operatorname{curl}(F) = (0,0,2)$, a constant field, and on S^2 the outward normal is $\nu(x,y,z) = (x,y,z)$ itself, so $\langle \operatorname{curl}(F), \nu\rangle = 2z$.

Directly. Using colatitude $\varphi \in [0, \arccos(-h)]$ (so $z = \cos\varphi > -h$) and $d\,\mathrm{vol}_2 = \sin\varphi\, d\varphi\, d\theta$,

$$\int_S 2z\, d\,\mathrm{vol}_2 = \int_0^{2\pi}\int_0^{\arccos(-h)} 2\cos\varphi\sin\varphi\, d\varphi\, d\theta = 2\pi\Big[-\tfrac{1}{2}\cos(2\varphi)\Big]_0^{\arccos(-h)} = 2\pi(1-h^2),$$

using $\cos(2\arccos(-h)) = 2h^2 - 1$.

Via Stokes. ∂S is the circle $x^2+y^2 = 1-h^2$, $z=-h$, of radius $r := \sqrt{1-h^2}$, oriented (right-hand rule, outward normal) by $\gamma(\theta) := (r\cos\theta, r\sin\theta, -h)$, $\theta \in [0, 2\pi]$. Then $\gamma'(\theta) = (-r\sin\theta, r\cos\theta, 0)$ and $F(\gamma(\theta)) = (-r\sin\theta, r\cos\theta, 0)$, so

$$\int_{\partial S} F\cdot d\gamma = \int_0^{2\pi} \left(r^2\sin^2\theta + r^2\cos^2\theta\right) d\theta = 2\pi r^2 = 2\pi(1-h^2),$$

agreeing with the direct computation.

Via Gauss. Since $\operatorname{div}(\operatorname{curl} F) \equiv 0$ (Proposition 22.b.12), the flux of $\operatorname{curl}(F)$ through any surface with boundary ∂S is the same. Replace S by the flat disk $D := \{x^2+y^2 \le 1-h^2,\ z=-h\}$, with upward normal $(0,0,1)$: $S \cup (-D)$ (the cap together with the disk, reversed) bounds the solid cap Ω, and the divergence theorem gives $\int_S \operatorname{curl}(F)\cdot\nu\, d\,\mathrm{vol}_2 - \int_D \operatorname{curl}(F)\cdot(0,0,1)\, d\,\mathrm{vol}_2 = \int_\Omega \operatorname{div}(\operatorname{curl} F)\, d\,\mathrm{vol}_3 = 0$. Since $\langle \operatorname{curl} F, (0,0,1)\rangle \equiv 2$,

$$\int_S \operatorname{curl}(F)\cdot\nu\, d\,\mathrm{vol}_2 = \int_D 2\, d\,\mathrm{vol}_2 = 2\cdot\mathrm{vol}_2(D) = 2\pi(1-h^2),$$

the area of a disk of radius $\sqrt{1-h^2}$ times 2, again the same answer. Q.E.D.

Remark 26.d.37 (*One theorem, four costumes*): Every integral theorem in this part of the course is the single identity $\int_M d\omega = \int_{\partial M}\omega$ of Theorem 26.d.31, for a different choice of the dimension m of M and the degree of ω. Collected in one place:

Theorem	M	m	ω	$d\omega$
FTC	$[a,b]\subset\mathbb{R}$	1	f (0-form)	$f'\,dx$
Green (Theorem 26.b.5)	$\Omega\subseteq\mathbb{R}^2$	2	$P\,dx + Q\,dy$	$(\partial_x Q - \partial_y P)\,dx\wedge dy$
Classical Stokes (Theorem 25.c.15)	surface $S\subset\mathbb{R}^3$	2	ω_F	$\omega^2_{\operatorname{curl} F}$
Divergence (Theorem 23.b.5)	$\Omega\subseteq\mathbb{R}^n$	n	ω_F^{n-1}	$(\operatorname{div} F)\,dx^1\wedge\cdots\wedge dx^n$

Each row's $\int_M d\omega = \int_{\partial M}\omega$ reads off as exactly that row's theorem. The top row is Section 23.a's $n=1$ computation seen from the other side: there, f played the role of a 1-dimensional vector field and the identity was reached via the divergence theorem; here f is instead a 0-form and the same identity is the $m=1$ case of Theorem 26.d.31 directly. These are two routes to the same fact, matching Example 24.b.16's remark that $df = \langle\nabla f, \cdot\rangle$ is exactly the shape that makes this specialization work. The bottom row uses $d\omega_F^{n-1} = (\operatorname{div} F)\,dx^1\wedge\cdots\wedge dx^n$ (Example 24.b.16, item **(c)**) together with the identification of $\int_{\partial\Omega}\omega_F^{n-1}$ with the flux $\int_{\partial\Omega}\langle F,\nu\rangle\, d\,\mathrm{vol}_{n-1}$ already noted in Chapter 25.

The pattern is worth stating in words, because it is easy to lose under the notation: a k-dimensional derivative measured **inside** a region always equals the corresponding $(k-1)$-dimensional quantity measured on its **boundary**. Differentiation and taking a boundary are, in this precise sense, the same operation viewed from two sides. And $d^2 = 0$ (Theorem 24.b.15**(c)**) is the mirror image of the fact that a boundary has no boundary: for compact M, Stokes' theorem applied twice gives $\int_M d(d\omega) = \int_{\partial M} d\omega = \int_{\partial(\partial M)}\omega = 0$, since $\partial(\partial M) = \emptyset$.

Exercise 26.d.38 (*Curl identity and Green's formula area preservation*): Let $\Omega \subseteq \mathbb{R}^2$ be a bounded C^1 domain, and let $\Phi = (\Phi_1, \Phi_2) \in C^2(\overline{\Omega}, \mathbb{R}^2)$ be a vector-valued mapping, not assumed to be a diffeomorphism.

(a) Consider the vector field $V \in C^1(\overline{\Omega}, \mathbb{R}^2)$ defined by

$$V := (\Phi_1 \partial_1 \Phi_2, \Phi_1 \partial_2 \Phi_2)^{\mathsf{T}} = \Phi_1 \nabla \Phi_2.$$

Prove that $\operatorname{curl} V(x) = \det \mathrm{J}\,\Phi(x)$ for all $x \in \overline{\Omega}$.

(b) Let $\Psi \in C^2(\overline{\Omega}, \mathbb{R}^2)$ be another mapping such that $\Phi(x) = \Psi(x)$ for all $x \in \partial\Omega$. Prove that

$$\int_\Omega \det \mathrm{J}\,\Phi(x)\,dx = \int_\Omega \det \mathrm{J}\,\Psi(x)\,dx.$$

For the remaining parts, take $\Omega = B_1$ to be the open unit ball of $\mathbb{R}^2$ and assume that $\Phi(x) = x$ for every $x \in \partial B_1$.

(c) Show that $\displaystyle\int_{B_1} |\det \mathrm{J}\,\Phi| \geq \pi$.

(d) Show that $\Phi(\overline{B_1})$ cannot be contained in the boundary ∂B_1.

(e) Show that $0 \in \Phi(\overline{B_1})$.

(f) **(Bonus.)** Show that $\overline{B_1} \subseteq \Phi(\overline{B_1})$.

Each part may be used in the later ones, even without proof.

Official hints: ***(b):*** *use Green's formula (i.e. 2D Stokes) on the plane.* ***(e):*** *consider* $\widetilde{\Phi}(x) := \Phi(x)/|\Phi(x)|$.

Info: *This exercise is the whole of Problem 3 of the exam of 21 August 2024 (Prof. Joaquim Serra), worth 15 points plus 6 bonus points for the last part.*

Solution: *See solution on Page 407*

Exercise 26.d.39 (*Constructing a vector potential for a divergence-free field*): Let $B \in C^1(\mathbb{R}^3, \mathbb{R}^3)$ be a continuously differentiable vector field on $\mathbb{R}^3$ satisfying $\operatorname{div} B = 0$. A vector field $A \in C^1(\mathbb{R}^3, \mathbb{R}^3)$ is called a "*vector potential*" for B if $\operatorname{curl} A = B$.

(a) Show that if A is a vector potential for B, then for any scalar function $\psi \in C^2(\mathbb{R}^3)$, the gauge-transformed field $\widetilde{A} := A + \nabla\psi$ is also a vector potential for B.

(b) To construct a vector potential, make the ansatz $A(x,y,z) = (A_1(x,y,z), A_2(x,y,z), 0)^{\mathsf{T}}$. Show that $\operatorname{curl} A = B$ is equivalent to the system of differential equations

$$-\partial_z A_2 = B_1, \qquad \partial_z A_1 = B_2, \qquad \partial_x A_2 - \partial_y A_1 = B_3.$$

Prove that this system always admits an explicit global solution on $\mathbb{R}^3$ given by

$$A_1(x,y,z) := \int_0^z B_2(x,y,s)\,ds, \qquad A_2(x,y,z) := -\int_0^z B_1(x,y,s)\,ds + \int_0^x B_3(t,y,0)\,dt.$$

(c) Apply this construction to obtain an explicit vector potential A for the divergence-free field

$$B(x,y,z) := (2xy,\ -y^2 + 2yz,\ -z^2)^{\mathsf{T}}.$$

Verify directly that $\operatorname{div} B = 0$ and that $\operatorname{curl} A = B$.

Info: *This exercise is adapted from the mock examination for Analysis II, Spring Semester 2024 (Prof. Joaquim Serra), Exercise 25, part 2.*

Solution: *See solution on Page 414*

Area via Green's formula (Section 26.e)

Stokes' theorem turns an integral over a region into one over its boundary. Read in that direction it computes fluxes and circulations, which is what the previous sections did with it. Read in the other

direction it says something initially surprising: the area of a region is determined by its boundary curve alone, and can be computed without ever looking inside.

Example 26.e.40 (*Area as a boundary functional*): Suppose $\gamma : [0, L) \to \mathbb{R}^2$ traces the boundary of a connected C^∞-domain $U \subseteq \mathbb{R}^2$ in the positive direction. Express the area of U as a functional of γ.

By definition, $A(U) = \int_U dx\, dy = \int_U 1\, dx \wedge dy$. By Green's formula (Theorem 26.b.5, in its 1-form guise $\int_\Omega d\omega = \int_{\partial\Omega} \omega$),

$$\int_U 1\, dx \wedge dy = \int_0^L \langle F \circ \gamma(t), \gamma'(t) \rangle\, dt$$

for any $F : \overline{U} \to \mathbb{R}^2$ satisfying $\partial_x F^2 - \partial_y F^1 = 1$. This is satisfied by $F(x, y) = \frac{1}{2}(-y, x)$, giving the "*shoelace formula*" (German: „*Gaußsche Trapezformel*")

$$A(\gamma) = \frac{1}{2} \int_0^L \left[\gamma_1(t)\gamma_2'(t) - \gamma_2(t)\gamma_1'(t)\right] dt.$$

Remark 26.e.41 (*The choice of F is free, and worth exploiting*): Any F with $\partial_x F^2 - \partial_y F^1 = 1$ serves, and the three standard choices give three different-looking formulas for the same number:

$$F = \tfrac{1}{2}(-y, x) \rightsquigarrow \tfrac{1}{2} \oint (x\, dy - y\, dx), \qquad F = (0, x) \rightsquigarrow \oint x\, dy, \qquad F = (-y, 0) \rightsquigarrow -\oint y\, dx.$$

The symmetric first choice is the one to quote, but on a particular curve one of the other two is often much cheaper. If γ_1 is constant on part of the boundary, dx vanishes there and the third form kills that piece outright.

Example 26.e.42 (*Ellipse*): The path $\gamma : [0, 2\pi) \to \mathbb{R}^2$, $t \mapsto (a \cos t, b \sin t)$, traces the boundary of an ellipse. By Example 26.e.40,

$$A(\gamma) = \frac{1}{2} \int_0^{2\pi} \left[a \cos t \cdot b \cos t - b \sin t \cdot (-a \sin t)\right] dt = \frac{ab}{2} \int_0^{2\pi} dt = \pi ab.$$

For the circle $a = b =: r$, this gives the familiar $A = \pi r^2$.

Remark 26.e.43 (*Area yes, perimeter no*): Unlike its area, the perimeter of an ellipse has no closed form: it is a complete elliptic integral of the second kind, not an elementary function of a and b.

Exercise 26.e.44 (*Work along paths and Stokes* (important)): Let $F : \mathbb{R}^3 \setminus (\{(0, 0)\} \times \mathbb{R}) \to \mathbb{R}^3$ be the vector field defined by

$$F(x, y, z) = \left(\frac{2(xz + y)}{x^2 + y^2},\ \frac{2(yz - x)}{x^2 + y^2},\ \log(x^2 + y^2) \right)$$

and let $\gamma_1, \gamma_2, \gamma_3 : [0, 2\pi] \to \mathbb{R}^3$ be the paths

$$\gamma_1(t) = (\cos t, \sin t, 0), \qquad \gamma_2(t) = (2 \cos t, 2 \sin t, 2), \qquad \gamma_3(t) = (3 \cos t + 6, 3 \sin t, 3).$$

(a) Describe the domain of definition of F. Is it connected? Is it simply connected?

(b) Compute the rotation $\operatorname{curl}(F)$ of F and the path integral $\int_{\gamma_1} F \cdot d\gamma_1$ of F along γ_1. Is F conservative?

(c) Explain how the equation $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2$ follows from Stokes' theorem.

(d) Does Stokes' theorem imply that $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_3} F \cdot d\gamma_3$ holds?

Info: *This exercise is Problem 12.3 of Problem Sheet 12, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 405*

Exercise 26.e.45 (*True or False (flux on a punctured domain)* (important)): Let $U := \mathbb{R}^3 \setminus \{0\}$ and $F \in C^1(U, \mathbb{R}^3)$. For $r > 0$ set

$$I_r := \int_{\partial B_r} \langle F, n \rangle \, dS$$

where n denotes the outward normal. Which of the following statements are always true?

(a) If F is divergence-free, then $I_r = 0$ for all $r > 0$.
(b) If F is divergence-free, then the value of I_r does not depend on r.
(c) If F is curl-free, then the value of I_r does not depend on r.
(d) If $F = \operatorname{curl}(G)$ for a vector field $G \in C^1(U, \mathbb{R}^3)$ and $|F(x)| \leq C/|x|$, then $I_r = 0$ for all $r > 0$.

Info: *This exercise is Problem 12.4 of Problem Sheet 12, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 406*

Exercise 26.e.46 (*Conservative vector fields* (important)): For which values of $\lambda \in \mathbb{R}$ is the vector field $F : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$F(x, y) = \begin{pmatrix} \lambda x e^y \\ (y + 1 + x^2) e^y \end{pmatrix}$$

conservative? Determine a potential for F for these values.

Info: *This exercise is Problem 12.5 of Problem Sheet 12, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 406*

Exercise 26.e.47 (*Multiple choice (signs in the flux through a rectangle)* (important)): Let $Q = [a, b] \times [c, d] \subseteq \mathbb{R}^2$ and $f \in C(Q, \mathbb{R}^2)$. Let n be an outward normal field on Q and

$$I_b = \int_{[a,b]\times\{c\}} \langle f, n \rangle, \qquad I_t = \int_{[a,b]\times\{d\}} \langle f, n \rangle, \qquad I_l = \int_{\{a\}\times[c,d]} \langle f, n \rangle, \qquad I_r = \int_{\{b\}\times[c,d]} \langle f, n \rangle.$$

Which of the following expressions corresponds to the outer flux of f through ∂Q?

(a) $I_b - I_t + I_r - I_l$
(b) $I_b + I_t + I_r + I_l$
(c) $I_b + I_t - I_r - I_l$
(d) $-I_b + I_t - I_r + I_l$

Info: *This exercise is Problem 12.8 of Problem Sheet 12, Analysis II, Spring Semester 2026.*

Solution: *See solution on Page 407*

Solutions (Section 26.f)

Not every solution in this chapter is collected here. Exercise 26.b.17 (Page 392), Exercise 26.d.35 (Page 400) and Exercise 26.d.36 (Page 400) are solved where they appear.

Proof of Exercise 26.b.4:

The concatenation is an admissible path. First check that $\gamma * \sigma$ is defined at all. The two formulas overlap at $t = \frac{1}{2}$, where they give $\sigma(1)$ and $\gamma(0)$; these agree by hypothesis, so $\gamma * \sigma$ is a well-defined continuous map $[0, 1] \to U$. It is piecewise C^1 but in general **not** C^1: at $t = \frac{1}{2}$ the one-sided derivatives are $2\sigma'(1)$ and $2\gamma'(0)$, and nothing forces those to agree. This is exactly why Definition 26.b.3 is stated for piecewise C^1 paths, and the single exceptional point does not affect any of the integrals below.

The computation. Split the defining integral at $t = \frac{1}{2}$:

$$\int_{\gamma*\sigma} F \cdot d(\gamma*\sigma) = \int_0^1 F\big((\gamma*\sigma)(t)\big) \cdot (\gamma*\sigma)'(t)\,dt = \underbrace{\int_0^{1/2} \cdots}_{=:A} + \underbrace{\int_{1/2}^1 \cdots}_{=:B}.$$

On $(0, \frac{1}{2})$ we have $(\gamma*\sigma)(t) = \sigma(2t)$, so by the chain rule $(\gamma*\sigma)'(t) = 2\sigma'(2t)$ and

$$A = \int_0^{1/2} F\big(\sigma(2t)\big) \cdot 2\sigma'(2t)\,dt = \int_0^1 F\big(\sigma(s)\big) \cdot \sigma'(s)\,ds = \int_\sigma F \cdot d\sigma,$$

substituting $s := 2t$, $ds = 2\,dt$, which carries $t \in [0, \frac{1}{2}]$ onto $s \in [0,1]$. On $(\frac{1}{2}, 1)$ we have $(\gamma*\sigma)(t) = \gamma(2t-1)$ and $(\gamma*\sigma)'(t) = 2\gamma'(2t-1)$, so with $s := 2t-1$, again $ds = 2\,dt$ and $t \in [\frac{1}{2}, 1]$ onto $s \in [0,1]$,

$$B = \int_{1/2}^1 F\big(\gamma(2t-1)\big) \cdot 2\gamma'(2t-1)\,dt = \int_0^1 F\big(\gamma(s)\big) \cdot \gamma'(s)\,ds = \int_\gamma F \cdot d\gamma.$$

Adding A and B gives the claim. Note that F enters only through its continuity; the hypothesis $F \in C^1$ is more than the statement needs.

In terms of work. The work done against a force field over a journey is the sum of the work done over its legs. Nothing about the field is required for this (not conservativity, not even path independence), because it is a statement about the **path**, not the field: the integral does not know that the journey was made in two stages.

Remark 26.f.48 (*The factor 2 is the whole proof, and it is not a coincidence*): Each leg is traversed at double speed, which multiplies γ' by 2; and each leg occupies half the parameter interval, which divides dt by 2. The two cancel exactly.

That cancellation is not special to this substitution. It is "*reparametrisation invariance*": for any C^1 increasing bijection $\varphi : [a,b] \to [0,1]$, the substitution $s = \varphi(t)$ gives $\int_{\gamma\circ\varphi} F \cdot d(\gamma\circ\varphi) = \int_\gamma F \cdot d\gamma$, by the same one-line chain rule and change of variable. The line integral depends on the curve and the direction of travel, not on the schedule, which is what licenses writing $\int_\gamma F$ for a **curve** at all, as the rest of this chapter does. Reverse the direction and the sign flips, since φ is then decreasing.

Q.E.D.

Solution of Exercise 26.e.44:

(a) The domain is $\{(x,y,z) \in \mathbb{R}^3 : (x,y) \neq (0,0)\}$, i.e. $\mathbb{R}^3$ with the z-axis removed. It is connected (path-connected: any two points can be joined by a path avoiding the axis), but not simply connected: γ_1, for instance, which winds once around the axis, cannot be contracted to a point without crossing it. (It is also not convex: the segment from $(-1,0,0)$ to $(1,0,0)$ leaves the domain.)

(b) Writing $F = (F^1, F^2, F^3)$ with $F^1 = 2(xz+y)/(x^2+y^2)$, $F^2 = 2(yz-x)/(x^2+y^2)$, $F^3 = \log(x^2+y^2)$, a direct computation of each pair of mixed partials gives

$$\partial_y F^3 = \partial_z F^2 = \frac{2y}{x^2+y^2}, \qquad \partial_z F^1 = \partial_x F^3 = \frac{2x}{x^2+y^2},$$
$$\partial_x F^2 = \partial_y F^1 = \frac{2(x^2-y^2-2xyz)}{(x^2+y^2)^2},$$

so $\operatorname{curl} F = (\partial_y F^3 - \partial_z F^2,\ \partial_z F^1 - \partial_x F^3,\ \partial_x F^2 - \partial_y F^1) = (0,0,0)$: F is curl-free. On γ_1, $x = \cos t$, $y = \sin t$, $z = 0$, $x^2+y^2 = 1$, so $F(\gamma_1(t)) = (2\sin t, -2\cos t, 0)$ and $\gamma_1'(t) = (-\sin t, \cos t, 0)$, giving

$$\int_{\gamma_1} F \cdot d\gamma_1 = \int_0^{2\pi} \big(2\sin t \cdot (-\sin t) + (-2\cos t)\cdot \cos t\big)\,dt = \int_0^{2\pi} -2\,dt = -4\pi.$$

Since this is nonzero on a closed loop, F is **not** conservative. It is curl-free, but its domain is not simply connected (part **(a)**), so curl-free does not force conservative here.

(c) Let $S := \{(x,y,z) : \sqrt{x^2+y^2} = 1 + z/2,\ 0 \le z \le 2\}$, the cone swept out by linearly interpolating the radius from γ_1 ($z = 0$, radius 1) to γ_2 ($z = 2$, radius 2), parametrized by $H(s,t) := \big((1+s)\cos t, (1+s)\sin t, 2s\big)$ for $(s,t) \in [0,1] \times [0, 2\pi]$. Since the radius $1 + s \ge 1 > 0$ throughout, S never meets the z-axis and lies entirely in the domain of F; its oriented boundary (outward normal) is γ_2 minus γ_1. Since $\operatorname{curl} F = 0$ on S by part **(b)**, Theorem 25.c.15 gives

$$\int_{\gamma_2} F \cdot d\gamma_2 - \int_{\gamma_1} F \cdot d\gamma_1 = \int_{\partial S} F \cdot ds = \int_S \operatorname{curl} F \cdot d\vec{n} = 0,$$

i.e. $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2$.

(d) No. The disk $D := \{(x,y,z) : (x-6)^2 + y^2 \le 9,\ z = 3\}$, oriented upward, has oriented boundary exactly γ_3, and D lies entirely in the domain of F (it does not meet the z-axis, since its center $(6,0,3)$ is at distance 6 from the axis and its radius is only 3). By Theorem 25.c.15 with $\operatorname{curl} F = 0$,

$$\int_{\gamma_3} F \cdot d\gamma_3 = \int_{\partial D} F \cdot ds = \int_D \operatorname{curl} F \cdot d\vec{n} = 0.$$

Since $\int_{\gamma_1} F \cdot d\gamma_1 = -4\pi \ne 0 = \int_{\gamma_3} F \cdot d\gamma_3$, the two do **not** agree, and Stokes' theorem gives no reason to expect they would: unlike the pair (γ_1, γ_2), there is no surface joining γ_1 and γ_3 that stays inside the domain, since γ_1 winds around the excluded axis and γ_3 does not.

Q.E.D.

Solution of Exercise 26.e.45:

(a) False. The Newton potential's gradient $F(x) = \nabla(1/|x|) = -x|x|^{-3}$ is divergence-free on U (it is V_{-3} up to sign, from Exercise 23.b.13, with $n = 3$), yet on ∂B_r the normal is $n = x/r$ and $\langle F, n\rangle = -|x|^2/r^4 = -1/r^2$, so

$$I_r = \int_{\partial B_r} F \cdot n\, dS = -\frac{1}{r^2}\operatorname{vol}_2(\partial B_r) = -4\pi \ne 0,$$

a constant, exactly as part **(b)** predicts.

(b) True. Applying Theorem 23.b.5 on the shell $B_{r_2} \setminus B_{r_1} \Subset U$ for $0 < r_1 < r_2$, whose boundary is ∂B_{r_2} (outward) together with ∂B_{r_1} with reversed orientation,

$$0 = \int_{B_{r_2}\setminus B_{r_1}} \operatorname{div} F\, dx = I_{r_2} - I_{r_1},$$

so I_r is independent of r.

(c) False. Take $F(x) = x = \frac{1}{2}\nabla|x|^2$, which is curl-free (being a gradient). Then $\operatorname{div} F \equiv 3$, so by Theorem 23.b.5, $I_r = \int_{B_r} 3\, dx = 3\operatorname{vol}_3(B_r) = 4\pi r^3$, which depends on r.

(d) True. If $F = \operatorname{curl} G$ then $\operatorname{div} F = \operatorname{div}(\operatorname{curl} G) = 0$ (a standard vector identity), so part (b) applies and I_r is independent of r. On the other hand,

$$|I_r| \le \int_{\partial B_r} |\langle F, n\rangle|\, dS \le \frac{C}{r}\operatorname{vol}_2(\partial B_r) = \frac{C}{r}\cdot 4\pi r^2 = 4\pi C r \longrightarrow 0 \quad \text{as } r \to 0^+.$$

Since I_r is constant in r and this constant is a limit of values tending to 0, $I_r = 0$ for all $r > 0$.

Q.E.D.

Solution of Exercise 26.e.46:

Writing $F = (F^1, F^2)$ with $F^1 = \lambda x e^y$, $F^2 = (y + 1 + x^2)e^y$, the integrability condition is $\partial_y F^1 = \partial_x F^2$:

$$\partial_y F^1 = \lambda x e^y, \qquad \partial_x F^2 = 2x e^y.$$

Since $\mathbb{R}^2$ is convex (so integrability is equivalent to conservativity here), F is conservative if and only if $\lambda = 2$.

For $\lambda = 2$, seek f with $\partial_x f = F^1 = 2xe^y$: integrating in x, $f(x,y) = x^2e^y + g(y)$ for some function g. Then $\partial_y f = x^2e^y + g'(y)$ must equal $F^2 = (y+1+x^2)e^y$, so $g'(y) = (y+1)e^y$; integrating by parts, $\int (y+1)e^y\,dy = ye^y - e^y + e^y = ye^y$ (up to an additive constant). Hence

$$f(x,y) = \boxed{(x^2+y)e^y} \qquad \text{(unique up to an additive constant),}$$

which one checks directly satisfies $\nabla f = F$. Q.E.D.

Solution of Exercise 26.e.47:

The four quantities I_b, I_t, I_l, I_r are already defined using the same outward normal field n on each of the four edges of ∂Q, so the total outer flux is simply their sum, with no relative sign corrections needed: $\boxed{I_b + I_t + I_r + I_l}$, answer **(b)**. Q.E.D.

Proof of Exercise 26.d.38:

(a) Write $V = (V_1, V_2)^\mathsf{T} = (\Phi_1\partial_1\Phi_2, \Phi_1\partial_2\Phi_2)^\mathsf{T}$. The scalar curl in the plane is $\operatorname{curl} V = \partial_1 V_2 - \partial_2 V_1$, so the product rule gives

$$\begin{aligned}\operatorname{curl} V &= \partial_1(\Phi_1\partial_2\Phi_2) - \partial_2(\Phi_1\partial_1\Phi_2)\\ &= (\partial_1\Phi_1\partial_2\Phi_2 + \Phi_1\partial_1\partial_2\Phi_2) - (\partial_2\Phi_1\partial_1\Phi_2 + \Phi_1\partial_2\partial_1\Phi_2).\end{aligned}$$

Since $\Phi \in C^2(\overline{\Omega}, \mathbb{R}^2)$, Schwarz's theorem (Theorem 11.b.13) gives $\partial_1\partial_2\Phi_2 = \partial_2\partial_1\Phi_2$, and the two second-derivative terms cancel. Consequently,

$$\operatorname{curl} V = \partial_1\Phi_1\partial_2\Phi_2 - \partial_2\Phi_1\partial_1\Phi_2 = \det\begin{pmatrix}\partial_1\Phi_1 & \partial_2\Phi_1\\ \partial_1\Phi_2 & \partial_2\Phi_2\end{pmatrix} = \det \mathrm{J}\,\Phi.$$

Note that the hypothesis $\Phi \in C^2$ is doing real work here: without equality of mixed partials the identity is simply false.

(b) Write $V_\Phi := \Phi_1\nabla\Phi_2$ and $V_\Psi := \Psi_1\nabla\Psi_2$. By part **(a)**, $\operatorname{curl} V_\Phi = \det \mathrm{J}\,\Phi$ and $\operatorname{curl} V_\Psi = \det \mathrm{J}\,\Psi$, so Green's theorem (Theorem 26.b.5) turns both integrals into boundary integrals:

$$\int_\Omega \det \mathrm{J}\,\Phi\,dx = \int_{\partial\Omega} V_\Phi \cdot \tau\,dL, \qquad \int_\Omega \det \mathrm{J}\,\Psi\,dx = \int_{\partial\Omega} V_\Psi \cdot \tau\,dL.$$

It remains to see that the two boundary integrands agree. Fix a point of $\partial\Omega$ and let τ be the unit tangent there. Since $\Phi = \Psi$ on all of $\partial\Omega$, the two maps agree along the boundary curve, and therefore so do their derivatives **in the direction** τ:

$$\nabla\Phi_2 \cdot \tau = \partial_\tau\Phi_2 = \partial_\tau\Psi_2 = \nabla\Psi_2 \cdot \tau.$$

(The full gradients need not agree, since nothing constrains the normal derivative; but the line integral only ever sees the tangential part.) Together with $\Phi_1 = \Psi_1$ on $\partial\Omega$ this gives $V_\Phi \cdot \tau = V_\Psi \cdot \tau$ pointwise on $\partial\Omega$, and hence

$$\int_\Omega \det \mathrm{J}\,\Phi\,dx = \int_\Omega \det \mathrm{J}\,\Psi\,dx.$$

Remark 26.f.49: The statement is a genuinely striking one: $\int_\Omega \det \mathrm{J}\,\Phi$ is an interior quantity, the signed area swept out by Φ, yet it depends on nothing but the boundary values of Φ. That is the same phenomenon as a conservative field having path-independent work, one dimension up.

From here on $\Omega = B_1$ and $\Phi(x) = x$ on ∂B_1. The four remaining parts form a chain, each one feeding the next, and the single computation they all run on is the one carried out in part **(d)**: a map taking values in the unit circle has vanishing Jacobian determinant.

(c) The identity map $\Psi := \mathrm{id}$ is C^2 on $\overline{B_1}$ and agrees with Φ on ∂B_1, which is exactly the hypothesis of **(b)**. Since $\mathrm{J}(\mathrm{id}) = I_2$ has determinant 1,

$$\int_{B_1} \det \mathrm{J}\,\Phi = \int_{B_1} \det \mathrm{J}(\mathrm{id}) = \int_{B_1} 1 = \mathrm{vol}_2(B_1) = \pi.$$

The integral of $|\det \mathrm{J}\,\Phi|$ dominates the absolute value of the integral of $\det \mathrm{J}\,\Phi$, so

$$\int_{B_1} |\det \mathrm{J}\,\Phi| \;\geq\; \left| \int_{B_1} \det \mathrm{J}\,\Phi \right| = \pi.$$

Note what has been proved: Φ may fold the disk over itself arbitrarily, but it can never do so in a way that makes the total unsigned area it sweeps out drop below π.

(d) Suppose, for contradiction, that $\Phi(\overline{B_1}) \subseteq \partial B_1$, so that $|\Phi(x)|^2 = 1$ for every $x \in \overline{B_1}$. Differentiating this identity with respect to x_j gives $2\sum_{i=1}^{2} \Phi_i\, \partial_j \Phi_i = 0$ for $j = 1, 2$, which in matrix form reads

$$\Phi(x)^{\mathsf{T}}\, \mathrm{J}\,\Phi(x) = 0 \qquad \text{for all } x \in \overline{B_1}.$$

The row vector $\Phi(x)^{\mathsf{T}}$ is non-zero, since $|\Phi(x)| = 1$, so $\mathrm{J}\,\Phi(x)$ has a non-trivial left kernel and is therefore singular. Hence $\det \mathrm{J}\,\Phi \equiv 0$ on $\overline{B_1}$ and $\int_{B_1} |\det \mathrm{J}\,\Phi| = 0$, contradicting **(c)**.

(e) Suppose $0 \notin \Phi(\overline{B_1})$. Then $|\Phi|$ is a continuous, strictly positive function on the compact set $\overline{B_1}$, hence bounded below by some $c > 0$, and

$$\widetilde{\Phi} := \frac{\Phi}{|\Phi|}$$

is C^2 on $\overline{B_1}$: it is a composition of $\Phi \in C^2$ with the map $y \mapsto y/|y|$, which is smooth on $\{|y| \geq c\}$. By construction $|\widetilde{\Phi}| \equiv 1$, so $\widetilde{\Phi}(\overline{B_1}) \subseteq \partial B_1$. Moreover $\widetilde{\Phi}(x) = x/|x| = x$ for $x \in \partial B_1$, since $\Phi(x) = x$ and $|x| = 1$ there. So $\widetilde{\Phi}$ satisfies every hypothesis that **(d)** places on Φ, and its image **is** contained in ∂B_1. That is exactly what **(d)** forbids, so no such Φ exists and $0 \in \Phi(\overline{B_1})$.

(f) Let $p \in \overline{B_1}$. If $|p| = 1$ then $\Phi(p) = p$ and there is nothing to prove, so assume $|p| < 1$ and suppose $p \notin \Phi(\overline{B_1})$.

Exactly as in **(e)**, $|\Phi - p|$ is bounded below by some $c > 0$ on the compact $\overline{B_1}$, so

$$G := \frac{\Phi - p}{|\Phi - p|} \;\in\; C^2(\overline{B_1}, \mathbb{R}^2), \qquad |G| \equiv 1.$$

The computation in **(d)** applies verbatim to G and gives $\det \mathrm{J}\,G \equiv 0$. Feeding that into **(a)** and Green's theorem (Theorem 26.b.5),

$$0 = \int_{B_1} \det \mathrm{J}\,G = \int_{\partial B_1} G_1 \nabla G_2 \cdot \tau \, dL. \tag{26.1}$$

Now evaluate that boundary integral. On ∂B_1 we have $\Phi(x) = x$, so $G(x) = (x - p)/|x - p|$. Parametrise ∂B_1 by $x(t) := (\cos t, \sin t)$ and write $c(t) := x(t) - p$, a circle of radius 1 centred at $-p$. Since $G(x(t)) = c(t)/|c(t)|$ lies on the unit circle, it admits a C^1 angle φ with

$$G(x(t)) = \big(\cos\varphi(t),\ \sin\varphi(t)\big).$$

Then $\frac{d}{dt} G_2(x(t)) = \varphi'(t)\cos\varphi(t)$, so the integrand of (26.1) is $\cos^2\varphi(t)\,\varphi'(t)$ and

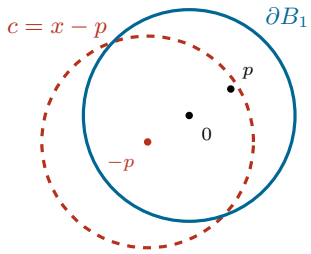
$$\int_{\partial B_1} G_1 \nabla G_2 \cdot \tau \, dL = \int_0^{2\pi} \cos^2\varphi\, \varphi' \, dt = \int_{\varphi(0)}^{\varphi(2\pi)} \cos^2\varphi \, d\varphi = \left[\frac{\varphi}{2} + \frac{\sin 2\varphi}{4} \right]_{\varphi(0)}^{\varphi(2\pi)}.$$

The loop closes up, so $\varphi(2\pi) - \varphi(0) = 2\pi N$ for an integer N, and $\sin 2\varphi$ is 2π-periodic so its two contributions cancel. The boundary integral is therefore πN, and (26.1) forces $N = 0$.

But $N = 1$, and the angle field of Exercise 26.b.18 is what shows it. Writing $c(t) = \rho(t)\big(\cos\varphi(t), \sin\varphi(t)\big)$ with $\rho := |c| > 0$, and taking $Y := (-x_2, x_1)/(x_1^2 + x_2^2)$, we have $Y(c) = \rho^{-1}(-\sin\varphi, \cos\varphi)$ and

$$\langle Y(c), \dot{c} \rangle = \frac{1}{\rho} \Big\langle (-\sin\varphi, \cos\varphi),\ \dot{\rho}\,(\cos\varphi, \sin\varphi) + \rho\dot{\varphi}\,(-\sin\varphi, \cos\varphi) \Big\rangle = \dot{\varphi},$$

the $\dot\rho$ term dropping out because the two vectors are orthogonal. Integrating, $\int_c Y\cdot dc = \varphi(2\pi)-\varphi(0) = 2\pi N$: the angle field measures exactly the total turning. Now c is the unit circle centred at $-p$, and $|-p|<1$ puts the origin strictly inside it, so $\int_c Y\cdot dc = 2\pi$.



$|p|<1$, so the origin lies inside c:
the loop winds once about it, and $N = 1$

To see the value 2π, apply Green's theorem to the region between c and a small circle about the origin: Y is closed there, so the region contributes nothing and the two circles carry the same integral, and on a circle centred at the origin the integrand is identically 1, as computed in the solution of Exercise 26.b.18. Hence $N=1$, contradicting $N=0$, and $p\in\Phi(\overline{B_1})$ after all.

Remark 26.f.50 (*What the chain actually proves*): Read backwards, the four parts are one theorem: a C^2 map of the closed disk that fixes the circle pointwise must hit every point of the disk. It cannot retract the disk onto its own boundary, which is **(d)**, and that single fact is what **(e)** and **(f)** both exploit, each time by dividing by a norm to manufacture a map whose image **is** in the circle. This is Brouwer's fixed point theorem in disguise, in the plane and in its smooth form, and the whole apparatus is two applications of Green's formula.

The integer N is where the topology finally enters. Everything before it is calculus, and calculus alone cannot produce a contradiction, since 0 and πN are both perfectly good numbers. What breaks the argument open is that N is forced to be a **whole number** by the loop closing up, and forced to be 1 by where p sits. An integer that must equal both 0 and 1 is the only genuinely non-analytic step in the proof.

Q.E.D.

Solution of Exercise 26.b.18:

Begin with the computation every part relies on. Writing $X=(P,Q)$ with $P=-y/(x^2+y^2)$ and $Q=x/(x^2+y^2)$,

$$\partial_x Q = \frac{(x^2+y^2)-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad \partial_y P = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2},$$

so $\operatorname{curl} X = \partial_x Q - \partial_y P = 0$ throughout U. The field is closed. Whether it is also a gradient is precisely what the domain decides.

(a) False. Suppose such a u existed. Then the work of X around any closed loop in U would vanish. But along the unit circle $\sigma(t) := (\cos t, \sin t)$, $t\in[0,2\pi]$, we have $X(\sigma(t)) = (-\sin t, \cos t) = \dot\sigma(t)$, so the integrand is $|\dot\sigma|^2 = 1$ and

$$\int_\sigma X\cdot d\sigma = \int_0^{2\pi} 1\,dt = 2\pi \neq 0.$$

So no potential exists on U. Being closed is not enough; U is not simply connected, and this field is the standard witness that Corollary 26.c.25 genuinely needs that hypothesis.

(b) True. The half-plane V is convex, hence simply connected, so a closed field on it is exact (Corollary 26.c.25). Explicitly, $v(x,y) := \arctan(y/x)$ is C^1 on V (note $x > 0$ there, so the quotient is defined and smooth), and

$$\nabla v = \frac{1}{1+y^2/x^2}\left(-\frac{y}{x^2},\, \frac{1}{x}\right) = \frac{1}{x^2+y^2}\,(-y,\, x) = X.$$

The potential is the polar angle, which is single-valued on the half-plane but not on U: that is **(a)** and **(b)** in one sentence.

(c) False. Injectivity on $[0,1)$ makes γ a simple closed curve, but it says nothing about **where** that curve runs. Two things can go wrong. First, the loop need not enclose the origin at all: the circle of radius $\frac{1}{2}$ about $\left(0, \frac{3}{2}\right)$ is a simple closed curve in U passing through $(0,1)$, and since it bounds a disc on which X is defined and closed (every point of that disc is at distance at least 1 from the origin), Green's theorem (Theorem 26.b.5) gives $\int_\gamma X \cdot d\gamma = 0$. Second, even a loop that does enclose the origin may be traversed clockwise, giving -2π. The correct general statement is that the integral equals 2π times the winding number of γ about the origin, and *"simple closed curve"* pins that number down only up to sign, and only once the origin is known to be inside.

Remark 26.f.51: This one field carries most of the chapter's content. It is closed but not exact, so it separates the two notions; the failure is measured by an integer, the winding number; and the obstruction is a property of the **domain**, not of the field. Restrict it to any simply connected piece of U, as in **(b)**, and a potential appears at once.

Q.E.D.

Solution of Exercise 26.b.19:

Both fields are closed on U, as the exam states and as one checks in a line: $X_2 = 0$ and X_1 does not depend on x_2, so both mixed partials vanish, and the computation for Y is the one carried out for Exercise 26.b.18, where Y appears under the name X. Being closed is where the two fields stop agreeing, and the six items are really two questions asked three times over.

(a) True. Take $u(x_1,x_2) := -\cos(x_1)$, which is smooth on all of $\mathbb{R}^2$ and has $\nabla u = (\sin x_1,\, 0) = X$. Restricting it to U gives the required potential. The puncture is irrelevant here, since X extends smoothly across the origin.

(b) False. This is Y's defining feature. If such a v existed, the work of Y around every closed loop in U would vanish, and item **(d)** exhibits one where it does not.

(c) True. X is a gradient on U by **(a)**, so by Proposition 26.b.14 its work along a path depends only on the endpoints: $\int_c X \cdot dc = u(c(b)) - u(c(a))$ for any C^1 path $c : [a,b] \to U$. Both γ and σ are closed, so both differences vanish.

(d) False. On the unit circle $Y(\sigma(t)) = (-\sin t,\, \cos t) = \dot\sigma(t)$, so the integrand is $|\dot\sigma|^2 = 1$ and

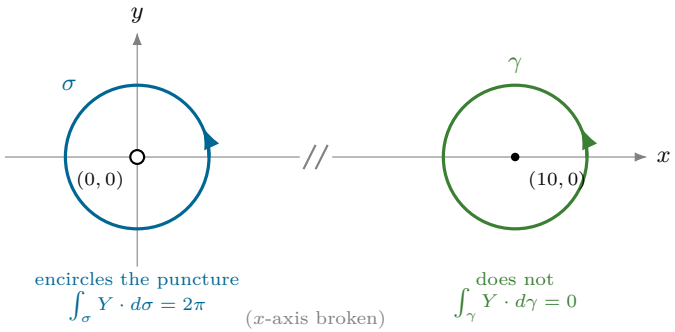
$$\int_\sigma Y \cdot d\sigma = \int_0^{2\pi} 1\, dt = 2\pi \neq 0.$$

(e) True. The curve γ is the unit circle centred at $(10,0)$, so its first coordinate satisfies $x_1 = 10 + \cos t \geq 9 > 0$. It therefore lies in the open right half-plane $V := \{x_1 > 0\}$, which is convex and hence simply connected. On V the closed field Y is exact (Corollary 26.c.25, with potential the polar angle $\arctan(x_2/x_1)$), so **(c)**'s argument applies verbatim and the integral vanishes.

(f) True. Same reason as **(c)**: X is exact on U and σ is closed. It does not matter that σ encircles the puncture, because X has no puncture to encircle.

Remark 26.f.52 (*Reading the four integrals as a table*): Items **(c)**–**(f)** are the four combinations

of two fields with two loops, and the single non-zero entry is the pair (Y, σ). The two loops are what to look at:



That is the whole point of the block:

- **The field alone does not decide.** Y gives 2π on σ and 0 on γ.
- **The loop alone does not decide.** σ gives 2π for Y and 0 for X.
- **It takes both:** a field that is closed without being exact, **and** a loop that encircles the hole responsible. Either one on its own gives zero.

The two circles are chosen so that this is visible without any computation. They are congruent, so they cannot be told apart by shape, size or orientation; they differ only in **where** they sit relative to the puncture.

Q.E.D.

Solution of Exercise 26.b.20:

Take $U := \mathbb{R}^3$ and

$$V(x, y, z) := (x,\ y,\ z)^{\mathsf{T}} = \nabla u(x, y, z), \qquad u(x, y, z) := \tfrac{1}{2}\big(x^2 + y^2 + z^2\big).$$

This V is a polynomial, hence C^∞, it is a gradient by construction, and it is not constant. The single-coordinate field $V(x, y, z) = (x, 0, 0)^{\mathsf{T}} = \nabla(x^2/2)$ serves equally well, and is the shortest answer that meets every clause.

Remark 26.f.53 (*Why the question is easier than it looks*): The clause to watch is *"nonconstant"*, which rules out the reflex answer $V \equiv e_1$. Beyond that, no verification is needed at all: writing down a potential **first** and differentiating it makes conservativeness true by construction, whereas starting from a field and hunting for a potential means solving three equations. On a box problem where only the final answer is graded, that is the difference between one line and half a page.

Q.E.D.

Solution to Exercise 26.c.29:

(a) U is **open**: it is the preimage of the open set $\mathbb{R} \setminus \{0\}$ under the continuous projection $(x, y, z) \mapsto y$.

U is **not connected**. The two sets $U^+ := \{y > 0\}$ and $U^- := \{y < 0\}$ are open, non-empty and disjoint, and their union is U, which is exactly a disconnection (Definition 8.a.1).

U is therefore **not simply connected** either, since simple connectedness includes path-connectedness. It is worth recording what **is** true: each of U^+ and U^- is an open half-space, hence convex, hence simply connected.

(b) Yes. Writing $E := e^{x+\frac{1}{y}}$ for brevity, so that $\partial_1 E = E$ and $\partial_2 E = -E/y^2$, we compute

$$\partial_2 F_1 = \partial_y\big((1+x)E\big) = -\frac{(1+x)}{y^2}\,E,$$
$$\partial_1 F_2 = \partial_x\left(-\frac{x}{y^2}E\right) = -\frac{1}{y^2}E - \frac{x}{y^2}E = -\frac{(1+x)}{y^2}\,E,$$

which agree. The remaining pairs are trivial: F_1 and F_2 do not depend on z and F_3 depends on nothing else, so $\partial_3 F_1 = 0 = \partial_1 F_3$ and $\partial_3 F_2 = 0 = \partial_2 F_3$. Hence F satisfies the integrability conditions (Lemma 26.b.12) on all of U.

(c) **Yes**, F is conservative, and the cleanest justification is to produce a potential outright, which is **(d)**. The structural reason is that U has two connected components, each of which is convex and therefore simply connected, so Corollary 26.c.25 applies **on each component separately** and supplies a potential there. Being conservative is a condition checked component by component, since a path in U never leaves the component it starts in. Consequently, the two local potentials together make F conservative on U.

(d) Yes. Take

$$u(x,y,z) := x\,e^{x+\frac{1}{y}} + g(z), \qquad (x,y,z) \in U,$$

which is well defined because $y \neq 0$ on U, and C^2 because g is. Then

$$\partial_x u = e^{x+\frac{1}{y}} + x\,e^{x+\frac{1}{y}} = (1+x)e^{x+\frac{1}{y}} = F_1, \qquad \partial_y u = -\frac{x}{y^2}e^{x+\frac{1}{y}} = F_2, \qquad \partial_z u = g'(z) = F_3,$$

so $\nabla u = F$ on U.

(e) First, γ does take values in U: its second component is $2 + t\sin(2\pi t)$, and $|t\sin(2\pi t)| \leq t \leq 1$ for $t \in [0,1]$, so that component is at least $1 > 0$. The path stays in the component U^+ throughout.

Its endpoints are

$$\gamma(0) = (0,\,2,\,1), \qquad \gamma(1) = \big(\cos(2\pi),\ 2+\sin(2\pi),\ 1-1+1\big) = (1,\,2,\,1).$$

By Proposition 26.b.14 the work integral of a conservative field depends only on the endpoints, so with the potential from **(d)**,

$$\int_\gamma F \cdot d\gamma = u(1,2,1) - u(0,2,1) = \Big(1\cdot e^{1+\frac{1}{2}} + g(1)\Big) - \Big(0\cdot e^{0+\frac{1}{2}} + g(1)\Big) = e^{3/2}.$$

Remark 26.f.54 (*Not simply connected does not mean not conservative*): Read this exercise against Example 26.c.28 and the pair says something the corollary alone does not. There, $U = \mathbb{R}^2\setminus\{0\}$ fails to be simply connected and the field really is not conservative. Here, $U = \{y \neq 0\}$ also fails to be simply connected, and the field is conservative anyway. Corollary 26.c.25 is a **sufficient** condition, so when its hypothesis fails the correct conclusion is that the theorem is silent, never that the field has no potential. Answering **(c)** with *"no, because U is not simply connected"* is the mistake the problem is built to catch.

The two failures are also of quite different kinds. $\mathbb{R}^2\setminus\{0\}$ is connected but has a hole, and it is the hole that a loop can wind around and accumulate work. $\{y \neq 0\}$ has no holes at all; it merely falls into two pieces, each as simple as a region can be. Disconnection is the harmless failure, because no path can exploit it, and that is why the component-by-component argument in **(c)** works. The unhelpful thing about the phrase *"simply connected"* is that it bundles the two together.

Finally, note that the answer to **(e)** does not involve g at all. The path returns to the height $z = 1$ it started at, so the $g(z)$ term of the potential cancels, and no property of g beyond differentiability was ever needed.

Q.E.D.

Solution of Exercise 26.b.21:

The answer is **(c)**, the integral is 0.

Parametrise the circle. It lies in the plane $x = 0$, has radius 1 and centre the origin, and must run counterclockwise as seen from the positive x-axis, which for the right-handed frame (y, z) means increasing angle from the y-axis towards the z-axis:

$$\gamma(t) = (0,\ \cos t,\ \sin t), \qquad t \in [0, 2\pi], \qquad \gamma'(t) = (0,\ -\sin t,\ \cos t).$$

Now evaluate the field on the curve. Since $x = 0$ along γ, the middle component of V vanishes identically there, and $V(\gamma(t)) = (1,\ 0,\ -2)$ is **constant**. Hence

$$V(\gamma(t)) \cdot \gamma'(t) = 1 \cdot 0 + 0 \cdot (-\sin t) + (-2)\cos t = -2\cos t,$$

and by Definition 26.b.3

$$\int_\gamma V \cdot d\gamma = \int_0^{2\pi} -2\cos t\, dt = \big[-2\sin t\big]_0^{2\pi} = 0.$$

Remark 26.f.55 (*Two ways to see the zero without integrating*): Neither requires the parametrisation, and both are worth having.

The field is constant on the curve. A constant field is a gradient field, of the linear function $\varphi(x, y, z) := x - 2z$, and by Proposition 26.b.14 the work of a gradient field around a **closed** path is zero. The only reason V itself is not a gradient field is its x component in the second slot, and that component is exactly what the plane $x = 0$ switches off.

Stokes' theorem. We have $\operatorname{curl} V = (0, 0, 1)^\mathsf{T}$, which is constant. Take the flat unit disc D in the plane $x = 0$ bounded by γ; its unit normal is $\pm(1, 0, 0)^\mathsf{T}$, so $\operatorname{curl} V \cdot \nu = 0$ pointwise on D and the classical Stokes theorem (Theorem 25.c.15) gives $\int_\gamma V \cdot d\gamma = \int_D \operatorname{curl} V \cdot \nu = 0$. The curl is non-zero, and the surface is oriented so as not to see it.

The trap the question sets is to compute $\operatorname{curl} V \neq 0$, conclude *"not conservative"*, and then guess a non-zero answer. Non-conservative fields have zero work around plenty of individual loops; what fails is only that the work is zero around **every** loop.

Q.E.D.

Solution of Exercise 26.c.30:

The correct answer is **(d)**, the guarded one.

- **(a) False.** This is the standing counterexample of this chapter. On $G := \mathbb{R}^2 \setminus \{0\}$ the angle field of Example 26.c.28 is irrotational, and its work around the unit circle is $2\pi \neq 0$. Vanishing curl gives path independence only **locally**, and a closed curve can encircle a hole.
- **(b) False**, and it is the option most likely to be chosen. The correct hypothesis is "*simply connected*", not connected: Corollary 26.c.25 needs every loop to be contractible, and $\mathbb{R}^2 \setminus \{0\}$ is connected, indeed path-connected, while having no potential for the field above. Connectedness only buys uniqueness of a potential up to an additive constant, once one exists.
- **(c) False.** Curl and divergence are independent of one another; nothing links them. The field $V(x, y) := (x, y)^\mathsf{T}$ has $\operatorname{curl} V = 0$ and $\operatorname{div} V = 2$.
- **(d) True**, and it is true because it claims almost nothing: **may** be. It is satisfied by any irrotational field on a simply connected G, for instance $G = \mathbb{R}^2$, where Corollary 26.c.25 produces a potential outright.

Remark 26.f.56 (*Why the correct answer is the weakest one*): The question is unusual in that the true option is the one asserting the least, and a reader who reads *"it may be that"* as a hedge rather than a claim will pass over it. The design is deliberate: options **(a)** and **(b)** are the two standard over-readings of $\operatorname{curl} V = 0$ (that it forces closed line integrals to vanish, and that connectedness is

the missing topological hypothesis), and **(c)** tests whether the two differential operators have been kept apart. Under a hypothesis on G this weak, nothing stronger than **(d)** survives.

Q.E.D.

Solution to Exercise 26.d.39:

(a) By linearity of the curl and the fundamental vector identity $\operatorname{curl}(\nabla\psi) \equiv 0$ (Proposition 22.b.12) for any C^2 function ψ:

$$\operatorname{curl} \widetilde{A} = \operatorname{curl}(A + \nabla\psi) = \operatorname{curl} A + \operatorname{curl}(\nabla\psi) = B + 0 = B.$$

This ambiguity $A \mapsto A + \nabla\psi$ is the classical "*gauge freedom*" in electromagnetism.

(b) For the ansatz $A = (A_1, A_2, 0)^{\mathsf{T}}$, the components of $\operatorname{curl} A$ are:

$$\operatorname{curl} A = \begin{pmatrix} \partial_y 0 - \partial_z A_2 \\ \partial_z A_1 - \partial_x 0 \\ \partial_x A_2 - \partial_y A_1 \end{pmatrix} = \begin{pmatrix} -\partial_z A_2 \\ \partial_z A_1 \\ \partial_x A_2 - \partial_y A_1 \end{pmatrix}.$$

Equating this with $B = (B_1, B_2, B_3)^{\mathsf{T}}$ gives the three partial differential equations

$$-\partial_z A_2 = B_1, \qquad \partial_z A_1 = B_2, \qquad \partial_x A_2 - \partial_y A_1 = B_3.$$

The proposed functions are

$$A_1(x,y,z) = \int_0^z B_2(x,y,s)\,ds, \qquad A_2(x,y,z) = -\int_0^z B_1(x,y,s)\,ds + \int_0^x B_3(t,y,0)\,dt.$$

Differentiating with respect to z by the fundamental theorem of calculus:

$$\partial_z A_1(x,y,z) = B_2(x,y,z), \qquad \partial_z A_2(x,y,z) = -B_1(x,y,z),$$

so $-\partial_z A_2 = B_1$ and $\partial_z A_1 = B_2$ hold identically. Now compute the third component $\partial_x A_2 - \partial_y A_1$:

$$\partial_x A_2(x,y,z) = -\int_0^z \partial_x B_1(x,y,s)\,ds + B_3(x,y,0),$$

$$\partial_y A_1(x,y,z) = \int_0^z \partial_y B_2(x,y,s)\,ds.$$

Subtracting them:

$$\partial_x A_2 - \partial_y A_1 = -\int_0^z \big(\partial_x B_1(x,y,s) + \partial_y B_2(x,y,s)\big)\,ds + B_3(x,y,0).$$

Since $\operatorname{div} B = \partial_x B_1 + \partial_y B_2 + \partial_z B_3 = 0$, we have $\partial_x B_1 + \partial_y B_2 = -\partial_z B_3$. Substituting this into the integral:

$$\begin{aligned} \partial_x A_2 - \partial_y A_1 &= -\int_0^z \big(-\partial_s B_3(x,y,s)\big)\,ds + B_3(x,y,0) \\ &= \int_0^z \partial_s B_3(x,y,s)\,ds + B_3(x,y,0) \\ &= \big[B_3(x,y,z) - B_3(x,y,0)\big] + B_3(x,y,0) = B_3(x,y,z). \end{aligned}$$

Thus all three component equations are satisfied, so $\operatorname{curl} A = B$ on all of $\mathbb{R}^3$.

(c) We are given $B(x,y,z) = (2xy,\ -y^2 + 2yz,\ -z^2)^{\mathsf{T}}$. First verify divergence-free:

$$\operatorname{div} B = \partial_x(2xy) + \partial_y(-y^2 + 2yz) + \partial_z(-z^2) = 2y + (-2y + 2z) - 2z = 0.$$

Now apply the formulas from part **(b)**:

$$A_1(x,y,z) = \int_0^z B_2(x,y,s)\,ds = \int_0^z (-y^2 + 2ys)\,ds = -y^2 z + yz^2,$$

$$A_2(x,y,z) = -\int_0^z B_1(x,y,s)\,ds + \int_0^x B_3(t,y,0)\,dt = -\int_0^z 2xy\,ds + \int_0^x 0\,dt = -2xyz,$$

$$A_3(x,y,z) = 0.$$

Thus $A(x,y,z) = (-y^2 z + yz^2,\ -2xyz,\ 0)^{\mathsf{T}}$. Direct verification of $\operatorname{curl} A$:

$$(\operatorname{curl} A)_1 = \partial_y 0 - \partial_z(-2xyz) = 2xy = B_1,$$
$$(\operatorname{curl} A)_2 = \partial_z(-y^2 z + yz^2) - \partial_x 0 = -y^2 + 2yz = B_2,$$
$$(\operatorname{curl} A)_3 = \partial_x(-2xyz) - \partial_y(-y^2 z + yz^2) = -2yz - (-2yz + z^2) = -z^2 = B_3.$$

Therefore $\operatorname{curl} A = B$.

Q.E.D.

Ordinary Differential Equations (Appendix A)

AI-Note A.1: This appendix compiles the material on Ordinary Differential Equations (ODEs). During the semester, this topic was covered in supplementary sessions.

Differential equations sit outside the main line of the course, but they draw on several of its tools, which is why they are collected here rather than left out. The first three sections are about solving. An n-th order equation is reduced to a first-order system, a linear system with constant coefficients is solved by diagonalising (or, failing that, by the Jordan form), and an inhomogeneous equation is reduced to a homogeneous one plus a single particular solution. The last section asks the prior question of when a solution exists and is unique at all. Its answer, the Picard–Lindelöf theorem, rests on the Lipschitz continuity of Section 5.b and the Banach fixed point theorem of Section 6.c.

Turning an n-th order ODE into an $n \times n$ system (Section A.a)

Given an n-th order linear ODE, we can rewrite it as a first-order problem in higher dimensions. Consider the example equation:

$$y(t) + y'(t) + 2y''(t) = f(t).$$

We define $u_1 := y$, which gives:

$$u_1 + u_1' + 2u_1'' = f.$$

Letting $u_2 := u_1'$, we can rewrite the second derivative as $u_1'' = u_2'$. The system becomes:

$$\begin{aligned} u_1 + u_2 + 2u_2' &= f \\ u_2 &= u_1'. \end{aligned}$$

Solving for the derivatives u_1' and u_2', we get:

$$\begin{aligned} u_1' &= u_2 \\ u_2' &= -\frac{1}{2}u_1 - \frac{1}{2}u_2 + \frac{1}{2}f. \end{aligned}$$

In matrix notation, this is:

$$\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1/2 & -1/2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} 0 \\ f/2 \end{pmatrix},$$

or more compactly,

$$u'(t) = Au(t) + F(t).$$

This reduction to a first-order system can be done with any n-th order ODE. We therefore focus on solving systems of the form $u'(t) = Au(t) + F(t)$.

Solving linear systems of ODEs (Section A.b)

Diagonalizable matrices (Subsection A.b.1)

We first look at the homogeneous case where $F \equiv 0$, so $u'(t) = Au(t)$. Suppose the matrix A is diagonalizable, meaning we can write $A = PDP^{-1}$ where D is a diagonal matrix and P is invertible:

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Then we have:

$$\begin{aligned} u'(t) &= Au(t) \\ \iff u'(t) &= PDP^{-1}u(t) \\ \iff P^{-1}u'(t) &= DP^{-1}u(t) \\ \iff (P^{-1}u(t))' &= D(P^{-1}u(t)). \end{aligned}$$

By defining a new variable $v(t) := P^{-1}u(t)$, the system uncouples:

$$v'(t) = Dv(t) \iff \begin{pmatrix} v_1'(t) \\ \vdots \\ v_n'(t) \end{pmatrix} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} v_1(t) \\ \vdots \\ v_n(t) \end{pmatrix}.$$

This yields n independent 1-dimensional equations $v_i'(t) = \lambda_i v_i(t)$, which have the solutions:

$$v_i(t) = C_i e^{\lambda_i t} \quad \text{for } i \in \{1, \dots, n\}.$$

The solution to the original system is then recovered by $u(t) = Pv(t)$. By diagonalizing, we were able to uncouple the $n \times n$ system and solve the scalar equations without a problem.

Non-diagonalizable matrices (Subsection A.b.2)

What if A is not diagonalizable? Then we have to use the Jordan Normal Form: $A = PJP^{-1}$. Following the same substitution $v(t) = P^{-1}u(t)$, we obtain $v'(t) = Jv(t)$. We now focus on each Jordan block in J. For a block with eigenvalue λ_i, the system looks like:

$$\begin{pmatrix} v_j'(t) \\ \vdots \\ v_{j+d}'(t) \end{pmatrix} = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix} \begin{pmatrix} v_j(t) \\ \vdots \\ v_{j+d}(t) \end{pmatrix}.$$

This translates to the equations:

$$\begin{aligned} v_j'(t) &= \lambda_i v_j(t) + v_{j+1}(t) \\ &\vdots \\ v_{j+d}'(t) &= \lambda_i v_{j+d}(t). \end{aligned}$$

We first solve for $v_{j+d}(t)$ from the last equation. Then we plug it into the preceding equation to find $v_{j+d-1}(t)$, and continue this process upwards.

Since this back-substitution can be time-consuming, it is useful to use matrix exponentials. For a system $u'(t) = Au(t)$, the formal solution is:

$$u(t) = e^{tA}u_0,$$

where u_0 is given by the initial conditions. Evaluating the matrix exponential e^{tA} provides a direct way to solve the system.

Remark A.b.1 (*Calculating the matrix exponential in practice*): When computing $\exp(A) = I + \sum_{m=1}^{\infty} \frac{A^m}{m!}$, the most common strategies are:

- **(a) Look for a pattern:** Sometimes $A^2 = A$, or $A^3 = 2A$, or A is nilpotent ($A^n = 0$ for $n \geq k$). In these cases, the infinite series can be simplified directly.
- **(b) Diagonalization:** If A is diagonalizable ($A = BDB^{-1}$), then $\exp(A) = B\exp(D)B^{-1}$, where $\exp(D)$ is the diagonal matrix of exponentials.
- **(c) Jordan Normal Form (JNF):** If A is not diagonalizable, one can use the JNF, though this is often more tedious and less frequently required.

Example A.b.2 (*Solving a 2×2 system using matrix exponentials*): Consider the system given by

$$u_1(t) + 2u_2(t) = u_1'(t)$$
$$2u_1(t) + u_2(t) = u_2'(t).$$

This can be written as $u'(t) = Au(t)$ with $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. The eigenvalues of A are found via $\det(A - \lambda I) = (1-\lambda)^2 - 4 = 0$, giving $\lambda_1 = 3$ and $\lambda_2 = -1$. The corresponding eigenvectors are $w_1 = (1,1)^\mathsf{T}$ and $w_2 = (1,-1)^\mathsf{T}$. Since A is diagonalizable, we have $A = PDP^{-1}$ with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$. The general solution is then given by:

$$u(t) = \sum_{i=1}^{2} C_i e^{\lambda_i t} w_i = C_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Now consider a non-diagonalizable example where the system is already in Jordan normal form, $u'(t) = Ju(t)$ with $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Using the matrix exponential, the solution is $u(t) = e^{tJ}u(0)$. We can split $J = D + N$ where $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is diagonal and $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent ($N^2 = 0$). Since D and N commute, $e^{tJ} = e^{tD}e^{tN}$. We calculate:

$$e^{tD} = \begin{pmatrix} e^t & 0 \\ 0 & e^t \end{pmatrix}, \qquad e^{tN} = I + tN = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Multiplying these gives:

$$e^{tJ} = \begin{pmatrix} e^t & te^t \\ 0 & e^t \end{pmatrix}.$$

Thus, the solution is $u_1(t) = e^t u_1(0) + te^t u_2(0)$ and $u_2(t) = e^t u_2(0)$.

Example A.b.3 (*Solving an initial value problem*): Consider the initial value problem:

$$y_1' = 3y_1 - 2y_2,$$
$$y_2' = -y_1 + 2y_2,$$

with initial conditions $y_1(0) = 1$ and $y_2(0) = 0$. The system is $y' = Ay$ with $A = \begin{pmatrix} 3 & -2 \\ -1 & 2 \end{pmatrix}$. We find the eigenvalues:

$$\det(A - \lambda I) = (3-\lambda)(2-\lambda) - 2 = \lambda^2 - 5\lambda + 4 = (\lambda - 4)(\lambda - 1) = 0.$$

The eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 1$. The corresponding eigenvectors are: For $\lambda_1 = 4$: $\begin{pmatrix} -1 & -2 \\ -1 & -2 \end{pmatrix} v = 0 \implies v_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$. For $\lambda_2 = 1$: $\begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix} v = 0 \implies v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The general solution is $y(t) = C_1 e^{4t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + C_2 e^{t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Using the initial conditions $y(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$-2C_1 + C_2 = 1$$
$$C_1 + C_2 = 0 \implies C_2 = -C_1.$$

Substituting this gives $-3C_1 = 1$, so $C_1 = -1/3$ and $C_2 = 1/3$. The particular solution is:

$$y(t) = -\frac{1}{3}e^{4t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \frac{1}{3}e^{t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2e^{4t} + e^t \\ -e^{4t} + e^t \end{pmatrix}.$$

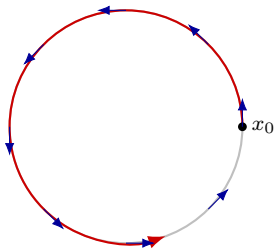
Autonomous systems and phase portraits (Subsection A.b.3)

A system $u'(t) = F(u(t))$ in which F does not depend on t explicitly is called "*autonomous*" (German: „*autonom*"). Its flow can then be pictured directly as a vector field on the state space, independent of time: at each point x, the arrow $F(x)$ shows the instantaneous velocity of the trajectory passing through x. The three examples below (two linear, one not) are the standard pictures.

Example A.b.4 (*A pure rotation*): The system $u_1' = -u_2$, $u_2' = u_1$ is $u' = Au$ with $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The unique solution with $u(0) = x_0$ is

$$u(t) = e^{tA}x_0 = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} x_0,$$

a rigid rotation of x_0 around the origin at unit angular speed: F is the velocity field of a fluid rotating rigidly counterclockwise, and every trajectory is a circle centred at the origin.

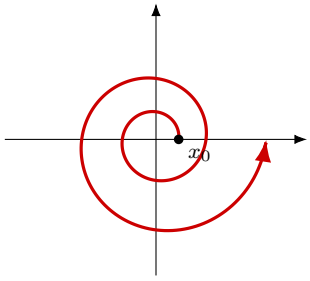


$u(t) = e^{tA}x_0$, circular trajectories

Example A.b.5 (*A rotating spiral*): The system $u_1' = u_1 - u_2$, $u_2' = u_1 + u_2$ is $u' = Au$ with $A = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) = \mathrm{id} + B$, where $B := \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is the generator of Example A.b.4. Since id and B commute, $e^{tA} = e^{t\,\mathrm{id}}e^{tB}$, so

$$u(t) = e^{tA}x_0 = e^t \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} x_0 :$$

the same rotation as before, now with an exponentially growing radius. The trajectory is an outward spiral.



$u(t) = e^{tA}x_0$: rotation combined with e^t growth

Example A.b.6 (*A nonlinear system with a limit cycle*): Consider the nonlinear autonomous system

$$u' = u - v - (u^2 + v^2)u, \qquad v' = u + v - (u^2 + v^2)v,$$

with initial value $(u(0), v(0)) = (x_0, y_0)$. If $(x_0, y_0) = (0, 0)$, the constant function $(u, v) \equiv (0, 0)$ is a solution, and Theorem A.d.13 (the right-hand side is C^1, Exercise A.d.16) shows it is the **only** one.

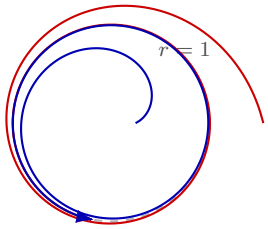
For $(x_0, y_0) \neq (0,0)$, switch to polar coordinates $u = r\cos\varphi$, $v = r\sin\varphi$ with $r^2 = u^2 + v^2$. Differentiating $r^2 = u^2 + v^2$ and substituting the system,

$$(r^2)' = 2uu' + 2vv' = 2\big(u^2 - uv - (u^2+v^2)u^2\big) + 2\big(uv + v^2 - (u^2+v^2)v^2\big) = 2(r^2 - r^4),$$

so, since $(r^2)' = 2rr'$, the radius satisfies the autonomous scalar equation

$$r' = r - r^3, \qquad r(0) = \sqrt{x_0^2 + y_0^2}.$$

Separating variables and solving gives $r(t) = \exp(t)/\sqrt{A + \exp(2t)}$ for $A := 1/(x_0^2 + y_0^2) - 1 \in (-1, \infty)$, and $r(t) \to 1$ as $t \to \infty$ **regardless of the initial value**. Therefore the circle $r = 1$ is a "*limit cycle*" (German: „*Grenzzykel*"), attracting every trajectory except the equilibrium at the origin. For the angle, $\varphi' = (uv' - vu')/r^2$, and substituting the system the r^2-terms cancel: $uv' - vu' = u(u+v) - v(u-v) = u^2 + v^2$, so $\varphi' \equiv 1$ away from the origin. The motion is therefore a steady rotation combined with the radial approach to $r = 1$: trajectories starting outside the unit circle spiral in, and those starting inside spiral out, both converging to the same circular orbit.



qualitative sketch: outside (red) spirals in, inside (blue) spirals out,
both approaching the limit cycle $r = 1$

Exercise A.b.7 (*True or False* (what the spectrum decides)): Let A be a real $n \times n$ matrix. Decide whether each of the following statements about the solutions of the linear system $\dot{x} = Ax$ is true or false, with justification.

(a) Every solution $x : \mathbb{R} \to \mathbb{R}^n$ with $\sup_{t>0}|x(t)| = \infty$ grows exponentially as $t \to \infty$.

(b) There is a non-trivial constant solution $x : \mathbb{R} \to \mathbb{R}^n$ if and only if $\det(A) = 0$.

(c) If A is diagonalisable over $\mathbb{R}$ and there exists a non-trivial bounded solution $x : \mathbb{R} \to \mathbb{R}^n$, then 0 is an eigenvalue of A.

(d) If A is diagonalisable over $\mathbb{C}$ and there exists a non-trivial bounded solution $x : \mathbb{R} \to \mathbb{R}^n$, then 0 is an eigenvalue of A.

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *(undated) in* `old_exams/fs2023/`*, Problem 8. That paper names no lecturer. "Bounded" is read, as the original intends, as bounded on all of* $\mathbb{R}$.

Solution to Exercise A.b.7:

(a) **False.** A Jordan block with eigenvalue 0 produces polynomial growth and no exponential at all. Take

$$A := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad x(t) := (t,\, 1)^{\mathsf{T}},$$

which satisfies $\dot{x} = (1,0)^{\mathsf{T}} = Ax$. It is unbounded, and it grows linearly, so it is not bounded below by any $ce^{\varepsilon t}$ with $\varepsilon > 0$. This is exactly the situation Appendix A.b.2 is about: the polynomial factors that a non-trivial Jordan block contributes are invisible to the eigenvalues alone.

(b) True. A constant function $x \equiv v$ solves the system precisely when $0 = \dot{x} = Av$, so a non-trivial constant solution exists if and only if $\ker A \neq \{0\}$, which is if and only if $\det A = 0$.

(c) True. Diagonalise A over $\mathbb{R}$ with eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ and a basis of eigenvectors $v_1, \ldots, v_n$. Every solution is $x(t) = \sum_k c_k e^{\lambda_k t} v_k$. If some $c_k \neq 0$ with $\lambda_k \neq 0$, then $|x(t)| \to \infty$ either as $t \to +\infty$ (if $\lambda_k > 0$) or as $t \to -\infty$ (if $\lambda_k < 0$), because the eigenvectors are independent and no other term can cancel that one. So a bounded non-trivial solution must have $c_k \neq 0$ only for $\lambda_k = 0$, and in particular 0 is an eigenvalue.

(d) False, and the reason is that a complex eigenvalue on the imaginary axis is bounded without being zero. Take the rotation

$$A := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

with eigenvalues $\pm i$, hence diagonalisable over $\mathbb{C}$ and not over $\mathbb{R}$. Every solution runs around a circle at unit speed (Example A.b.4), so all of them are bounded, while $\det A = 1 \neq 0$ and 0 is not an eigenvalue.

Remark A.b.8 (*The real dividing line is the real part*): Parts **(c)** and **(d)** differ in one word and have opposite answers, which is a signal that neither is stating the real criterion. For a diagonalisable A the true statement is that a solution $\sum_k c_k e^{\lambda_k t} v_k$ is bounded on all of $\mathbb{R}$ exactly when every λ_k with $c_k \neq 0$ has $\operatorname{Re} \lambda_k = 0$, because $|e^{\lambda t}| = e^{t \operatorname{Re} \lambda}$ and the oscillation contributes nothing to the size. Part **(c)** is then true only because a **real** eigenvalue with zero real part has no choice but to be 0.

Part **(a)** is the same observation from the other side: it is the **diagonalisability** that is doing the work in **(c)** and **(d)**, and dropping it lets a Jordan block manufacture growth that the spectrum does not predict. Together the four say that the spectrum governs the long-run behaviour only once the Jordan structure is known to be trivial.

Q.E.D.

Inhomogeneous linear ODEs (Section A.c)

Given the n-th order linear inhomogeneous ODE

$$Ly(x) = b(x), \qquad \text{where } L = \sum_{k=0}^{n} a_k \frac{d^k}{dx^k},$$

the general solution is always of the form

$$y(x) = y_{\text{hom}}(x) + y_{\text{part}}(x).$$

The homogeneous solution $y_{\text{hom}}(x)$ fulfills $Ly_{\text{hom}}(x) = 0$, or $\sum_{k=0}^{n} a_k y_{\text{hom}}^{(k)}(x) = 0$. This is solved using the general Ansatz $y_{\text{hom}}(x) = Ce^{rx}$, which leads to the characteristic equation:

$$\sum_{k=0}^{n} a_k C r^k e^{rx} = 0 \implies \sum_{k=0}^{n} a_k r^k = 0,$$

yielding roots $r_1, \ldots, r_n$ (assuming they are all distinct).

The particular solution $y_{\text{part}}(x)$ solves $Ly_{\text{part}}(x) = b(x)$ and is found using a fitting Ansatz, often an *"educated guess"* based on the form of the inhomogeneous term $b(x)$.

Example A.c.9 (*Solving an inhomogeneous linear ODE*): Consider the second-order ODE:

$$y'' - 4y' + 13y = xe^x.$$

Inhomogeneous Term $b(x)$	**Ansatz for** $y_{\text{part}}(x)$
$\sum_{i=0}^{m} b_i x^i$	$\sum_{i=0}^{m} A_i x^i$
$e^{\alpha x} \sum_{i=0}^{m} b_i x^i$	$e^{\alpha x} \sum_{i=0}^{m} A_i x^i$
$\sin(\omega x) \sum_{i=0}^{m} b_i x^i + \cos(\omega x) \sum_{i=0}^{m} c_i x^i$	$\sin(\omega x) \sum_{i=0}^{m} A_i x^i + \cos(\omega x) \sum_{i=0}^{m} B_i x^i$
$\sinh(\omega x) \sum_{i=0}^{m} b_i x^i + \cosh(\omega x) \sum_{i=0}^{m} c_i x^i$	$\sinh(\omega x) \sum_{i=0}^{m} A_i x^i + \cosh(\omega x) \sum_{i=0}^{m} B_i x^i$
$e^{\alpha x} \sin(\omega x) \sum_{i=0}^{m} b_i x^i + e^{\alpha x} \cos(\omega x) \sum_{i=0}^{m} c_i x^i$	$e^{\alpha x} \sin(\omega x) \sum_{i=0}^{m} A_i x^i + e^{\alpha x} \cos(\omega x) \sum_{i=0}^{m} B_i x^i$

Table A.1: Standard Ansätze for solving inhomogeneous linear ODEs. **Special case:** If a part of the chosen function for $y_{\text{part}}(x)$ is already present in the solution of the homogeneous problem, the Ansatz is additionally multiplied by x.

First, we solve the homogeneous problem $y'' - 4y' + 13y = 0$. The characteristic polynomial is $\lambda^2 - 4\lambda + 13 = 0$, with roots:

$$\lambda_{1,2} = \frac{4 \pm \sqrt{16 - 52}}{2} = 2 \pm 3i.$$

The homogeneous solution is therefore:

$$y_{\text{hom}}(x) = e^{2x}\big(C_1 \cos(3x) + C_2 \sin(3x)\big).$$

Next, we find a particular solution for the inhomogeneous term $b(x) = xe^x$. According to Table A.1, we use the Ansatz $y_{\text{part}}(x) = (Ax+B)e^x$. Since $\lambda = 1$ is not a root of the characteristic polynomial, we do not need to multiply by an extra factor of x. We compute the derivatives:

$$y'_{\text{part}}(x) = Ae^x + (Ax + B)e^x = (Ax + A + B)e^x,$$
$$y''_{\text{part}}(x) = Ae^x + (Ax + A + B)e^x = (Ax + 2A + B)e^x.$$

Plugging these into the original ODE yields:

$$\begin{aligned}(Ax + 2A + B)e^x - 4(Ax + A + B)e^x + 13(Ax + B)e^x &= xe^x \\ \big((A - 4A + 13A)x + (2A + B - 4A - 4B + 13B)\big)e^x &= xe^x \\ (10Ax - 2A + 10B)e^x &= xe^x.\end{aligned}$$

Comparing coefficients, we get $10A = 1 \implies A = \frac{1}{10}$, and $-2A + 10B = 0 \implies 10B = \frac{2}{10} \implies B = \frac{1}{50}$. The particular solution is $y_{\text{part}}(x) = \left(\frac{1}{10}x + \frac{1}{50}\right)e^x$. The general solution is then $y(x) = y_{\text{hom}}(x) + y_{\text{part}}(x)$.

Exercise A.c.10 (*Three linear ODEs*):

(a) Determine the general solution on $I = \mathbb{R}$ of $y'' - 4y' + 4y = \sin t$.

(b) Determine the solution on $I = \mathbb{R}$ of the initial value problem $y'' - y = t$, $y(0) = 1$, $y'(0) = 3$.

(c) Find a solution on $I = (0, \infty)$ of the initial value problem $y' - \left(\frac{4}{t} + 1\right)y = t^4$, $y(1) = 1$.

Proof of Exercise A.c.10:

(a) The characteristic polynomial $r^2 - 4r + 4 = (r-2)^2$ has the double root $r = 2$, so $y_{\text{hom}}(t) = (C_1 + C_2 t)e^{2t}$. For the particular solution, try $y_{\text{part}}(t) := A \sin t + B \cos t$ (Table A.1; since $\pm i\omega = \pm i$ are not roots of the characteristic polynomial, no extra factor of t is needed). Substituting,

$$y''_{\text{part}} - 4y'_{\text{part}} + 4y_{\text{part}} = (3A + 4B)\sin t + (3B - 4A)\cos t \overset{!}{=} \sin t,$$

so $3A + 4B = 1$ and $-4A + 3B = 0$, giving $B = \frac{4}{3}A$ and then $A = \frac{3}{25}$, $B = \frac{4}{25}$. The general solution is

$$y(t) = (C_1 + C_2 t)e^{2t} + \tfrac{3}{25}\sin t + \tfrac{4}{25}\cos t.$$

(b) The characteristic polynomial $r^2 - 1$ has roots $r = \pm 1$, so $y_{\text{hom}}(t) = C_1 e^t + C_2 e^{-t}$. Since $r = 0$ is not a root, try $y_{\text{part}}(t) := At + B$; then $y''_{\text{part}} - y_{\text{part}} = -At - B \overset{!}{=} t$ gives $A = -1$, $B = 0$, so $y_{\text{part}}(t) = -t$. The general solution is $y(t) = C_1 e^t + C_2 e^{-t} - t$, with $y'(t) = C_1 e^t - C_2 e^{-t} - 1$. The initial conditions give $C_1 + C_2 = 1$ and $C_1 - C_2 = 4$, so $C_1 = \frac{5}{2}$, $C_2 = -\frac{3}{2}$, and

$$y(t) = \tfrac{5}{2} e^t - \tfrac{3}{2} e^{-t} - t.$$

(c) This is a first-order linear ODE, $y' + p(t)y = t^4$ with $p(t) := -\left(\frac{4}{t} + 1\right)$, so the standard integrating factor is $\mu(t) := \exp\left(\int p(t)\,dt\right) = \exp\left(-4\ln t - t\right) = t^{-4}e^{-t}$.

AI-Note A.2: The script's own hint suggests $u(t) := e^{H(t)}y(t)$ with $H'(t) = \frac{4}{t} + 1$, giving a **positive** exponent. Checking it directly, $u' = e^H(H'y + y')$, and substituting $y' = H'y + t^4$ (from the ODE) gives $u' = e^H(2H'y + t^4)$, which still contains y. The substitution therefore does not decouple. The sign must be negative, $u(t) := e^{-H(t)}y(t) = t^{-4}e^{-t}y(t)$: then $u' = e^{-H}(y' - H'y) = e^{-H}\left(y' - (\frac{4}{t} + 1)y\right) = e^{-H}t^4$ directly from the ODE, which **does** decouple. Used below.

So with $u(t) := t^{-4}e^{-t}y(t)$,

$$u'(t) = t^{-4}e^{-t}t^4 = e^{-t} \qquad \Longrightarrow \qquad u(t) = -e^{-t} + C$$

for a constant C. Hence $y(t) = t^4 e^t\left(C - e^{-t}\right) = Ct^4e^t - t^4$. The initial condition $y(1) = 1$ gives $Ce - 1 = 1$, i.e. $C = 2/e$, so

$$y(t) = 2t^4e^{t-1} - t^4, \qquad t \in (0, \infty).$$

Q.E.D.

Example A.c.11 (*Pure resonance of a pumped harmonic oscillator*)**:** Consider the forced harmonic oscillator $y : [0, \infty) \to \mathbb{R}$,

$$y''(t) + ky(t) = \sin(\omega t) \quad \text{for } t > 0, \qquad y(0) = y'(0) = 0,$$

with $k > 0$. The homogeneous equation $z'' + kz = 0$ has general solution $z(t) = C_1\cos(\sqrt{k}t) + C_2\sin(\sqrt{k}t)$. For $\omega \neq \sqrt{k}$, the Ansatz $y_{\text{part}}(t) := A\sin(\omega t) + B\cos(\omega t)$ gives, on substitution, $A(k-\omega^2) = 1$ and $B(k-\omega^2) = 0$, so $B = 0$ and $A = 1/(k-\omega^2)$, giving $y_{\text{part}}(t) = \sin(\omega t)/(k-\omega^2)$. Imposing $y(0) = y'(0) = 0$ on the general solution $y = z + y_{\text{part}}$ fixes $C_1 = 0$ and $C_2 = -\omega/\left(\sqrt{k}(k-\omega^2)\right)$, giving the complete solution, for $\omega \neq \sqrt{k}$,

$$y(t) = \frac{1}{\sqrt{k} + \omega}\left(\frac{\sin(\sqrt{k}t)}{\sqrt{k}} - \frac{\sin(\sqrt{k}t) - \sin(\omega t)}{\sqrt{k} - \omega}\right).$$

Taking the limit $\omega \to \sqrt{k}$ ("*resonance*": the forcing frequency matches the system's natural frequency) gives

$$y(t) = \frac{1}{2\sqrt{k}}\left(\frac{\sin(\sqrt{k}t)}{\sqrt{k}} - t\cos(\sqrt{k}t)\right).$$

The term $t\cos(\sqrt{k}t)$ shows the amplitude of the oscillation now grows **linearly** in t, without bound. This is the phenomenon behind bridges failing under marching troops and, per popular-science lore not verified here, opera singers shattering glasses.

Remark A.c.12: This is the concrete instance of the *"special case"* noted in the caption of Table A.1: the naive Ansatz $A\sin(\omega t) + B\cos(\omega t)$ fails exactly when $\omega = \sqrt{k}$, because $\sin(\sqrt{k}t)$ and $\cos(\sqrt{k}t)$ already solve the homogeneous equation. The fix is the same one the table prescribes. Multiplying the Ansatz by t is visibly what appears in the limiting solution above.

Duhamel's Principle (Subsection A.c.1)

For a system of inhomogeneous linear ODEs $u'(t) = Au(t) + f(t)$, the solution can be constructed by combining the homogeneous solution and a particular solution obtained via Duhamel's principle:

$$u(t) = \underbrace{e^{At}u(0)}_{\text{homogeneous solution}} + \underbrace{\int_0^t e^{(t-s)A} f(s)\,ds}_{\text{inhomogeneous part}}.$$

This elegantly generalizes the 1-dimensional case where the solution to $u'(t) = au(t) + f(t)$ is given by $u(t) = e^{at}u(0) + \int_0^t e^{a(t-s)} f(s)\,ds$.

The Cauchy–Lipschitz / Picard–Lindelöf theorem (Section A.d)

The Cauchy–Lipschitz theorem (also known as the Picard–Lindelöf theorem) guarantees the existence and uniqueness of solutions to first-order initial value problems, provided the right-hand side is Lipschitz continuous with respect to the state variable. For an equation of the form $y'(x) = F(x, y(x))$, one first puts the differential equation in normal form to determine if the theorem applies, which requires checking the properties of F.

Theorem A.d.13 (*Picard–Lindelöf, local existence and uniqueness*): Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be an open set, $(t_0, x_0) \in U$, and $F : U \to \mathbb{R}^n$ be continuous. If F is locally Lipschitz continuous in x, i.e.,

$$\forall (t_1, x_1) \in U \quad \exists \varepsilon > 0, L \geq 0 \text{ such that}$$
$$\forall (t, x_2), (t, x_3) \in B_\varepsilon(t_1, x_1) \cap U, \quad |F(t, x_2) - F(t, x_3)| \leq L|x_2 - x_3|,$$

then there exists an interval $(a, b) \subseteq \mathbb{R}$ with $t_0 \in (a, b)$ such that a differentiable function $u : (a, b) \to \mathbb{R}^n$ exists which satisfies $(t, u(t)) \in U$ for all $t \in (a, b)$, and u is a solution to the initial value problem:

$$\begin{cases} u'(t) = F(t, u(t)) \\ u(t_0) = x_0. \end{cases}$$

Furthermore, for any other solution $v : (\tilde{a}, \tilde{b}) \to \mathbb{R}^n$ of the same initial value problem, we have $(\tilde{a}, \tilde{b}) \subseteq (a, b)$ and $u|_{(\tilde{a},\tilde{b})} = v$. As $t \to a$ or $t \to b$, the limit $\lim_{t\to a}(t, u(t))$ (respectively $\lim_{t\to b}(t, u(t))$) either does not exist or does not lie in U.

A simplified, highly practical formulation is the following:

Corollary A.d.14 (*Picard–Lindelöf, simplified version*): Let $I \subseteq \mathbb{R}$ be a compact interval, $\Omega \subseteq \mathbb{R}^n$ an open interval (or open set), $(t_0, y_0) \in I \times \Omega$, and $f : I \times \Omega \to \mathbb{R}^n$ continuous. Suppose f satisfies the Lipschitz condition

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2| \qquad \forall t \in I \text{ and } \forall y_1, y_2 \in \Omega.$$

Then there exists a $\delta > 0$ such that the initial value problem $u' = f(t, u)$ with $u(t_0) = y_0$ has a unique solution $u : I \cap [t_0 - \delta, t_0 + \delta] \to \mathbb{R}^n$. If $\Omega = \mathbb{R}^n$, one can choose the domain to be the entire interval I.

Example A.d.15 (*Applying Picard–Lindelöf*): Fix $T > 0$, let $I := [0, T]$ and $\Omega := \mathbb{R}$, and consider the function $f : I \times \Omega \to \mathbb{R}$ given by $f(t, y) = \frac{y}{1+t^2}$. We want to prove that the initial value problem

$$u'(t) - \frac{u(t)}{1 + t^2} = 0, \qquad u(0) = y_0$$

has a unique solution $u : I \to \mathbb{R}$, for whatever $y_0 \in \mathbb{R}$ is prescribed.

We rewrite the ODE as $u'(t) = f(t, u(t))$. To apply the Picard–Lindelöf theorem, we check if f satisfies the Lipschitz condition in y:

$$|f(t, y_1) - f(t, y_2)| = \left| \frac{y_1}{1 + t^2} - \frac{y_2}{1 + t^2} \right| = \frac{1}{1 + t^2}|y_1 - y_2| \leq 1 \cdot |y_1 - y_2|.$$

Since $\frac{1}{1+t^2} \le 1$ for all $t \in I$, the function f is Lipschitz continuous in y with Lipschitz constant $L = 1$, uniformly in t. Because $\Omega = \mathbb{R}$, Corollary A.d.14 delivers a unique solution on the whole of I, not merely near $t = 0$. As T was arbitrary and any two of these solutions agree on the shorter interval by the same uniqueness, they patch together to a unique solution on $[0, \infty)$, which separating variables identifies as $u(t) = y_0 e^{\arctan t}$.

AI-Note A.3: Note where the initial condition is doing its work. Picard–Lindelöf never says that a differential equation has a unique solution; that is false for every equation with a one-parameter family of solutions, this one included. What it says is that an initial value problem does. The distinction is easy to lose because the Lipschitz estimate, which is the part requiring effort, does not mention y_0 at all.

Exercise A.d.16 (*C^1 in space implies locally Lipschitz*): Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and $F : U \to \mathbb{R}^n$ be continuous, such that the partial derivatives $\partial_{x_k} F$ for $k \in \{1, \dots, n\}$ exist and are continuous on U. Show that F is locally Lipschitz in x and thus satisfies the hypotheses of Theorem A.d.13.

Proof of Exercise A.d.16:

Fix $(t_1, x_1) \in U$. Since U is open, choose $\varepsilon > 0$ with the closed ball $\overline{B_\varepsilon(t_1, x_1)} \subseteq U$; this closed ball is compact. Each $\partial_{x_k} F$ is continuous, hence bounded on this compact set, say by M_k; set $L := \sum_{k=1}^n M_k$.

For $(t, x_2), (t, x_3) \in B_\varepsilon(t_1, x_1)$, the straight segment $s \mapsto x_3 + s(x_2 - x_3)$, $s \in [0, 1]$, stays inside the ball (which is convex), so $g(s) := F(t,\, x_3 + s(x_2 - x_3))$ is differentiable on $[0, 1]$ with $g'(s) = \sum_{k=1}^n \partial_{x_k} F(t, x_3 + s(x_2 - x_3))\,(x_2 - x_3)_k$ by the chain rule. Hence

$$\begin{aligned} |F(t, x_2) - F(t, x_3)| = |g(1) - g(0)| &= \left| \int_0^1 g'(s)\, ds \right| \\ &\le \int_0^1 \sum_{k=1}^n |\partial_{x_k} F(t, x_3 + s(x_2 - x_3))|\, |(x_2 - x_3)_k|\, ds \\ &\le \sum_{k=1}^n M_k\, |(x_2 - x_3)_k| \le \Big(\sum_{k=1}^n M_k \Big) |x_2 - x_3| = L |x_2 - x_3|, \end{aligned}$$

using $|(x_2 - x_3)_k| \le |x_2 - x_3|$ in the last step. So F is L-Lipschitz in x on $B_\varepsilon(t_1, x_1) \cap U$, and since $(t_1, x_1) \in U$ was arbitrary, F is locally Lipschitz in x. Q.E.D.

Remark A.d.17: This is the standard shortcut for checking the hypothesis of Theorem A.d.13 in practice: one essentially never verifies the Lipschitz estimate by hand, but instead checks that F is C^1 in the spatial variables. That is exactly the regularity most vector fields one writes down already have.

Lemma A.d.18 (*Grönwall's inequality*): Let $u \in C^1([a, b])$ satisfy $u'(t) \le \beta(t) u(t)$ for all $t \in [a, b]$, where $\beta \in C^0([a, b])$. Then

$$u(t) \;\le\; u(a) \exp\!\left(\int_a^t \beta(s)\, ds \right) \qquad \text{for all } t \in [a, b].$$

Proof:

Let $v(t) := \exp\big(\int_a^t \beta(s)\, ds \big) > 0$, so $v'(t) = \beta(t) v(t)$, and set $f(t) := u(t)/v(t) \in C^1([a, b])$. Then

$$f'(t) = \frac{u'(t) v(t) - u(t) v'(t)}{v(t)^2} \le \frac{\beta(t) u(t) v(t) - u(t) \beta(t) v(t)}{v(t)^2} = 0,$$

using the hypothesis $u'(t) \le \beta(t) u(t)$ in the numerator. So f is decreasing on $[a, b]$, giving $u(t)/v(t) = f(t) \le f(a) = u(a)/v(a) = u(a)$, i.e. $u(t) \le u(a) v(t)$, which is the claim. Q.E.D.

Remark A.d.19 (*What Grönwall is for*): Grönwall's inequality is the standard tool for turning a **differential** bound into an **integral** one. The official script uses it to give a clean uniqueness proof for linear ODEs: if z solves an order-m linear equation $Lz = 0$, the quantity $u(t) := z(t)^2 + z'(t)^2 + \cdots + z^{(m-1)}(t)^2$ satisfies a bound $u'(t) \le Cu(t)$ for a constant C depending only on the coefficients of L, and Grönwall with $u(t_0) = 0$ (zero initial data) forces $u \equiv 0$, hence $z \equiv 0$. This is the *"zero in, zero out"* principle behind uniqueness for linear ODEs. The same device, run on $u(t) := |v_1(t) - v_2(t)|^2$ for two solutions v_1, v_2 of a Lipschitz system agreeing at one time, gives an alternative route to the uniqueness half of Theorem A.d.13.

Picard iteration (Subsection A.d.1)

The proof of Theorem A.d.13 is a Banach fixed-point argument (Theorem 6.c.19), and fixed-point arguments are **constructive**: the sequence of approximations built along the way converges to the solution. So Picard–Lindelöf does not merely assert that a solution exists. It hands over a procedure for computing one.

The starting point is that an initial value problem can be rewritten as an integral equation. Integrating $y'(s) = F(s, y(s))$ from t_0 to t and using $y(t_0) = u_0$,

$$y'(t) = F(t, y(t)), \quad y(t_0) = u_0 \qquad \Longleftrightarrow \qquad y(t) = u_0 + \int_{t_0}^{t} F\big(s, y(s)\big)\, ds.$$

The right-hand side exhibits y as a fixed point of the operator sending a function to $u_0 + \int_{t_0}^{\cdot} F(s, \cdot)\, ds$, and the Lipschitz hypothesis is exactly what makes that operator a contraction on a small enough time interval.

Definition A.d.20 (*Picard iteration*): Starting from the constant function $y_0(t) :\equiv u_0$, the "*Picard iteration*" (German: „*Picard-Iteration*") is the sequence

$$y_n(t) := u_0 + \int_{t_0}^{t} F\big(s,\, y_{n-1}(s)\big)\, ds, \qquad n \ge 1.$$

Under the hypotheses of Theorem A.d.13, $(y_n)_{n\in\mathbb{N}}$ converges uniformly to the solution on a short enough interval $[t_0 - h, t_0 + h]$, one on which the iterates stay in a box around (t_0, u_0) where F is bounded and Lipschitz. When F is Lipschitz in the state variable on all of $I \times \mathbb{R}^n$, as in the last sentence of Corollary A.d.14, it converges on the whole compact interval I. Beyond that there is no guarantee: convergence on the whole maximal interval of existence can fail.

AI-Example A.d.21 (*Picard iteration recovers a known solution*): Take $y' = y$ with $y(0) = 1$, so $F(t, y) = y$, $t_0 = 0$ and $u_0 = 1$. Then $y_0 \equiv 1$ and

$$y_1(t) = 1 + \int_0^t 1\, ds = 1 + t,$$
$$y_2(t) = 1 + \int_0^t (1 + s)\, ds = 1 + t + \tfrac{t^2}{2},$$
$$y_3(t) = 1 + \int_0^t \Big(1 + s + \tfrac{s^2}{2}\Big) ds = 1 + t + \tfrac{t^2}{2} + \tfrac{t^3}{6}.$$

An induction gives $y_n(t) = \sum_{k=0}^{n} t^k/k!$, the n-th partial sum of the exponential series. The iteration converges to e^t, which is indeed the solution. Pleasingly, it reconstructs the Taylor series of e^t one term per step, since each integration raises the degree by exactly one.

Theorem A.d.22 (*Peano existence theorem*): Let $I := [t_0 - a,\ t_0 + a] \subseteq \mathbb{R}$ and $B := \overline{B_b(x_0)} \subseteq \mathbb{R}^n$, and let $f : I \times B \to \mathbb{R}^n$ be **continuous**, with no Lipschitz condition in x assumed. Put $M := \max_{I\times B} |f|$, which is finite by compactness, and $h := \min\{a,\ b/M\}$. Then the initial value problem

$$\dot{x}(t) = f\big(t, x(t)\big), \qquad x(t_0) = x_0$$

has at least one C^1 solution $x : [t_0 - h,\ t_0 + h] \to B$.

Remark A.d.23 (*Peano versus Picard–Lindelöf*): Dropping the Lipschitz hypothesis costs exactly uniqueness and nothing else. The standard witness is $\dot{x} = \sqrt{|x|}$ with $x(0) = 0$: the right-hand side is continuous everywhere but not Lipschitz at $x = 0$, since $\sqrt{s}/s \to \infty$ as $s \downarrow 0$. Both

$$x(t) \equiv 0 \qquad \text{and} \qquad x(t) = \tfrac{1}{4}t^2 \quad (t \geq 0)$$

solve it, as do the infinitely many solutions that wait at 0 until an arbitrary time $\tau > 0$ and only then take off along $\frac{1}{4}(t - \tau)^2$. Peano's theorem still applies and still delivers a solution; it simply cannot say which one.

Note also where the two theorems get their machinery. Picard–Lindelöf is a fixed-point argument. It runs the Banach fixed-point theorem (Theorem 6.c.19) on the integral operator, and the Lipschitz constant is precisely what makes that operator a contraction. Peano has no contraction to work with and instead argues by compactness, extracting a convergent subsequence from a family of approximate solutions. The theorem licensing that extraction is Ascoli–Arzelà (Theorem 7.d.54), whose equicontinuity hypothesis is supplied by the bound $|f| \leq M$: it makes all the approximate solutions Lipschitz with the same constant M. That is also why it gives no uniqueness: a compactness argument produces **a** limit, never **the** limit.

Dependence on the initial condition (Subsection A.d.2)

Picard–Lindelöf produces, for each starting point, one solution. In applications one rarely has the starting point exactly: it is measured, and therefore wrong by a little. So the question is not only whether a solution exists but whether it **moves continuously** when the initial condition is perturbed, and more sharply whether it is differentiable in that data.

Write $u(t; x_0)$ for the value at time t of the solution of

$$u' = F(t, u), \qquad u(0) = x_0,$$

so that $u(\,\cdot\,; x_0)$ is the trajectory through x_0 and $u(t;\,\cdot\,)$ records where each starting point has been carried by time t.

Theorem A.d.24 (*Differentiability in the initial condition*): Let $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and convex and let $F : \Omega \to \mathbb{R}^n$ be continuous and C^1 in its spatial variable. Then $x_0 \mapsto u(t; x_0)$ is continuously differentiable, on the set of x_0 for which the solution exists up to time t.

Remark A.d.25 (*Why this is more than a technicality*): The hypothesis is barely stronger than the one Picard–Lindelöf already needs (C^1 in space rather than locally Lipschitz in space), and the conclusion is what makes the whole subject usable. It says the **flow** $x_0 \mapsto u(t; x_0)$ is a C^1 map, so the trajectories of an ODE do not merely exist individually but fit together smoothly into a family. Two consequences worth naming:

- **Errors stay controlled.** A measurement error ε in x_0 produces an error in $u(t; x_0)$ that is $O(\varepsilon)$ for fixed t, rather than something uncontrolled. The constant does typically grow with t. Differentiability at each fixed time is not a statement about long-run stability, and chaotic systems are exactly the ones where that constant grows fast.
- **The derivative itself solves an ODE.** Differentiating $u' = F(t, u)$ in x_0 gives that $J(t) := \partial u(t; x_0)/\partial x_0$ satisfies the linear system $J' = \mathrm{J}_u F(t, u(t; x_0))\, J$ with $J(0) = \mathrm{id}_n$, the "*variational equation*". So the sensitivity is computable without ever solving the original equation twice.

Higher-order linear ODEs and Cauchy–Lipschitz counterexamples (Subsection A.d.3)

Exercise A.d.26 (*True or False* (which functions can be solutions at all?)): Which of the following functions $\mathbb{R} \to \mathbb{R}$ can occur as solutions of a homogeneous linear ordinary differential equation of

third order with constant real coefficients? Decide each, with justification.

(a) $x \mapsto e^x\big(\cos(x) + \sin(x) - 1\big)$

(b) $x \mapsto x^2 e^x$

(c) $x \mapsto x\cos(x)$

(d) $x \mapsto x\cosh(x)$

Info: *This exercise is taken from the mock examination* ***Probeprüfung Analysis II*** *of 31 January 2023, Problem 8. That paper names no lecturer.*

Solution to Exercise A.d.26:

The solution space of a homogeneous linear equation $Ly = 0$ of order m with constant real coefficients is spanned by the functions

$$x \mapsto x^j e^{\mu x}, \qquad 0 \le j < m_\mu,$$

one family for each root μ of the characteristic polynomial p, where m_μ is its multiplicity; real coefficients force the non-real roots to come in conjugate pairs, whose real combinations are $x^j e^{ax}\cos(bx)$ and $x^j e^{ax}\sin(bx)$ for $\mu = a \pm ib$. So the question is purely arithmetic: **read off which roots the given function needs, with what multiplicities, and check whether they fit in a polynomial of degree three.**

(a) **True.** The three summands $e^x\cos x$, $e^x \sin x$ and $-e^x$ require the roots $1 \pm i$, which cost two, and the root 1, which costs one. Three in total, and the polynomial

$$p(\lambda) = (\lambda - 1)\big((\lambda - 1)^2 + 1\big)$$

has real coefficients and is exactly of degree three.

(b) **True.** The function $x^2 e^x$ needs the root 1 with multiplicity at least 3, which costs three and no more: $p(\lambda) = (\lambda - 1)^3$ works, and the function is one of the three basis solutions.

(c) **False.** Here $x\cos(x)$ needs $\pm i$ with multiplicity at least 2, so p is divisible by $(\lambda^2 + 1)^2$, of degree four. Three is not enough.

(d) **False**, for the same reason with real roots. Writing $x\cosh(x) = \frac{1}{2}\big(xe^x + xe^{-x}\big)$ exhibits both xe^x and xe^{-x}, so p is divisible by $(\lambda - 1)^2(\lambda + 1)^2$, again of degree four.

Remark A.d.27 (*The order is a budget*): Every one of the four verdicts came from counting, and the count is the same one every time: a factor x^j multiplying an exponential costs $j+1$ roots, and a genuinely oscillatory factor costs twice what its real counterpart would, because the root and its conjugate must both be paid for. The order of the equation is the total budget.

Read that way, the two false cases are false for one reason: $x\cos(x)$ and $x\cosh(x)$ both demand a repeated **pair**, at four. And the contrast between **(b)** and **(c)** is the whole lesson, since $x^2 e^x$ and $x\cos x$ look comparably complicated and are not: one repeated real root is cheap, one repeated complex pair is not.

Q.E.D.

Exercise A.d.28 (*Third-order linear ODE with constant coefficients*): For a parameter $\lambda > 0$ and a continuous function $f : [0, \infty) \to \mathbb{R}$, consider the third-order linear differential equation

$$y^{(3)}(t) + 2\lambda y''(t) + \lambda^2 y'(t) = f(t), \qquad y(0) = y'(0) = y''(0) = 0. \tag{A.1}$$

(a) Find the general solution of the associated homogeneous equation $y^{(3)} + 2\lambda y'' + \lambda^2 y' = 0$.

(b) For $\lambda = 1$ and $f(t) = e^{-t}$, find the explicit solution $y(t)$ of (A.1).

(c) Show that there exists a smooth impulse response function $h : [0, \infty) \to \mathbb{R}$ such that for any continuous f, the solution of (A.1) is given by the convolution integral

$$y(t) = \int_0^t f(s)h(t - s)\, ds,$$

and determine $h(t)$ explicitly for general $\lambda > 0$.

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Short Problem 2. Part 3 of the original, which asks for the explicit solution when $f(t) = \cos(\lambda t)$, is omitted here.*

Proof of Exercise A.d.28:

(a) The characteristic polynomial of the homogeneous equation is $P(r) = r^3 + 2\lambda r^2 + \lambda^2 r = r(r+\lambda)^2 = 0$. Its roots are $r_1 = 0$ (simple) and $r_2 = -\lambda$ (double root). The general homogeneous solution is:

$$y_h(t) = C_1 + C_2 e^{-\lambda t} + C_3 t e^{-\lambda t}, \qquad C_1, C_2, C_3 \in \mathbb{R}.$$

(b) For $\lambda = 1$ and $f(t) = e^{-t}$, the characteristic root $r = -1$ is a double root of $P(r)$, so the trial particular solution is $y_p(t) = At^2 e^{-t}$. Differentiating y_p:

$$y_p'(t) = A(2t - t^2)e^{-t},$$
$$y_p''(t) = A(2 - 4t + t^2)e^{-t},$$
$$y_p^{(3)}(t) = A(-6 + 6t - t^2)e^{-t}.$$

Plugging into $y_p^{(3)} + 2y_p'' + y_p' = e^{-t}$, the terms in t and t^2 cancel and only the constants survive:

$$A\Big[(-6 + 6t - t^2) + 2(2 - 4t + t^2) + (2t - t^2)\Big]e^{-t} = -2Ae^{-t}.$$

Therefore $-2A = 1$, so $A = -\frac{1}{2}$ and $y_p(t) = -\frac{1}{2}t^2 e^{-t}$. Setting initial conditions on $y(t) = C_1 + C_2 e^{-t} + C_3 t e^{-t} - \frac{1}{2}t^2 e^{-t}$:

- $y(0) = C_1 + C_2 = 0$.
- $y'(0) = -C_2 + C_3 = 0$, and therefore $C_3 = C_2$.
- $y''(0) = C_2 - 2C_3 - 1 = 0$, which reduces to $-C_2 = 1$, so $C_2 = -1$.

Thus $C_1 = 1$, $C_2 = -1$, $C_3 = -1$. The explicit solution is:

$$y(t) = 1 - e^{-t} - te^{-t} - \frac{1}{2}t^2 e^{-t} = 1 - \left(1 + t + \frac{1}{2}t^2\right)e^{-t}.$$

This is consistent with **(c)**: for $\lambda = 1$ the impulse response is $h(t) = 1 - e^{-t} - te^{-t}$, and $\int_0^t e^{-s} h(t-s)\,ds$ reproduces exactly this y.

(c) By Duhamel's principle, $h(t)$ is the solution of the homogeneous equation $h^{(3)} + 2\lambda h'' + \lambda^2 h' = 0$ with initial conditions $h(0) = 0, h'(0) = 0, h''(0) = 1$. Writing $h(t) = A + Be^{-\lambda t} + Cte^{-\lambda t}$:

- $h(0) = A + B = 0 \implies B = -A$.
- $h'(0) = -\lambda B + C = 0 \implies C = \lambda B = -\lambda A$.
- $h''(0) = \lambda^2 B - 2\lambda C = -\lambda^2 A + 2\lambda^2 A = \lambda^2 A = 1 \implies A = \frac{1}{\lambda^2}$.

Thus $B = -\frac{1}{\lambda^2}$ and $C = -\frac{1}{\lambda}$. The impulse response is:

$$h(t) = \frac{1}{\lambda^2}\left(1 - e^{-\lambda t} - \lambda t e^{-\lambda t}\right).$$

By linear ODE theory, $y(t) = \int_0^t f(s)h(t-s)\,ds$ satisfies (A.1).

Q.E.D.

Exercise A.d.29 (*Cauchy–Lipschitz non-uniqueness counterexample*): Let $\alpha > 1$. Consider the initial value problem

$$u'(t) = F(t, u(t)), \quad t > 0, \qquad u(0) = 0.$$

Give an explicit continuous function $F \in C^0(\mathbb{R}^2)$ such that both $u(t) \equiv 0$ and $u(t) = t^\alpha$ are solutions for $t \geq 0$, and explain why this does not contradict the Cauchy–Lipschitz Theorem (Theorem A.d.13).

Info: *This exercise is taken from the exam of 13 February 2025 (Prof. Joaquim Serra), Box Problem 4. The request to explain why Cauchy–Lipschitz is not contradicted is added here; the original asks only for F.*

Proof of Exercise A.d.29:

Differentiating $u(t) = t^\alpha$ gives $u'(t) = \alpha t^{\alpha-1} = \alpha(t^\alpha)^{(\alpha-1)/\alpha} = \alpha(u(t))^{(\alpha-1)/\alpha}$. Thus, define $F(t,u) := \alpha|u|^{(\alpha-1)/\alpha}$. Since $\alpha > 1$, $\beta := \frac{\alpha-1}{\alpha} \in (0,1)$, so $F(t,u) = \alpha|u|^\beta$ is continuous on $\mathbb{R}^2$. Both functions do solve the problem:

- **The zero solution:** $u_1(t) \equiv 0$ has $u_1'(t) = 0$, and $F(t,0) = 0$. Hence u_1 solves the initial value problem.
- **The power solution:** $u_2(t) = t^\alpha$ has $u_2'(t) = \alpha t^{\alpha-1}$, and $F(t,u_2(t)) = \alpha(t^\alpha)^{(\alpha-1)/\alpha} = \alpha t^{\alpha-1}$ for $t \geq 0$. Since $u_2(0) = 0$, it too solves the initial value problem.

Why does this not contradict Cauchy–Lipschitz? The theorem asks for F to be locally Lipschitz in its second argument, and this F is not. Indeed,

$$\frac{\partial F}{\partial u}(t,u) = \alpha\beta|u|^{\beta-1} = (\alpha-1)|u|^{-1/\alpha} \longrightarrow \infty \quad \text{as } u \to 0.$$

The hypothesis therefore fails at the initial point $(0,0)$, which is exactly where the two solutions separate. Note that continuity alone still buys existence (Peano); it is only uniqueness that is lost. **Q.E.D.**

Exercise A.d.30 (*Normal form first, then the theorem, then the solution*): Put each of the following differential equations into normal form: that is, in each case find an open set $U \subseteq \mathbb{R}^2$ and a function $F : U \to \mathbb{R}$ with

$$y'(x) = F\big(x, y(x)\big), \qquad (x,y) \in U \subseteq \mathbb{R}^2.$$

Then explain in each case whether the Cauchy–Lipschitz–Picard theorem (Theorem A.d.13) applies. Finally, find an explicit formula for the solution $y(x)$ for a general initial datum.

(a) $(x^2 - x)\,y' = y^2 + y$.
(b) $y' + e^y = 1$.
(c) $x\,y' = 1 - y^2$.
(d) $y'x^2 = y^2 + yx + x^2$.

Info: *This exercise is Problem 13.4 of Problem Sheet 13, Analysis II, Spring Semester 2026. The FS 2024 edition of the course sets the same problem on its own Sheet 13.*

Solution of Exercise A.d.30:

One observation covers the *"does the theorem apply"* question in all four cases, so make it once. In each part F is a rational or elementary expression that is C^∞ on the open set U where its denominator does not vanish. In particular $\partial_y F$ exists and is continuous on U, so F is locally Lipschitz in y by Exercise A.d.16, and Theorem A.d.13 applies at **every** point of U. The interesting question is therefore not whether the theorem applies but **what U is**, because U is what the excluded points carve up.

(a) $y' = F(x,y) := \dfrac{y^2+y}{x^2-x}$, with $U = \{(x,y) : x \notin \{0,1\}\}$, three open vertical strips. The theorem applies on each, and no solution can be continued across $x = 0$ or $x = 1$.
Separating variables, and using $\frac{1}{y(y+1)} = \frac{1}{y} - \frac{1}{y+1}$ and $\frac{1}{x(x-1)} = \frac{1}{x-1} - \frac{1}{x}$,

$$\ln\left|\frac{y}{y+1}\right| = \ln\left|\frac{x-1}{x}\right| + c \quad\Longrightarrow\quad \frac{y}{y+1} = K\,\frac{x-1}{x},$$

which rearranges to

$$y(x) = \frac{K(x-1)}{(1-K)x + K}.$$

The two constant solutions $y \equiv 0$ and $y \equiv -1$ are the roots of $y^2 + y$. The first is $K = 0$; the second is the limit $K \to \infty$ and is not of the displayed form.

(b) $y' = F(x,y) := 1 - e^y$, with $U = \mathbb{R}^2$. The theorem applies everywhere.
Here $\frac{1}{1-e^y} = 1 + \frac{e^y}{1-e^y}$, so $\int \frac{dy}{1-e^y} = y - \ln|1 - e^y|$ and

$$\ln\left|\frac{e^y}{1-e^y}\right| = x + c \quad\Longrightarrow\quad \frac{e^y}{1-e^y} = Ae^x \quad\Longrightarrow\quad e^y = \frac{Ae^x}{1+Ae^x},$$

that is

$$y(x) = x + \ln|A| - \ln\left|1 + Ae^x\right|,$$

together with the constant solution $y \equiv 0$, the root of $1 - e^y$. The sign of A selects the branch: $A > 0$ gives the solutions lying below the equilibrium, defined for every x, and $A < 0$ those lying above it, defined for $x > -\ln|A|$ and blowing up as x descends to that endpoint. In both cases $y \to 0$ as $x \to \infty$, from below in the first and from above in the second, which the formula shows directly, since $\ln|1 + Ae^x| - x - \ln|A| \to 0$.

(c) $y' = F(x,y) := \dfrac{1-y^2}{x}$, with $U = \{(x,y) : x \neq 0\}$, two open half-planes. The theorem applies on each.
With $\frac{1}{1-y^2} = \frac{1}{2}\left(\frac{1}{1-y} + \frac{1}{1+y}\right)$,

$$\tfrac{1}{2}\ln\left|\frac{1+y}{1-y}\right| = \ln|x| + c \quad\Longrightarrow\quad \frac{1+y}{1-y} = Ax^2 \quad\Longrightarrow\quad y(x) = \frac{Ax^2 - 1}{Ax^2 + 1},$$

plus the constant solutions $y \equiv \pm 1$, the roots of $1 - y^2$.

(d) $y' = F(x,y) := \dfrac{y^2 + yx + x^2}{x^2} = \left(\dfrac{y}{x}\right)^2 + \dfrac{y}{x} + 1$, with $U = \{(x,y) : x \neq 0\}$ again.
F depends on (x,y) only through $v := y/x$, so substitute $y = vx$, $y' = v + xv'$:

$$v + xv' = v^2 + v + 1 \quad\Longrightarrow\quad x\,v' = 1 + v^2 \quad\Longrightarrow\quad \arctan v = \ln|x| + c,$$

whence

$$y(x) = x\,\tan\bigl(\ln|x| + c\bigr).$$

Remark A.d.31 (*Solutions can stop inside U*): It is tempting to think that a solution stops only where the **equation** stops, at the points the normal form had to exclude. Part **(a)** already refutes this. With $K = 2$ its solution is $y = 2(x-1)/(2-x)$, which starts in the strip $x > 1$ and blows up as $x \to 2$, at a point of U that nothing in the equation singles out. The same happens in **(c)** for $A < 0$, at $x = |A|^{-1/2}$, and in **(b)** for $A < 0$, at $x = -\ln|A|$. Part **(d)** shows it in its purest form. There F is perfectly well behaved on the whole half-plane $x > 0$ and has no constant solution, and yet **every** solution $x\tan(\ln x + c)$ blows up at both ends of the interval on which $\ln x + c$ runs through $(-\pi/2, \pi/2)$. Every maximal interval is bounded and lies strictly inside $(0, \infty)$.

This is exactly the last clause of Theorem A.d.13 in action: as x tends to an endpoint of the maximal interval, $(x, y(x))$ must leave every compact subset of U, and here it does so by the y coordinate running to infinity rather than by the x coordinate reaching a forbidden value. Local existence is genuinely local, and no amount of regularity of F makes it global. Compare Exercise A.d.29, which loses uniqueness instead of existence, and for the opposite reason: there the hypothesis fails, here it holds.

Q.E.D.

Exercise A.d.32 (*Single choice: which hypothesis makes the solution unique?*): Consider the initial value problem

$$\frac{dy}{dx} = f(x,y), \qquad y \in \mathbb{R}, \qquad y(0) = y_0.$$

Under exactly one of the following conditions does this problem have a unique (at least local) solution. Which?

(a) f is continuous and bounded on a sufficiently small neighbourhood of $(0, y_0)$.

(b) f is continuous and bounded on a sufficiently small neighbourhood of $(0, y_0)$, and there is a real number L with $|f(x, y_2) - f(x, y_1)| \leq L|y_2 - y_1|$ for all arguments in that neighbourhood.

(c) f is continuous and bounded on a sufficiently small neighbourhood of $(0, y_0)$, and there is a real number L with $|f(x_2, y) - f(x_1, y)| \leq L|x_2 - x_1|$ for all arguments in that neighbourhood.

(d) f is bounded and Riemann integrable on a sufficiently small neighbourhood of $(0, y_0)$.

Info: *This exercise is taken from the Analysis II examination of August 2025 (Prof. Laura Kobel-Keller), Question 2.*

Solution of Exercise A.d.32:

The correct answer is **(b)**: it is exactly the hypothesis of Theorem A.d.13, with the Lipschitz estimate taken in the state variable y and uniformly in x.

(a) **False.** Continuity alone gives **existence**, by Peano's theorem, and nothing more. Exercise A.d.29 is the standard witness: $u' = \alpha|u|^{(\alpha-1)/\alpha}$ with $u(0) = 0$ has a continuous, locally bounded right-hand side and at least two solutions through the origin. Uniqueness is the half of Picard–Lindelöf that the Lipschitz condition pays for.

(b) **True**, as above.

(c) **False**, and this is the option the question exists for. The estimate is taken in the **wrong variable**: it constrains how f varies with the independent variable x while y is held fixed, which is a regularity statement about the coefficient functions and has nothing to do with uniqueness. The counterexample of **(a)** again applies, since $f(x, y) = \alpha|y|^{(\alpha-1)/\alpha}$ does not depend on x at all and is therefore Lipschitz in x with $L = 0$, while the initial value problem still has two solutions.

(d) **False.** Riemann integrability is weaker than continuity, so this hypothesis is weaker than **(a)**, which already fails. It does not even guarantee existence.

Remark A.d.33 (*Which variable, and why*): The distinction between **(b)** and **(c)** is the one to carry away, and the proof of Theorem A.d.13 explains it in one line. That proof runs the contraction

$$(\Phi u)(x) := y_0 + \int_0^x f\big(s, u(s)\big)\, ds$$

on a space of functions, and the estimate needed is $|(\Phi u)(x) - (\Phi v)(x)| \leq \int_0^x |f(s, u(s)) - f(s, v(s))|\, ds$. The two arguments compared differ in the **second** slot and agree in the first, because the integration variable s is the same on both sides. So the only estimate the argument can use is one in y, and control in x is never asked for; continuity in x is enough to make the integral exist.

Note also that **(b)**'s condition is **uniform** in x over the neighbourhood, which is what the local Lipschitz condition of Theorem A.d.13 amounts to once the neighbourhood has been shrunk, and what Exercise A.d.16 delivers from continuity of $\partial_y f$.

Q.E.D.

Glossary — English and German terms (Appendix B)

The lecture and the exercise sheets for this course exist in both English and German, and the two vocabularies do not always line up in the obvious way. Note that „*Abschluss*“ is the closure and not a “*conclusion*”, „*Rand*“ is the boundary and not a “*rim*”, and „*Satz von Schwarz*“ is what English-language texts often call Clairaut’s theorem. Each term below is mirrored once in the main text, at the point where it is first introduced; this appendix collects those pairs in one place so the document can be used alongside German material.

The **Ch.** column gives the chapter in which the term is first introduced; a letter there is an appendix.

A–E

English	German	Ch.
accumulation point	Häufungspunkt	4
antisymmetric k-linear form	antisymmetrische k-lineare Form	24
autonomous	autonom	A
boundary	Rand	3
bounded	beschränkt	7
bump function	Höckerfunktion	25
centroid	Flächenschwerpunkt	20
chain rule	Kettenregel	10
closed	abgeschlossen	3
closure	Abschluss	3
compact	kompakt	7
compactly supported	kompakt getragen	25
complete metric space	vollständiger metrischer Raum	6
completion	Vervollständigung	6
concatenation (of paths)	Verkettung	26
cone over Ω	Kegel über Ω	20
connected	zusammenhängend	8
connected component	Zusammenhangskomponente	8
continuity equation	Kontinuitätsgleichung	23
continuous	stetig	5
contraction	Kontraktion	6
convex	konvex	14
critical point	kritischer Punkt	12
cuspidal cubic	Neilsche Parabel	13
degenerate (critical point)	entartet	12
diffeomorphism	Diffeomorphismus	15
differentiable	differenzierbar	9
differential (the)	das Differential	10
differential p-form	Differentialform	24
directional derivative	Richtungsableitung	10
disconnected	unzusammenhängend	8
dyadic cube	dyadischer Würfel	19
dyadic set	dyadische Menge	19
dyadic step functions	dyadische Treppenfunktionen	19
equicontinuous	gleichgradig stetig	7
equivalent (of norms)	äquivalent	2
eventually	schliesslich	4

F–L

English	German	Ch.
figure eight	Lemniskate	C
fixed point	Fixpunkt	6
flux	Fluss	23
Frobenius norm	Frobeniusnorm	2
Gabriel's horn	Torricellis Trompete	21
geometric mean	geometrisches Mittel	2
gradient	Gradient	9
Gram determinant	Gram-Determinante	21
harmonic	harmonisch	23
Hessian matrix	Hesse-Matrix	11
Hölder's inequality	Höldersche Ungleichung	1
homeomorphism	Homöomorphismus	15
homotopic	homotop	8
homotopy	Homotopie	8
Hurwitz criterion	Hurwitz-Kriterium	12
immersion	Immersion	17
improper integral	uneigentliches Integral	19
inner product / scalar product	Skalarprodukt	2
interior	Inneres	3
isometry	Isometrie	21
Jacobi matrix	Jacobi-Matrix	9
Jordan measurable	Jordan-messbar	19
Jordan measure	Jordan-Mass	19
Lagrangian	Lagrange-Funktion	13
Laplacian	Laplace-Operator	23
Lebesgue measurable	Lebesgue-messbar	19
Lebesgue measure	Lebesgue-Mass	19
Lebesgue number	Lebesgue-Zahl	7
Lebesgue outer measure	äusseres Lebesgue-Mass	19
Lebesgue's covering lemma	Lebesguesches Überdeckungslemma	7
length	Länge	21
level set	Niveaumenge	13
limit cycle	Grenzzykel	A
Lipschitz continuous	Lipschitz-stetig	5
local minimum	lokales Minimum	12
locally Lipschitz	lokal Lipschitz-stetig	7

M–P

English	German	Ch.
mean value inequality	Mittelwertsatz-Ungleichung	11
metric	Metrik	2
metric space	metrischer Raum	2
multi-index	Multiindex	11
negative definite	negativ definit	12
negative semidefinite	negativ semidefinit	14
norm	Norm	2
normal space	Normalraum	17
open	offen	3
open cover	offene Überdeckung	7
orientable	orientierbar	25
outward normal derivative	Aussennormalableitung	23
partial derivative	partielle Ableitung	9
path	Weg	8
path-connected	wegzusammenhängend	8
Picard iteration	Picard-Iteration	A
positive definite	positiv definit	12
positive semidefinite	positiv semidefinit	14
pseudo-metric	Pseudometrik	2
pullback	Zurückziehung	25

Q–Z

English	German	Ch.
saddle point	Sattelpunkt	12
scalar field	Skalarfeld	10
Schwarz's theorem	Satz von Schwarz	11
sequence	Folge	1
shadow price	Schattenpreis	13
shoelace formula	Gaußsche Trapezformel	26
σ-algebra	σ-Algebra	19
simply connected	einfach zusammenhängend	8
sink	Senke	22
smooth partition of unity	glatte Zerlegung der Eins	25
source	Quelle	22
space	Raum	2
submanifold	Untermannigfaltigkeit	17
submultiplicative	submultiplikativ	2
support (of a function)	Träger	25
supremum norm	Supremumsnorm	7
tangent space	Tangentialraum	17
totally bounded	total beschränkt	7
uniform convergence	gleichmässige Konvergenz	4
uniformly continuous	gleichmässig stetig	5
vector field (C^k)	Vektorfeld	22
velocity (of a curve)	Geschwindigkeit	9
wedge product	Dachprodukt	24

AI-Note B.1: Three entries in this table are worth flagging: two where the German and English names genuinely diverge rather than merely translate, and one where both languages hide a distinction.

- **„*Satz von Schwarz*“** is the symmetry of mixed partial derivatives (Theorem 11.b.13). English texts more often call it **Clairaut's theorem**; both names are used in this document, with Schwarz first because that is what the lecture uses.
- **„*Neilsche Parabel*“** names the curve $y^2 = x^3$ after William Neile. The English **cuspidal cubic** is descriptive rather than eponymous, so searching for one name will not find the other.
- **„*gleichmässig*“** carries both English senses. It is *“uniform”* in **uniform convergence** and in **uniformly continuous**, which are different notions (Exercise 4.a.14, Definition 5.a.4). The German gives no hint that these are distinct, and neither does the English.

Repetition Quizzes (Appendix C)

The first quiz below revisits everything up to the middle of the course: differentiability, the chain rule, extrema, the implicit function theorem, submanifolds. It ranges across Chapters 1 to 17 and belongs to no single chapter, so it is collected here. A second quiz follows it, covering the integration half of the course in the same format, so that the two together span the whole document.

The mid-semester quiz (Section C.a)

AI-Note C.1: The questions in this section are by Prof. Serra and Prof. Lang, from the mid-semester repetition session. Each answer gives an explicit **True**/**False** verdict.

Question C.a.1: Let $U := \{(x,y) \in \mathbb{R}^2 : 1 \le x^2+y^2 \le 4\}$ and $f(x,y) := \sin(xy) - y^4$. Which of the following holds?

(a) $f(U) = [a,b]$ for some $a < b$.

(b) $f(U)$ is disconnected and closed with two connected components.

Answer C.a.2: (a) is true, (b) is false. U is compact and connected and f is continuous, so $f(U) \subseteq \mathbb{R}$ is compact (Item **(a)**) and connected (Theorem 8.a.7). The connected subsets of $\mathbb{R}$ are exactly the intervals, and a compact one is closed and bounded, so $f(U) = [a,b]$.

AI-Note C.2: Worth noting how little computation this needs: f is never evaluated. The annulus U is connected (it is **not** simply connected, but that is a different property and irrelevant here), and the two preservation theorems do the rest. Recognising that a question is asking about preserved properties rather than about values is most of the work.

Question C.a.3: Let $X \subseteq \mathbb{R}^2$ with $X \neq \mathbb{R}^2$ and $X \neq \emptyset$. If X is complete, is it closed?

Answer C.a.4: True. Let $(a_n)_{n=0}^{\infty}$ be a sequence in X converging to some $a \in \mathbb{R}^2$. Then (a_n) is Cauchy, and by completeness it converges to a point of X; by uniqueness of limits that point is a. So X is sequentially closed, hence closed (Proposition 4.a.7).

AI-Note C.3: The limit is taken in $\mathbb{R}^2$, and completeness of X is what forces it back into X. The converse also holds here, but only because the ambient space $\mathbb{R}^2$ is itself complete: a closed subset of a complete space is complete. In a non-complete ambient space closedness gives nothing. Note that $\mathbb{Q}$ is closed in $\mathbb{Q}$ and not complete.

Question C.a.5: Let $X \subseteq \mathbb{R}^2$ as above. If X is not open, is it closed?

Answer C.a.6: False.

AI-Note C.4: The half-open square $X := [0,1]\times(0,1)$ from Example 4.a.12 is a counterexample, being neither open nor closed. *"Open"* and *"closed"* are not opposites and not exhaustive: $\emptyset$ and $\mathbb{R}^2$ are both, most sets are neither, and the excluded values $X \neq \emptyset$, $X \neq \mathbb{R}^2$ in the hypothesis are there precisely to rule out the two clopen cases.

Question C.a.7: Let

$$f(x,y,z) := \frac{xy^2}{x^2+y^2+z^2} \quad \text{for } (x,y,z) \neq 0, \qquad f(0,0,0) := 0.$$

Is $f \in C^1(\mathbb{R}^3)$?

Answer C.a.8: False. We compute

$$\partial_x f = \frac{y^2}{x^2+y^2+z^2} - 2\,\frac{x^2y^2}{(x^2+y^2+z^2)^2}.$$

In spherical coordinates $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$, every power of r cancels:

$$\partial_x f = \sin^2\theta\sin^2\varphi - 2\sin^4\theta\cos^2\varphi\sin^2\varphi.$$

The limit $\lim_{r\to 0}\partial_x f(r,\theta,\varphi)$ is therefore ill-defined, since it depends on the direction (θ,φ) along which the origin is approached. Consequently $\partial_x f$ is not sequentially continuous at 0.

AI-Note C.5: The cancellation of r is the whole argument, and it is worth naming the reason: $\partial_x f$ is "*homogeneous of degree 0*", so it is constant along each ray from the origin and takes a different constant on different rays. Compare AI-Example 10.b.10, where the same trick (restrict to a ray, watch the parameter cancel) exposed a function whose directional derivatives all existed without it being differentiable.

Question C.a.9: Let $f \in C^\infty(\mathbb{R}^3)$ and $\alpha \in \mathbb{R}$ be such that

$$\nabla f(0) = (0,0,0), \qquad \operatorname{Hess} f(0) = \begin{pmatrix} 1 & \alpha & 0 \\ \alpha & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Which of the following is **false**?

- **(a)** For all $\alpha \in \mathbb{R}$, 0 is not a local maximum point of f.
- **(b)** For $\alpha > \sqrt{2}$, 0 is a saddle point.
- **(c)** For $\alpha \neq \pm\sqrt{2}$, $\nabla f : \mathbb{R}^3 \to \mathbb{R}^3$ is a local diffeomorphism around 0.
- **(d)** For $\alpha < \sqrt{2}$, 0 is a local minimum point of f.

Answer C.a.10: (d) is the false one.

(a) is true, since $1 > 0$ is clearly an eigenvalue of $\operatorname{Hess} f(0)$ for any choice of α (the third standard basis vector is an eigenvector), so $\operatorname{Hess} f(0)$ is never negative semidefinite and 0 can never be a local maximum.

(b) is true. Let λ_1, λ_2 and $\lambda_3 = 1$ be the eigenvalues of $\operatorname{Hess} f(0)$. Then

$$\lambda_1\lambda_2 = \det\big(\operatorname{Hess} f(0)\big) = 2 - \alpha^2 < 0 \quad \text{for } \alpha > \sqrt{2},$$

so $\operatorname{Hess} f(0)$ has both positive and negative eigenvalues, making it indefinite, and 0 is a saddle point by Theorem 12.b.11.

(c) is true. For these values $\det(\operatorname{Hess} f(0)) \neq 0$, and therefore the inverse function theorem (Theorem 15.a.1) applies. Note that $\operatorname{Hess} f(0) = D(\nabla f)_0$: the Hessian of f **is** the differential of the gradient map.

(d) is false, because the same reasoning as in **(b)** gives that for $\alpha < -\sqrt{2} < \sqrt{2}$ we again have $2 - \alpha^2 < 0$, so 0 is a saddle point of f, not a local minimum. The statement is true only on $-\sqrt{2} < \alpha < \sqrt{2}$.

AI-Note C.6: The negative root is the whole trap of this question. Part **(c)** has to exclude $\alpha = -\sqrt{2}$ as well as $\alpha = \sqrt{2}$, since $\det(\operatorname{Hess} f(0)) = 2 - \alpha^2$ vanishes at both, and part **(d)** is false for exactly that reason.

Question C.a.11: Let $V := \{(x,y) \in \mathbb{R}^2 : y^4 + y^2 = x^3 + x\}$. Decide:

- **(a)** V is a graph over y in a neighbourhood of $(0,0)$.
- **(b)** V is a graph over x in a neighbourhood of $(0,0)$.

Answer C.a.12: (a) True. Notice that $V = f^{-1}\{0\}$ for $f(x,y) := y^4 + y^2 - x^3 - x$. Then

$$\mathrm{J}\,f(x,y) = (\partial_x f,\ \partial_y f) = (-3x^2 - 1,\ 4y^3 + 2y), \qquad \mathrm{J}\,f(0,0) = (-1,\ 0).$$

Since $\partial_x f(0,0) = -1 \neq 0$, the implicit function theorem (Theorem 16.a.1) applies with x as the variable to solve for: in a neighbourhood of $(0,0)$,

$$(x,y) \in V \iff f(x,y) = 0 \iff x = x(y).$$

The theorem does **not** apply for y, since $\partial_y f(0,0) = 0$.

(b) False. Solving the defining equation for y^2 by the quadratic formula gives

$$y^2 = -\tfrac{1}{2} + \sqrt{\tfrac{1}{4} + x^3 + x},$$

which has no solution for $x \in (-\varepsilon, 0)$: there $x^3 + x < 0$, so the square root is smaller than $\frac{1}{2}$ and the right-hand side is negative, while $y^2 \geq 0$. So V contains no points at all just to the left of the origin, and cannot be a graph over x there.

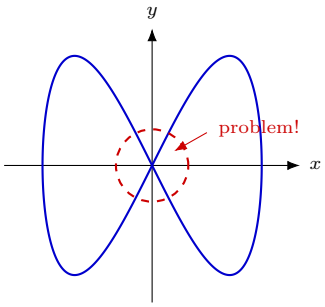
AI-Note C.7: The two parts are settled by different means, and the asymmetry is the point. Part **(a)** is decided by **checking a derivative**; part **(b)** needs an **explicit computation** showing the set is empty on one side. That is because the implicit function theorem yields no negative results: failure of its hypothesis proves nothing on its own, exactly as for the regular value theorem in AI-Example 17.a.6.

Question C.a.13: Let $\gamma : I \to \mathbb{R}^2$ be injective and smooth, with I an open interval. Is $\gamma(I)$ then a smooth submanifold of $\mathbb{R}^2$?

Answer C.a.14: False. Consider

$$\gamma : (0, 2\pi) \to \mathbb{R}^2, \qquad t \mapsto \big(\sin t,\ \sin 2t\big),$$

whose image is a "*figure eight*" (German: „*Lemniskate*"). This γ is injective and smooth, but there is no graphical representation of $\mathrm{im}(\gamma)$ around $(0,0)$.



AI-Note C.8: Two things are worth extracting. First, how injectivity survives a self-crossing at all: the two parameter ends $t \to 0^+$ and $t \to 2\pi^-$ both send $\gamma(t) \to (0,0)$, but neither **attains** it, so the image passes through the origin only once (via $t = \pi$) while being approached from two further directions.

Second, this locates precisely which hypothesis of Theorem 17.a.9 fails. It is not injectivity, and not injectivity of $D\gamma_t$; it is the requirement that the parametrization be a **homeomorphism** onto its image. Here γ^{-1} is discontinuous at $(0,0)$: points of the image arbitrarily close to the origin have parameters near 0 and near 2π, far apart in I. That the resulting set is not a submanifold can then be confirmed by the component count of AI-Example 17.a.6. Near the origin, deleting the crossing point leaves four arcs, while deleting a point from an interval leaves two. (Globally the punctured figure eight has only two components, the two loops; the count must be local.)

The quiz ends here, and with it the differential-topology half of the course.

A second quiz: the integration half (Section C.b)

The mid-semester quiz stops where integration begins, so on its own it tests Chapters 1 to 17 and leaves Chapters 19 to 26 untested. The seven questions below close that gap, in the same format and at the same level: each is answerable from a definition or a theorem statement, and none needs a long computation. They are best attempted with the book shut.

AI-Note C.9: These seven questions and their answers were generated by Claude Opus 5 for these notes. Unlike Appendix C.a, they are not from the course, and no claim is made that they resemble what an examiner would ask. What they do reproduce is the shape of the mid-semester quiz: each one is built around a hypothesis that is easy to forget, and the answer says which hypothesis it is.

Question C.b.15: Every bounded **open** subset of $\mathbb{R}^n$ is Jordan measurable. True or false?

Answer C.b.16: False. By Theorem 19.a.6, Jordan measurability is a condition on the boundary, not on the interior, and an open set can have a boundary of positive outer measure. Example 19.a.9 is the standard witness: a union of shrinking intervals around an enumeration of $\mathbb{Q} \cap [0,1]$, open and bounded, with $\mu_{\text{in}} \leq \frac{1}{2}$ and $\mu_{\text{out}} = 1$.

AI-Note C.10: The instinct to be corrected is that *"nice"* should mean open or closed. It does not: a closed set can fail just as badly. With U the example above, $[0,1] \setminus U$ is closed and bounded, and it is not Jordan measurable either, since otherwise $U = [0,1] \setminus ([0,1] \setminus U)$ would be. Meanwhile $\overline{U} = [0,1]$ is Jordan measurable while U is not. What Jordan measurability actually asks is that the boundary be thin, and **thin** is a size condition that topology alone never supplies.

Question C.b.17: Let $\Phi : U \to V$ be a C^1-diffeomorphism, let $A \subseteq U$ be Jordan measurable with $\overline{A} \subseteq U$, and let $f : \Phi(A) \to \mathbb{R}$ be continuous. Which of the following hold?

(a) $\displaystyle\int_{\Phi(A)} f(y)\,dy = \int_A f\big(\Phi(x)\big)\,\big|\det \mathrm{J}\,\Phi(x)\big|\,dx$.

(b) $\displaystyle\int_{\Phi(A)} f(y)\,dy = \int_A f\big(\Phi(x)\big)\,\big|\det \mathrm{J}\,\Phi(x)\big|^{-1}\,dx$.

(c) Both, since $\big|\det \mathrm{J}\,\Phi\big(\Phi^{-1}(y)\big)\big| = \big|\det \mathrm{J}\,\Phi^{-1}(y)\big|$.

Answer C.b.18: (a) is true; (b) and (c) are false. Statement **(a)** is Theorem 20.a.1 with Φ moved to the other side, which is where the determinant stops being divided and starts being multiplied. Statement **(c)** misstates the inverse function theorem: differentiating $\Phi^{-1} \circ \Phi = \mathrm{id}$ gives

$$\mathrm{J}\,\Phi^{-1}(y) = \Big(\mathrm{J}\,\Phi\big(\Phi^{-1}(y)\big)\Big)^{-1}, \qquad \text{so} \qquad \det \mathrm{J}\,\Phi^{-1}(y) = \frac{1}{\det \mathrm{J}\,\Phi\big(\Phi^{-1}(y)\big)}.$$

The two determinants are reciprocals, not equals, and it is exactly that reciprocal which turns **(a)** into the form stated in Theorem 20.a.1. Since **(a)** is true and the two right-hand sides differ whenever $|\det \mathrm{J}\,\Phi| \neq 1$, **(b)** is false.

AI-Note C.11: The reliable way to settle which side carries the determinant is a one-line dimensional check, and it is worth doing every time rather than memorising the formula. Take $f \equiv 1$ and $\Phi(x) := 2x$ on $\mathbb{R}^n$. The left-hand side is $\mathrm{vol}_n(2A) = 2^n \,\mathrm{vol}_n(A)$, and $|\det \mathrm{J}\,\Phi| = 2^n$, so only **(a)** balances.

Question C.b.19: Let $\phi : B \to \mathbb{R}^n$ and $\psi : \widetilde{B} \to \mathbb{R}^n$ be two parametrizations of the same m-dimensional surface patch, and let vol_m be computed from each by Definition 21.b.6. Do the two computations agree?

Answer C.b.20: True, and the Gram determinant is what makes it work. Write $\psi = \phi \circ \Psi$ for the transition diffeomorphism $\Psi := \phi^{-1} \circ \psi$. By the chain rule $D\psi = (D\phi \circ \Psi)\, D\Psi$, and therefore

$$D\psi^{\mathsf{T}} D\psi = D\Psi^{\mathsf{T}} \big(D\phi^{\mathsf{T}} D\phi\big) D\Psi, \qquad \sqrt{\det\left(D\psi^{\mathsf{T}} D\psi\right)} = \left|\det D\Psi\right| \sqrt{\det\left(D\phi^{\mathsf{T}} D\phi\right)} \circ \Psi,$$

using $\det(D\Psi^{\mathsf{T}} M\, D\Psi) = (\det D\Psi)^2 \det M$ and taking the square root, which is where the absolute value appears. That factor $|\det D\Psi|$ is precisely what Theorem 20.a.1 consumes when the integral over $\widetilde{B}$ is rewritten as an integral over B, so the two values coincide.

AI-Note C.12: This is the reason Definition 21.b.6 is a definition of the surface at all, rather than of the map used to describe it. Note also which ingredient did the work: the square root of a Gram determinant is homogeneous of degree one in the parametrization, so it transforms with $|\det D\Psi|$, exactly the weight that change of variables cancels. The signed version of the same statement is the integral of an m-form, which transforms with $\det D\Psi$ instead and therefore **does** depend on the parametrization, through its orientation (see Definition 25.b.4).

Question C.b.21: Let $U \subseteq \mathbb{R}^3$ be open and $F \in C^1(U, \mathbb{R}^3)$ with $\operatorname{curl} F \equiv 0$ on U. Which of the following holds?

(a) F is conservative on U.
(b) F is conservative on U provided U is simply connected.
(c) F is conservative on every ball contained in U.

Answer C.b.22: (a) is false; (b) and (c) are true. By Lemma 26.b.12 the vanishing of $\operatorname{curl} F$ is **necessary** for F to be conservative, and Corollary 26.c.25 makes it sufficient once the domain is simply connected, which gives **(b)**. A ball is convex, so any two paths in it with the same endpoints are homotopic through the straight-line homotopy $H(s,t) := (1-s)\gamma_0(t) + s\gamma_1(t)$; it is simply connected, and **(c)** follows from **(b)**. It shows that the failure in **(a)** can only ever be global.

For **(a)** itself, take $U := \mathbb{R}^3 \setminus \{x = y = 0\}$, the complement of the z-axis, and

$$F(x,y,z) := \left(\frac{-y}{x^2+y^2},\ \frac{x}{x^2+y^2},\ 0 \right).$$

Then $\operatorname{curl} F = 0$ on U, but the line integral of F around the unit circle in the plane $z = 0$ is $2\pi \neq 0$, so no potential exists by Proposition 26.b.14.

AI-Note C.13: The field above is the angle form of AI-Exercise 24.b.18 carried into $\mathbb{R}^3$ and given a third component of zero, and its failure has the same cause: the domain has a hole, so the polar angle it wants as a potential cannot be defined continuously all the way round. The three parts of this question are worth reading as one statement. Vanishing curl is a **local** condition, being conservative is a **global** one, and the gap between them is measured by the topology of U and by nothing else.

Question C.b.23: Gauss' theorem (Theorem 23.b.5) requires the boundary $\partial\Omega$ to be connected. True or false?

Answer C.b.24: False. Nothing in the hypotheses mentions connectedness of the boundary, and the theorem is routinely used on domains where it fails. Take the spherical shell $\Omega := \{x \in \mathbb{R}^3 : 1 < |x| < 2\}$, whose boundary is the disjoint union of two spheres, and the field $F(x) := x/|x|^3$, which is C^1 on a neighbourhood of $\overline{\Omega}$ and has $\operatorname{div} F = 0$ there. On the outer sphere the exterior normal is $\nu = x/2$ and $\langle F, \nu \rangle = \frac{1}{4}$, contributing $\frac{1}{4} \cdot 16\pi = 4\pi$. On the inner sphere the exterior normal of Ω points **towards** the origin, so $\nu = -x$ and $\langle F, \nu \rangle = -1$, contributing -4π. The two add to $0 = \int_\Omega \operatorname{div} F$, as the theorem requires.

AI-Note C.14: The step that has to be got right on a domain like this is not the theorem but the normal: on an inner boundary component the exterior normal of Ω points into the hole. Getting it backwards produces 8π instead of 0, and the error is invisible in the algebra.

It is also worth comparing with Exercise 23.b.13. The same field, integrated over the solid ball instead of the shell, gives a non-zero boundary integral against a vanishing divergence integral, and there the resolution is that F is singular at the origin. Excising a small ball around the singularity turns the ball into a shell, and the contradiction into the computation above.

Question C.b.25: The Möbius strip is a 2-dimensional submanifold of $\mathbb{R}^3$. Is it orientable?

Answer C.b.26: No. An orientation in the sense of Definition 25.c.13 is an atlas whose transition maps all have positive Jacobian determinant, and for a surface in $\mathbb{R}^3$ that is equivalent to a continuous choice of unit normal. Transporting a normal once around the core circle of the strip returns it reversed, so no continuous choice exists.

AI-Note C.15: This is the same fact used earlier for a different purpose. In Section 17.b it was the reason the Möbius strip is not $g^{-1}\{0\}$ for any single g with $\nabla g \neq 0$: such a g would hand over the global normal $\nabla g/|\nabla g|$ that the strip does not have. Non-orientability is therefore not an exotic pathology bolted on at the end of the course. It is the obstruction that was already forcing implicit descriptions to be local.

Question C.b.27: Let M be a compact orientable m-manifold without boundary.

(a) Is $\int_M d\omega = 0$ for every $(m-1)$-form ω?

(b) The standard area form σ on $\mathbb{S}^2$ satisfies $\int_{\mathbb{S}^2} \sigma = 4\pi$. Is σ exact?

Answer C.b.28: (a) Yes. Theorem 26.d.31 gives $\int_M d\omega = \int_{\partial M} \omega$, and $\partial M = \emptyset$, so the right-hand side is an integral over the empty set.

(b) No, and part **(a)** is the proof. If $\sigma = d\omega$ for some 1-form ω, then $\int_{\mathbb{S}^2} \sigma = \int_{\mathbb{S}^2} d\omega = 0$ by **(a)**, contradicting $\int_{\mathbb{S}^2} \sigma = 4\pi \neq 0$. Since σ is a top-degree form on a 2-manifold, $d\sigma$ is a 3-form on a 2-manifold and vanishes automatically, so σ is closed. It is therefore closed and not exact.

AI-Note C.16: Part **(b)** is worth holding onto, because differentiation cannot detect it: $d\sigma = 0$ is all a local computation reveals, and the obstruction is visible only to an **integral**. The angle form of AI-Exercise 24.b.18 was closed and not exact for the same reason, one dimension down, and there too the witness was a number: 2π around the puncture, 4π over the sphere. One difference is instructive. The punctured plane is not simply connected, but the sphere is, so Corollary 26.c.25 cannot be what fails here: it is a statement about 1-forms, and simple connectedness says nothing about whether a closed 2-form is exact.

Problem Sheet Index (Appendix D)

The problem sheets are not reproduced as blocks anywhere in this document. Every problem is quoted verbatim from the official sheet and placed next to the material it tests, with its solution in the **Solutions** section of the chapter that hosts it. That is the right arrangement for reading the notes and the wrong one for answering *"what is on sheet 7?"*, so this appendix restores the sheet view.

All statements are quoted from `exercises/ExN_Analysis2_eng.pdf`; attribution: Prof. Joaquim Serra, D-MATH, ETH Zürich.

Notation D.1 (*Priority key*): From sheet 4 onwards each problem carries a priority, taken from the colour-coded *Recommended exercises* pages of Corsin Nick's class notes: blue = **important**, orange = **semi-important**, red = **optional**. The official sheets additionally mark harder problems with **(*)**. Each problem carries its tag in its own title throughout this document; the tables below collect them, together with the notes quoted from the same pages.

AI-Note D.1: Sheets 1–3 have no priority page, so no problem on them is singled out, and all of them are reproduced and solved.

Sheets 1–3 (Section D.a)

Problem	Topic	Now in
1.1	Recap questions	Section 1.a
1.2	From equations to drawings	Section 1.c
1.3	From drawings to equations	Section 1.c
1.4	Young's inequality	Section 1.b
2.1	Metric spaces	Section 2.b
2.2	Multiple choice (closure)	Section 3.c
2.3	Multiple choice (distance from a set)	Section 3.c
2.4	Boundary, interior	Section 3.c
2.5	Product of metric spaces	Section 2.b
2.6	Continuity of the distance	Section 5.a
2.7	Continuity of the composition	Section 5.a
3.1	Lipschitz or not	Section 5.b
3.2	Problems at the origin	Section 5.a
3.3	Open, closed, complete and compact	Section 7.c
3.4	Multiple choice (open, complete, compact)	Section 7.c
3.5	Multiple choice (fixed points)	Section 6.c
3.6	Multiple choice (uniform continuity)	Section 5.b
3.7	Cauchy–Schwarz	Section 2.c
3.8	Inner product and norm	Section 2.c
3.9	All norms are equivalent	Section 2.c
3.10	Hilbert–Schmidt norm	Section 2.c
3.11	(*) Uniform continuity and moduli	Section 5.b

Sheet 4 (Section D.b)

Problem	Priority	Note	Now in
4.1	important	"Use the chain rule"	Section 9.b
4.2	important	"Recall that path connected implies connected"	Section 8.b
4.3	semi-important (parts 1, 2, 3, 5); **parts 4 and 6 important**	"For 3., use Cauchy–Schwarz in $\mathbb{R}^n$ applied to convenient vectors."	Section 2.c
4.4	optional	—	Section 2.c
4.5	optional	—	Section 11.a
4.6	important (parts 1, 2, 3)	—	Section 10.d

Sheet 5 (Section D.c)

Problem	Priority	Note	Now in
5.1	important	"Recall the linearity of the differential and that $C^1 \Rightarrow$ differentiable"	Section 10.b
5.2	important	"Compare with the example from the script and recall that continuous functions on compact sets attain their extrema."	Section 13.a
5.3	important	"There is no difference between $x \to 0$ and $\lvert x\rvert \to 0$."	Section 11.b
5.4	semi-important	"Use the definition of directional derivative, $\partial_v f(x) = \frac{\partial}{\partial t}\big\rvert_{t=0} f(x+tv)$. Solve a few of these, but not all."	Section 10.b
5.5	optional	—	Section 11.b
5.6	optional	"Interesting, but irrelevant for the course"	Section 11.b
5.7	optional	"Interesting for physicists, will be covered in classical mechanics."	Section 10.d

Sheet 6 (Section D.d)

Problem	Priority	Note	Now in
6.1	unmarked	—	not transcribed
6.2	important	"This is what we covered on Friday. Find a Lemma in your linear algebra notes which relates the determinant and trace to the eigenvalues."	Section 12.b
6.3	semi-important	"By using $\alpha+\beta+\gamma=\pi$ for the angles, you can express this as an optimization problem in two variables."	not transcribed
6.4	unmarked	—	not transcribed
6.5	optional	"See 'data analysis'."	not transcribed
6.6	optional	"Do two or three, not all of them."	not transcribed
6.7	important	—	Section 14.a
6.8	important	—	Section 12.b
6.9	important	—	Section 12.b
6.10–6.12	important	"These are good exercises, but how many optimization problems do you really need?"	Section 13.a, Section 12.b

AI-Note D.2: Sheet 6 is incomplete. Only the problems tagged **important** (6.2 and 6.7–6.12) were ever transcribed, and 6.1 and 6.3–6.6 have not been. Everything that **is** transcribed is solved. This is a gap in the source material, not a casualty of the restructure.

6.1 is the exception, and it is deliberate. It asks for two things about a convex $f \in C^2(\mathbb{R}^n)$: that a critical point is exactly a global minimiser, and for a non-negative example whose infimum is not attained. Both are already in the document: Exercise 14.a.6**(c)** and Example 14.a.10, the latter with precisely the required $f(x) = e^{x_1+x_2}$. Transcribing 6.1 would print the same two questions a second time, so the row above stays *"not transcribed"* while the mathematics is covered.

Sheet 7 (Section D.e)

Problem	Priority	Note	Now in
7.1	important	"First think of a diffeomorphism $(a,b) \leftrightarrow \mathbb{R}$ and adjust it such that $(a,b)=(0,1)$."	Section 15.a
7.2–7.4	unmarked	—	not transcribed
7.5	important	"This is a great exercise to help with understanding of IFT."	Section 16.a
7.6	important	"You will need these coordinates very often over the next weeks."	Section 17.c
7.7	semi-important	"In **1)**, only the Möbius strip should be enough."	not transcribed
7.8	semi-important	"Do **1)** and **2)**. The others will be in the lecture."	not transcribed
7.9	unmarked	—	not transcribed

AI-Note D.3: Like sheet 6, sheet 7 is only partly transcribed. The three tagged **important**, namely 7.1, 7.5 and 7.6, are in the document and solved. Problems 7.2–7.4 and 7.7–7.9 were never transcribed.

Sheet 8 (Section D.f)

Problem	Priority	Now in
8.1	—	Section 19.a
8.2	important	Section 19.a
8.3	—	not transcribed
8.4	important	Section 19.a
8.5	important	Section 20.a
8.6–8.8	—	not transcribed

Sheet 9 (Section D.g)

Only **9.1** and **9.6** carry an explicit colour marker on the priority page; **9.2** is annotated with a hint but no clearly distinguishable colour, so it is listed as unmarked below even though a hint is quoted for it. **9.3–9.5** and **9.7** carry neither colour nor hint.

Problem	Priority	Note	Now in
9.1	important	(hint on coordinates, quoted with the problem)	Section 21.b
9.2	unmarked	(hint on the domain of part 2, quoted with the problem)	Section 21.c
9.3–9.5	unmarked	—	not transcribed
9.6	semi-important	"This may become important in later courses. $\langle \cdot, \cdot \rangle$ is the standard inner prod."	not transcribed
9.7	unmarked	—	not transcribed

AI-Note D.4: Only 9.1 is tagged important, but its companion 9.2 carries an explicit worked hint (unlike 9.3–9.5 and 9.7, which carry nothing at all), so both are reproduced and solved. 9.3–9.7 were never transcribed.

Sheet 10 (Section D.h)

Problem	Priority	Note	Now in
10.1	important	"Use spherical coordinates with $r \equiv 1$."	Section 21.b
10.2	important	(worked parametrization of the torus, quoted with the problem)	Section 21.b
10.3	important	"1.)–4.), 5.)"	Section 21.c
10.4–10.6	semi-important	—	not transcribed

AI-Note D.5: For **10.3**, the note *"1.)–4.), 5.)"* appears to flag that the starred sub-part **5.** (the harder *"if and only if"* characterization of a bounded C^1 domain, marked (*) on the official sheet) is worth attempting alongside the more routine parts 1–4, not skipped as the (*) marker might suggest. 10.4–10.6 (tractrix, logarithmic spiral, moment of inertia of an ellipse) were never transcribed.

Sheet 11 (Section D.i)

Problem	Priority	Now in
11.1	important	Section 22.b
11.2–11.4	semi-important	not transcribed
11.5	important	Section 23.b
11.6–11.9	optional	not transcribed

AI-Note D.6: Problems 11.2–11.4 and 11.6–11.9 were never transcribed, consistent with the standing decision to follow the priority marking. Between them they cover improper-integral classification, a standard flux computation, a cylinder's fluxes and volumes, the heat kernel, a divergence-theorem reality check, a multiple-choice question on Lipschitz continuity, and differential-form computations.

Sheet 12 (Section D.j)

Problem	Priority	Now in
12.1	semi-important	Section 26.b
12.2	optional	not transcribed
12.3	important	Section 26.e
12.4	important	Section 26.e
12.5	important	Section 26.e
12.6–12.7	semi-important	not transcribed
12.8	important	Section 26.e
12.9	important	Section 25.b

AI-Note D.7: 12.2 (existence of a potential via a path integral, building the tool used implicitly in Exercise 26.e.46), 12.6 (a Stokes reality check structurally identical to Exercise 26.e.44 part **(c)**) and 12.7 (four standard curl/div product-rule identities) have not been transcribed. 12.1 is Exercise 26.b.4, placed beside Definition 26.b.3 rather than with the other sheet-12 problems, because it is a property of that definition and is used implicitly from there onwards.

Sheet 13 (Section D.k)

Problem	What it asks	Now in
13.1	An inhomogeneous second-order initial value problem, solved and its solution shown unique	not transcribed
13.2	Three constant-coefficient homogeneous equations, and what changes over $\mathbb{C}$	not transcribed
13.3	Three non-homogeneous problems, the last by the integrating-factor trick	not transcribed
13.4	Four equations put in normal form, tested against Cauchy–Lipschitz, then solved	Appendix A.d
13.5	Multiple choice: when the solution set is a vector space; bounded solutions of $\dot{x} = Ax$; convergence of the Picard iteration	not transcribed
13.6	True or false	not transcribed

AI-Note D.8: Sheet 13 is the course's ODE sheet, and its material belongs to Appendix A rather than to any chapter of the main text.

Only 13.4 is transcribed, as Exercise A.d.30. It was chosen because it is the one problem on the sheet that is not a computation: it asks in each case **whether** the existence and uniqueness theorem applies before asking for a solution, which is the question Appendix A.d exists to answer. 13.1–13.3 are solution techniques that Appendix A.b and Appendix A.c already cover, and 13.5 asks about the long-time behaviour of $\dot{x} = Ax$, which this document does not develop, and about convergence of the Picard iteration, which Appendix A.d.1 treats only briefly.